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In USA and Canada Contact

SAI GLOBAL, 20 Carlson Court, Etobicoke, ON

M9W 7K6, Canada

+1 416 401 8730. Email: i2iSupport-US@saiglobal.com

In Europe Contact

SAI GLOBAL, Heron House, Davy Avenue, Knowlhill, Milton Keynes, MK5 8HJ, UK

+44 203 327 3140. Email: i2iSupport-EMEA@saiglobal.com

In Asia/Pacific Contact

SAI Global Ltd, Level 37, 680 George Street, Sydney, NSW 2000, Australia

+61 131 242. Email: i2iSupport-APAC@saiglobal.com

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UNE-EN IEC 61970-600-2:2021

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Interfaz de programación de aplicación para sistemas de gestión de energía (EMS-API). Parte 600-2: Especificación de intercambio de modelo de red común (CGMES). Especificación de perfiles de intercambio (Ratificada por la Asociación Española de Normalización en septiembre de 2021.)

Asociación Española
de Normalización
Génova, 6 - 28004 Madrid
915 294 900
info@une.org
www.une.org

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Energy management system application program interface (EMS-API) - Part 600-2: Common Grid Model Exchange Specification (CGMES) - Exchange profiles specification (Endorsed by Asociación Española de Normalización in September of 2021.)

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NORME EUROPÉENNE

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Interface de programmation d'application pour système de gestion d'énergie (EMS-API) - Partie 600-1: Spécification pour l'échange de modèle de réseau commun (CGMES) - Spécification de profils d'échange (IEC 61970-600-2:2021)

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IEC 61970-501:2006 NOTE Harmonized as EN 61970-501:2006 (not modified)

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Annex ZA

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Normative references to international publications with their corresponding European publications

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 Where an International Publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

NOTE 2 Up-to-date information on the latest versions of the European Standards listed in this annex is available here: www.cenelec.eu.

Publication	Year	Title	EN/HD	Year
IEC 61968-11	-	Application integration at electric utilities - System interfaces for distribution management - Part 11: Common information model (CIM) extensions for distribution	EN 61968-11	-
IEC 61968-13	-	Application integration at electric utilities - System interfaces for distribution management - Part 13: Common distribution power system model profiles	EN IEC 61968-13	-
IEC 61970-301	2020	Energy management system application program interface (EMS-API) - Part 301: Common information model (CIM) base	EN IEC 61970-301	2020
IEC 61970-302	2018	Energy management system application program interface (EMS-API) - Part 302: Common information model (CIM) dynamics	EN IEC 61970-302	2018
IEC 61970-452	2017	Energy management system application program interface (EMS-API) - Part 452: CIM static transmission network model profiles	EN 61970-452	2017
IEC 61970-453	2014	Energy management system application program interface (EMS-API) - Part 453: Diagram layout profile	EN 61970-453	2014
IEC 61970-456	2018	Energy management system application program interface (EMS-API) - Part 456: Solved power system state profiles	EN IEC 61970-456	2018
IEC 61970-600–1	2021	Energy management system application program interface (EMS-API) - Part 600–1: Common Grid Model Exchange Standard (CGMES) - Structure and rules	EN IEC 61970-600–1	2021

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IEC Central Office
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CH-1211 Geneva 20
Switzerland

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

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ENERGY MANAGEMENT SYSTEM APPLICATION PROGRAM INTERFACE (EMS-API) –

Part 600-2: Common Grid Model Exchange Standard (CGMES) – Exchange profiles specification

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International Standard IEC 61970-600-2 has been prepared by IEC technical committee 57: Power systems management and associated information exchange.

This first edition cancels and replaces IEC TS 61970-600-2 published in 2017. This edition constitutes a technical revision.

The text of this international standard is based on the following documents:

FDIS	Report on voting
57/2361/FDIS	57/2373/RVD

Full information on the voting for the approval of this international standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

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The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
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ENERGY MANAGEMENT SYSTEM APPLICATION PROGRAM INTERFACE (EMS-API) –

Part 600-2: Common Grid Model Exchange Standard (CGMES) – Exchange profiles specification

1 Scope

This part of IEC 61970-600 defines the profiles included in the Common Grid Model Exchange Standard (CGMES) that are based on IEC 61970-450-series and IEC 61968-13 profiles. This document refers to the IEC 61970-450-series and IEC 61968-13 profiles only in cases where they are identical. If the referenced profile is not yet published, this document includes the profile definition and related constraints' definitions. In the case where a CGMES profile makes restriction on the referenced profile, the restriction is defined in this document.

The equipment boundary profile (EQBD) is the only profile that is not part of IEC 61970-450-series and IEC 61968-13 profiles. This profile is deprecated as modifications have been made to align between EQBP and the equipment profile (EQ). Although the updated EQBD is addressing the requirement that boundary also can be located inside a substation, which will be the case for many Distribution System Operators (DSOs), additional information would need to be exchanged. For instance, system integrity protection schemes, that can be shared by multiple utility would require another way of boundary handling. In this document EQBD is included in CGMES only to create better backwards compatibility with previous version of the CGMES.

The machine-readable documentation that supports model driven development of the profiles defined in this part are generated as Resource Description Framework Schema (RDFS) according to IEC 61970-501:2006 (with some extension) and IEC 61970-501:ED2 when published.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61968-11, *Application integration at electric utilities – System interfaces for distribution management – Part 11: Common information model (CIM) extensions for distribution*

IEC 61968-13, *Application integration at electric utilities – System interfaces for distribution management – Part 13: CIM RDF Model exchange format for distribution*

IEC 61970-301:2020, *Energy management system application program interface (EMS-API) – Part 301: Common information model (CIM) base*

IEC 61970-302:2018, *Energy management system application program interface (EMS-API) – Part 302: Common information model (CIM) dynamics*

IEC 61970-452:2017, *Energy management system application program interface (EMS-API) – Part 452: CIM static transmission network model profiles*

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IEC 61970-453:2014, *Energy management system application program interface (EMS-API) – Part 453: Diagram layout profile*

IEC 61970-456:2018, *Energy management system application program interface (EMS-API) – Part 456: Solved power system state profiles*

IEC 61970-600-1:2020, *Energy management system application program interface (EMS-API) – Part 600-1: Common Grid Model Exchange Standard (CGMES) – Structure and rules*

3 Terms, definitions and abbreviated terms

For the purposes of this document, the following terms, definitions and abbreviated terms apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

NOTE For definitions which are not specified in the CGMES the definitions in the IEC CIM 61970 standards shall be applied.

3.1 Terms and definitions

3.1.1
boundary point
BP
connection point between two Model Authority Sets, that has been agreed on by both relevant Model Authority

[SOURCE: IEC 61970-600-1]

3.1.2
boundary set
set containing all boundary points necessary for a merge model

[SOURCE: IEC 61970-600-1]

3.1.3
distribution
specific representation of a dataset

Note 1 to entry: A dataset might be available in multiple serializations that may differ in various ways, including natural language, media-type or format, schematic organization, temporal and spatial resolution. The level of detail in the distribution is defined by one or more profiles to which the dataset conforms.

[SOURCE: IEC 61970-600-1]

3.1.4
Common Grid Model Exchange Standard
CGMES
collection of standards defined in IEC 61970-600 series that support the exchange of power system models (e.g. individual grid model or common grid model) between model authorities (TSOs, DSOs, etc.) for the purpose of coordinated set of services to be performed on the same model according to legislation or general data exchanges in the frame of system operation, system development or utilities' projects

[SOURCE: IEC 61970-600-1]

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3.1.5
European Extensions
collection of classes, attributes and associations, which either extend or are defined in the standard IEC CIM model (IEC 61970-300 series, IEC 91968-11 and IEC 62325-300 series)

Note 1 to entry: The European extensions aim at satisfying requirements by the European legislation hence not necessarily applicable to other continents. The worldwide adoption of these extensions may not be exactly the same as the defined extension.

[SOURCE: IEC 61970-600-1]

3.1.6
profile
data model to describe instance file for exchange of CIM data

Note 1 to entry: A profile is a subset of classes, associations and attributes needed to accomplish a specific type of interface and based upon a CIM data model. Profiles may impose stricter rules on original classes and associations. A profile is usually converted to schema (XSD, RDF, OWL, etc.) that can be used to create, read and validate instance files for data exchange.

Note 2 to entry: This term may be used to define either the semantic model for an instance data payload or the syntactic schema for an instance data payload. A profile may be expressed in XSD, RDF, and/or OWL files. An instance data conforming to a profile can be tested in exchanges between applications. A profile is necessary in order to “use” the canonical model.

[SOURCE: IEC 61970-600-1]

3.2 Abbreviated terms

BP	Boundary point
CIM	Common Information Model (electricity)
CGMES	Common Grid Model Exchange Standard
DL	Diagram Layout profile
DSO	Distribution System Operator
DY	Dynamics profile
ENTSO-E	European Network of Transmission System Operators for Electricity
EQ	Core Equipment profile
EQBD	Equipment boundary profile
GL	Geographical Location profile
HVDC	High Voltage Direct Current
IEC	The International Electrotechnical Commission
IOP	Interoperability Test
ISO	Independent System Operator
MAS	Model Authority Set
mRID	CIM Master Resource Identifier
OCL	Object Constraint Language
OP	Operation profile
OWL	Web Ontology Language
RCC	Regional Coordination Centres
RDF	Resource Description Framework
RDFS	RDF Schema
SC	Short-Circuit profile
SHACL	Shapes Constraint Language

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SSH	Steady State Hypothesis profile
SV	State Variables profile
TP	Topology profile
TSO	Transmission System Operator
URI	Uniform Resource Identifier
UUID	Universally Unique Identifier
XML	Extensible Markup Language
XSD	XML Schema Definition

4 Relation with other CIM standards

The CGMES profiles are relying on the specification given in the relevant IEC 61970-450 series and IEC 61970-13 profiles. The CGMES profiles can include extensions of the vocabulary (add classes, attributes and/or associations) or provide additional constrains. The goal is that a valid instance dataset based on a CGMES profile will also be a valid instance dataset for of the relevant IEC 61970-450 series and IEC 61970-13 profile. When possible CGMES will include the relevant IEC 61970-450 series and IEC 61970-13 profile as is, without making any extension or additional constraints.

In this part edition all profiles are included as CGMES European (EU) profile as shown in Figure 1.

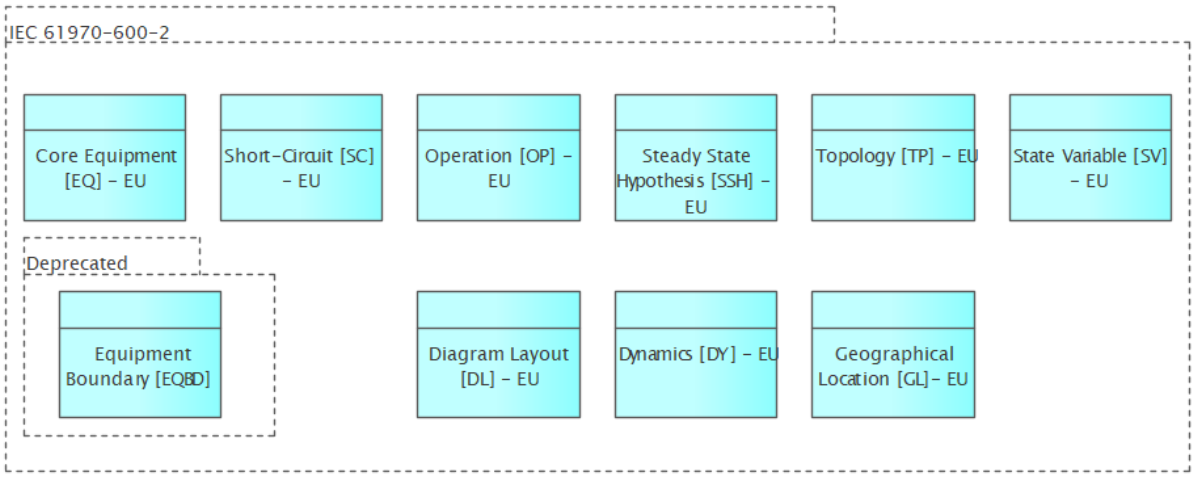

Figure 1 – CGMES profiles

The content of the profiles is defined in Clauses 6 to 15. This includes also constrains that are part of the relevant standards. Any CGMES profile constrain is also included in the Clause 16. However, the Clauses do not include any guidance and deeper explanation this is expected to be provided in the relevant standard document.

Therefore, this part is relying on this guidance and requirements. When implementing application support for CGMES it shall be taken into consideration the guidance that these standards are providing. Regarding CGMES conformity this part will supersede the relevant standard for the items that are explicitly stated.

For instance, IEC 61970-452 defines Core equipment profile, Operation profile and Short circuit profile. This standard has clauses on “General requirements”, “Transformer modelling”, “Voltage or active power regulation”, etc. which applications conforming to CGMES shall comply with.

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The remainder of this clause is informative.

To achieve the goal of making CGMES profiles an extension to existing profile when possible it is expected that upcoming edition of this part will only refer to the relevant IEC 61970-450 series profile. The expectation for the next edition is illustrated in Figure 2.

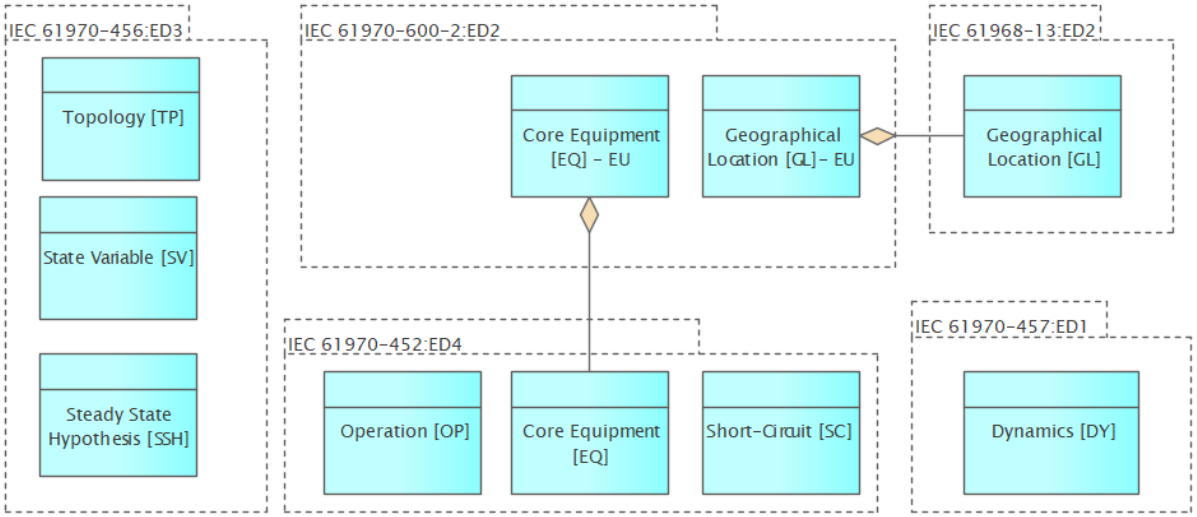

Figure 2 – Expected content of next edition

4.1 Deprecation

A deprecated item is retained in the present edition, but it is expected to be removed from future editions. An item that has been deprecated is not guaranteed to be retained in future editions, but using an item marked as deprecated should be avoided, if possible.

5 Version information for the CGMES profiles

5.1 Core equipment profile

Title: Core Equipment Vocabulary

Keyword: EQ

Description: This vocabulary is describing the core equipment profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/CoreEquipment-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/EquipmentCore/3/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:9fd00587-84d4-4910-bbd4-e003ed7e0821

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5.2 Operation profile

Title: Operation Vocabulary

Keyword: OP

Description: This vocabulary is describing the operation profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/Operation-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/EquipmentOperation/3/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:b603c770-6469-45da-a569-551394116c97

5.3 Short-circuit profile

Title: Short Circuit Vocabulary

Keyword: SC

Description: This vocabulary is describing the short circuit profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/ShortCircuit-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/EquipmentShortCircuit/3/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:97938d45-bb5f-4424-b64d-219c1f76ab31

5.4 Steady state hypothesis profile

Title: Steady State Hypothesis Vocabulary

Keyword: SSH

Description: This vocabulary is describing the steady state hypothesis profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/SteadyStateHypothesis-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/SteadyStateHypothesis/1/1

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Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:5aabb9bd-8fe3-4f3c-a0fa-f6902e767ce5

5.5 Topology profile

Title: Topology Vocabulary

Keyword: TP

Description: This vocabulary is describing the topology profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/Topology-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/Topology/4/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:2c5559fa-12b4-445a-b92a-958c90628cee

5.6 State variables profile

Title: State Variables Vocabulary

Keyword: SV

Description: This vocabulary is describing the state variables profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/StateVariables-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/StateVariables/4/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:767279cf-b7f6-4693-a5a1-7c7c0ead64c4

5.7 Diagram layout profile

Title: Diagram Layout Vocabulary

Keyword: DL

Description: This vocabulary is describing the diagram layout profile from IEC 61970-453.

Version IRI: http://iec.ch/TC57/ns/CIM/DiagramLayout-EU/3.0

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Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/DiagramLayout/3/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:7b269643-3da3-4dd0-83e8-58d7f13ee3bc

5.8 Geographical location profile

Title: Geographical Location Vocabulary

Keyword: GL

Description: This vocabulary is describing the geographical location profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/GeographicalLocation-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/GeographicalLocation/2/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:862c1230-be19-4a01-bd6f-1a6c03d213ff

5.9 Dynamics profile

Title: Dynamics Vocabulary

Keyword: DY

Description: This vocabulary is describing the dynamics profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/Dynamics-EU/1.0

Version info: 1.0.0

Prior version:

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|urn:iso:std:iec:61970-302:ed-1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:ba96da07-eca3-4259-b96c-cf5b51baa082

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5.10 Equipment boundary profile

Title: Equipment Boundary Vocabulary

Keyword: EQBD

Description: This vocabulary is describing the equipment boundary profile from IEC 61970-600-2.

Version IRI: http://iec.ch/TC57/ns/CIM/EquipmentBoundary-EU/3.0

Version info: 3.0.0

Prior version: http://entsoe.eu/CIM/EquipmentBoundary/3/1

Conforms to: urn:iso:std:iec:61970-600-2:ed-1|urn:iso:std:iec:61970-301:ed-7:amd1|file://iec61970cim17v40_iec61968cim13v13a_iec62325cim03v17a.eap|urn:iso:std:iec:61970-401:draft:ed-1|urn:iso:std:iec:61970-501:draft:ed-2

Identifier: urn:uuid:1e79354a-ccc2-434b-bab3-3dc4a08df5ea

6 Package CoreEquipmentProfile

6.1 General

This is the IEC 61970-452 core equipment profile.

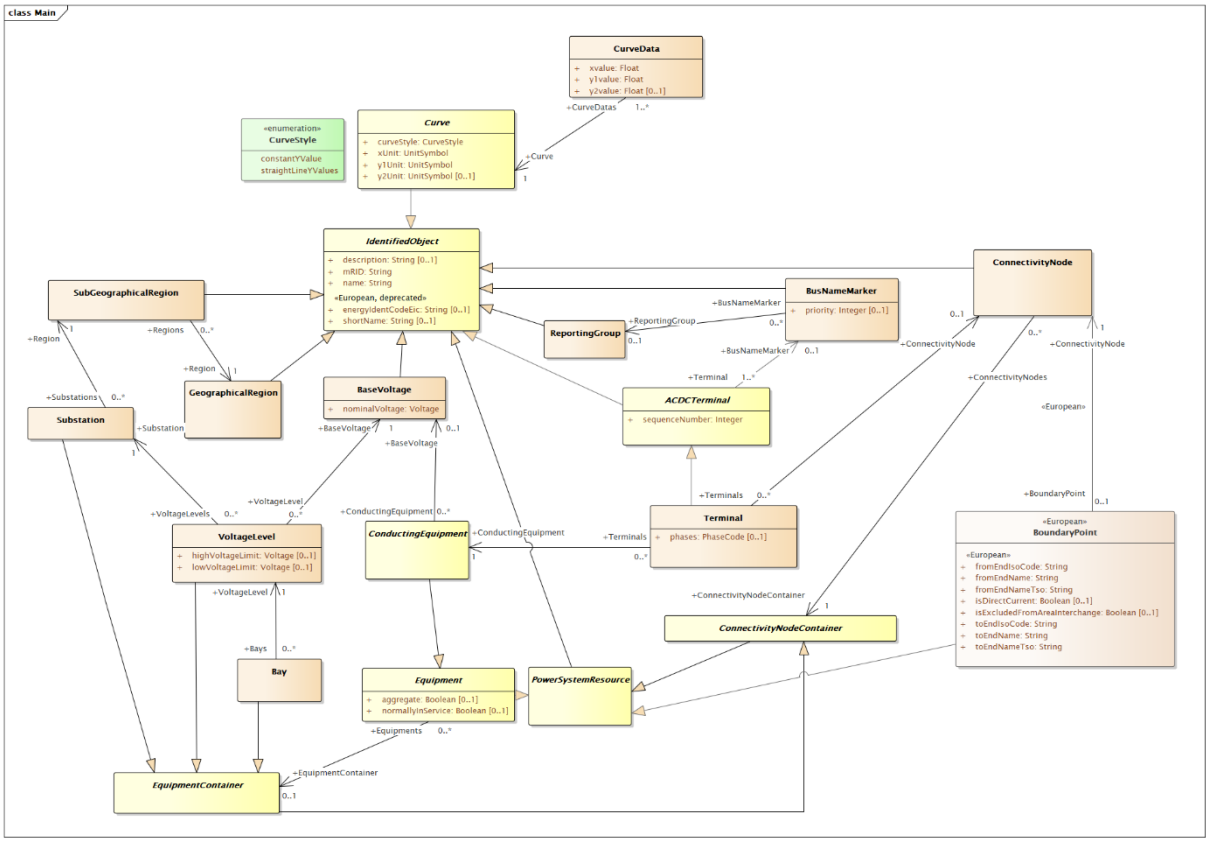

Figure 3 – Class diagram CoreEquipmentProfile::Main

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Figure 3: The main diagram of the equipment core profile.

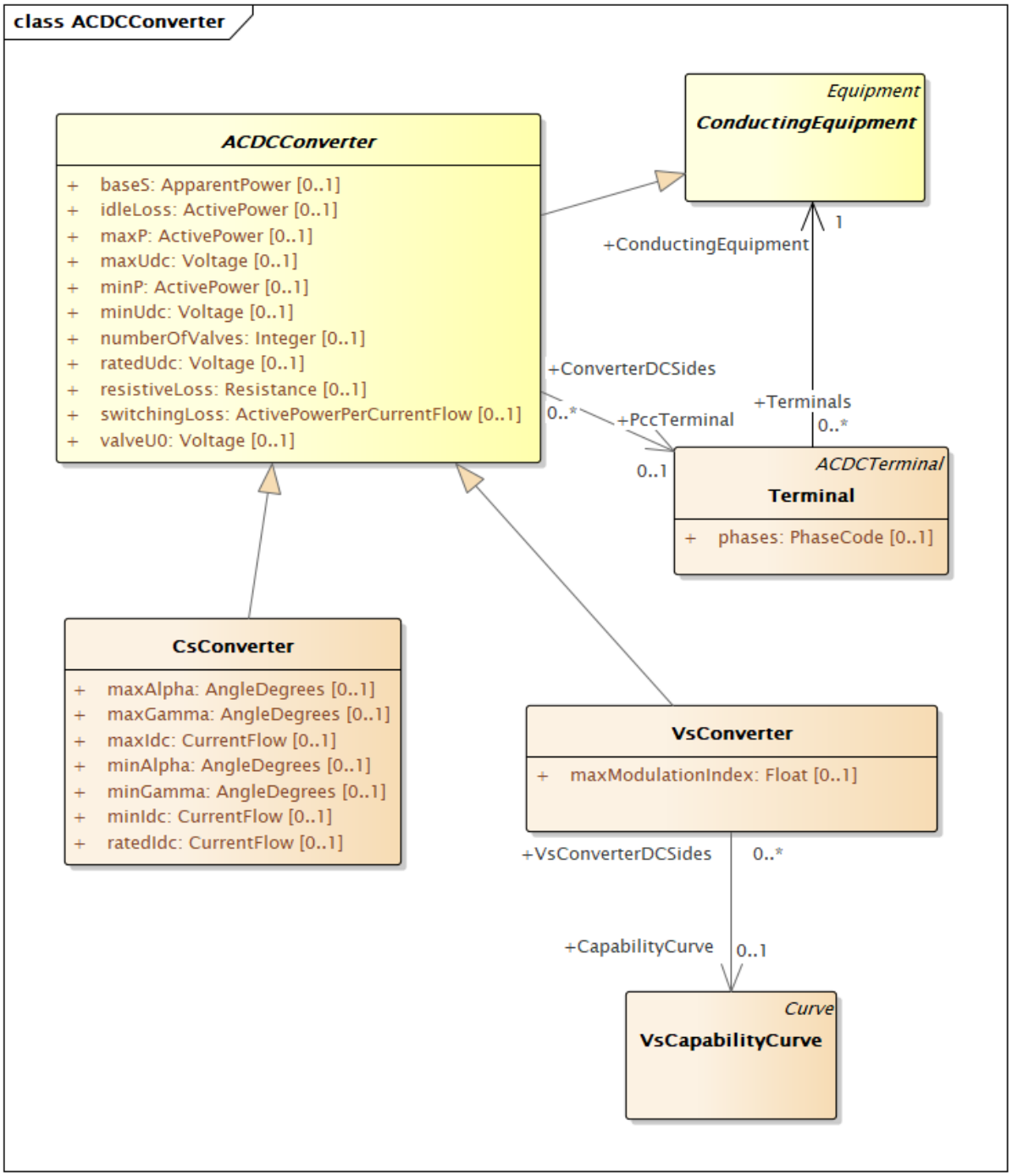

Figure 4 – Class diagram CoreEquipmentProfile::ACDCConverter

Figure 4: This diagram shows the classes used to model DC converters.

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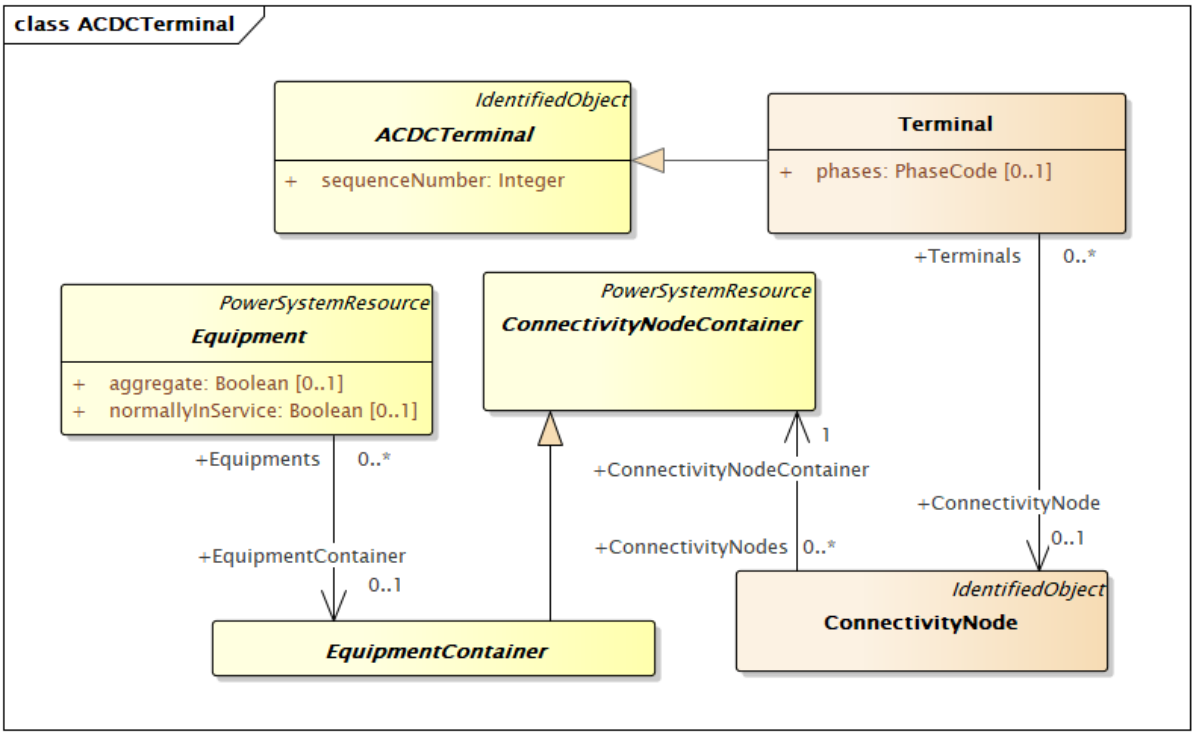

Figure 5 – Class diagram CoreEquipmentProfile::ACDCTerminal

Figure 5: The diagram shows the ACDCTerminal and related classes.

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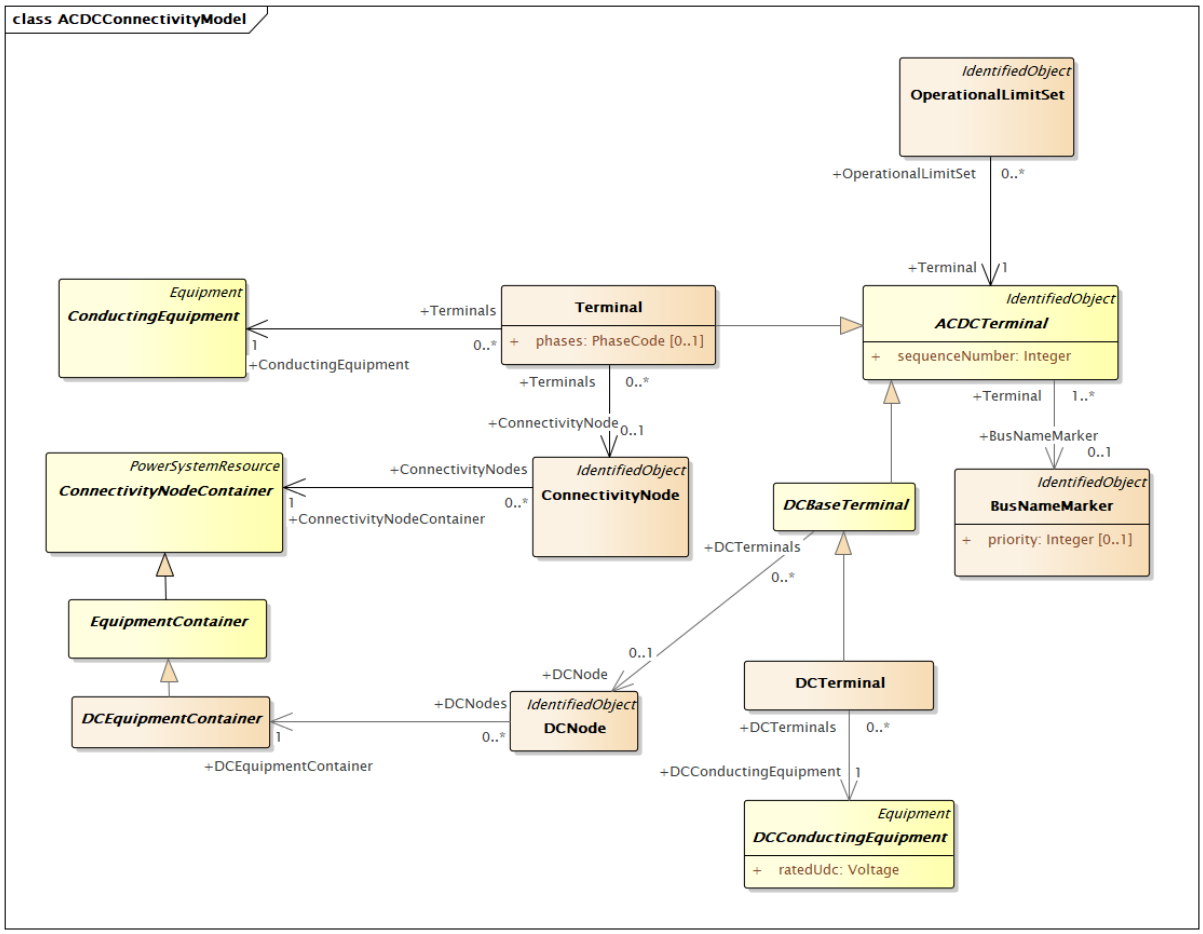

Figure 6 – Class diagram CoreEquipmentProfile::ACDCConnectivityModel

Figure 6: This diagram shows the connectivity and topology model for both AC and DC networks.

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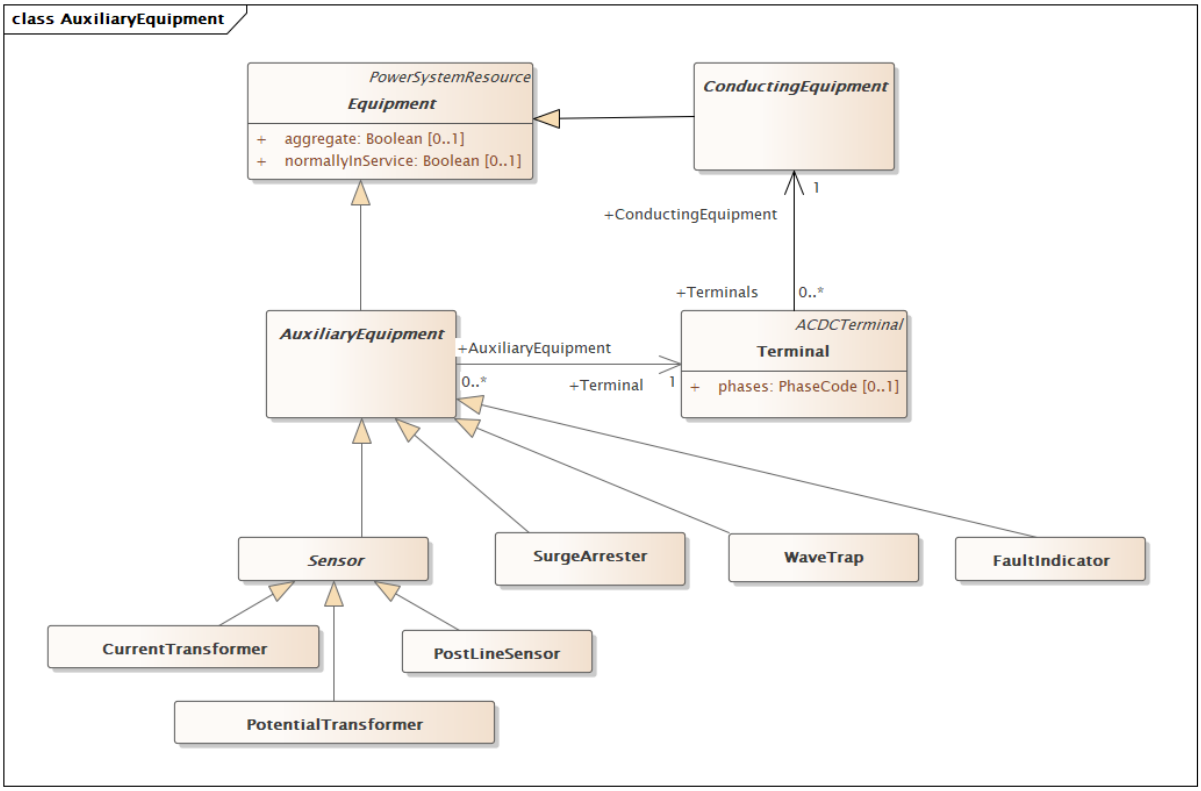

Figure 7 – Class diagram CoreEquipmentProfile::AuxiliaryEquipment

Figure 7: The diagram shows main classes related to auxiliary equipment.

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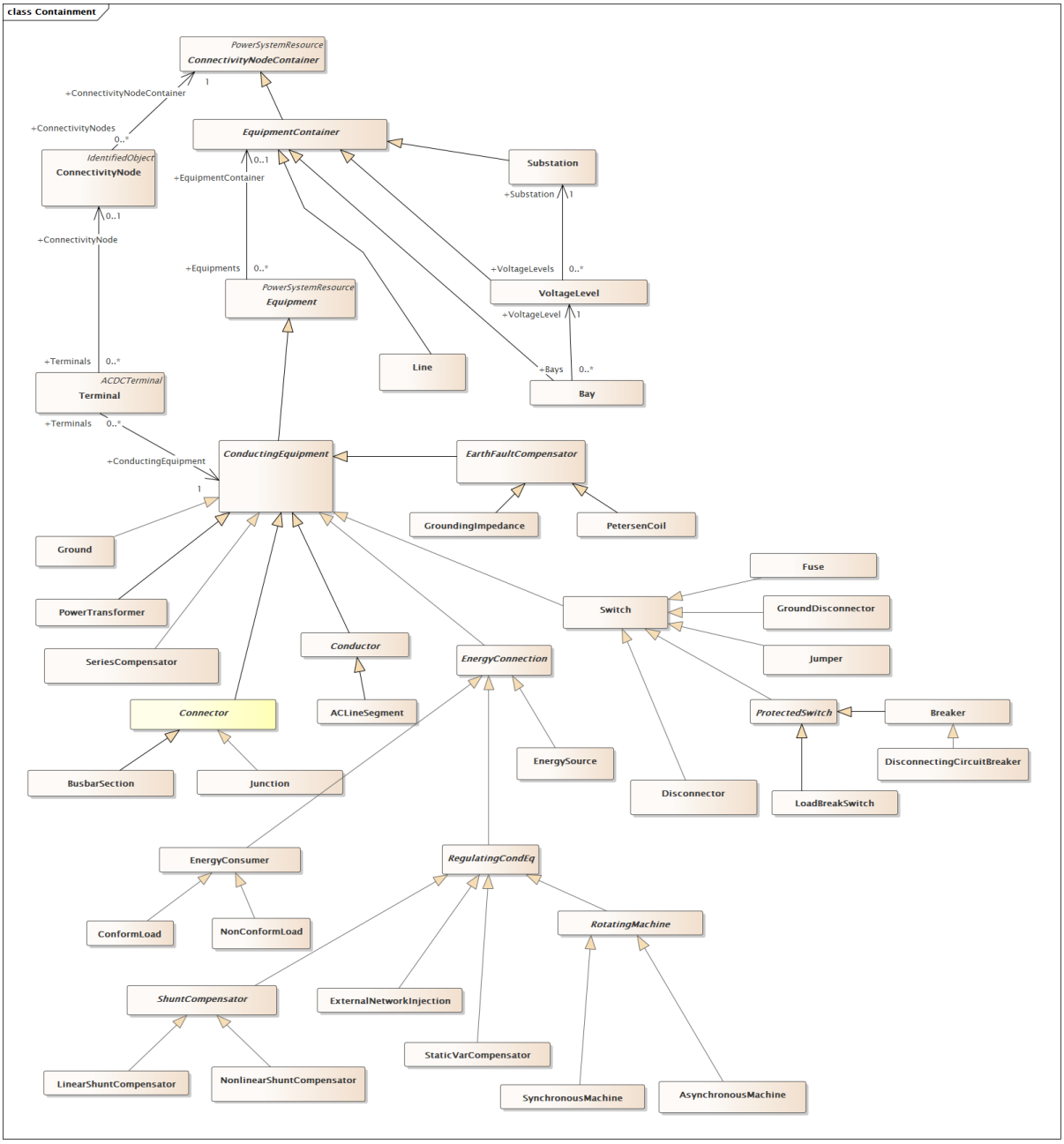

Figure 8 – Class diagram CoreEquipmentProfile::Containment

Figure 8: The diagram shows the main classes related to containment.

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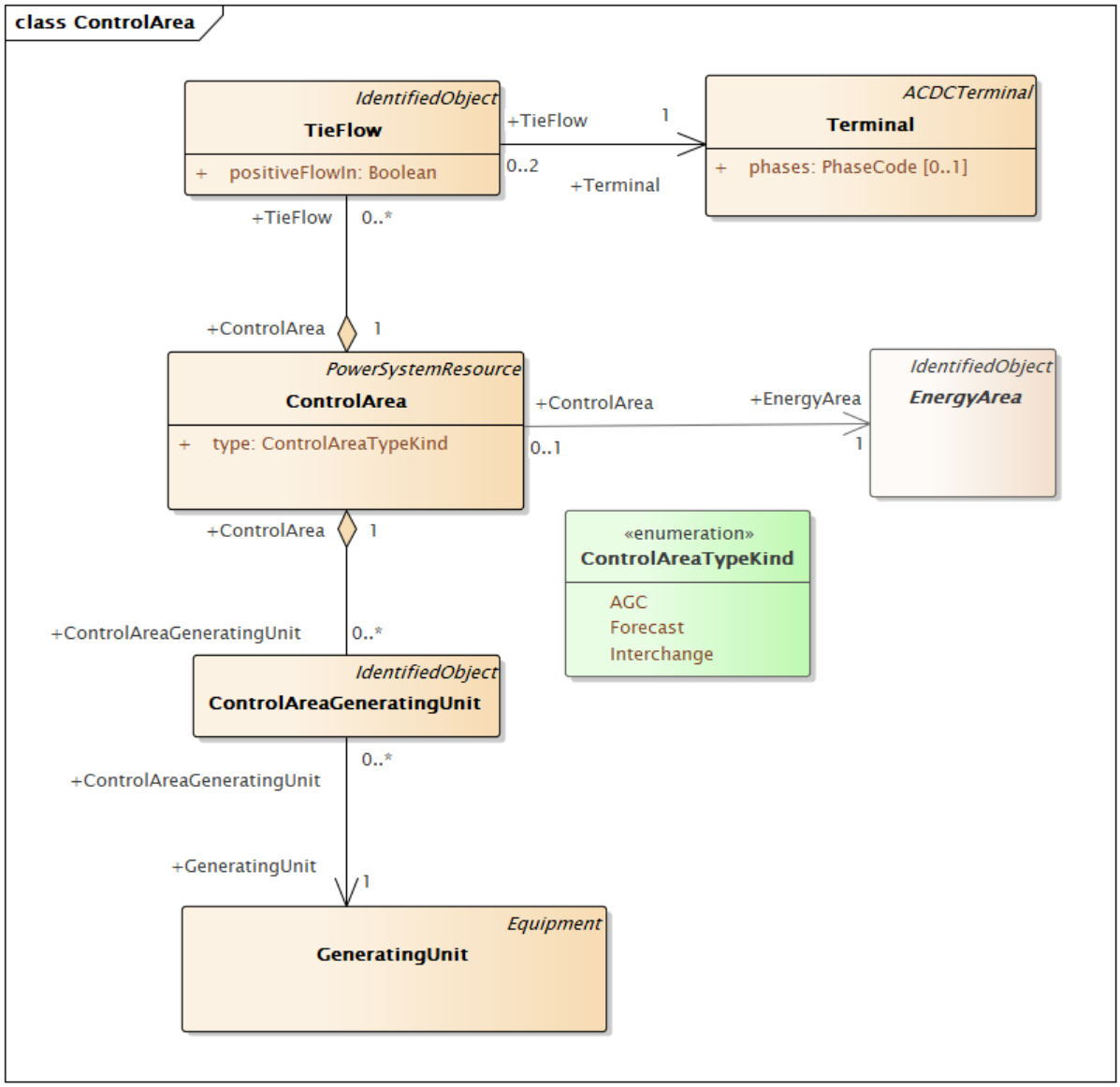

Figure 9 – Class diagram CoreEquipmentProfile::ControlArea

Figure 9: This diagram shows control area specification and some related classes. The Terminal to AnalogValue linkages are shown for clarity in understanding the control area specification. The GeneratingUnit to Terminal linkages are also shown to illustrate how generation flows are specifically tied to the network.

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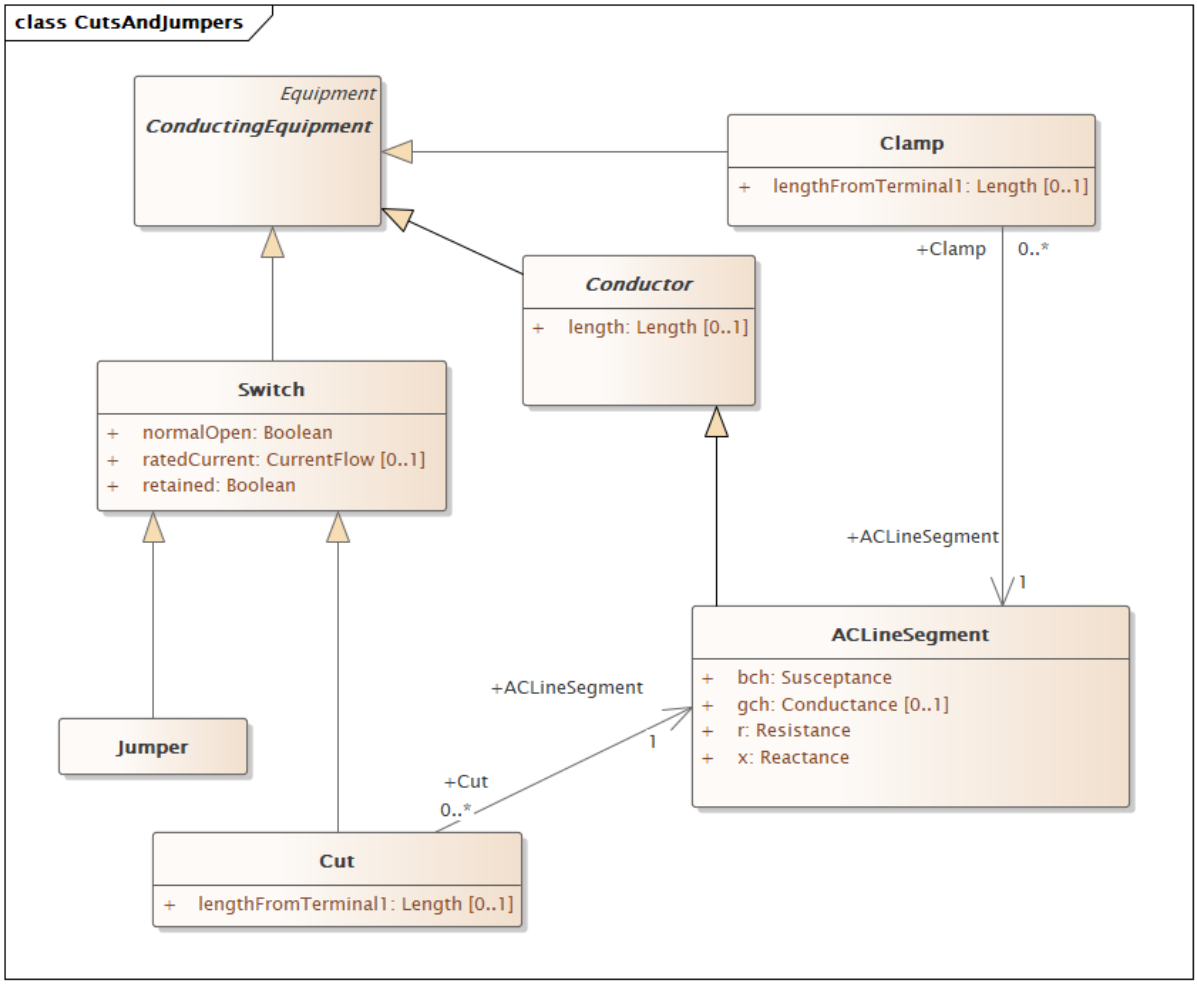

Figure 10 – Class diagram CoreEquipmentProfile::CutsAndJumpers

Figure 10: The diagram shows classes related to cuts and jumpers.

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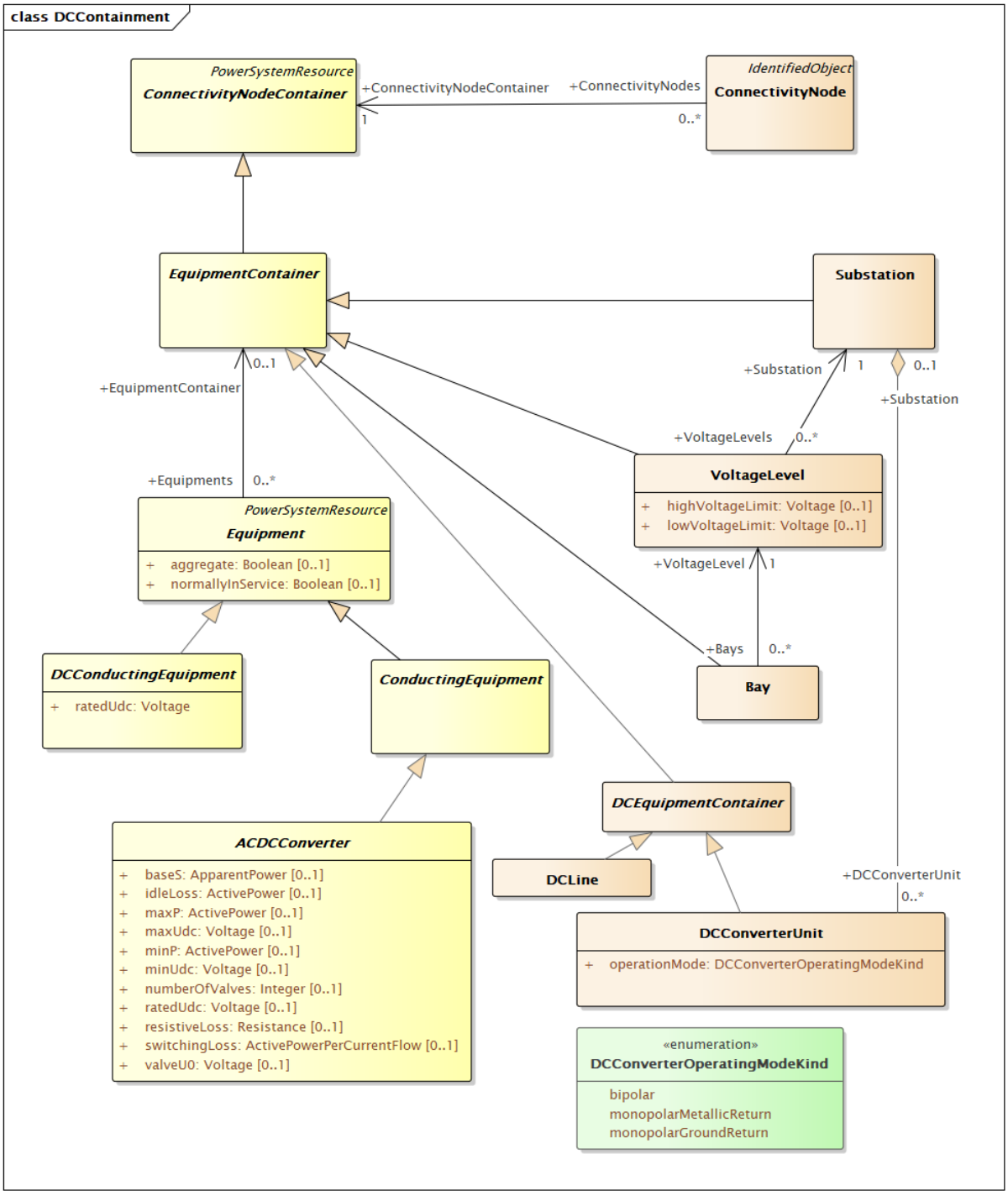

Figure 11 – Class diagram CoreEquipmentProfile::DCContainment

Figure 11: The diagram shows the DC containment model.

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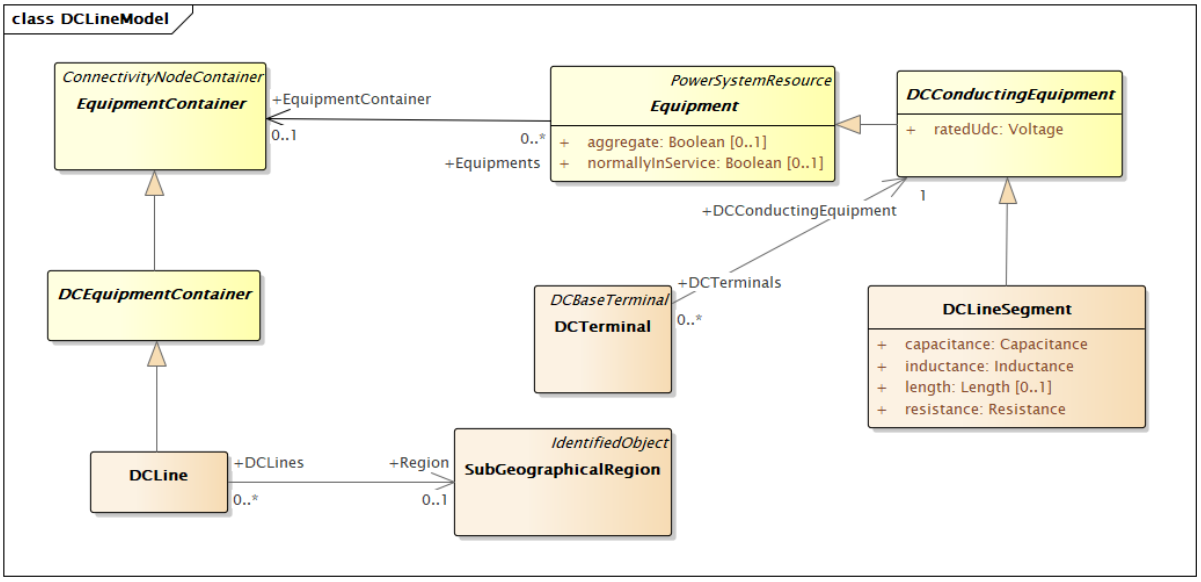

Figure 12 – Class diagram CoreEquipmentProfile::DCLineModel

Figure 12: The diagram shows the DC line model.

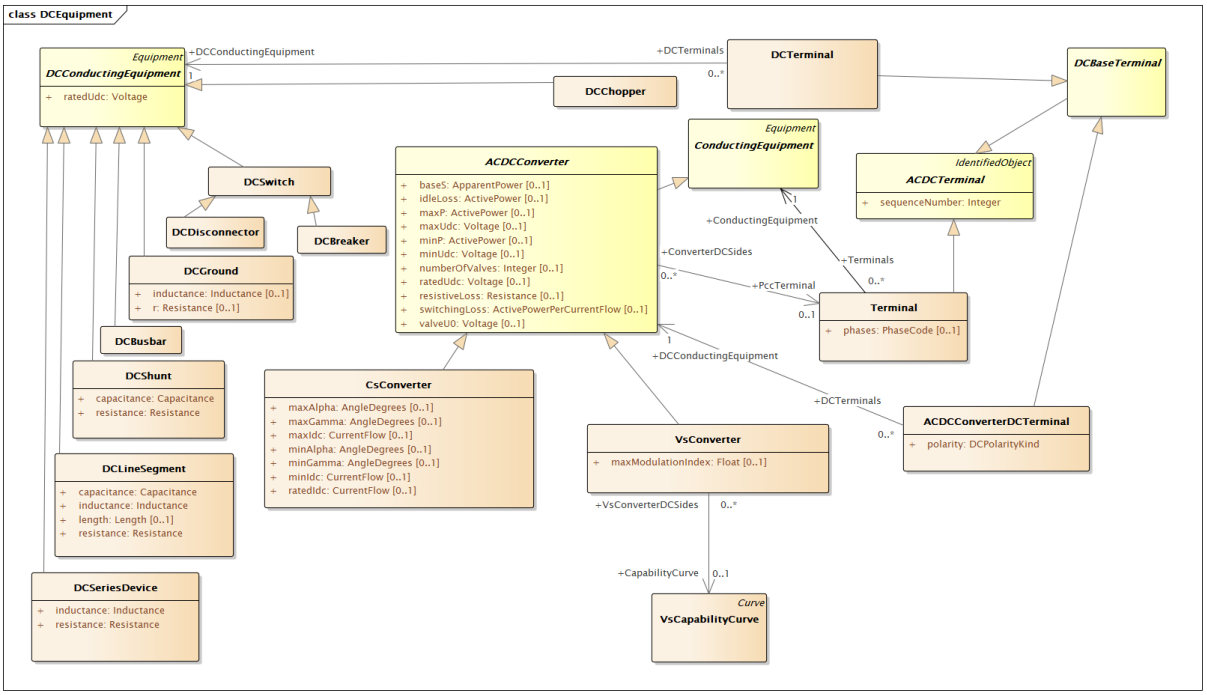

Figure 13 – Class diagram CoreEquipmentProfile::DCEquipment

Figure 13: The diagram shows the DC equipment model.

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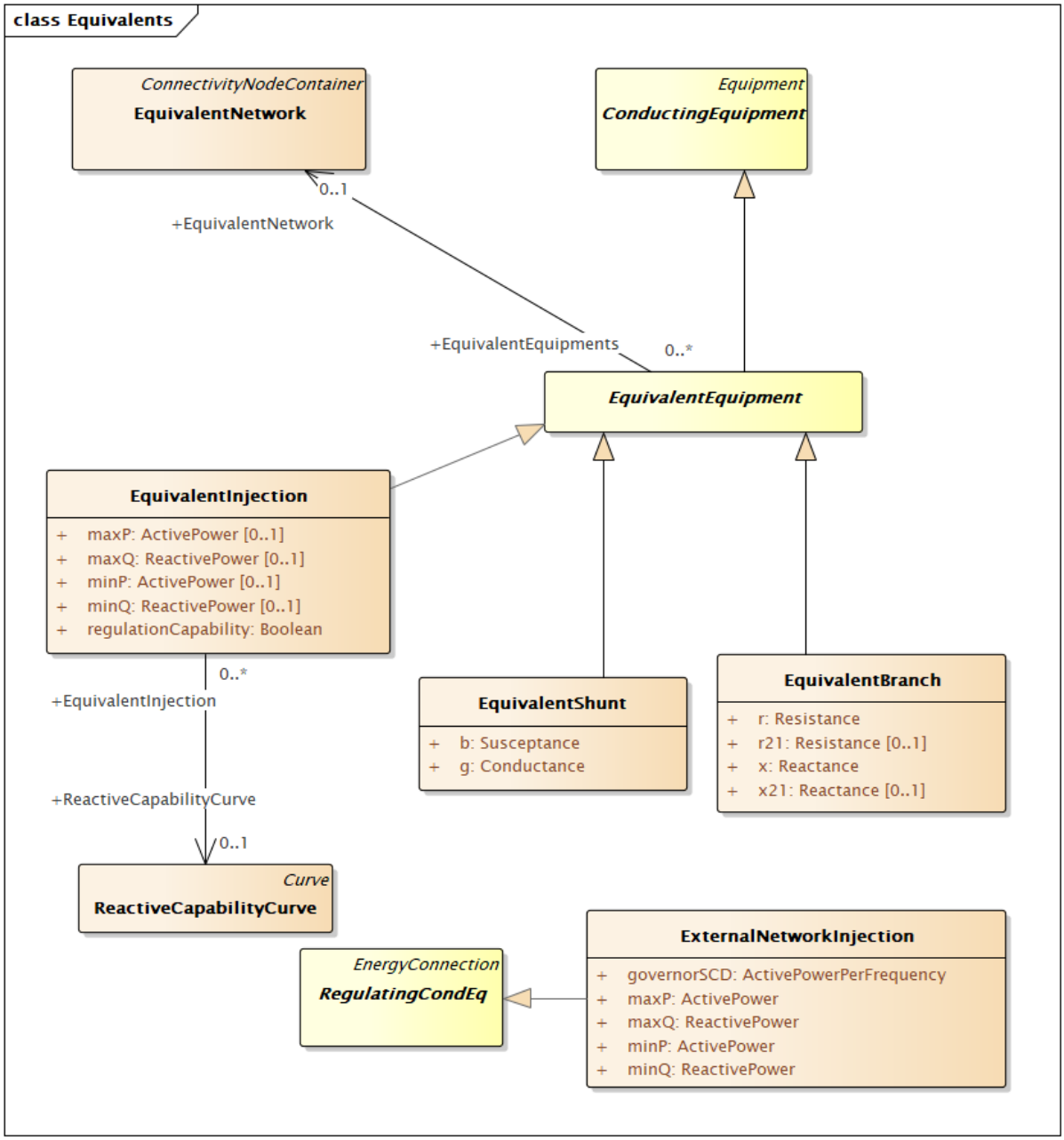

Figure 14 – Class diagram CoreEquipmentProfile::Equivalents

Figure 14: The diagram shows classes related to equivalents.

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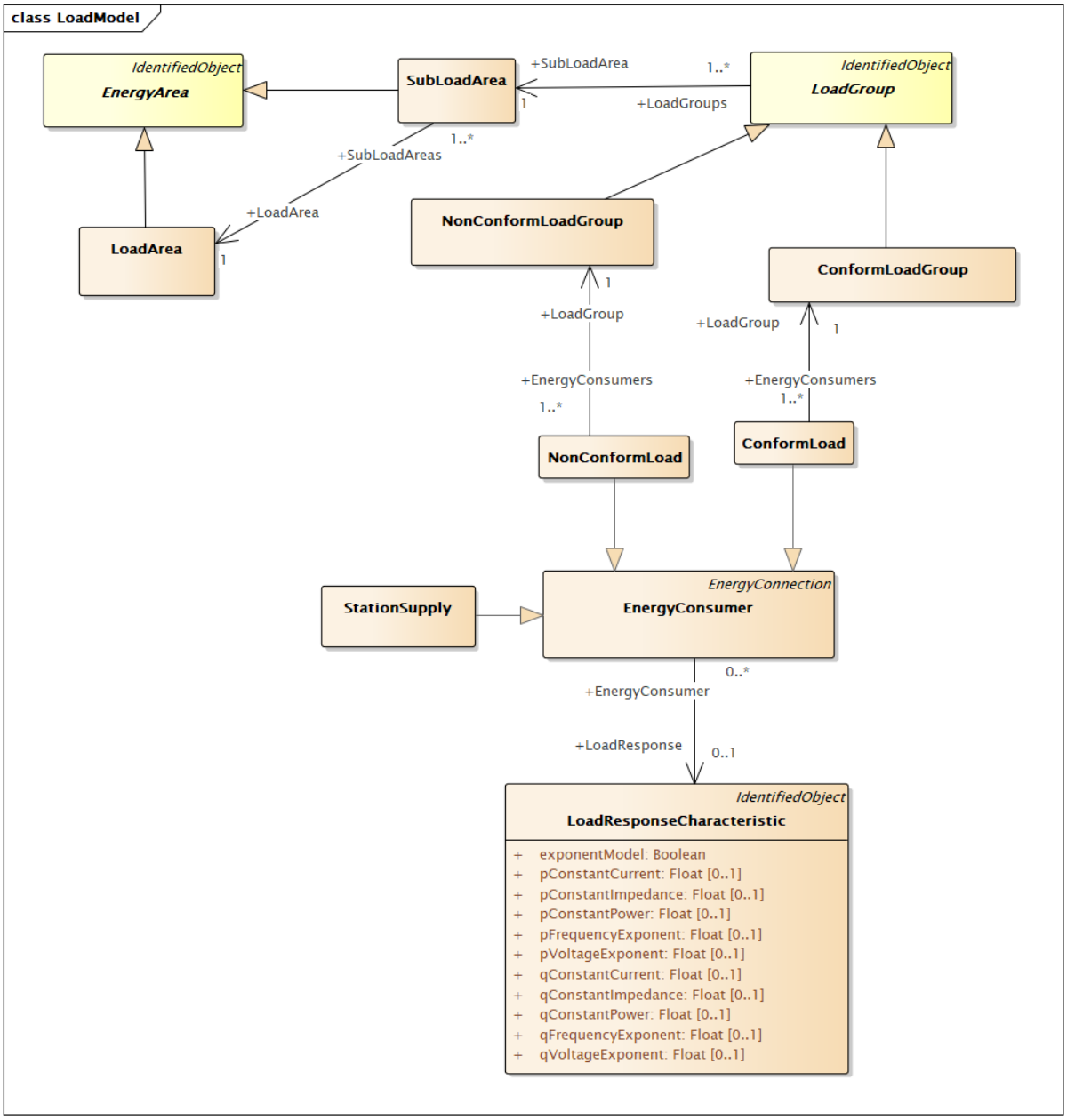

Figure 15 – Class diagram CoreEquipmentProfile::LoadModel

Figure 15: The diagram shows classes related to the load model.

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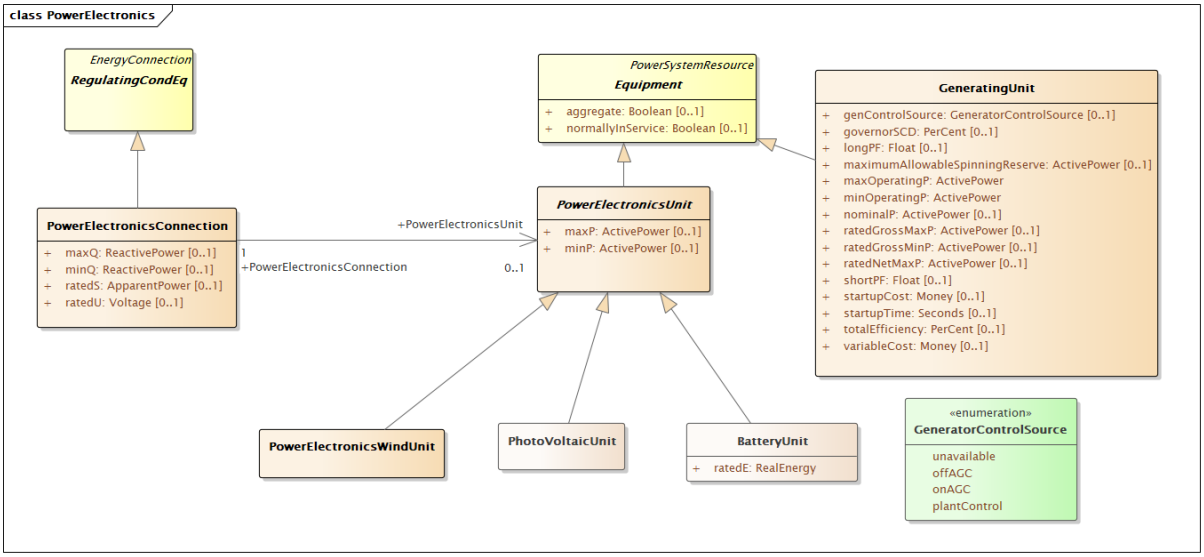

Figure 16 – Class diagram CoreEquipmentProfile::PowerElectronics

Figure 16: This diagram shows power electronics classes and inheritance.

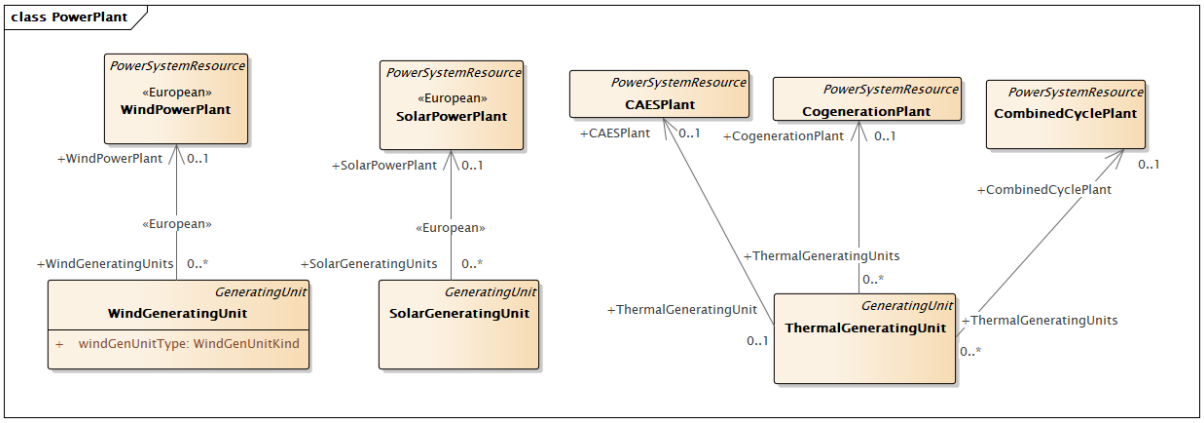

Figure 17 – Class diagram CoreEquipmentProfile::PowerPlant

Figure 17: The diagram shows the power plant model.

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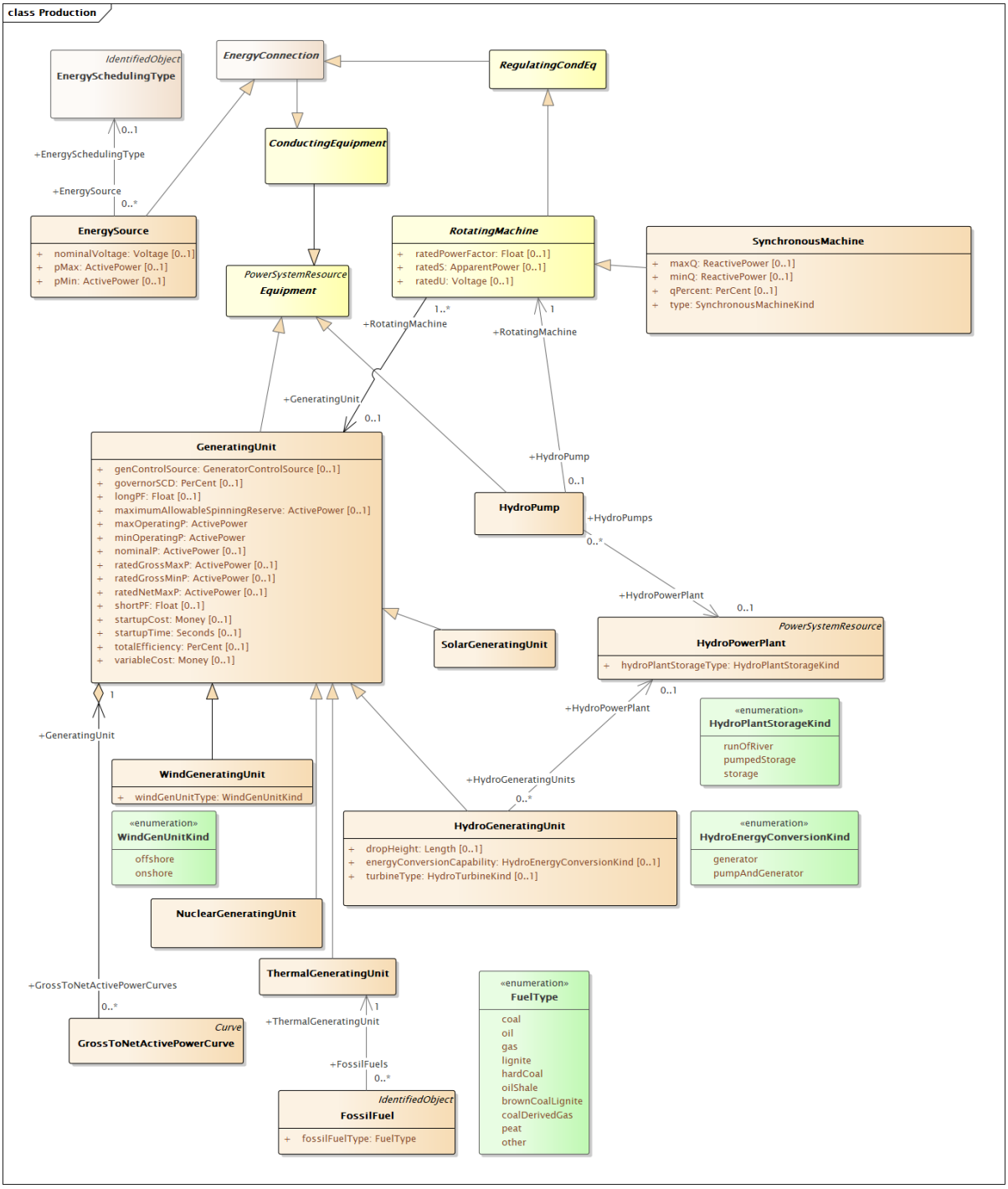

Figure 18 – Class diagram CoreEquipmentProfile::Production

Figure 18: The diagram shows production related classes.

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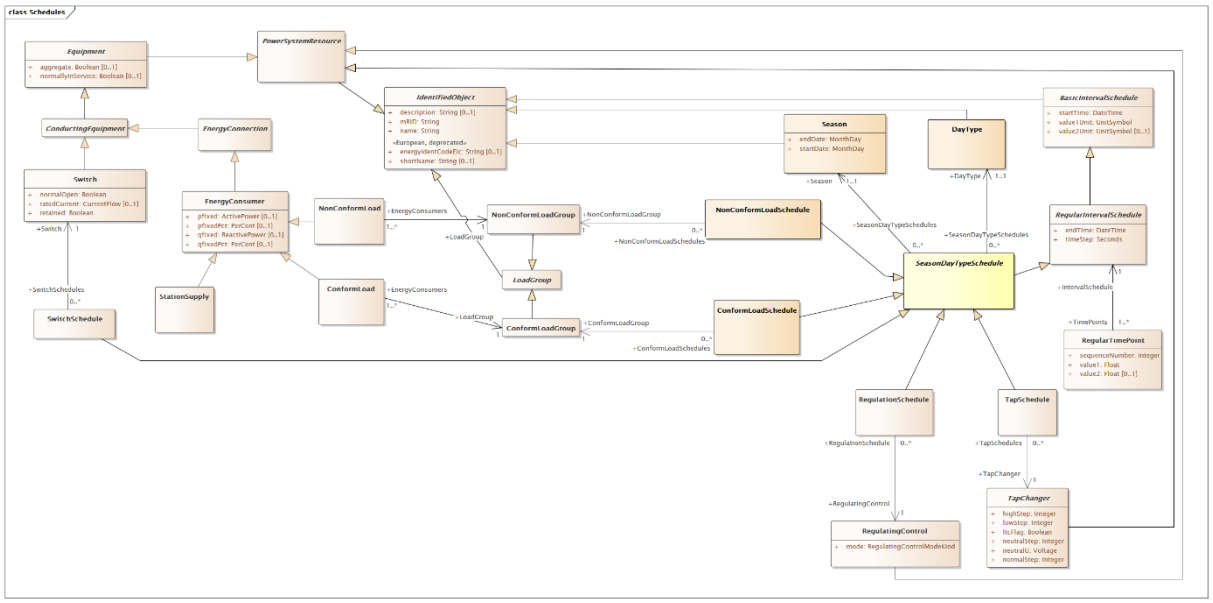

Figure 19 – Class diagram CoreEquipmentProfile::Schedules

Figure 19: The diagram shows main classes related to schedules.

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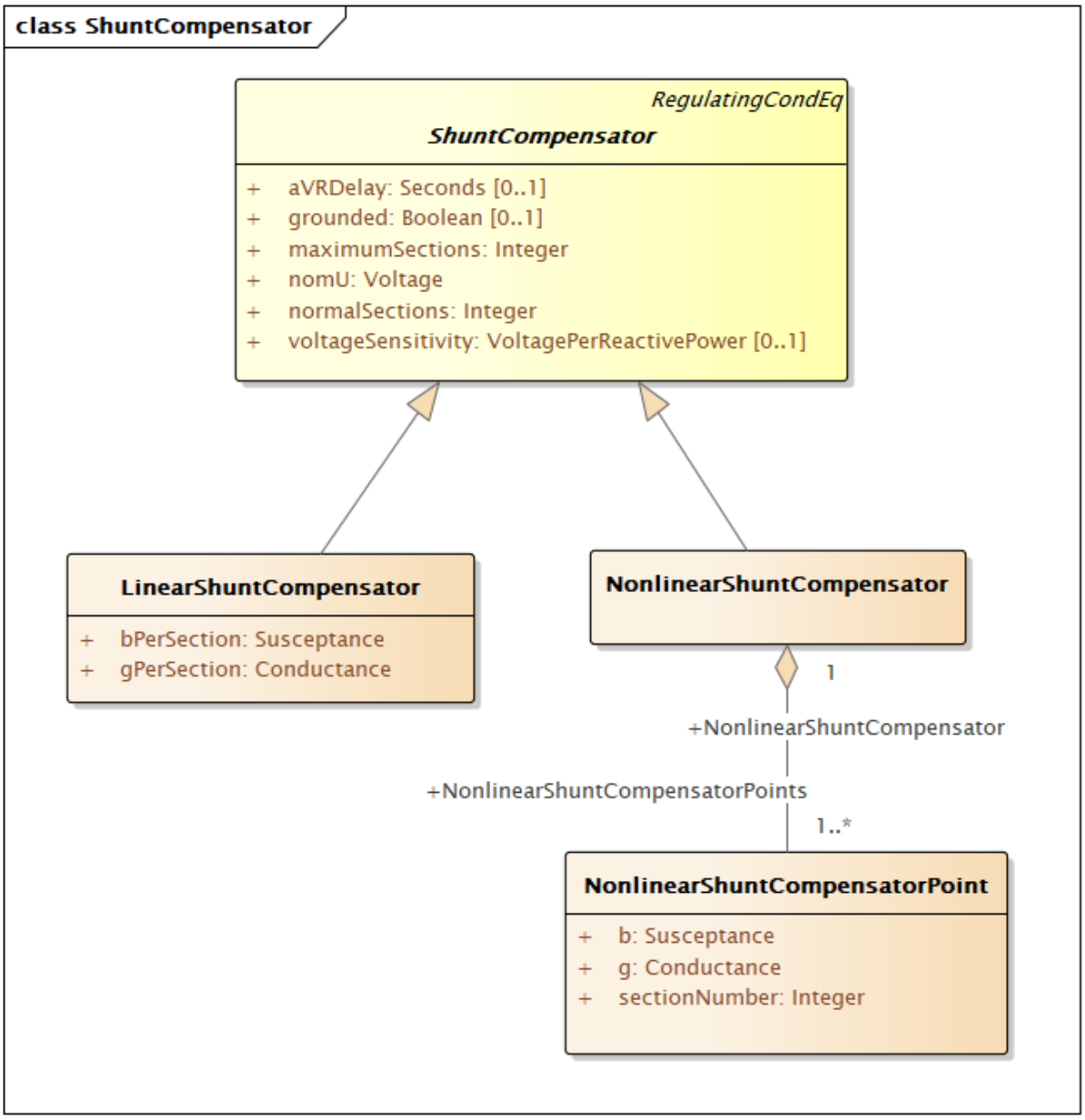

Figure 20 – Class diagram CoreEquipmentProfile::ShuntCompensator

Figure 20: Shows the shunt compensator inheritance structure.

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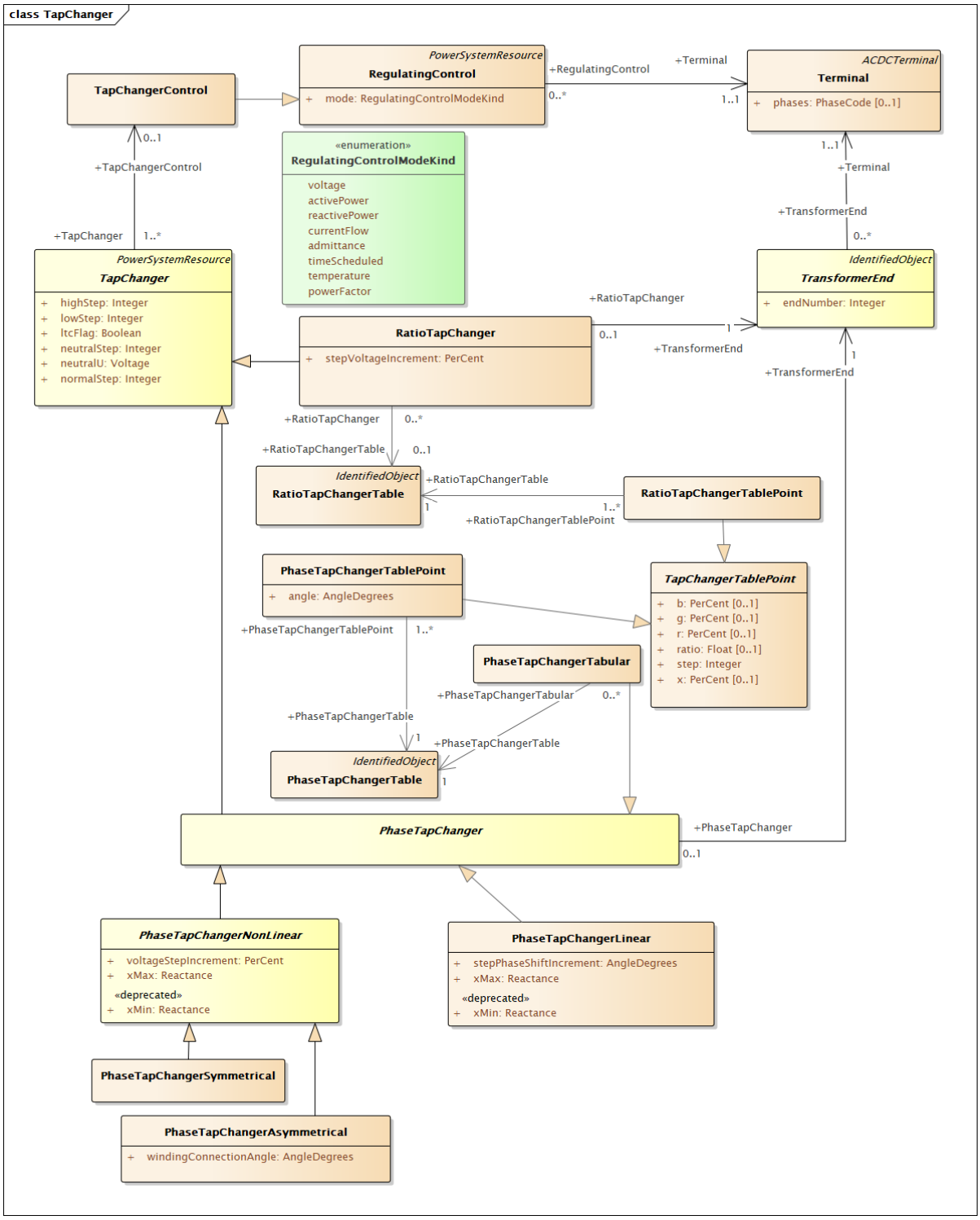

Figure 21 – Class diagram CoreEquipmentProfile::TapChanger

Figure 21: This diagram shows all classes related to the transformer tap model.

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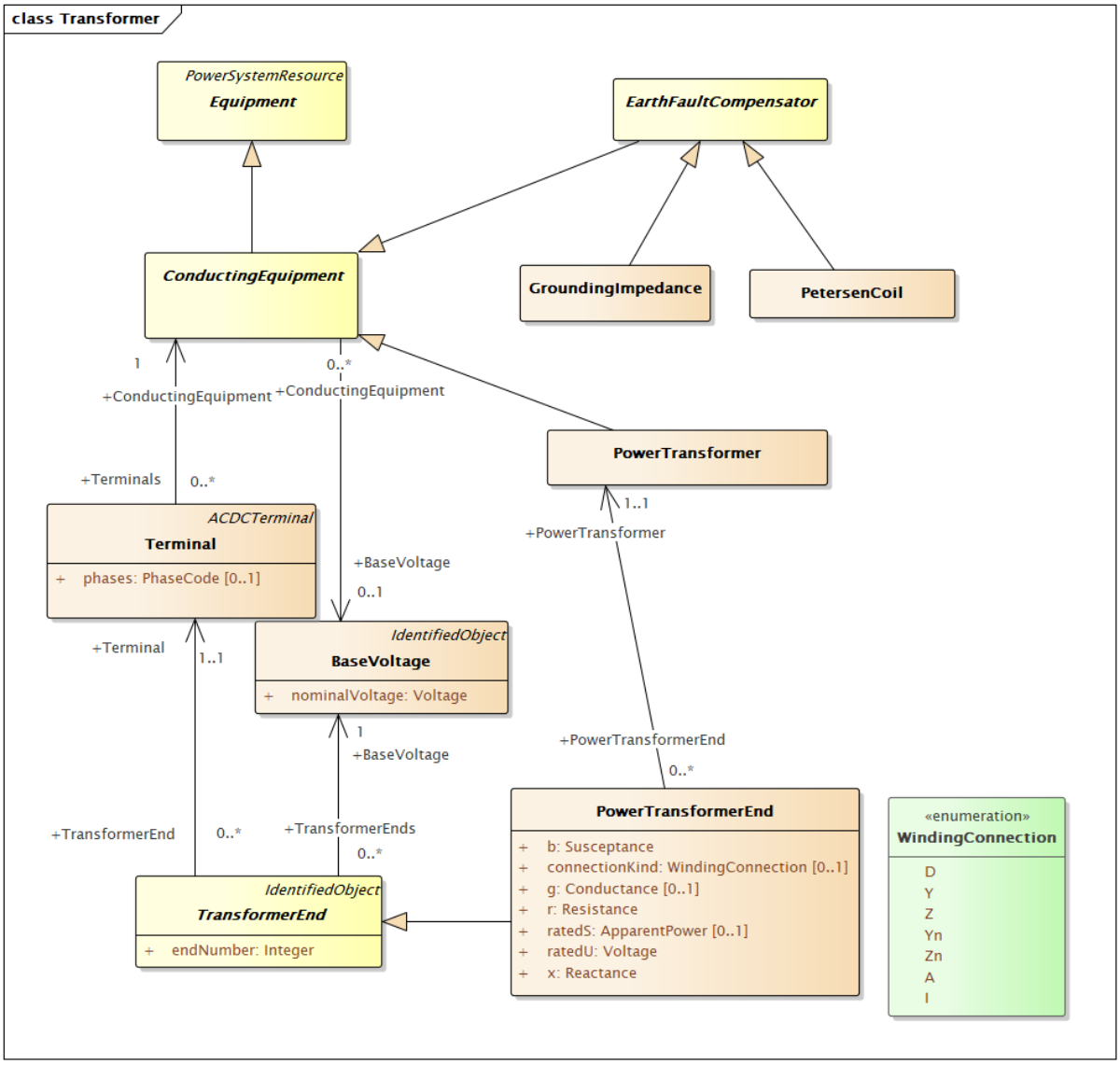

Figure 22 – Class diagram CoreEquipmentProfile::Transformer

Figure 22: This diagram shows classes related to the core transformer model, but does not show the tap changer details.

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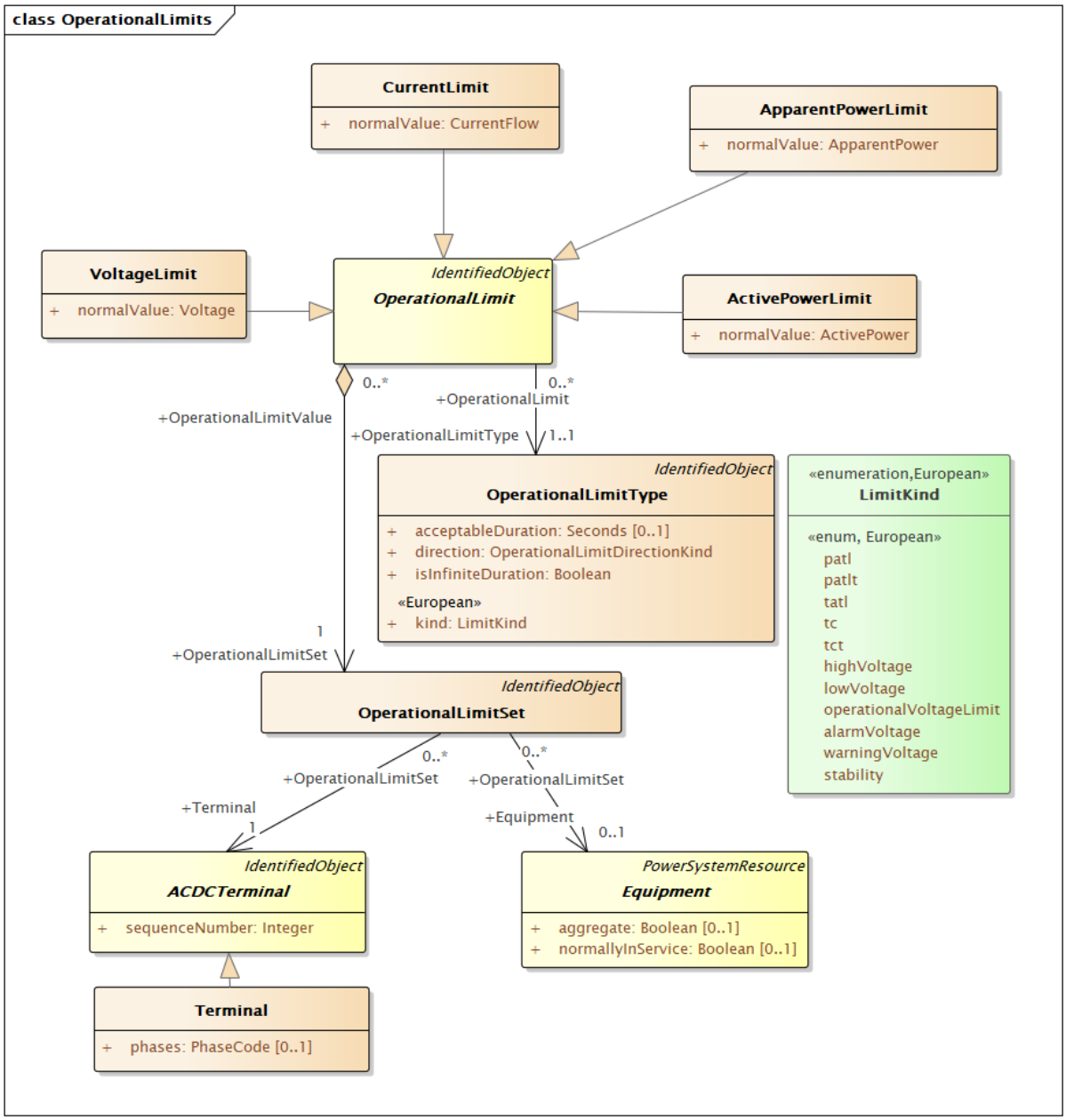

Figure 23 – Class diagram CoreEquipmentProfile::OperationalLimits

Figure 23: This diagram shows operational limits as they tie back into the core model.

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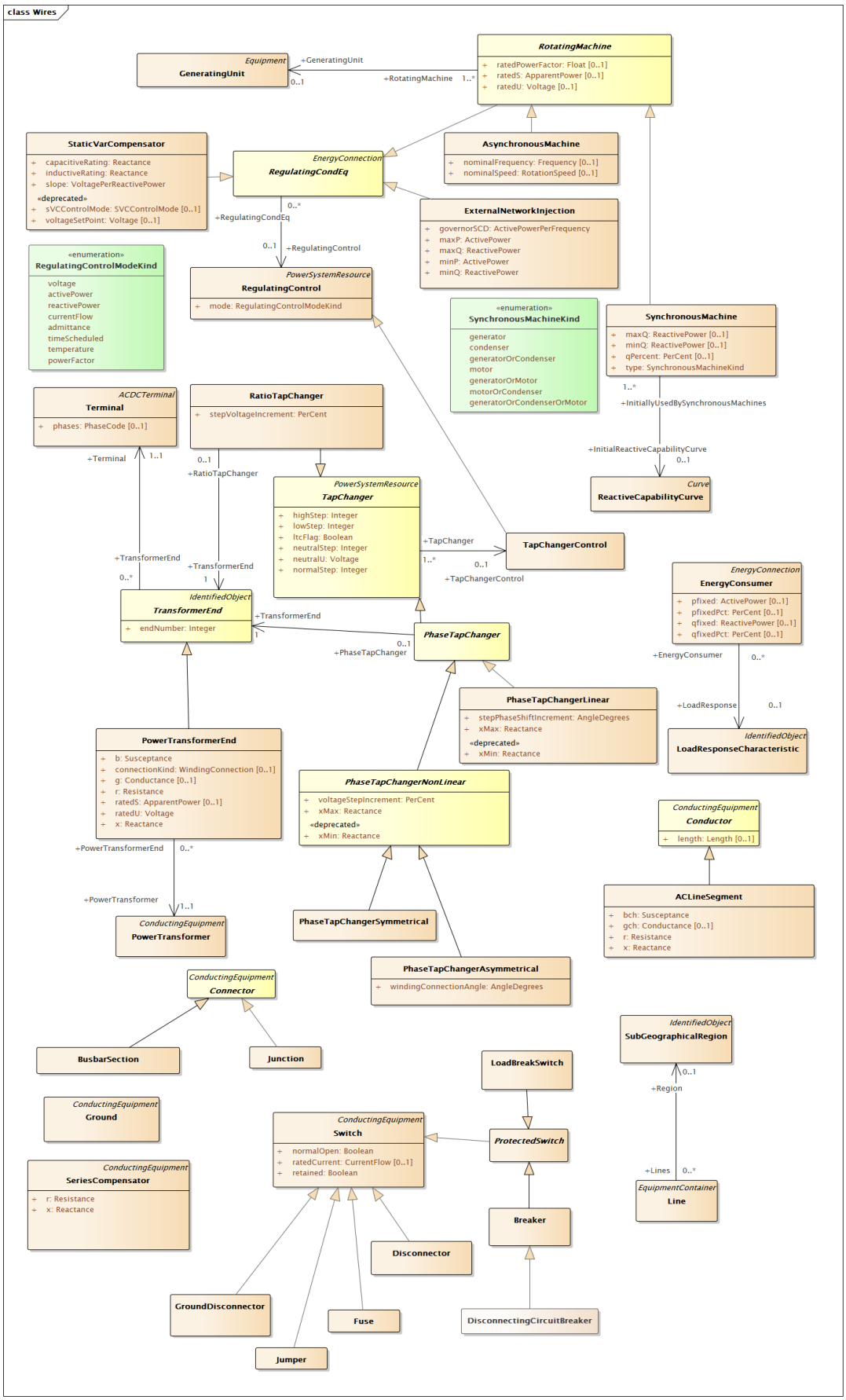

Figure 24 – Class diagram CoreEquipmentProfile::Wires

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Figure 24: The diagram shows the main classes from the wires package in the canonical CIM.

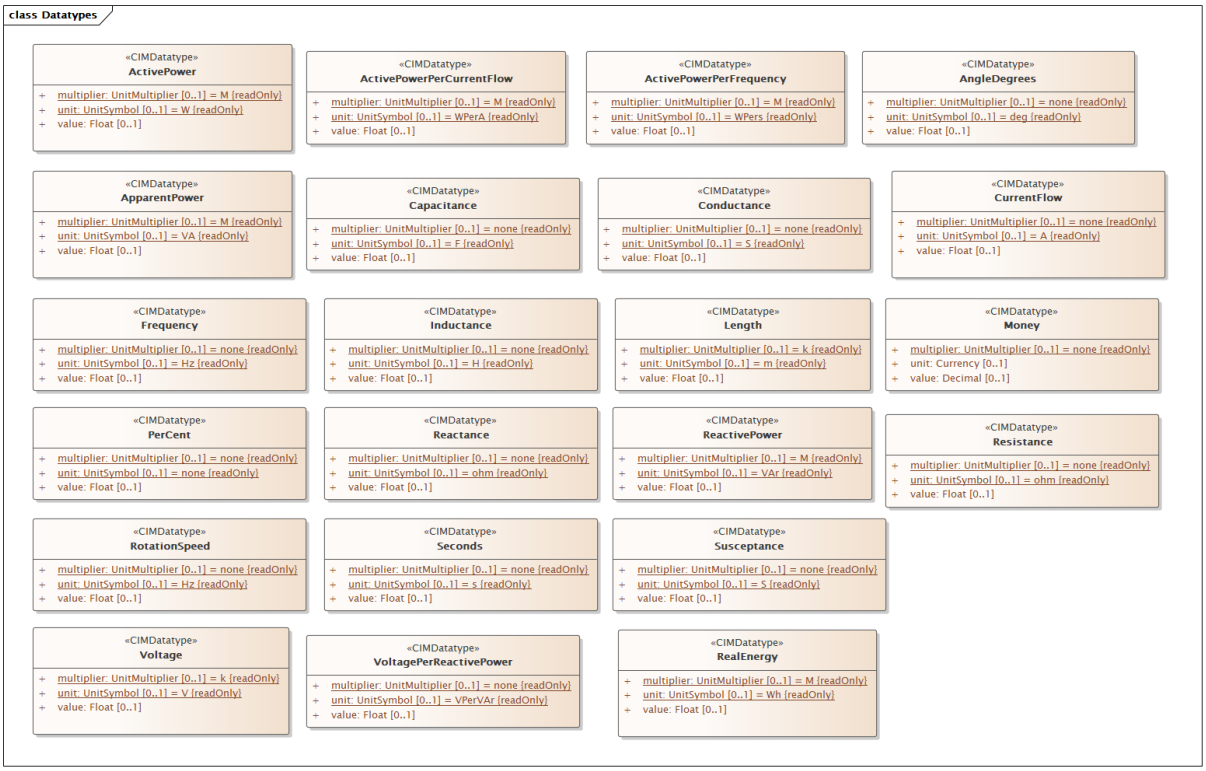

Figure 25 – Class diagram CoreEquipmentProfile::Datatypes

Figure 25: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

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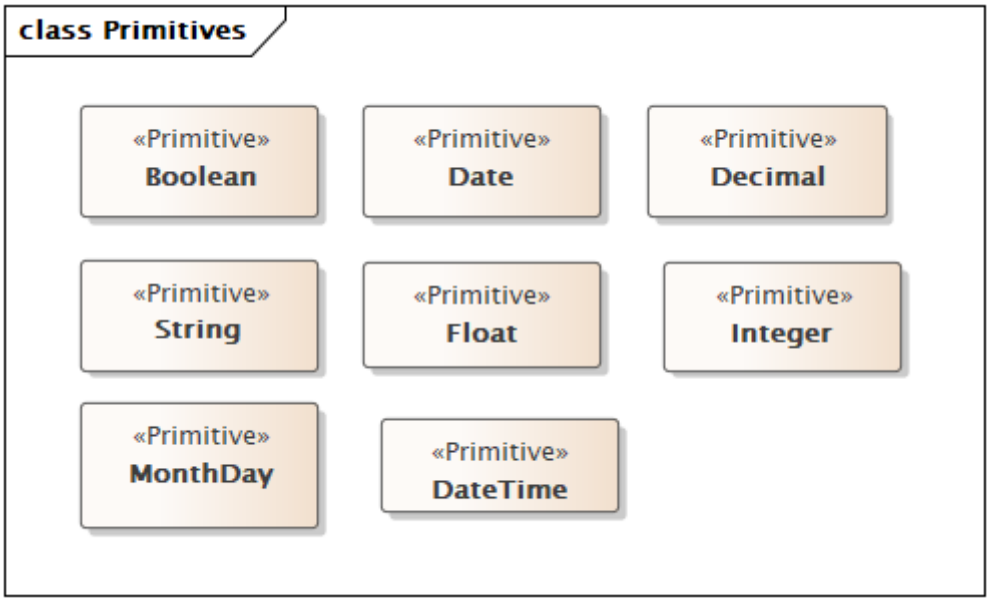

Figure 26 – Class diagram CoreEquipmentProfile::Primitives

Figure 26: The diagram shows datatypes that are used by classes in the profile.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

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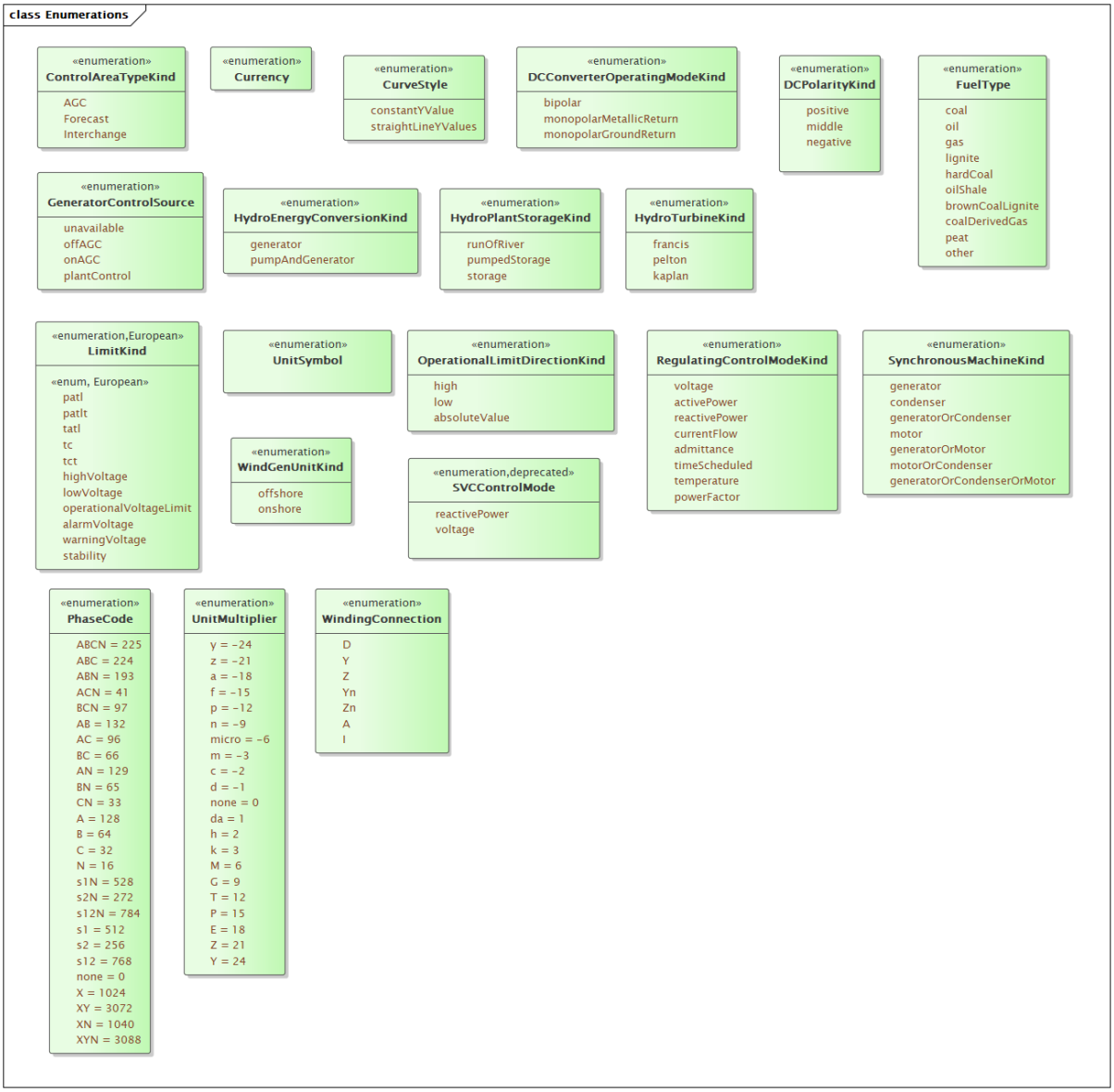

Figure 27 – Class diagram CoreEquipmentProfile::Enumerations

Figure 27: The diagram shows enumerations that are used by classes in the profile.

6.2 (abstract) EarthFaultCompensator

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conducting equipment used to represent a connection to ground which is typically used to compensate earth faults. An earth fault compensator device modelled with a single terminal implies a second terminal solidly connected to ground. If two terminals are modelled, the ground is not assumed and normal connection rules apply.

Table 1 shows all attributes of EarthFaultCompensator.

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Table 1 – Attributes of CoreEquipmentProfile::EarthFaultCompensator

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 2 shows all association ends of EarthFaultCompensator with other classes.

Table 2 – Association ends of CoreEquipmentProfile:: EarthFaultCompensator with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.3 (abstract) EnergyArea

Inheritance path = IdentifiedObject

Describes an area having energy production or consumption. Specializations are intended to support the load allocation function as typically required in energy management systems or planning studies to allocate hypothesized load levels to individual load points for power flow analysis. Often the energy area can be linked to both measured and forecast load levels.

Table 3 shows all attributes of EnergyArea.

Table 3 – Attributes of CoreEquipmentProfile::EnergyArea

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.4 (abstract) EnergyConnection

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection of energy generation or consumption on the power system model.

Table 4 shows all attributes of EnergyConnection.

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Table 4 – Attributes of CoreEquipmentProfile::EnergyConnection

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 5 shows all association ends of EnergyConnection with other classes.

Table 5 – Association ends of CoreEquipmentProfile:: EnergyConnection with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.5 EnergyConsumer

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Generic user of energy – a point of consumption on the power system model.

EnergyConsumer.pfixed, .qfixed, .pfixedPct and .qfixedPct have meaning only if there is no LoadResponseCharacteristic associated with EnergyConsumer or if LoadResponseCharacteristic.exponentModel is set to False.

Table 6 shows all attributes of EnergyConsumer.

Table 6 – Attributes of CoreEquipmentProfile::EnergyConsumer

name	mult	type	description
pfixed	0..1	ActivePower	Active power of the load that is a fixed quantity and does not vary as load group value varies. Load sign convention is used, i.e. positive sign means flow out from a node.
pfixedPct	0..1	PerCent	Fixed active power as a percentage of load group fixed active power. Used to represent the time-varying components. Load sign convention is used, i.e. positive sign means flow out from a node.
qfixed	0..1	ReactivePower	Reactive power of the load that is a fixed quantity and does not vary as load group value varies. Load sign convention is used, i.e. positive sign means flow out from a node.

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name	mult	type	description
qfixedPct	0..1	PerCent	Fixed reactive power as a percentage of load group fixed reactive power. Used to represent the time-varying components. Load sign convention is used, i.e. positive sign means flow out from a node.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 7 shows all association ends of EnergyConsumer with other classes.

Table 7 – Association ends of CoreEquipmentProfile:: EnergyConsumer with other classes

mult from	name	mult to	type	description
0..*	LoadResponse	0..1	LoadResponseCharacteristic	The load response characteristic of this load. If missing, this load is assumed to be constant power.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.6 EnergySchedulingType

Inheritance path = IdentifiedObject

Used to define the type of generation for scheduling purposes.

Table 8 shows all attributes of EnergySchedulingType.

Table 8 – Attributes of CoreEquipmentProfile::EnergySchedulingType

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.7 TapSchedule

Inheritance path = SeasonDayTypeSchedule: RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

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A pre-established pattern over time for a tap step.

Table 9 shows all attributes of TapSchedule.

Table 9 – Attributes of CoreEquipmentProfile::TapSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 10 shows all association ends of TapSchedule with other classes.

Table 10 – Association ends of CoreEquipmentProfile::TapSchedule with other classes

mult from	name	mult to	type	description
0..*	TapChanger	1..1	TapChanger	A TapSchedule is associated with a TapChanger.
0..*	DayType	1..1	DayType	inherited from: SeasonDayTypeSchedule
0..*	Season	1..1	Season	inherited from: SeasonDayTypeSchedule

6.8 EnergySource

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Table 11 shows all attributes of EnergySource.

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Table 11 – Attributes of CoreEquipmentProfile::EnergySource

name	mult	type	description
nominalVoltage	0..1	Voltage	Phase-to-phase nominal voltage.
pMin	0..1	ActivePower	This is the minimum active power that can be produced by the source. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
pMax	0..1	ActivePower	This is the maximum active power that can be produced by the source. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 12 shows all association ends of EnergySource with other classes.

Table 12 – Association ends of CoreEquipmentProfile::EnergySource with other classes

mult from	name	mult to	type	description
0..*	EnergySchedulingType	0..1	EnergySchedulingType	Energy Scheduling Type of an Energy Source.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.9 (abstract) Equipment

Inheritance path = PowerSystemResource: IdentifiedObject

The parts of a power system that are physical devices, electronic or mechanical.

Table 13 shows all attributes of Equipment.

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Table 13 – Attributes of CoreEquipmentProfile::Equipment

name	mult	type	description
aggregate	0..1	Boolean	The aggregate flag provides an alternative way of representing an aggregated (equivalent) element. It is applicable in cases when the dedicated classes for equivalent equipment do not have all of the attributes necessary to represent the required level of detail. In case the flag is set to “true” the single instance of equipment represents multiple pieces of equipment that have been modelled together as an aggregate equivalent obtained by a network reduction procedure. Examples would be power transformers or synchronous machines operating in parallel modelled as a single aggregate power transformer or aggregate synchronous machine. The attribute is not used for EquivalentBranch, EquivalentShunt and EquivalentInjection.
normallyInService	0..1	Boolean	Specifies the availability of the equipment under normal operating conditions. True means the equipment is available for topology processing, which determines if the equipment is energized or not. False means that the equipment is treated by network applications as if it is not in the model.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 14 shows all association ends of Equipment with other classes.

Table 14 – Association ends of CoreEquipmentProfile::Equipment with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	Container of this equipment.

6.10 (abstract) EquipmentContainer

Inheritance path = ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A modelling construct to provide a root class for containing equipment.

Table 15 shows all attributes of EquipmentContainer.

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Table 15 – Attributes of CoreEquipmentProfile::EquipmentContainer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.11 (abstract) ACDCConverter

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Table 16 shows all attributes of ACDCConverter.

Table 16 – Attributes of CoreEquipmentProfile::ACDCConverter

name	mult	type	description
baseS	0..1	ApparentPower	Base apparent power of the converter pole. The attribute shall be a positive value.
idleLoss	0..1	ActivePower	Active power loss in pole at no power transfer. It is converter's configuration data used in power flow. The attribute shall be a positive value.
maxUdc	0..1	Voltage	The maximum voltage on the DC side at which the converter should operate. It is converter's configuration data used in power flow. The attribute shall be a positive value.
minUdc	0..1	Voltage	The minimum voltage on the DC side at which the converter should operate. It is converter's configuration data used in power flow. The attribute shall be a positive value.
numberOfValves	0..1	Integer	Number of valves in the converter. Used in loss calculations.
ratedUdc	0..1	Voltage	Rated converter DC voltage, also called UdN. The attribute shall be a positive value. It is converter's configuration data used in power flow. For instance a bipolar HVDC link with value 200 kV has a 400kV difference between the dc lines.
resistiveLoss	0..1	Resistance	It is converter's configuration data used in power flow. Refer to poleLossP. The attribute shall be a positive value.
switchingLoss	0..1	ActivePowerPerCurrentFlow	Switching losses, relative to the base apparent power 'baseS'. Refer to poleLossP. The attribute shall be a positive value.
valveU0	0..1	Voltage	Valve threshold voltage, also called Uvalve. Forward voltage drop when the valve is conducting. Used in loss calculations, i.e. the switchLoss depend on numberOfValves * valveU0.
maxP	0..1	ActivePower	Maximum active power limit. The value is overwritten by values of VsCapabilityCurve, if present.

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name	mult	type	description
minP	0..1	ActivePower	Minimum active power limit. The value is overwritten by values of VsCapabilityCurve, if present.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 17 shows all association ends of ACDCConverter with other classes.

Table 17 – Association ends of CoreEquipmentProfile:: ACDCConverter with other classes

mult from	name	mult to	type	description
0..*	PccTerminal	0..1	Terminal	Point of common coupling terminal for this converter DC side. It is typically the terminal on the power transformer (or switch) closest to the AC network.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.12 ACDCConverterDCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Table 18 shows all attributes of ACDCConverterDCTerminal.

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Table 18 – Attributes of CoreEquipmentProfile::ACDCConverterDCTerminal

name	mult	type	description
polarity	1..1	DCPolarityKind	Represents the normal network polarity condition. Depending on the converter configuration the value shall be set as follows: – For a monopole with two converter terminals use DCPolarityKind “positive” and “negative”. – For a bi-pole or symmetric monopole with three converter terminals use DCPolarityKind “positive”, “middle” and “negative”.
sequenceNumber	1..1	Integer	inherited from: ACDCTerminal
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 19 shows all association ends of ACDCConverterDCTerminal with other classes.

Table 19 – Association ends of CoreEquipmentProfile:: ACDCConverterDCTerminal with other classes

mult from	name	mult to	type	description
0..*	DCConductingEquipment	1..1	ACDCConverter	A DC converter terminal belong to an DC converter.
0..*	DCNode	0..1	DCNode	inherited from: DCBaseTerminal
1..*	BusNameMarker	0..1	BusNameMarker	inherited from: ACDCTerminal

6.13 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 20 shows all attributes of ACDCTerminal.

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Table 20 – Attributes of CoreEquipmentProfile::ACDCTerminal

name	mult	type	description
sequenceNumber	1..1	Integer	The orientation of the terminal connections for a multiple terminal conducting equipment. The sequence numbering starts with 1 and additional terminals should follow in increasing order. The first terminal is the "starting point" for a two terminal branch.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 21 shows all association ends of ACDCTerminal with other classes.

Table 21 – Association ends of CoreEquipmentProfile:: ACDCTerminal with other classes

mult from	name	mult to	type	description
1..*	BusNameMarker	0..1	BusNameMarker	The bus name marker used to name the bus (topological node).

6.14 ACLineSegment

Inheritance path = Conductor: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A wire or combination of wires, with consistent electrical characteristics, building a single electrical system, used to carry alternating current between points in the power system.

For symmetrical, transposed three phase lines, it is sufficient to use attributes of the line segment, which describe impedances and admittances for the entire length of the segment. Additionally impedances can be computed by using length and associated per length impedances.

The BaseVoltage at the two ends of ACLineSegments in a Line shall have the same BaseVoltage.nominalVoltage. However, boundary lines may have slightly different BaseVoltage.nominalVoltages and variation is allowed. Larger voltage difference in general requires use of an equivalent branch.

Table 22 shows all attributes of ACLineSegment.

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Table 22 – Attributes of CoreEquipmentProfile::ACLineSegment

name	mult	type	description
bch	1..1	Susceptance	Positive sequence shunt (charging) susceptance, uniformly distributed, of the entire line section. This value represents the full charging over the full length of the line.
gch	0..1	Conductance	Positive sequence shunt (charging) conductance, uniformly distributed, of the entire line section.
r	1..1	Resistance	Positive sequence series resistance of the entire line section.
x	1..1	Reactance	Positive sequence series reactance of the entire line section.
length	0..1	Length	inherited from: Conductor
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 23 shows all association ends of ACLineSegment with other classes.

Table 23 – Association ends of CoreEquipmentProfile:: ACLineSegment with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.15 ActivePowerLimit

Inheritance path = OperationalLimit: IdentifiedObject

Limit on active power flow.

Table 24 shows all attributes of ActivePowerLimit.

Table 24 – Attributes of CoreEquipmentProfile::ActivePowerLimit

name	mult	type	description
normalValue	1..1	ActivePower	The normal value of active power limit. The attribute shall be a positive value or zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 25 shows all association ends of ActivePowerLimit with other classes.

Table 25 – Association ends of CoreEquipmentProfile:: ActivePowerLimit with other classes

mult from	name	mult to	type	description
0..*	OperationalLimitType	1..1	OperationalLimitType	inherited from: OperationalLimit
0..*	OperationalLimitSet	1..1	OperationalLimitSet	inherited from: OperationalLimit

6.16 ApparentPowerLimit

Inheritance path = OperationalLimit: IdentifiedObject

Apparent power limit.

Table 26 shows all attributes of ApparentPowerLimit.

Table 26 – Attributes of CoreEquipmentProfile::ApparentPowerLimit

name	mult	type	description
normalValue	1..1	ApparentPower	The normal apparent power limit. The attribute shall be a positive value or zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 27 shows all association ends of ApparentPowerLimit with other classes.

Table 27 – Association ends of CoreEquipmentProfile:: ApparentPowerLimit with other classes

mult from	name	mult to	type	description
0..*	OperationalLimitType	1..1	OperationalLimitType	inherited from: OperationalLimit
0..*	OperationalLimitSet	1..1	OperationalLimitSet	inherited from: OperationalLimit

6.17 AsynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

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A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g. squirrel-cage induction machine.

Table 28 shows all attributes of AsynchronousMachine.

Table 28 – Attributes of CoreEquipmentProfile::AsynchronousMachine

name	mult	type	description
nominalFrequency	0..1	Frequency	Nameplate data indicates if the machine is 50 Hz or 60 Hz.
nominalSpeed	0..1	RotationSpeed	Nameplate data. Depends on the slip and number of pole pairs.
ratedPowerFactor	0..1	Float	inherited from: RotatingMachine
ratedS	0..1	ApparentPower	inherited from: RotatingMachine
ratedU	0..1	Voltage	inherited from: RotatingMachine
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 29 shows all association ends of AsynchronousMachine with other classes.

Table 29 – Association ends of CoreEquipmentProfile:: AsynchronousMachine with other classes

mult from	name	mult to	type	description
1..*	GeneratingUnit	0..1	GeneratingUnit	inherited from: RotatingMachine
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.18 (abstract) AuxiliaryEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

AuxiliaryEquipment describe equipment that is not performing any primary functions but support for the equipment performing the primary function.

AuxiliaryEquipment is attached to primary equipment via an association with Terminal.

Table 30 shows all attributes of AuxiliaryEquipment.

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Table 30 – Attributes of CoreEquipmentProfile::AuxiliaryEquipment

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 31 shows all association ends of AuxiliaryEquipment with other classes.

Table 31 – Association ends of CoreEquipmentProfile:: AuxiliaryEquipment with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	The Terminal at the equipment where the AuxiliaryEquipment is attached.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.19 BaseVoltage

Inheritance path = IdentifiedObject

Defines a system base voltage which is referenced.

Table 32 shows all attributes of BaseVoltage.

Table 32 – Attributes of CoreEquipmentProfile::BaseVoltage

name	mult	type	description
nominalVoltage	1..1	Voltage	The power system resource's base voltage. Shall be a positive value and not zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.20 (abstract) BasicIntervalSchedule

Inheritance path = IdentifiedObject

Schedule of values at points in time.

Table 33 shows all attributes of BasicIntervalSchedule.

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Table 33 – Attributes of CoreEquipmentProfile::BasicIntervalSchedule

name	mult	type	description
startTime	1..1	DateTime	The time for the first time point. The value can be a time of day, not a specific date.
value1Unit	1..1	UnitSymbol	Value1 units of measure.
value2Unit	0..1	UnitSymbol	Value2 units of measure.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.21 BatteryUnit

Inheritance path = PowerElectronicsUnit: Equipment: PowerSystemResource: IdentifiedObject

An electrochemical energy storage device.

Table 34 shows all attributes of BatteryUnit.

Table 34 – Attributes of CoreEquipmentProfile::BatteryUnit

name	mult	type	description
ratedE	1..1	RealEnergy	Full energy storage capacity of the battery. The attribute shall be a positive value.
maxP	0..1	ActivePower	inherited from: PowerElectronicsUnit
minP	0..1	ActivePower	inherited from: PowerElectronicsUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 35 shows all association ends of BatteryUnit with other classes.

Table 35 – Association ends of CoreEquipmentProfile::BatteryUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.22 Bay

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A collection of power system resources (within a given substation) including conducting equipment, protection relays, measurements, and telemetry. A bay typically represents a physical grouping related to modularization of equipment.

Table 36 shows all attributes of Bay.

Table 36 – Attributes of CoreEquipmentProfile::Bay

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 37 shows all association ends of Bay with other classes.

Table 37 – Association ends of CoreEquipmentProfile::Bay with other classes

mult from	name	mult to	type	description
0..*	VoltageLevel	1..1	VoltageLevel	The voltage level containing this bay.

6.23 (European) BoundaryPoint

Inheritance path = PowerSystemResource: IdentifiedObject

Designates a connection point at which one or more model authority sets shall connect to. The location of the connection point as well as other properties are agreed between organisations responsible for the interconnection, hence all attributes of the class represent this agreement. It is primarily used in a boundary model authority set which can contain one or many BoundaryPoint-s among other Equipment-s and their connections.

Table 38 shows all attributes of BoundaryPoint.

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Table 38 – Attributes of CoreEquipmentProfile::BoundaryPoint

name	mult	type	description
fromEndIsoCode	1..1	String	(European) The ISO code of the region which the "From" side of the Boundary point belongs to or it is connected to. The ISO code is a two-character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.
fromEndName	1..1	String	(European) A human readable name with length of the string 64 characters maximum. It covers the following two cases: – if the Boundary point is placed on a tie-line, it is the name (IdentifiedObject.name) of the substation at which the "From" side of the tie-line is connected to. – if the Boundary point is placed in a substation, it is the name (IdentifiedObject.name) of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc.) at which the "From" side of the Boundary point is connected to.
fromEndNameTso	1..1	String	(European) Identifies the name of the transmission system operator, distribution system operator or other entity at which the "From" side of the interconnection is connected to. The length of the string is 64 characters maximum.
toEndIsoCode	1..1	String	(European) The ISO code of the region which the "To" side of the Boundary point belongs to or is connected to. The ISO code is a two-character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.
toEndName	1..1	String	(European) A human readable name with length of the string 64 characters maximum. It covers the following two cases: – if the Boundary point is placed on a tie-line, it is the name (IdentifiedObject.name) of the substation at which the "To" side of the tie-line is connected to. – if the Boundary point is placed in a substation, it is the name (IdentifiedObject.name) of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc.) at which the "To" side of the Boundary point is connected to.
toEndNameTso	1..1	String	(European) Identifies the name of the transmission system operator, distribution system operator or other entity at which the "To" side of the interconnection is connected to. The length of the string is 64 characters maximum.
isDirectCurrent	0..1	Boolean	(European) If true, this boundary point is a point of common coupling (PCC) of a direct current (DC) interconnection, otherwise the interconnection is AC (default).
isExcludedFromAreaInterchange	0..1	Boolean	(European) If true, this boundary point is on the interconnection that is excluded from control area interchange calculation and consequently has no related tie flows. Otherwise, the interconnection is included in control area interchange and a TieFlow is required at all sides of the boundary point (default).
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 39 shows all association ends of BoundaryPoint with other classes.

Table 39 – Association ends of CoreEquipmentProfile:: BoundaryPoint with other classes

mult from	name	mult to	type	description
0..1	ConnectivityNode	1..1	ConnectivityNode	(European) The connectivity node that is designated as a boundary point.

6.24 Breaker

Inheritance path = ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions and also making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions e.g. those of short circuit.

Table 40 shows all attributes of Breaker.

Table 40 – Attributes of CoreEquipmentProfile::Breaker

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 41 shows all association ends of Breaker with other classes.

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Table 41 – Association ends of CoreEquipmentProfile::Breaker with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.25 BusbarSection

Inheritance path = Connector: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation.

Voltage measurements are typically obtained from voltage transformers that are connected to busbar sections. A bus bar section may have many physical terminals but for analysis is modelled with exactly one logical terminal.

Table 42 shows all attributes of BusbarSection.

Table 42 – Attributes of CoreEquipmentProfile::BusbarSection

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 43 shows all association ends of BusbarSection with other classes.

Table 43 – Association ends of CoreEquipmentProfile:: BusbarSection with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.26 BusNameMarker

Inheritance path = IdentifiedObject

Used to apply user standard names to TopologicalNodes. Associated with one or more terminals that are normally connected with the bus name. The associated terminals are normally connected by non-retained switches. For a ring bus station configuration, all BusbarSection terminals in the ring are typically associated. For a breaker and a half scheme, both BusbarSections would normally be associated. For a ring bus, all BusbarSections would normally be associated. For a "straight" busbar configuration, normally only the main terminal at the BusbarSection would be associated.

Table 44 shows all attributes of BusNameMarker.

Table 44 – Attributes of CoreEquipmentProfile::BusNameMarker

name	mult	type	description
priority	0..1	Integer	Priority of bus name marker for use as topology bus name. Use 0 for do not care. Use 1 for highest priority. Use 2 as priority is less than 1 and so on.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 45 shows all association ends of BusNameMarker with other classes.

Table 45 – Association ends of CoreEquipmentProfile:: BusNameMarker with other classes

mult from	name	mult to	type	description
0..*	ReportingGroup	0..1	ReportingGroup	The reporting group to which this bus name marker belongs.

6.27 CAESPlant

Inheritance path = PowerSystemResource: IdentifiedObject

Compressed air energy storage plant.

Table 46 shows all attributes of CAESPlant.

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Table 46 – Attributes of CoreEquipmentProfile::CAESPlant

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.28 Clamp

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A Clamp is a galvanic connection at a line segment where other equipment is connected. A Clamp does not cut the line segment.

A Clamp is ConductingEquipment and has one Terminal with an associated ConnectivityNode. Any other ConductingEquipment can be connected to the Clamp ConnectivityNode.

Table 47 shows all attributes of Clamp.

Table 47 – Attributes of CoreEquipmentProfile::Clamp

name	mult	type	description
lengthFromTerminal1	0..1	Length	The length to the place where the clamp is located starting from side one of the line segment, i.e. the line segment terminal with sequence number equal to 1.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 48 shows all association ends of Clamp with other classes.

Table 48 – Association ends of CoreEquipmentProfile::Clamp with other classes

mult from	name	mult to	type	description
0..*	ACLineSegment	1..1	ACLineSegment	The line segment to which the clamp is connected.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.29 ConformLoadSchedule

Inheritance path = SeasonDayTypeSchedule: RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

A curve of load versus time (X-axis) showing the active power values (Y1-axis) and reactive power (Y2-axis) for each unit of the period covered. This curve represents a typical pattern of load over the time period for a given day type and season.

Table 49 shows all attributes of ConformLoadSchedule.

Table 49 – Attributes of CoreEquipmentProfile::ConformLoadSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 50 shows all association ends of ConformLoadSchedule with other classes.

Table 50 – Association ends of CoreEquipmentProfile:: ConformLoadSchedule with other classes

mult from	name	mult to	type	description
0..*	ConformLoadGroup	1..1	ConformLoadGroup	The ConformLoadGroup where the ConformLoadSchedule belongs.
0..*	DayType	1..1	DayType	inherited from: SeasonDayTypeSchedule
0..*	Season	1..1	Season	inherited from: SeasonDayTypeSchedule

6.30 CogenerationPlant

Inheritance path = PowerSystemResource: IdentifiedObject

A set of thermal generating units for the production of electrical energy and process steam (usually from the output of the steam turbines). The steam sendout is typically used for industrial purposes or for municipal heating and cooling.

Table 51 shows all attributes of CogenerationPlant.

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Table 51 – Attributes of CoreEquipmentProfile::CogenerationPlant

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.31 CombinedCyclePlant

Inheritance path = PowerSystemResource: IdentifiedObject

A set of combustion turbines and steam turbines where the exhaust heat from the combustion turbines is recovered to make steam for the steam turbines, resulting in greater overall plant efficiency.

Table 52 shows all attributes of CombinedCyclePlant.

Table 52 – Attributes of CoreEquipmentProfile::CombinedCyclePlant

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.32 (abstract) ConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Table 53 shows all attributes of ConductingEquipment.

Table 53 – Attributes of CoreEquipmentProfile::ConductingEquipment

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 54 shows all association ends of ConductingEquipment with other classes.

Table 54 – Association ends of CoreEquipmentProfile:: ConductingEquipment with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	Base voltage of this conducting equipment. Use only when there is no voltage level container used and only one base voltage applies. For example, not used for transformers.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.33 (abstract) Conductor

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Combination of conducting material with consistent electrical characteristics, building a single electrical system, used to carry current between points in the power system.

Table 55 shows all attributes of Conductor.

Table 55 – Attributes of CoreEquipmentProfile::Conductor

name	mult	type	description
length	0..1	Length	Segment length for calculating line section capabilities.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 56 shows all association ends of Conductor with other classes.

Table 56 – Association ends of CoreEquipmentProfile::Conductor with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.34 ConformLoad

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

ConformLoad represent loads that follow a daily load change pattern where the pattern can be used to scale the load with a system load.

Table 57 shows all attributes of ConformLoad.

Table 57 – Attributes of CoreEquipmentProfile::ConformLoad

name	mult	type	description
pfixed	0..1	ActivePower	inherited from: EnergyConsumer
pfixedPct	0..1	PerCent	inherited from: EnergyConsumer
qfixed	0..1	ReactivePower	inherited from: EnergyConsumer
qfixedPct	0..1	PerCent	inherited from: EnergyConsumer
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 58 shows all association ends of ConformLoad with other classes.

Table 58 – Association ends of CoreEquipmentProfile::ConformLoad with other classes

mult from	name	mult to	type	description
1..*	LoadGroup	1..1	ConformLoadGroup	Group of this ConformLoad.
0..*	LoadResponse	0..1	LoadResponseCharacteristic	inherited from: EnergyConsumer
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.35 ConformLoadGroup

Inheritance path = LoadGroup: IdentifiedObject

A group of loads conforming to an allocation pattern.

Table 59 shows all attributes of ConformLoadGroup.

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Table 59 – Attributes of CoreEquipmentProfile::ConformLoadGroup

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 60 shows all association ends of ConformLoadGroup with other classes.

Table 60 – Association ends of CoreEquipmentProfile:: ConformLoadGroup with other classes

mult from	name	mult to	type	description
1..*	SubLoadArea	1..1	SubLoadArea	inherited from: LoadGroup

6.36 ConnectivityNode

Inheritance path = IdentifiedObject

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

Table 61 shows all attributes of ConnectivityNode.

Table 61 – Attributes of CoreEquipmentProfile::ConnectivityNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 62 shows all association ends of ConnectivityNode with other classes.

Table 62 – Association ends of CoreEquipmentProfile:: ConnectivityNode with other classes

mult from	name	mult to	type	description
0..*	ConnectivityNodeContainer	1..1	ConnectivityNodeContainer	Container of this connectivity node.

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6.37 (abstract) ConnectivityNodeContainer

Inheritance path = PowerSystemResource: IdentifiedObject

A base class for all objects that may contain connectivity nodes or topological nodes.

Table 63 shows all attributes of ConnectivityNodeContainer.

Table 63 – Attributes of CoreEquipmentProfile::ConnectivityNodeContainer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.38 (abstract) Connector

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation and are modelled with a single logical terminal.

Table 64 shows all attributes of Connector.

Table 64 – Attributes of CoreEquipmentProfile::Connector

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 65 shows all association ends of Connector with other classes.

Table 65 – Association ends of CoreEquipmentProfile::Connector with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.39 ControlArea

Inheritance path = PowerSystemResource: IdentifiedObject

A control area is a grouping of generating units and/or loads and a cutset of tie lines (as terminals) which may be used for a variety of purposes including automatic generation control, power flow solution area interchange control specification, and input to load forecasting. All generation and load within the area defined by the terminals on the border are considered in the area interchange control. Note that any number of overlapping control area specifications can be superimposed on the physical model. The following general principles apply to ControlArea:

1) The control area orientation for net interchange is positive for an import, negative for an export.
2) The control area net interchange is determined by summing flows in Terminals. The Terminals are identified by creating a set of TieFlow objects associated with a ControlArea object. Each TieFlow object identifies one Terminal.
3) In a single network model, a tie between two control areas must be modelled in both control area specifications, such that the two representations of the tie flow sum to zero.
4) The normal orientation of Terminal flow is positive for flow into the conducting equipment that owns the Terminal. (i.e. flow from a bus into a device is positive.) However, the orientation of each flow in the control area specification must align with the control area convention, i.e. import is positive. If the orientation of the Terminal flow referenced by a TieFlow is positive into the control area, then this is confirmed by setting TieFlow.positiveFlowIn flag TRUE. If not, the orientation must be reversed by setting the TieFlow.positiveFlowIn flag FALSE.

Table 66 shows all attributes of ControlArea.

Table 66 – Attributes of CoreEquipmentProfile::ControlArea

name	mult	type	description
type	1..1	ControlAreaTypeKind	The primary type of control area definition used to determine if this is used for automatic generation control, for planning interchange control, or other purposes. A control area specified with primary type of automatic generation control could still be forecast and used as an interchange area in power flow analysis.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 67 shows all association ends of ControlArea with other classes.

Table 67 – Association ends of CoreEquipmentProfile::ControlArea with other classes

mult from	name	mult to	type	description
0..1	EnergyArea	1..1	EnergyArea	The energy area that is forecast from this control area specification.

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6.40 ControlAreaGeneratingUnit

Inheritance path = IdentifiedObject

A control area generating unit. This class is needed so that alternate control area definitions may include the same generating unit. It should be noted that only one instance within a control area should reference a specific generating unit.

Table 68 shows all attributes of ControlAreaGeneratingUnit.

Table 68 – Attributes of CoreEquipmentProfile::ControlAreaGeneratingUnit

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 69 shows all association ends of ControlAreaGeneratingUnit with other classes.

Table 69 – Association ends of CoreEquipmentProfile:: ControlAreaGeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	ControlArea	1..1	ControlArea	The parent control area for the generating unit specifications.
0..*	GeneratingUnit	1..1	GeneratingUnit	The generating unit specified for this control area. Note that a control area should include a GeneratingUnit only once.

6.41 CsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

DC side of the current source converter (CSC).

The firing angle controls the DC voltage at the converter, both for rectifier and inverter. The difference between the DC voltages of the rectifier and inverter determines the DC current. The extinction angle is used to limit the DC voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source DC line. Higher level controls are built on top, e.g. DC voltage, DC current and active power. From a steady state perspective it is sufficient to specify the wanted active power transfer (ACDCConverter.targetPpcc) and the control functions will set the DC voltage, dc current, firing angle, transformer tap position and number of connected filters to meet this. Therefore attributes targetAlpha and targetGamma are not applicable in this case.

The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses is a function of the DC voltage and DC current.

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The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10 to 18 degrees.

The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17 to 20 degrees.

Table 70 shows all attributes of CsConverter.

Table 70 – Attributes of CoreEquipmentProfile::CsConverter

name	mult	type	description
maxAlpha	0..1	AngleDegrees	Maximum firing angle. It is converter's configuration data used in power flow. The attribute shall be a positive value.
maxGamma	0..1	AngleDegrees	Maximum extinction angle. It is converter's configuration data used in power flow. The attribute shall be a positive value.
maxIdc	0..1	CurrentFlow	The maximum direct current (Id) on the DC side at which the converter should operate. It is converter's configuration data use in power flow. The attribute shall be a positive value.
minAlpha	0..1	AngleDegrees	Minimum firing angle. It is converter's configuration data used in power flow. The attribute shall be a positive value.
minGamma	0..1	AngleDegrees	Minimum extinction angle. It is converter's configuration data used in power flow. The attribute shall be a positive value.
minIdc	0..1	CurrentFlow	The minimum direct current (Id) on the DC side at which the converter should operate. It is converter's configuration data used in power flow. The attribute shall be a positive value.
ratedIdc	0..1	CurrentFlow	Rated converter DC current, also called IdN. The attribute shall be a positive value. It is converter's configuration data used in power flow.
baseS	0..1	ApparentPower	inherited from: ACDCConverter
idleLoss	0..1	ActivePower	inherited from: ACDCConverter
maxUdc	0..1	Voltage	inherited from: ACDCConverter
minUdc	0..1	Voltage	inherited from: ACDCConverter
numberOfValves	0..1	Integer	inherited from: ACDCConverter
ratedUdc	0..1	Voltage	inherited from: ACDCConverter
resistiveLoss	0..1	Resistance	inherited from: ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	inherited from: ACDCConverter
valveU0	0..1	Voltage	inherited from: ACDCConverter
maxP	0..1	ActivePower	inherited from: ACDCConverter
minP	0..1	ActivePower	inherited from: ACDCConverter
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 71 shows all association ends of CsConverter with other classes.

Table 71 – Association ends of CoreEquipmentProfile::CsConverter with other classes

mult from	name	mult to	type	description
0..*	PccTerminal	0..1	Terminal	inherited from: ACDCConverter
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.42 CurrentLimit

Inheritance path = OperationalLimit: IdentifiedObject

Operational limit on current.

Table 72 shows all attributes of CurrentLimit.

Table 72 – Attributes of CoreEquipmentProfile::CurrentLimit

name	mult	type	description
normalValue	1..1	CurrentFlow	The normal value for limit on current flow. The attribute shall be a positive value or zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 73 shows all association ends of CurrentLimit with other classes.

Table 73 – Association ends of CoreEquipmentProfile::CurrentLimit with other classes

mult from	name	mult to	type	description
0..*	OperationalLimitType	1..1	OperationalLimitType	inherited from: OperationalLimit
0..*	OperationalLimitSet	1..1	OperationalLimitSet	inherited from: OperationalLimit

6.43 CurrentTransformer

Inheritance path = Sensor: AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

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Instrument transformer used to measure electrical qualities of the circuit that is being protected and/or monitored. Typically used as current transducer for the purpose of metering or protection. A typical secondary current rating would be 5A.

Table 74 shows all attributes of CurrentTransformer.

Table 74 – Attributes of CoreEquipmentProfile::CurrentTransformer

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 75 shows all association ends of CurrentTransformer with other classes.

Table 75 – Association ends of CoreEquipmentProfile:: CurrentTransformer with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.44 (abstract) Curve

Inheritance path = IdentifiedObject

A multi-purpose curve or functional relationship between an independent variable (X-axis) and dependent (Y-axis) variables.

Table 76 shows all attributes of Curve.

Table 76 – Attributes of CoreEquipmentProfile::Curve

name	mult	type	description
curveStyle	1..1	CurveStyle	The style or shape of the curve.
xUnit	1..1	UnitSymbol	The X-axis units of measure.
y1Unit	1..1	UnitSymbol	The Y1-axis units of measure.
y2Unit	0..1	UnitSymbol	The Y2-axis units of measure.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.45 CurveData root class

Multi-purpose data points for defining a curve. The use of this generic class is discouraged if a more specific class can be used to specify the X and Y axis values along with their specific data types.

Table 77 shows all attributes of CurveData.

Table 77 – Attributes of CoreEquipmentProfile::CurveData

name	mult	type	description
xvalue	1..1	Float	The data value of the X-axis variable, depending on the X-axis units.
y1value	1..1	Float	The data value of the first Y-axis variable, depending on the Y-axis units.
y2value	0..1	Float	The data value of the second Y-axis variable (if present), depending on the Y-axis units.

Table 78 shows all association ends of CurveData with other classes.

Table 78 – Association ends of CoreEquipmentProfile::CurveData with other classes

mult from	name	mult to	type	description
1..*	Curve	1..1	Curve	The curve of this curve data point.

6.46 Cut

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A cut separates a line segment into two parts. The cut appears as a switch inserted between these two parts and connects them together. As the cut is normally open there is no galvanic connection between the two line segment parts. But it is possible to close the cut to get galvanic connection.

The cut terminals are oriented towards the line segment terminals with the same sequence number. Hence the cut terminal with sequence number equal to 1 is oriented to the line segment's terminal with sequence number equal to 1.

The cut terminals also act as connection points for jumpers and other equipment, e.g. a mobile generator. To enable this, connectivity nodes are placed at the cut terminals. Once the connectivity nodes are in place any conducting equipment can be connected at them.

Table 79 shows all attributes of Cut.

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Table 79 – Attributes of CoreEquipmentProfile::Cut

name	mult	type	description
lengthFromTerminal1	0..1	Length	The length to the place where the cut is located starting from side one of the cut line segment, i.e. the line segment Terminal with sequenceNumber equal to 1.
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 80 shows all association ends of Cut with other classes.

Table 80 – Association ends of CoreEquipmentProfile::Cut with other classes

mult from	name	mult to	type	description
0..*	ACLineSegment	1..1	ACLineSegment	The line segment to which the cut is applied.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.47 DayType

Inheritance path = IdentifiedObject

Group of similar days. For example it could be used to represent weekdays, weekend, or holidays.

Table 81 shows all attributes of DayType.

Table 81 – Attributes of CoreEquipmentProfile::DayType

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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6.48 (abstract) DCBaseTerminal

Inheritance path = ACDCTerminal: IdentifiedObject

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model requires that DC connections are distinct from AC connections.

Table 82 shows all attributes of DCBaseTerminal.

Table 82 – Attributes of CoreEquipmentProfile::DCBaseTerminal

name	mult	type	description
sequenceNumber	1..1	Integer	inherited from: ACDCTerminal
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 83 shows all association ends of DCBaseTerminal with other classes.

Table 83 – Association ends of CoreEquipmentProfile:: DCBaseTerminal with other classes

mult from	name	mult to	type	description
0..*	DCNode	0..1	DCNode	The DC connectivity node to which this DC base terminal connects with zero impedance.
1..*	BusNameMarker	0..1	BusNameMarker	inherited from: ACDCTerminal

6.49 DCBreaker

Inheritance path = DCSwitch: DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A breaker within a DC system.

Table 84 shows all attributes of DCBreaker.

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Table 84 – Attributes of CoreEquipmentProfile::DCBreaker

name	mult	type	description
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 85 shows all association ends of DCBreaker with other classes.

Table 85 – Association ends of CoreEquipmentProfile::DCBreaker with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.50 DCBusbar

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A busbar within a DC system.

Table 86 shows all attributes of DCBusbar.

Table 86 – Attributes of CoreEquipmentProfile::DCBusbar

name	mult	type	description
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 87 shows all association ends of DCBusbar with other classes.

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Table 87 – Association ends of CoreEquipmentProfile::DCBusbar with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.51 DCChopper

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Low resistance equipment used in the internal DC circuit to balance voltages. It has typically positive and negative pole terminals and a ground.

Table 88 shows all attributes of DCChopper.

Table 88 – Attributes of CoreEquipmentProfile::DCChopper

name	mult	type	description
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 89 shows all association ends of DCChopper with other classes.

Table 89 – Association ends of CoreEquipmentProfile::DCChopper with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.52 (abstract) DCConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the DC power system that are designed to carry current or that are conductively connected through DC terminals.

Table 90 shows all attributes of DCConductingEquipment.

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Table 90 – Attributes of CoreEquipmentProfile::DCConductingEquipment

name	mult	type	description
ratedUdc	1..1	Voltage	Rated DC device voltage. The attribute shall be a positive value. It is configuration data used in power flow.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 91 shows all association ends of DCConductingEquipment with other classes.

Table 91 – Association ends of CoreEquipmentProfile:: DCConductingEquipment with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.53 DCConverterUnit

Inheritance path = DCEquipmentContainer: EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

Indivisible operative unit comprising all equipment between the point of common coupling on the AC side and the point of common coupling – DC side, essentially one or more converters, together with one or more converter transformers, converter control equipment, essential protective and switching devices and auxiliaries, if any, used for conversion.

Table 92 shows all attributes of DCConverterUnit.

Table 92 – Attributes of CoreEquipmentProfile::DCConverterUnit

name	mult	type	description
operationMode	1..1	DCConverterOperatingModeKind	The operating mode of an HVDC bipole (bipolar, monopolar metallic return, etc).
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 93 shows all association ends of DCConverterUnit with other classes.

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Table 93 – Association ends of CoreEquipmentProfile:: DCConverterUnit with other classes

mult from	name	mult to	type	description
0..*	Substation	0..1	Substation	The containing substation of the DC converter unit.

6.54 DCDisconnector

Inheritance path = DCSwitch: DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A disconnector within a DC system.

Table 94 shows all attributes of DCDisconnector.

Table 94 – Attributes of CoreEquipmentProfile::DCDisconnector

name	mult	type	description
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 95 shows all association ends of DCDisconnector with other classes.

Table 95 – Association ends of CoreEquipmentProfile:: DCDisconnector with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.55 (abstract) DCEquipmentContainer

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A modelling construct to provide a root class for containment of DC as well as AC equipment. The class differ from the EquipmentContaner for AC in that it may also contain DCNode-s. Hence it can contain both AC and DC equipment.

Table 96 shows all attributes of DCEquipmentContainer.

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Table 96 – Attributes of CoreEquipmentProfile::DCEquipmentContainer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.56 DCGround

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A ground within a DC system.

Table 97 shows all attributes of DCGround.

Table 97 – Attributes of CoreEquipmentProfile::DCGround

name	mult	type	description
inductance	0..1	Inductance	Inductance to ground.
r	0..1	Resistance	Resistance to ground.
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 98 shows all association ends of DCGround with other classes.

Table 98 – Association ends of CoreEquipmentProfile::DCGround with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.57 DCLine

Inheritance path = DCEquipmentContainer: EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

Overhead lines and/or cables connecting two or more HVDC substations.

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Table 99 shows all attributes of DCLine.

Table 99 – Attributes of CoreEquipmentProfile::DCLine

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 100 shows all association ends of DCLine with other classes.

Table 100 – Association ends of CoreEquipmentProfile::DCLine with other classes

mult from	name	mult to	type	description
0..*	Region	0..1	SubGeographicalRegion	The SubGeographicalRegion containing the DC line.

6.58 DCLineSegment

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A wire or combination of wires not insulated from one another, with consistent electrical characteristics, used to carry direct current between points in the DC region of the power system.

Table 101 shows all attributes of DCLineSegment.

Table 101 – Attributes of CoreEquipmentProfile::DCLineSegment

name	mult	type	description
capacitance	1..1	Capacitance	Capacitance of the DC line segment. Significant for cables only.
inductance	1..1	Inductance	Inductance of the DC line segment. Negligible compared with DCSeriesDevice used for smoothing.
resistance	1..1	Resistance	Resistance of the DC line segment.
length	0..1	Length	Segment length for calculating line section capabilities.
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 102 shows all association ends of DCLineSegment with other classes.

Table 102 – Association ends of CoreEquipmentProfile:: DCLineSegment with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.59 DCNode

Inheritance path = IdentifiedObject

DC nodes are points where terminals of DC conducting equipment are connected together with zero impedance.

Table 103 shows all attributes of DCNode.

Table 103 – Attributes of CoreEquipmentProfile::DCNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 104 shows all association ends of DCNode with other classes.

Table 104 – Association ends of CoreEquipmentProfile::DCNode with other classes

mult from	name	mult to	type	description
0..*	DCEquipmentContainer	1..1	DCEquipmentContainer	The DC container for the DC nodes.

6.60 RegulationSchedule

Inheritance path = SeasonDayTypeSchedule: RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

A pre-established pattern over time for a controlled variable, e.g. busbar voltage.

Table 105 shows all attributes of RegulationSchedule.

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Table 105 – Attributes of CoreEquipmentProfile::RegulationSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 106 shows all association ends of RegulationSchedule with other classes.

Table 106 – Association ends of CoreEquipmentProfile:: RegulationSchedule with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	1..1	RegulatingControl	Regulating controls that have this schedule.
0..*	DayType	1..1	DayType	inherited from: SeasonDayTypeSchedule
0..*	Season	1..1	Season	inherited from: SeasonDayTypeSchedule

6.61 DCSeriesDevice

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A series device within the DC system, typically a reactor used for filtering or smoothing. Needed for transient and short circuit studies.

Table 107 shows all attributes of DCSeriesDevice.

Table 107 – Attributes of CoreEquipmentProfile::DCSeriesDevice

name	mult	type	description
inductance	1..1	Inductance	Inductance of the device.
resistance	1..1	Resistance	Resistance of the DC device.
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 108 shows all association ends of DCSeriesDevice with other classes.

Table 108 – Association ends of CoreEquipmentProfile:: DCSeriesDevice with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.62 DCShunt

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A shunt device within the DC system, typically used for filtering. Needed for transient and short circuit studies.

Table 109 shows all attributes of DCShunt.

Table 109 – Attributes of CoreEquipmentProfile::DCShunt

name	mult	type	description
capacitance	1..1	Capacitance	Capacitance of the DC shunt.
resistance	1..1	Resistance	Resistance of the DC device.
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 110 shows all association ends of DCShunt with other classes.

Table 110 – Association ends of CoreEquipmentProfile::DCShunt with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.63 DCSwitch

Inheritance path = DCConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A switch within the DC system.

Table 111 shows all attributes of DCSwitch.

Table 111 – Attributes of CoreEquipmentProfile::DCSwitch

name	mult	type	description
ratedUdc	1..1	Voltage	inherited from: DCConductingEquipment
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 112 shows all association ends of DCSwitch with other classes.

Table 112 – Association ends of CoreEquipmentProfile::DCSwitch with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.64 DCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

An electrical connection point to generic DC conducting equipment.

Table 113 shows all attributes of DCTerminal.

Table 113 – Attributes of CoreEquipmentProfile::DCTerminal

name	mult	type	description
sequenceNumber	1..1	Integer	inherited from: ACDCTerminal
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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Table 114 shows all association ends of DCTerminal with other classes.

Table 114 – Association ends of CoreEquipmentProfile::DCTerminal with other classes

mult from	name	mult to	type	description
0..*	DCConductingEquipment	1..1	DCConductingEquipment	An DC terminal belong to a DC conducting equipment.
0..*	DCNode	0..1	DCNode	inherited from: DCBaseTerminal
1..*	BusNameMarker	0..1	BusNameMarker	inherited from: ACDCTerminal

6.65 Disconnector

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A manually operated or motor operated mechanical switching device used for changing the connections in a circuit, or for isolating a circuit or equipment from a source of power. It is required to open or close circuits when negligible current is broken or made.

Table 115 shows all attributes of Disconnector.

Table 115 – Attributes of CoreEquipmentProfile::Disconnector

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 116 shows all association ends of Disconnector with other classes.

Table 116 – Association ends of CoreEquipmentProfile:: Disconnector with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.66 DisconnectingCircuitBreaker

Inheritance path = Breaker: ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A circuit breaking device including disconnecting function, eliminating the need for separate disconnectors.

Table 117 shows all attributes of DisconnectingCircuitBreaker.

Table 117 – Attributes of CoreEquipmentProfile::DisconnectingCircuitBreaker

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 118 shows all association ends of DisconnectingCircuitBreaker with other classes.

Table 118 – Association ends of CoreEquipmentProfile:: DisconnectingCircuitBreaker with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.67 EquivalentBranch

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent branches. In cases where a transformer phase shift is modelled and the EquivalentBranch is spanning the same nodes, the impedance quantities for the EquivalentBranch shall consider the needed phase shift.

Table 119 shows all attributes of EquivalentBranch.

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Table 119 – Attributes of CoreEquipmentProfile::EquivalentBranch

name	mult	type	description
r	1..1	Resistance	Positive sequence series resistance of the reduced branch.
r21	0..1	Resistance	Resistance from terminal sequence 2 to terminal sequence 1. Used for steady state power flow. This attribute is optional and represent unbalanced network such as off-nominal phase shifter. If only EquivalentBranch.r is given, then EquivalentBranch.r21 is assumed equal to EquivalentBranch.r. Usage rule: EquivalentBranch is a result of network reduction prior to the data exchange.
x	1..1	Reactance	Positive sequence series reactance of the reduced branch.
x21	0..1	Reactance	Reactance from terminal sequence 2 to terminal sequence 1. Used for steady state power flow. This attribute is optional and represents an unbalanced network such as off-nominal phase shifter. If only EquivalentBranch.x is given, then EquivalentBranch.x21 is assumed equal to EquivalentBranch.x. Usage rule: EquivalentBranch is a result of network reduction prior to the data exchange.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 120 shows all association ends of EquivalentBranch with other classes.

Table 120 – Association ends of CoreEquipmentProfile:: EquivalentBranch with other classes

mult from	name	mult to	type	description
0..*	EquivalentNetwork	0..1	EquivalentNetwork	inherited from: EquivalentEquipment
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.68 (abstract) EquivalentEquipment

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent objects that are the result of a network reduction. The class is the base for equivalent objects of different types.

Table 121 shows all attributes of EquivalentEquipment.

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Table 121 – Attributes of CoreEquipmentProfile::EquivalentEquipment

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 122 shows all association ends of EquivalentEquipment with other classes.

Table 122 – Association ends of CoreEquipmentProfile:: EquivalentEquipment with other classes

mult from	name	mult to	type	description
0..*	EquivalentNetwork	0..1	EquivalentNetwork	The equivalent where the reduced model belongs.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.69 EquivalentInjection

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents equivalent injections (generation or load). Voltage regulation is allowed only at the point of connection.

Table 123 shows all attributes of EquivalentInjection.

Table 123 – Attributes of CoreEquipmentProfile::EquivalentInjection

name	mult	type	description
maxP	0..1	ActivePower	Maximum active power of the injection.
maxQ	0..1	ReactivePower	Maximum reactive power of the injection. Used for modelling of infeed for load flow exchange. Not used for short circuit modelling. If maxQ and minQ are not used ReactiveCapabilityCurve can be used.
minP	0..1	ActivePower	Minimum active power of the injection.
minQ	0..1	ReactivePower	Minimum reactive power of the injection. Used for modelling of infeed for load flow exchange. Not used for short circuit modelling. If maxQ and minQ are not used ReactiveCapabilityCurve can be used.

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name	mult	type	description
regulationCapability	1..1	Boolean	Specifies whether or not the EquivalentInjection has the capability to regulate the local voltage. If true the EquivalentInjection can regulate. If false the EquivalentInjection cannot regulate. ReactiveCapabilityCurve can only be associated with EquivalentInjection if the flag is true.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 124 shows all association ends of EquivalentInjection with other classes.

Table 124 – Association ends of CoreEquipmentProfile:: EquivalentInjection with other classes

mult from	name	mult to	type	description
0..*	ReactiveCapabilityCurve	0..1	ReactiveCapabilityCurve	The reactive capability curve used by this equivalent injection.
0..*	EquivalentNetwork	0..1	EquivalentNetwork	inherited from: EquivalentEquipment
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.70 EquivalentNetwork

Inheritance path = ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A class that groups electrical equivalents, including internal nodes, of a network that has been reduced. The ConnectivityNodes contained in the equivalent are intended to reflect internal nodes of the equivalent. The boundary Connectivity nodes where the equivalent connects outside itself are not contained by the equivalent.

Table 125 shows all attributes of EquivalentNetwork.

Table 125 – Attributes of CoreEquipmentProfile::EquivalentNetwork

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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6.71 EquivalentShunt

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent shunts.

Table 126 shows all attributes of EquivalentShunt.

Table 126 – Attributes of CoreEquipmentProfile::EquivalentShunt

name	mult	type	description
b	1..1	Susceptance	Positive sequence shunt susceptance.
g	1..1	Conductance	Positive sequence shunt conductance.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 127 shows all association ends of EquivalentShunt with other classes.

Table 127 – Association ends of CoreEquipmentProfile:: EquivalentShunt with other classes

mult from	name	mult to	type	description
0..*	EquivalentNetwork	0..1	EquivalentNetwork	inherited from: EquivalentEquipment
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.72 ExternalNetworkInjection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents the external network and it is used for IEC 60909 calculations.

Table 128 shows all attributes of ExternalNetworkInjection.

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Table 128 – Attributes of CoreEquipmentProfile::ExternalNetworkInjection

name	mult	type	description
governorSCD	1..1	ActivePowerPerFrequency	Power Frequency Bias. This is the change in power injection divided by the change in frequency and negated. A positive value of the power frequency bias provides additional power injection upon a drop in frequency.
maxP	1..1	ActivePower	Maximum active power of the injection.
maxQ	1..1	ReactivePower	Maximum reactive power limit. It is used for modelling of infeed for load flow exchange and not for short circuit modelling.
minP	1..1	ActivePower	Minimum active power of the injection.
minQ	1..1	ReactivePower	Minimum reactive power limit. It is used for modelling of infeed for load flow exchange and not for short circuit modelling.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 129 shows all association ends of ExternalNetworkInjection with other classes.

Table 129 – Association ends of CoreEquipmentProfile:: ExternalNetworkInjection with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.73 FaultIndicator

Inheritance path = AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

A FaultIndicator is typically only an indicator (which may or may not be remotely monitored), and not a piece of equipment that actually initiates a protection event. It is used for FLISR (Fault Location, Isolation and Restoration) purposes, assisting with the dispatch of crews to "most likely" part of the network (i.e. assists with determining circuit section where the fault most likely happened).

Table 130 shows all attributes of FaultIndicator.

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Table 130 – Attributes of CoreEquipmentProfile::FaultIndicator

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 131 shows all association ends of FaultIndicator with other classes.

Table 131 – Association ends of CoreEquipmentProfile:: FaultIndicator with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.74 FossilFuel

Inheritance path = IdentifiedObject

The fossil fuel consumed by the non-nuclear thermal generating unit. For example, coal, oil, gas, etc. These are the specific fuels that the generating unit can consume.

Table 132 shows all attributes of FossilFuel.

Table 132 – Attributes of CoreEquipmentProfile::FossilFuel

name	mult	type	description
fossilFuelType	1..1	FuelType	The type of fossil fuel, such as coal, oil, or gas.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 133 shows all association ends of FossilFuel with other classes.

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Table 133 – Association ends of CoreEquipmentProfile::FossilFuel with other classes

mult from	name	mult to	type	description
0..*	ThermalGeneratingUnit	1..1	ThermalGeneratingUnit	A thermal generating unit may have one or more fossil fuels.

6.75 Fuse

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An overcurrent protective device with a circuit opening fusible part that is heated and severed by the passage of overcurrent through it. A fuse is considered a switching device because it breaks current.

Table 134 shows all attributes of Fuse.

Table 134 – Attributes of CoreEquipmentProfile::Fuse

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 135 shows all association ends of Fuse with other classes.

Table 135 – Association ends of CoreEquipmentProfile::Fuse with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.76 GeneratingUnit

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

A single or set of synchronous machines for converting mechanical power into alternating-current power. For example, individual machines within a set may be defined for scheduling purposes while a single control signal is derived for the set. In this case there would be a GeneratingUnit for each member of the set and an additional GeneratingUnit corresponding to the set.

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Table 136 shows all attributes of GeneratingUnit.

Table 136 – Attributes of CoreEquipmentProfile::GeneratingUnit

name	mult	type	description
genControlSource	0..1	GeneratorControlSource	The source of controls for a generating unit. Defines the control status of the generating unit.
governorSCD	0..1	PerCent	Governor Speed Changer Droop. This is the change in generator power output divided by the change in frequency normalized by the nominal power of the generator and the nominal frequency and expressed in percent and negated. A positive value of speed change droop provides additional generator output upon a drop in frequency.
longPF	0..1	Float	Generating unit long term economic participation factor.
maximumAllowableSpinningReserve	0..1	ActivePower	Maximum allowable spinning reserve. Spinning reserve will never be considered greater than this value regardless of the current operating point.
maxOperatingP	1..1	ActivePower	This is the maximum operating active power limit the dispatcher can enter for this unit.
minOperatingP	1..1	ActivePower	This is the minimum operating active power limit the dispatcher can enter for this unit.
nominalP	0..1	ActivePower	The nominal power of the generating unit. Used to give precise meaning to percentage based attributes such as the governor speed change droop (governorSCD attribute). The attribute shall be a positive value equal to or less than RotatingMachine.ratedS.
ratedGrossMaxP	0..1	ActivePower	The unit's gross rated maximum capacity (book value). The attribute shall be a positive value.
ratedGrossMinP	0..1	ActivePower	The gross rated minimum generation level which the unit can safely operate at while delivering power to the transmission grid. The attribute shall be a positive value.
ratedNetMaxP	0..1	ActivePower	The net rated maximum capacity determined by subtracting the auxiliary power used to operate the internal plant machinery from the rated gross maximum capacity. The attribute shall be a positive value.
shortPF	0..1	Float	Generating unit short term economic participation factor.
startupCost	0..1	Money	The initial startup cost incurred for each start of the GeneratingUnit.
variableCost	0..1	Money	The variable cost component of production per unit of ActivePower.
startupTime	0..1	Seconds	Time it takes to get the unit on-line, from the time that the prime mover mechanical power is applied.
totalEfficiency	0..1	PerCent	The efficiency of the unit in converting the fuel into electrical energy.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 137 shows all association ends of GeneratingUnit with other classes.

Table 137 – Association ends of CoreEquipmentProfile:: GeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.77 GeographicalRegion

Inheritance path = IdentifiedObject

A geographical region of a power system network model.

Table 138 shows all attributes of GeographicalRegion.

Table 138 – Attributes of CoreEquipmentProfile::GeographicalRegion

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.78 GrossToNetActivePowerCurve

Inheritance path = Curve: IdentifiedObject

Relationship between the generating unit's gross active power output on the X-axis (measured at the terminals of the machine(s)) and the generating unit's net active power output on the Y-axis (based on utility-defined measurements at the power station). Station service loads, when modelled, should be treated as non-conforming bus loads. There may be more than one curve, depending on the auxiliary equipment that is in service.

Table 139 shows all attributes of GrossToNetActivePowerCurve.

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Table 139 – Attributes of CoreEquipmentProfile::GrossToNetActivePowerCurve

name	mult	type	description
curveStyle	1..1	CurveStyle	inherited from: Curve
xUnit	1..1	UnitSymbol	inherited from: Curve
y1Unit	1..1	UnitSymbol	inherited from: Curve
y2Unit	0..1	UnitSymbol	inherited from: Curve
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 140 shows all association ends of GrossToNetActivePowerCurve with other classes.

Table 140 – Association ends of CoreEquipmentProfile:: GrossToNetActivePowerCurve with other classes

mult from	name	mult to	type	description
0..*	GeneratingUnit	1..1	GeneratingUnit	A generating unit may have a gross active power to net active power curve, describing the losses and auxiliary power requirements of the unit.

6.79 Ground

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A point where the system is grounded used for connecting conducting equipment to ground. The power system model can have any number of grounds.

Table 141 shows all attributes of Ground.

Table 141 – Attributes of CoreEquipmentProfile::Ground

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 142 shows all association ends of Ground with other classes.

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Table 142 – Association ends of CoreEquipmentProfile::Ground with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.80 GroundDisconnector

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A manually operated or motor operated mechanical switching device used for isolating a circuit or equipment from ground.

Table 143 shows all attributes of GroundDisconnector.

Table 143 – Attributes of CoreEquipmentProfile::GroundDisconnector

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 144 shows all association ends of GroundDisconnector with other classes.

Table 144 – Association ends of CoreEquipmentProfile:: GroundDisconnector with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.81 GroundingImpedance

Inheritance path = EarthFaultCompensator: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A fixed impedance device used for grounding.

Table 145 shows all attributes of GroundingImpedance.

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Table 145 – Attributes of CoreEquipmentProfile::GroundingImpedance

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 146 shows all association ends of GroundingImpedance with other classes.

Table 146 – Association ends of CoreEquipmentProfile:: GroundingImpedance with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.82 HydroGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A generating unit whose prime mover is a hydraulic turbine (e.g., Francis, Pelton, Kaplan).

Table 147 shows all attributes of HydroGeneratingUnit.

Table 147 – Attributes of CoreEquipmentProfile::HydroGeneratingUnit

name	mult	type	description
energyConversionCapability	0..1	HydroEnergyConversionKind	Energy conversion capability for generating.
dropHeight	0..1	Length	The height water drops from the reservoir mid-point to the turbine.
turbineType	0..1	HydroTurbineKind	Type of turbine.
genControlSource	0..1	GeneratorControlSource	inherited from: GeneratingUnit
governorSCD	0..1	PerCent	inherited from: GeneratingUnit
longPF	0..1	Float	inherited from: GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	inherited from: GeneratingUnit
maxOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
minOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
nominalP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMinP	0..1	ActivePower	inherited from: GeneratingUnit
ratedNetMaxP	0..1	ActivePower	inherited from: GeneratingUnit

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name	mult	type	description
shortPF	0..1	Float	inherited from: GeneratingUnit
startupCost	0..1	Money	inherited from: GeneratingUnit
variableCost	0..1	Money	inherited from: GeneratingUnit
startupTime	0..1	Seconds	inherited from: GeneratingUnit
totalEfficiency	0..1	PerCent	inherited from: GeneratingUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 148 shows all association ends of HydroGeneratingUnit with other classes.

Table 148 – Association ends of CoreEquipmentProfile:: HydroGeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	HydroPowerPlant	0..1	HydroPowerPlant	The hydro generating unit belongs to a hydro power plant.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.83 HydroPowerPlant

Inheritance path = PowerSystemResource: IdentifiedObject

A hydro power station which can generate or pump. When generating, the generator turbines receive water from an upper reservoir. When pumping, the pumps receive their water from a lower reservoir.

Table 149 shows all attributes of HydroPowerPlant.

Table 149 – Attributes of CoreEquipmentProfile::HydroPowerPlant

name	mult	type	description
hydroPlantStorageType	1..1	HydroPlantStorageKind	The type of hydro power plant water storage.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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6.84 HydroPump

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

A synchronous motor-driven pump, typically associated with a pumped storage plant.

Table 150 shows all attributes of HydroPump.

Table 150 – Attributes of CoreEquipmentProfile::HydroPump

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 151 shows all association ends of HydroPump with other classes.

Table 151 – Association ends of CoreEquipmentProfile::HydroPump with other classes

mult from	name	mult to	type	description
0..1	RotatingMachine	1..1	RotatingMachine	The synchronous machine drives the turbine which moves the water from a low elevation to a higher elevation. The direction of machine rotation for pumping may or may not be the same as for generating.
0..*	HydroPowerPlant	0..1	HydroPowerPlant	The hydro pump may be a member of a pumped storage plant or a pump for distributing water.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.85 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 152 shows all attributes of IdentifiedObject.

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Table 152 – Attributes of CoreEquipmentProfile::IdentifiedObject

name	mult	type	description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
energyIdentCodeEic	0..1	String	(deprecated,European) The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName	0..1	String	(deprecated,European) The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

6.86 Jumper

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A short section of conductor with negligible impedance which can be manually removed and replaced if the circuit is de-energized. Note that zero-impedance branches can potentially be modelled by other equipment types.

Table 153 shows all attributes of Jumper.

Table 153 – Attributes of CoreEquipmentProfile::Jumper

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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Table 154 shows all association ends of Jumper with other classes.

Table 154 – Association ends of CoreEquipmentProfile::Jumper with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.87 Junction

Inheritance path = Connector: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A point where one or more conducting equipments are connected with zero resistance.

Table 155 shows all attributes of Junction.

Table 155 – Attributes of CoreEquipmentProfile::Junction

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 156 shows all association ends of Junction with other classes.

Table 156 – Association ends of CoreEquipmentProfile::Junction with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.88 Line

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

Contains equipment beyond a substation belonging to a power transmission line.

Table 157 shows all attributes of Line.

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Table 157 – Attributes of CoreEquipmentProfile::Line

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 158 shows all association ends of Line with other classes.

Table 158 – Association ends of CoreEquipmentProfile::Line with other classes

mult from	name	mult to	type	description
0..*	Region	0..1	SubGeographicalRegion	The sub-geographical region of the line.

6.89 LinearShuntCompensator

Inheritance path = ShuntCompensator: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A linear shunt compensator has banks or sections with equal admittance values.

Table 159 shows all attributes of LinearShuntCompensator.

Table 159 – Attributes of CoreEquipmentProfile::LinearShuntCompensator

name	mult	type	description
bPerSection	1..1	Susceptance	Positive sequence shunt (charging) susceptance per section.
gPerSection	1..1	Conductance	Positive sequence shunt (charging) conductance per section.
aVRDelay	0..1	Seconds	inherited from: ShuntCompensator
grounded	0..1	Boolean	inherited from: ShuntCompensator
maximumSections	1..1	Integer	inherited from: ShuntCompensator
nomU	1..1	Voltage	inherited from: ShuntCompensator
normalSections	1..1	Integer	inherited from: ShuntCompensator
voltageSensitivity	0..1	VoltagePerReactivePower	inherited from: ShuntCompensator
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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Table 160 shows all association ends of LinearShuntCompensator with other classes.

Table 160 – Association ends of CoreEquipmentProfile:: LinearShuntCompensator with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.90 LoadArea

Inheritance path = EnergyArea: IdentifiedObject

The class is the root or first level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

Table 161 shows all attributes of LoadArea.

Table 161 – Attributes of CoreEquipmentProfile::LoadArea

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.91 LoadBreakSwitch

Inheritance path = ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A mechanical switching device capable of making, carrying, and breaking currents under normal operating conditions.

Table 162 shows all attributes of LoadBreakSwitch.

Table 162 – Attributes of CoreEquipmentProfile::LoadBreakSwitch

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 163 shows all association ends of LoadBreakSwitch with other classes.

Table 163 – Association ends of CoreEquipmentProfile:: LoadBreakSwitch with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.92 (abstract) LoadGroup

Inheritance path = IdentifiedObject

The class is the third level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

Table 164 shows all attributes of LoadGroup.

Table 164 – Attributes of CoreEquipmentProfile::LoadGroup

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 165 shows all association ends of LoadGroup with other classes.

Table 165 – Association ends of CoreEquipmentProfile::LoadGroup with other classes

mult from	name	mult to	type	description
1..*	SubLoadArea	1..1	SubLoadArea	The SubLoadArea where the Loadgroup belongs.

6.93 LoadResponseCharacteristic

Inheritance path = IdentifiedObject

Models the characteristic response of the load demand due to changes in system conditions such as voltage and frequency. It is not related to demand response.

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If LoadResponseCharacteristic.exponentModel is True, the exponential voltage or frequency dependent models are specified and used as to calculate active and reactive power components of the load model.

The equations to calculate active and reactive power components of the load model are internal to the power flow calculation, hence they use different quantities depending on the use case of the data exchange.

The equations for exponential voltage dependent load model injected power are:

pInjection = Pnominal* (Voltage/cim:BaseVoltage.nominalVoltage) **
cim:LoadResponseCharacteristic.pVoltageExponent

qInjection = Qnominal* (Voltage/cim:BaseVoltage.nominalVoltage) **
cim:LoadResponseCharacteristic.qVoltageExponent

Where:

1) * means "multiply" and ** is "raised to power of";
2) Pnominal and Qnominal represent the active power and reactive power at nominal voltage as any load described by the voltage exponential model shall be given at nominal voltage. This means that EnergyConsumer.p and EnergyConsumer.q are at nominal voltage.
3) After power flow is solved:
   – pInjection and qInjection correspond to SvPowerflow.p and SvPowerflow.q respectively.
   – Voltage corresponds to SvVoltage.v at the TopologicalNode where the load is connected.

Table 166 shows all attributes of LoadResponseCharacteristic.

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Table 166 – Attributes of CoreEquipmentProfile::LoadResponseCharacteristic

name	mult	type	description
exponentModel	1..1	Boolean	Indicates the exponential voltage dependency model is to be used. If false, the coefficient model is to be used. The exponential voltage dependency model consist of the attributes: – pVoltageExponent – qVoltageExponent – pFrequencyExponent – qFrequencyExponent. The coefficient model consist of the attributes: – pConstantImpedance – pConstantCurrent – pConstantPower – qConstantImpedance – qConstantCurrent – qConstantPower. The sum of pConstantImpedance, pConstantCurrent and pConstantPower shall equal 1. The sum of qConstantImpedance, qConstantCurrent and qConstantPower shall equal 1.
pConstantCurrent	0..1	Float	Portion of active power load modelled as constant current.
pConstantImpedance	0..1	Float	Portion of active power load modelled as constant impedance.
pConstantPower	0..1	Float	Portion of active power load modelled as constant power.
pFrequencyExponent	0..1	Float	Exponent of per unit frequency effecting active power.
pVoltageExponent	0..1	Float	Exponent of per unit voltage effecting real power.
qConstantCurrent	0..1	Float	Portion of reactive power load modelled as constant current.
qConstantImpedance	0..1	Float	Portion of reactive power load modelled as constant impedance.
qConstantPower	0..1	Float	Portion of reactive power load modelled as constant power.
qFrequencyExponent	0..1	Float	Exponent of per unit frequency effecting reactive power.
qVoltageExponent	0..1	Float	Exponent of per unit voltage effecting reactive power.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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6.94 NonConformLoad

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

NonConformLoad represents loads that do not follow a daily load change pattern and whose changes are not correlated with the daily load change pattern.

Table 167 shows all attributes of NonConformLoad.

Table 167 – Attributes of CoreEquipmentProfile::NonConformLoad

name	mult	type	description
pfixed	0..1	ActivePower	inherited from: EnergyConsumer
pfixedPct	0..1	PerCent	inherited from: EnergyConsumer
qfixed	0..1	ReactivePower	inherited from: EnergyConsumer
qfixedPct	0..1	PerCent	inherited from: EnergyConsumer
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 168 shows all association ends of NonConformLoad with other classes.

Table 168 – Association ends of CoreEquipmentProfile:: NonConformLoad with other classes

mult from	name	mult to	type	description
1..*	LoadGroup	1..1	NonConformLoadGroup	Group of this ConformLoad.
0..*	LoadResponse	0..1	LoadResponseCharacteristic	inherited from: EnergyConsumer
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.95 NonConformLoadGroup

Inheritance path = LoadGroup: IdentifiedObject

Loads that do not follow a daily and seasonal load variation pattern.

Table 169 shows all attributes of NonConformLoadGroup.

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Table 169 – Attributes of CoreEquipmentProfile::NonConformLoadGroup

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 170 shows all association ends of NonConformLoadGroup with other classes.

Table 170 – Association ends of CoreEquipmentProfile:: NonConformLoadGroup with other classes

mult from	name	mult to	type	description
1..*	SubLoadArea	1..1	SubLoadArea	inherited from: LoadGroup

6.96 NonConformLoadSchedule

Inheritance path = SeasonDayTypeSchedule: RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

An active power (Y1-axis) and reactive power (Y2-axis) schedule (curves) versus time (X-axis) for non-conforming loads, e.g., large industrial load or power station service (where modelled).

Table 171 shows all attributes of NonConformLoadSchedule.

Table 171 – Attributes of CoreEquipmentProfile::NonConformLoadSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 172 shows all association ends of NonConformLoadSchedule with other classes.

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Table 172 – Association ends of CoreEquipmentProfile:: NonConformLoadSchedule with other classes

mult from	name	mult to	type	description
0..*	NonConformLoadGroup	1..1	NonConformLoadGroup	The NonConformLoadGroup where the NonConformLoadSchedule belongs.
0..*	DayType	1..1	DayType	inherited from: SeasonDayTypeSchedule
0..*	Season	1..1	Season	inherited from: SeasonDayTypeSchedule

6.97 NonlinearShuntCompensator

Inheritance path = ShuntCompensator: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A non linear shunt compensator has bank or section admittance values that differ. The attributes g, b, g0 and b0 of the associated NonlinearShuntCompensatorPoint describe the total conductance and admittance of a NonlinearShuntCompensatorPoint at a section number specified by NonlinearShuntCompensatorPoint.sectionNumber.

Table 173 shows all attributes of NonlinearShuntCompensator.

Table 173 – Attributes of CoreEquipmentProfile::NonlinearShuntCompensator

name	mult	type	description
aVRDelay	0..1	Seconds	inherited from: ShuntCompensator
grounded	0..1	Boolean	inherited from: ShuntCompensator
maximumSections	1..1	Integer	inherited from: ShuntCompensator
nomU	1..1	Voltage	inherited from: ShuntCompensator
normalSections	1..1	Integer	inherited from: ShuntCompensator
voltageSensitivity	0..1	VoltagePerReactivePower	inherited from: ShuntCompensator
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 174 shows all association ends of NonlinearShuntCompensator with other classes.

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Table 174 – Association ends of CoreEquipmentProfile:: NonlinearShuntCompensator with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.98 NonlinearShuntCompensatorPoint root class

A non linear shunt compensator bank or section admittance value. The number of NonlinearShuntCompenstorPoint instances associated with a NonlinearShuntCompensator shall be equal to ShuntCompensator.maximumSections. ShuntCompensator.sections shall only be set to one of the NonlinearShuntCompenstorPoint.sectionNumber. There is no interpolation between NonlinearShuntCompenstorPoint-s.

Table 175 shows all attributes of NonlinearShuntCompensatorPoint.

Table 175 – Attributes of CoreEquipmentProfile::NonlinearShuntCompensatorPoint

name	mult	type	description
b	1..1	Susceptance	Positive sequence shunt (charging) susceptance per section.
g	1..1	Conductance	Positive sequence shunt (charging) conductance per section.
sectionNumber	1..1	Integer	The number of the section.

Table 176 shows all association ends of NonlinearShuntCompensatorPoint with other classes.

Table 176 – Association ends of CoreEquipmentProfile:: NonlinearShuntCompensatorPoint with other classes

mult from	name	mult to	type	description
1..*	NonlinearShuntCompensator	1..1	NonlinearShuntCompensator	Non-linear shunt compensator owning this point.

6.99 NuclearGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A nuclear generating unit.

Table 177 shows all attributes of NuclearGeneratingUnit.

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Table 177 – Attributes of CoreEquipmentProfile::NuclearGeneratingUnit

name	mult	type	description
genControlSource	0..1	GeneratorControlSource	inherited from: GeneratingUnit
governorSCD	0..1	PerCent	inherited from: GeneratingUnit
longPF	0..1	Float	inherited from: GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	inherited from: GeneratingUnit
maxOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
minOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
nominalP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMinP	0..1	ActivePower	inherited from: GeneratingUnit
ratedNetMaxP	0..1	ActivePower	inherited from: GeneratingUnit
shortPF	0..1	Float	inherited from: GeneratingUnit
startupCost	0..1	Money	inherited from: GeneratingUnit
variableCost	0..1	Money	inherited from: GeneratingUnit
startupTime	0..1	Seconds	inherited from: GeneratingUnit
totalEfficiency	0..1	PerCent	inherited from: GeneratingUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 178 shows all association ends of NuclearGeneratingUnit with other classes.

Table 178 – Association ends of CoreEquipmentProfile:: NuclearGeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.100 (abstract) OperationalLimit

Inheritance path = IdentifiedObject

A value and normal value associated with a specific kind of limit.

The sub class value and normalValue attributes vary inversely to the associated OperationalLimitType.acceptableDuration (acceptableDuration for short).

If a particular piece of equipment has multiple operational limits of the same kind (apparent power, current, etc.), the limit with the greatest acceptableDuration shall have the smallest limit value and the limit with the smallest acceptableDuration shall have the largest limit value.

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NOTE A large current can only be allowed to flow through a piece of equipment for a short duration without causing damage, but a lesser current can be allowed to flow for a longer duration.

Table 179 shows all attributes of OperationalLimit.

Table 179 – Attributes of CoreEquipmentProfile::OperationalLimit

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 180 shows all association ends of OperationalLimit with other classes.

Table 180 – Association ends of CoreEquipmentProfile:: OperationalLimit with other classes

mult from	name	mult to	type	description
0..*	OperationalLimitType	1..1	OperationalLimitType	The limit type associated with this limit.
0..*	OperationalLimitSet	1..1	OperationalLimitSet	The limit set to which the limit values belong.

6.101 OperationalLimitSet

Inheritance path = IdentifiedObject

A set of limits associated with equipment. Sets of limits might apply to a specific temperature, or season for example. A set of limits may contain different severities of limit levels that would apply to the same equipment. The set may contain limits of different types such as apparent power and current limits or high and low voltage limits that are logically applied together as a set.

Table 181 shows all attributes of OperationalLimitSet.

Table 181 – Attributes of CoreEquipmentProfile::OperationalLimitSet

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 182 shows all association ends of OperationalLimitSet with other classes.

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Table 182 – Association ends of CoreEquipmentProfile:: OperationalLimitSet with other classes

mult from	name	mult to	type	description
0..*	Equipment	0..1	Equipment	The equipment to which the limit set applies.
0..*	Terminal	1..1	ACDCTerminal	The terminal where the operational limit set apply.

6.102 OperationalLimitType

Inheritance path = IdentifiedObject

The operational meaning of a category of limits.

Table 183 shows all attributes of OperationalLimitType.

Table 183 – Attributes of CoreEquipmentProfile::OperationalLimitType

name	mult	type	description
acceptableDuration	0..1	Seconds	The nominal acceptable duration of the limit. Limits are commonly expressed in terms of the time limit for which the limit is normally acceptable. The actual acceptable duration of a specific limit may depend on other local factors such as temperature or wind speed. The attribute has meaning only if the flag isInfiniteDuration is set to false, hence it shall not be exchanged when isInfiniteDuration is set to true.
direction	1..1	OperationalLimitDirectionKind	The direction of the limit.
isInfiniteDuration	1..1	Boolean	Defines if the operational limit type has infinite duration. If true, the limit has infinite duration. If false, the limit has definite duration which is defined by the attribute acceptableDuration.
kind	1..1	LimitKind	(European) Types of limits defined in the ENTSO-E Operational Handbook Policy 3.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.103 PetersenCoil

Inheritance path = EarthFaultCompensator: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A variable impedance device normally used to offset line charging during single line faults in an ungrounded section of network.

Table 184 shows all attributes of PetersenCoil.

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Table 184 – Attributes of CoreEquipmentProfile::PetersenCoil

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 185 shows all association ends of PetersenCoil with other classes.

Table 185 – Association ends of CoreEquipmentProfile::PetersenCoil with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.104 (abstract) PhaseTapChanger

Inheritance path = TapChanger: PowerSystemResource: IdentifiedObject

A transformer phase shifting tap model that controls the phase angle difference across the power transformer and potentially the active power flow through the power transformer. This phase tap model may also impact the voltage magnitude.

Table 186 shows all attributes of PhaseTapChanger.

Table 186 – Attributes of CoreEquipmentProfile::PhaseTapChanger

name	mult	type	description
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 187 shows all association ends of PhaseTapChanger with other classes.

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Table 187 – Association ends of CoreEquipmentProfile:: PhaseTapChanger with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	Transformer end to which this phase tap changer belongs.
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.105 PhaseTapChangerAsymmetrical

Inheritance path = PhaseTapChangerNonLinear: PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes the tap model for an asymmetrical phase shifting transformer in which the difference voltage vector adds to the in-phase winding. The out-of-phase winding is the transformer end where the tap changer is located. The angle between the in-phase and out-of-phase windings is named the winding connection angle. The phase shift depends on both the difference voltage magnitude and the winding connection angle.

Table 188 shows all attributes of PhaseTapChangerAsymmetrical.

Table 188 – Attributes of CoreEquipmentProfile::PhaseTapChangerAsymmetrical

name	mult	type	description
windingConnectionAngle	1..1	AngleDegrees	The phase angle between the in-phase winding and the out-of -phase winding used for creating phase shift. The out-of-phase winding produces what is known as the difference voltage. Setting this angle to 90 degrees is not the same as a symmetrical transformer. The attribute can only be multiples of 30 degrees. The allowed range is -150 degrees to 150 degrees excluding 0.
voltageStepIncrement	1..1	PerCent	inherited from: PhaseTapChangerNonLinear
xMax	1..1	Reactance	inherited from: PhaseTapChangerNonLinear
xMin	1..1	Reactance	(deprecated) inherited from: PhaseTapChangerNonLinear
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 189 shows all association ends of PhaseTapChangerAsymmetrical with other classes.

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Table 189 – Association ends of CoreEquipmentProfile:: PhaseTapChangerAsymmetrical with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	inherited from: PhaseTapChanger
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.106 PhaseTapChangerLinear

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes a tap changer with a linear relation between the tap step and the phase angle difference across the transformer. This is a mathematical model that is an approximation of a real phase tap changer.

The phase angle is computed as stepPhaseShiftIncrement times the tap position.

The voltage magnitude of both sides is the same.

Table 190 shows all attributes of PhaseTapChangerLinear.

Table 190 – Attributes of CoreEquipmentProfile::PhaseTapChangerLinear

name	mult	type	description
stepPhaseShiftIncrement	1..1	AngleDegrees	Phase shift per step position. A positive value indicates a positive angle variation from the Terminal at the PowerTransformerEnd, where the TapChanger is located, into the transformer. The actual phase shift increment might be more accurately computed from the symmetrical or asymmetrical models or a tap step table lookup if those are available.
xMax	1..1	Reactance	The reactance depends on the tap position according to a "u" shaped curve. The maximum reactance (xMax) appears at the low and high tap positions. Depending on the “u” curve the attribute can be either higher or lower than PowerTransformerEnd.x.
xMin	1..1	Reactance	(deprecated) The reactance depends on the tap position according to a "u" shaped curve. The minimum reactance (xMin) appears at the mid tap position. PowerTransformerEnd.x shall be consistent with PhaseTapChangerLinear.xMin and PhaseTapChangerNonLinear.xMin. In case of inconsistency, PowerTransformerEnd.x shall be used.
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 191 shows all association ends of PhaseTapChangerLinear with other classes.

Table 191 – Association ends of CoreEquipmentProfile:: PhaseTapChangerLinear with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	inherited from: PhaseTapChanger
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.107 (abstract) PhaseTapChangerNonLinear

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

The non-linear phase tap changer describes the non-linear behaviour of a phase tap changer. This is a base class for the symmetrical and asymmetrical phase tap changer models. The details of these models can be found in IEC 61970-301.

Table 192 shows all attributes of PhaseTapChangerNonLinear.

Table 192 – Attributes of CoreEquipmentProfile::PhaseTapChangerNonLinear

name	mult	type	description
voltageStepIncrement	1..1	PerCent	The voltage step increment on the out of phase winding (the PowerTransformerEnd where the TapChanger is located) specified in percent of rated voltage of the PowerTransformerEnd. A positive value means a positive voltage variation from the Terminal at the PowerTransformerEnd, where the TapChanger is located, into the transformer. When the increment is negative, the voltage decreases when the tap step increases.
xMax	1..1	Reactance	The reactance depends on the tap position according to a "u" shaped curve. The maximum reactance (xMax) appears at the low and high tap positions. Depending on the “u” curve the attribute can be either higher or lower than PowerTransformerEnd.x.
xMin	1..1	Reactance	(deprecated) The reactance depend on the tap position according to a "u" shaped curve. The minimum reactance (xMin) appear at the mid tap position. PowerTransformerEnd.x shall be consistent with PhaseTapChangerLinear.xMin and PhaseTapChangerNonLinear.xMin. In case of inconsistency, PowerTransformerEnd.x shall be used.
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger

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name	mult	type	description
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 193 shows all association ends of PhaseTapChangerNonLinear with other classes.

Table 193 – Association ends of CoreEquipmentProfile:: PhaseTapChangerNonLinear with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	inherited from: PhaseTapChanger
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.108 PhaseTapChangerSymmetrical

Inheritance path = PhaseTapChangerNonLinear: PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes a symmetrical phase shifting transformer tap model in which the voltage magnitude of both sides is the same. The difference voltage magnitude is the base in an equal-sided triangle where the sides corresponds to the primary and secondary voltages. The phase angle difference corresponds to the top angle and can be expressed as twice the arctangent of half the total difference voltage.

Table 194 shows all attributes of PhaseTapChangerSymmetrical.

Table 194 – Attributes of CoreEquipmentProfile::PhaseTapChangerSymmetrical

name	mult	type	description
voltageStepIncrement	1..1	PerCent	inherited from: PhaseTapChangerNonLinear
xMax	1..1	Reactance	inherited from: PhaseTapChangerNonLinear
xMin	1..1	Reactance	(deprecated) inherited from: PhaseTapChangerNonLinear
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 195 shows all association ends of PhaseTapChangerSymmetrical with other classes.

Table 195 – Association ends of CoreEquipmentProfile:: PhaseTapChangerSymmetrical with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	inherited from: PhaseTapChanger
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.109 PhaseTapChangerTable

Inheritance path = IdentifiedObject

Describes a tabular curve for how the phase angle difference and impedance varies with the tap step.

Table 196 shows all attributes of PhaseTapChangerTable.

Table 196 – Attributes of CoreEquipmentProfile::PhaseTapChangerTable

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.110 PhaseTapChangerTablePoint

Inheritance path = TapChangerTablePoint

Describes each tap step in the phase tap changer tabular curve.

Table 197 shows all attributes of PhaseTapChangerTablePoint.

Table 197 – Attributes of CoreEquipmentProfile::PhaseTapChangerTablePoint

name	mult	type	description
angle	1..1	AngleDegrees	The angle difference in degrees. A positive value indicates a positive angle variation from the Terminal at the PowerTransformerEnd, where the TapChanger is located, into the transformer.
b	0..1	PerCent	inherited from: TapChangerTablePoint
g	0..1	PerCent	inherited from: TapChangerTablePoint
r	0..1	PerCent	inherited from: TapChangerTablePoint

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name	mult	type	description
ratio	0..1	Float	inherited from: TapChangerTablePoint
step	1..1	Integer	inherited from: TapChangerTablePoint
x	0..1	PerCent	inherited from: TapChangerTablePoint

Table 198 shows all association ends of PhaseTapChangerTablePoint with other classes.

Table 198 – Association ends of CoreEquipmentProfile:: PhaseTapChangerTablePoint with other classes

mult from	name	mult to	type	description
1..*	PhaseTapChangerTable	1..1	PhaseTapChangerTable	The table of this point.

6.111 PhaseTapChangerTabular

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes a tap changer with a table defining the relation between the tap step and the phase angle difference across the transformer.

Table 199 shows all attributes of PhaseTapChangerTabular.

Table 199 – Attributes of CoreEquipmentProfile::PhaseTapChangerTabular

name	mult	type	description
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 200 shows all association ends of PhaseTapChangerTabular with other classes.

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Table 200 – Association ends of CoreEquipmentProfile:: PhaseTapChangerTabular with other classes

mult from	name	mult to	type	description
0..*	PhaseTapChangerTable	1..1	PhaseTapChangerTable	The phase tap changer table for this phase tap changer.
0..1	TransformerEnd	1..1	TransformerEnd	inherited from: PhaseTapChanger
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.112 PhotoVoltaicUnit

Inheritance path = PowerElectronicsUnit: Equipment: PowerSystemResource: IdentifiedObject

A photovoltaic device or an aggregation of such devices.

Table 201 shows all attributes of PhotoVoltaicUnit.

Table 201 – Attributes of CoreEquipmentProfile::PhotoVoltaicUnit

name	mult	type	description
maxP	0..1	ActivePower	inherited from: PowerElectronicsUnit
minP	0..1	ActivePower	inherited from: PowerElectronicsUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 202 shows all association ends of PhotoVoltaicUnit with other classes.

Table 202 – Association ends of CoreEquipmentProfile:: PhotoVoltaicUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.113 PostLineSensor

Inheritance path = Sensor: AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

A sensor used mainly in overhead distribution networks as the source of both current and voltage measurements.

Table 203 shows all attributes of PostLineSensor.

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Table 203 – Attributes of CoreEquipmentProfile::PostLineSensor

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 204 shows all association ends of PostLineSensor with other classes.

Table 204 – Association ends of CoreEquipmentProfile:: PostLineSensor with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.114 PotentialTransformer

Inheritance path = Sensor: AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

Instrument transformer (also known as Voltage Transformer) used to measure electrical qualities of the circuit that is being protected and/or monitored. Typically used as voltage transducer for the purpose of metering, protection, or sometimes auxiliary substation supply. A typical secondary voltage rating would be 120V.

Table 205 shows all attributes of PotentialTransformer.

Table 205 – Attributes of CoreEquipmentProfile::PotentialTransformer

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 206 shows all association ends of PotentialTransformer with other classes.

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Table 206 – Association ends of CoreEquipmentProfile:: PotentialTransformer with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.115 PowerElectronicsConnection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection to the AC network for energy production or consumption that uses power electronics rather than rotating machines.

Table 207 shows all attributes of PowerElectronicsConnection.

Table 207 – Attributes of CoreEquipmentProfile::PowerElectronicsConnection

name	mult	type	description
maxQ	0..1	ReactivePower	Maximum reactive power limit. This is the maximum (nameplate) limit for the unit.
minQ	0..1	ReactivePower	Minimum reactive power limit for the unit. This is the minimum (nameplate) limit for the unit.
ratedS	0..1	ApparentPower	Nameplate apparent power rating for the unit. The attribute shall have a positive value.
ratedU	0..1	Voltage	Rated voltage (nameplate data, Ur in IEC 60909-0). It is primarily used for short circuit data exchange according to IEC 60909. The attribute shall be a positive value.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 208 shows all association ends of PowerElectronicsConnection with other classes.

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Table 208 – Association ends of CoreEquipmentProfile:: PowerElectronicsConnection with other classes

mult from	name	mult to	type	description
1..1	PowerElectronicsUnit	0..1	PowerElectronicsUnit	An AC network connection may have several power electronics units connecting through it.
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.116 (abstract) PowerElectronicsUnit

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

A generating unit or battery or aggregation that connects to the AC network using power electronics rather than rotating machines.

Table 209 shows all attributes of PowerElectronicsUnit.

Table 209 – Attributes of CoreEquipmentProfile::PowerElectronicsUnit

name	mult	type	description
maxP	0..1	ActivePower	Maximum active power limit. This is the maximum (nameplate) limit for the unit.
minP	0..1	ActivePower	Minimum active power limit. This is the minimum (nameplate) limit for the unit.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 210 shows all association ends of PowerElectronicsUnit with other classes.

Table 210 – Association ends of CoreEquipmentProfile:: PowerElectronicsUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.117 PowerElectronicsWindUnit

Inheritance path = PowerElectronicsUnit: Equipment: PowerSystemResource: IdentifiedObject

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A wind generating unit that connects to the AC network with power electronics rather than rotating machines or an aggregation of such units.

Table 211 shows all attributes of PowerElectronicsWindUnit.

Table 211 – Attributes of CoreEquipmentProfile::PowerElectronicsWindUnit

name	mult	type	description
maxP	0..1	ActivePower	inherited from: PowerElectronicsUnit
minP	0..1	ActivePower	inherited from: PowerElectronicsUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 212 shows all association ends of PowerElectronicsWindUnit with other classes.

Table 212 – Association ends of CoreEquipmentProfile:: PowerElectronicsWindUnit with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.118 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 213 shows all attributes of PowerSystemResource.

Table 213 – Attributes of CoreEquipmentProfile::PowerSystemResource

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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6.119 PowerTransformer

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An electrical device consisting of two or more coupled windings, with or without a magnetic core, for introducing mutual coupling between electric circuits. Transformers can be used to control voltage and phase shift (active power flow).

A power transformer may be composed of separate transformer tanks that need not be identical.

A power transformer can be modelled with or without tanks and is intended for use in both balanced and unbalanced representations. A power transformer typically has two terminals, but may have one (grounding), three or more terminals.

The inherited association ConductingEquipment.BaseVoltage should not be used. The association from TransformerEnd to BaseVoltage should be used instead.

Table 214 shows all attributes of PowerTransformer.

Table 214 – Attributes of CoreEquipmentProfile::PowerTransformer

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 215 shows all association ends of PowerTransformer with other classes.

Table 215 – Association ends of CoreEquipmentProfile:: PowerTransformer with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.120 PowerTransformerEnd

Inheritance path = TransformerEnd: IdentifiedObject

A PowerTransformerEnd is associated with each Terminal of a PowerTransformer.

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The impedance values r, r0, x, and x0 of a PowerTransformerEnd represents a star equivalent as follows.

1) for a two Terminal PowerTransformer the high voltage (TransformerEnd.endNumber=1) PowerTransformerEnd has non zero values on r, r0, x, and x0 while the low voltage (TransformerEnd.endNumber=2) PowerTransformerEnd has zero values for r, r0, x, and x0. Parameters are always provided, even if the PowerTransformerEnds have the same rated voltage. In this case, the parameters are provided at the PowerTransformerEnd which has TransformerEnd.endNumber equal to 1.
2) for a three Terminal PowerTransformer the three PowerTransformerEnds represent a star equivalent with each leg in the star represented by r, r0, x, and x0 values.
3) For a three Terminal transformer each PowerTransformerEnd shall have g, g0, b and b0 values corresponding to the no load losses distributed on the three PowerTransformerEnds. The total no load loss shunt impedances may also be placed at one of the PowerTransformerEnds, preferably the end numbered 1, having the shunt values on end 1. This is the preferred way.
4) for a PowerTransformer with more than three Terminals the PowerTransformerEnd impedance values cannot be used. Instead use the TransformerMeshImpedance or split the transformer into multiple PowerTransformers.

Each PowerTransformerEnd must be contained by a PowerTransformer. Because a PowerTransformerEnd (or any other object) can not be contained by more than one parent, a PowerTransformerEnd can not have an association to an EquipmentContainer (Substation, VoltageLevel, etc).

Table 216 shows all attributes of PowerTransformerEnd.

Table 216 – Attributes of CoreEquipmentProfile::PowerTransformerEnd

name	mult	type	description
b	1..1	Susceptance	Magnetizing branch susceptance (B mag). The value can be positive or negative.
connectionKind	0..1	WindingConnection	Kind of connection.
ratedS	0..1	ApparentPower	Normal apparent power rating. The attribute shall be a positive value. For a two-winding transformer the values for the high and low voltage sides shall be identical.
g	0..1	Conductance	Magnetizing branch conductance.
ratedU	1..1	Voltage	Rated voltage: phase-phase for three-phase windings, and either phase-phase or phase-neutral for single-phase windings. A high voltage side, as given by TransformerEnd.endNumber, shall have a ratedU that is greater than or equal to ratedU for the lower voltage sides. The attribute shall be a positive value.
r	1..1	Resistance	Resistance (star-model) of the transformer end. The attribute shall be equal to or greater than zero for non-equivalent transformers.
x	1..1	Reactance	Positive sequence series reactance (star-model) of the transformer end.
endNumber	1..1	Integer	inherited from: TransformerEnd
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 217 shows all association ends of PowerTransformerEnd with other classes.

Table 217 – Association ends of CoreEquipmentProfile:: PowerTransformerEnd with other classes

mult from	name	mult to	type	description
0..*	PowerTransformer	1..1	PowerTransformer	The power transformer of this power transformer end.
0..*	BaseVoltage	1..1	BaseVoltage	inherited from: TransformerEnd
0..*	Terminal	1..1	Terminal	inherited from: TransformerEnd

6.121 (abstract) ProtectedSwitch

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A ProtectedSwitch is a switching device that can be operated by ProtectionEquipment.

Table 218 shows all attributes of ProtectedSwitch.

Table 218 – Attributes of CoreEquipmentProfile::ProtectedSwitch

name	mult	type	description
normalOpen	1..1	Boolean	inherited from: Switch
ratedCurrent	0..1	CurrentFlow	inherited from: Switch
retained	1..1	Boolean	inherited from: Switch
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 219 shows all association ends of ProtectedSwitch with other classes.

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Table 219 – Association ends of CoreEquipmentProfile:: ProtectedSwitch with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.122 RatioTapChanger

Inheritance path = TapChanger: PowerSystemResource: IdentifiedObject

A tap changer that changes the voltage ratio impacting the voltage magnitude but not the phase angle across the transformer.

Angle sign convention (general): Positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer).

Table 220 shows all attributes of RatioTapChanger.

Table 220 – Attributes of CoreEquipmentProfile::RatioTapChanger

name	mult	type	description
stepVoltageIncrement	1..1	PerCent	Tap step increment, in per cent of rated voltage of the power transformer end, per step position. When the increment is negative, the voltage decreases when the tap step increases.
highStep	1..1	Integer	inherited from: TapChanger
lowStep	1..1	Integer	inherited from: TapChanger
ltcFlag	1..1	Boolean	inherited from: TapChanger
neutralStep	1..1	Integer	inherited from: TapChanger
neutralU	1..1	Voltage	inherited from: TapChanger
normalStep	1..1	Integer	inherited from: TapChanger
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 221 shows all association ends of RatioTapChanger with other classes.

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Table 221 – Association ends of CoreEquipmentProfile:: RatioTapChanger with other classes

mult from	name	mult to	type	description
0..1	TransformerEnd	1..1	TransformerEnd	Transformer end to which this ratio tap changer belongs.
0..*	RatioTapChangerTable	0..1	RatioTapChangerTable	The tap ratio table for this ratio tap changer.
1..*	TapChangerControl	0..1	TapChangerControl	inherited from: TapChanger

6.123 RatioTapChangerTable

Inheritance path = IdentifiedObject

Describes a curve for how the voltage magnitude and impedance varies with the tap step.

Table 222 shows all attributes of RatioTapChangerTable.

Table 222 – Attributes of CoreEquipmentProfile::RatioTapChangerTable

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.124 RatioTapChangerTablePoint

Inheritance path = TapChangerTablePoint

Describes each tap step in the ratio tap changer tabular curve.

Table 223 shows all attributes of RatioTapChangerTablePoint.

Table 223 – Attributes of CoreEquipmentProfile::RatioTapChangerTablePoint

name	mult	type	description
b	0..1	PerCent	inherited from: TapChangerTablePoint
g	0..1	PerCent	inherited from: TapChangerTablePoint
r	0..1	PerCent	inherited from: TapChangerTablePoint
ratio	0..1	Float	inherited from: TapChangerTablePoint
step	1..1	Integer	inherited from: TapChangerTablePoint
x	0..1	PerCent	inherited from: TapChangerTablePoint

Table 224 shows all association ends of RatioTapChangerTablePoint with other classes.

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Table 224 – Association ends of CoreEquipmentProfile:: RatioTapChangerTablePoint with other classes

mult from	name	mult to	type	description
1..*	RatioTapChangerTable	1..1	RatioTapChangerTable	Table of this point.

6.125 ReactiveCapabilityCurve

Inheritance path = Curve: IdentifiedObject

Reactive power rating envelope versus the synchronous machine's active power, in both the generating and motoring modes. For each active power value there is a corresponding high and low reactive power limit value. Typically there will be a separate curve for each coolant condition, such as hydrogen pressure. The Y1 axis values represent reactive minimum and the Y2 axis values represent reactive maximum.

Table 225 shows all attributes of ReactiveCapabilityCurve.

Table 225 – Attributes of CoreEquipmentProfile::ReactiveCapabilityCurve

name	mult	type	description
curveStyle	1..1	CurveStyle	inherited from: Curve
xUnit	1..1	UnitSymbol	inherited from: Curve
y1Unit	1..1	UnitSymbol	inherited from: Curve
y2Unit	0..1	UnitSymbol	inherited from: Curve
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.126 (abstract) RegulatingCondEq

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Table 226 shows all attributes of RegulatingCondEq.

Table 226 – Attributes of CoreEquipmentProfile::RegulatingCondEq

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 227 shows all association ends of RegulatingCondEq with other classes.

Table 227 – Association ends of CoreEquipmentProfile:: RegulatingCondEq with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	The regulating control scheme in which this equipment participates.
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.127 RegulatingControl

Inheritance path = PowerSystemResource: IdentifiedObject

Specifies a set of equipment that works together to control a power system quantity such as voltage or flow.

Remote bus voltage control is possible by specifying the controlled terminal located at some place remote from the controlling equipment.

The specified terminal shall be associated with the connectivity node of the controlled point. The most specific subtype of RegulatingControl shall be used in case such equipment participate in the control, e.g. TapChangerControl for tap changers.

For flow control, load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

The attribute minAllowedTargetValue and maxAllowedTargetValue are required in the following cases:

– For a power generating module operated in power factor control mode to specify maximum and minimum power factor values;
– Whenever it is necessary to have an off center target voltage for the tap changer regulator. For instance, due to long cables to off shore wind farms and the need to have a simpler setup at the off shore transformer platform, the voltage is controlled from the land at the connection point for the off shore wind farm. Since there usually is a voltage rise along the cable, there is typical and overvoltage of up 3 to 4 kV compared to the on shore station. Thus in normal operation the tap changer on the on shore station is operated with a target set point, which is in the lower parts of the dead band.

The attributes minAllowedTargetValue and maxAllowedTargetValue are not related to the attribute targetDeadband and thus they are not treated as an alternative of the targetDeadband. They are needed due to limitations in the local substation controller. The attribute targetDeadband is used to prevent the power flow from move the tap position in circles (hunting) that is to be used regardless of the attributes minAllowedTargetValue and maxAllowedTargetValue.

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Table 228 shows all attributes of RegulatingControl.

Table 228 – Attributes of CoreEquipmentProfile::RegulatingControl

name	mult	type	description
mode	1..1	RegulatingControlModeKind	The regulating control mode presently available. This specification allows for determining the kind of regulation without need for obtaining the units from a schedule.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 229 shows all association ends of RegulatingControl with other classes.

Table 229 – Association ends of CoreEquipmentProfile:: RegulatingControl with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	The terminal associated with this regulating control. The terminal is associated instead of a node, since the terminal could connect into either a topological node or a connectivity node. Sometimes it is useful to model regulation at a terminal of a bus bar object.

6.128 RegularTimePoint root class

Time point for a schedule where the time between the consecutive points is constant.

Table 230 shows all attributes of RegularTimePoint.

Table 230 – Attributes of CoreEquipmentProfile::RegularTimePoint

name	mult	type	description
sequenceNumber	1..1	Integer	The position of the regular time point in the sequence. Note that time points don't have to be sequential, i.e. time points may be omitted. The actual time for a RegularTimePoint is computed by multiplying the associated regular interval schedule's time step with the regular time point sequence number and adding the associated schedules start time. To specify values for the start time, use sequence number 0. The sequence number cannot be negative.
value1	1..1	Float	The first value at the time. The meaning of the value is defined by the derived type of the associated schedule.
value2	0..1	Float	The second value at the time. The meaning of the value is defined by the derived type of the associated schedule.

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Table 231 shows all association ends of RegularTimePoint with other classes.

Table 231 – Association ends of CoreEquipmentProfile:: RegularTimePoint with other classes

mult from	name	mult to	type	description
1..*	IntervalSchedule	1..1	RegularIntervalSchedule	Regular interval schedule containing this time point.

6.129 (abstract) RegularIntervalSchedule

Inheritance path = BasicIntervalSchedule: IdentifiedObject

The schedule has time points where the time between them is constant.

Table 232 shows all attributes of RegularIntervalSchedule.

Table 232 – Attributes of CoreEquipmentProfile::RegularIntervalSchedule

name	mult	type	description
timeStep	1..1	Seconds	The time between each pair of subsequent regular time points in sequence order.
endTime	1..1	DateTime	The time for the last time point. The value can be a time of day, not a specific date.
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.130 ReportingGroup

Inheritance path = IdentifiedObject

A reporting group is used for various ad-hoc groupings used for reporting.

Table 233 shows all attributes of ReportingGroup.

Table 233 – Attributes of CoreEquipmentProfile::ReportingGroup

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.131 (abstract) RotatingMachine

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine which may be used as a generator or motor.

Table 234 shows all attributes of RotatingMachine.

Table 234 – Attributes of CoreEquipmentProfile::RotatingMachine

name	mult	type	description
ratedPowerFactor	0..1	Float	Power factor (nameplate data). It is primarily used for short circuit data exchange according to IEC 60909. The attribute cannot be a negative value.
ratedS	0..1	ApparentPower	Nameplate apparent power rating for the unit. The attribute shall have a positive value.
ratedU	0..1	Voltage	Rated voltage (nameplate data, Ur in IEC 60909-0). It is primarily used for short circuit data exchange according to IEC 60909. The attribute shall be a positive value.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 235 shows all association ends of RotatingMachine with other classes.

Table 235 – Association ends of CoreEquipmentProfile:: RotatingMachine with other classes

mult from	name	mult to	type	description
1..*	GeneratingUnit	0..1	GeneratingUnit	A synchronous machine may operate as a generator and as such becomes a member of a generating unit.
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.132 Season

Inheritance path = IdentifiedObject

A specified time period of the year.

Table 236 shows all attributes of Season.

Table 236 – Attributes of CoreEquipmentProfile::Season

name	mult	type	description
endDate	1..1	MonthDay	Date season ends.
startDate	1..1	MonthDay	Date season starts.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.133 (abstract) Sensor

Inheritance path = AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class describe devices that transform a measured quantity into signals that can be presented at displays, used in control or be recorded.

Table 237 shows all attributes of Sensor.

Table 237 – Attributes of CoreEquipmentProfile::Sensor

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 238 shows all association ends of Sensor with other classes.

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Table 238 – Association ends of CoreEquipmentProfile::Sensor with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.134 (abstract) SeasonDayTypeSchedule

Inheritance path = RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

A time schedule covering a 24-hour period, with curve data for a specific type of season and day.

Table 239 shows all attributes of SeasonDayTypeSchedule.

Table 239 – Attributes of CoreEquipmentProfile::SeasonDayTypeSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 240 shows all association ends of SeasonDayTypeSchedule with other classes.

Table 240 – Association ends of CoreEquipmentProfile:: SeasonDayTypeSchedule with other classes

mult from	name	mult to	type	description
0..*	DayType	1..1	DayType	DayType for the Schedule.
0..*	Season	1..1	Season	Season for the Schedule.

6.135 SeriesCompensator

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A Series Compensator is a series capacitor or reactor or an AC transmission line without charging susceptance. It is a two terminal device.

Table 241 shows all attributes of SeriesCompensator.

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Table 241 – Attributes of CoreEquipmentProfile::SeriesCompensator

name	mult	type	description
r	1..1	Resistance	Positive sequence resistance.
x	1..1	Reactance	Positive sequence reactance.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 242 shows all association ends of SeriesCompensator with other classes.

Table 242 – Association ends of CoreEquipmentProfile:: SeriesCompensator with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.136 (abstract) ShuntCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for bPerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Table 243 shows all attributes of ShuntCompensator.

Table 243 – Attributes of CoreEquipmentProfile::ShuntCompensator

name	mult	type	description
aVRDelay	0..1	Seconds	An automatic voltage regulation delay (AVRDelay) which is the time delay from a change in voltage to when the capacitor is allowed to change state. This filters out temporary changes in voltage.
grounded	0..1	Boolean	Used for Yn and Zn connections. True if the neutral is solidly grounded.
maximumSections	1..1	Integer	The maximum number of sections that may be switched in.
nomU	1..1	Voltage	The voltage at which the nominal reactive power may be calculated. This should normally be within 10 % of the voltage at which the capacitor is connected to the network.

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name	mult	type	description
normalSections	1..1	Integer	The normal number of sections switched in. The value shall be between zero and ShuntCompensator.maximumSections.
voltageSensitivity	0..1	VoltagePerReactivePower	Voltage sensitivity required for the device to regulate the bus voltage, in voltage/reactive power.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 244 shows all association ends of ShuntCompensator with other classes.

Table 244 – Association ends of CoreEquipmentProfile:: ShuntCompensator with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.137 SolarGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A solar thermal generating unit, connected to the grid by means of a rotating machine. This class does not represent photovoltaic (PV) generation.

Table 245 shows all attributes of SolarGeneratingUnit.

Table 245 – Attributes of CoreEquipmentProfile::SolarGeneratingUnit

name	mult	type	description
genControlSource	0..1	GeneratorControlSource	inherited from: GeneratingUnit
governorSCD	0..1	PerCent	inherited from: GeneratingUnit
longPF	0..1	Float	inherited from: GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	inherited from: GeneratingUnit
maxOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
minOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
nominalP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMinP	0..1	ActivePower	inherited from: GeneratingUnit
ratedNetMaxP	0..1	ActivePower	inherited from: GeneratingUnit

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name	mult	type	description
shortPF	0..1	Float	inherited from: GeneratingUnit
startupCost	0..1	Money	inherited from: GeneratingUnit
variableCost	0..1	Money	inherited from: GeneratingUnit
startupTime	0..1	Seconds	inherited from: GeneratingUnit
totalEfficiency	0..1	PerCent	inherited from: GeneratingUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 246 shows all association ends of SolarGeneratingUnit with other classes.

Table 246 – Association ends of CoreEquipmentProfile:: SolarGeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	SolarPowerPlant	0..1	SolarPowerPlant	(European) A solar power plant may have solar generating units.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.138 (European) SolarPowerPlant

Inheritance path = PowerSystemResource: IdentifiedObject

Solar power plant.

Table 247 shows all attributes of SolarPowerPlant.

Table 247 – Attributes of CoreEquipmentProfile::SolarPowerPlant

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.139 StaticVarCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

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A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.

The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Table 248 shows all attributes of StaticVarCompensator.

Table 248 – Attributes of CoreEquipmentProfile::StaticVarCompensator

name	mult	type	description
capacitiveRating	1..1	Reactance	Capacitive reactance at maximum capacitive reactive power. Shall always be positive.
inductiveRating	1..1	Reactance	Inductive reactance at maximum inductive reactive power. Shall always be negative.
slope	1..1	VoltagePerReactivePower	The characteristics slope of an SVC defines how the reactive power output changes in proportion to the difference between the regulated bus voltage and the voltage setpoint. The attribute shall be a positive value or zero.
sVCControlMode	0..1	SVCControlMode	(deprecated) SVC control mode.
voltageSetPoint	0..1	Voltage	(deprecated) The reactive power output of the SVC is proportional to the difference between the voltage at the regulated bus and the voltage setpoint. When the regulated bus voltage is equal to the voltage setpoint, the reactive power output is zero.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 249 shows all association ends of StaticVarCompensator with other classes.

Table 249 – Association ends of CoreEquipmentProfile:: StaticVarCompensator with other classes

mult from	name	mult to	type	description
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

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6.140 StationSupply

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Station supply with load derived from the station output.

Table 250 shows all attributes of StationSupply.

Table 250 – Attributes of CoreEquipmentProfile::StationSupply

name	mult	type	description
pfixed	0..1	ActivePower	inherited from: EnergyConsumer
pfixedPct	0..1	PerCent	inherited from: EnergyConsumer
qfixed	0..1	ReactivePower	inherited from: EnergyConsumer
qfixedPct	0..1	PerCent	inherited from: EnergyConsumer
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 251 shows all association ends of StationSupply with other classes.

Table 251 – Association ends of CoreEquipmentProfile:: StationSupply with other classes

mult from	name	mult to	type	description
0..*	LoadResponse	0..1	LoadResponseCharacteristic	inherited from: EnergyConsumer
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.141 SubGeographicalRegion

Inheritance path = IdentifiedObject

A subset of a geographical region of a power system network model.

Table 252 shows all attributes of SubGeographicalRegion.

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Table 252 – Attributes of CoreEquipmentProfile::SubGeographicalRegion

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 253 shows all association ends of SubGeographicalRegion with other classes.

Table 253 – Association ends of CoreEquipmentProfile:: SubGeographicalRegion with other classes

mult from	name	mult to	type	description
0..*	Region	1..1	GeographicalRegion	The geographical region which this sub-geographical region is within.

6.142 SubLoadArea

Inheritance path = EnergyArea: IdentifiedObject

The class is the second level in a hierarchical structure for grouping of loads for the purpose of load flow load scaling.

Table 254 shows all attributes of SubLoadArea.

Table 254 – Attributes of CoreEquipmentProfile::SubLoadArea

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 255 shows all association ends of SubLoadArea with other classes.

Table 255 – Association ends of CoreEquipmentProfile:: SubLoadArea with other classes

mult from	name	mult to	type	description
1..*	LoadArea	1..1	LoadArea	The LoadArea where the SubLoadArea belongs.

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6.143 Substation

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A collection of equipment for purposes other than generation or utilization, through which electric energy in bulk is passed for the purposes of switching or modifying its characteristics.

Table 256 shows all attributes of Substation.

Table 256 – Attributes of CoreEquipmentProfile::Substation

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 257 shows all association ends of Substation with other classes.

Table 257 – Association ends of CoreEquipmentProfile::Substation with other classes

mult from	name	mult to	type	description
0..*	Region	1..1	SubGeographicalRegion	The SubGeographicalRegion containing the substation.

6.144 SurgeArrester

Inheritance path = AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

Shunt device, installed on the network, usually in the proximity of electrical equipment in order to protect the said equipment against transient voltage transients caused by lightning or switching activity.

Table 258 shows all attributes of SurgeArrester.

Table 258 – Attributes of CoreEquipmentProfile::SurgeArrester

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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Table 259 shows all association ends of SurgeArrester with other classes.

Table 259 – Association ends of CoreEquipmentProfile:: SurgeArrester with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.145 Switch

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches. The ACDCTerminal.connected at the two sides of the switch shall not be considered for assessing switch connectivity, i.e. only Switch.open, .normalOpen and .locked are relevant.

Table 260 shows all attributes of Switch.

Table 260 – Attributes of CoreEquipmentProfile::Switch

name	mult	type	description
normalOpen	1..1	Boolean	The attribute is used in cases when no Measurement for the status value is present. If the Switch has a status measurement the Discrete.normalValue is expected to match with the Switch.normalOpen.
ratedCurrent	0..1	CurrentFlow	The maximum continuous current carrying capacity in amps governed by the device material and construction. The attribute shall be a positive value.
retained	1..1	Boolean	Branch is retained in the topological solution. The flow through retained switches will normally be calculated in power flow.
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 261 shows all association ends of Switch with other classes.

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Table 261 – Association ends of CoreEquipmentProfile::Switch with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.146 SwitchSchedule

Inheritance path = SeasonDayTypeSchedule: RegularIntervalSchedule: BasicIntervalSchedule: IdentifiedObject

A schedule of switch positions. If RegularTimePoint.value1 is 0, the switch is open. If 1, the switch is closed.

Table 262 shows all attributes of SwitchSchedule.

Table 262 – Attributes of CoreEquipmentProfile::SwitchSchedule

name	mult	type	description
timeStep	1..1	Seconds	inherited from: RegularIntervalSchedule
endTime	1..1	DateTime	inherited from: RegularIntervalSchedule
startTime	1..1	DateTime	inherited from: BasicIntervalSchedule
value1Unit	1..1	UnitSymbol	inherited from: BasicIntervalSchedule
value2Unit	0..1	UnitSymbol	inherited from: BasicIntervalSchedule
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 263 shows all association ends of SwitchSchedule with other classes.

Table 263 – Association ends of CoreEquipmentProfile:: SwitchSchedule with other classes

mult from	name	mult to	type	description
0..*	Switch	1..1	Switch	A SwitchSchedule is associated with a Switch.
0..*	DayType	1..1	DayType	inherited from: SeasonDayTypeSchedule
0..*	Season	1..1	Season	inherited from: SeasonDayTypeSchedule

6.147 SynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

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An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Table 264 shows all attributes of SynchronousMachine.

Table 264 – Attributes of CoreEquipmentProfile::SynchronousMachine

name	mult	type	description
maxQ	0..1	ReactivePower	Maximum reactive power limit. This is the maximum (nameplate) limit for the unit.
minQ	0..1	ReactivePower	Minimum reactive power limit for the unit.
qPercent	0..1	PerCent	Part of the coordinated reactive control that comes from this machine. The attribute is used as a participation factor not necessarily summing up to 100% for the participating devices in the control.
type	1..1	SynchronousMachineKind	Modes that this synchronous machine can operate in.
ratedPowerFactor	0..1	Float	inherited from: RotatingMachine
ratedS	0..1	ApparentPower	inherited from: RotatingMachine
ratedU	0..1	Voltage	inherited from: RotatingMachine
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 265 shows all association ends of SynchronousMachine with other classes.

Table 265 – Association ends of CoreEquipmentProfile:: SynchronousMachine with other classes

mult from	name	mult to	type	description
1..*	InitialReactiveCapabilityCurve	0..1	ReactiveCapabilityCurve	The default reactive capability curve for use by a synchronous machine.
1..*	GeneratingUnit	0..1	GeneratingUnit	inherited from: RotatingMachine
0..*	RegulatingControl	0..1	RegulatingControl	inherited from: RegulatingCondEq
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.148 (abstract) TapChanger

Inheritance path = PowerSystemResource: IdentifiedObject

Mechanism for changing transformer winding tap positions.

Table 266 shows all attributes of TapChanger.

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Table 266 – Attributes of CoreEquipmentProfile::TapChanger

name	mult	type	description
highStep	1..1	Integer	Highest possible tap step position, advance from neutral. The attribute shall be greater than lowStep.
lowStep	1..1	Integer	Lowest possible tap step position, retard from neutral.
ltcFlag	1..1	Boolean	Specifies whether or not a TapChanger has load tap changing capabilities.
neutralStep	1..1	Integer	The neutral tap step position for this winding. The attribute shall be equal to or greater than lowStep and equal or less than highStep. It is the step position where the voltage is neutralU when the other terminals of the transformer are at the ratedU. If there are other tap changers on the transformer those taps are kept constant at their neutralStep.
neutralU	1..1	Voltage	Voltage at which the winding operates at the neutral tap setting. It is the voltage at the terminal of the PowerTransformerEnd associated with the tap changer when all tap changers on the transformer are at their neutralStep position. Normally neutralU of the tap changer is the same as ratedU of the PowerTransformerEnd, but it can differ in special cases such as when the tapping mechanism is separate from the winding more common on lower voltage transformers. This attribute is not relevant for PhaseTapChangerAsymmetrical, PhaseTapChangerSymmetrical and PhaseTapChangerLinear.
normalStep	1..1	Integer	The tap step position used in "normal" network operation for this winding. For a "Fixed" tap changer indicates the current physical tap setting. The attribute shall be equal to or greater than lowStep and equal to or less than highStep.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 267 shows all association ends of TapChanger with other classes.

Table 267 – Association ends of CoreEquipmentProfile::TapChanger with other classes

mult from	name	mult to	type	description
1..*	TapChangerControl	0..1	TapChangerControl	The regulating control scheme in which this tap changer participates.

6.149 TapChangerControl

Inheritance path = RegulatingControl: PowerSystemResource: IdentifiedObject

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Describes behaviour specific to tap changers, e.g. how the voltage at the end of a line varies with the load level and compensation of the voltage drop by tap adjustment.

Table 268 shows all attributes of TapChangerControl.

Table 268 – Attributes of CoreEquipmentProfile::TapChangerControl

name	mult	type	description
mode	1..1	RegulatingControlModeKind	inherited from: RegulatingControl
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 269 shows all association ends of TapChangerControl with other classes.

Table 269 – Association ends of CoreEquipmentProfile:: TapChangerControl with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: RegulatingControl

6.150 (abstract) TapChangerTablePoint root class

Describes each tap step in the tabular curve.

Table 270 shows all attributes of TapChangerTablePoint.

Table 270 – Attributes of CoreEquipmentProfile::TapChangerTablePoint

name	mult	type	description
b	0..1	PerCent	The magnetizing branch susceptance deviation as a percentage of nominal value. The actual susceptance is calculated as follows: calculated magnetizing susceptance = b(nominal) * (1 + b(from this class)/100). The b(nominal) is defined as the static magnetizing susceptance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.
g	0..1	PerCent	The magnetizing branch conductance deviation as a percentage of nominal value. The actual conductance is calculated as follows: calculated magnetizing conductance = g(nominal) * (1 + g(from this class)/100). The g(nominal) is defined as the static magnetizing conductance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.

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name	mult	type	description
r	0..1	PerCent	The resistance deviation as a percentage of nominal value. The actual reactance is calculated as follows: calculated resistance = r(nominal) * (1 + r(from this class)/100). The r(nominal) is defined as the static resistance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.
ratio	0..1	Float	The voltage at the tap step divided by rated voltage of the transformer end having the tap changer. Hence this is a value close to one. For example, if the ratio at step 1 is 1.01, and the rated voltage of the transformer end is 110kV, then the voltage obtained by setting the tap changer to step 1 to is 111.1kV.
step	1..1	Integer	The tap step.
x	0..1	PerCent	The series reactance deviation as a percentage of nominal value. The actual reactance is calculated as follows: calculated reactance = x(nominal) * (1 + x(from this class)/100). The x(nominal) is defined as the static series reactance on the associated power transformer end or ends. This model assumes the star impedance (pi model) form.

6.151 Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 271 shows all attributes of Terminal.

Table 271 – Attributes of CoreEquipmentProfile::Terminal

name	mult	type	description
phases	0..1	PhaseCode	Represents the normal network phasing condition. If the attribute is missing, three phases (ABC) shall be assumed, except for terminals of grounding classes (specializations of EarthFaultCompensator, GroundDisconnector, and Ground) which will be assumed to be N. Therefore, phase code ABCN is explicitly declared when needed, e.g. for star point grounding equipment. The phase code on terminals connecting same ConnectivityNode or same TopologicalNode as well as for equipment between two terminals shall be consistent.
sequenceNumber	1..1	Integer	inherited from: ACDCTerminal
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 272 shows all association ends of Terminal with other classes.

Table 272 – Association ends of CoreEquipmentProfile::Terminal with other classes

mult from	name	mult to	type	description
0..*	ConductingEquipment	1..1	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be connected to other conducting equipment terminals via connectivity nodes or topological nodes.
0..*	ConnectivityNode	0..1	ConnectivityNode	The connectivity node to which this terminal connects with zero impedance.
1..*	BusNameMarker	0..1	BusNameMarker	inherited from: ACDCTerminal

6.152 ThermalGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A generating unit whose prime mover could be a steam turbine, combustion turbine, or diesel engine.

Table 273 shows all attributes of ThermalGeneratingUnit.

Table 273 – Attributes of CoreEquipmentProfile::ThermalGeneratingUnit

name	mult	type	description
genControlSource	0..1	GeneratorControlSource	inherited from: GeneratingUnit
governorSCD	0..1	PerCent	inherited from: GeneratingUnit
longPF	0..1	Float	inherited from: GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	inherited from: GeneratingUnit
maxOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
minOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
nominalP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMinP	0..1	ActivePower	inherited from: GeneratingUnit
ratedNetMaxP	0..1	ActivePower	inherited from: GeneratingUnit
shortPF	0..1	Float	inherited from: GeneratingUnit
startupCost	0..1	Money	inherited from: GeneratingUnit
variableCost	0..1	Money	inherited from: GeneratingUnit
startupTime	0..1	Seconds	inherited from: GeneratingUnit
totalEfficiency	0..1	PerCent	inherited from: GeneratingUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment

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name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 274 shows all association ends of ThermalGeneratingUnit with other classes.

Table 274 – Association ends of CoreEquipmentProfile:: ThermalGeneratingUnit with other classes

mult from	name	mult to	type	description
0..1	CAESPlant	0..1	CAESPlant	A thermal generating unit may be a member of a compressed air energy storage plant.
0..*	CogenerationPlant	0..1	CogenerationPlant	A thermal generating unit may be a member of a cogeneration plant.
0..*	CombinedCyclePlant	0..1	CombinedCyclePlant	A thermal generating unit may be a member of a combined cycle plant.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.153 TieFlow

Inheritance path = IdentifiedObject

Defines the structure (in terms of location and direction) of the net interchange constraint for a control area. This constraint may be used by either AGC or power flow.

Table 275 shows all attributes of TieFlow.

Table 275 – Attributes of CoreEquipmentProfile::TieFlow

name	mult	type	description
positiveFlowIn	1..1	Boolean	Specifies the sign of the tie flow associated with a control area. True if positive flow into the terminal (load convention) is also positive flow into the control area. See the description of ControlArea for further explanation of how TieFlow.positiveFlowIn is used.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 276 shows all association ends of TieFlow with other classes.

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Table 276 – Association ends of CoreEquipmentProfile::TieFlow with other classes

mult from	name	mult to	type	description
0..*	ControlArea	1..1	ControlArea	The control area of the tie flows.
0..2	Terminal	1..1	Terminal	The terminal to which this tie flow belongs.

6.154 (abstract) TransformerEnd

Inheritance path = IdentifiedObject

A conducting connection point of a power transformer. It corresponds to a physical transformer winding terminal. In earlier CIM versions, the TransformerWinding class served a similar purpose, but this class is more flexible because it associates to terminal but is not a specialization of ConductingEquipment.

Table 277 shows all attributes of TransformerEnd.

Table 277 – Attributes of CoreEquipmentProfile::TransformerEnd

name	mult	type	description
endNumber	1..1	Integer	Number for this transformer end, corresponding to the end's order in the power transformer vector group or phase angle clock number. Highest voltage winding should be 1. Each end within a power transformer should have a unique subsequent end number. Note the transformer end number need not match the terminal sequence number.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 278 shows all association ends of TransformerEnd with other classes.

Table 278 – Association ends of CoreEquipmentProfile:: TransformerEnd with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	1..1	BaseVoltage	Base voltage of the transformer end. This is essential for PU calculation.
0..*	Terminal	1..1	Terminal	Terminal of the power transformer to which this transformer end belongs.

6.155 VoltageLevel

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

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A collection of equipment at one common system voltage forming a switchgear. The equipment typically consists of breakers, busbars, instrumentation, control, regulation and protection devices as well as assemblies of all these.

Table 279 shows all attributes of VoltageLevel.

Table 279 – Attributes of CoreEquipmentProfile::VoltageLevel

name	mult	type	description
highVoltageLimit	0..1	Voltage	The bus bar's high voltage limit. The limit applies to all equipment and nodes contained in a given VoltageLevel. It is not required that it is exchanged in pair with lowVoltageLimit. It is preferable to use operational VoltageLimit, which prevails, if present.
lowVoltageLimit	0..1	Voltage	The bus bar's low voltage limit. The limit applies to all equipment and nodes contained in a given VoltageLevel. It is not required that it is exchanged in pair with highVoltageLimit. It is preferable to use operational VoltageLimit, which prevails, if present.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 280 shows all association ends of VoltageLevel with other classes.

Table 280 – Association ends of CoreEquipmentProfile::VoltageLevel with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	1..1	BaseVoltage	The base voltage used for all equipment within the voltage level.
0..*	Substation	1..1	Substation	The substation of the voltage level.

6.156 VoltageLimit

Inheritance path = OperationalLimit: IdentifiedObject

Operational limit applied to voltage.

The use of operational VoltageLimit is preferred instead of limits defined at VoltageLevel. The operational VoltageLimits are used, if present.

Table 281 shows all attributes of VoltageLimit.

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Table 281 – Attributes of CoreEquipmentProfile::VoltageLimit

name	mult	type	description
normalValue	1..1	Voltage	The normal limit on voltage. High or low limit nature of the limit depends upon the properties of the operational limit type. The attribute shall be a positive value or zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 282 shows all association ends of VoltageLimit with other classes.

Table 282 – Association ends of CoreEquipmentProfile::VoltageLimit with other classes

mult from	name	mult to	type	description
0..*	OperationalLimitType	1..1	OperationalLimitType	inherited from: OperationalLimit
0..*	OperationalLimitSet	1..1	OperationalLimitSet	inherited from: OperationalLimit

6.157 VsCapabilityCurve

Inheritance path = Curve: IdentifiedObject

The P-Q capability curve for a voltage source converter, with P on X-axis and Qmin and Qmax on Y1-axis and Y2-axis.

Table 283 shows all attributes of VsCapabilityCurve.

Table 283 – Attributes of CoreEquipmentProfile::VsCapabilityCurve

name	mult	type	description
curveStyle	1..1	CurveStyle	inherited from: Curve
xUnit	1..1	UnitSymbol	inherited from: Curve
y1Unit	1..1	UnitSymbol	inherited from: Curve
y2Unit	0..1	UnitSymbol	inherited from: Curve
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.158 VsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

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DC side of the voltage source converter (VSC).

Table 284 shows all attributes of VsConverter.

Table 284 – Attributes of CoreEquipmentProfile::VsConverter

name	mult	type	description
maxModulationIndex	0..1	Float	The maximum quotient between the AC converter voltage (Uc) and DC voltage (Ud). A factor typically less than 1. It is converter's configuration data used in power flow.
baseS	0..1	ApparentPower	inherited from: ACDCConverter
idleLoss	0..1	ActivePower	inherited from: ACDCConverter
maxUdc	0..1	Voltage	inherited from: ACDCConverter
minUdc	0..1	Voltage	inherited from: ACDCConverter
numberOfValves	0..1	Integer	inherited from: ACDCConverter
ratedUdc	0..1	Voltage	inherited from: ACDCConverter
resistiveLoss	0..1	Resistance	inherited from: ACDCConverter
switchingLoss	0..1	ActivePowerPerCurrentFlow	inherited from: ACDCConverter
valveU0	0..1	Voltage	inherited from: ACDCConverter
maxP	0..1	ActivePower	inherited from: ACDCConverter
minP	0..1	ActivePower	inherited from: ACDCConverter
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 285 shows all association ends of VsConverter with other classes.

Table 285 – Association ends of CoreEquipmentProfile::VsConverter with other classes

mult from	name	mult to	type	description
0..*	CapabilityCurve	0..1	VsCapabilityCurve	Capability curve of this converter.
0..*	PccTerminal	0..1	Terminal	inherited from: ACDCConverter
0..*	BaseVoltage	0..1	BaseVoltage	inherited from: ConductingEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.159 WaveTrap

Inheritance path = AuxiliaryEquipment: Equipment: PowerSystemResource: IdentifiedObject

Line traps are devices that impede high frequency power line carrier signals yet present a negligible impedance at the main power frequency.

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Table 286 shows all attributes of WaveTrap.

Table 286 – Attributes of CoreEquipmentProfile::WaveTrap

name	mult	type	description
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 287 shows all association ends of WaveTrap with other classes.

Table 287 – Association ends of CoreEquipmentProfile::WaveTrap with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	inherited from: AuxiliaryEquipment
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.160 WindGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A wind driven generating unit, connected to the grid by means of a rotating machine. May be used to represent a single turbine or an aggregation.

Table 288 shows all attributes of WindGeneratingUnit.

Table 288 – Attributes of CoreEquipmentProfile::WindGeneratingUnit

name	mult	type	description
windGenUnitType	1..1	WindGenUnitKind	The kind of wind generating unit.
genControlSource	0..1	GeneratorControlSource	inherited from: GeneratingUnit
governorSCD	0..1	PerCent	inherited from: GeneratingUnit
longPF	0..1	Float	inherited from: GeneratingUnit
maximumAllowableSpinningReserve	0..1	ActivePower	inherited from: GeneratingUnit
maxOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
minOperatingP	1..1	ActivePower	inherited from: GeneratingUnit
nominalP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMaxP	0..1	ActivePower	inherited from: GeneratingUnit
ratedGrossMinP	0..1	ActivePower	inherited from: GeneratingUnit
ratedNetMaxP	0..1	ActivePower	inherited from: GeneratingUnit
shortPF	0..1	Float	inherited from: GeneratingUnit
startupCost	0..1	Money	inherited from: GeneratingUnit

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name	mult	type	description
variableCost	0..1	Money	inherited from: GeneratingUnit
startupTime	0..1	Seconds	inherited from: GeneratingUnit
totalEfficiency	0..1	PerCent	inherited from: GeneratingUnit
aggregate	0..1	Boolean	inherited from: Equipment
normallyInService	0..1	Boolean	inherited from: Equipment
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 289 shows all association ends of WindGeneratingUnit with other classes.

Table 289 – Association ends of CoreEquipmentProfile:: WindGeneratingUnit with other classes

mult from	name	mult to	type	description
0..*	WindPowerPlant	0..1	WindPowerPlant	(European) A wind power plant may have wind generating units.
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

6.161 (European) WindPowerPlant

Inheritance path = PowerSystemResource: IdentifiedObject

Wind power plant.

Table 290 shows all attributes of WindPowerPlant.

Table 290 – Attributes of CoreEquipmentProfile::WindPowerPlant

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

6.162 ControlAreaTypeKind enumeration

The type of control area.

Table 291 shows all literals of ControlAreaTypeKind.

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Table 291 – Literals of CoreEquipmentProfile::ControlAreaTypeKind

literal	value	description
AGC		Used for automatic generation control.
Forecast		Used for load forecast.
Interchange		Used for interchange specification or control.

6.163 Currency enumeration

Monetary currencies. ISO 4217 standard including 3-character currency code.

Table 292 shows all literals of Currency.

Table 292 – Literals of CoreEquipmentProfile::Currency

literal	value	description
AED	784	United Arab Emirates dirham.
AFN	971	Afghan afghani.
ALL	008	Albanian lek.
AMD	051	Armenian dram.
ANG	532	Netherlands Antillean guilder.
AOA	973	Angolan kwanza.
ARS	032	Argentine peso.
AUD	036	Australian dollar.
AWG	533	Aruban florin.
AZN	944	Azerbaijani manat.
BAM	977	Bosnia and Herzegovina convertible mark.
BBD	052	Barbados dollar.
BDT	050	Bangladeshi taka.
BGN	975	Bulgarian lev.
BHD	048	Bahraini dinar.
BIF	108	Burundian franc.
BMD	060	Bermudian dollar (customarily known as Bermuda dollar).
BND	096	Brunei dollar.
BOB	068	Boliviano.
BOV	984	Bolivian Mvdol (funds code).
BRL	986	Brazilian real.
BSD	044	Bahamian dollar.
BTN	064	Bhutanese ngultrum.
BWP	072	Botswana pula.
BYR	974	Belarusian ruble.
BZD	084	Belize dollar.
CAD	124	Canadian dollar.
CDF	976	Congolese franc.
CHF	756	Swiss franc.
CLF	990	Unidad de Fomento (funds code), Chile.

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literal	value	description
CLP	152	Chilean peso.
CNY	156	Chinese yuan.
COP	170	Colombian peso.
COU	970	Unidad de Valor Real.
CRC	188	Costa Rican colon.
CUC	931	Cuban convertible peso.
CUP	192	Cuban peso.
CVE	132	Cape Verde escudo.
CZK	203	Czech koruna.
DJF	262	Djiboutian franc.
DKK	208	Danish krone.
DOP	214	Dominican peso.
DZD	012	Algerian dinar.
EEK	233	Estonian kroon.
EGP	818	Egyptian pound.
ERN	232	Eritrean nakfa.
ETB	230	Ethiopian birr.
EUR	978	Euro.
FJD	242	Fiji dollar.
FKP	238	Falkland Islands pound.
GBP	826	Pound sterling.
GEL	981	Georgian lari.
GHS	936	Ghanaian cedi.
GIP	929	Gibraltar pound.
GMD	270	Gambian dalasi.
GNF	324	Guinean franc.
GTQ	320	Guatemalan quetzal.
GYD	328	Guyanese dollar.
HKD	344	Hong Kong dollar.
HNL	340	Honduran lempira.
HRK	191	Croatian kuna.
HTG	332	Haitian gourde.
HUF	348	Hungarian forint.
IDR	360	Indonesian rupiah.
ILS	376	Israeli new sheqel.
INR	356	Indian rupee.
IQD	368	Iraqi dinar.
IRR	364	Iranian rial.
ISK	352	Icelandic króna.
JMD	388	Jamaican dollar.
JOD	400	Jordanian dinar.
JPY	392	Japanese yen.
KES	404	Kenyan shilling.
KGS	417	Kyrgyzstani som.

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literal	value	description
KHR	116	Cambodian riel.
KMF	174	Comoro franc.
KPW	408	North Korean won.
KRW	410	South Korean won.
KWD	414	Kuwaiti dinar.
KYD	136	Cayman Islands dollar.
KZT	398	Kazakhstani tenge.
LAK	418	Lao kip.
LBP	422	Lebanese pound.
LKR	144	Sri Lanka rupee.
LRD	430	Liberian dollar.
LSL	426	Lesotho loti.
LTL	440	Lithuanian litas.
LVL	428	Latvian lats.
LYD	434	Libyan dinar.
MAD	504	Moroccan dirham.
MDL	498	Moldovan leu.
MGA	969	Malagasy ariary.
MKD	807	Macedonian denar.
MMK	104	Myanma kyat.
MNT	496	Mongolian tugrik.
MOP	446	Macanese pataca.
MRO	478	Mauritanian ouguiya.
MUR	480	Mauritian rupee.
MVR	462	Maldivian rufiyaa.
MWK	454	Malawian kwacha.
MXN	484	Mexican peso.
MYR	458	Malaysian ringgit.
MZN	943	Mozambican metical.
NAD	516	Namibian dollar.
NGN	566	Nigerian naira.
NIO	558	Cordoba oro.
NOK	578	Norwegian krone.
NPR	524	Nepalese rupee.
NZD	554	New Zealand dollar.
OMR	512	Omani rial.
PAB	590	Panamanian balboa.
PEN	604	Peruvian nuevo sol.
PGK	598	Papua New Guinean kina.
PHP	608	Philippine peso.
PKR	586	Pakistani rupee.
PLN	985	Polish zloty.
PYG	600	Paraguayan guaraní.
QAR	634	Qatari rial.

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literal	value	description
RON	946	Romanian new leu.
RSD	941	Serbian dinar.
RUB	643	Russian rouble.
RWF	646	Rwandan franc.
SAR	682	Saudi riyal.
SBD	090	Solomon Islands dollar.
SCR	690	Seychelles rupee.
SDG	938	Sudanese pound.
SEK	752	Swedish krona/kronor.
SGD	702	Singapore dollar.
SHP	654	Saint Helena pound.
SLL	694	Sierra Leonean leone.
SOS	706	Somali shilling.
SRD	968	Surinamese dollar.
STD	678	São Tomé and Príncipe dobra.
SYP	760	Syrian pound.
SZL	748	Lilangeni.
THB	764	Thai baht.
TJS	972	Tajikistani somoni.
TMT	934	Turkmenistani manat.
TND	788	Tunisian dinar.
TOP	776	Tongan pa'anga.
TRY	949	Turkish lira.
TTD	780	Trinidad and Tobago dollar.
TWD	901	New Taiwan dollar.
TZS	834	Tanzanian shilling.
UAH	980	Ukrainian hryvnia.
UGX	800	Ugandan shilling.
USD	840	United States dollar.
UYU	858	Uruguayan peso.
UZS	860	Uzbekistan som.
VEF	937	Venezuelan bolívar fuerte.
VND	704	Vietnamese Dong.
VUV	548	Vanuatu vatu.
WST	882	Samoan tala.
XAF	950	CFA franc BEAC.
XCD	951	East Caribbean dollar.
XOF	952	CFA Franc BCEAO.
XPF	953	CFP franc.
YER	886	Yemeni rial.
ZAR	710	South African rand.
ZMK	894	Zambian kwacha.
ZWL	932	Zimbabwe dollar.

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6.164 CurveStyle enumeration

Style or shape of curve.

Table 293 shows all literals of CurveStyle.

Table 293 – Literals of CoreEquipmentProfile::CurveStyle

literal	value	description
constantYValue		The Y-axis values are assumed constant until the next curve point and prior to the first curve point.
straightLineYValues		The Y-axis values are assumed to be a straight line between values. Also known as linear interpolation.

6.165 DCConverterOperatingModeKind enumeration

The operating mode of an HVDC bipole.

Table 294 shows all literals of DCConverterOperatingModeKind.

Table 294 – Literals of CoreEquipmentProfile::DCConverterOperatingModeKind

literal	value	description
bipolar		Bipolar operation.
monopolarMetallicReturn		Monopolar operation with metallic return.
monopolarGroundReturn		Monopolar operation with ground return.

6.166 DCPolarityKind enumeration

Polarity for DC circuits.

Table 295 shows all literals of DCPolarityKind.

Table 295 – Literals of CoreEquipmentProfile::DCPolarityKind

literal	value	description
positive		Positive pole. The converter terminal is intended to operate at a positive voltage relative the midpoint or negative terminal.
middle		Middle pole. The converter terminal is the midpoint in a bipolar or symmetric monopole configuration. The midpoint can be grounded and/or have a metallic return.
negative		Negative pole. The converter terminal is intended to operate at a negative voltage relative the midpoint or positive terminal.

6.167 FuelType enumeration

Type of fuel.

Table 296 shows all literals of FuelType.

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Table 296 – Literals of CoreEquipmentProfile::FuelType

literal	value	description
coal		Generic coal, not including lignite type.
oil		Oil.
gas		Natural gas.
lignite		The fuel is lignite coal. Note that this is a special type of coal, so the other enum of coal is reserved for hard coal types or if the exact type of coal is not known.
hardCoal		Hard coal.
oilShale		Oil Shale.
brownCoalLignite		Brown coal lignite.
coalDerivedGas		Coal derived gas.
peat		Peat.
other		Any fuel type not included in the rest of the enumerated value.

6.168 GeneratorControlSource enumeration

The source of controls for a generating unit.

Table 297 shows all literals of GeneratorControlSource.

Table 297 – Literals of CoreEquipmentProfile::GeneratorControlSource

literal	value	description
unavailable		Not available.
offAGC		Off of automatic generation control (AGC).
onAGC		On automatic generation control (AGC).
plantControl		Plant is controlling.

6.169 HydroEnergyConversionKind enumeration

Specifies the capability of the hydro generating unit to convert energy as a generator or pump.

Table 298 shows all literals of HydroEnergyConversionKind.

Table 298 – Literals of CoreEquipmentProfile::HydroEnergyConversionKind

literal	value	description
generator		Able to generate power, but not able to pump water for energy storage.
pumpAndGenerator		Able to both generate power and pump water for energy storage.

6.170 HydroPlantStorageKind enumeration

The type of hydro power plant.

Table 299 shows all literals of HydroPlantStorageKind.

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Table 299 – Literals of CoreEquipmentProfile::HydroPlantStorageKind

literal	value	description
runOfRiver		Run of river.
pumpedStorage		Pumped storage.
storage		Storage.

6.171 HydroTurbineKind enumeration

Type of turbine.

Table 300 shows all literals of HydroTurbineKind.

Table 300 – Literals of CoreEquipmentProfile::HydroTurbineKind

literal	value	description
francis		Francis.
pelton		Pelton.
kaplan		Kaplan.

6.172 (European) LimitKind enumeration

Limit kinds.

Table 301 shows all literals of LimitKind.

Table 301 – Literals of CoreEquipmentProfile::LimitKind

literal	value	description
patl		(European) The Permanent Admissible Transmission Loading (PATL) is the loading in amperes, MVA or MW that can be accepted by a network branch for an unlimited duration without any risk for the material. The OperationnalLimitType.isInfiniteDuration is set to true. There shall be only one OperationalLimitType of kind PATL per OperationalLimitSet if the PATL is ApparentPowerLimit, ActivePowerLimit, or CurrentLimit for a given Terminal or Equipment.
patlt		(European) Permanent Admissible Transmission Loading Threshold (PATLT) is a value in engineering units defined for PATL and calculated using a percentage less than 100 % of the PATL type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.

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literal	value	description
tatl		(European) Temporarily Admissible Transmission Loading (TATL) which is the loading in amperes, MVA or MW that can be accepted by a branch for a certain limited duration. The TATL can be defined in different ways: – as a fixed percentage of the PATL for a given time (for example, 115% of the PATL that can be accepted during 15 minutes), – pairs of TATL type and Duration calculated for each line taking into account its particular configuration and conditions of functioning (for example, it can define a TATL acceptable during 20 minutes and another one acceptable during 10 minutes). Such a definition of TATL can depend on the initial operating conditions of the network element (sag situation of a line). The duration attribute can be used to define several TATL limit types. Hence multiple TATL limit values may exist having different durations.
tc		(European) Tripping Current (TC) is the ultimate intensity without any delay. It is defined as the threshold the line will trip without any possible remedial actions. The tripping of the network element is ordered by protections against short circuits or by overload protections, but in any case, the activation delay of these protections is not compatible with the reaction delay of an operator (less than one minute). The duration is always zero if the OperationalLimitType.acceptableDuration is exchanged. Only one limit value exists for the TC type.
tct		(European) Tripping Current Threshold (TCT) is a value in engineering units defined for TC and calculated using a percentage less than 100 % of the TC type intended to alert operators of an arising condition. The percentage should be given in the name of the OperationalLimitSet. The aceptableDuration is another way to express the severity of the limit.
highVoltage		(European) Referring to the rating of the equipments, a voltage too high can lead to accelerated ageing or the destruction of the equipment. This limit type may or may not have duration.
lowVoltage		(European) A too low voltage can disturb the normal operation of some protections and transformer equipped with on-load tap changers, electronic power devices or can affect the behaviour of the auxiliaries of generation units. This limit type may or may not have duration.
operationalVoltageLimit		(European) Operational voltage limit.
alarmVoltage		(European) Voltage alarm.
warningVoltage		(European) Voltage warning.
stability		(European) Stability.

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6.173 OperationalLimitDirectionKind enumeration

The direction attribute describes the side of a limit that is a violation.

Table 302 shows all literals of OperationalLimitDirectionKind.

Table 302 – Literals of CoreEquipmentProfile::OperationalLimitDirectionKind

literal	value	description
high		High means that a monitored value above the limit value is a violation. If applied to a terminal flow, the positive direction is into the terminal.
low		Low means a monitored value below the limit is a violation. If applied to a terminal flow, the positive direction is into the terminal.
absoluteValue		An absoluteValue limit means that a monitored absolute value above the limit value is a violation.

6.174 PhaseCode enumeration

An unordered enumeration of phase identifiers. Allows designation of phases for both transmission and distribution equipment, circuits and loads. The enumeration, by itself, does not describe how the phases are connected together or connected to ground. Ground is not explicitly denoted as a phase.

Residential and small commercial loads are often served from single-phase, or split-phase, secondary circuits. For the example of s12N, phases 1 and 2 refer to hot wires that are 180 degrees out of phase, while N refers to the neutral wire. Through single-phase transformer connections, these secondary circuits may be served from one or two of the primary phases A, B, and C. For three-phase loads, use the A, B, C phase codes instead of s12N.

The integer values are from IEC 61968-9 to support revenue metering applications.

Table 303 shows all literals of PhaseCode.

Table 303 – Literals of CoreEquipmentProfile::PhaseCode

literal	value	description
ABCN	225	Phases A, B, C, and N.
ABC	224	Phases A, B, and C.
ABN	193	Phases A, B, and neutral.
ACN	41	Phases A, C and neutral.
BCN	97	Phases B, C, and neutral.
AB	132	Phases A and B.
AC	96	Phases A and C.
BC	66	Phases B and C.
AN	129	Phases A and neutral.
BN	65	Phases B and neutral.
CN	33	Phases C and neutral.
A	128	Phase A.
B	64	Phase B.
C	32	Phase C.

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literal	value	description
N	16	Neutral phase.
s1N	528	Secondary phase 1 and neutral.
s2N	272	Secondary phase 2 and neutral.
s12N	784	Secondary phases 1, 2, and neutral.
s1	512	Secondary phase 1.
s2	256	Secondary phase 2.
s12	768	Secondary phase 1 and 2.
none	0	No phases specified.
X	1024	Unknown non-neutral phase.
XY	3072	Two unknown non-neutral phases.
XN	1040	Unknown non-neutral phase plus neutral.
XYN	3088	Two unknown non-neutral phases plus neutral.

6.175 RegulatingControlModeKind enumeration

The kind of regulation model. For example regulating voltage, reactive power, active power, etc.

Table 304 shows all literals of RegulatingControlModeKind.

Table 304 – Literals of CoreEquipmentProfile::RegulatingControlModeKind

literal	value	description
voltage		Voltage is specified.
activePower		Active power is specified.
reactivePower		Reactive power is specified.
currentFlow		Current flow is specified.
admittance		Admittance is specified.
timeScheduled		Control switches on/off by time of day. The times may change on the weekend, or in different seasons.
temperature		Control switches on/off based on the local temperature (i.e., a thermostat).
powerFactor		Power factor is specified.

6.176 SynchronousMachineKind enumeration

Synchronous machine type.

Table 305 shows all literals of SynchronousMachineKind.

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Table 305 – Literals of CoreEquipmentProfile::SynchronousMachineKind

literal	value	description
generator		Indicates the synchronous machine can operate as a generator.
condenser		Indicates the synchronous machine can operate as a condenser.
generatorOrCondenser		Indicates the synchronous machine can operate as a generator or as a condenser.
motor		Indicates the synchronous machine can operate as a motor.
generatorOrMotor		Indicates the synchronous machine can operate as a generator or as a motor.
motorOrCondenser		Indicates the synchronous machine can operate as a motor or as a condenser.
generatorOrCondenserOrMotor		Indicates the synchronous machine can operate as a generator or as a condenser or as a motor.

6.177 (deprecated) SVCControlMode enumeration

Static VAr Compensator control mode.

Table 306 shows all literals of SVCControlMode.

Table 306 – Literals of CoreEquipmentProfile::SVCControlMode

literal	value	description
reactivePower		Reactive power control.
voltage		Voltage control.

6.178 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 307 shows all literals of UnitMultiplier.

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Table 307 – Literals of CoreEquipmentProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

6.179 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

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The integer values are used for harmonization with IEC 61850.

Table 308 shows all literals of UnitSymbol.

Table 308 – Literals of CoreEquipmentProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.

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literal	value	description
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).

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literal	value	description
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres (kg·m²) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μg/g’.

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literal	value	description
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.

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literal	value	description
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133.3 Pa).
M	217	Length, nautical miles (1 M = 1852 m).
kn	219	Speed, knots (1 kn = 1852/3600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

6.180 WindGenUnitKind enumeration

Kind of wind generating unit.

Table 309 shows all literals of WindGenUnitKind.

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Table 309 – Literals of CoreEquipmentProfile::WindGenUnitKind

literal	value	description
offshore		The wind generating unit is located offshore.
onshore		The wind generating unit is located onshore.

6.181 WindingConnection enumeration

Winding connection type.

Table 310 shows all literals of WindingConnection.

Table 310 – Literals of CoreEquipmentProfile::WindingConnection

literal	value	description
D		Delta.
Y		Wye.
Z		ZigZag.
Yn		Wye, with neutral brought out for grounding.
Zn		ZigZag, with neutral brought out for grounding.
A		Autotransformer common winding.
I		Independent winding, for single-phase connections.

6.182 ActivePower datatype

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Table 311 shows all attributes of ActivePower.

Table 311 – Attributes of CoreEquipmentProfile::ActivePower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=W)

6.183 ActivePowerPerCurrentFlow datatype

Active power variation with current flow.

Table 312 shows all attributes of ActivePowerPerCurrentFlow.

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Table 312 – Attributes of CoreEquipmentProfile::ActivePowerPerCurrentFlow

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=WPerA)
value	0..1	Float	

6.184 ActivePowerPerFrequency datatype

Active power variation with frequency.

Table 313 shows all attributes of ActivePowerPerFrequency.

Table 313 – Attributes of CoreEquipmentProfile::ActivePowerPerFrequency

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=WPers)
value	0..1	Float	

6.185 AngleDegrees datatype

Measurement of angle in degrees.

Table 314 shows all attributes of AngleDegrees.

Table 314 – Attributes of CoreEquipmentProfile::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

6.186 ApparentPower datatype

Product of the RMS value of the voltage and the RMS value of the current.

Table 315 shows all attributes of ApparentPower.

Table 315 – Attributes of CoreEquipmentProfile::ApparentPower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=VA)

6.187 Capacitance datatype

Capacitive part of reactance (imaginary part of impedance), at rated frequency.

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Table 316 shows all attributes of Capacitance.

Table 316 – Attributes of CoreEquipmentProfile::Capacitance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=F)
multiplier	0..1	UnitMultiplier	(const=none)

6.188 Conductance datatype

Factor by which voltage must be multiplied to give corresponding power lost from a circuit. Real part of admittance.

Table 317 shows all attributes of Conductance.

Table 317 – Attributes of CoreEquipmentProfile::Conductance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=S)
multiplier	0..1	UnitMultiplier	(const=none)

6.189 CurrentFlow datatype

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Table 318 shows all attributes of CurrentFlow.

Table 318 – Attributes of CoreEquipmentProfile::CurrentFlow

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=A)

6.190 Frequency datatype

Cycles per second.

Table 319 shows all attributes of Frequency.

Table 319 – Attributes of CoreEquipmentProfile::Frequency

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=Hz)
multiplier	0..1	UnitMultiplier	(const=none)

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6.191 Inductance datatype

Inductive part of reactance (imaginary part of impedance), at rated frequency.

Table 320 shows all attributes of Inductance.

Table 320 – Attributes of CoreEquipmentProfile::Inductance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=H)
multiplier	0..1	UnitMultiplier	(const=none)

6.192 Length datatype

Unit of length. It shall be a positive value or zero.

Table 321 shows all attributes of Length.

Table 321 – Attributes of CoreEquipmentProfile::Length

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=m)
multiplier	0..1	UnitMultiplier	(const=k)

6.193 Money datatype

Amount of money.

Table 322 shows all attributes of Money.

Table 322 – Attributes of CoreEquipmentProfile::Money

name	mult	type	description
unit	0..1	Currency	
multiplier	0..1	UnitMultiplier	(const=none)
value	0..1	Decimal	

6.194 PerCent datatype

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Table 323 shows all attributes of PerCent.

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Table 323 – Attributes of CoreEquipmentProfile::PerCent

name	mult	type	description
value	0..1	Float	Normally 0 to 100 on a defined base.
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

6.195 Reactance datatype

Reactance (imaginary part of impedance), at rated frequency.

Table 324 shows all attributes of Reactance.

Table 324 – Attributes of CoreEquipmentProfile::Reactance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=ohm)
multiplier	0..1	UnitMultiplier	(const=none)

6.196 ReactivePower datatype

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Table 325 shows all attributes of ReactivePower.

Table 325 – Attributes of CoreEquipmentProfile::ReactivePower

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=VAr)
multiplier	0..1	UnitMultiplier	(const=M)

6.197 RealEnergy datatype

Real electrical energy.

Table 326 shows all attributes of RealEnergy.

Table 326 – Attributes of CoreEquipmentProfile::RealEnergy

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=Wh)
value	0..1	Float	

6.198 Resistance datatype

Resistance (real part of impedance).

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Table 327 shows all attributes of Resistance.

Table 327 – Attributes of CoreEquipmentProfile::Resistance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=ohm)
multiplier	0..1	UnitMultiplier	(const=none)

6.199 RotationSpeed datatype

Number of revolutions per second.

Table 328 shows all attributes of RotationSpeed.

Table 328 – Attributes of CoreEquipmentProfile::RotationSpeed

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=Hz)
value	0..1	Float	

6.200 Seconds datatype

Time, in seconds.

Table 329 shows all attributes of Seconds.

Table 329 – Attributes of CoreEquipmentProfile::Seconds

name	mult	type	description
value	0..1	Float	Time, in seconds
unit	0..1	UnitSymbol	(const=s)
multiplier	0..1	UnitMultiplier	(const=none)

6.201 Susceptance datatype

Imaginary part of admittance.

Table 330 shows all attributes of Susceptance.

Table 330 – Attributes of CoreEquipmentProfile::Susceptance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=S)
multiplier	0..1	UnitMultiplier	(const=none)

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6.202 Voltage datatype

Electrical voltage, can be both AC and DC.

Table 331 shows all attributes of Voltage.

Table 331 – Attributes of CoreEquipmentProfile::Voltage

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=k)
unit	0..1	UnitSymbol	(const=V)

6.203 VoltagePerReactivePower datatype

Voltage variation with reactive power.

Table 332 shows all attributes of VoltagePerReactivePower.

Table 332 – Attributes of CoreEquipmentProfile::VoltagePerReactivePower

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=VPerVAr)
multiplier	0..1	UnitMultiplier	(const=none)

6.204 Boolean primitive

A type with the value space "true" and "false".

6.205 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

6.206 DateTime primitive

Date and time as "yyyy-mm-ddThh:mm:ss.sss", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddThh:mm:ss.sssZ". A local timezone relative UTC is specified as "yyyy-mm-ddThh:mm:ss.sss-hh:mm". The second component (shown here as "ss.sss") could have any number of digits in its fractional part to allow any kind of precision beyond seconds.

6.207 Decimal primitive

Decimal is the base-10 notational system for representing real numbers.

6.208 Float primitive

A floating point number. The range is unspecified and not limited.

6.209 Integer primitive

An integer number. The range is unspecified and not limited.

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6.210 MonthDay primitive

MonthDay format as "--mm-dd", which conforms with XSD data type gMonthDay.

6.211 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

7 Package OperationProfile

7.1 General

This is the IEC 61970-452 operation profile.

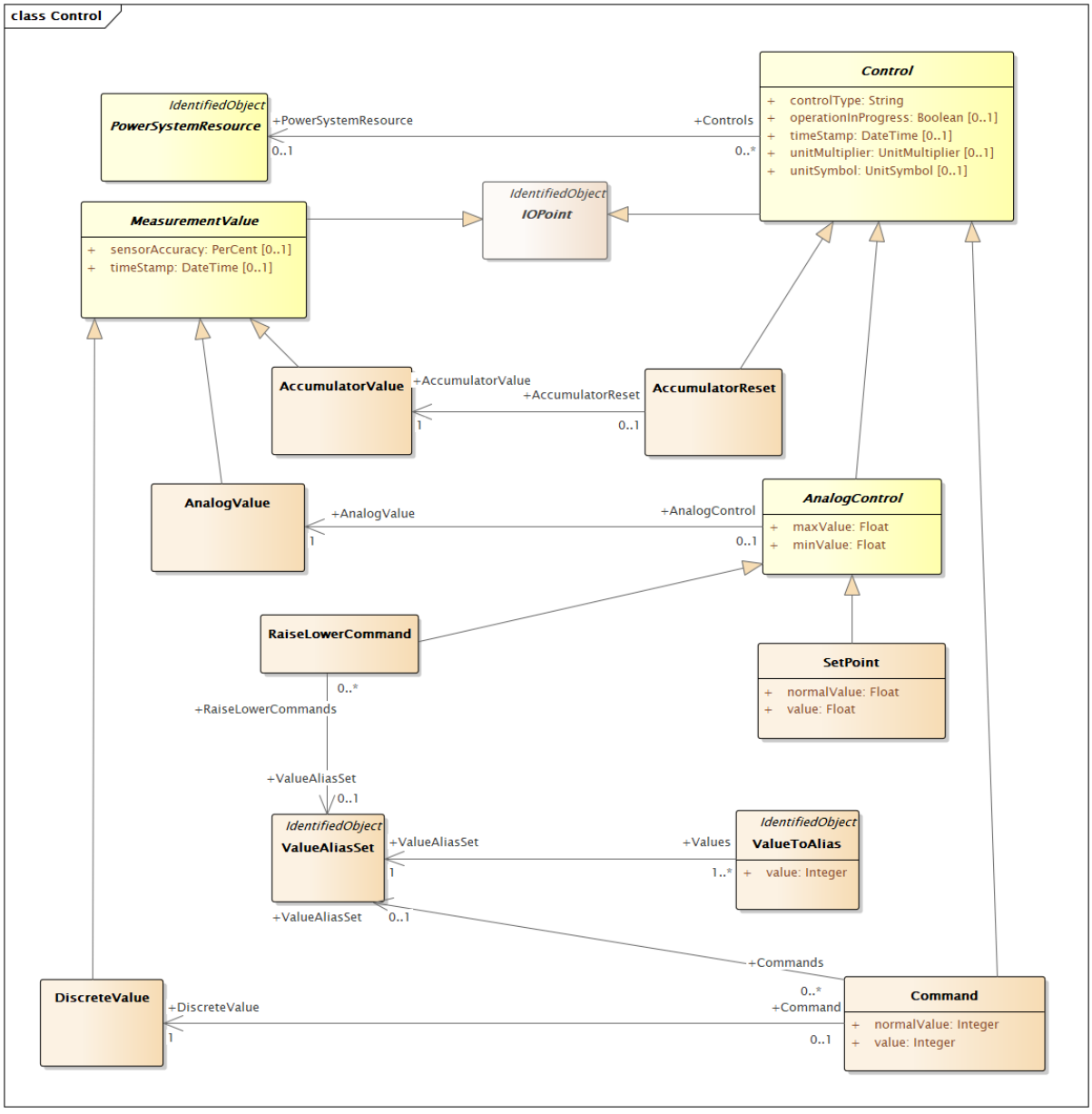

Figure 28 – Class diagram OperationProfile::Control

Figure 28: This diagram shows control related classes included in the Meas package.

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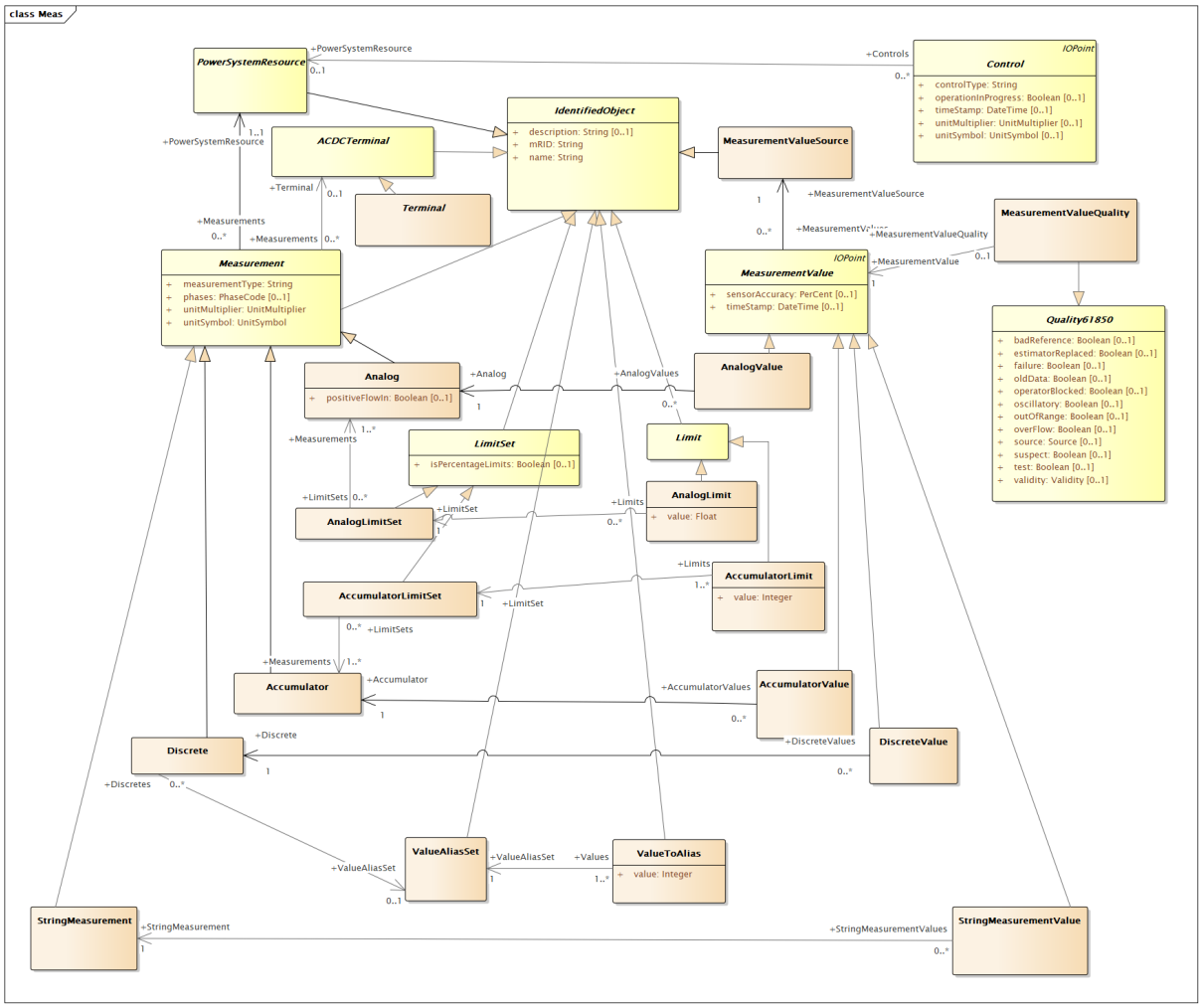

Figure 29 – Class diagram OperationProfile::Meas

Figure 29: This diagram shows measurement related classes included in the Meas package.

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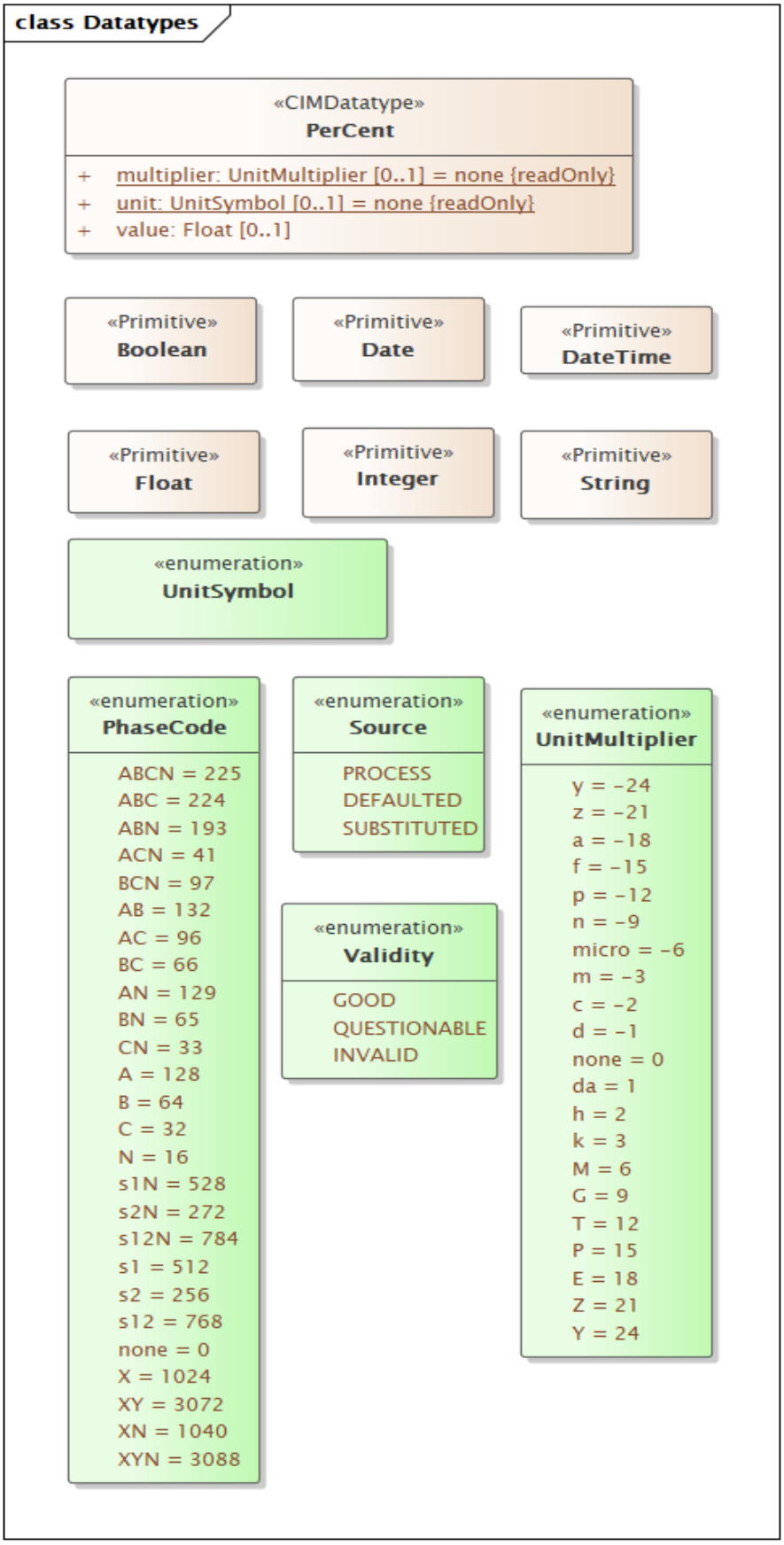

Figure 30 – Class diagram OperationProfile::Datatypes

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Figure 30: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

7.2 Accumulator

Inheritance path = Measurement: IdentifiedObject

Accumulator represents an accumulated (counted) Measurement, e.g. an energy value.

Table 333 shows all attributes of Accumulator.

Table 333 – Attributes of OperationProfile::Accumulator

name	mult	type	description
measurementType	1..1	String	inherited from: Measurement
phases	0..1	PhaseCode	inherited from: Measurement
unitMultiplier	1..1	UnitMultiplier	inherited from: Measurement
unitSymbol	1..1	UnitSymbol	inherited from: Measurement
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 334 shows all association ends of Accumulator with other classes.

Table 334 – Association ends of OperationProfile::Accumulator with other classes

mult from	name	mult to	type	description
0..*	Terminal	0..1	ACDCTerminal	inherited from: Measurement
0..*	PowerSystemResource	1..1	PowerSystemResource	inherited from: Measurement

7.3 AccumulatorLimit

Inheritance path = Limit: IdentifiedObject

Limit values for Accumulator measurements.

Table 335 shows all attributes of AccumulatorLimit.

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Table 335 – Attributes of OperationProfile::AccumulatorLimit

name	mult	type	description
value	1..1	Integer	The value to supervise against. The value is positive.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 336 shows all association ends of AccumulatorLimit with other classes.

Table 336 – Association ends of OperationProfile::AccumulatorLimit with other classes

mult from	name	mult to	type	description
1..*	LimitSet	1..1	AccumulatorLimitSet	The set of limits.

7.4 AccumulatorLimitSet

Inheritance path = LimitSet: IdentifiedObject

An AccumulatorLimitSet specifies a set of Limits that are associated with an Accumulator measurement.

Table 337 shows all attributes of AccumulatorLimitSet.

Table 337 – Attributes of OperationProfile::AccumulatorLimitSet

name	mult	type	description
isPercentageLimits	0..1	Boolean	inherited from: LimitSet
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 338 shows all association ends of AccumulatorLimitSet with other classes.

Table 338 – Association ends of OperationProfile:: AccumulatorLimitSet with other classes

mult from	name	mult to	type	description
0..*	Measurements	1..*	Accumulator	The Measurements using the LimitSet.

7.5 AccumulatorReset

Inheritance path = Control: IOPoint: IdentifiedObject

This command resets the counter value to zero.

Table 339 shows all attributes of AccumulatorReset.

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Table 339 – Attributes of OperationProfile::AccumulatorReset

name	mult	type	description
controlType	1..1	String	inherited from: Control
operationInProgress	0..1	Boolean	inherited from: Control
timeStamp	0..1	DateTime	inherited from: Control
unitMultiplier	0..1	UnitMultiplier	inherited from: Control
unitSymbol	0..1	UnitSymbol	inherited from: Control
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 340 shows all association ends of AccumulatorReset with other classes.

Table 340 – Association ends of OperationProfile::AccumulatorReset with other classes

mult from	name	mult to	type	description
0..1	AccumulatorValue	1..1	AccumulatorValue	The accumulator value that is reset by the command.
0..*	PowerSystemResource	0..1	PowerSystemResource	inherited from: Control

7.6 AccumulatorValue

Inheritance path = MeasurementValue: IOPoint: IdentifiedObject

AccumulatorValue represents an accumulated (counted) MeasurementValue.

Table 341 shows all attributes of AccumulatorValue.

Table 341 – Attributes of OperationProfile::AccumulatorValue

name	mult	type	description
timeStamp	0..1	DateTime	inherited from: MeasurementValue
sensorAccuracy	0..1	PerCent	inherited from: MeasurementValue
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 342 shows all association ends of AccumulatorValue with other classes.

Table 342 – Association ends of OperationProfile::AccumulatorValue with other classes

mult from	name	mult to	type	description
0..*	Accumulator	1..1	Accumulator	Measurement to which this value is connected.
0..*	MeasurementValueSource	1..1	MeasurementValueSource	inherited from: MeasurementValue

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7.7 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 343 shows all attributes of ACDCTerminal.

Table 343 – Attributes of OperationProfile::ACDCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.8 Analog

Inheritance path = Measurement: IdentifiedObject

Analog represents an analog Measurement.

Table 344 shows all attributes of Analog.

Table 344 – Attributes of OperationProfile::Analog

name	mult	type	description
positiveFlowIn	0..1	Boolean	If true then this measurement is an active power, reactive power or current with the convention that a positive value measured at the Terminal means power is flowing into the related PowerSystemResource.
measurementType	1..1	String	inherited from: Measurement
phases	0..1	PhaseCode	inherited from: Measurement
unitMultiplier	1..1	UnitMultiplier	inherited from: Measurement
unitSymbol	1..1	UnitSymbol	inherited from: Measurement
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 345 shows all association ends of Analog with other classes.

Table 345 – Association ends of OperationProfile::Analog with other classes

mult from	name	mult to	type	description
0..*	Terminal	0..1	ACDCTerminal	inherited from: Measurement
0..*	PowerSystemResource	1..1	PowerSystemResource	inherited from: Measurement

7.9 (abstract) AnalogControl

Inheritance path = Control: IOPoint: IdentifiedObject

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An analog control used for supervisory control.

Table 346 shows all attributes of AnalogControl.

Table 346 – Attributes of OperationProfile::AnalogControl

name	mult	type	description
maxValue	1..1	Float	Normal value range maximum for any of the Control.value. Used for scaling, e.g. in bar graphs.
minValue	1..1	Float	Normal value range minimum for any of the Control.value. Used for scaling, e.g. in bar graphs.
controlType	1..1	String	inherited from: Control
operationInProgress	0..1	Boolean	inherited from: Control
timeStamp	0..1	DateTime	inherited from: Control
unitMultiplier	0..1	UnitMultiplier	inherited from: Control
unitSymbol	0..1	UnitSymbol	inherited from: Control
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 347 shows all association ends of AnalogControl with other classes.

Table 347 – Association ends of OperationProfile::AnalogControl with other classes

mult from	name	mult to	type	description
0..1	AnalogValue	1..1	AnalogValue	The MeasurementValue that is controlled.
0..*	PowerSystemResource	0..1	PowerSystemResource	inherited from: Control

7.10 AnalogLimit

Inheritance path = Limit: IdentifiedObject

Limit values for Analog measurements.

Table 348 shows all attributes of AnalogLimit.

Table 348 – Attributes of OperationProfile::AnalogLimit

name	mult	type	description
value	1..1	Float	The value to supervise against.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 349 shows all association ends of AnalogLimit with other classes.

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Table 349 – Association ends of OperationProfile::AnalogLimit with other classes

mult from	name	mult to	type	description
0..*	LimitSet	1..1	AnalogLimitSet	The set of limits.

7.11 AnalogLimitSet

Inheritance path = LimitSet: IdentifiedObject

An AnalogLimitSet specifies a set of Limits that are associated with an Analog measurement.

Table 350 shows all attributes of AnalogLimitSet.

Table 350 – Attributes of OperationProfile::AnalogLimitSet

name	mult	type	description
isPercentageLimits	0..1	Boolean	inherited from: LimitSet
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 351 shows all association ends of AnalogLimitSet with other classes.

Table 351 – Association ends of OperationProfile::AnalogLimitSet with other classes

mult from	name	mult to	type	description
0..*	Measurements	1..*	Analog	The Measurements using the LimitSet.

7.12 AnalogValue

Inheritance path = MeasurementValue: IOPoint: IdentifiedObject

AnalogValue represents an analog MeasurementValue.

Table 352 shows all attributes of AnalogValue.

Table 352 – Attributes of OperationProfile::AnalogValue

name	mult	type	description
timeStamp	0..1	DateTime	inherited from: MeasurementValue
sensorAccuracy	0..1	PerCent	inherited from: MeasurementValue
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 353 shows all association ends of AnalogValue with other classes.

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Table 353 – Association ends of OperationProfile::AnalogValue with other classes

mult from	name	mult to	type	description
0..*	Analog	1..1	Analog	Measurement to which this value is connected.
0..*	MeasurementValueSource	1..1	MeasurementValueSource	inherited from: MeasurementValue

7.13 Command

Inheritance path = Control: IOPoint: IdentifiedObject

A Command is a discrete control used for supervisory control.

Table 354 shows all attributes of Command.

Table 354 – Attributes of OperationProfile::Command

name	mult	type	description
normalValue	1..1	Integer	Normal value for Control.value e.g. used for percentage scaling.
value	1..1	Integer	The value representing the actuator output.
controlType	1..1	String	inherited from: Control
operationInProgress	0..1	Boolean	inherited from: Control
timeStamp	0..1	DateTime	inherited from: Control
unitMultiplier	0..1	UnitMultiplier	inherited from: Control
unitSymbol	0..1	UnitSymbol	inherited from: Control
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 355 shows all association ends of Command with other classes.

Table 355 – Association ends of OperationProfile::Command with other classes

mult from	name	mult to	type	description
0..*	ValueAliasSet	0..1	ValueAliasSet	The ValueAliasSet used for translation of a Control value to a name.
0..1	DiscreteValue	1..1	DiscreteValue	The MeasurementValue that is controlled.
0..*	PowerSystemResource	0..1	PowerSystemResource	inherited from: Control

7.14 (abstract) Control

Inheritance path = IOPoint: IdentifiedObject

Control is used for supervisory/device control. It represents control outputs that are used to change the state in a process, e.g. close or open breaker, a set point value or a raise lower command.

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Table 356 shows all attributes of Control.

Table 356 – Attributes of OperationProfile::Control

name	mult	type	description
controlType	1..1	String	Specifies the type of Control. For example, this specifies if the Control represents BreakerOpen, BreakerClose, GeneratorVoltageSetPoint, GeneratorRaise, GeneratorLower, etc.
operationInProgress	0..1	Boolean	Indicates that a client is currently sending control commands that has not completed.
timeStamp	0..1	DateTime	The last time a control output was sent.
unitMultiplier	0..1	UnitMultiplier	The unit multiplier of the controlled quantity.
unitSymbol	0..1	UnitSymbol	The unit of measure of the controlled quantity.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 357 shows all association ends of Control with other classes.

Table 357 – Association ends of OperationProfile::Control with other classes

mult from	name	mult to	type	description
0..*	PowerSystemResource	0..1	PowerSystemResource	Regulating device governed by this control output.

7.15 Discrete

Inheritance path = Measurement: IdentifiedObject

Discrete represents a discrete Measurement, i.e. a Measurement representing discrete values, e.g. a Breaker position.

Table 358 shows all attributes of Discrete.

Table 358 – Attributes of OperationProfile::Discrete

name	mult	type	description
measurementType	1..1	String	inherited from: Measurement
phases	0..1	PhaseCode	inherited from: Measurement
unitMultiplier	1..1	UnitMultiplier	inherited from: Measurement
unitSymbol	1..1	UnitSymbol	inherited from: Measurement
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 359 shows all association ends of Discrete with other classes.

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Table 359 – Association ends of OperationProfile::Discrete with other classes

mult from	name	mult to	type	description
0..*	ValueAliasSet	0..1	ValueAliasSet	The ValueAliasSet used for translation of a MeasurementValue.value to a name.
0..*	Terminal	0..1	ACDCTerminal	inherited from: Measurement
0..*	PowerSystemResource	1..1	PowerSystemResource	inherited from: Measurement

7.16 DiscreteValue

Inheritance path = MeasurementValue: IOPoint: IdentifiedObject

DiscreteValue represents a discrete MeasurementValue.

Table 360 shows all attributes of DiscreteValue.

Table 360 – Attributes of OperationProfile::DiscreteValue

name	mult	type	description
timeStamp	0..1	DateTime	inherited from: MeasurementValue
sensorAccuracy	0..1	PerCent	inherited from: MeasurementValue
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 361 shows all association ends of DiscreteValue with other classes.

Table 361 – Association ends of OperationProfile::DiscreteValue with other classes

mult from	name	mult to	type	description
0..*	Discrete	1..1	Discrete	Measurement to which this value is connected.
0..*	MeasurementValueSource	1..1	MeasurementValueSource	inherited from: MeasurementValue

7.17 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 362 shows all attributes of IdentifiedObject.

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Table 362 – Attributes of OperationProfile::IdentifiedObject

name	mult	type	description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

7.18 (abstract) IOPoint

Inheritance path = IdentifiedObject

The class describe a measurement or control value. The purpose is to enable having attributes and associations common for measurement and control.

Table 363 shows all attributes of IOPoint.

Table 363 – Attributes of OperationProfile::IOPoint

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.19 (abstract) Limit

Inheritance path = IdentifiedObject

Specifies one limit value for a Measurement. A Measurement typically has several limits that are kept together by the LimitSet class. The actual meaning and use of a Limit instance (i.e., if it is an alarm or warning limit or if it is a high or low limit) is not captured in the Limit class. However the name of a Limit instance may indicate both meaning and use.

Table 364 shows all attributes of Limit.

Table 364 – Attributes of OperationProfile::Limit

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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7.20 (abstract) LimitSet

Inheritance path = IdentifiedObject

Specifies a set of Limits that are associated with a Measurement. A Measurement may have several LimitSets corresponding to seasonal or other changing conditions. The condition is captured in the name and description attributes. The same LimitSet may be used for several Measurements. In particular percentage limits are used this way.

Table 365 shows all attributes of LimitSet.

Table 365 – Attributes of OperationProfile::LimitSet

name	mult	type	description
isPercentageLimits	0..1	Boolean	Tells if the limit values are in percentage of normalValue or the specified Unit for Measurements and Controls.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.21 (abstract) Measurement

Inheritance path = IdentifiedObject

A Measurement represents any measured, calculated or non-measured non-calculated quantity. Any piece of equipment may contain Measurements, e.g. a substation may have temperature measurements and door open indications, a transformer may have oil temperature and tank pressure measurements, a bay may contain a number of power flow measurements and a Breaker may contain a switch status measurement.

The PSR – Measurement association is intended to capture this use of Measurement and is included in the naming hierarchy based on EquipmentContainer. The naming hierarchy typically has Measurements as leaves, e.g. Substation-VoltageLevel-Bay-Switch-Measurement.

Some Measurements represent quantities related to a particular sensor location in the network, e.g. a voltage transformer (VT) or potential transformer (PT) at a busbar or a current transformer (CT) at the bar between a breaker and an isolator. The sensing position is not captured in the PSR – Measurement association. Instead it is captured by the Measurement – Terminal association that is used to define the sensing location in the network topology. The location is defined by the connection of the Terminal to ConductingEquipment.

If both a Terminal and PSR are associated, and the PSR is of type ConductingEquipment, the associated Terminal should belong to that ConductingEquipment instance.

When the sensor location is needed both Measurement-PSR and Measurement-Terminal are used. The Measurement-Terminal association is never used alone.

Table 366 shows all attributes of Measurement.

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Table 366 – Attributes of OperationProfile::Measurement

name	mult	type	description
measurementType	1..1	String	Specifies the type of measurement. For example, this specifies if the measurement represents an indoor temperature, outdoor temperature, bus voltage, line flow, etc. When the measurementType is set to "Specialization", the type of Measurement is defined in more detail by the specialized class which inherits from Measurement.
phases	0..1	PhaseCode	Indicates to which phases the measurement applies and avoids the need to use 'measurementType' to also encode phase information (which would explode the types). The phase information in Measurement, along with 'measurementType' and 'phases' uniquely defines a Measurement for a device, based on normal network phase. Their meaning will not change when the computed energizing phasing is changed due to jumpers or other reasons. If the attribute is missing three phases (ABC) shall be assumed.
unitMultiplier	1..1	UnitMultiplier	The unit multiplier of the measured quantity.
unitSymbol	1..1	UnitSymbol	The unit of measure of the measured quantity.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 367 shows all association ends of Measurement with other classes.

Table 367 – Association ends of OperationProfile::Measurement with other classes

mult from	name	mult to	type	description
0..*	Terminal	0..1	ACDCTerminal	One or more measurements may be associated with a terminal in the network.
0..*	PowerSystemResource	1..1	PowerSystemResource	The power system resource that contains the measurement.

7.22 (abstract) MeasurementValue

Inheritance path = IOPoint: IdentifiedObject

The current state for a measurement. A state value is an instance of a measurement from a specific source. Measurements can be associated with many state values, each representing a different source for the measurement.

Table 368 shows all attributes of MeasurementValue.

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Table 368 – Attributes of OperationProfile::MeasurementValue

name	mult	type	description
timeStamp	0..1	DateTime	The time when the value was last updated.
sensorAccuracy	0..1	PerCent	The limit, expressed as a percentage of the sensor maximum, that errors will not exceed when the sensor is used under reference conditions.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 369 shows all association ends of MeasurementValue with other classes.

Table 369 – Association ends of OperationProfile:: MeasurementValue with other classes

mult from	name	mult to	type	description
0..*	MeasurementValueSource	1..1	MeasurementValueSource	A reference to the type of source that updates the MeasurementValue, e.g. SCADA, CCLink, manual, etc. User conventions for the names of sources are contained in the introduction to IEC 61970-301.

7.23 MeasurementValueQuality

Inheritance path = Quality61850

Measurement quality flags. Bits 0-10 are defined for substation automation in IEC 61850-7-3. Bits 11 to 15 are reserved for future expansion by that document. Bits 16 to 31 are reserved for EMS applications.

Table 370 shows all attributes of MeasurementValueQuality.

Table 370 – Attributes of OperationProfile::MeasurementValueQuality

name	mult	type	description
badReference	0..1	Boolean	inherited from: Quality61850
estimatorReplaced	0..1	Boolean	inherited from: Quality61850
failure	0..1	Boolean	inherited from: Quality61850
oldData	0..1	Boolean	inherited from: Quality61850
operatorBlocked	0..1	Boolean	inherited from: Quality61850
oscillatory	0..1	Boolean	inherited from: Quality61850
outOfRange	0..1	Boolean	inherited from: Quality61850
overFlow	0..1	Boolean	inherited from: Quality61850
source	0..1	Source	inherited from: Quality61850
suspect	0..1	Boolean	inherited from: Quality61850
test	0..1	Boolean	inherited from: Quality61850
validity	0..1	Validity	inherited from: Quality61850

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Table 371 shows all association ends of MeasurementValueQuality with other classes.

Table 371 – Association ends of OperationProfile:: MeasurementValueQuality with other classes

mult from	name	mult to	type	description
0..1	MeasurementValue	1..1	MeasurementValue	A MeasurementValue has a MeasurementValueQuality associated with it.

7.24 MeasurementValueSource

Inheritance path = IdentifiedObject

MeasurementValueSource describes the alternative sources updating a MeasurementValue. User conventions for how to use the MeasurementValueSource attributes are defined in IEC 61970-301.

Table 372 shows all attributes of MeasurementValueSource.

Table 372 – Attributes of OperationProfile::MeasurementValueSource

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.25 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 373 shows all attributes of PowerSystemResource.

Table 373 – Attributes of OperationProfile::PowerSystemResource

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.26 (abstract) Quality61850 root class

Quality flags in this class are as defined in IEC 61850, except for estimatorReplaced, which has been included in this class for convenience.

Table 374 shows all attributes of Quality61850.

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Table 374 – Attributes of OperationProfile::Quality61850

name	mult	type	description
badReference	0..1	Boolean	Measurement value may be incorrect due to a reference being out of calibration.
estimatorReplaced	0..1	Boolean	Value has been replaced by State Estimator. estimatorReplaced is not an IEC61850 quality bit but has been put in this class for convenience.
failure	0..1	Boolean	This identifier indicates that a supervision function has detected an internal or external failure, e.g. communication failure.
oldData	0..1	Boolean	Measurement value is old and possibly invalid, as it has not been successfully updated during a specified time interval.
operatorBlocked	0..1	Boolean	Measurement value is blocked and hence unavailable for transmission.
oscillatory	0..1	Boolean	To prevent some overload of the communication it is sensible to detect and suppress oscillating (fast changing) binary inputs. If a signal changes in a defined time twice in the same direction (from 0 to 1 or from 1 to 0) then oscillation is detected and the detail quality identifier "oscillatory" is set. If it is detected a configured numbers of transient changes could be passed by. In this time the validity status "questionable" is set. If after this defined numbers of changes the signal is still in the oscillating state the value shall be set either to the opposite state of the previous stable value or to a defined default value. In this case the validity status "questionable" is reset and "invalid" is set as long as the signal is oscillating. If it is configured such that no transient changes should be passed by then the validity status "invalid" is set immediately in addition to the detail quality identifier "oscillatory" (used for status information only).
outOfRange	0..1	Boolean	Measurement value is beyond a predefined range of value.
overFlow	0..1	Boolean	Measurement value is beyond the capability of being represented properly. For example, a counter value overflows from maximum count back to a value of zero.
source	0..1	Source	Source gives information related to the origin of a value. The value may be acquired from the process, defaulted or substituted.
suspect	0..1	Boolean	A correlation function has detected that the value is not consistent with other values. Typically set by a network State Estimator.
test	0..1	Boolean	Measurement value is transmitted for test purposes.
validity	0..1	Validity	Validity of the measurement value.

7.27 RaiseLowerCommand

Inheritance path = AnalogControl: Control: IOPoint: IdentifiedObject

An analog control that increases or decreases a set point value with pulses. Unless otherwise specified, one pulse moves the set point by one.

Table 375 shows all attributes of RaiseLowerCommand.

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Table 375 – Attributes of OperationProfile::RaiseLowerCommand

name	mult	type	description
maxValue	1..1	Float	inherited from: AnalogControl
minValue	1..1	Float	inherited from: AnalogControl
controlType	1..1	String	inherited from: Control
operationInProgress	0..1	Boolean	inherited from: Control
timeStamp	0..1	DateTime	inherited from: Control
unitMultiplier	0..1	UnitMultiplier	inherited from: Control
unitSymbol	0..1	UnitSymbol	inherited from: Control
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 376 shows all association ends of RaiseLowerCommand with other classes.

Table 376 – Association ends of OperationProfile:: RaiseLowerCommand with other classes

mult from	name	mult to	type	description
0..*	ValueAliasSet	0..1	ValueAliasSet	The ValueAliasSet used for translation of a Control value to a name.
0..1	AnalogValue	1..1	AnalogValue	inherited from: AnalogControl
0..*	PowerSystemResource	0..1	PowerSystemResource	inherited from: Control

7.28 SetPoint

Inheritance path = AnalogControl: Control: IOPoint: IdentifiedObject

An analog control that issues a set point value.

Table 377 shows all attributes of SetPoint.

Table 377 – Attributes of OperationProfile::SetPoint

name	mult	type	description
normalValue	1..1	Float	Normal value for Control.value e.g. used for percentage scaling.
value	1..1	Float	The value representing the actuator output.
maxValue	1..1	Float	inherited from: AnalogControl
minValue	1..1	Float	inherited from: AnalogControl
controlType	1..1	String	inherited from: Control
operationInProgress	0..1	Boolean	inherited from: Control
timeStamp	0..1	DateTime	inherited from: Control
unitMultiplier	0..1	UnitMultiplier	inherited from: Control
unitSymbol	0..1	UnitSymbol	inherited from: Control
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	1..1	String	inherited from: IdentifiedObject

Table 378 shows all association ends of SetPoint with other classes.

Table 378 – Association ends of OperationProfile::SetPoint with other classes

mult from	name	mult to	type	description
0..1	AnalogValue	1..1	AnalogValue	inherited from: AnalogControl
0..*	PowerSystemResource	0..1	PowerSystemResource	inherited from: Control

7.29 StringMeasurement

Inheritance path = Measurement: IdentifiedObject

StringMeasurement represents a measurement with values of type string.

Table 379 shows all attributes of StringMeasurement.

Table 379 – Attributes of OperationProfile::StringMeasurement

name	mult	type	description
measurementType	1..1	String	inherited from: Measurement
phases	0..1	PhaseCode	inherited from: Measurement
unitMultiplier	1..1	UnitMultiplier	inherited from: Measurement
unitSymbol	1..1	UnitSymbol	inherited from: Measurement
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 380 shows all association ends of StringMeasurement with other classes.

Table 380 – Association ends of OperationProfile:: StringMeasurement with other classes

mult from	name	mult to	type	description
0..*	Terminal	0..1	ACDCTerminal	inherited from: Measurement
0..*	PowerSystemResource	1..1	PowerSystemResource	inherited from: Measurement

7.30 StringMeasurementValue

Inheritance path = MeasurementValue: IOPoint: IdentifiedObject

StringMeasurementValue represents a measurement value of type string.

Table 381 shows all attributes of StringMeasurementValue.

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Table 381 – Attributes of OperationProfile::StringMeasurementValue

name	mult	type	description
timeStamp	0..1	DateTime	inherited from: MeasurementValue
sensorAccuracy	0..1	PerCent	inherited from: MeasurementValue
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 382 shows all association ends of StringMeasurementValue with other classes.

Table 382 – Association ends of OperationProfile:: StringMeasurementValue with other classes

mult from	name	mult to	type	description
0..*	StringMeasurement	1..1	StringMeasurement	Measurement to which this value is connected.
0..*	MeasurementValueSource	1..1	MeasurementValueSource	inherited from: MeasurementValue

7.31 (abstract) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 383 shows all attributes of Terminal.

Table 383 – Attributes of OperationProfile::Terminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.32 ValueAliasSet

Inheritance path = IdentifiedObject

Describes the translation of a set of values into a name and is intendend to facilitate custom translations. Each ValueAliasSet has a name, description etc. A specific Measurement may represent a discrete state like Open, Closed, Intermediate etc. This requires a translation from the MeasurementValue.value number to a string, e.g. 0->"Invalid", 1->"Open", 2->"Closed", 3->"Intermediate". Each ValueToAlias member in ValueAliasSet.Value describe a mapping for one particular value to a name.

Table 384 shows all attributes of ValueAliasSet.

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Table 384 – Attributes of OperationProfile::ValueAliasSet

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

7.33 ValueToAlias

Inheritance path = IdentifiedObject

Describes the translation of one particular value into a name, e.g. 1 as "Open".

Table 385 shows all attributes of ValueToAlias.

Table 385 – Attributes of OperationProfile::ValueToAlias

name	mult	type	description
value	1..1	Integer	The value that is mapped.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 386 shows all association ends of ValueToAlias with other classes.

Table 386 – Association ends of OperationProfile::ValueToAlias with other classes

mult from	name	mult to	type	description
1..*	ValueAliasSet	1..1	ValueAliasSet	The ValueAliasSet having the ValueToAlias mappings.

7.34 PhaseCode enumeration

An unordered enumeration of phase identifiers. Allows designation of phases for both transmission and distribution equipment, circuits and loads. The enumeration, by itself, does not describe how the phases are connected together or connected to ground. Ground is not explicitly denoted as a phase.

Residential and small commercial loads are often served from single-phase, or split-phase, secondary circuits. For the example of s12N, phases 1 and 2 refer to hot wires that are 180 degrees out of phase, while N refers to the neutral wire. Through single-phase transformer connections, these secondary circuits may be served from one or two of the primary phases A, B, and C. For three-phase loads, use the A, B, C phase codes instead of s12N.

The integer values are from IEC 61968-9 to support revenue metering applications.

Table 387 shows all literals of PhaseCode.

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Table 387 – Literals of OperationProfile::PhaseCode

literal	value	description
ABCN	225	Phases A, B, C, and N.
ABC	224	Phases A, B, and C.
ABN	193	Phases A, B, and neutral.
ACN	41	Phases A, C and neutral.
BCN	97	Phases B, C, and neutral.
AB	132	Phases A and B.
AC	96	Phases A and C.
BC	66	Phases B and C.
AN	129	Phases A and neutral.
BN	65	Phases B and neutral.
CN	33	Phases C and neutral.
A	128	Phase A.
B	64	Phase B.
C	32	Phase C.
N	16	Neutral phase.
s1N	528	Secondary phase 1 and neutral.
s2N	272	Secondary phase 2 and neutral.
s12N	784	Secondary phases 1, 2, and neutral.
s1	512	Secondary phase 1.
s2	256	Secondary phase 2.
s12	768	Secondary phase 1 and 2.
none	0	No phases specified.
X	1024	Unknown non-neutral phase.
XY	3072	Two unknown non-neutral phases.
XN	1040	Unknown non-neutral phase plus neutral.
XYN	3088	Two unknown non-neutral phases plus neutral.

7.35 Source enumeration

Source gives information related to the origin of a value.

Table 388 shows all literals of Source.

Table 388 – Literals of OperationProfile::Source

literal	value	description
PROCESS		The value is provided by input from the process I/O or being calculated from some function.
DEFAULTED		The value contains a default value.
SUBSTITUTED		The value is provided by input of an operator or by an automatic source.

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7.36 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 389 shows all literals of UnitMultiplier.

Table 389 – Literals of OperationProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

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7.37 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

Table 390 shows all literals of UnitSymbol.

Table 390 – Literals of OperationProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).

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literal	value	description
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).

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literal	value	description
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg·m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier

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literal	value	description
		“d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µg/g’.
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.

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literal	value	description
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133.3 Pa).
M	217	Length, nautical miles (1 M = 1852 m).
kn	219	Speed, knots (1 kn = 1852/3600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).

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literal	value	description
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

7.38 Validity enumeration

Validity for MeasurementValue.

Table 391 shows all literals of Validity.

Table 391 – Literals of OperationProfile::Validity

literal	value	description
GOOD		The value is marked good if no abnormal condition of the acquisition function or the information source is detected.
QUESTIONABLE		The value is marked questionable if a supervision function detects an abnormal behaviour, however the value could still be valid. The client is responsible for determining whether or not values marked "questionable" should be used.
INVALID		The value is marked invalid when a supervision function recognises abnormal conditions of the acquisition function or the information source (missing or non-operating updating devices). The value is not defined under this condition. The mark invalid is used to indicate to the client that the value may be incorrect and shall not be used.

7.39 PerCent datatype

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Table 392 shows all attributes of PerCent.

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Table 392 – Attributes of OperationProfile::PerCent

name	mult	type	description
value	0..1	Float	Normally 0 to 100 on a defined base.
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

7.40 Boolean primitive

A type with the value space "true" and "false".

7.41 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

7.42 DateTime primitive

Date and time as "yyyy-mm-ddThh:mm:ss.sss", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddThh:mm:ss.sssZ". A local timezone relative UTC is specified as "yyyy-mm-ddThh:mm:ss.sss-hh:mm". The second component (shown here as "ss.sss") could have any number of digits in its fractional part to allow any kind of precision beyond seconds.

7.43 Float primitive

A floating point number. The range is unspecified and not limited.

7.44 Integer primitive

An integer number. The range is unspecified and not limited.

7.45 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

8 Package ShortCircuitProfile

8.1 General

This is the IEC 61970-452 short circuit profile.

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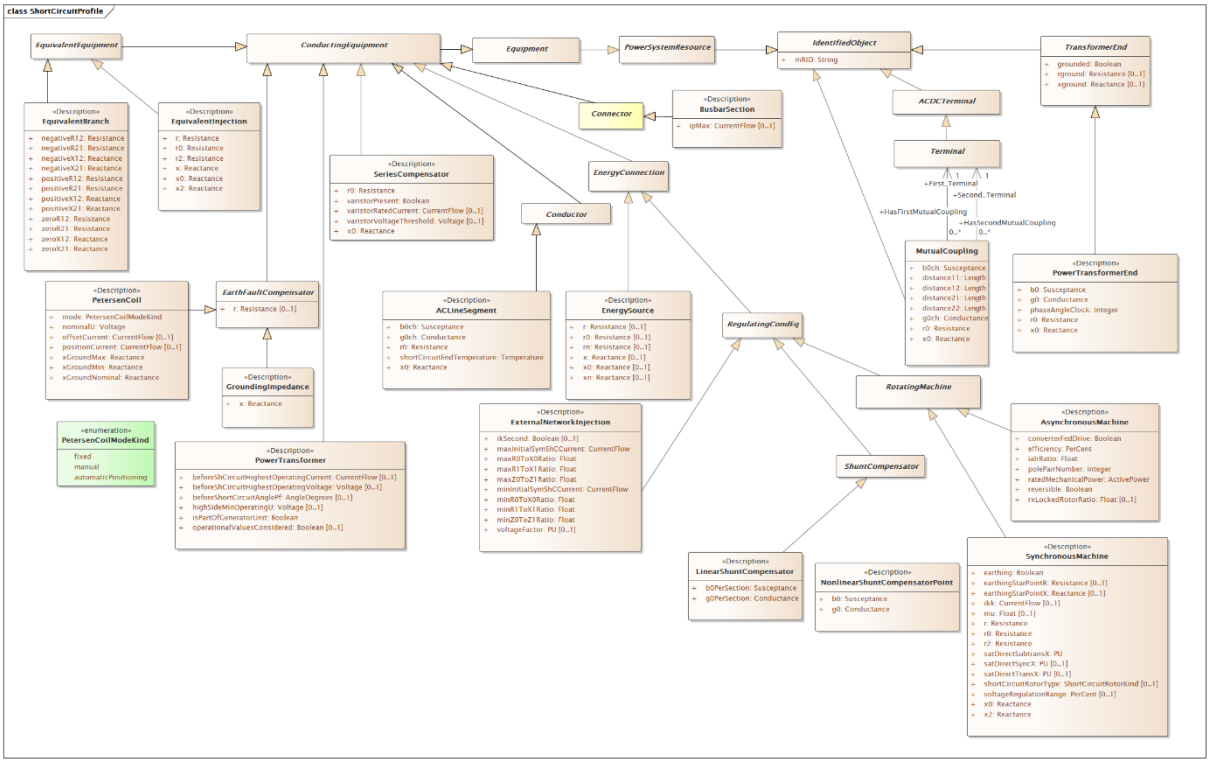

Figure 31 – Class diagram ShortCircuitProfile::ShortCircuitProfile

Figure 31: The diagram shows all classes related to equipment short circuit profile.

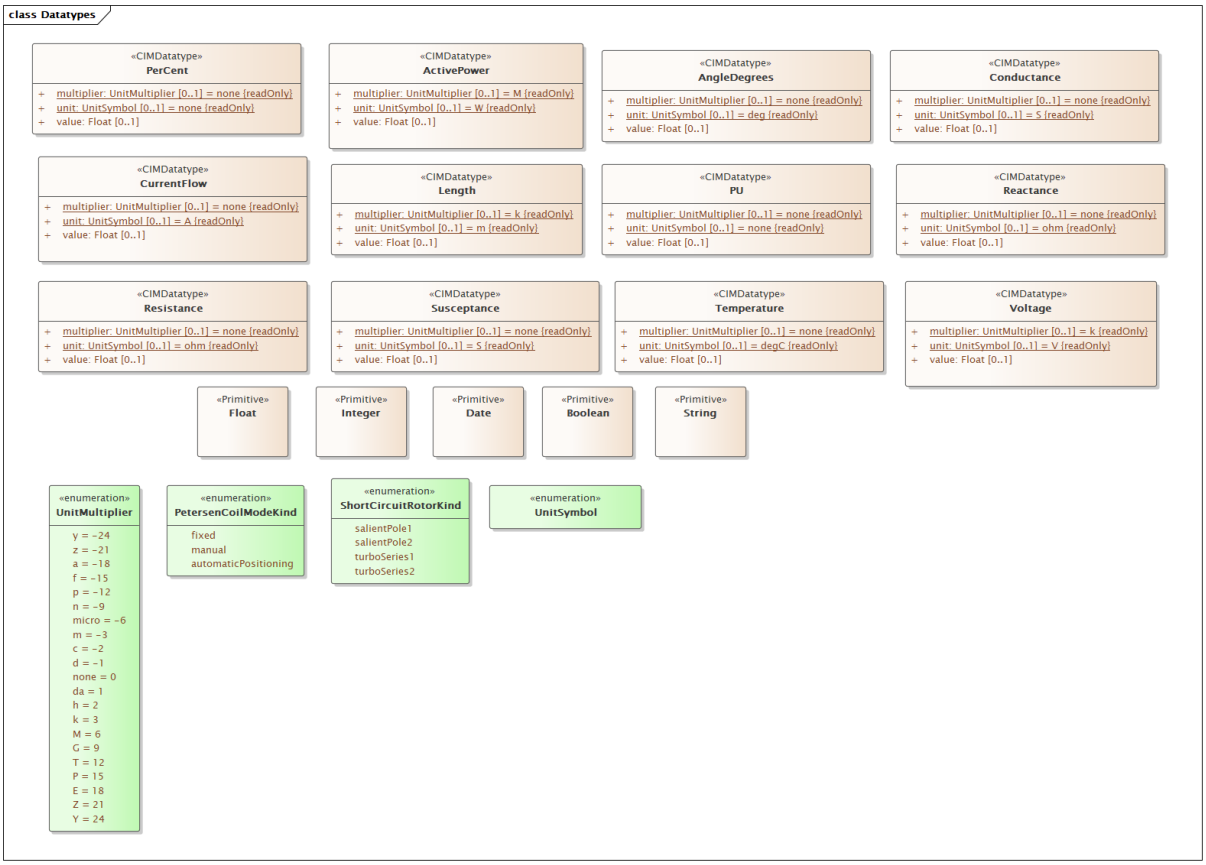

Figure 32 – Class diagram ShortCircuitProfile::Datatypes

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Figure 32: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

8.2 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 393 shows all attributes of ACDCTerminal.

Table 393 – Attributes of ShortCircuitProfile::ACDCTerminal

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.3 (Description) ACLineSegment

Inheritance path = Conductor: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A wire or combination of wires, with consistent electrical characteristics, building a single electrical system, used to carry alternating current between points in the power system.

For symmetrical, transposed three phase lines, it is sufficient to use attributes of the line segment, which describe impedances and admittances for the entire length of the segment. Additionally impedances can be computed by using length and associated per length impedances.

The BaseVoltage at the two ends of ACLineSegments in a Line shall have the same BaseVoltage.nominalVoltage. However, boundary lines may have slightly different BaseVoltage.nominalVoltages and variation is allowed. Larger voltage difference in general requires use of an equivalent branch.

Table 394 shows all attributes of ACLineSegment.

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Table 394 – Attributes of ShortCircuitProfile::ACLineSegment

name	mult	type	description
b0ch	1..1	Susceptance	Zero sequence shunt (charging) susceptance, uniformly distributed, of the entire line section.
g0ch	1..1	Conductance	Zero sequence shunt (charging) conductance, uniformly distributed, of the entire line section.
r0	1..1	Resistance	Zero sequence series resistance of the entire line section.
shortCircuitEndTemperature	1..1	Temperature	Maximum permitted temperature at the end of SC for the calculation of minimum short-circuit currents. Used for short circuit data exchange according to IEC 60909.
x0	1..1	Reactance	Zero sequence series reactance of the entire line section.
mRID	1..1	String	inherited from: IdentifiedObject

8.4 (Description) AsynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g. squirrel-cage induction machine.

Table 395 shows all attributes of AsynchronousMachine.

Table 395 – Attributes of ShortCircuitProfile::AsynchronousMachine

name	mult	type	description
converterFedDrive	1..1	Boolean	Indicates whether the machine is a converter fed drive. Used for short circuit data exchange according to IEC 60909.
efficiency	1..1	PerCent	Efficiency of the asynchronous machine at nominal operation as a percentage. Indicator for converter drive motors. Used for short circuit data exchange according to IEC 60909.
iaIrRatio	1..1	Float	Ratio of locked-rotor current to the rated current of the motor (Ia/Ir). Used for short circuit data exchange according to IEC 60909.
polePairNumber	1..1	Integer	Number of pole pairs of stator. Used for short circuit data exchange according to IEC 60909.
ratedMechanicalPower	1..1	ActivePower	Rated mechanical power (Pr in IEC 60909-0). Used for short circuit data exchange according to IEC 60909.
reversible	1..1	Boolean	Indicates for converter drive motors if the power can be reversible. Used for short circuit data exchange according to IEC 60909.
rxLockedRotorRatio	0..1	Float	Locked rotor ratio (R/X). Used for short circuit data exchange according to IEC 60909.
mRID	1..1	String	inherited from: IdentifiedObject

8.5 (Description) BusbarSection

Inheritance path = Connector: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

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A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation.

Voltage measurements are typically obtained from voltage transformers that are connected to busbar sections. A bus bar section may have many physical terminals but for analysis is modelled with exactly one logical terminal.

Table 396 shows all attributes of BusbarSection.

Table 396 – Attributes of ShortCircuitProfile::BusbarSection

name	mult	type	description
ipMax	0..1	CurrentFlow	Maximum allowable peak short-circuit current of busbar (Ipmax in IEC 60909-0). Mechanical limit of the busbar in the substation itself. Used for short circuit data exchange according to IEC 60909.
mRID	1..1	String	inherited from: IdentifiedObject

8.6 (abstract) ConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Table 397 shows all attributes of ConductingEquipment.

Table 397 – Attributes of ShortCircuitProfile::ConductingEquipment

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.7 (abstract) Conductor

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Combination of conducting material with consistent electrical characteristics, building a single electrical system, used to carry current between points in the power system.

Table 398 shows all attributes of Conductor.

Table 398 – Attributes of ShortCircuitProfile::Conductor

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.8 (abstract) Connector

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation and are modelled with a single logical terminal.

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Table 399 shows all attributes of Connector.

Table 399 – Attributes of ShortCircuitProfile::Connector

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.9 (abstract) EarthFaultCompensator

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conducting equipment used to represent a connection to ground which is typically used to compensate earth faults. An earth fault compensator device modelled with a single terminal implies a second terminal solidly connected to ground. If two terminals are modelled, the ground is not assumed and normal connection rules apply.

Table 400 shows all attributes of EarthFaultCompensator.

Table 400 – Attributes of ShortCircuitProfile::EarthFaultCompensator

name	mult	type	description
r	0..1	Resistance	Nominal resistance of device.
mRID	1..1	String	inherited from: IdentifiedObject

8.10 (abstract) EnergyConnection

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection of energy generation or consumption on the power system model.

Table 401 shows all attributes of EnergyConnection.

Table 401 – Attributes of ShortCircuitProfile::EnergyConnection

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.11 (Description) EnergySource

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Table 402 shows all attributes of EnergySource.

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Table 402 – Attributes of ShortCircuitProfile::EnergySource

name	mult	type	description
r	0..1	Resistance	Positive sequence Thevenin resistance.
r0	0..1	Resistance	Zero sequence Thevenin resistance.
rn	0..1	Resistance	Negative sequence Thevenin resistance.
x	0..1	Reactance	Positive sequence Thevenin reactance.
x0	0..1	Reactance	Zero sequence Thevenin reactance.
xn	0..1	Reactance	Negative sequence Thevenin reactance.
mRID	1..1	String	inherited from: IdentifiedObject

8.12 (abstract) Equipment

Inheritance path = PowerSystemResource: IdentifiedObject

The parts of a power system that are physical devices, electronic or mechanical.

Table 403 shows all attributes of Equipment.

Table 403 – Attributes of ShortCircuitProfile::Equipment

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.13 (Description) EquivalentBranch

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent branches. In cases where a transformer phase shift is modelled and the EquivalentBranch is spanning the same nodes, the impedance quantities for the EquivalentBranch shall consider the needed phase shift.

Table 404 shows all attributes of EquivalentBranch.

Table 404 – Attributes of ShortCircuitProfile::EquivalentBranch

name	mult	type	description
negativeR12	1..1	Resistance	Negative sequence series resistance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.
negativeR21	1..1	Resistance	Negative sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.

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name	mult	type	description
negativeX12	1..1	Reactance	Negative sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
negativeX21	1..1	Reactance	Negative sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
positiveR12	1..1	Resistance	Positive sequence series resistance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.
positiveR21	1..1	Resistance	Positive sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.
positiveX12	1..1	Reactance	Positive sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
positiveX21	1..1	Reactance	Positive sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
zeroR12	1..1	Resistance	Zero sequence series resistance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. EquivalentBranch is a result of network reduction prior to the data exchange.
zeroR21	1..1	Resistance	Zero sequence series resistance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
zeroX12	1..1	Reactance	Zero sequence series reactance from terminal sequence 1 to terminal sequence 2. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
zeroX21	1..1	Reactance	Zero sequence series reactance from terminal sequence 2 to terminal sequence 1. Used for short circuit data exchange according to IEC 60909. Usage: EquivalentBranch is a result of network reduction prior to the data exchange.
mRID	1..1	String	inherited from: IdentifiedObject

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8.14 (abstract) EquivalentEquipment

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent objects that are the result of a network reduction. The class is the base for equivalent objects of different types.

Table 405 shows all attributes of EquivalentEquipment.

Table 405 – Attributes of ShortCircuitProfile::EquivalentEquipment

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.15 (Description) EquivalentInjection

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents equivalent injections (generation or load). Voltage regulation is allowed only at the point of connection.

Table 406 shows all attributes of EquivalentInjection.

Table 406 – Attributes of ShortCircuitProfile::EquivalentInjection

name	mult	type	description
r	1..1	Resistance	Positive sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
r0	1..1	Resistance	Zero sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
r2	1..1	Resistance	Negative sequence resistance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
x	1..1	Reactance	Positive sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
x0	1..1	Reactance	Zero sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
x2	1..1	Reactance	Negative sequence reactance. Used to represent Extended-Ward (IEC 60909). Usage: Extended-Ward is a result of network reduction prior to the data exchange.
mRID	1..1	String	inherited from: IdentifiedObject

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8.16 (Description) ExternalNetworkInjection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents the external network and it is used for IEC 60909 calculations.

Table 407 shows all attributes of ExternalNetworkInjection.

Table 407 – Attributes of ShortCircuitProfile::ExternalNetworkInjection

name	mult	type	description
ikSecond	0..1	Boolean	Indicates whether initial symmetrical short-circuit current and power have been calculated according to IEC (Ik"). Used only if short circuit calculations are done according to superposition method.
maxInitialSymShCCurrent	1..1	CurrentFlow	Maximum initial symmetrical short-circuit currents (Ik" max) in A (Ik" = Sk"/(SQRT(3) Un)). Used for short circuit data exchange according to IEC 60909.
maxR0ToX0Ratio	1..1	Float	Maximum ratio of zero sequence resistance of Network Feeder to its zero sequence reactance (R(0)/X(0) max). Used for short circuit data exchange according to IEC 60909.
maxR1ToX1Ratio	1..1	Float	Maximum ratio of positive sequence resistance of Network Feeder to its positive sequence reactance (R(1)/X(1) max). Used for short circuit data exchange according to IEC 60909.
maxZ0ToZ1Ratio	1..1	Float	Maximum ratio of zero sequence impedance to its positive sequence impedance (Z(0)/Z(1) max). Used for short circuit data exchange according to IEC 60909.
minInitialSymShCCurrent	1..1	CurrentFlow	Minimum initial symmetrical short-circuit currents (Ik" min) in A (Ik" = Sk"/(SQRT(3) Un)). Used for short circuit data exchange according to IEC 60909.
minR0ToX0Ratio	1..1	Float	Indicates whether initial symmetrical short-circuit current and power have been calculated according to IEC (Ik"). Used for short circuit data exchange according to IEC 6090.
minR1ToX1Ratio	1..1	Float	Minimum ratio of positive sequence resistance of Network Feeder to its positive sequence reactance (R(1)/X(1) min). Used for short circuit data exchange according to IEC 60909.
minZ0ToZ1Ratio	1..1	Float	Minimum ratio of zero sequence impedance to its positive sequence impedance (Z(0)/Z(1) min). Used for short circuit data exchange according to IEC 60909.
voltageFactor	0..1	PU	Voltage factor in pu, which was used to calculate short-circuit current Ik" and power Sk". Used only if short circuit calculations are done according to superposition method.
mRID	1..1	String	inherited from: IdentifiedObject

8.17 (Description) GroundingImpedance

Inheritance path = EarthFaultCompensator: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A fixed impedance device used for grounding.

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Table 408 shows all attributes of GroundingImpedance.

Table 408 – Attributes of ShortCircuitProfile::GroundingImpedance

name	mult	type	description
x	1..1	Reactance	Reactance of device.
r	0..1	Resistance	inherited from: EarthFaultCompensator
mRID	1..1	String	inherited from: IdentifiedObject

8.18 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 409 shows all attributes of IdentifiedObject.

Table 409 – Attributes of ShortCircuitProfile::IdentifiedObject

name	mult	type	description
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

8.19 (Description) LinearShuntCompensator

Inheritance path = ShuntCompensator: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A linear shunt compensator has banks or sections with equal admittance values.

Table 410 shows all attributes of LinearShuntCompensator.

Table 410 – Attributes of ShortCircuitProfile::LinearShuntCompensator

name	mult	type	description
b0PerSection	1..1	Susceptance	Zero sequence shunt (charging) susceptance per section.
g0PerSection	1..1	Conductance	Zero sequence shunt (charging) conductance per section.
mRID	1..1	String	inherited from: IdentifiedObject

8.20 MutualCoupling

Inheritance path = IdentifiedObject

This class represents the zero sequence line mutual coupling.

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Table 411 shows all attributes of MutualCoupling.

Table 411 – Attributes of ShortCircuitProfile::MutualCoupling

name	mult	type	description
b0ch	1..1	Susceptance	Zero sequence mutual coupling shunt (charging) susceptance, uniformly distributed, of the entire line section.
distance11	1..1	Length	Distance to the start of the coupled region from the first line's terminal having sequence number equal to 1.
distance12	1..1	Length	Distance to the end of the coupled region from the first line's terminal with sequence number equal to 1.
distance21	1..1	Length	Distance to the start of coupled region from the second line's terminal with sequence number equal to 1.
distance22	1..1	Length	Distance to the end of coupled region from the second line's terminal with sequence number equal to 1.
g0ch	1..1	Conductance	Zero sequence mutual coupling shunt (charging) conductance, uniformly distributed, of the entire line section.
r0	1..1	Resistance	Zero sequence branch-to-branch mutual impedance coupling, resistance.
x0	1..1	Reactance	Zero sequence branch-to-branch mutual impedance coupling, reactance.
mRID	1..1	String	inherited from: IdentifiedObject

Table 412 shows all association ends of MutualCoupling with other classes.

Table 412 – Association ends of ShortCircuitProfile::MutualCoupling with other classes

mult from	name	mult to	type	description
0..*	Second_Terminal	1..1	Terminal	The starting terminal for the calculation of distances along the second branch of the mutual coupling.
0..*	First_Terminal	1..1	Terminal	The starting terminal for the calculation of distances along the first branch of the mutual coupling. Normally MutualCoupling would only be used for terminals of AC line segments. The first and second terminals of a mutual coupling should point to different AC line segments.

8.21 (Description) NonlinearShuntCompensatorPoint root class

A non linear shunt compensator bank or section admittance value. The number of NonlinearShuntCompenstorPoint instances associated with a NonlinearShuntCompensator shall be equal to ShuntCompensator.maximumSections. ShuntCompensator.sections shall only be set to one of the NonlinearShuntCompenstorPoint.sectionNumber. There is no interpolation between NonlinearShuntCompenstorPoint-s.

Table 413 shows all attributes of NonlinearShuntCompensatorPoint.

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Table 413 – Attributes of ShortCircuitProfile::NonlinearShuntCompensatorPoint

name	mult	type	description
b0	1..1	Susceptance	Zero sequence shunt (charging) susceptance per section.
g0	1..1	Conductance	Zero sequence shunt (charging) conductance per section.

8.22 (Description) PetersenCoil

Inheritance path = EarthFaultCompensator: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A variable impedance device normally used to offset line charging during single line faults in an ungrounded section of network.

Table 414 shows all attributes of PetersenCoil.

Table 414 – Attributes of ShortCircuitProfile::PetersenCoil

name	mult	type	description
mode	1..1	PetersenCoilModeKind	The mode of operation of the Petersen coil.
nominalU	1..1	Voltage	The nominal voltage for which the coil is designed.
offsetCurrent	0..1	CurrentFlow	The offset current that the Petersen coil controller is operating from the resonant point. This is normally a fixed amount for which the controller is configured and could be positive or negative. Typically 0 to 60 A depending on voltage and resonance conditions.
positionCurrent	0..1	CurrentFlow	The control current used to control the Petersen coil also known as the position current. Typically in the range of 20 mA to 200 mA.
xGroundMax	1..1	Reactance	The maximum reactance.
xGroundMin	1..1	Reactance	The minimum reactance.
xGroundNominal	1..1	Reactance	The nominal reactance. This is the operating point (normally over compensation) that is defined based on the resonance point in the healthy network condition. The impedance is calculated based on nominal voltage divided by position current.
r	0..1	Resistance	inherited from: EarthFaultCompensator
mRID	1..1	String	inherited from: IdentifiedObject

8.23 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 415 shows all attributes of PowerSystemResource.

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Table 415 – Attributes of ShortCircuitProfile::PowerSystemResource

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.24 (Description) PowerTransformer

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An electrical device consisting of two or more coupled windings, with or without a magnetic core, for introducing mutual coupling between electric circuits. Transformers can be used to control voltage and phase shift (active power flow).

A power transformer may be composed of separate transformer tanks that need not be identical.

A power transformer can be modelled with or without tanks and is intended for use in both balanced and unbalanced representations. A power transformer typically has two terminals, but may have one (grounding), three or more terminals.

The inherited association ConductingEquipment.BaseVoltage should not be used. The association from TransformerEnd to BaseVoltage should be used instead.

Table 416 shows all attributes of PowerTransformer.

Table 416 – Attributes of ShortCircuitProfile::PowerTransformer

name	mult	type	description
beforeShCircuitHighestOperatingCurrent	0..1	CurrentFlow	The highest operating current (Ib in IEC 60909-0) before short circuit (depends on network configuration and relevant reliability philosophy). It is used for calculation of the impedance correction factor KT defined in IEC 60909-0.
beforeShCircuitHighestOperatingVoltage	0..1	Voltage	The highest operating voltage (Ub in IEC 60909-0) before short circuit. It is used for calculation of the impedance correction factor KT defined in IEC 60909-0. This is worst case voltage on the low side winding (3.7.1 of IEC 60909:2001). Used to define operating conditions.
beforeShortCircuitAnglePf	0..1	AngleDegrees	The angle of power factor before short circuit (phib in IEC 60909-0). It is used for calculation of the impedance correction factor KT defined in IEC 60909-0. This is the worst case power factor. Used to define operating conditions.
highSideMinOperatingU	0..1	Voltage	The minimum operating voltage (uQmin in IEC 60909-0) at the high voltage side (Q side) of the unit transformer of the power station unit. A value well established from long-term operating experience of the system. It is used for calculation of the impedance correction factor KG defined in IEC 60909-0.
isPartOfGeneratorUnit	1..1	Boolean	Indicates whether the machine is part of a power station unit. Used for short circuit data exchange according to IEC 60909. It has an impact on how the correction factors are calculated for transformers, since the transformer is not necessarily part of a synchronous machine and generating unit. It is not always possible to derive this information from the model. This is why the attribute is necessary.

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name	mult	type	description
operationalValuesConsidered	0..1	Boolean	It is used to define if the data (other attributes related to short circuit data exchange) defines long term operational conditions or not. Used for short circuit data exchange according to IEC 60909.
mRID	1..1	String	inherited from: IdentifiedObject

8.25 (Description) PowerTransformerEnd

Inheritance path = TransformerEnd: IdentifiedObject

A PowerTransformerEnd is associated with each Terminal of a PowerTransformer.

The impedance values r, r0, x, and x0 of a PowerTransformerEnd represents a star equivalent as follows.

1) for a two Terminal PowerTransformer the high voltage (TransformerEnd.endNumber=1) PowerTransformerEnd has non zero values on r, r0, x, and x0 while the low voltage (TransformerEnd.endNumber=2) PowerTransformerEnd has zero values for r, r0, x, and x0. Parameters are always provided, even if the PowerTransformerEnds have the same rated voltage. In this case, the parameters are provided at the PowerTransformerEnd which has TransformerEnd.endNumber equal to 1.
2) for a three Terminal PowerTransformer the three PowerTransformerEnds represent a star equivalent with each leg in the star represented by r, r0, x, and x0 values.
3) For a three Terminal transformer each PowerTransformerEnd shall have g, g0, b and b0 values corresponding to the no load losses distributed on the three PowerTransformerEnds. The total no load loss shunt impedances may also be placed at one of the PowerTransformerEnds, preferably the end numbered 1, having the shunt values on end 1. This is the preferred way.
4) for a PowerTransformer with more than three Terminals the PowerTransformerEnd impedance values cannot be used. Instead use the TransformerMeshImpedance or split the transformer into multiple PowerTransformers.

Each PowerTransformerEnd must be contained by a PowerTransformer. Because a PowerTransformerEnd (or any other object) can not be contained by more than one parent, a PowerTransformerEnd can not have an association to an EquipmentContainer (Substation, VoltageLevel, etc).

Table 417 shows all attributes of PowerTransformerEnd.

Table 417 – Attributes of ShortCircuitProfile::PowerTransformerEnd

name	mult	type	description
b0	1..1	Susceptance	Zero sequence magnetizing branch susceptance.
phaseAngleClock	1..1	Integer	Terminal voltage phase angle displacement where 360 degrees are represented with clock hours. The valid values are 0 to 11. For example, for the secondary side end of a transformer with vector group code of 'Dyn11', specify the connection kind as wye with neutral and specify the phase angle of the clock as 11. The clock value of the transformer end number specified as 1, is assumed to be zero. Note the transformer end number is not assumed to be the same as the terminal sequence number.
g0	1..1	Conductance	Zero sequence magnetizing branch conductance (star-model).

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name	mult	type	description
r0	1..1	Resistance	Zero sequence series resistance (star-model) of the transformer end.
x0	1..1	Reactance	Zero sequence series reactance of the transformer end.
rground	0..1	Resistance	inherited from: TransformerEnd
grounded	1..1	Boolean	inherited from: TransformerEnd
xground	0..1	Reactance	inherited from: TransformerEnd
mRID	1..1	String	inherited from: IdentifiedObject

8.26 (abstract) RegulatingCondEq

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Table 418 shows all attributes of RegulatingCondEq.

Table 418 – Attributes of ShortCircuitProfile::RegulatingCondEq

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.27 (abstract) RotatingMachine

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine which may be used as a generator or motor.

Table 419 shows all attributes of RotatingMachine.

Table 419 – Attributes of ShortCircuitProfile::RotatingMachine

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.28 (Description) SeriesCompensator

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A Series Compensator is a series capacitor or reactor or an AC transmission line without charging susceptance. It is a two terminal device.

Table 420 shows all attributes of SeriesCompensator.

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Table 420 – Attributes of ShortCircuitProfile::SeriesCompensator

name	mult	type	description
r0	1..1	Resistance	Zero sequence resistance.
x0	1..1	Reactance	Zero sequence reactance.
varistorPresent	1..1	Boolean	Describe if a metal oxide varistor (mov) for over voltage protection is configured in parallel with the series compensator. It is used for short circuit calculations.
varistorRatedCurrent	0..1	CurrentFlow	The maximum current the varistor is designed to handle at specified duration. It is used for short circuit calculations and exchanged only if SeriesCompensator.varistorPresent is true. The attribute shall be a positive value.
varistorVoltageThreshold	0..1	Voltage	The dc voltage at which the varistor starts conducting. It is used for short circuit calculations and exchanged only if SeriesCompensator.varistorPresent is true.
mRID	1..1	String	inherited from: IdentifiedObject

8.29 (abstract) ShuntCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for bPerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Table 421 shows all attributes of ShuntCompensator.

Table 421 – Attributes of ShortCircuitProfile::ShuntCompensator

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.30 (Description) SynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Table 422 shows all attributes of SynchronousMachine.

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Table 422 – Attributes of ShortCircuitProfile::SynchronousMachine

name	mult	type	description
earthing	1..1	Boolean	Indicates whether or not the generator is earthed. Used for short circuit data exchange according to IEC 60909.
earthingStarPointR	0..1	Resistance	Generator star point earthing resistance (Re). Used for short circuit data exchange according to IEC 60909.
earthingStarPointX	0..1	Reactance	Generator star point earthing reactance (Xe). Used for short circuit data exchange according to IEC 60909.
ikk	0..1	CurrentFlow	Steady-state short-circuit current (in A for the profile) of generator with compound excitation during 3-phase short circuit. – Ikk=0: Generator with no compound excitation. – Ikk<>0: Generator with compound excitation. Ikk is used to calculate the minimum steady-state short-circuit current for generators with compound excitation. (4.6.1.2 in IEC 60909-0:2001). Used only for single fed short circuit on a generator. (4.3.4.2. in IEC 60909-0:2001).
mu	0..1	Float	Factor to calculate the breaking current (Section 4.5.2.1 in IEC 60909-0:2001). Used only for single fed short circuit on a generator (4.3.4.2. in IEC 60909-0:2001).
x0	1..1	Reactance	Zero sequence reactance of the synchronous machine.
r0	1..1	Resistance	Zero sequence resistance of the synchronous machine.
x2	1..1	Reactance	Negative sequence reactance.
r2	1..1	Resistance	Negative sequence resistance.
r	1..1	Resistance	Equivalent resistance (RG) of generator. RG is considered for the calculation of all currents, except for the calculation of the peak current ip. Used for short circuit data exchange according to IEC 60909.
satDirectSubtransX	1..1	PU	Direct-axis subtransient reactance saturated, also known as Xd"sat.
satDirectSyncX	0..1	PU	Direct-axes saturated synchronous reactance (xdsat); reciprocal of short-circuit ration. Used for short circuit data exchange, only for single fed short circuit on a generator. (4.3.4.2. in IEC 60909-0:2001).
satDirectTransX	0..1	PU	Saturated Direct-axis transient reactance. The attribute is primarily used for short circuit calculations according to ANSI.
shortCircuitRotorType	0..1	ShortCircuitRotorKind	Type of rotor, used by short circuit applications, only for single fed short circuit according to IEC 60909.
voltageRegulationRange	0..1	PerCent	Range of generator voltage regulation (PG in IEC 60909-0) used for calculation of the impedance correction factor KG defined in IEC 60909-0. This attribute is used to describe the operating voltage of the generating unit.
mRID	1..1	String	inherited from: IdentifiedObject

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8.31 (abstract) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 423 shows all attributes of Terminal.

Table 423 – Attributes of ShortCircuitProfile::Terminal

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

8.32 (abstract) TransformerEnd

Inheritance path = IdentifiedObject

A conducting connection point of a power transformer. It corresponds to a physical transformer winding terminal. In earlier CIM versions, the TransformerWinding class served a similar purpose, but this class is more flexible because it associates to terminal but is not a specialization of ConductingEquipment.

Table 424 shows all attributes of TransformerEnd.

Table 424 – Attributes of ShortCircuitProfile::TransformerEnd

name	mult	type	description
rground	0..1	Resistance	(for Yn and Zn connections) Resistance part of neutral impedance where 'grounded' is true.
grounded	1..1	Boolean	(for Yn and Zn connections) True if the neutral is solidly grounded.
xground	0..1	Reactance	(for Yn and Zn connections) Reactive part of neutral impedance where 'grounded' is true.
mRID	1..1	String	inherited from: IdentifiedObject

8.33 PetersenCoilModeKind enumeration

The mode of operation for a Petersen coil.

Table 425 shows all literals of PetersenCoilModeKind.

Table 425 – Literals of ShortCircuitProfile::PetersenCoilModeKind

literal	value	description
fixed		Fixed position.
manual		Manual positioning.
automaticPositioning		Automatic positioning.

8.34 ShortCircuitRotorKind enumeration

Type of rotor, used by short circuit applications.

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Table 426 shows all literals of ShortCircuitRotorKind.

Table 426 – Literals of ShortCircuitProfile::ShortCircuitRotorKind

literal	value	description
salientPole1		Salient pole 1 in IEC 60909.
salientPole2		Salient pole 2 in IEC 60909.
turboSeries1		Turbo Series 1 in IEC 60909.
turboSeries2		Turbo series 2 in IEC 60909.

8.35 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 427 shows all literals of UnitMultiplier.

Table 427 – Literals of ShortCircuitProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.

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literal	value	description
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

8.36 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

Table 428 shows all literals of UnitSymbol.

Table 428 – Literals of ShortCircuitProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.

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literal	value	description
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.
m2	41	Area in square metres (m^2.
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.

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literal	value	description
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg·m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.

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literal	value	description
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µg/g’.
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.

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literal	value	description
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86 400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1 000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133,3 Pa).

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literal	value	description
M	217	Length, nautical miles (1 M = 1 852 m).
kn	219	Speed, knots (1 kn = 1 852/3 600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

8.37 ActivePower datatype

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Table 429 shows all attributes of ActivePower.

Table 429 – Attributes of ShortCircuitProfile::ActivePower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=W)

8.38 AngleDegrees datatype

Measurement of angle in degrees.

Table 430 shows all attributes of AngleDegrees.

Table 430 – Attributes of ShortCircuitProfile::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

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8.39 Conductance datatype

Factor by which voltage must be multiplied to give corresponding power lost from a circuit. Real part of admittance.

Table 431 shows all attributes of Conductance.

Table 431 – Attributes of ShortCircuitProfile::Conductance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=S)
multiplier	0..1	UnitMultiplier	(const=none)

8.40 CurrentFlow datatype

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Table 432 shows all attributes of CurrentFlow.

Table 432 – Attributes of ShortCircuitProfile::CurrentFlow

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=A)

8.41 Length datatype

Unit of length. It shall be a positive value or zero.

Table 433 shows all attributes of Length.

Table 433 – Attributes of ShortCircuitProfile::Length

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=m)
multiplier	0..1	UnitMultiplier	(const=k)

8.42 PerCent datatype

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Table 434 shows all attributes of PerCent.

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Table 434 – Attributes of ShortCircuitProfile::PerCent

name	mult	type	description
value	0..1	Float	Normally 0 to 100 on a defined base.
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

8.43 PU datatype

Per Unit – a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Table 435 shows all attributes of PU.

Table 435 – Attributes of ShortCircuitProfile::PU

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

8.44 Reactance datatype

Reactance (imaginary part of impedance), at rated frequency.

Table 436 shows all attributes of Reactance.

Table 436 – Attributes of ShortCircuitProfile::Reactance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=ohm)
multiplier	0..1	UnitMultiplier	(const=none)

8.45 Resistance datatype

Resistance (real part of impedance).

Table 437 shows all attributes of Resistance.

Table 437 – Attributes of ShortCircuitProfile::Resistance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=ohm)
multiplier	0..1	UnitMultiplier	(const=none)

8.46 Susceptance datatype

Imaginary part of admittance.

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Table 438 shows all attributes of Susceptance.

Table 438 – Attributes of ShortCircuitProfile::Susceptance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=S)
multiplier	0..1	UnitMultiplier	(const=none)

8.47 Temperature datatype

Value of temperature in degrees Celsius.

Table 439 shows all attributes of Temperature.

Table 439 – Attributes of ShortCircuitProfile::Temperature

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=degC)
value	0..1	Float	

8.48 Voltage datatype

Electrical voltage, can be both AC and DC.

Table 440 shows all attributes of Voltage.

Table 440 – Attributes of ShortCircuitProfile::Voltage

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=k)
unit	0..1	UnitSymbol	(const=V)

8.49 Boolean primitive

A type with the value space "true" and "false".

8.50 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

8.51 Float primitive

A floating point number. The range is unspecified and not limited.

8.52 Integer primitive

An integer number. The range is unspecified and not limited.

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8.53 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

9 Package SteadyStateHypothesisProfile

9.1 General

A steady state hypothesis dataset which conforms to the IEC 61970-456 steady state hypothesis profile contains all objects required to exchange input parameters to be able to perform load flow simulations. Due to the nature of the SSH profile, all objects in a Steady State Hypothesis instance file should have persistent mRIDs.

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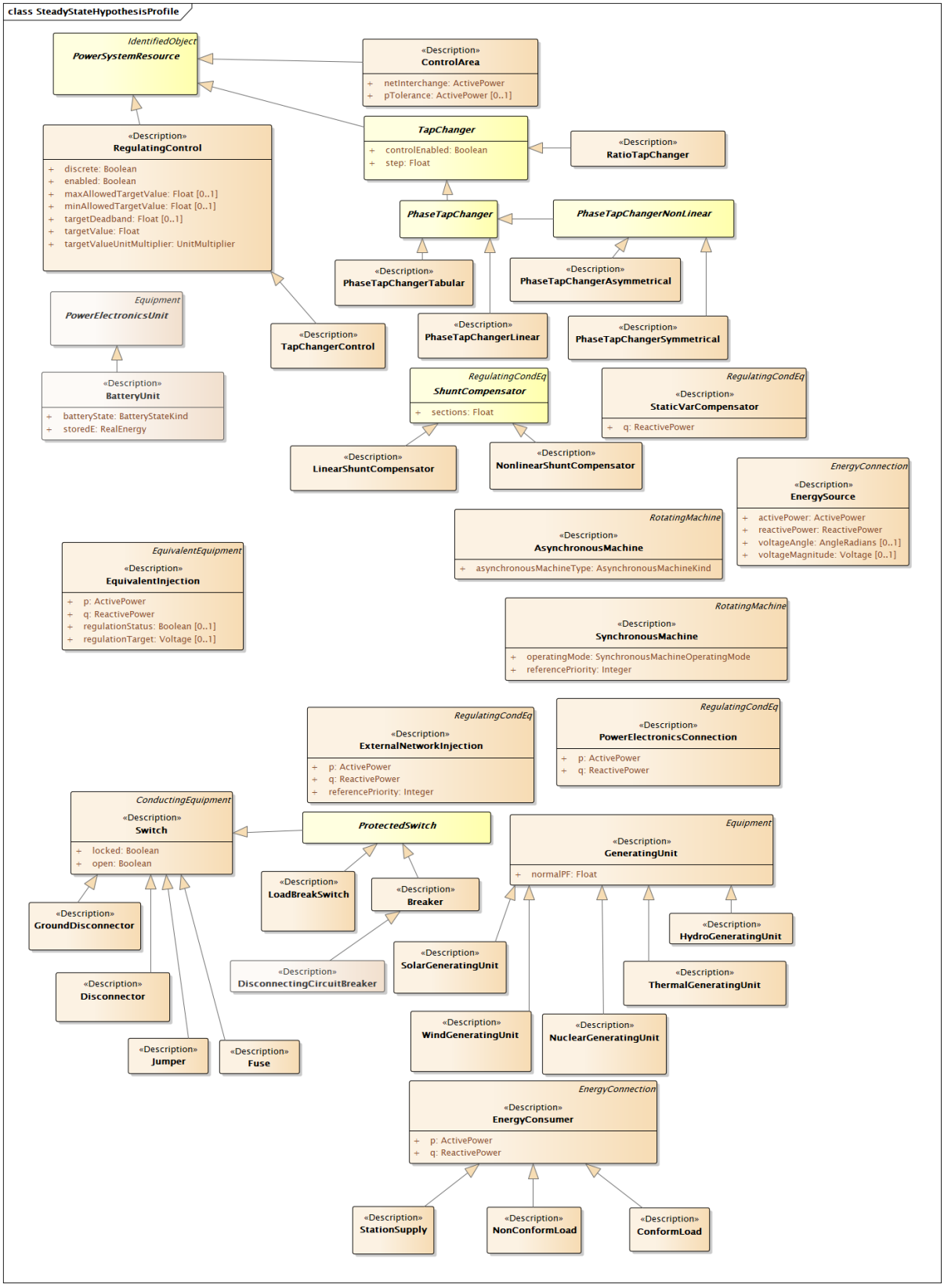

Figure 33 – Class diagram SteadyStateHypothesisProfile::SteadyStateHypothesisProfile

Figure 33: The diagram illustrates most of the classes in steady state hypothesis profile.

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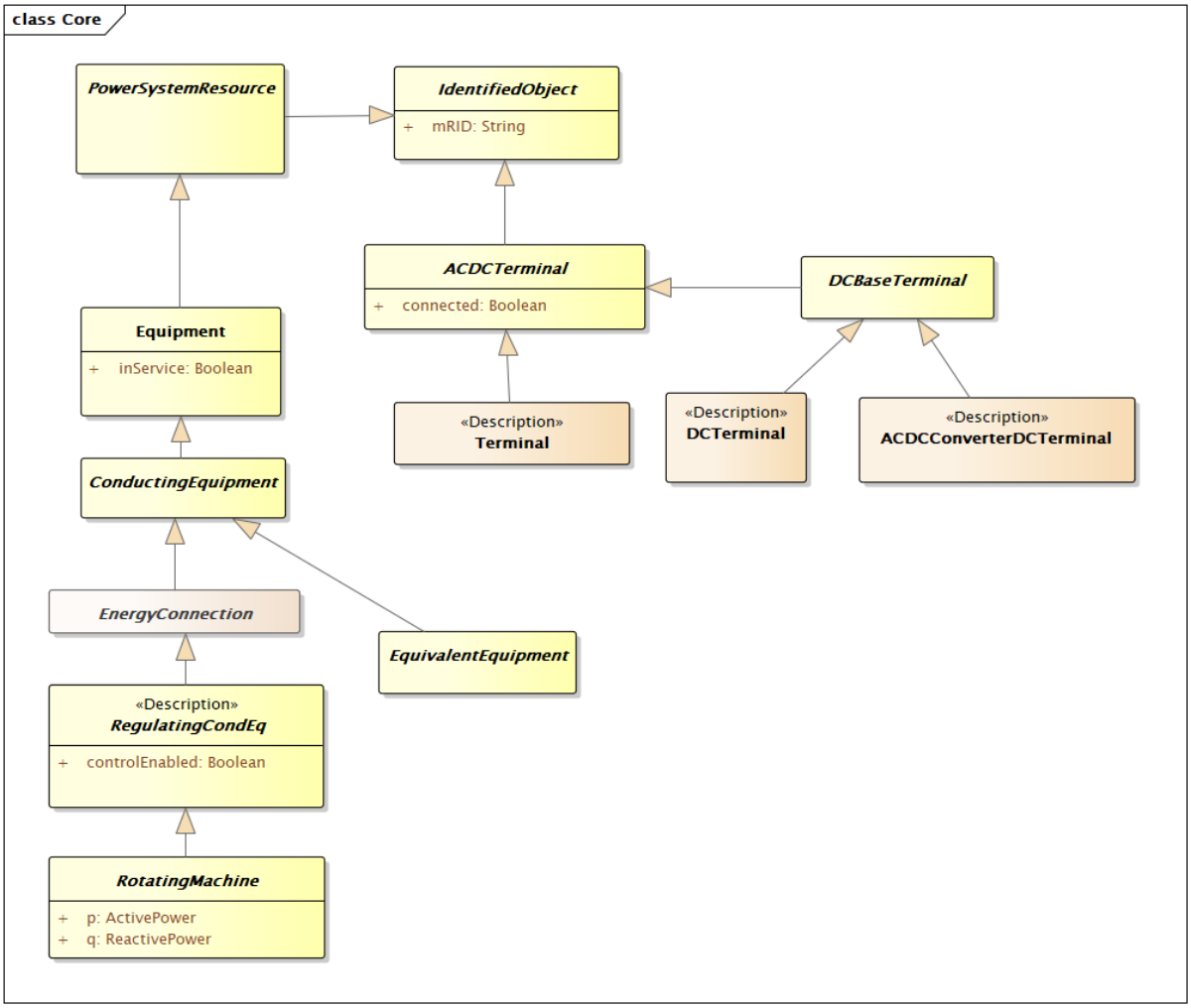

Figure 34 – Class diagram SteadyStateHypothesisProfile::Core

Figure 34: The diagram shows main classes from Core package in the canonical CIM used in this profile.

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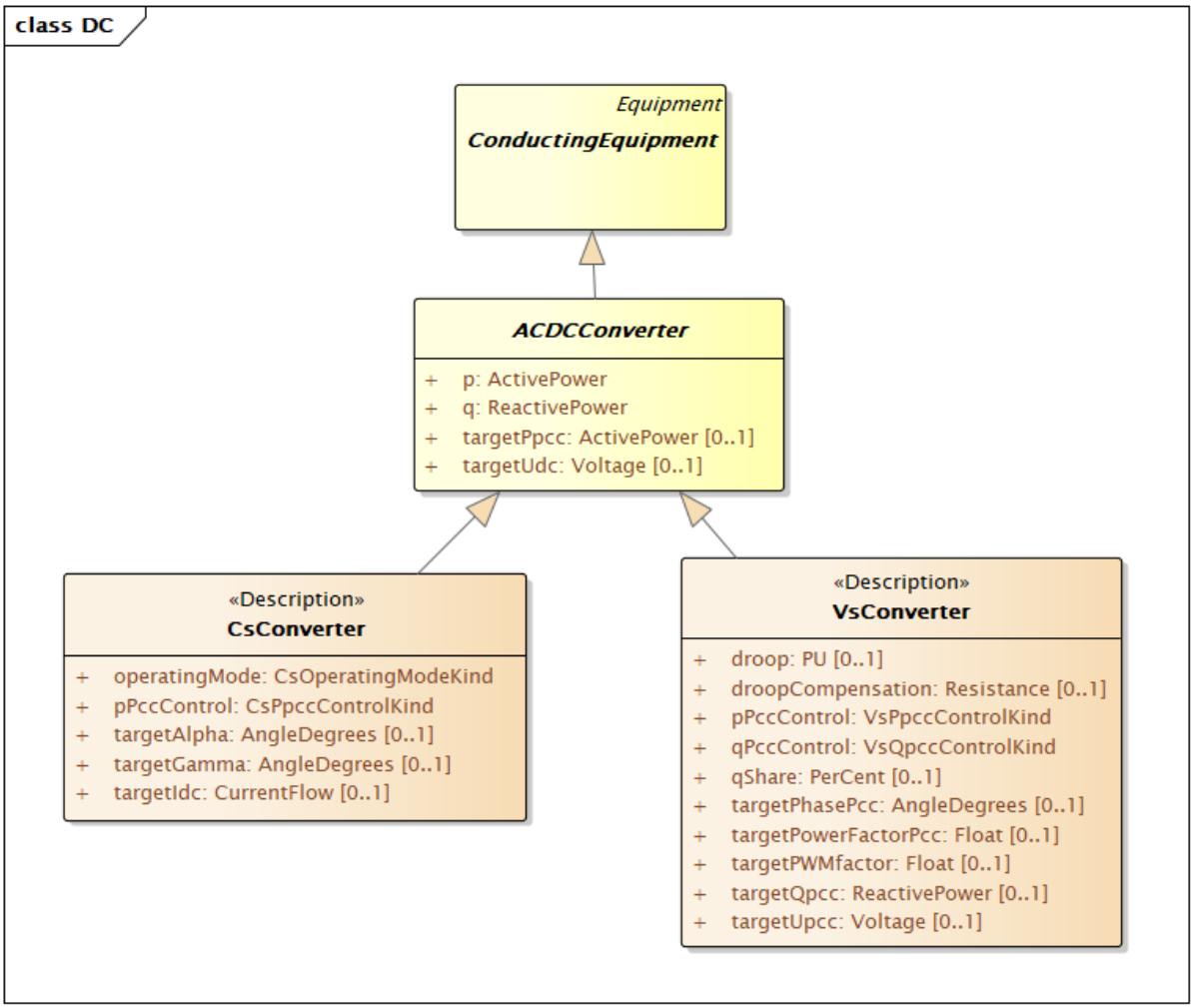

Figure 35 – Class diagram SteadyStateHypothesisProfile::DC

Figure 35: The diagram shows DC related classes included in the profile.

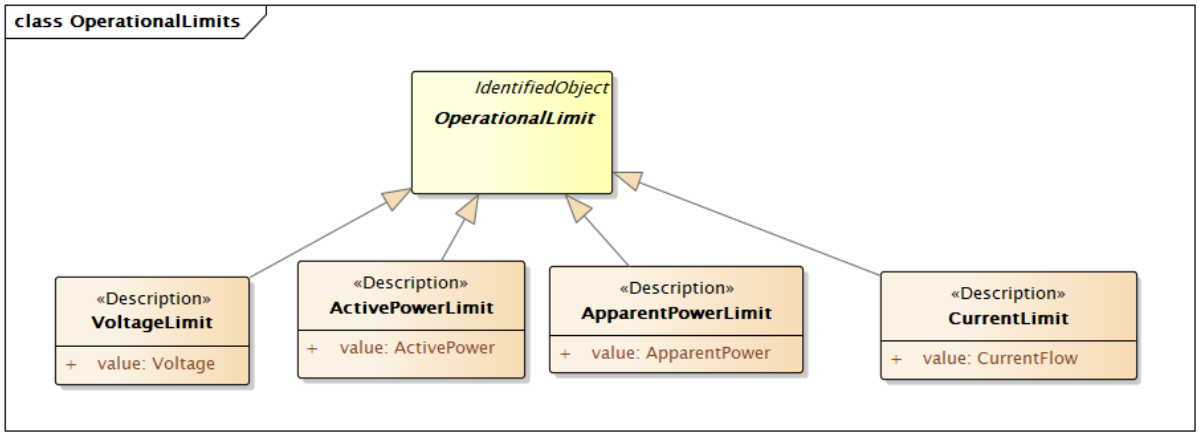

Figure 36 – Class diagram SteadyStateHypothesisProfile::OperationalLimits

Figure 36: This diagram shows operational limits as they tie back into the core model.

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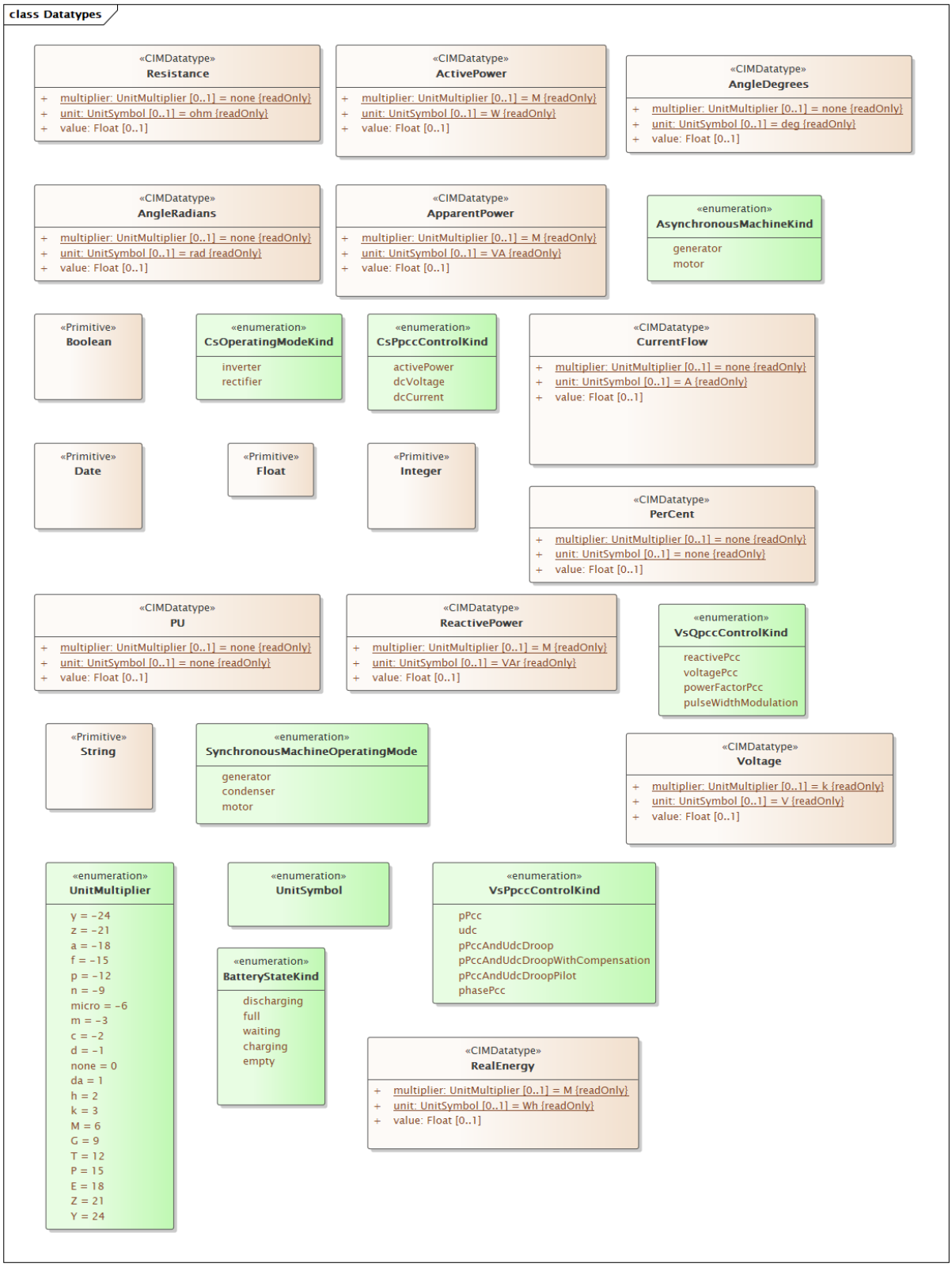

Figure 37 – Class diagram SteadyStateHypothesisProfile::Datatypes

Figure 37: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.

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- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

9.2 (abstract) ACDCConverter

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Table 441 shows all attributes of ACDCConverter.

Table 441 – Attributes of SteadyStateHypothesisProfile::ACDCConverter

name	mult	type	description
p	1..1	ActivePower	Active power at the point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.
q	1..1	ReactivePower	Reactive power at the point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution in the case a simplified power flow model is used.
targetPpcc	0..1	ActivePower	Real power injection target in AC grid, at point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node.
targetUdc	0..1	Voltage	Target value for DC voltage magnitude. The attribute shall be a positive value.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.3 (Description) ACDCConverterDCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Table 442 shows all attributes of ACDCConverterDCTerminal.

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Table 442 – Attributes of SteadyStateHypothesisProfile::ACDCConverterDCTerminal

name	mult	type	description
connected	1..1	Boolean	inherited from: ACDCTerminal
mRID	1..1	String	inherited from: IdentifiedObject

9.4 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 443 shows all attributes of ACDCTerminal.

Table 443 – Attributes of SteadyStateHypothesisProfile::ACDCTerminal

name	mult	type	description
connected	1..1	Boolean	The connected status is related to a bus-branch model and the topological node to terminal relation. True implies the terminal is connected to the related topological node and false implies it is not. In a bus-branch model, the connected status is used to tell if equipment is disconnected without having to change the connectivity described by the topological node to terminal relation. A valid case is that conducting equipment can be connected in one end and open in the other. In particular for an AC line segment, where the reactive line charging can be significant, this is a relevant case.
mRID	1..1	String	inherited from: IdentifiedObject

9.5 (Description) ActivePowerLimit

Inheritance path = OperationalLimit: IdentifiedObject

Limit on active power flow.

Table 444 shows all attributes of ActivePowerLimit.

Table 444 – Attributes of SteadyStateHypothesisProfile::ActivePowerLimit

name	mult	type	description
value	1..1	ActivePower	Value of active power limit. The attribute shall be a positive value or zero.
mRID	1..1	String	inherited from: IdentifiedObject

9.6 (Description) ApparentPowerLimit

Inheritance path = OperationalLimit: IdentifiedObject

Apparent power limit.

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Table 445 shows all attributes of ApparentPowerLimit.

Table 445 – Attributes of SteadyStateHypothesisProfile::ApparentPowerLimit

name	mult	type	description
value	1..1	ApparentPower	The apparent power limit. The attribute shall be a positive value or zero.
mRID	1..1	String	inherited from: IdentifiedObject

9.7 (Description) AsynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g. squirrel-cage induction machine.

Table 446 shows all attributes of AsynchronousMachine.

Table 446 – Attributes of SteadyStateHypothesisProfile::AsynchronousMachine

name	mult	type	description
asynchronousMachineType	1..1	AsynchronousMachineKind	Indicates the type of Asynchronous Machine (motor or generator).
p	1..1	ActivePower	inherited from: RotatingMachine
q	1..1	ReactivePower	inherited from: RotatingMachine
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.8 (Description) BatteryUnit

Inheritance path = PowerElectronicsUnit: Equipment: PowerSystemResource: IdentifiedObject

An electrochemical energy storage device.

Table 447 shows all attributes of BatteryUnit.

Table 447 – Attributes of SteadyStateHypothesisProfile::BatteryUnit

name	mult	type	description
batteryState	1..1	BatteryStateKind	The current state of the battery (charging, full, etc.).
storedE	1..1	RealEnergy	Amount of energy currently stored. The attribute shall be a positive value or zero and lower than BatteryUnit.ratedE.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

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9.9 (Description) Breaker

Inheritance path = ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A mechanical switching device capable of making, carrying, and breaking currents under normal circuit conditions and also making, carrying for a specified time, and breaking currents under specified abnormal circuit conditions e.g. those of short circuit.

Table 448 shows all attributes of Breaker.

Table 448 – Attributes of SteadyStateHypothesisProfile::Breaker

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.10 (abstract) ConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Table 449 shows all attributes of ConductingEquipment.

Table 449 – Attributes of SteadyStateHypothesisProfile::ConductingEquipment

name	mult	type	description
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.11 (Description) ConformLoad

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

ConformLoad represent loads that follow a daily load change pattern where the pattern can be used to scale the load with a system load.

Table 450 shows all attributes of ConformLoad.

Table 450 – Attributes of SteadyStateHypothesisProfile::ConformLoad

name	mult	type	description
p	1..1	ActivePower	inherited from: EnergyConsumer
q	1..1	ReactivePower	inherited from: EnergyConsumer
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

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9.12 (Description) ControlArea

Inheritance path = PowerSystemResource: IdentifiedObject

A control area is a grouping of generating units and/or loads and a cutset of tie lines (as terminals) which may be used for a variety of purposes including automatic generation control, power flow solution area interchange control specification, and input to load forecasting. All generation and load within the area defined by the terminals on the border are considered in the area interchange control. Note that any number of overlapping control area specifications can be superimposed on the physical model. The following general principles apply to ControlArea:

1) The control area orientation for net interchange is positive for an import, negative for an export.
2) The control area net interchange is determined by summing flows in Terminals. The Terminals are identified by creating a set of TieFlow objects associated with a ControlArea object. Each TieFlow object identifies one Terminal.
3) In a single network model, a tie between two control areas must be modelled in both control area specifications, such that the two representations of the tie flow sum to zero.
4) The normal orientation of Terminal flow is positive for flow into the conducting equipment that owns the Terminal. (i.e. flow from a bus into a device is positive.) However, the orientation of each flow in the control area specification must align with the control area convention, i.e. import is positive. If the orientation of the Terminal flow referenced by a TieFlow is positive into the control area, then this is confirmed by setting TieFlow.positiveFlowIn flag TRUE. If not, the orientation must be reversed by setting the TieFlow.positiveFlowIn flag FALSE.

Table 451 shows all attributes of ControlArea.

Table 451 – Attributes of SteadyStateHypothesisProfile::ControlArea

name	mult	type	description
netInterchange	1..1	ActivePower	The specified positive net interchange into the control area, i.e. positive sign means flow into the area.
pTolerance	0..1	ActivePower	Active power net interchange tolerance. The attribute shall be a positive value or zero.
mRID	1..1	String	inherited from: IdentifiedObject

9.13 (Description) CsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

DC side of the current source converter (CSC).

The firing angle controls the dc voltage at the converter, both for rectifier and inverter. The difference between the dc voltages of the rectifier and inverter determines the dc current. The extinction angle is used to limit the dc voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source dc line. Higher level controls are built on top, e.g. dc voltage, dc current and active power. From a steady state perspective it is sufficient to specify the wanted active power transfer (ACDCConverter.targetPpcc) and the control functions will set the dc voltage, dc current, firing angle, transformer tap position and number of connected filters to meet this. Therefore attributes targetAlpha and targetGamma are not applicable in this case.

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The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses is a function of the dc voltage and dc current.

The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10-18 degrees.

The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17-20 degrees.

Table 452 shows all attributes of CsConverter.

Table 452 – Attributes of SteadyStateHypothesisProfile::CsConverter

name	mult	type	description
operatingMode	1..1	CsOperatingModeKind	Indicates whether the DC pole is operating as an inverter or as a rectifier. It is converter's control variable used in power flow.
pPccControl	1..1	CsPpccControlKind	Kind of active power control.
targetAlpha	0..1	AngleDegrees	Target firing angle. It is converter's control variable used in power flow. It is only applicable for rectifier if continuous tap changer control is used. Allowed values are within the range minAlpha<=targetAlpha<=maxAlpha. The attribute shall be a positive value.
targetGamma	0..1	AngleDegrees	Target extinction angle. It is converter's control variable used in power flow. It is only applicable for inverter if continuous tap changer control is used. Allowed values are within the range minGamma<=targetGamma<=maxGamma. The attribute shall be a positive value.
targetIdc	0..1	CurrentFlow	DC current target value. It is converter's control variable used in power flow. The attribute shall be a positive value.
p	1..1	ActivePower	inherited from: ACDCConverter
q	1..1	ReactivePower	inherited from: ACDCConverter
targetPpcc	0..1	ActivePower	inherited from: ACDCConverter
targetUdc	0..1	Voltage	inherited from: ACDCConverter
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.14 (Description) CurrentLimit

Inheritance path = OperationalLimit: IdentifiedObject

Operational limit on current.

Table 453 shows all attributes of CurrentLimit.

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Table 453 – Attributes of SteadyStateHypothesisProfile::CurrentLimit

name	mult	type	description
value	1..1	CurrentFlow	Limit on current flow. The attribute shall be a positive value or zero.
mRID	1..1	String	inherited from: IdentifiedObject

9.15 (abstract) DCBaseTerminal

Inheritance path = ACDCTerminal: IdentifiedObject

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model requires that DC connections are distinct from AC connections.

Table 454 shows all attributes of DCBaseTerminal.

Table 454 – Attributes of SteadyStateHypothesisProfile::DCBaseTerminal

name	mult	type	description
connected	1..1	Boolean	inherited from: ACDCTerminal
mRID	1..1	String	inherited from: IdentifiedObject

9.16 (Description) DCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

An electrical connection point to generic DC conducting equipment.

Table 455 shows all attributes of DCTerminal.

Table 455 – Attributes of SteadyStateHypothesisProfile::DCTerminal

name	mult	type	description
connected	1..1	Boolean	inherited from: ACDCTerminal
mRID	1..1	String	inherited from: IdentifiedObject

9.17 (Description) Disconnector

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A manually operated or motor operated mechanical switching device used for changing the connections in a circuit, or for isolating a circuit or equipment from a source of power. It is required to open or close circuits when negligible current is broken or made.

Table 456 shows all attributes of Disconnector.

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Table 456 – Attributes of SteadyStateHypothesisProfile::Disconnector

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.18 (Description) DisconnectingCircuitBreaker

Inheritance path = Breaker: ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A circuit breaking device including disconnecting function, eliminating the need for separate disconnectors.

Table 457 shows all attributes of DisconnectingCircuitBreaker.

Table 457 – Attributes of SteadyStateHypothesisProfile::DisconnectingCircuitBreaker

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.19 (abstract) EnergyConnection

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection of energy generation or consumption on the power system model.

Table 458 shows all attributes of EnergyConnection.

Table 458 – Attributes of SteadyStateHypothesisProfile::EnergyConnection

name	mult	type	description
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.20 (Description) EnergyConsumer

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Generic user of energy – a point of consumption on the power system model.

EnergyConsumer.pfixed, .qfixed, .pfixedPct and .qfixedPct have meaning only if there is no LoadResponseCharacteristic associated with EnergyConsumer or if LoadResponseCharacteristic.exponentModel is set to False.

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Table 459 shows all attributes of EnergyConsumer.

Table 459 – Attributes of SteadyStateHypothesisProfile::EnergyConsumer

name	mult	type	description
p	1..1	ActivePower	Active power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power of the load. Load sign convention is used, i.e. positive sign means flow out from a node. For voltage dependent loads the value is at rated voltage. Starting value for a steady state solution.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.21 (Description) EnergySource

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A generic equivalent for an energy supplier on a transmission or distribution voltage level.

Table 460 shows all attributes of EnergySource.

Table 460 – Attributes of SteadyStateHypothesisProfile::EnergySource

name	mult	type	description
activePower	1..1	ActivePower	High voltage source active injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
reactivePower	1..1	ReactivePower	High voltage source reactive injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
voltageAngle	0..1	AngleRadians	Phase angle of a-phase open circuit used when voltage characteristics need to be imposed at the node associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network. The attribute shall be a positive value or zero.
voltageMagnitude	0..1	Voltage	Phase-to-phase open circuit voltage magnitude used when voltage characteristics need to be imposed at the node associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network. The attribute shall be a positive value or zero.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

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9.22 Equipment

Inheritance path = PowerSystemResource: IdentifiedObject

The parts of a power system that are physical devices, electronic or mechanical.

Table 461 shows all attributes of Equipment.

Table 461 – Attributes of SteadyStateHypothesisProfile::Equipment

name	mult	type	description
inService	1..1	Boolean	Specifies the availability of the equipment. True means the equipment is available for topology processing, which determines if the equipment is energized or not. False means that the equipment is treated by network applications as if it is not in the model.
mRID	1..1	String	inherited from: IdentifiedObject

9.23 (abstract) EquivalentEquipment

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

The class represents equivalent objects that are the result of a network reduction. The class is the base for equivalent objects of different types.

Table 462 shows all attributes of EquivalentEquipment.

Table 462 – Attributes of SteadyStateHypothesisProfile::EquivalentEquipment

name	mult	type	description
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.24 (Description) EquivalentInjection

Inheritance path = EquivalentEquipment: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents equivalent injections (generation or load). Voltage regulation is allowed only at the point of connection.

Table 463 shows all attributes of EquivalentInjection.

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Table 463 – Attributes of SteadyStateHypothesisProfile::EquivalentInjection

name	mult	type	description
regulationStatus	0..1	Boolean	Specifies the regulation status of the EquivalentInjection. True is regulating. False is not regulating.
regulationTarget	0..1	Voltage	The target voltage for voltage regulation. The attribute shall be a positive value.
p	1..1	ActivePower	Equivalent active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
q	1..1	ReactivePower	Equivalent reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.25 (Description) ExternalNetworkInjection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

This class represents the external network and it is used for IEC 60909 calculations.

Table 464 shows all attributes of ExternalNetworkInjection.

Table 464 – Attributes of SteadyStateHypothesisProfile::ExternalNetworkInjection

name	mult	type	description
referencePriority	1..1	Integer	Priority of unit for use as powerflow voltage phase angle reference bus selection. 0 = don t care (default) 1 = highest priority. 2 is less than 1 and so on.
p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for steady state solutions.
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.26 (Description) Fuse

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An overcurrent protective device with a circuit opening fusible part that is heated and severed by the passage of overcurrent through it. A fuse is considered a switching device because it breaks current.

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Table 465 shows all attributes of Fuse.

Table 465 – Attributes of SteadyStateHypothesisProfile::Fuse

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.27 (Description) GeneratingUnit

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

A single or set of synchronous machines for converting mechanical power into alternating-current power. For example, individual machines within a set may be defined for scheduling purposes while a single control signal is derived for the set. In this case there would be a GeneratingUnit for each member of the set and an additional GeneratingUnit corresponding to the set.

Table 466 shows all attributes of GeneratingUnit.

Table 466 – Attributes of SteadyStateHypothesisProfile::GeneratingUnit

name	mult	type	description
normalPF	1..1	Float	Generating unit economic participation factor. The sum of the participation factors across generating units does not have to sum to one. It is used for representing distributed slack participation factor. The attribute shall be a positive value or zero.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.28 (Description) GroundDisconnector

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A manually operated or motor operated mechanical switching device used for isolating a circuit or equipment from ground.

Table 467 shows all attributes of GroundDisconnector.

Table 467 – Attributes of SteadyStateHypothesisProfile::GroundDisconnector

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

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9.29 (Description) HydroGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A generating unit whose prime mover is a hydraulic turbine (e.g., Francis, Pelton, Kaplan).

Table 468 shows all attributes of HydroGeneratingUnit.

Table 468 – Attributes of SteadyStateHypothesisProfile::HydroGeneratingUnit

name	mult	type	description
normalPF	1..1	Float	inherited from: GeneratingUnit
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.30 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 469 shows all attributes of IdentifiedObject.

Table 469 – Attributes of SteadyStateHypothesisProfile::IdentifiedObject

name	mult	type	description
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

9.31 (Description) Jumper

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A short section of conductor with negligible impedance which can be manually removed and replaced if the circuit is de-energized. Note that zero-impedance branches can potentially be modelled by other equipment types.

Table 470 shows all attributes of Jumper.

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Table 470 – Attributes of SteadyStateHypothesisProfile::Jumper

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.32 (Description) LinearShuntCompensator

Inheritance path = ShuntCompensator: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A linear shunt compensator has banks or sections with equal admittance values.

Table 471 shows all attributes of LinearShuntCompensator.

Table 471 – Attributes of SteadyStateHypothesisProfile::LinearShuntCompensator

name	mult	type	description
sections	1..1	Float	inherited from: ShuntCompensator
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.33 (Description) LoadBreakSwitch

Inheritance path = ProtectedSwitch: Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A mechanical switching device capable of making, carrying, and breaking currents under normal operating conditions.

Table 472 shows all attributes of LoadBreakSwitch.

Table 472 – Attributes of SteadyStateHypothesisProfile::LoadBreakSwitch

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.34 (Description) NonConformLoad

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

NonConformLoad represents loads that do not follow a daily load change pattern and whose changes are not correlated with the daily load change pattern.

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Table 473 shows all attributes of NonConformLoad.

Table 473 – Attributes of SteadyStateHypothesisProfile::NonConformLoad

name	mult	type	description
p	1..1	ActivePower	inherited from: EnergyConsumer
q	1..1	ReactivePower	inherited from: EnergyConsumer
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.35 (Description) NonlinearShuntCompensator

Inheritance path = ShuntCompensator: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A non linear shunt compensator has bank or section admittance values that differ. The attributes g, b, g0 and b0 of the associated NonlinearShuntCompensatorPoint describe the total conductance and admittance of a NonlinearShuntCompensatorPoint at a section number specified by NonlinearShuntCompensatorPoint.sectionNumber.

Table 474 shows all attributes of NonlinearShuntCompensator.

Table 474 – Attributes of SteadyStateHypothesisProfile::NonlinearShuntCompensator

name	mult	type	description
sections	1..1	Float	inherited from: ShuntCompensator
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.36 (Description) NuclearGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A nuclear generating unit.

Table 475 shows all attributes of NuclearGeneratingUnit.

Table 475 – Attributes of SteadyStateHypothesisProfile::NuclearGeneratingUnit

name	mult	type	description
normalPF	1..1	Float	inherited from: GeneratingUnit
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.37 (abstract) OperationalLimit

Inheritance path = IdentifiedObject

A value and normal value associated with a specific kind of limit.

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The sub class value and normalValue attributes vary inversely to the associated OperationalLimitType.acceptableDuration (acceptableDuration for short).

If a particular piece of equipment has multiple operational limits of the same kind (apparent power, current, etc.), the limit with the greatest acceptableDuration shall have the smallest limit value and the limit with the smallest acceptableDuration shall have the largest limit value. Note: A large current can only be allowed to flow through a piece of equipment for a short duration without causing damage, but a lesser current can be allowed to flow for a longer duration.

Table 476 shows all attributes of OperationalLimit.

Table 476 – Attributes of SteadyStateHypothesisProfile::OperationalLimit

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

9.38 (abstract) PhaseTapChanger

Inheritance path = TapChanger: PowerSystemResource: IdentifiedObject

A transformer phase shifting tap model that controls the phase angle difference across the power transformer and potentially the active power flow through the power transformer. This phase tap model may also impact the voltage magnitude.

Table 477 shows all attributes of PhaseTapChanger.

Table 477 – Attributes of SteadyStateHypothesisProfile::PhaseTapChanger

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.39 (Description) PhaseTapChangerAsymmetrical

Inheritance path = PhaseTapChangerNonLinear: PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes the tap model for an asymmetrical phase shifting transformer in which the difference voltage vector adds to the in-phase winding. The out-of-phase winding is the transformer end where the tap changer is located. The angle between the in-phase and out-of-phase windings is named the winding connection angle. The phase shift depends on both the difference voltage magnitude and the winding connection angle.

Table 478 shows all attributes of PhaseTapChangerAsymmetrical.

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Table 478 – Attributes of SteadyStateHypothesisProfile:: PhaseTapChangerAsymmetrical

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.40 (Description) PhaseTapChangerLinear

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes a tap changer with a linear relation between the tap step and the phase angle difference across the transformer. This is a mathematical model that is an approximation of a real phase tap changer.

The phase angle is computed as stepPhaseShiftIncrement times the tap position.

The voltage magnitude of both sides is the same.

Table 479 shows all attributes of PhaseTapChangerLinear.

Table 479 – Attributes of SteadyStateHypothesisProfile::PhaseTapChangerLinear

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.41 (abstract) PhaseTapChangerNonLinear

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

The non-linear phase tap changer describes the non-linear behaviour of a phase tap changer. This is a base class for the symmetrical and asymmetrical phase tap changer models. The details of these models can be found in IEC 61970-301.

Table 480 shows all attributes of PhaseTapChangerNonLinear.

Table 480 – Attributes of SteadyStateHypothesisProfile::PhaseTapChangerNonLinear

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.42 (Description) PhaseTapChangerSymmetrical

Inheritance path = PhaseTapChangerNonLinear: PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

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Describes a symmetrical phase shifting transformer tap model in which the voltage magnitude of both sides is the same. The difference voltage magnitude is the base in an equal-sided triangle where the sides corresponds to the primary and secondary voltages. The phase angle difference corresponds to the top angle and can be expressed as twice the arctangent of half the total difference voltage.

Table 481 shows all attributes of PhaseTapChangerSymmetrical.

Table 481 – Attributes of SteadyStateHypothesisProfile::PhaseTapChangerSymmetrical

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.43 (Description) PhaseTapChangerTabular

Inheritance path = PhaseTapChanger: TapChanger: PowerSystemResource: IdentifiedObject

Describes a tap changer with a table defining the relation between the tap step and the phase angle difference across the transformer.

Table 482 shows all attributes of PhaseTapChangerTabular.

Table 482 – Attributes of SteadyStateHypothesisProfile::PhaseTapChangerTabular

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.44 (Description) PowerElectronicsConnection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection to the AC network for energy production or consumption that uses power electronics rather than rotating machines.

Table 483 shows all attributes of PowerElectronicsConnection.

Table 483 – Attributes of SteadyStateHypothesisProfile::PowerElectronicsConnection

name	mult	type	description
p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq

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name	mult	type	description
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.45 (abstract) PowerElectronicsUnit

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

A generating unit or battery or aggregation that connects to the AC network using power electronics rather than rotating machines.

Table 484 shows all attributes of PowerElectronicsUnit.

Table 484 – Attributes of SteadyStateHypothesisProfile::PowerElectronicsUnit

name	mult	type	description
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.46 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 485 shows all attributes of PowerSystemResource.

Table 485 – Attributes of SteadyStateHypothesisProfile::PowerSystemResource

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

9.47 (abstract) ProtectedSwitch

Inheritance path = Switch: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A ProtectedSwitch is a switching device that can be operated by ProtectionEquipment.

Table 486 shows all attributes of ProtectedSwitch.

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Table 486 – Attributes of SteadyStateHypothesisProfile::ProtectedSwitch

name	mult	type	description
open	1..1	Boolean	inherited from: Switch
locked	1..1	Boolean	inherited from: Switch
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.48 (Description) RatioTapChanger

Inheritance path = TapChanger: PowerSystemResource: IdentifiedObject

A tap changer that changes the voltage ratio impacting the voltage magnitude but not the phase angle across the transformer.

Angle sign convention (general): Positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer).

Table 487 shows all attributes of RatioTapChanger.

Table 487 – Attributes of SteadyStateHypothesisProfile::RatioTapChanger

name	mult	type	description
controlEnabled	1..1	Boolean	inherited from: TapChanger
step	1..1	Float	inherited from: TapChanger
mRID	1..1	String	inherited from: IdentifiedObject

9.49 (abstract,Description) RegulatingCondEq

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Table 488 shows all attributes of RegulatingCondEq.

Table 488 – Attributes of SteadyStateHypothesisProfile::RegulatingCondEq

name	mult	type	description
controlEnabled	1..1	Boolean	Specifies the regulation status of the equipment. True is regulating, false is not regulating.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.50 (Description) RegulatingControl

Inheritance path = PowerSystemResource: IdentifiedObject

Specifies a set of equipment that works together to control a power system quantity such as voltage or flow.

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Remote bus voltage control is possible by specifying the controlled terminal located at some place remote from the controlling equipment.

The specified terminal shall be associated with the connectivity node of the controlled point. The most specific subtype of RegulatingControl shall be used in case such equipment participate in the control, e.g. TapChangerControl for tap changers.

For flow control, load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

The attribute minAllowedTargetValue and maxAllowedTargetValue are required in the following cases:

– For a power generating module operated in power factor control mode to specify maximum and minimum power factor values;
– Whenever it is necessary to have an off center target voltage for the tap changer regulator. For instance, due to long cables to off shore wind farms and the need to have a simpler setup at the off shore transformer platform, the voltage is controlled from the land at the connection point for the off shore wind farm. Since there usually is a voltage rise along the cable, there is typical and overvoltage of up 3-4 kV compared to the on shore station. Thus in normal operation the tap changer on the on shore station is operated with a target set point, which is in the lower parts of the dead band.

The attributes minAllowedTargetValue and maxAllowedTargetValue are not related to the attribute targetDeadband and thus they are not treated as an alternative of the targetDeadband. They are needed due to limitations in the local substation controller. The attribute targetDeadband is used to prevent the power flow from move the tap position in circles (hunting) that is to be used regardless of the attributes minAllowedTargetValue and maxAllowedTargetValue.

Table 489 shows all attributes of RegulatingControl.

Table 489 – Attributes of SteadyStateHypothesisProfile::RegulatingControl

name	mult	type	description
discrete	1..1	Boolean	The regulation is performed in a discrete mode. This applies to equipment with discrete controls, e.g. tap changers and shunt compensators.
enabled	1..1	Boolean	The flag tells if regulation is enabled.
targetDeadband	0..1	Float	This is a deadband used with discrete control to avoid excessive update of controls like tap changers and shunt compensator banks while regulating. The units of those appropriate for the mode. The attribute shall be a positive value or zero. If RegulatingControl.discrete is set to "false", the RegulatingControl.targetDeadband is to be ignored. Note that for instance, if the targetValue is 100 kV and the targetDeadband is 2 kV the range is from 99 to 101 kV.
targetValue	1..1	Float	The target value specified for case input. This value can be used for the target value without the use of schedules. The value has the units appropriate to the mode attribute.
targetValueUnitMultiplier	1..1	UnitMultiplier	Specify the multiplier for used for the targetValue.
maxAllowedTargetValue	0..1	Float	Maximum allowed target value (RegulatingControl.targetValue).
minAllowedTargetValue	0..1	Float	Minimum allowed target value (RegulatingControl.targetValue).

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject

9.51 (abstract) RotatingMachine

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine which may be used as a generator or motor.

Table 490 shows all attributes of RotatingMachine.

Table 490 – Attributes of SteadyStateHypothesisProfile::RotatingMachine

name	mult	type	description
p	1..1	ActivePower	Active power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.52 (abstract) ShuntCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for bPerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

Table 491 shows all attributes of ShuntCompensator.

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Table 491 – Attributes of SteadyStateHypothesisProfile::ShuntCompensator

name	mult	type	description
sections	1..1	Float	Shunt compensator sections in use. Starting value for steady state solution. The attribute shall be a positive value or zero. Non integer values are allowed to support continuous variables. The reasons for continuous value are to support study cases where no discrete shunt compensators has yet been designed, a solutions where a narrow voltage band force the sections to oscillate or accommodate for a continuous solution as input. For LinearShuntConpensator the value shall be between zero and ShuntCompensator.maximumSections. At value zero the shunt compensator conductance and admittance is zero. Linear interpolation of conductance and admittance between the previous and next integer section is applied in case of non-integer values. For NonlinearShuntCompensator-s shall only be set to one of the NonlinearShuntCompenstorPoint.sectionNumber. There is no interpolation between NonlinearShuntCompenstorPoint-s.
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.53 (Description) SolarGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A solar thermal generating unit, connected to the grid by means of a rotating machine. This class does not represent photovoltaic (PV) generation.

Table 492 shows all attributes of SolarGeneratingUnit.

Table 492 – Attributes of SteadyStateHypothesisProfile::SolarGeneratingUnit

name	mult	type	description
normalPF	1..1	Float	inherited from: GeneratingUnit
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.54 (Description) StaticVarCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.

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The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Table 493 shows all attributes of StaticVarCompensator.

Table 493 – Attributes of SteadyStateHypothesisProfile::StaticVarCompensator

name	mult	type	description
q	1..1	ReactivePower	Reactive power injection. Load sign convention is used, i.e. positive sign means flow out from a node. Starting value for a steady state solution.
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.55 (Description) StationSupply

Inheritance path = EnergyConsumer: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Station supply with load derived from the station output.

Table 494 shows all attributes of StationSupply.

Table 494 – Attributes of SteadyStateHypothesisProfile::StationSupply

name	mult	type	description
p	1..1	ActivePower	inherited from: EnergyConsumer
q	1..1	ReactivePower	inherited from: EnergyConsumer
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.56 (Description) Switch

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches. The ACDCTerminal.connected at the two sides of the switch shall not be considered for assessing switch connectivity, i.e. only Switch.open, .normalOpen and .locked are relevant.

Table 495 shows all attributes of Switch.

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Table 495 – Attributes of SteadyStateHypothesisProfile::Switch

name	mult	type	description
open	1..1	Boolean	The attribute tells if the switch is considered open when used as input to topology processing.
locked	1..1	Boolean	If true, the switch is locked. The resulting switch state is a combination of locked and Switch.open attributes as follows: – locked=true and Switch.open=true. The resulting state is open and locked; – locked=false and Switch.open=true. The resulting state is open; – locked=false and Switch.open=false. The resulting state is closed.
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.57 (Description) SynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Table 496 shows all attributes of SynchronousMachine.

Table 496 – Attributes of SteadyStateHypothesisProfile::SynchronousMachine

name	mult	type	description
operatingMode	1..1	SynchronousMachineOperatingMode	Current mode of operation.
referencePriority	1..1	Integer	Priority of unit for use as powerflow voltage phase angle reference bus selection. 0 = don t care (default) 1 = highest priority. 2 is less than 1 and so on.
p	1..1	ActivePower	inherited from: RotatingMachine
q	1..1	ReactivePower	inherited from: RotatingMachine
controlEnabled	1..1	Boolean	inherited from: RegulatingCondEq
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.58 (abstract) TapChanger

Inheritance path = PowerSystemResource: IdentifiedObject

Mechanism for changing transformer winding tap positions.

Table 497 shows all attributes of TapChanger.

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Table 497 – Attributes of SteadyStateHypothesisProfile::TapChanger

name	mult	type	description
controlEnabled	1..1	Boolean	Specifies the regulation status of the equipment. True is regulating, false is not regulating.
step	1..1	Float	Tap changer position. Starting step for a steady state solution. Non integer values are allowed to support continuous tap variables. The reasons for continuous value are to support study cases where no discrete tap changer has yet been designed, a solution where a narrow voltage band forces the tap step to oscillate or to accommodate for a continuous solution as input. The attribute shall be equal to or greater than lowStep and equal to or less than highStep.
mRID	1..1	String	inherited from: IdentifiedObject

9.59 (Description) TapChangerControl

Inheritance path = RegulatingControl: PowerSystemResource: IdentifiedObject

Describes behaviour specific to tap changers, e.g. how the voltage at the end of a line varies with the load level and compensation of the voltage drop by tap adjustment.

Table 498 shows all attributes of TapChangerControl.

Table 498 – Attributes of SteadyStateHypothesisProfile::TapChangerControl

name	mult	type	description
discrete	1..1	Boolean	inherited from: RegulatingControl
enabled	1..1	Boolean	inherited from: RegulatingControl
targetDeadband	0..1	Float	inherited from: RegulatingControl
targetValue	1..1	Float	inherited from: RegulatingControl
targetValueUnitMultiplier	1..1	UnitMultiplier	inherited from: RegulatingControl
maxAllowedTargetValue	0..1	Float	inherited from: RegulatingControl
minAllowedTargetValue	0..1	Float	inherited from: RegulatingControl
mRID	1..1	String	inherited from: IdentifiedObject

9.60 (Description) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 499 shows all attributes of Terminal.

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Table 499 – Attributes of SteadyStateHypothesisProfile::Terminal

name	mult	type	description
connected	1..1	Boolean	inherited from: ACDCTerminal
mRID	1..1	String	inherited from: IdentifiedObject

9.61 (Description) ThermalGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A generating unit whose prime mover could be a steam turbine, combustion turbine, or diesel engine.

Table 500 shows all attributes of ThermalGeneratingUnit.

Table 500 – Attributes of SteadyStateHypothesisProfile::ThermalGeneratingUnit

name	mult	type	description
normalPF	1..1	Float	inherited from: GeneratingUnit
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.62 (Description) VoltageLimit

Inheritance path = OperationalLimit: IdentifiedObject

Operational limit applied to voltage.

The use of operational VoltageLimit is preferred instead of limits defined at VoltageLevel. The operational VoltageLimits are used, if present.

Table 501 shows all attributes of VoltageLimit.

Table 501 – Attributes of SteadyStateHypothesisProfile::VoltageLimit

name	mult	type	description
value	1..1	Voltage	Limit on voltage. High or low limit nature of the limit depends upon the properties of the operational limit type. The attribute shall be a positive value or zero.
mRID	1..1	String	inherited from: IdentifiedObject

9.63 (Description) VsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

DC side of the voltage source converter (VSC).

Table 502 shows all attributes of VsConverter.

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Table 502 – Attributes of SteadyStateHypothesisProfile::VsConverter

name	mult	type	description
droop	0..1	PU	Droop constant. The pu value is obtained as D [kV/MW] x Sb / Ubdc. The attribute shall be a positive value.
droopCompensation	0..1	Resistance	Compensation constant. Used to compensate for voltage drop when controlling voltage at a distant bus. The attribute shall be a positive value.
pPccControl	1..1	VsPpccControlKind	Kind of control of real power and/or DC voltage.
qPccControl	1..1	VsQpccControlKind	Kind of reactive power control.
qShare	0..1	PerCent	Reactive power sharing factor among parallel converters on Uac control. The attribute shall be a positive value or zero.
targetQpcc	0..1	ReactivePower	Reactive power injection target in AC grid, at point of common coupling. Load sign convention is used, i.e. positive sign means flow out from a node.
targetUpcc	0..1	Voltage	Voltage target in AC grid, at point of common coupling. The attribute shall be a positive value.
targetPowerFactorPcc	0..1	Float	Power factor target at the AC side, at point of common coupling. The attribute shall be a positive value.
targetPhasePcc	0..1	AngleDegrees	Phase target at AC side, at point of common coupling. The attribute shall be a positive value.
targetPWMfactor	0..1	Float	Magnitude of pulse-modulation factor. The attribute shall be a positive value.
p	1..1	ActivePower	inherited from: ACDCConverter
q	1..1	ReactivePower	inherited from: ACDCConverter
targetPpcc	0..1	ActivePower	inherited from: ACDCConverter
targetUdc	0..1	Voltage	inherited from: ACDCConverter
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.64 (Description) WindGeneratingUnit

Inheritance path = GeneratingUnit: Equipment: PowerSystemResource: IdentifiedObject

A wind driven generating unit, connected to the grid by means of a rotating machine. May be used to represent a single turbine or an aggregation.

Table 503 shows all attributes of WindGeneratingUnit.

Table 503 – Attributes of SteadyStateHypothesisProfile::WindGeneratingUnit

name	mult	type	description
normalPF	1..1	Float	inherited from: GeneratingUnit
inService	1..1	Boolean	inherited from: Equipment
mRID	1..1	String	inherited from: IdentifiedObject

9.65 AsynchronousMachineKind enumeration

Kind of Asynchronous Machine.

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Table 504 shows all literals of AsynchronousMachineKind.

Table 504 – Literals of SteadyStateHypothesisProfile::AsynchronousMachineKind

literal	value	description
generator		The Asynchronous Machine is a generator.
motor		The Asynchronous Machine is a motor.

9.66 BatteryStateKind enumeration

The state of the battery unit.

Table 505 shows all literals of BatteryStateKind.

Table 505 – Literals of SteadyStateHypothesisProfile::BatteryStateKind

literal	value	description
discharging		Stored energy is decreasing.
full		Unable to charge, and not discharging.
waiting		Neither charging nor discharging, but able to do so.
charging		Stored energy is increasing.
empty		Unable to discharge, and not charging.

9.67 CsOperatingModeKind enumeration

Operating mode for HVDC line operating as Current Source Converter.

Table 506 shows all literals of CsOperatingModeKind.

Table 506 – Literals of SteadyStateHypothesisProfile::CsOperatingModeKind

literal	value	description
inverter		Operating as inverter, which is the power receiving end.
rectifier		Operating as rectifier, which is the power sending end.

9.68 CsPpccControlKind enumeration

Active power control modes for HVDC line operating as Current Source Converter.

Table 507 shows all literals of CsPpccControlKind.

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Table 507 – Literals of SteadyStateHypothesisProfile::CsPpccControlKind

literal	value	description
activePower		Control is active power control at AC side, at point of common coupling. Target is provided by ACDCConverter.targetPpcc.
dcVoltage		Control is DC voltage with target value provided by ACDCConverter.targetUdc.
dcCurrent		Control is DC current with target value provided by CsConverter.targetIdc.

9.69 SynchronousMachineOperatingMode enumeration

Synchronous machine operating mode.

Table 508 shows all literals of SynchronousMachineOperatingMode.

Table 508 – Literals of SteadyStateHypothesisProfile::SynchronousMachineOperatingMode

literal	value	description
generator		Operating as generator.
condenser		Operating as condenser.
motor		Operating as motor.

9.70 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 509 shows all literals of UnitMultiplier.

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Table 509 – Literals of SteadyStateHypothesisProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

9.71 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

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The integer values are used for harmonization with IEC 61850.

Table 510 shows all literals of UnitSymbol.

Table 510 – Literals of SteadyStateHypothesisProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.

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literal	value	description
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).

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literal	value	description
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg·m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μg/g’.

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literal	value	description
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.

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literal	value	description
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133.3 Pa).
M	217	Length, nautical miles (1 M = 1852 m).
kn	219	Speed, knots (1 kn = 1852/3600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

9.72 VsPpccControlKind enumeration

Types applicable to the control of real power and/or DC voltage by voltage source converter.

Table 511 shows all literals of VsPpccControlKind.

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Table 511 – Literals of SteadyStateHypothesisProfile::VsPpccControlKind

literal	value	description
pPcc		Control is real power at point of common coupling. The target value is provided by ACDCConverter.targetPpcc.
udc		Control is DC voltage with target value provided by ACDCConverter.targetUdc.
pPccAndUdcDroop		Control is active power at point of common coupling and local DC voltage, with the droop. Target values are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.
pPccAndUdcDroopWithCompensation		Control is active power at point of common coupling and compensated DC voltage, with the droop. Compensation factor is the resistance, as an approximation of the DC voltage of a common (real or virtual) node in the DC network. Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc, VsConverter.droop and VsConverter.droopCompensation.
pPccAndUdcDroopPilot		Control is active power at point of common coupling and the pilot DC voltage, with the droop. The mode is used for Multi Terminal High Voltage DC (MTDC) systems where multiple HVDC Substations are connected to the HVDC transmission lines. The pilot voltage is then used to coordinate the control the DC voltage across the HVDC substations. Targets are provided by ACDCConverter.targetPpcc, ACDCConverter.targetUdc and VsConverter.droop.
phasePcc		Control is phase at point of common coupling. Target is provided by VsConverter.targetPhasePcc.

9.73 VsQpccControlKind enumeration

Kind of reactive power control at point of common coupling for a voltage source converter.

Table 512 shows all literals of VsQpccControlKind.

Table 512 – Literals of SteadyStateHypothesisProfile::VsQpccControlKind

literal	value	description
reactivePcc		Control is reactive power at point of common coupling. Target is provided by VsConverter.targetQpcc.
voltagePcc		Control is voltage at point of common coupling. Target is provided by VsConverter.targetUpcc.
powerFactorPcc		Control is power factor at point of common coupling. Target is provided by VsConverter.targetPowerFactorPcc.
pulseWidthModulation		No explicit control. Pulse-modulation factor is directly set in magnitude (VsConverter.targetPWMfactor) and phase (VsConverter.targetPhasePcc).

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9.74 ActivePower datatype

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Table 513 shows all attributes of ActivePower.

Table 513 – Attributes of SteadyStateHypothesisProfile::ActivePower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=W)

9.75 AngleDegrees datatype

Measurement of angle in degrees.

Table 514 shows all attributes of AngleDegrees.

Table 514 – Attributes of SteadyStateHypothesisProfile::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

9.76 AngleRadians datatype

Phase angle in radians.

Table 515 shows all attributes of AngleRadians.

Table 515 – Attributes of SteadyStateHypothesisProfile::AngleRadians

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=rad)
multiplier	0..1	UnitMultiplier	(const=none)

9.77 ApparentPower datatype

Product of the RMS value of the voltage and the RMS value of the current.

Table 516 shows all attributes of ApparentPower.

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Table 516 – Attributes of SteadyStateHypothesisProfile::ApparentPower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=VA)

9.78 CurrentFlow datatype

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Table 517 shows all attributes of CurrentFlow.

Table 517 – Attributes of SteadyStateHypothesisProfile::CurrentFlow

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=A)

9.79 PerCent datatype

Percentage on a defined base. For example, specify as 100 to indicate at the defined base.

Table 518 shows all attributes of PerCent.

Table 518 – Attributes of SteadyStateHypothesisProfile::PerCent

name	mult	type	description
value	0..1	Float	Normally 0 to 100 on a defined base.
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

9.80 PU datatype

Per Unit – a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Table 519 shows all attributes of PU.

Table 519 – Attributes of SteadyStateHypothesisProfile::PU

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

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9.81 ReactivePower datatype

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Table 520 shows all attributes of ReactivePower.

Table 520 – Attributes of SteadyStateHypothesisProfile::ReactivePower

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=VAr)
multiplier	0..1	UnitMultiplier	(const=M)

9.82 RealEnergy datatype

Real electrical energy.

Table 521 shows all attributes of RealEnergy.

Table 521 – Attributes of SteadyStateHypothesisProfile::RealEnergy

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=Wh)
value	0..1	Float	

9.83 Resistance datatype

Resistance (real part of impedance).

Table 522 shows all attributes of Resistance.

Table 522 – Attributes of SteadyStateHypothesisProfile::Resistance

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=ohm)
multiplier	0..1	UnitMultiplier	(const=none)

9.84 Voltage datatype

Electrical voltage, can be both AC and DC.

Table 523 shows all attributes of Voltage.

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Table 523 – Attributes of SteadyStateHypothesisProfile::Voltage

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=k)
unit	0..1	UnitSymbol	(const=V)

9.85 Boolean primitive

A type with the value space "true" and "false".

9.86 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

9.87 Float primitive

A floating point number. The range is unspecified and not limited.

9.88 Integer primitive

An integer number. The range is unspecified and not limited.

9.89 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

10 Package TopologyProfile

10.1 General

A topology dataset which conforms to the IEC 61970-456 topology profile contains all objects defined in a topology profile and includes data for topology information relating to a given exchange.

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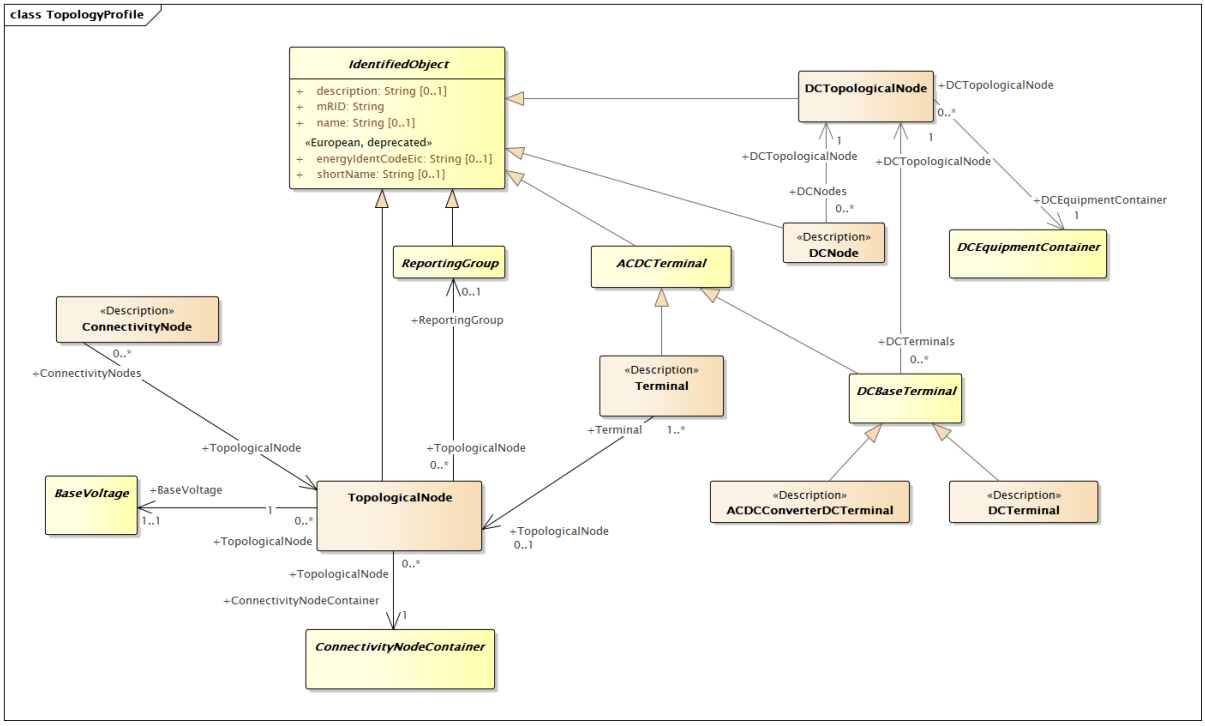

Figure 38 – Class diagram TopologyProfile::TopologyProfile

Figure 38: The diagram shows topology profile.

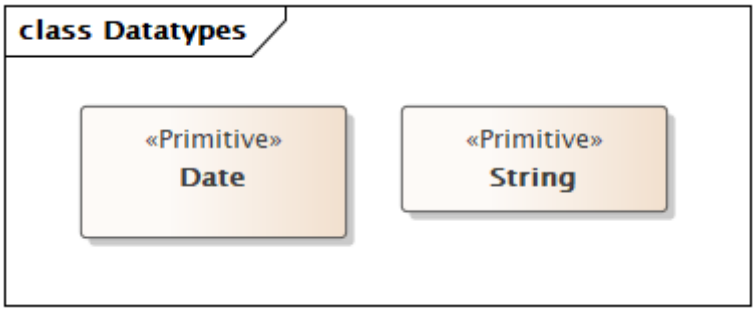

Figure 39 – Class diagram TopologyProfile::Datatypes

Figure 39: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

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10.2 (Description) ACDCConverterDCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

A DC electrical connection point at the AC/DC converter. The AC/DC converter is electrically connected also to the AC side. The AC connection is inherited from the AC conducting equipment in the same way as any other AC equipment. The AC/DC converter DC terminal is separate from generic DC terminal to restrict the connection with the AC side to AC/DC converter and so that no other DC conducting equipment can be connected to the AC side.

Table 524 shows all attributes of ACDCConverterDCTerminal.

Table 524 – Attributes of TopologyProfile::ACDCConverterDCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 525 shows all association ends of ACDCConverterDCTerminal with other classes.

Table 525 – Association ends of TopologyProfile:: ACDCConverterDCTerminal with other classes

mult from	name	mult to	type	description
0..*	DCTopologicalNode	1..1	DCTopologicalNode	inherited from: DCBaseTerminal

10.3 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 526 shows all attributes of ACDCTerminal.

Table 526 – Attributes of TopologyProfile::ACDCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

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10.4 (abstract) BaseVoltage root class

Defines a system base voltage which is referenced.

10.5 (Description) ConnectivityNode root class

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

Table 527 shows all association ends of ConnectivityNode with other classes.

Table 527 – Association ends of TopologyProfile::ConnectivityNode with other classes

mult from	name	mult to	type	description
0..*	TopologicalNode	1..1	TopologicalNode	The topological node to which this connectivity node is assigned. May depend on the current state of switches in the network.

10.6 (abstract) ConnectivityNodeContainer root class

A base class for all objects that may contain connectivity nodes or topological nodes.

10.7 (abstract) DCBaseTerminal

Inheritance path = ACDCTerminal: IdentifiedObject

An electrical connection point at a piece of DC conducting equipment. DC terminals are connected at one physical DC node that may have multiple DC terminals connected. A DC node is similar to an AC connectivity node. The model requires that DC connections are distinct from AC connections.

Table 528 shows all attributes of DCBaseTerminal.

Table 528 – Attributes of TopologyProfile::DCBaseTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 529 shows all association ends of DCBaseTerminal with other classes.

Table 529 – Association ends of TopologyProfile::DCBaseTerminal with other classes

mult from	name	mult to	type	description
0..*	DCTopologicalNode	1..1	DCTopologicalNode	See association end Terminal.TopologicalNode.

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10.8 (abstract) DCEquipmentContainer root class

A modelling construct to provide a root class for containment of DC as well as AC equipment. The class differ from the EquipmentContaner for AC in that it may also contain DCNode-s. Hence it can contain both AC and DC equipment.

10.9 (Description) DCNode

Inheritance path = IdentifiedObject

DC nodes are points where terminals of DC conducting equipment are connected together with zero impedance.

Table 530 shows all attributes of DCNode.

Table 530 – Attributes of TopologyProfile::DCNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 531 shows all association ends of DCNode with other classes.

Table 531 – Association ends of TopologyProfile::DCNode with other classes

mult from	name	mult to	type	description
0..*	DCTopologicalNode	1..1	DCTopologicalNode	The DC topological node to which this DC connectivity node is assigned. May depend on the current state of switches in the network.

10.10 (Description) DCTerminal

Inheritance path = DCBaseTerminal: ACDCTerminal: IdentifiedObject

An electrical connection point to generic DC conducting equipment.

Table 532 shows all attributes of DCTerminal.

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Table 532 – Attributes of TopologyProfile::DCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 533 shows all association ends of DCTerminal with other classes.

Table 533 – Association ends of TopologyProfile::DCTerminal with other classes

mult from	name	mult to	type	description
0..*	DCTopologicalNode	1..1	DCTopologicalNode	inherited from: DCBaseTerminal

10.11 DCTopologicalNode

Inheritance path = IdentifiedObject

DC bus.

Table 534 shows all attributes of DCTopologicalNode.

Table 534 – Attributes of TopologyProfile::DCTopologicalNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 535 shows all association ends of DCTopologicalNode with other classes.

Table 535 – Association ends of TopologyProfile:: DCTopologicalNode with other classes

mult from	name	mult to	type	description
0..*	DCEquipmentContainer	1..1	DCEquipmentContainer	The connectivity node container to which the topological node belongs.

10.12 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

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Table 536 shows all attributes of IdentifiedObject.

Table 536 – Attributes of TopologyProfile::IdentifiedObject

name	mult	type	description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
energyIdentCodeEic	0..1	String	(deprecated,European) The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName	0..1	String	(deprecated,European) The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

10.13 (Description) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 537 shows all attributes of Terminal.

Table 537 – Attributes of TopologyProfile::Terminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 538 shows all association ends of Terminal with other classes.

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Table 538 – Association ends of TopologyProfile::Terminal with other classes

mult from	name	mult to	type	description
1..*	TopologicalNode	0..1	TopologicalNode	The topological node associated with the terminal. This can be used as an alternative to the connectivity node path to topological node, thus making it unnecessary to model connectivity nodes in some cases. Note that the if connectivity nodes are in the model, this association would probably not be used as an input specification.

10.14 TopologicalNode

Inheritance path = IdentifiedObject

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).

For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

Table 539 shows all attributes of TopologicalNode.

Table 539 – Attributes of TopologyProfile::TopologicalNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 540 shows all association ends of TopologicalNode with other classes.

Table 540 – Association ends of TopologyProfile::TopologicalNode with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	1..1	BaseVoltage	The base voltage of the topological node.
0..*	ConnectivityNodeContainer	1..1	ConnectivityNodeContainer	The connectivity node container to which the topological node belongs.
0..*	ReportingGroup	0..1	ReportingGroup	The reporting group to which the topological node belongs.

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10.15 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

10.16 (abstract) ReportingGroup

Inheritance path = IdentifiedObject

A reporting group is used for various ad-hoc groupings used for reporting.

Table 541 shows all attributes of ReportingGroup.

Table 541 – Attributes of TopologyProfile::ReportingGroup

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

10.17 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

11 Package StateVariablesProfile

11.1 General

A state variables dataset which conforms to the IEC 61970-456 state variables profile contains all objects required to complete the specification of a steady-state solution.

A state variables dataset is always exchanged in full.

A state variables dataset of an assembled model contains state variables related objects for all model authority sets being part of the assembled model.

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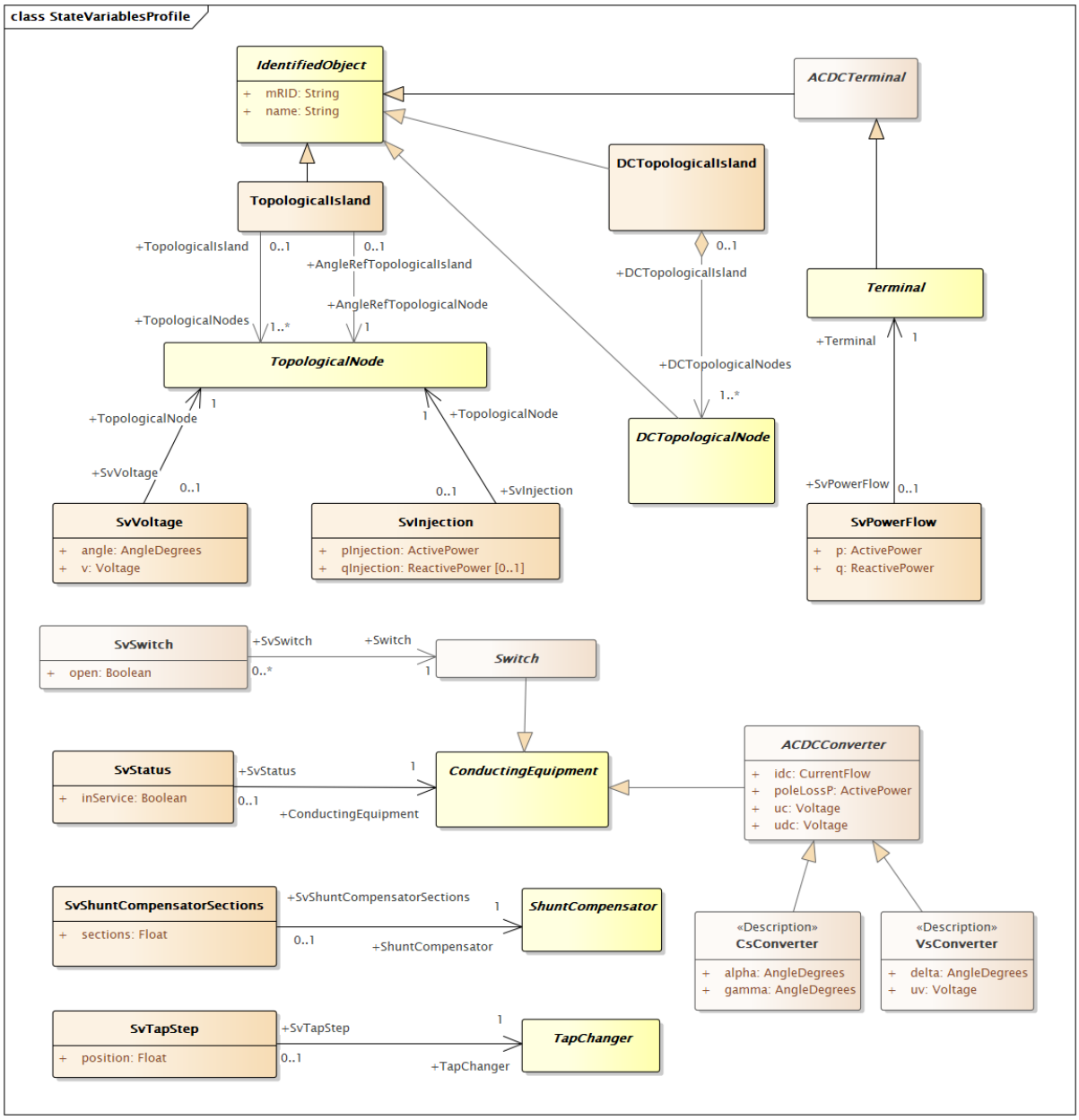

Figure 40 – Class diagram StateVariablesProfile::StateVariablesProfile

Figure 40: The diagram shows the state variables profile.

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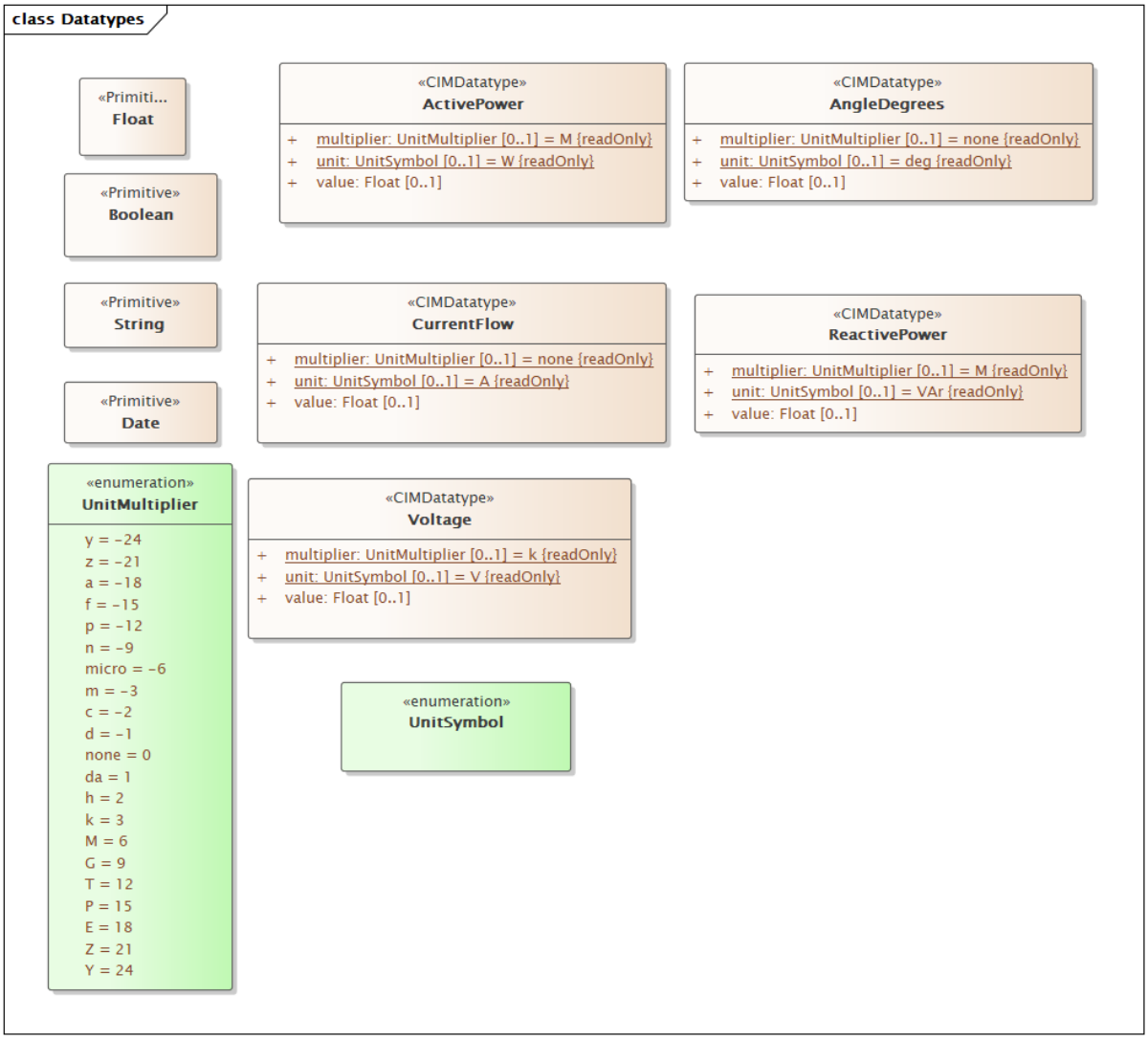

Figure 41 – Class diagram StateVariablesProfile::Datatypes

Figure 41: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

11.2 (abstract) ACDCConverter

Inheritance path = ConductingEquipment

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A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Table 542 shows all attributes of ACDCConverter.

Table 542 – Attributes of StateVariablesProfile::ACDCConverter

name	mult	type	description
idc	1..1	CurrentFlow	Converter DC current, also called Id. It is converter's state variable, result from power flow.
poleLossP	1..1	ActivePower	The active power loss at a DC Pole = idleLoss + switchingLoss*|Idc| + resitiveLoss*Idc^2. For lossless operation Pdc=Pac. For rectifier operation with losses Pdc=Pac-lossP. For inverter operation with losses Pdc=Pac+lossP. It is converter's state variable used in power flow. The attribute shall be a positive value.
uc	1..1	Voltage	Line-to-line converter voltage, the voltage at the AC side of the valve. It is converter's state variable, result from power flow. The attribute shall be a positive value.
udc	1..1	Voltage	Converter voltage at the DC side, also called Ud. It is converter's state variable, result from power flow. The attribute shall be a positive value.

11.3 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 543 shows all attributes of ACDCTerminal.

Table 543 – Attributes of StateVariablesProfile::ACDCTerminal

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

11.4 (Description) CsConverter

Inheritance path = ACDCConverter: ConductingEquipment

DC side of the current source converter (CSC).

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The firing angle controls the DC voltage at the converter, both for rectifier and inverter. The difference between the DC voltages of the rectifier and inverter determines the DC current. The extinction angle is used to limit the DC voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source dc line. Higher level controls are built on top, e.g. DC voltage, DC current and active power. From a steady state perspective it is sufficient to specify the wanted active power transfer (ACDCConverter.targetPpcc) and the control functions will set the DC voltage, DC current, firing angle, transformer tap position and number of connected filters to meet this. Therefore attributes targetAlpha and targetGamma are not applicable in this case.

The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses are a function of the DC voltage and DC current.

The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10 to 18 degrees.

The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17 to 20 degrees.

Table 544 shows all attributes of CsConverter.

Table 544 – Attributes of StateVariablesProfile::CsConverter

name	mult	type	description
alpha	1..1	AngleDegrees	Firing angle that determines the dc voltage at the converter dc terminal. Typical value between 10 degrees and 18 degrees for a rectifier. It is converter's state variable, result from power flow. The attribute shall be a positive value.
gamma	1..1	AngleDegrees	Extinction angle. It is used to limit the dc voltage at the inverter if needed. Typical value between 17 degrees and 20 degrees for an inverter. It is converter's state variable, result from power flow. The attribute shall be a positive value.
idc	1..1	CurrentFlow	inherited from: ACDCConverter
poleLossP	1..1	ActivePower	inherited from: ACDCConverter
uc	1..1	Voltage	inherited from: ACDCConverter
udc	1..1	Voltage	inherited from: ACDCConverter

11.5 (abstract) ConductingEquipment root class

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

11.6 DCTopologicalIsland

Inheritance path = IdentifiedObject

An electrically connected subset of the network. DC topological islands can change as the current network state changes, e.g. due to:

– disconnect switches or breakers changing state in a SCADA/EMS.

– manual creation, change or deletion of topological nodes in a planning tool.

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Only energised TopologicalNode-s shall be part of the topological island.

Table 545 shows all attributes of DCTopologicalIsland.

Table 545 – Attributes of StateVariablesProfile::DCTopologicalIsland

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 546 shows all association ends of DCTopologicalIsland with other classes.

Table 546 – Association ends of StateVariablesProfile:: DCTopologicalIsland with other classes

mult from	name	mult to	type	description
0..1	DCTopologicalNodes	1..*	DCTopologicalNode	The DC topological nodes in a DC topological island.

11.7 (abstract) DCTopologicalNode

Inheritance path = IdentifiedObject

DC bus.

Table 547 shows all attributes of DCTopologicalNode.

Table 547 – Attributes of StateVariablesProfile::DCTopologicalNode

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

11.8 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 548 shows all attributes of IdentifiedObject.

Table 548 – Attributes of StateVariablesProfile::IdentifiedObject

name	mult	type	description
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

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name	mult	type	description
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.

11.9 (abstract) ShuntCompensator root class

A shunt capacitor or reactor or switchable bank of shunt capacitors or reactors. A section of a shunt compensator is an individual capacitor or reactor. A negative value for bPerSection indicates that the compensator is a reactor. ShuntCompensator is a single terminal device. Ground is implied.

11.10 SvInjection root class

The SvInjection reports the calculated bus injection minus the sum of the terminal flows. The terminal flow is positive out from the bus (load sign convention) and bus injection has positive flow into the bus. SvInjection may have the remainder after state estimation or slack after power flow calculation.

Table 549 shows all attributes of SvInjection.

Table 549 – Attributes of StateVariablesProfile::SvInjection

name	mult	type	description
pInjection	1..1	ActivePower	The active power mismatch between calculated injection and initial injection. Positive sign means injection into the TopologicalNode (bus).
qInjection	0..1	ReactivePower	The reactive power mismatch between calculated injection and initial injection. Positive sign means injection into the TopologicalNode (bus).

Table 550 shows all association ends of SvInjection with other classes.

Table 550 – Association ends of StateVariablesProfile::SvInjection with other classes

mult from	name	mult to	type	description
0..1	TopologicalNode	1..1	TopologicalNode	The topological node associated with the flow injection state variable.

11.11 SvPowerFlow root class

State variable for power flow. Load convention is used for flow direction. This means flow out from the TopologicalNode into the equipment is positive.

Table 551 shows all attributes of SvPowerFlow.

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Table 551 – Attributes of StateVariablesProfile::SvPowerFlow

name	mult	type	description
p	1..1	ActivePower	The active power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.
q	1..1	ReactivePower	The reactive power flow. Load sign convention is used, i.e. positive sign means flow out from a TopologicalNode (bus) into the conducting equipment.

Table 552 shows all association ends of SvPowerFlow with other classes.

Table 552 – Association ends of StateVariablesProfile::SvPowerFlow with other classes

mult from	name	mult to	type	description
0..1	Terminal	1..1	Terminal	The terminal associated with the power flow state variable.

11.12 SvShuntCompensatorSections root class

State variable for the number of sections in service for a shunt compensator.

Table 553 shows all attributes of SvShuntCompensatorSections.

Table 553 – Attributes of StateVariablesProfile::SvShuntCompensatorSections

name	mult	type	description
sections	1..1	Float	The number of sections in service as a continuous variable. The attribute shall be a positive value or zero. To get integer value scale with ShuntCompensator.bPerSection.

Table 554 shows all association ends of SvShuntCompensatorSections with other classes.

Table 554 – Association ends of StateVariablesProfile:: SvShuntCompensatorSections with other classes

mult from	name	mult to	type	description
0..1	ShuntCompensator	1..1	ShuntCompensator	The shunt compensator for which the state applies.

11.13 SvStatus root class

State variable for status.

Table 555 shows all attributes of SvStatus.

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Table 555 – Attributes of StateVariablesProfile::SvStatus

name	mult	type	description
inService	1..1	Boolean	The in service status as a result of topology processing. It indicates if the equipment is considered as energized by the power flow. It reflects if the equipment is connected within a solvable island. It does not necessarily reflect whether or not the island was solved by the power flow.

Table 556 shows all association ends of SvStatus with other classes.

Table 556 – Association ends of StateVariablesProfile::SvStatus with other classes

mult from	name	mult to	type	description
0..1	ConductingEquipment	1..1	ConductingEquipment	The conducting equipment associated with the status state variable.

11.14 SvSwitch root class

State variable for switch.

Table 557 shows all attributes of SvSwitch.

Table 557 – Attributes of StateVariablesProfile::SvSwitch

name	mult	type	description
open	1..1	Boolean	The attribute tells if the computed state of the switch is considered open.

Table 558 shows all association ends of SvSwitch with other classes.

Table 558 – Association ends of StateVariablesProfile::SvSwitch with other classes

mult from	name	mult to	type	description
0..*	Switch	1..1	Switch	The switch associated with the switch state.

11.15 SvTapStep root class

State variable for transformer tap step.

Table 559 shows all attributes of SvTapStep.

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Table 559 – Attributes of StateVariablesProfile::SvTapStep

name	mult	type	description
position	1..1	Float	The floating point tap position. This is not the tap ratio, but rather the tap step position as defined by the related tap changer model and normally is constrained to be within the range of minimum and maximum tap positions.

Table 560 shows all association ends of SvTapStep with other classes.

Table 560 – Association ends of StateVariablesProfile::SvTapStep with other classes

mult from	name	mult to	type	description
0..1	TapChanger	1..1	TapChanger	The tap changer associated with the tap step state.

11.16 SvVoltage root class

State variable for voltage.

Table 561 shows all attributes of SvVoltage.

Table 561 – Attributes of StateVariablesProfile::SvVoltage

name	mult	type	description
angle	1..1	AngleDegrees	The voltage angle of the topological node complex voltage with respect to system reference.
v	1..1	Voltage	The voltage magnitude at the topological node. The attribute shall be a positive value.

Table 562 shows all association ends of SvVoltage with other classes.

Table 562 – Association ends of StateVariablesProfile::SvVoltage with other classes

mult from	name	mult to	type	description
0..1	TopologicalNode	1..1	TopologicalNode	The topological node associated with the voltage state.

11.17 (abstract) Switch

Inheritance path = ConductingEquipment

A generic device designed to close, or open, or both, one or more electric circuits. All switches are two terminal devices including grounding switches. The ACDCTerminal.connected at the two sides of the switch shall not be considered for assessing switch connectivity, i.e. only Switch.open, .normalOpen and .locked are relevant.

11.18 (abstract) TapChanger root class

Mechanism for changing transformer winding tap positions.

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11.19 (abstract) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 563 shows all attributes of Terminal.

Table 563 – Attributes of StateVariablesProfile::Terminal

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

11.20 TopologicalIsland

Inheritance path = IdentifiedObject

An electrically connected subset of the network. Topological islands can change as the current network state changes, e.g. due to:

– disconnect switches or breakers changing state in a SCADA/EMS.
– manual creation, change or deletion of topological nodes in a planning tool.

Only energised TopologicalNode-s shall be part of the topological island.

Table 564 shows all attributes of TopologicalIsland.

Table 564 – Attributes of StateVariablesProfile::TopologicalIsland

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

Table 565 shows all association ends of TopologicalIsland with other classes.

Table 565 – Association ends of StateVariablesProfile:: TopologicalIsland with other classes

mult from	name	mult to	type	description
0..1	AngleRefTopologicalNode	1..1	TopologicalNode	The angle reference for the island. Normally there is one TopologicalNode that is selected as the angle reference for each island. Other reference schemes exist, so the association is typically optional.
0..1	TopologicalNodes	1..*	TopologicalNode	A topological node belongs to a topological island.

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11.21 (abstract) TopologicalNode root class

For a detailed substation model a topological node is a set of connectivity nodes that, in the current network state, are connected together through any type of closed switches, including jumpers. Topological nodes change as the current network state changes (i.e., switches, breakers, etc. change state).

For a planning model, switch statuses are not used to form topological nodes. Instead they are manually created or deleted in a model builder tool. Topological nodes maintained this way are also called "busses".

11.22 (Description) VsConverter

Inheritance path = ACDCConverter: ConductingEquipment

DC side of the voltage source converter (VSC).

Table 566 shows all attributes of VsConverter.

Table 566 – Attributes of StateVariablesProfile::VsConverter

name	mult	type	description
delta	1..1	AngleDegrees	Angle between VsConverter.uv and ACDCConverter.uc. It is converter's state variable used in power flow. The attribute shall be a positive value or zero.
uv	1..1	Voltage	Line-to-line voltage on the valve side of the converter transformer. It is converter's state variable, result from power flow. The attribute shall be a positive value.
idc	1..1	CurrentFlow	inherited from: ACDCConverter
poleLossP	1..1	ActivePower	inherited from: ACDCConverter
uc	1..1	Voltage	inherited from: ACDCConverter
udc	1..1	Voltage	inherited from: ACDCConverter

11.23 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

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Table 567 shows all literals of UnitMultiplier.

Table 567 – Literals of StateVariablesProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

11.24 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet

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instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

Table 568 shows all literals of UnitSymbol.

Table 568 – Literals of StateVariablesProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.

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literal	value	description
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.

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literal	value	description
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg·m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.

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literal	value	description
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µg/g’.
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.

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literal	value	description
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133,3 Pa).
M	217	Length, nautical miles (1 M = 1852 m).
kn	219	Speed, knots (1 kn = 1 852/3 600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

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11.25 ActivePower datatype

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Table 569 shows all attributes of ActivePower.

Table 569 – Attributes of StateVariablesProfile::ActivePower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=W)

11.26 AngleDegrees datatype

Measurement of angle in degrees.

Table 570 shows all attributes of AngleDegrees.

Table 570 – Attributes of StateVariablesProfile::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

11.27 CurrentFlow datatype

Electrical current with sign convention: positive flow is out of the conducting equipment into the connectivity node. Can be both AC and DC.

Table 571 shows all attributes of CurrentFlow.

Table 571 – Attributes of StateVariablesProfile::CurrentFlow

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=A)

11.28 ReactivePower datatype

Product of RMS value of the voltage and the RMS value of the quadrature component of the current.

Table 572 shows all attributes of ReactivePower.

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Table 572 – Attributes of StateVariablesProfile::ReactivePower

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=VAr)
multiplier	0..1	UnitMultiplier	(const=M)

11.29 Voltage datatype

Electrical voltage, can be both AC and DC.

Table 573 shows all attributes of Voltage.

Table 573 – Attributes of StateVariablesProfile::Voltage

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=k)
unit	0..1	UnitSymbol	(const=V)

11.30 Boolean primitive

A type with the value space "true" and "false".

11.31 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

11.32 Float primitive

A floating point number. The range is unspecified and not limited.

11.33 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

12 Package DiagramLayoutProfile

12.1 General

This standard specifies an exchange profile for schematics with the following characteristics.

A generic method for linking the diagram object to the domain data. Domain data and diagram layout data can be exchanged together with the domain data, or separately from each other with the assumption that domain data will have already been imported should diagram layout data be imported separately.

It does not require or imply a specific domain data format. Therefore, it supports domain data modelled according to the IEC 61970-301 Common information model (CIM) that is exchanged in the IEC 61970-501 format (CIM RDF schema). The diagram layout data references domain data in compliance with IEC TR 62541-1, *OPC Unified Architecture*).

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The intended usage of this standard is that the source of a diagram will use the standard to encode the layout of their diagram as they created it. It is then incumbent upon any receiver to supply the means of using that diagram layout in their system.

– In the simplest situation, if the source uses Switch object placements, the receiver creates a generic Switch template in the receiver's system that will be used to render Switches (i.e. there is no exchange of Switch templates).
– In more complex situations, a 1:1 correspondence may not be desirable, and receivers may have to create generic transformations in order to use exchanges.
– However, in both situations, it is expected that once an investment is made in a strategy for rendering, repeated imports may be carried out automatically.

Supports the exchange of diagram objects that have no relationship to domain data, i.e., pure static background objects.

Supports multiple representations of the same domain object in the same or different diagrams.

Allows the diagram to be used as the unit of exchange, providing a straightforward approach to partial exchange, or an exchange between systems that have a separate model and repository for diagram layout data.

Supports assignment of diagram objects to layers or other means, for showing or hiding information based on zoom level and/or user interest.

A generic method for proprietary extensions to enable full round tripping (export and import back into the same system) without information loss within a system, and without breaking the standard exchange format.

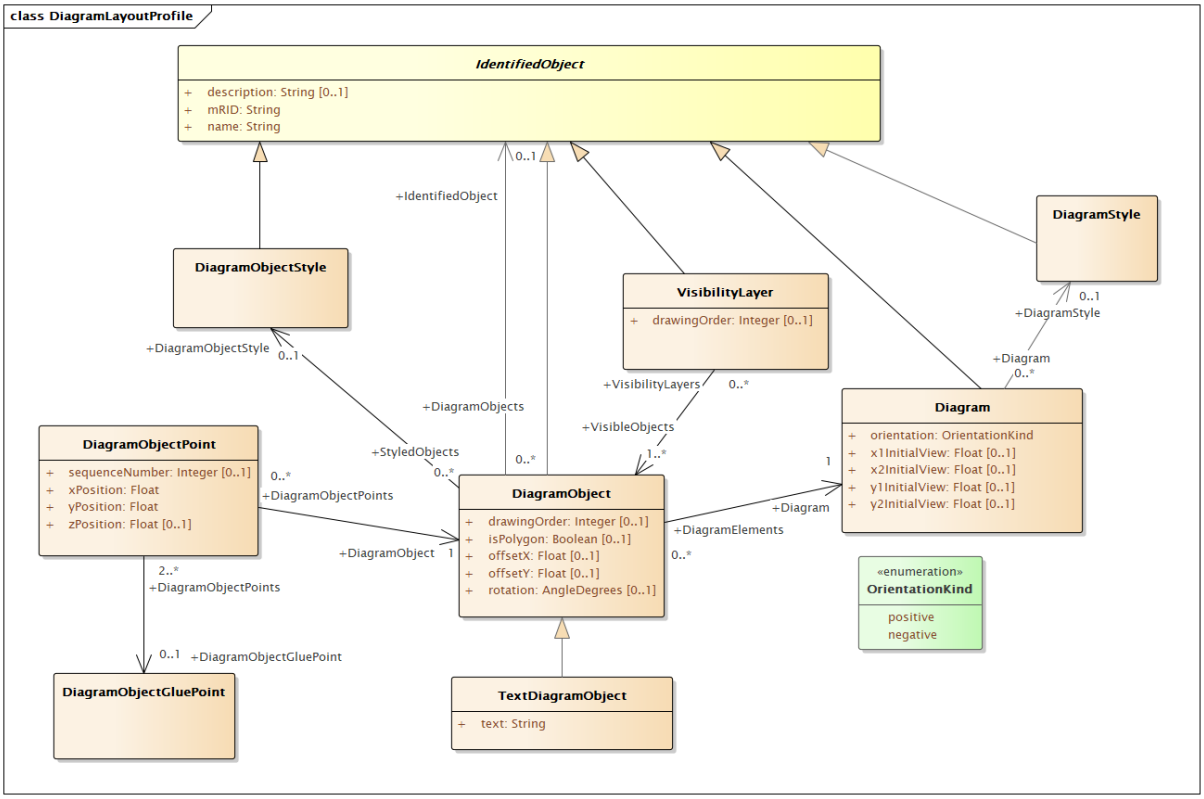

Figure 42 – Class diagram DiagramLayoutProfile::DiagramLayoutProfile

Figure 42: The diagram shows Diagram Layout Profile.

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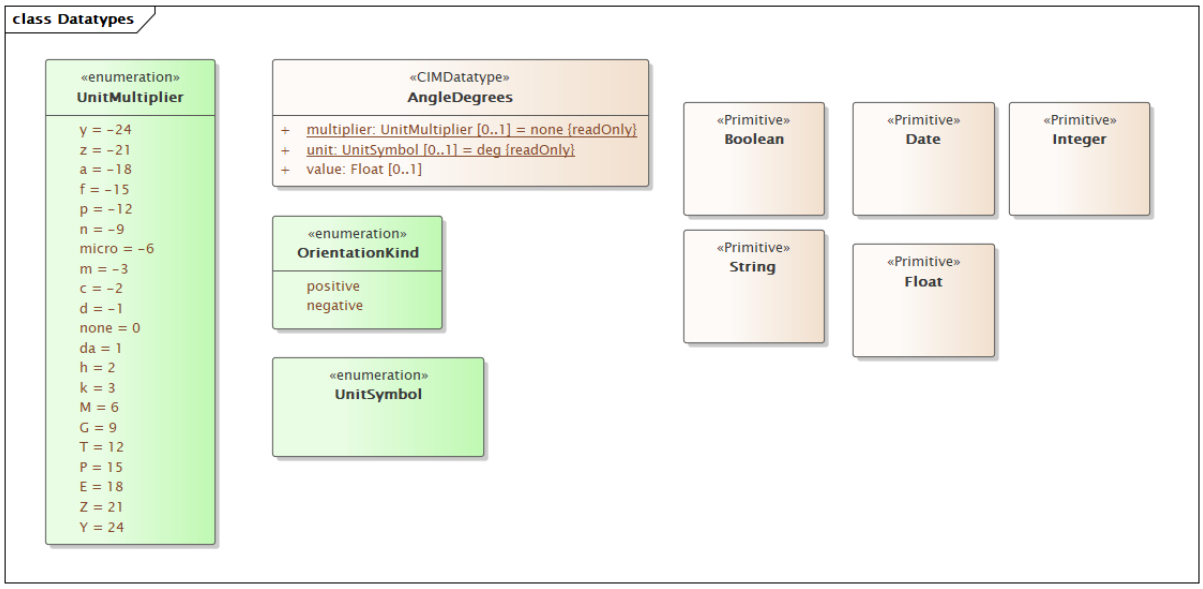

Figure 43 – Class diagram DiagramLayoutProfile::Datatypes

Figure 43: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

12.2 Diagram

Inheritance path = IdentifiedObject

The diagram being exchanged. The coordinate system is a standard Cartesian coordinate system and the orientation attribute defines the orientation. The initial view related attributes can be used to specify an initial view with the x,y coordinates of the diagonal points.

Table 574 shows all attributes of Diagram.

Table 574 – Attributes of DiagramLayoutProfile::Diagram

name	mult	type	description
orientation	1..1	OrientationKind	Coordinate system orientation of the diagram. A positive orientation gives standard “right-hand” orientation, with negative orientation indicating a “left-hand” orientation. For 2D diagrams, a positive orientation will result in X values increasing from left to right and Y values increasing from bottom to top. A negative orientation gives the “left-hand” orientation (favoured by computer graphics displays) with X values increasing from left to right and Y values increasing from top to bottom.

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name	mult	type	description
x1InitialView	0..1	Float	X coordinate of the first corner of the initial view.
x2InitialView	0..1	Float	X coordinate of the second corner of the initial view.
y1InitialView	0..1	Float	Y coordinate of the first corner of the initial view.
y2InitialView	0..1	Float	Y coordinate of the second corner of the initial view.
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
description	0..1	String	inherited from: IdentifiedObject

Table 575 shows all association ends of Diagram with other classes.

Table 575 – Association ends of DiagramLayoutProfile::Diagram with other classes

mult from	name	mult to	type	description
0..*	DiagramStyle	0..1	DiagramStyle	A Diagram may have a DiagramStyle.

12.3 DiagramObject

Inheritance path = IdentifiedObject

An object that defines one or more points in a given space. This object can be associated with anything that specializes IdentifiedObject. For single line diagrams such objects typically include such items as analog values, breakers, disconnectors, power transformers, and transmission lines.

Table 576 shows all attributes of DiagramObject.

Table 576 – Attributes of DiagramLayoutProfile::DiagramObject

name	mult	type	description
drawingOrder	0..1	Integer	The drawing order of this element. The higher the number, the later the element is drawn in sequence. This is used to ensure that elements that overlap are rendered in the correct order.
isPolygon	0..1	Boolean	Defines whether or not the diagram objects points define the boundaries of a polygon or the routing of a polyline. If this value is true then a receiving application should consider the first and last points to be connected.
offsetX	0..1	Float	The offset in the X direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the horizontal centre of the icon. -0,5 indicates it is offset by 50 % to the left and 0,5 indicates an offset of 50 % to the right.

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name	mult	type	description
offsetY	0..1	Float	The offset in the Y direction. This is used for defining the offset from centre for rendering an icon (the default is that a single point specifies the centre of the icon). The offset is in per-unit with 0 indicating there is no offset from the vertical centre of the icon. The offset direction is dependent on the orientation of the diagram, with -0,5 and 0,5 indicating an offset of +/- 50 % on the vertical axis.
rotation	0..1	AngleDegrees	Sets the angle of rotation of the diagram object. Zero degrees is pointing to the top of the diagram. Rotation is clockwise. DiagramObject.rotation=0 has the following meaning: The connection point of an element which has one terminal is pointing to the top side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the top side of the diagram. DiagramObject.rotation=90 has the following meaning: The connection point of an element which has one terminal is pointing to the right hand side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the right hand side of the diagram.
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
description	0..1	String	inherited from: IdentifiedObject

Table 577 shows all association ends of DiagramObject with other classes.

Table 577 – Association ends of DiagramLayoutProfile:: DiagramObject with other classes

mult from	name	mult to	type	description
0..*	Diagram	1..1	Diagram	A diagram object is part of a diagram.
0..*	DiagramObjectStyle	0..1	DiagramObjectStyle	A diagram object has a style associated that provides a reference for the style used in the originating system.
0..*	IdentifiedObject	0..1	IdentifiedObject	The domain object to which this diagram object is associated.

12.4 DiagramObjectGluePoint root class

This is used for grouping diagram object points from different diagram objects that are considered to be glued together in a diagram even if they are not at the exact same coordinates.

12.5 DiagramObjectPoint root class

A point in a given space defined by 3 coordinates and associated to a diagram object. The coordinates may be positive or negative as the origin does not have to be in the corner of a diagram.

Table 578 shows all attributes of DiagramObjectPoint.

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Table 578 – Attributes of DiagramLayoutProfile::DiagramObjectPoint

name	mult	type	description
sequenceNumber	0..1	Integer	The sequence position of the point, used for defining the order of points for diagram objects acting as a polyline or polygon with more than one point. The attribute shall be a positive value.
xPosition	1..1	Float	The X coordinate of this point.
yPosition	1..1	Float	The Y coordinate of this point.
zPosition	0..1	Float	The Z coordinate of this point.

Table 579 shows all association ends of DiagramObjectPoint with other classes.

Table 579 – Association ends of DiagramLayoutProfile:: DiagramObjectPoint with other classes

mult from	name	mult to	type	description
0..*	DiagramObject	1..1	DiagramObject	The diagram object with which the points are associated.
2..*	DiagramObjectGluePoint	0..1	DiagramObjectGluePoint	The 'glue' point to which this point is associated.

12.6 DiagramObjectStyle

Inheritance path = IdentifiedObject

A reference to a style used by the originating system for a diagram object. A diagram object style describes information such as line thickness, shape such as circle or rectangle etc, and colour.

Table 580 shows all attributes of DiagramObjectStyle.

Table 580 – Attributes of DiagramLayoutProfile::DiagramObjectStyle

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
description	0..1	String	inherited from: IdentifiedObject

12.7 DiagramStyle

Inheritance path = IdentifiedObject

The diagram style refers to a style used by the originating system for a diagram. A diagram style describes information such as schematic, geographic, etc.

Table 581 shows all attributes of DiagramStyle.

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Table 581 – Attributes of DiagramLayoutProfile::DiagramStyle

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
description	0..1	String	inherited from: IdentifiedObject

12.8 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 582 shows all attributes of IdentifiedObject.

Table 582 – Attributes of DiagramLayoutProfile::IdentifiedObject

name	mult	type	description
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

12.9 TextDiagramObject

Inheritance path = DiagramObject: IdentifiedObject

A diagram object for placing free-text or text derived from an associated domain object.

Table 583 shows all attributes of TextDiagramObject.

Table 583 – Attributes of DiagramLayoutProfile::TextDiagramObject

name	mult	type	description
text	1..1	String	The text that is displayed by this text diagram object.
drawingOrder	0..1	Integer	inherited from: DiagramObject
isPolygon	0..1	Boolean	inherited from: DiagramObject
offsetX	0..1	Float	inherited from: DiagramObject
offsetY	0..1	Float	inherited from: DiagramObject
rotation	0..1	AngleDegrees	inherited from: DiagramObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject

Table 584 shows all association ends of TextDiagramObject with other classes.

Table 584 – Association ends of DiagramLayoutProfile:: TextDiagramObject with other classes

mult from	name	mult to	type	description
0..*	Diagram	1..1	Diagram	inherited from: DiagramObject
0..*	DiagramObjectStyle	0..1	DiagramObjectStyle	inherited from: DiagramObject
0..*	IdentifiedObject	0..1	IdentifiedObject	inherited from: DiagramObject

12.10 VisibilityLayer

Inheritance path = IdentifiedObject

Layers are typically used for grouping diagram objects according to themes and scales. Themes are used to display or hide certain information (e.g., lakes, borders), while scales are used for hiding or displaying information depending on the current zoom level (hide text when it is too small to be read, or when it exceeds the screen size). This is also called de-cluttering.

CIM based graphics exchange supports an m:n relationship between diagram objects and layers. The importing system shall convert an m:n case into an appropriate 1:n representation if the importing system does not support m:n.

Table 585 shows all attributes of VisibilityLayer.

Table 585 – Attributes of DiagramLayoutProfile::VisibilityLayer

name	mult	type	description
drawingOrder	0..1	Integer	The drawing order for this layer. The higher the number, the later the layer and the objects within it are rendered.
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
description	0..1	String	inherited from: IdentifiedObject

Table 586 shows all association ends of VisibilityLayer with other classes.

Table 586 – Association ends of DiagramLayoutProfile:: VisibilityLayer with other classes

mult from	name	mult to	type	description
0..*	VisibleObjects	1..*	DiagramObject	A visibility layer can contain one or more diagram objects.

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12.11 OrientationKind enumeration

The orientation of the coordinate system with respect to top, left, and the coordinate number system.

Table 587 shows all literals of OrientationKind.

Table 587 – Literals of DiagramLayoutProfile::OrientationKind

literal	value	description
positive		For 2D diagrams, a positive orientation will result in X values increasing from left to right and Y values increasing from bottom to top. This is also known as a right hand orientation.
negative		For 2D diagrams, a negative orientation gives the left-hand orientation (favoured by computer graphics displays) with X values increasing from left to right and Y values increasing from top to bottom. This is also known as a left hand orientation.

12.12 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 588 shows all literals of UnitMultiplier.

Table 588 – Literals of DiagramLayoutProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.

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literal	value	description
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

12.13 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

Table 589 shows all literals of UnitSymbol.

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Table 589 – Literals of DiagramLayoutProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A × s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C × V = W·s).
N	32	Force in newtons (kg × m/s^2).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd × sr).
Wb	36	Magnetic flux in webers (V × s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).

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literal	value	description
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).

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literal	value	description
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg \cdot m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3 600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1 000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µg/g’.
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m^3.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.

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literal	value	description
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.

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literal	value	description
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133,3 Pa).
M	217	Length, nautical miles (1 M = 1 852 m).
kn	219	Speed, knots (1 kn = 1 852/3 600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

12.14 AngleDegrees datatype

Measurement of angle in degrees.

Table 590 shows all attributes of AngleDegrees.

Table 590 – Attributes of DiagramLayoutProfile::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

12.15 Boolean primitive

A type with the value space "true" and "false".

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12.16 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

12.17 Float primitive

A floating point number. The range is unspecified and not limited.

12.18 Integer primitive

An integer number. The range is unspecified and not limited.

12.19 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

13 Package GeographicalLocationProfile

13.1 General

This is the geographical location profile as in 61968-13.

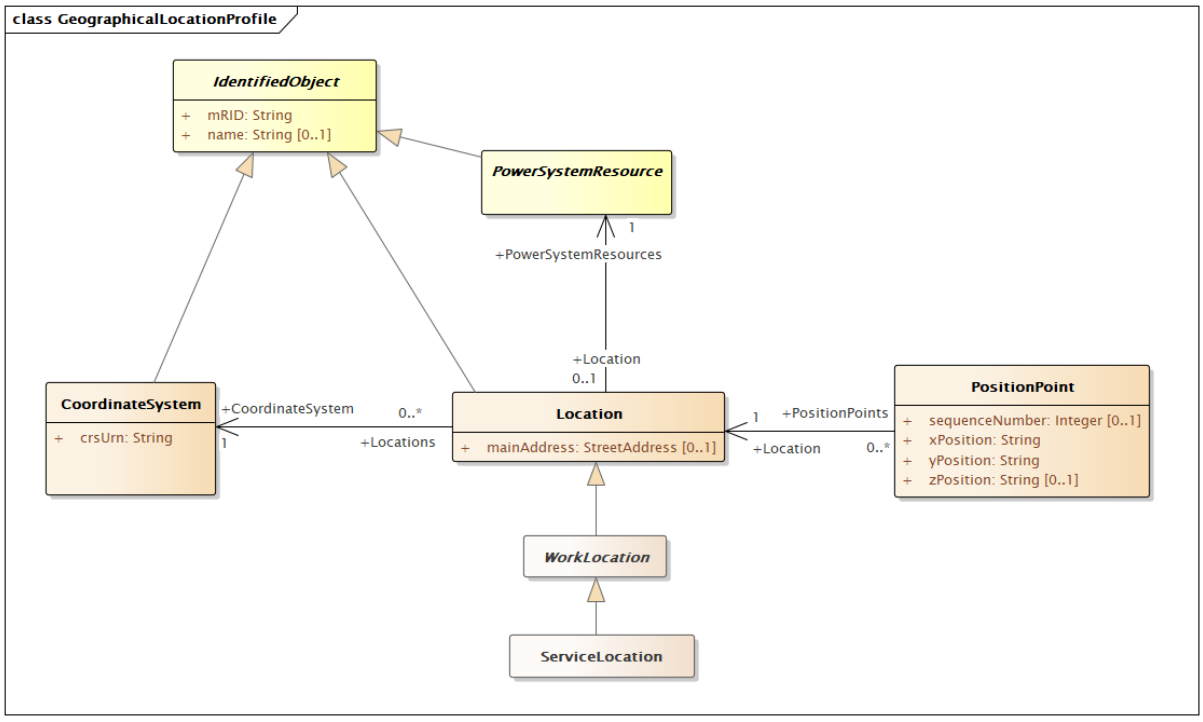

Figure 44 – Class diagram GeographicalLocationProfile::GeographicalLocationProfile

Figure 44: The diagram shows geographical location profile.

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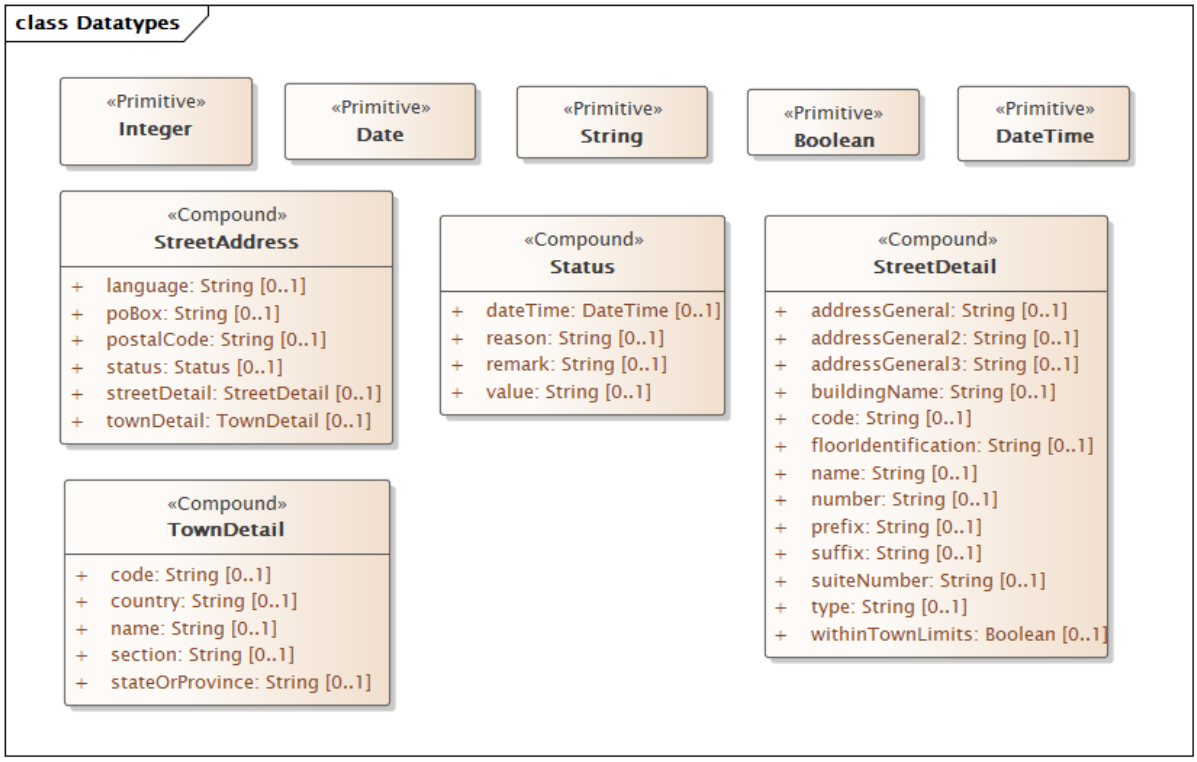

Figure 45 – Class diagram GeographicalLocationProfile::Datatypes

Figure 45: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

13.2 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 591 shows all attributes of IdentifiedObject.

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Table 591 – Attributes of GeographicalLocationProfile::IdentifiedObject

name	mult	type	description
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

13.3 ServiceLocation

Inheritance path = WorkLocation: Location: IdentifiedObject

A real estate location, commonly referred to as premises.

Table 592 shows all attributes of ServiceLocation.

Table 592 – Attributes of GeographicalLocationProfile::ServiceLocation

name	mult	type	description
mainAddress	0..1	StreetAddress	inherited from: Location
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 593 shows all association ends of ServiceLocation with other classes.

Table 593 – Association ends of GeographicalLocationProfile:: ServiceLocation with other classes

mult from	name	mult to	type	description
0..*	CoordinateSystem	1..1	CoordinateSystem	inherited from: Location
0..1	PowerSystemResources	1..1	PowerSystemResource	inherited from: Location

13.4 (abstract) WorkLocation

Inheritance path = Location: IdentifiedObject

Information about a particular location for various forms of work.

Table 594 shows all attributes of WorkLocation.

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Table 594 – Attributes of GeographicalLocationProfile::WorkLocation

name	mult	type	description
mainAddress	0..1	StreetAddress	inherited from: Location
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 595 shows all association ends of WorkLocation with other classes.

Table 595 – Association ends of GeographicalLocationProfile:: WorkLocation with other classes

mult from	name	mult to	type	description
0..*	CoordinateSystem	1..1	CoordinateSystem	inherited from: Location
0..1	PowerSystemResources	1..1	PowerSystemResource	inherited from: Location

13.5 CoordinateSystem

Inheritance path = IdentifiedObject

Coordinate reference system.

Table 596 shows all attributes of CoordinateSystem.

Table 596 – Attributes of GeographicalLocationProfile::CoordinateSystem

name	mult	type	description
crsUrn	1..1	String	A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:crs:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:crs:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties.
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

13.6 Location

Inheritance path = IdentifiedObject

The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time. It can be defined with one or more position points (coordinates) in a given coordinate system.

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Table 597 shows all attributes of Location.

Table 597 – Attributes of GeographicalLocationProfile::Location

name	mult	type	description
mainAddress	0..1	StreetAddress	Main address of the location.
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 598 shows all association ends of Location with other classes.

Table 598 – Association ends of GeographicalLocationProfile:: Location with other classes

mult from	name	mult to	type	description
0..*	CoordinateSystem	1..1	CoordinateSystem	Coordinate system used to describe position points of this location.
0..1	PowerSystemResources	1..1	PowerSystemResource	All power system resources at this location.

13.7 PositionPoint root class

Set of spatial coordinates that determine a point, defined in the coordinate system specified in 'Location.CoordinateSystem'. Use a single position point instance to describe a point-oriented location. Use a sequence of position points to describe a line-oriented object (physical location of non-point oriented objects like cables or lines), or area of an object (like a substation or a geographical zone – in this case, have first and last position point with the same values).

Table 599 shows all attributes of PositionPoint.

Table 599 – Attributes of GeographicalLocationProfile::PositionPoint

name	mult	type	description
sequenceNumber	0..1	Integer	Zero-relative sequence number of this point within a series of points.
xPosition	1..1	String	X axis position.
yPosition	1..1	String	Y axis position.
zPosition	0..1	String	(if applicable) Z axis position.

Table 600 shows all association ends of PositionPoint with other classes.

Table 600 – Association ends of GeographicalLocationProfile:: PositionPoint with other classes

mult from	name	mult to	type	description
0..*	Location	1..1	Location	Location described by this position point.

13.8 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

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A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 601 shows all attributes of PowerSystemResource.

Table 601 – Attributes of GeographicalLocationProfile::PowerSystemResource

name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

13.9 StreetAddress compound

General purpose street and postal address information.

Table 602 shows all attributes of StreetAddress.

Table 602 – Attributes of GeographicalLocationProfile::StreetAddress

name	mult	type	description
streetDetail	0..1	StreetDetail	Street detail.
townDetail	0..1	TownDetail	Town detail.
status	0..1	Status	Status of this address.
postalCode	0..1	String	Postal code for the address.
poBox	0..1	String	Post office box.
language	0..1	String	The language in which the address is specified, using ISO 639-1 two digit language code.

13.10 TownDetail compound

Town details, in the context of address.

Table 603 shows all attributes of TownDetail.

Table 603 – Attributes of GeographicalLocationProfile::TownDetail

name	mult	type	description
code	0..1	String	Town code.
section	0..1	String	Town section. For example, it is common for there to be 36 sections per township.
name	0..1	String	Town name.
stateOrProvince	0..1	String	Name of the state or province.
country	0..1	String	Name of the country.

13.11 Status compound

Current status information relevant to an entity.

Table 604 shows all attributes of Status.

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Table 604 – Attributes of GeographicalLocationProfile::Status

name	mult	type	description
value	0..1	String	Status value at 'dateTime'; prior status changes may have been kept in instances of activity records associated with the object to which this status applies.
dateTime	0..1	DateTime	Date and time for which status 'value' applies.
remark	0..1	String	Pertinent information regarding the current 'value', as free form text.
reason	0..1	String	Reason code or explanation for why an object went to the current status 'value'.

13.12 StreetDetail compound

Street details, in the context of address.

Table 605 shows all attributes of StreetDetail.

Table 605 – Attributes of GeographicalLocationProfile::StreetDetail

name	mult	type	description
number	0..1	String	Designator of the specific location on the street.
name	0..1	String	Name of the street.
suffix	0..1	String	Suffix to the street name. For example: North, South, East, West.
prefix	0..1	String	Prefix to the street name. For example: North, South, East, West.
type	0..1	String	Type of street. Examples include: street, circle, boulevard, avenue, road, drive, etc.
code	0..1	String	(if applicable) Utilities often make use of external reference systems, such as those of the town-planner's department or surveyor general's mapping system, that allocate global reference codes to streets.
buildingName	0..1	String	(if applicable) In certain cases the physical location of the place of interest does not have a direct point of entry from the street, but may be located inside a larger structure such as a building, complex, office block, apartment, etc.
suiteNumber	0..1	String	Number of the apartment or suite.
addressGeneral	0..1	String	First line of a free form address or some additional address information (for example a mail stop).
addressGeneral2	0..1	String	(if applicable) Second line of a free form address.
addressGeneral3	0..1	String	(if applicable) Third line of a free form address.
withinTownLimits	0..1	Boolean	True if this street is within the legal geographical boundaries of the specified town (default).
floorIdentification	0..1	String	The identification by name or number, expressed as text, of the floor in the building as part of this address.

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13.13 DateTime primitive

Date and time as "yyyy-mm-ddThh:mm:ss.sss", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddThh:mm:ss.sssZ". A local timezone relative UTC is specified as "yyyy-mm-ddThh:mm:ss.sss-hh:mm". The second component (shown here as "ss.sss") could have any number of digits in its fractional part to allow any kind of precision beyond seconds.

13.14 Boolean primitive

A type with the value space "true" and "false".

13.15 Integer primitive

An integer number. The range is unspecified and not limited.

13.16 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

13.17 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

14 Package DynamicsProfile

14.1 General

This is the IEC 61970-457 profile.

14.2 Package Base

14.2.1 General

Content of the base CIM published as IEC 61970-301.

14.2.2 Package Domain

14.2.2.1 General

The domain package defines primitive datatypes that are used by classes in other packages. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

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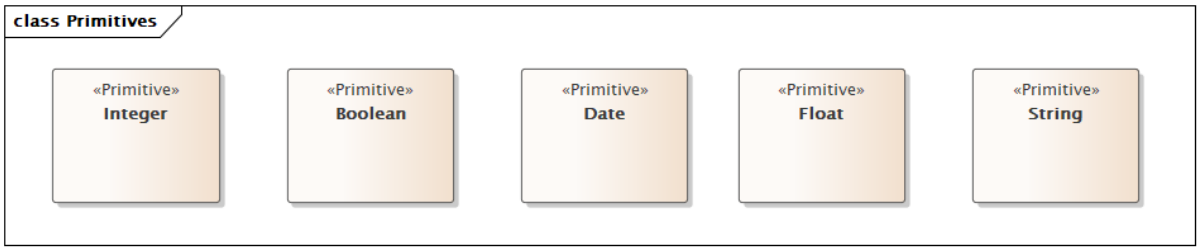

Figure 46 – Class diagram Domain::Primitives

Figure 46: This diagram lists the primitives defined in the CIM and used in this document.

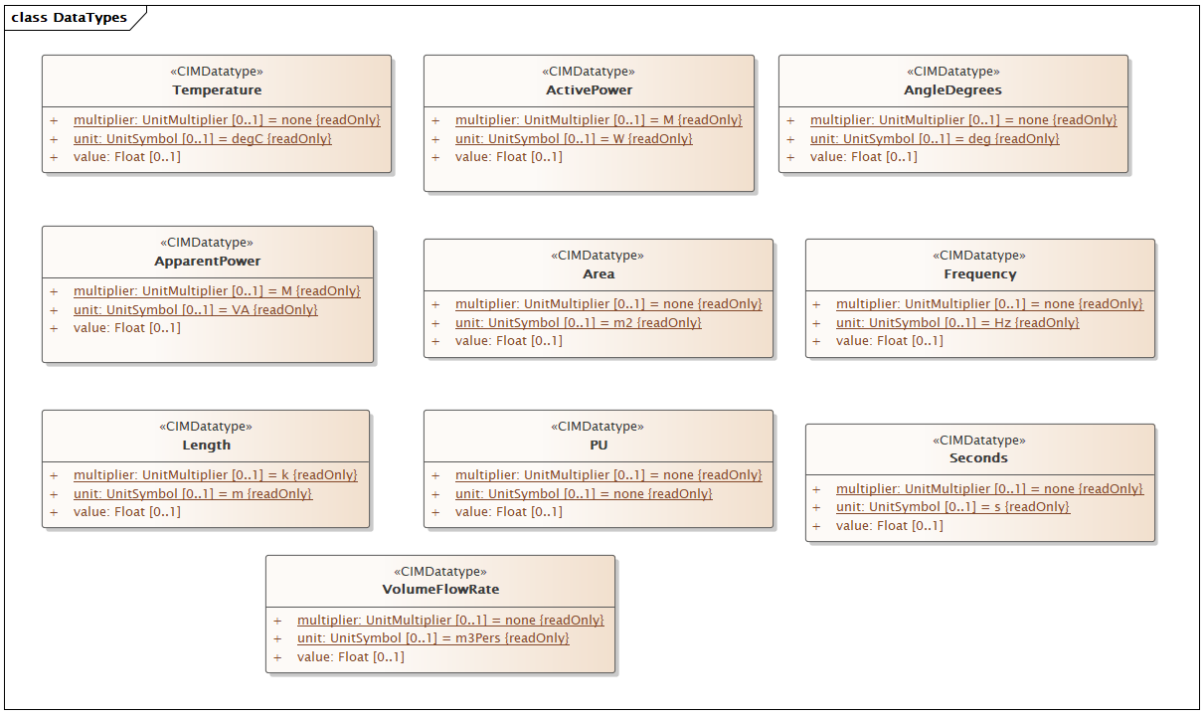

Figure 47 – Class diagram Domain::DataTypes

Figure 47: This diagram illustrates the data types supported by the CIM and used in this document.

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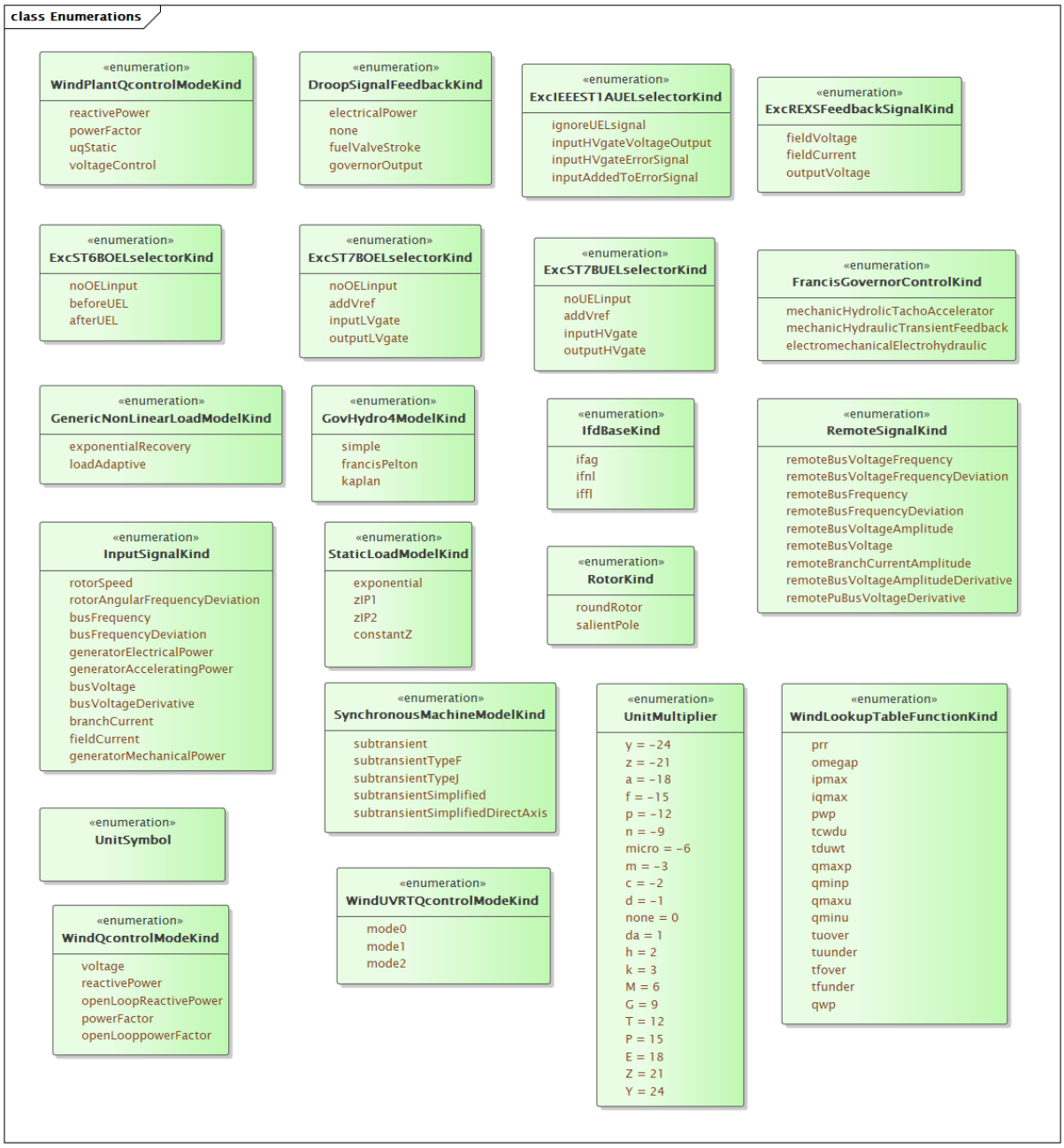

Figure 48 – Class diagram Domain::Enumerations

Figure 48: This diagram provides information on the enumerations defined in the CIM and used in this document.

14.2.2.2 ActivePower datatype

Product of RMS value of the voltage and the RMS value of the in-phase component of the current.

Table 606 shows all attributes of ActivePower.

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Table 606 – Attributes of Domain::ActivePower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=W)

14.2.2.3 AngleDegrees datatype

Measurement of angle in degrees.

Table 607 shows all attributes of AngleDegrees.

Table 607 – Attributes of Domain::AngleDegrees

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=deg)
multiplier	0..1	UnitMultiplier	(const=none)

14.2.2.4 ApparentPower datatype

Product of the RMS value of the voltage and the RMS value of the current.

Table 608 shows all attributes of ApparentPower.

Table 608 – Attributes of Domain::ApparentPower

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=M)
unit	0..1	UnitSymbol	(const=VA)

14.2.2.5 Area datatype

Area.

Table 609 shows all attributes of Area.

Table 609 – Attributes of Domain::Area

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=m2)
multiplier	0..1	UnitMultiplier	(const=none)

14.2.2.6 Frequency datatype

Cycles per second.

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Table 610 shows all attributes of Frequency.

Table 610 – Attributes of Domain::Frequency

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=Hz)
multiplier	0..1	UnitMultiplier	(const=none)

14.2.2.7 Length datatype

Unit of length. It shall be a positive value or zero.

Table 611 shows all attributes of Length.

Table 611 – Attributes of Domain::Length

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=m)
multiplier	0..1	UnitMultiplier	(const=k)

14.2.2.8 PU datatype

Per Unit – a positive or negative value referred to a defined base. Values typically range from -10 to +10.

Table 612 shows all attributes of PU.

Table 612 – Attributes of Domain::PU

name	mult	type	description
value	0..1	Float	
unit	0..1	UnitSymbol	(const=none)
multiplier	0..1	UnitMultiplier	(const=none)

14.2.2.9 Seconds datatype

Time, in seconds.

Table 613 shows all attributes of Seconds.

Table 613 – Attributes of Domain::Seconds

name	mult	type	description
value	0..1	Float	Time, in seconds
unit	0..1	UnitSymbol	(const=s)
multiplier	0..1	UnitMultiplier	(const=none)

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14.2.2.10 Temperature datatype

Value of temperature in degrees Celsius.

Table 614 shows all attributes of Temperature.

Table 614 – Attributes of Domain::Temperature

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=degC)
value	0..1	Float	

14.2.2.11 VolumeFlowRate datatype

Volume per time.

Table 615 shows all attributes of VolumeFlowRate.

Table 615 – Attributes of Domain::VolumeFlowRate

name	mult	type	description
multiplier	0..1	UnitMultiplier	(const=none)
unit	0..1	UnitSymbol	(const=m3Pers)
value	0..1	Float	

14.2.2.12 Boolean primitive

A type with the value space "true" and "false".

14.2.2.13 Float primitive

A floating point number. The range is unspecified and not limited.

14.2.2.14 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

14.2.2.15 Integer primitive

An integer number. The range is unspecified and not limited.

14.2.2.16 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

14.2.2.17 DroopSignalFeedbackKind enumeration

Governor droop signal feedback source.

Table 616 shows all literals of DroopSignalFeedbackKind.

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Table 616 – Literals of Domain::DroopSignalFeedbackKind

literal	value	description
electricalPower		Electrical power feedback (connection indicated as 1 in the block diagrams of models, e.g. GovCT1, GovCT2).
none		No droop signal feedback, is isochronous governor.
fuelValveStroke		Fuel valve stroke feedback (true stroke) (connection indicated as 2 in the block diagrams of model, e.g. GovCT1, GovCT2).
governorOutput		Governor output feedback (requested stroke) (connection indicated as 3 in the block diagrams of models, e.g. GovCT1, GovCT2).

14.2.2.18 ExcIEEEST1AUELselectorKind enumeration

Types of connections for the UEL input used in ExcIEEEST1A.

Table 617 shows all literals of ExcIEEEST1AUELselectorKind.

Table 617 – Literals of Domain::ExcIEEEST1AUELselectorKind

literal	value	description
ignoreUELsignal		Ignore UEL signal.
inputHVgateVoltageOutput		UEL input HV gate with voltage regulator output.
inputHVgateErrorSignal		UEL input HV gate with error signal.
inputAddedToErrorSignal		UEL input added to error signal.

14.2.2.19 ExcREXSFeedbackSignalKind enumeration

Types of rate feedback signals.

Table 618 shows all literals of ExcREXSFeedbackSignalKind.

Table 618 – Literals of Domain::ExcREXSFeedbackSignalKind

literal	value	description
fieldVoltage		The voltage regulator output voltage is used. It is the same as exciter field voltage.
fieldCurrent		The exciter field current is used.
outputVoltage		The output voltage of the exciter is used.

14.2.2.20 ExcST6BOELselectorKind enumeration

Types of connections for the OEL input used for static excitation systems type 6B.

Table 619 shows all literals of ExcST6BOELselectorKind.

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Table 619 – Literals of Domain::ExcST6BOELselectorKind

literal	value	description
noOELinput		No OEL input is used. Corresponds to OELin not = 1 and not = 2 on the ExcST6B diagram. Original ExcST6B model would have called this OELin = 0.
beforeUEL		The connection is before UEL. Corresponds to OELin = 1 on the ExcST6B diagram.
afterUEL		The connection is after UEL. Corresponds to OELin = 2 on the ExcST6B diagram.

14.2.2.21 ExcST7BOELselectorKind enumeration

Types of connections for the OEL input used for static excitation systems type 7B.

Table 620 shows all literals of ExcST7BOELselectorKind.

Table 620 – Literals of Domain::ExcST7BOELselectorKind

literal	value	description
noOELinput		No OEL input is used. Corresponds to OELin not = 1 and not = 2 and not = 3 on the ExcST7B diagram. Original ExcST7B model would have called this OELin = 0.
addVref		The signal is added to Vref. Corresponds to OELin = 1 on the ExcST7B diagram.
inputLVgate		The signal is connected into the input LVGate. Corresponds to OELin = 2 on the ExcST7B diagram.
outputLVgate		The signal is connected into the output LVGate. Corresponds to OELin = 3 on the ExcST7B diagram.

14.2.2.22 ExcST7BUELselectorKind enumeration

Types of connections for the UEL input used for static excitation systems type 7B.

Table 621 shows all literals of ExcST7BUELselectorKind.

Table 621 – Literals of Domain::ExcST7BUELselectorKind

literal	value	description
noUELinput		No UEL input is used. Corresponds to UELin not = 1 and not = 2 and not = 3 on the ExcST7B diagram. Original ExcST7B model would have called this UELin = 0.
addVref		The signal is added to Vref. Corresponds to UELin = 1 on the ExcST7B diagram.
inputHVgate		The signal is connected into the input HVGate. Corresponds to UELin = 2 on the ExcST7B diagram.
outputHVgate		The signal is connected into the output HVGate. Corresponds to UELin = 3 on the ExcST7B diagram.

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14.2.2.23 FrancisGovernorControlKind enumeration

Governor control flag for Francis hydro model.

Table 622 shows all literals of FrancisGovernorControlKind.

Table 622 – Literals of Domain::FrancisGovernorControlKind

literal	value	description
mechanicHydrolicTachoAccelerator		Mechanic-hydraulic regulator with tacho-accelerometer (Cflag = 1).
mechanicHydraulicTransientFeedback		Mechanic-hydraulic regulator with transient feedback (Cflag=2).
electromechanicalElectrohydraulic		Electromechanical and electrohydraulic regulator (Cflag=3).

14.2.2.24 GenericNonLinearLoadModelKind enumeration

Type of generic non-linear load model.

Table 623 shows all literals of GenericNonLinearLoadModelKind.

Table 623 – Literals of Domain::GenericNonLinearLoadModelKind

literal	value	description
exponentialRecovery		Exponential recovery model.
loadAdaptive		Load adaptive model.

14.2.2.25 GovHydro4ModelKind enumeration

Possible types of GovHydro4 models.

Table 624 shows all literals of GovHydro4ModelKind.

Table 624 – Literals of Domain::GovHydro4ModelKind

literal	value	description
simple		Simple model.
francisPelton		Francis or Pelton model.
kaplan		Kaplan model.

14.2.2.26 IfdBaseKind enumeration

Excitation base system mode.

Table 625 shows all literals of IfdBaseKind.

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Table 625 – Literals of Domain::IfdBaseKind

literal	value	description
ifag		Air gap line mode.
ifnl		No load system with saturation mode.
iffl		Full load system mode.

14.2.2.27 InputSignalKind enumeration

Types of input signals. In dynamics modelling, commonly represented by the j parameter.

Table 626 shows all literals of InputSignalKind.

Table 626 – Literals of Domain::InputSignalKind

literal	value	description
rotorSpeed		Input signal is rotor or shaft speed (angular frequency).
rotorAngularFrequencyDeviation		Input signal is rotor or shaft angular frequency deviation.
busFrequency		Input signal is bus voltage frequency. This could be a terminal frequency or remote frequency.
busFrequencyDeviation		Input signal is deviation of bus voltage frequency. This could be a terminal frequency deviation or remote frequency deviation.
generatorElectricalPower		Input signal is generator electrical power on rated S.
generatorAcceleratingPower		Input signal is generator accelerating power.
busVoltage		Input signal is bus voltage. This could be a terminal voltage or remote voltage.
busVoltageDerivative		Input signal is derivative of bus voltage. This could be a terminal voltage derivative or remote voltage derivative.
branchCurrent		Input signal is amplitude of remote branch current.
fieldCurrent		Input signal is generator field current.
generatorMechanicalPower		Input signal is generator mechanical power.

14.2.2.28 RemoteSignalKind enumeration

Type of input signal coming from remote bus.

Table 627 shows all literals of RemoteSignalKind.

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Table 627 – Literals of Domain::RemoteSignalKind

literal	value	description
remoteBusVoltageFrequency		Input is voltage frequency from remote terminal bus.
remoteBusVoltageFrequencyDeviation		Input is voltage frequency deviation from remote terminal bus.
remoteBusFrequency		Input is frequency from remote terminal bus.
remoteBusFrequencyDeviation		Input is frequency deviation from remote terminal bus.
remoteBusVoltageAmplitude		Input is voltage amplitude from remote terminal bus.
remoteBusVoltage		Input is voltage from remote terminal bus.
remoteBranchCurrentAmplitude		Input is branch current amplitude from remote terminal bus.
remoteBusVoltageAmplitudeDerivative		Input is branch current amplitude derivative from remote terminal bus.
remotePuBusVoltageDerivative		Input is PU voltage derivative from remote terminal bus.

14.2.2.29 RotorKind enumeration

Type of rotor on physical machine.

Table 628 shows all literals of RotorKind.

Table 628 – Literals of Domain::RotorKind

literal	value	description
roundRotor		Round rotor type of synchronous machine.
salientPole		Salient pole type of synchronous machine.

14.2.2.30 StaticLoadModelKind enumeration

Type of static load model.

Table 629 shows all literals of StaticLoadModelKind.

Table 629 – Literals of Domain::StaticLoadModelKind

literal	value	description
exponential		This model is an exponential representation of the load. Exponential equations for active and reactive power are used and the following attributes are required: kp1, kp2, kp3, kpf, ep1, ep2, ep3 kq1, kq2, kq3, kqf, eq1, eq2, eq3.
zIP1		This model integrates the frequency-dependent load (primarily motors). ZIP1 equations for active and reactive power are used and the following attributes are required: kp1, kp2, kp3, kpf kq1, kq2, kq3, kqf.

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literal	value	description
zIP2		This model separates the frequency-dependent load (primarily motors) from other load. ZIP2 equations for active and reactive power are used and the following attributes are required: kp1, kp2, kp3, kq4, kpf kq1, kq2, kq3, kq4, kqf.
constantZ		The load is represented as a constant impedance. ConstantZ equations are used for active and reactive power and no attributes are required.

14.2.2.31 SynchronousMachineModelKind enumeration

Type of synchronous machine model used in dynamic simulation applications.

Table 630 shows all literals of SynchronousMachineModelKind.

Table 630 – Literals of Domain::SynchronousMachineModelKind

literal	value	description
subtransient		Subtransient synchronous machine model.
subtransientTypeF		WECC type F variant of subtransient synchronous machine model.
subtransientTypeJ		WECC type J variant of subtransient synchronous machine model.
subtransientSimplified		Simplified version of subtransient synchronous machine model where magnetic coupling between the direct- and quadrature- axes is ignored.
subtransientSimplifiedDirectAxis		Simplified version of a subtransient synchronous machine model with no damper circuit on the direct-axis.

14.2.2.32 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 631 shows all literals of UnitMultiplier.

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Table 631 – Literals of Domain::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

14.2.2.33 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

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The integer values are used for harmonization with IEC 61850.

Table 632 shows all literals of UnitSymbol.

Table 632 – Literals of Domain::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A × s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N × m = C × V = W × s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd × sr).
Wb	36	Magnetic flux in webers (V × s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.

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literal	value	description
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).

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literal	value	description
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres ($kg·m^2$) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3 600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1 000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘μ’ to express a quantity such as ‘μg/g’.

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literal	value	description
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.

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literal	value	description
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86 400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1 000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133,3 Pa).
M	217	Length, nautical miles (1 M = 1 852 m).
kn	219	Speed, knots (1 kn = 1 852/3 600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

14.2.2.34 WindLookupTableFunctionKind enumeration

Function of the lookup table.

Table 633 shows all literals of WindLookupTableFunctionKind.

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Table 633 – Literals of Domain::WindLookupTableFunctionKind

literal	value	description
prr		Power versus speed change (negative slip) lookup table (prr(deltaomega)). It is used for the rotor resistance control model, IEC 61400-27-1:2015, 5.6.5.3.
omegap		Power vs. speed lookup table (omega(p)). It is used for the P control model type 3, IEC 61400-27-1:2015, 5.6.5.4.
ipmax		Lookup table for voltage dependency of active current limits (ipmax(uWT)). It is used for the current limitation model, IEC 61400-27-1:2015, 5.6.5.8.
iqmax		Lookup table for voltage dependency of reactive current limits (iqmax(uWT)). It is used for the current limitation model, IEC 61400-27-1:2015, 5.6.5.8.
pwp		Power vs. frequency lookup table (pWPbias(f)). It is used for the wind power plant frequency and active power control model, IEC 61400-27-1:2015, Annex D.
tcwdu		Crowbar duration versus voltage variation look-up table (TCW(du)). It is a case-dependent parameter. It is used for the type 3B generator set model, IEC 61400-27-1:2015, 5.6.3.3.
tduwt		Lookup table to determine the duration of the power reduction after a voltage dip, depending on the size of the voltage dip (Td(uWT)). It is a type-dependent parameter. It is used for the pitch control power model, IEC 61400-27-1:2015, 5.6.5.1.
qmaxp		Lookup table for active power dependency of reactive power maximum limit (qmaxp(p)). It is used for the QP and QU limitation model, IEC 61400-27-1:2015, 5.6.5.10.
qminp		Lookup table for active power dependency of reactive power minimum limit (qminp(p)). It is used for the QP and QU limitation model, IEC 61400-27-1:2015, 5.6.5.10.
qmaxu		Lookup table for voltage dependency of reactive power maximum limit (qmaxu(p)). It is used for the QP and QU limitation model, IEC 61400-27-1:2015, 5.6.5.10.
qminu		Lookup table for voltage dependency of reactive power minimum limit (qminu(p)). It is used for the QP and QU limitation model, IEC 61400-27-1:2015, 5.6.5.10.
tuover		Disconnection time versus over-voltage lookup table (Tuover(uWT)). It is used for the grid protection model, IEC 61400-27-1:2015, 5.6.6.
tuunder		Disconnection time versus under-voltage lookup table (Tuunder(uWT)). It is used for the grid protection model, IEC 61400-27-1:2015, 5.6.6.
tfover		Disconnection time versus over-frequency lookup table (Tfover(fWT)). It is used for the grid protection model, IEC 61400-27-1:2015, 5.6.6.
tfunder		Disconnection time versus under-frequency lookup table (Tfunder(fWT)). It is used for the grid protection model, IEC 61400-27-1:2015, 5.6.6.
qwp		Look up table for the UQ static mode (qWP(uerr)). It is used for the voltage and reactive power control model, IEC 61400-27-1:2015, Annex D.

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14.2.2.35 WindPlantQcontrolModeKind enumeration

Reactive power/voltage controller mode.

Table 634 shows all literals of WindPlantQcontrolModeKind.

Table 634 – Literals of Domain::WindPlantQcontrolModeKind

literal	value	description
reactivePower		Reactive power reference.
powerFactor		Power factor reference.
uqStatic		UQ static.
voltageControl		Voltage control.

14.2.2.36 WindQcontrolModeKind enumeration

General wind turbine Q control modes MqG.

Table 635 shows all literals of WindQcontrolModeKind.

Table 635 – Literals of Domain::WindQcontrolModeKind

literal	value	description
voltage		Voltage control (MqG equals 0).
reactivePower		Reactive power control (MqG equals 1).
openLoopReactivePower		Open loop reactive power control (only used with closed loop at plant level) (MqG equals 2).
powerFactor		Power factor control (MqG equals 3).
openLooppowerFactor		Open loop power factor control (MqG equals 4).

14.2.2.37 WindUVRTQcontrolModeKind enumeration

UVRT Q control modes MqUVRT.

Table 636 shows all literals of WindUVRTQcontrolModeKind.

Table 636 – Literals of Domain::WindUVRTQcontrolModeKind

literal	value	description
mode0		Voltage-dependent reactive current injection (MqUVRT equals 0).
mode1		Reactive current injection controlled as the pre-fault value plus an additional voltage dependent reactive current injection (MqUVRT equals 1).
mode2		Reactive current injection controlled as the pre-fault value plus an additional voltage-dependent reactive current injection during fault, and as the pre-fault value plus an additional constant reactive current injection post fault (MqUVRT equals 2).

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14.2.3 Package Core

14.2.3.1 General

Contains the core PowerSystemResource and ConductingEquipment entities shared by all applications plus common collections of those entities. Not all applications require all the Core entities. This package does not depend on any other package except the Domain package, but most of the other packages have associations and generalizations that depend on it.

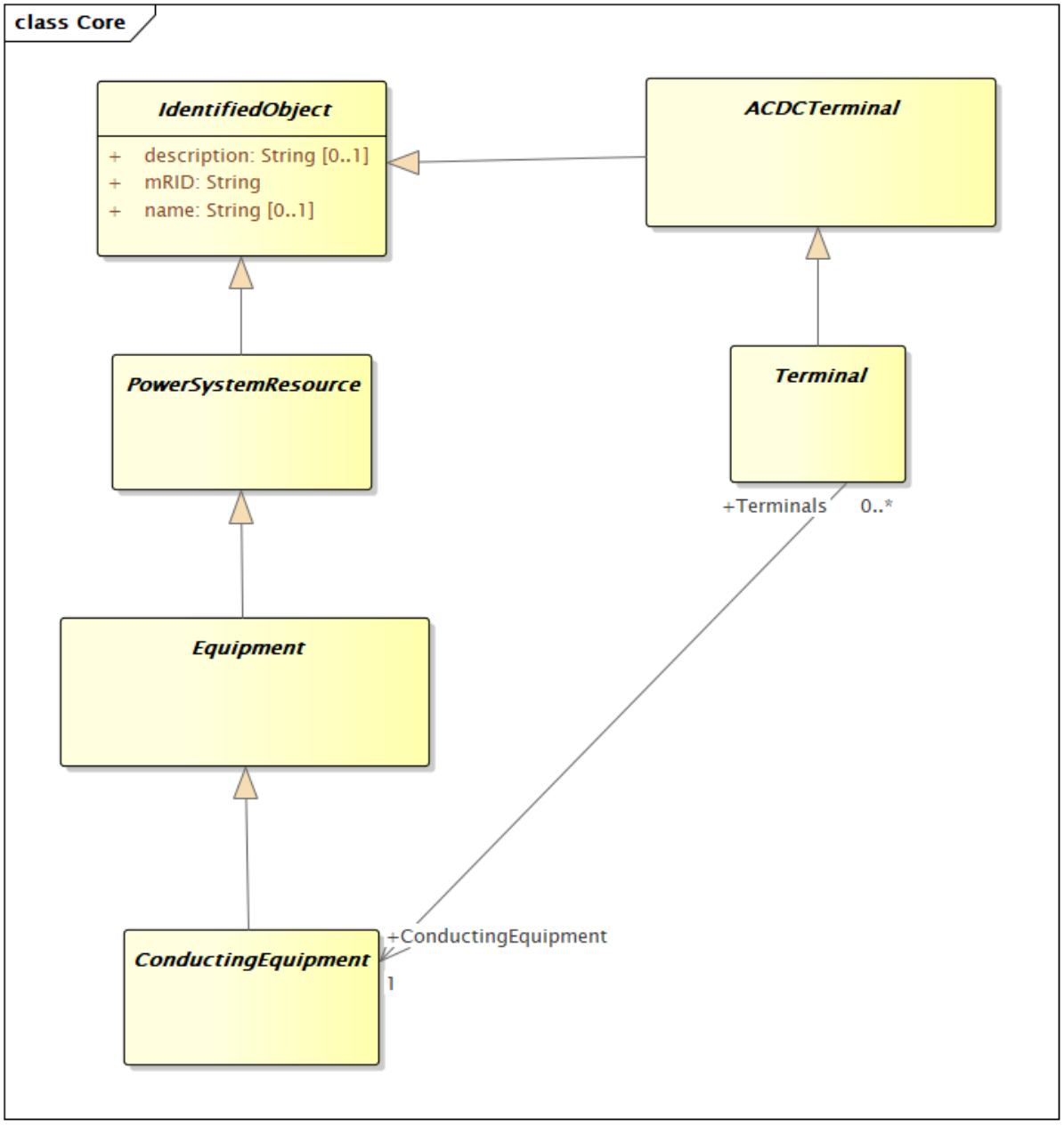

Figure 49 – Class diagram Core::Core

Figure 49: This diagram shows the main classes which are exchanged in other profiles, but have connection with the dynamics profile.

14.2.3.2 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

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An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 637 shows all attributes of ACDCTerminal.

Table 637 – Attributes of Core::ACDCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.3.3 (abstract) ConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Table 638 shows all attributes of ConductingEquipment.

Table 638 – Attributes of Core::ConductingEquipment

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.3.4 (abstract) Equipment

Inheritance path = PowerSystemResource: IdentifiedObject

The parts of a power system that are physical devices, electronic or mechanical.

Table 639 shows all attributes of Equipment.

Table 639 – Attributes of Core::Equipment

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.3.5 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 640 shows all attributes of IdentifiedObject.

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Table 640 – Attributes of Core::IdentifiedObject

name	mult	type	description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	0..1	String	The name is any free human readable and possibly non unique text naming the object.

14.2.3.6 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 641 shows all attributes of PowerSystemResource.

Table 641 – Attributes of Core::PowerSystemResource

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.3.7 (abstract) Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 642 shows all attributes of Terminal.

Table 642 – Attributes of Core::Terminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 643 shows all association ends of Terminal with other classes.

Table 643 – Association ends of Core::Terminal with other classes

mult from	name	mult to	type	description
0..*	ConductingEquipment	1..1	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be connected to other conducting equipment terminals via connectivity nodes or topological nodes.

14.2.4 Package Wires

14.2.4.1 General

An extension to the Core and Topology package that models information on the electrical characteristics of Transmission and Distribution networks. This package is used by network applications such as State Estimation, Load Flow and Optimal Power Flow.

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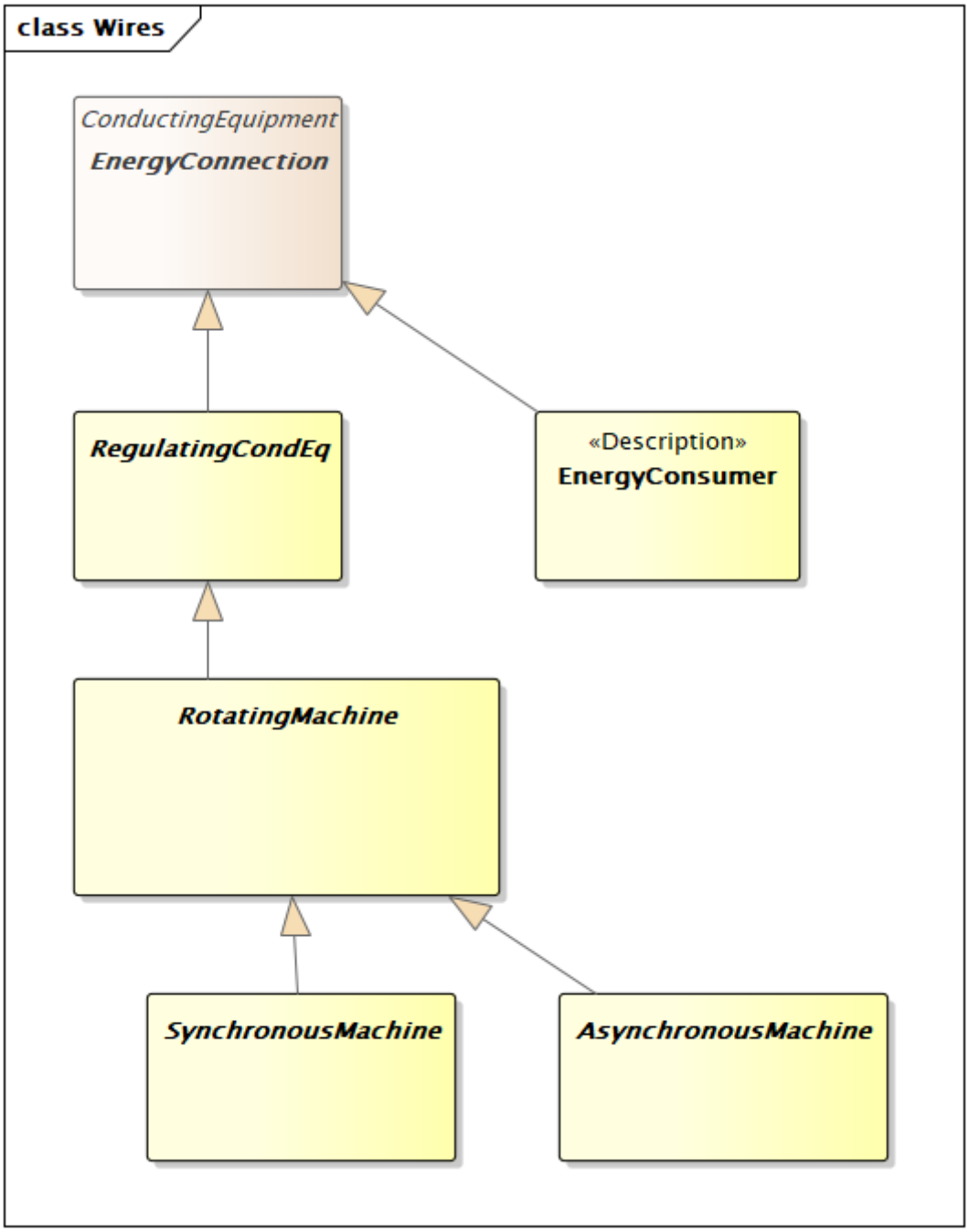

Figure 50 – Class diagram Wires::Wires

Figure 50: This diagram shows the main classes which are exchanged in other profiles, but have connection with the dynamics profile.

14.2.4.2 (abstract) AsynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine whose shaft rotates asynchronously with the electrical field. Also known as an induction machine with no external connection to the rotor windings, e.g. squirrel-cage induction machine.

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Table 644 shows all attributes of AsynchronousMachine.

Table 644 – Attributes of Wires::AsynchronousMachine

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.3 (abstract) EnergyConnection

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection of energy generation or consumption on the power system model.

Table 645 shows all attributes of EnergyConnection.

Table 645 – Attributes of Wires::EnergyConnection

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.4 (Description) EnergyConsumer

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

Generic user of energy – a point of consumption on the power system model.

EnergyConsumer.pfixed, .qfixed, .pfixedPct and .qfixedPct have meaning only if there is no LoadResponseCharacteristic associated with EnergyConsumer or if LoadResponseCharacteristic.exponentModel is set to False.

Table 646 shows all attributes of EnergyConsumer.

Table 646 – Attributes of Wires::EnergyConsumer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 647 shows all association ends of EnergyConsumer with other classes.

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Table 647 – Association ends of Wires::EnergyConsumer with other classes

mult from	name	mult to	type	description
0..*	LoadDynamics	0..1	LoadDynamics	Load dynamics model used to describe dynamic behaviour of this energy consumer.

14.2.4.5 (abstract) PowerElectronicsConnection

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A connection to the AC network for energy production or consumption that uses power electronics rather than rotating machines.

Table 648 shows all attributes of PowerElectronicsConnection.

Table 648 – Attributes of Wires::PowerElectronicsConnection

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.6 (abstract) RegulatingCondEq

Inheritance path = EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A type of conducting equipment that can regulate a quantity (i.e. voltage or flow) at a specific point in the network.

Table 649 shows all attributes of RegulatingCondEq.

Table 649 – Attributes of Wires::RegulatingCondEq

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.7 (abstract) RotatingMachine

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A rotating machine which may be used as a generator or motor.

Table 650 shows all attributes of RotatingMachine.

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Table 650 – Attributes of Wires::RotatingMachine

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.8 (abstract) StaticVarCompensator

Inheritance path = RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A facility for providing variable and controllable shunt reactive power. The SVC typically consists of a stepdown transformer, filter, thyristor-controlled reactor, and thyristor-switched capacitor arms.

The SVC may operate in fixed MVar output mode or in voltage control mode. When in voltage control mode, the output of the SVC will be proportional to the deviation of voltage at the controlled bus from the voltage setpoint. The SVC characteristic slope defines the proportion. If the voltage at the controlled bus is equal to the voltage setpoint, the SVC MVar output is zero.

Table 651 shows all attributes of StaticVarCompensator.

Table 651 – Attributes of Wires::StaticVarCompensator

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.4.9 (abstract) SynchronousMachine

Inheritance path = RotatingMachine: RegulatingCondEq: EnergyConnection: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

An electromechanical device that operates with shaft rotating synchronously with the network. It is a single machine operating either as a generator or synchronous condenser or pump.

Table 652 shows all attributes of SynchronousMachine.

Table 652 – Attributes of Wires::SynchronousMachine

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.5 Package DC

14.2.5.1 General

This package contains model for direct current equipment and controls.

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14.2.5.2 (abstract) ACDCConverter

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A unit with valves for three phases, together with unit control equipment, essential protective and switching devices, DC storage capacitors, phase reactors and auxiliaries, if any, used for conversion.

Table 653 shows all attributes of ACDCConverter.

Table 653 – Attributes of DC::ACDCConverter

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.5.3 (abstract) CsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

DC side of the current source converter (CSC).

The firing angle controls the dc voltage at the converter, both for rectifier and inverter. The difference between the dc voltages of the rectifier and inverter determines the dc current. The extinction angle is used to limit the dc voltage at the inverter, if needed, and is not used in active power control. The firing angle, transformer tap position and number of connected filters are the primary means to control a current source dc line. Higher level controls are built on top, e.g. dc voltage, dc current and active power. From a steady state perspective it is sufficient to specify the wanted active power transfer (ACDCConverter.targetPpcc) and the control functions will set the dc voltage, dc current, firing angle, transformer tap position and number of connected filters to meet this. Therefore attributes targetAlpha and targetGamma are not applicable in this case.

The reactive power consumed by the converter is a function of the firing angle, transformer tap position and number of connected filter, which can be approximated with half of the active power. The losses is a function of the dc voltage and dc current.

The attributes minAlpha and maxAlpha define the range of firing angles for rectifier operation between which no discrete tap changer action takes place. The range is typically 10-18 degrees.

The attributes minGamma and maxGamma define the range of extinction angles for inverter operation between which no discrete tap changer action takes place. The range is typically 17-20 degrees.

Table 654 shows all attributes of CsConverter.

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Table 654 – Attributes of DC::CsConverter

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.2.5.4 (abstract) VsConverter

Inheritance path = ACDCConverter: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

DC side of the voltage source converter (VSC).

Table 655 shows all attributes of VsConverter.

Table 655 – Attributes of DC::VsConverter

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3 Package Dynamics

14.3.1 General

The CIM dynamic model definitions reflect the most common IEEE or, in the case of wind models, IEC, representations of models as well as models included in some of the transient stability software widely used by utilities.

These dynamic models are intended to ensure interoperability between different vendors' software products currently in use by electric utility energy companies, utilities, Transmission System Operators (TSOs), Regional Transmission Organizations (RTOs), and Independent System Operators (ISOs). It is important to note that each vendor is free to select its own internal implementation of these models. Differences in vendor results, as long as they are within accepted engineering practice, caused by different internal representations, are acceptable.

Unless explicitly stated otherwise, the following modelling conventions are followed:

– limited integrators are of the non-windup type;
– it shall be possible to enter a time constant of zero (where it makes sense).

14.3.2 Package StandardInterconnections

14.3.2.1 General

This subclause describes the standard interconnections for various types of equipment. These interconnections are understood by the application programs and can be identified based on the presence of one of the key classes with a relationship to the static power flow model: SynchronousMachineDynamics, AsynchronousMachineDynamics, EnergyConsumerDynamics or WindTurbineType3or4Dynamics.

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The relationships between classes expressed in the interconnection diagrams are intended to support dynamic behaviour described by either standard models or user-defined models.

In the interconnection diagrams, boxes which are black in colour represent function blocks whose functionality can be provided by one of many standard models or by a user-defined model. Blue boxes represent specific standard models. A dashed box means that the function block or specific standard model is optional.

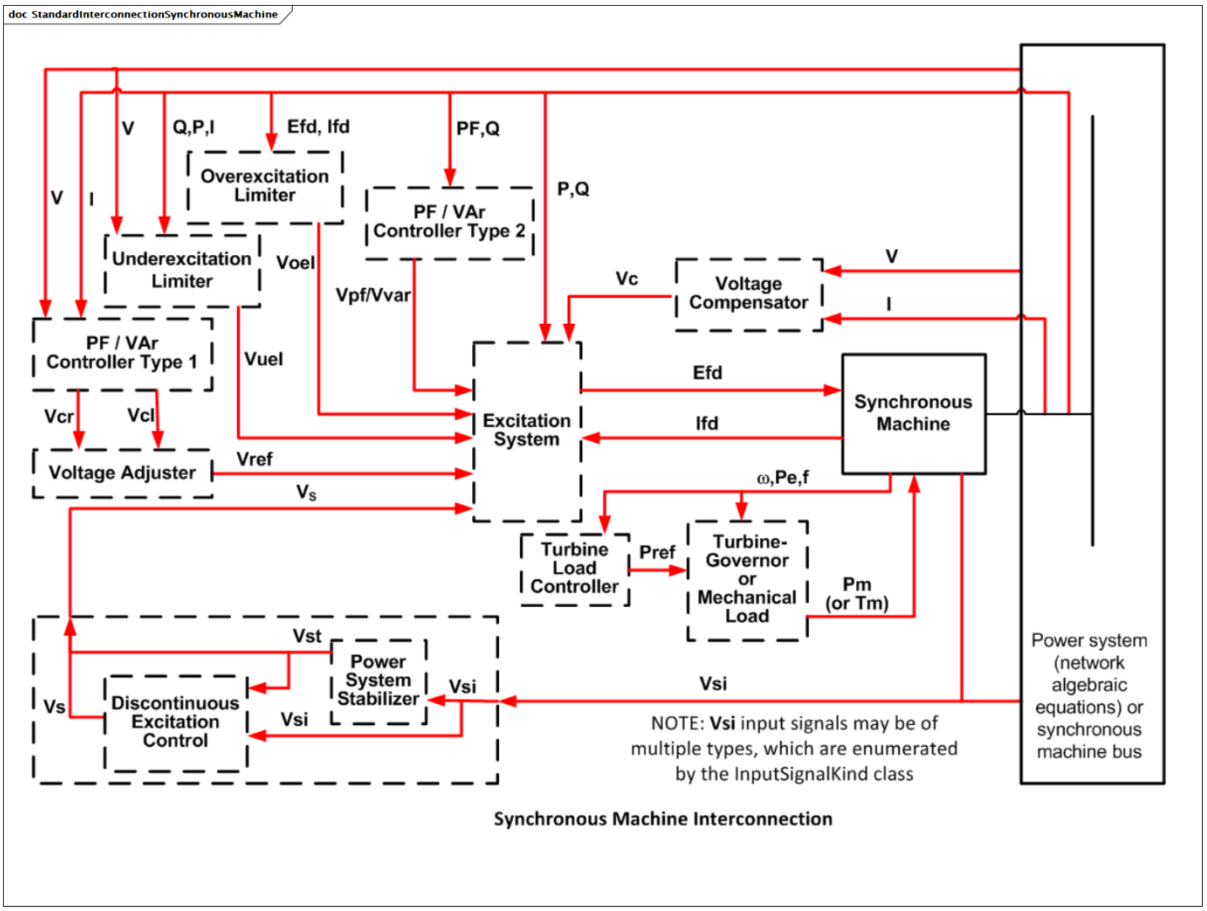

Figure 51 – StandardInterconnectionSynchronousMachine

Figure 51: A dynamic model that includes a single synchronous machine (generator, motor or condenser) and its related equipment models (function blocks) is associated with a synchronous machine in the static (power flow) model and has the standard interconnection shown above.

The interface between the dynamic synchronous machine and the network algebraic equations is application dependent. The variables used for this interface do not need to be identified since they are internal to the application program, well-defined and not necessary to be exchanged between applications.

The interfaces between different blocks in the above figure are indicative of the variety of input and output signals that are defined by the individual standard models. The figure provides information on most of the alternative interfaces, for instance, the turbine-governor (for generators) or the mechanical load (for motors) and the dynamic synchronous machine can be mechanical torque (Tm) or mechanical power (Pm).

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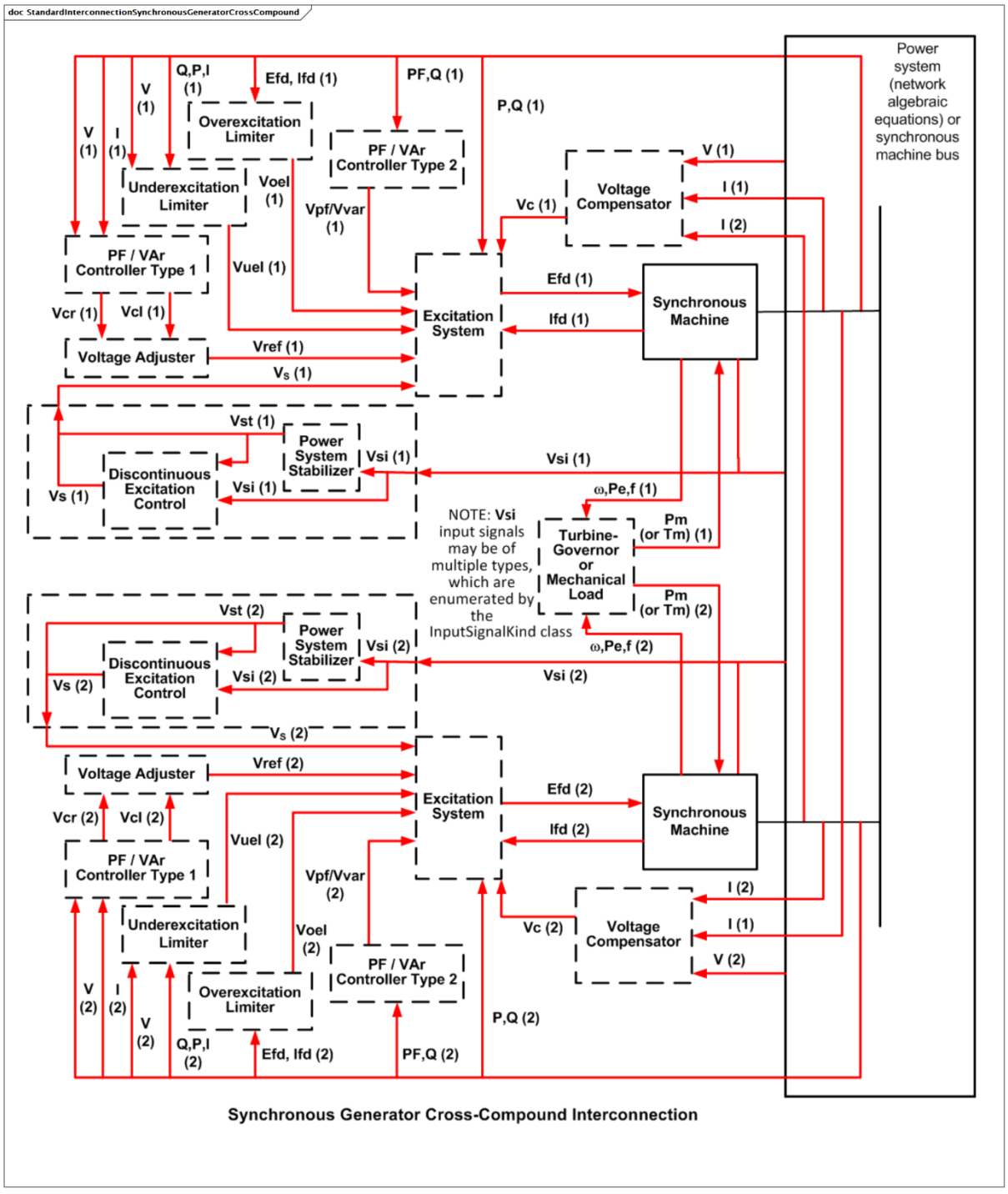

Figure 52 – StandardInterconnectionSynchronousGeneratorCrossCompound

Figure 52: A cross-compound dynamic model that includes multiple synchronous generators and their related equipment models (function blocks), is associated with multiple synchronous machines in the static (power flow) model and has the standard interconnection shown above.

Generator 2 is the second unit of a cross-compound pair of generators and is usually connected to the same terminal bus. A single turbine-governor model determines the mechanical power for both units.

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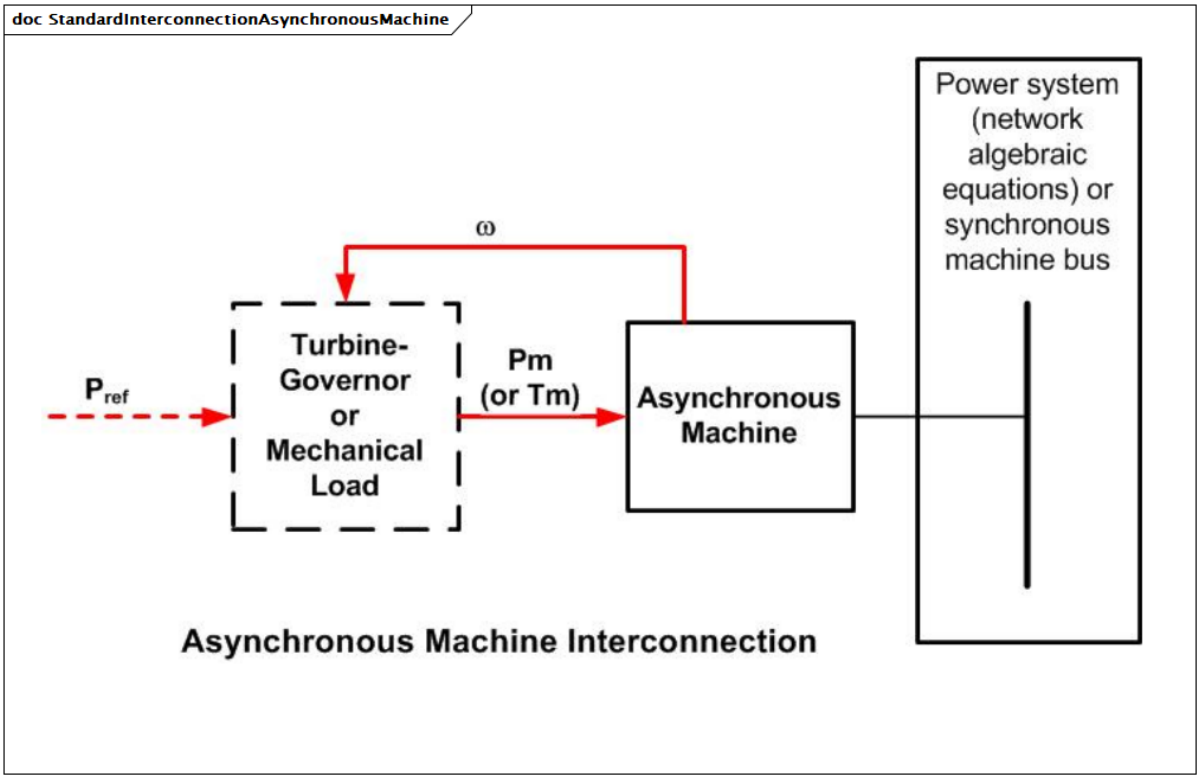

Figure 53 – StandardInterconnectionAsynchronousMachine

Figure 53: A dynamic model including an asynchronous machine (either a generator or motor) and its related equipment models (function blocks) is associated with an asynchronous machine in the static (power flow) model and has the standard interconnection shown above.

This interconnection is for a "squirrel-cage" induction machine or a wound-rotor induction machine with short-circuited field windings. Other models and a different interconnection are required for a wound-rotor machine with external connections to the field windings.

The interface between the dynamic asynchronous machine and the network algebraic equations is application-dependent.

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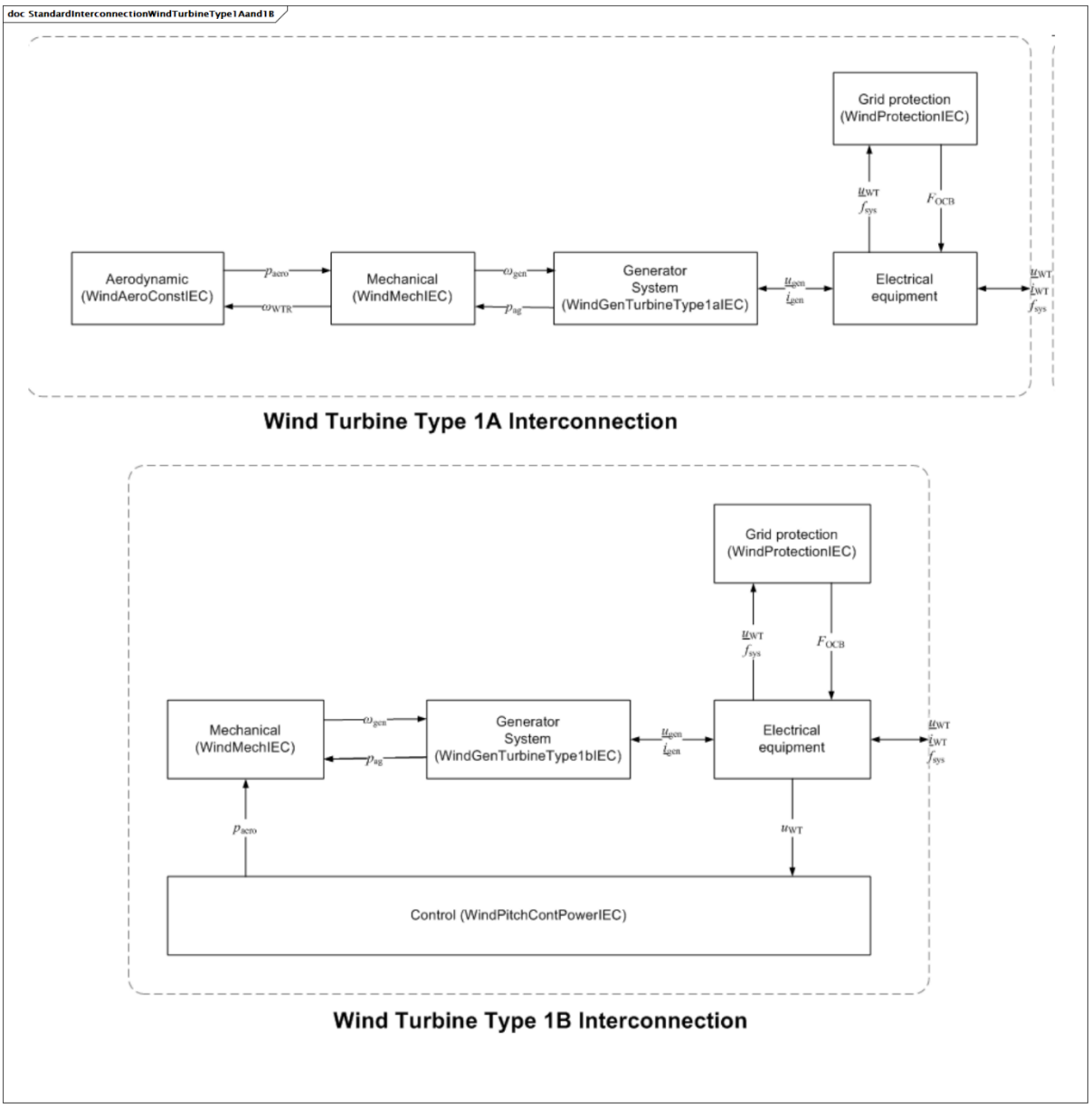

Figure 54 – StandardInterconnectionWindTurbineType1Aand1B

Figure 54: A dynamic model representing a type 1 wind turbine model uses asynchronous generators directly connected to the grid, i.e. without power converters. Most type 1 wind turbines have a soft-starter, but this is only active during start-up. There are two type 1 models:

– type 1A: wind turbines with fixed pitch angle;
– type 1B: wind turbines with under voltage ride-through pitch angle control.

The two standard interconnections are shown in the figure.

Reference: IEC 61400-27-1:2015, 5.5.2.

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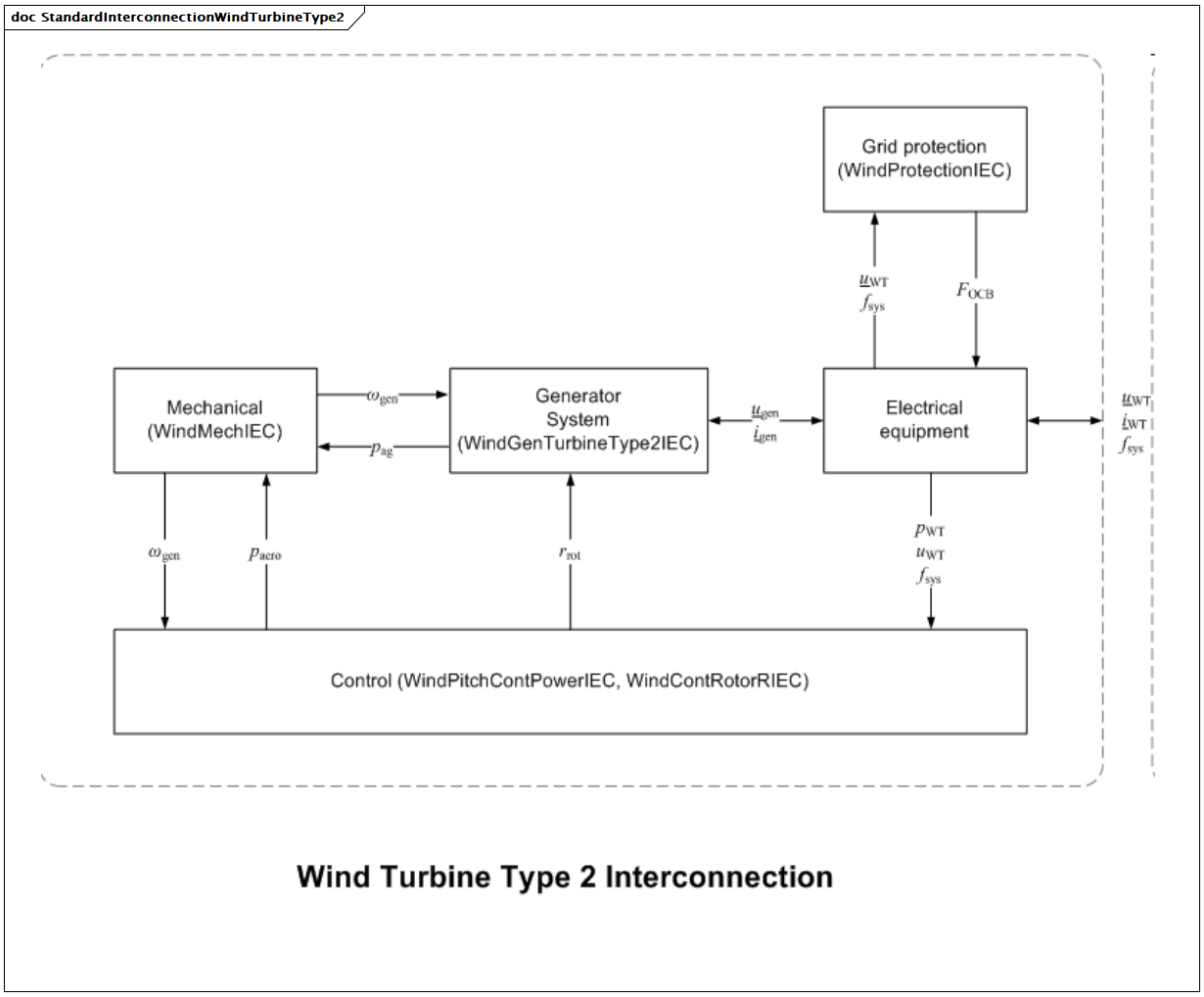

Figure 55 – StandardInterconnectionWindTurbineType2

Figure 55: A dynamic model representing a type 2 wind generator and its related standard equipment models is associated with an asynchronous machine in the static (power flow) model and has the standard interconnection shown in the figure.

The type 2 turbine is equipped with a variable rotor resistance (VRR) and therefore uses a variable rotor resistance asynchronous generator (VRRAG). Type 2 wind turbines are also normally equipped with blade angle control.

Reference: IEC 61400-27-1:2015, 5.5.3.

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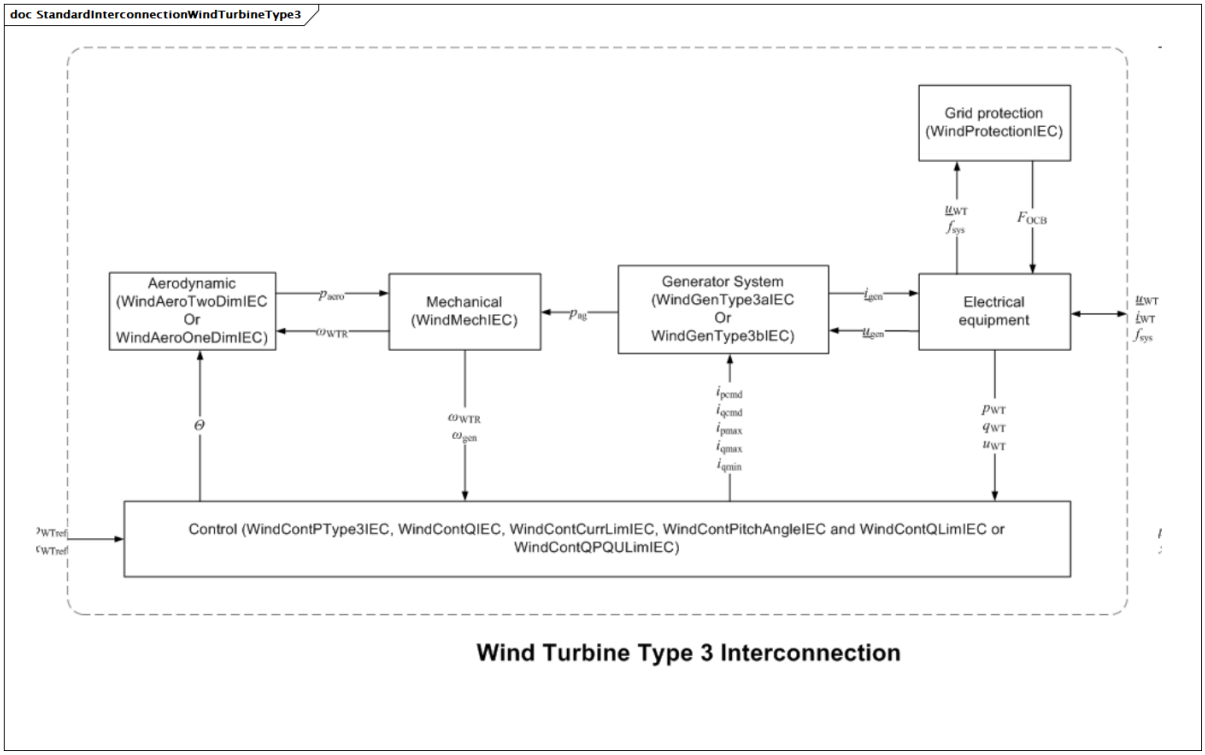

Figure 56 – StandardInterconnectionWindTurbineType3

Figure 56: A dynamic model representing a type 3 wind turbine which uses a doubly fed asynchronous generator (DFAG), where the stator is directly connected to the grid and the rotor is connected through a back-to-back power converter. It has the standard interconnection shown in the figure.

Reference: IEC 61400-27-1:2015, 5.5.4.

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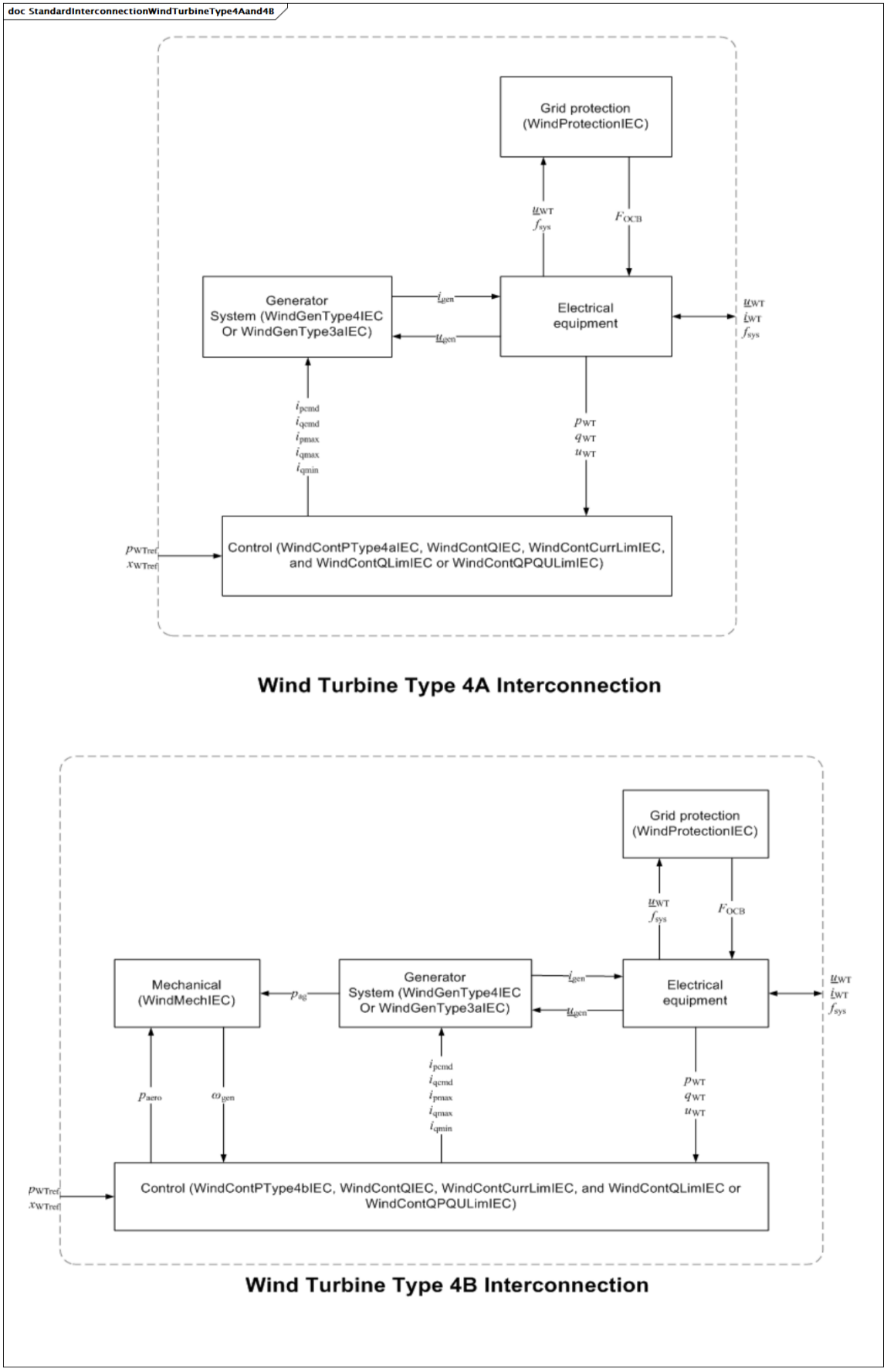

Figure 57 – StandardInterconnectionWindTurbineType4Aand4B

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Figure 57: A dynamic model representing a type 4 wind turbine which uses either synchronous generators (SG) or asynchronous generators (AG). Some type 4 wind turbines use direct drive synchronous generators, and therefore have no gearbox.

There are two type 4 models as follows:

– type 4A: a model neglecting the aerodynamic and mechanical parts and thus not simulating any power oscillations;
– type 4B: a model including a 2-mass mechanical model to replicate the power oscillations but assuming constant aerodynamic torque.

The standard interconnections for the two models are shown in the figure.

Reference: IEC 61400-27-1:2015, 5.5.5.

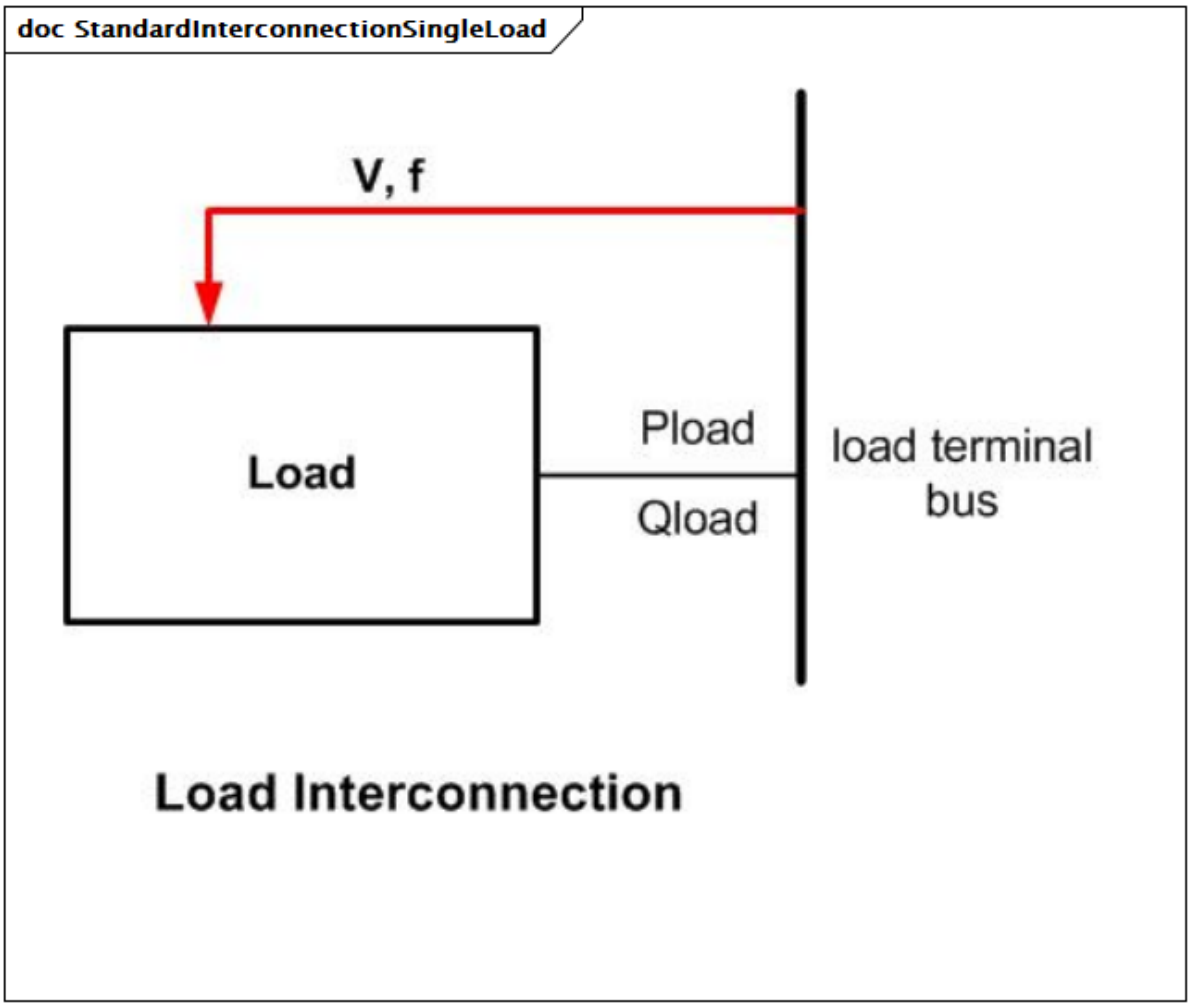

Figure 58 – StandardInterconnectionSingleLoad

Figure 58: A load model is associated with an energy consumer in the power flow data and has no standard interconnection with other models.

The P and Q consumption of the load are determined as a function of the magnitude and frequency of the voltage at the terminal bus as shown above.

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The interface between the model and the network algebraic equations is application-dependent. Dynamic (motor) load models generally interface with the network equations in a similar manner to motor models.

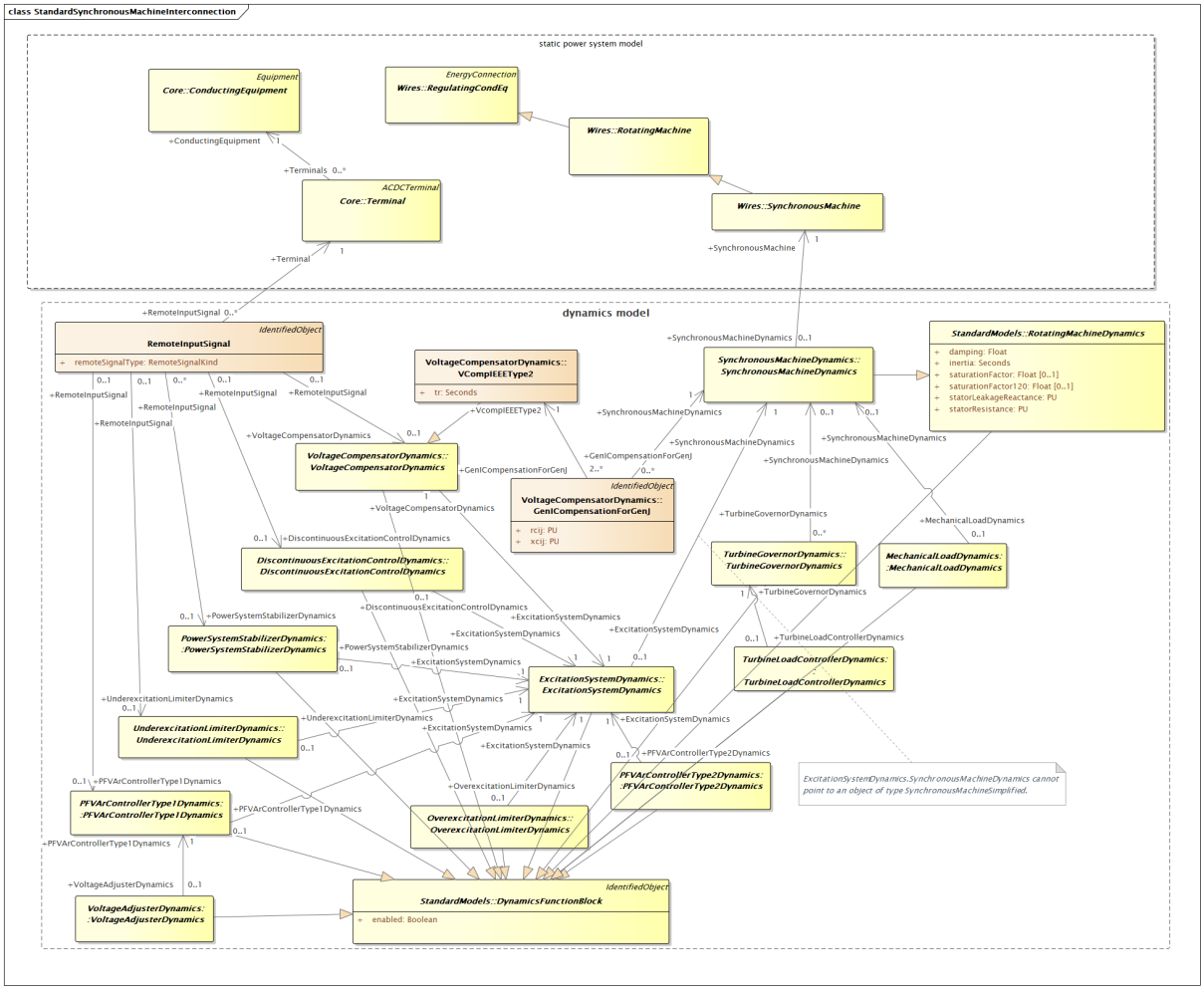

Figure 59 – Class diagram StandardInterconnections:: StandardSynchronousMachineInterconnection

Figure 59: This diagram shows the classes necessary for both of the two types of synchronous machine interconnection models: standard synchronous machine and standard synchronous generator cross-compound.

A standard synchronous machine interconnection has a single instance of a SynchronousMachineDynamics child class whereas a generator cross-compound connection has multiple instances of SynchronousMachineDynamics child classes related by a single instance of a TurbineGovernorDynamics child class.

A valid standard synchronous machine interconnection model reflects the following rules.

– It can have:

  + one instance of TurbineGovernorDynamics child class, but only if SynchronousMachine.operatingMode = generator;

  + one instance of MechanicalLoadDynamics child class, but only if SynchronousMachine.operatingMode = motor or SynchronousMachine.operatingMode = condenser.

– It cannot have any instance of GenICompensationForGenJ.

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A generator cross-compound interconnection model reflects the following rules.

– It can have:

  + multiple instances of SynchronousMachineDynamics child classes, each of which has its related SynchronousMachine.operatingMode = generator;

  + one instance of TurbineGovernorDynamics child class.

– It shall have two or more instances of GenICompensationForGenJ associated with each instance of a SynchronousMachineDynamics child class.

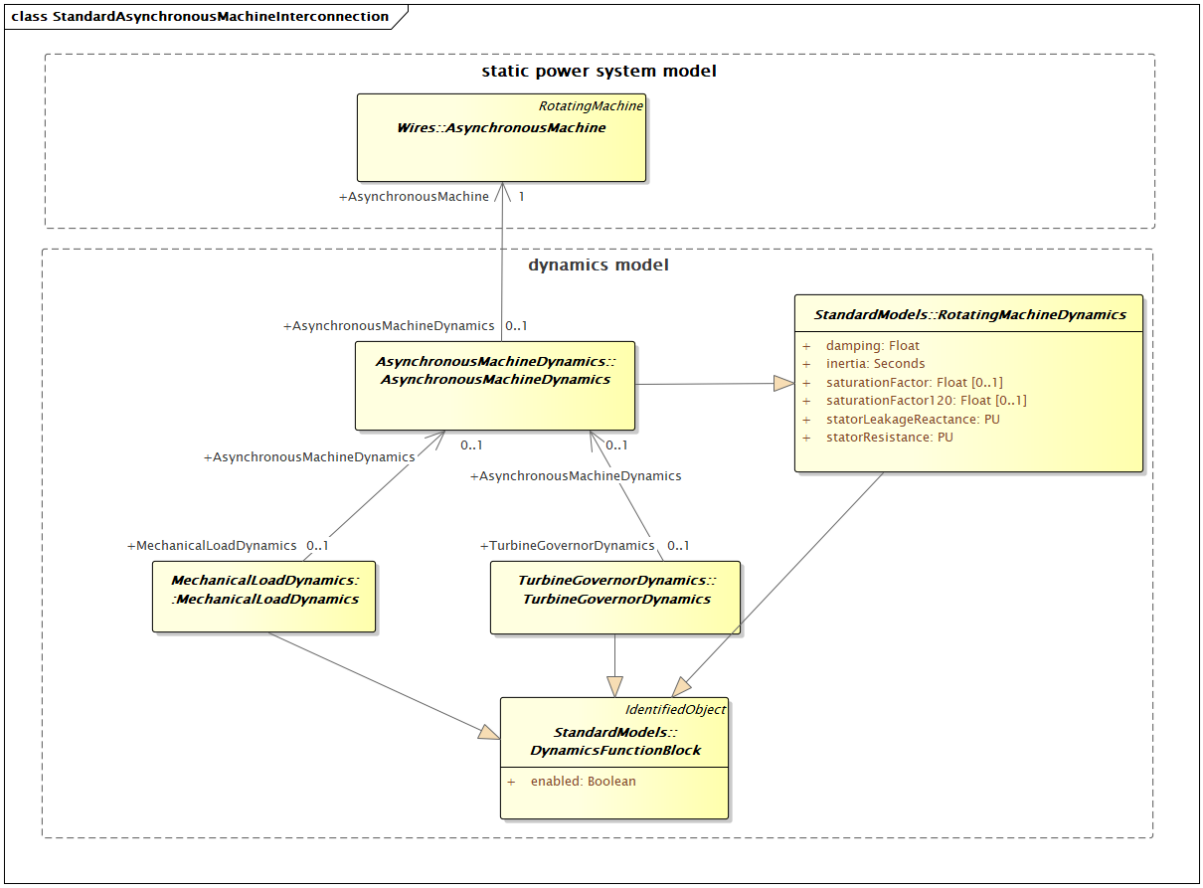

Figure 60 – Class diagram StandardInterconnections:: StandardAsynchronousMachineInterconnection

Figure 60: This diagram shows the classes necessary for the asynchronous machine interconnection model.

A standard asynchronous machine interconnection has a single instance of an AsynchronousMachineDynamics child class.

A valid standard asynchronous machine interconnection model also reflects the following rules. It can have:

– one instance of TurbineGovernorDynamics child class, but only if AsynchronousMachine.asynchronousMachineType = generator;

– one instance of MechanicalLoadDynamics child class, but only if AsynchronousMachine.asynchronousMachineType = motor.

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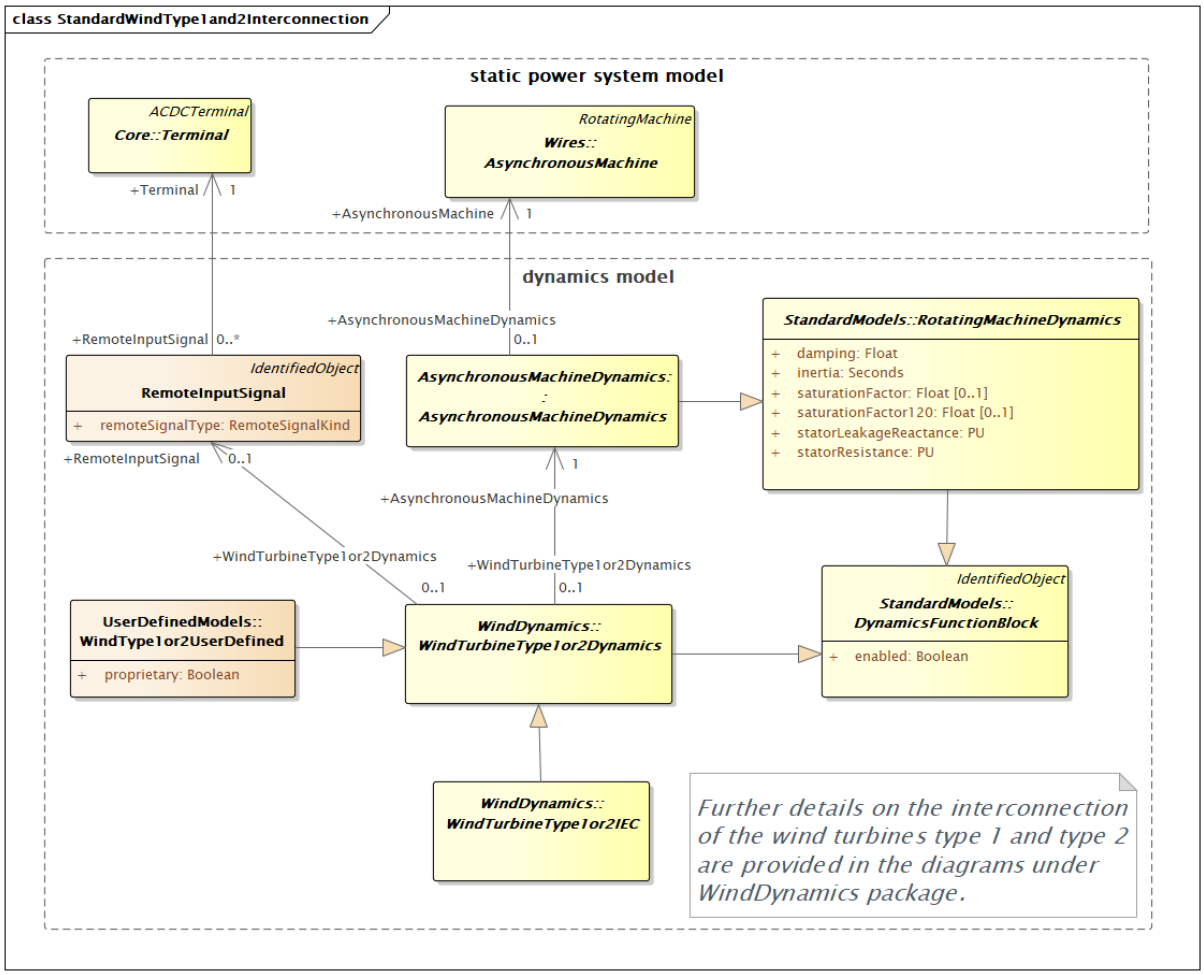

Figure 61 – Class diagram StandardInterconnections:: StandardWindType1and2Interconnection

Figure 61: This diagram shows the classes necessary for wind type 1 and type 2 interconnection models.

Standard IEC wind turbines type 1 and 2 interconnection models all have a single instance of an AsynchronousMachineDynamics child class and a single instance of the WindGenTurbineType1aIEC class, the WindGenTurbineType1bIEC class, or the WindGenTurbineType2IEC class as appropriate.

A valid IEC wind type 1A interconnection model also shall have one instance of each of the following classes (and no others):

– WindMechIEC;
– WindAeroConstIEC;
– WindProtectionIEC.

A valid IEC wind type 1B interconnection model also shall have one instance of each of the following classes (and no others):

– WindMechIEC;
– WindProtectionIEC;
– WindPitchContPowerIEC.

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A valid IEC wind type 2 interconnection model also shall have one instance of each of the following classes (and no others):

– WindMechIEC;
– WindContRotorRIEC;
– WindProtectionIEC;
– WindPitchContPowerIEC.

The WindProtectionIEC class and the WindContRotorRIEC class use the WindDynamicsLookupTable class.

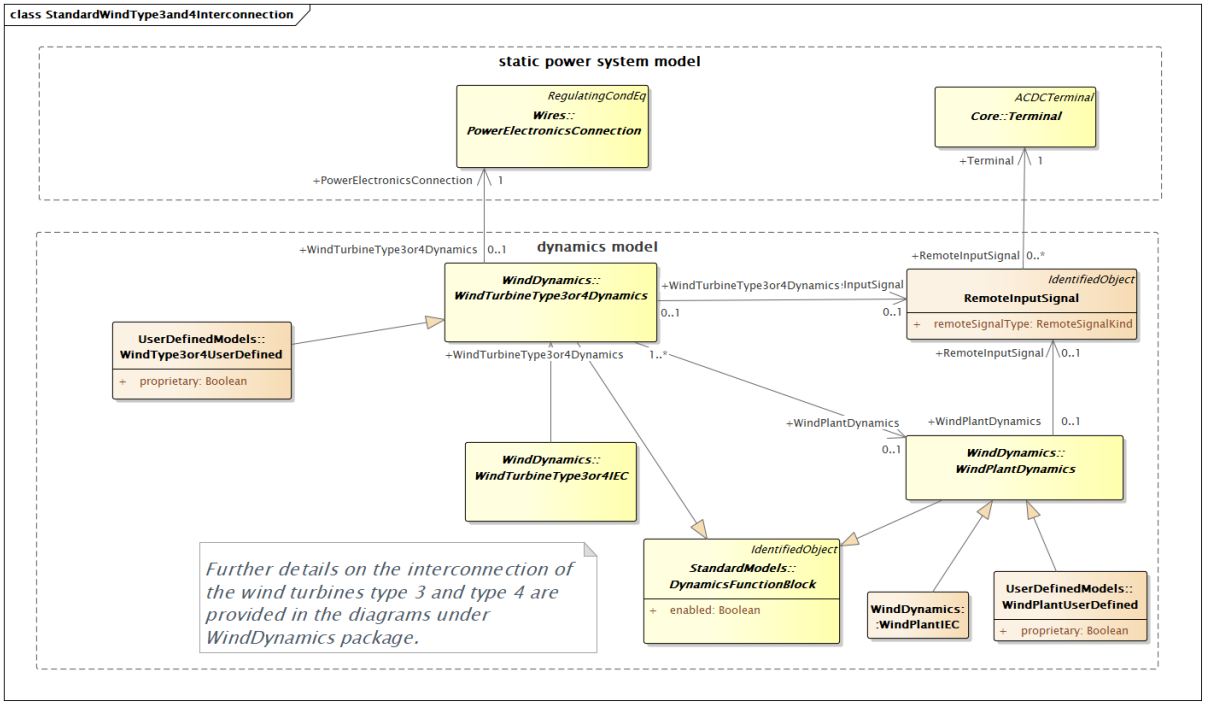

Figure 62 – Class diagram StandardInterconnections:: StandardWindType3and4Interconnection

Figure 62: This diagram shows the classes necessary for wind type 3 and type 4 interconnection models, including the wind plant model.

Standard IEC wind types 3, 4A, 4B and plant interconnection models all have a single instances as follows:

IEC wind type 3:

– WindTurbineType3IEC;
– WindAeroOneDimIEC or WindAeroTwoDimIEC;
– WindMechIEC;
– WindGenType3aIEC or WindGenType3bIEC;
– WindContPType3IEC;
– WindContQIEC;
– WindContCurrLimIEC;
– WindContQLimIEC or WindContQPQULimIEC;
– WindContPitchAngleIEC;

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– WindProtectionIEC.

IEC wind type 4A:

– WindTurbineType4aIEC;
– WindGenType4IEC or WindGenType3aIEC;
– WindContPType4aIEC;
– WindContQIEC;
– WindContCurrLimIEC;
– WindContQLimIEC or WindContQPQULimIEC;
– WindProtectionIEC.

IEC wind type 4B:

– WindTurbineType4bIEC;
– WindMechIEC;
– WindGenType4IEC or WindGenType3aIEC;
– WindContPType4bIEC;
– WindContQIEC;
– WindContCurrLimIEC;
– WindContQLimIEC or WindContQPQULimIEC;
– WindProtectionIEC.

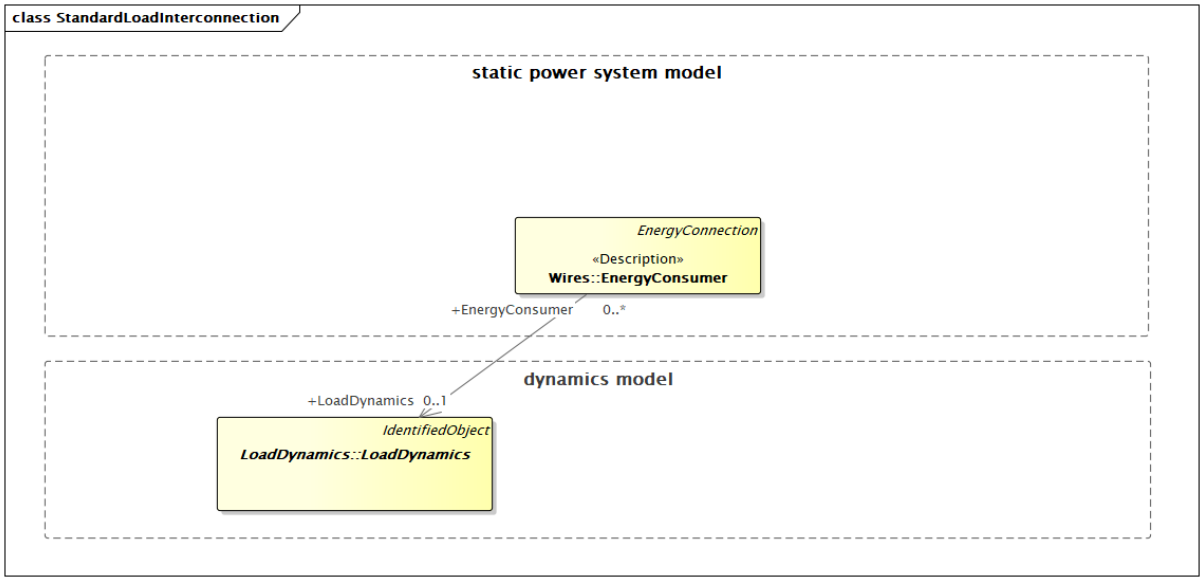

Figure 63 – Class diagram StandardInterconnections::StandardLoadInterconnection

Figure 63: This diagram shows the classes necessary for a load interconnection model.

A standard load interconnection model has a single instance of an EnergyConsumerDynamics child class plus a related instance of a LoadDynamics child class.

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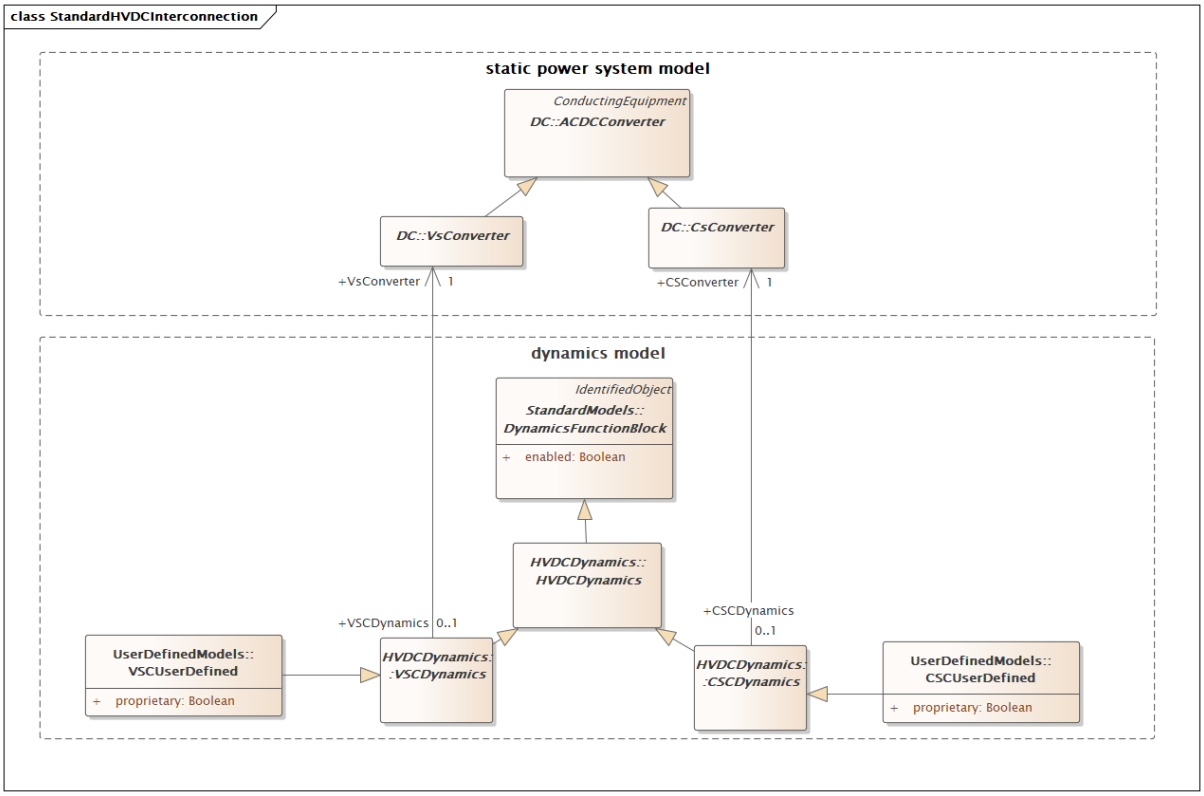

Figure 64 – Class diagram StandardInterconnections::StandardHVDCInterconnection

Figure 64: This diagram shows the classes necessary for a HVDC interconnection model.

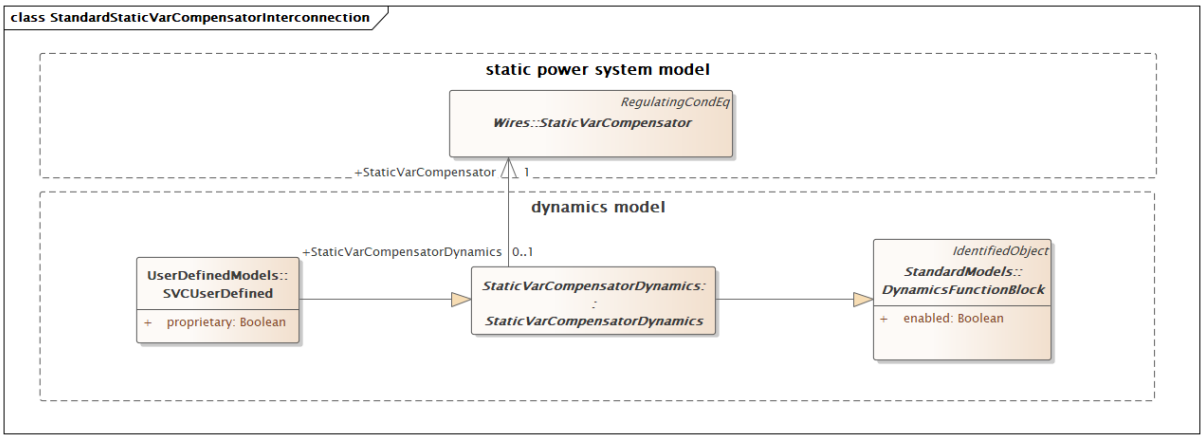

Figure 65 – Class diagram StandardInterconnections::
StandardStaticVarCompensatorInterconnection

Figure 65: This diagram shows the classes necessary for a static var compensator (SVC) interconnection model.

14.3.2.2 RemoteInputSignal

Inheritance path = IdentifiedObject

Supports connection to a terminal associated with a remote bus from which an input signal of a specific type is coming.

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Table 656 shows all attributes of RemoteInputSignal.

Table 656 – Attributes of StandardInterconnections::RemoteInputSignal

name	mult	type	description
remoteSignalType	1..1	RemoteSignalKind	Type of input signal.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 657 shows all association ends of RemoteInputSignal with other classes.

Table 657 – Association ends of StandardInterconnections:: RemoteInputSignal with other classes

mult from	name	mult to	type	description
0..*	Terminal	1..1	Terminal	Remote terminal with which this input signal is associated.
0..1	DiscontinuousExcitationControlDynamics	0..1	DiscontinuousExcitationControlDynamics	Discontinuous excitation control model using this remote input signal.
0..*	PowerSystemStabilizerDynamics	0..1	PowerSystemStabilizerDynamics	Power system stabilizer model using this remote input signal.
0..1	UnderexcitationLimiterDynamics	0..1	UnderexcitationLimiterDynamics	Underexcitation limiter model using this remote input signal.
0..1	PFVArControllerType1Dynamics	0..1	PFVArControllerType1Dynamics	Power factor or VAr controller type 1 model using this remote input signal.
0..1	VoltageCompensatorDynamics	0..1	VoltageCompensatorDynamics	Voltage compensator model using this remote input signal.

14.3.3 Package StandardModels

14.3.3.1 General

This subclause contains standard dynamic model specifications grouped into packages by standard function block (type of equipment being modelled).

In the CIM, standard dynamic models are expressed by means of a class named with the standard model name and attributes reflecting each of the parameters necessary to describe the behaviour of an instance of the standard model.

14.3.3.2 (abstract) DynamicsFunctionBlock

Inheritance path = IdentifiedObject

Abstract parent class for all Dynamics function blocks.

Table 658 shows all attributes of DynamicsFunctionBlock.

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Table 658 – Attributes of StandardModels::DynamicsFunctionBlock

name	mult	type	description
enabled	1..1	Boolean	Function block used indicator. true = use of function block is enabled false = use of function block is disabled.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.3 (abstract) RotatingMachineDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Abstract parent class for all synchronous and asynchronous machine standard models.

Table 659 shows all attributes of RotatingMachineDynamics.

Table 659 – Attributes of StandardModels::RotatingMachineDynamics

name	mult	type	description
damping	1..1	Float	Damping torque coefficient (D) (>= 0). A proportionality constant that, when multiplied by the angular velocity of the rotor poles with respect to the magnetic field (frequency), results in the damping torque. This value is often zero when the sources of damping torques (generator damper windings, load damping effects, etc.) are modelled in detail. Typical value = 0.
inertia	1..1	Seconds	Inertia constant of generator or motor and mechanical load (H) (> 0). This is the specification for the stored energy in the rotating mass when operating at rated speed. For a generator, this includes the generator plus all other elements (turbine, exciter) on the same shaft and has units of MW x s. For a motor, it includes the motor plus its mechanical load. Conventional units are PU on the generator MVA base, usually expressed as MW x s / MVA or just s. This value is used in the accelerating power reference frame for operator training simulator solutions. Typical value = 3.
saturationFactor	0..1	Float	Saturation factor at rated terminal voltage (S1) (>= 0). Not used by simplified model. Defined by defined by S(E1) in the SynchronousMachineSaturationParameters diagram. Typical value = 0,02.
saturationFactor120	0..1	Float	Saturation factor at 120 % of rated terminal voltage (S12) (>= RotatingMachineDynamics.saturationFactor). Not used by the simplified model, defined by S(E2) in the SynchronousMachineSaturationParameters diagram. Typical value = 0,12.
statorLeakageReactance	1..1	PU	Stator leakage reactance (Xl) (>= 0). Typical value = 0,15.
statorResistance	1..1	PU	Stator (armature) resistance (Rs) (>= 0). Typical value = 0,005.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	0..1	String	inherited from: IdentifiedObject

14.3.3.4 Package SynchronousMachineDynamics

14.3.3.4.1 General

For conventional power generating units (e.g., thermal, hydro, combustion turbine), a synchronous machine model represents the electrical characteristics of the generator and the mechanical characteristics of the turbine-generator rotational inertia. Large industrial motors or groups of similar motors can be represented by individual motor models which are represented as generators with negative active power in the static (power flow) data.

The interconnection with the electrical network equations can differ among simulation tools. The tool only needs to know the synchronous machine to establish the correct interconnection. The interconnection with the motor's equipment could also differ due to input and output signals required by standard models.

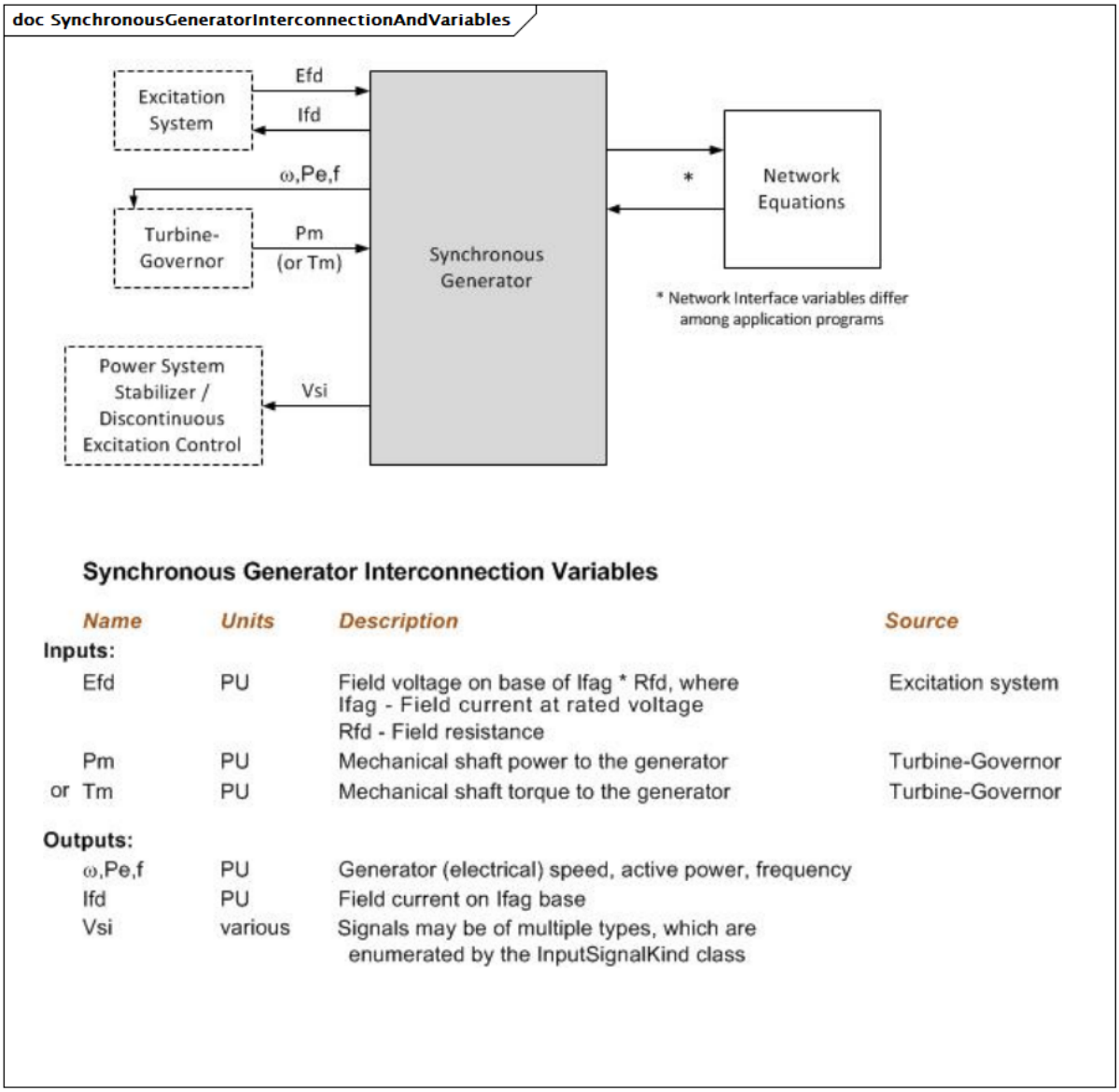

Figure 66 – SynchronousGeneratorInterconnectionAndVariables

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Figure 66: This diagram illustrates the inputs and outputs for a synchronous generator.

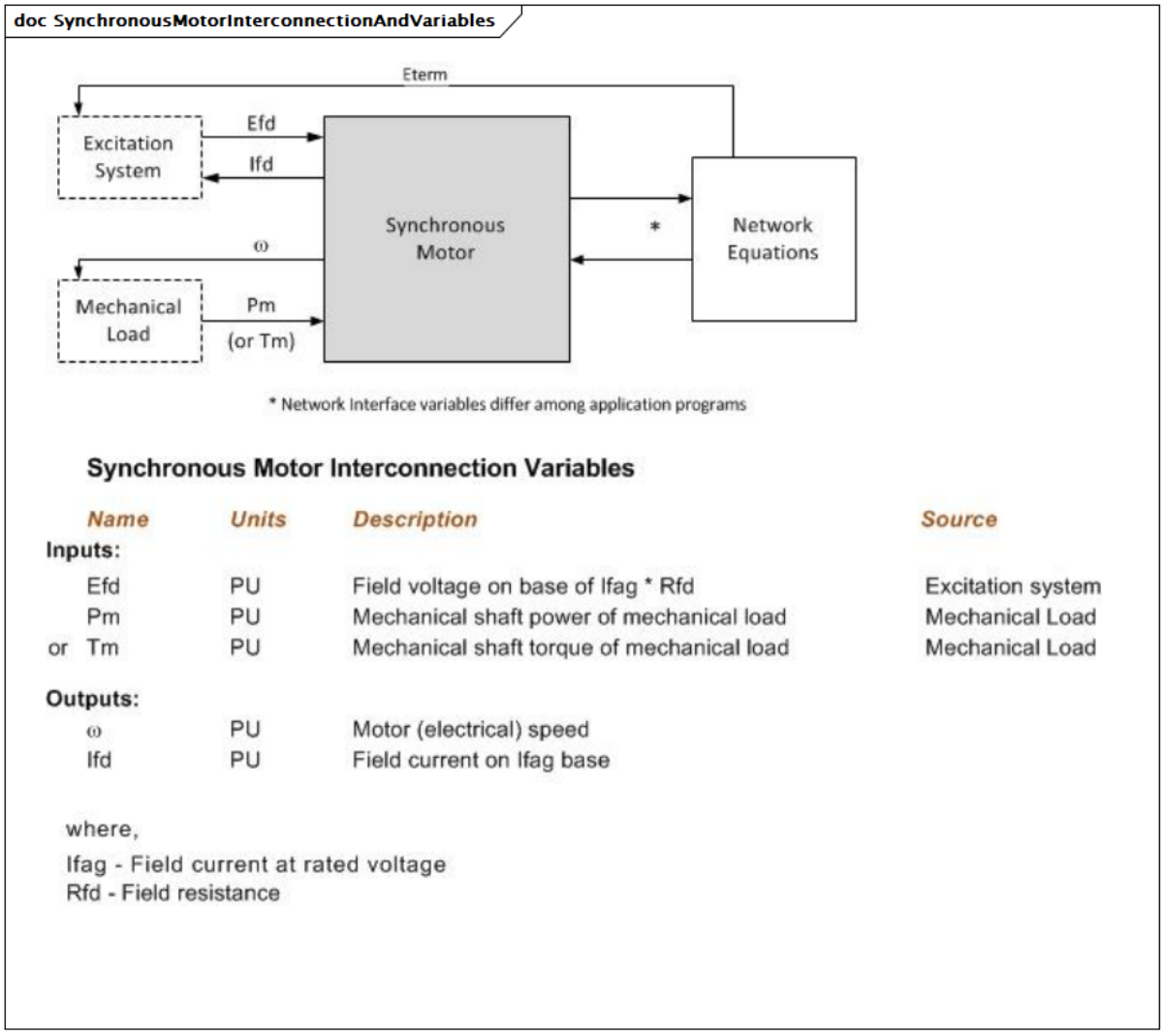

Figure 67 – SynchronousMotorInterconnectionAndVariables

Figure 67: This diagram illustrates the inputs and outputs for a synchronous motor.

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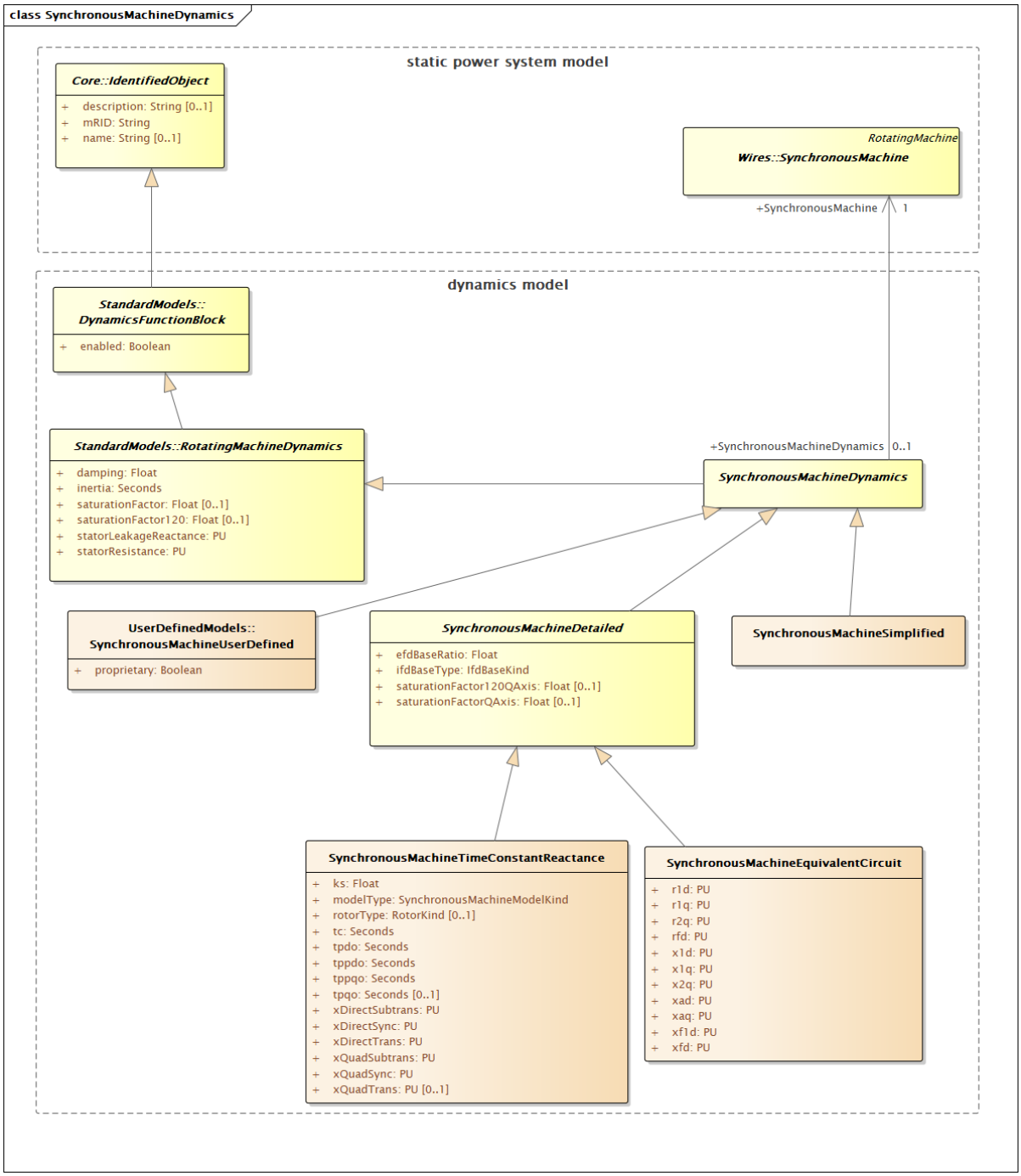

Figure 68 – Class diagram SynchronousMachineDynamics:: SynchronousMachineDynamics

Figure 68: This diagram shows static power system model classes and dynamics model classes related to the synchronous machine dynamics model. An association between the SynchronousMachineDynamics class and the SynchronousMachine class provides a means for relating the dynamics model to the static model.

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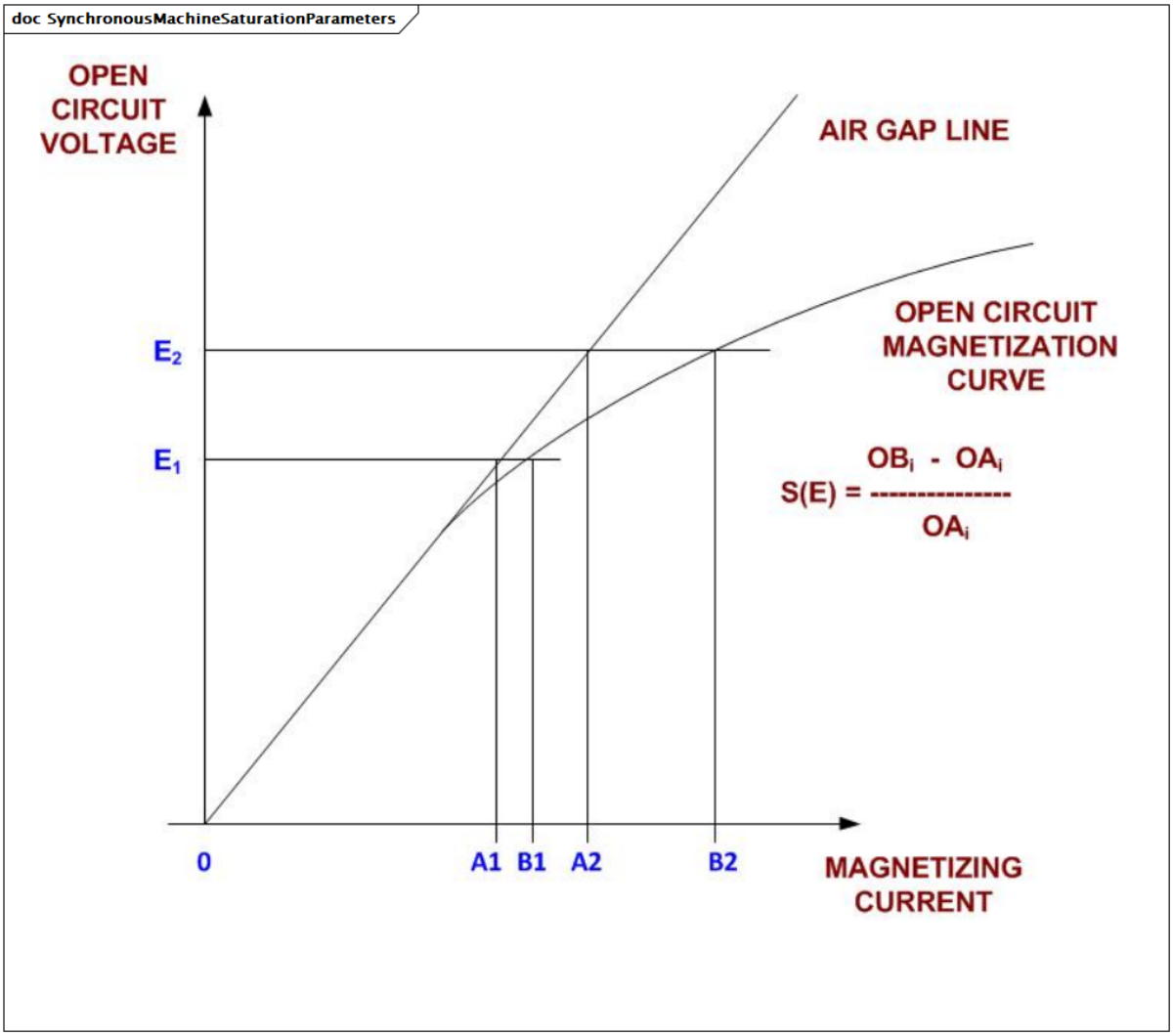

Figure 69 – SynchronousMachineSaturationParameters

Figure 69: This diagram shows the synchronous machine saturation parameters.

Parameter details:

1) The quantity OAi in amperes is normally called Ifag (which means field current at rated voltage, open circuit on the air gap (no saturation) line). OBi in amperes isnormally called Ifn (which means field current at rated voltage with no load on the machine).
2) Saturation factors for generators are given by RotatingMachineDynamics.saturationFactor and RotatingMachineDynamics.saturationFactor120.
3) For exciters, the values are given in the attributes for each standard excitation system model.

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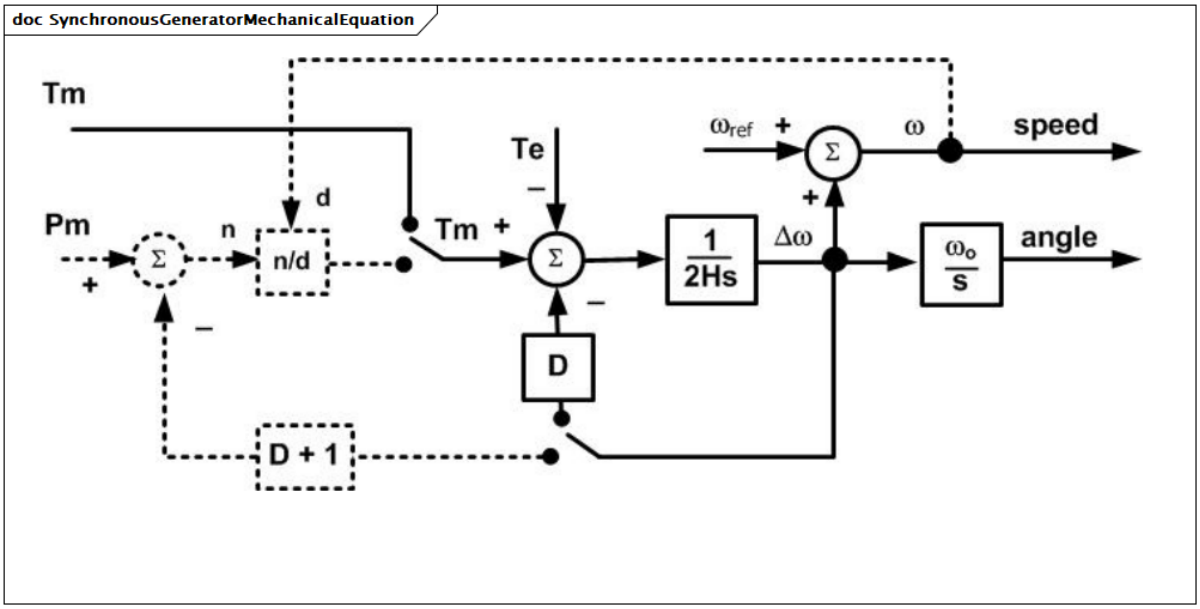

Figure 70 – SynchronousGeneratorMechanicalEquation

Figure 70: The mechanical equations for all the variations of the synchronous generator model are the same and can be represented by the SynchronousGeneratorMechanicalEquation block diagram.

Parameter details:

1) All variables are PU on generator MVA base except angle, which is in radians.
2) omega0 is the system synchronous frequency in radians per s. omegaref is the PU system frequency (or isolated area frequency), which can be constant at the initial frequency or vary during a simulation depending on the application program.
3) The input to the mechanical equations can be either Pm or Tm depending on the application program. Application programs using Pm as the input with the damping feedback subtracted from Pm will need to increase D by 1 as shown. In that case, the damping feedback to the torque summing point is not used.

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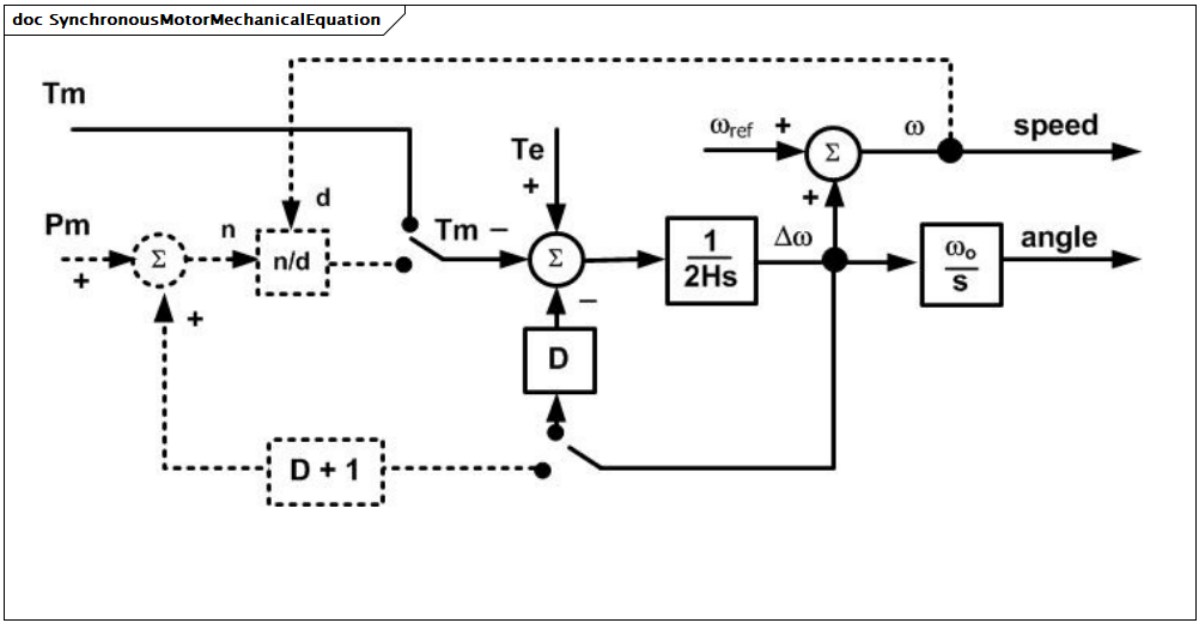

Figure 71 – SynchronousMotorMechanicalEquation

Figure 71: The mechanical equations for all variations of the synchronous motor model are the same and can be represented by the SynchronousMotorMechanicalEquation diagram.

Parameter details:

1) All variables are PU on motor MVA base except angle, which is in radians.
2) omega0 is the system synchronous frequency in radians per s. omegaref is the PU system frequency (or isolated area frequency), which can be constant at the initial frequency or vary during a simulation depending on the application program.

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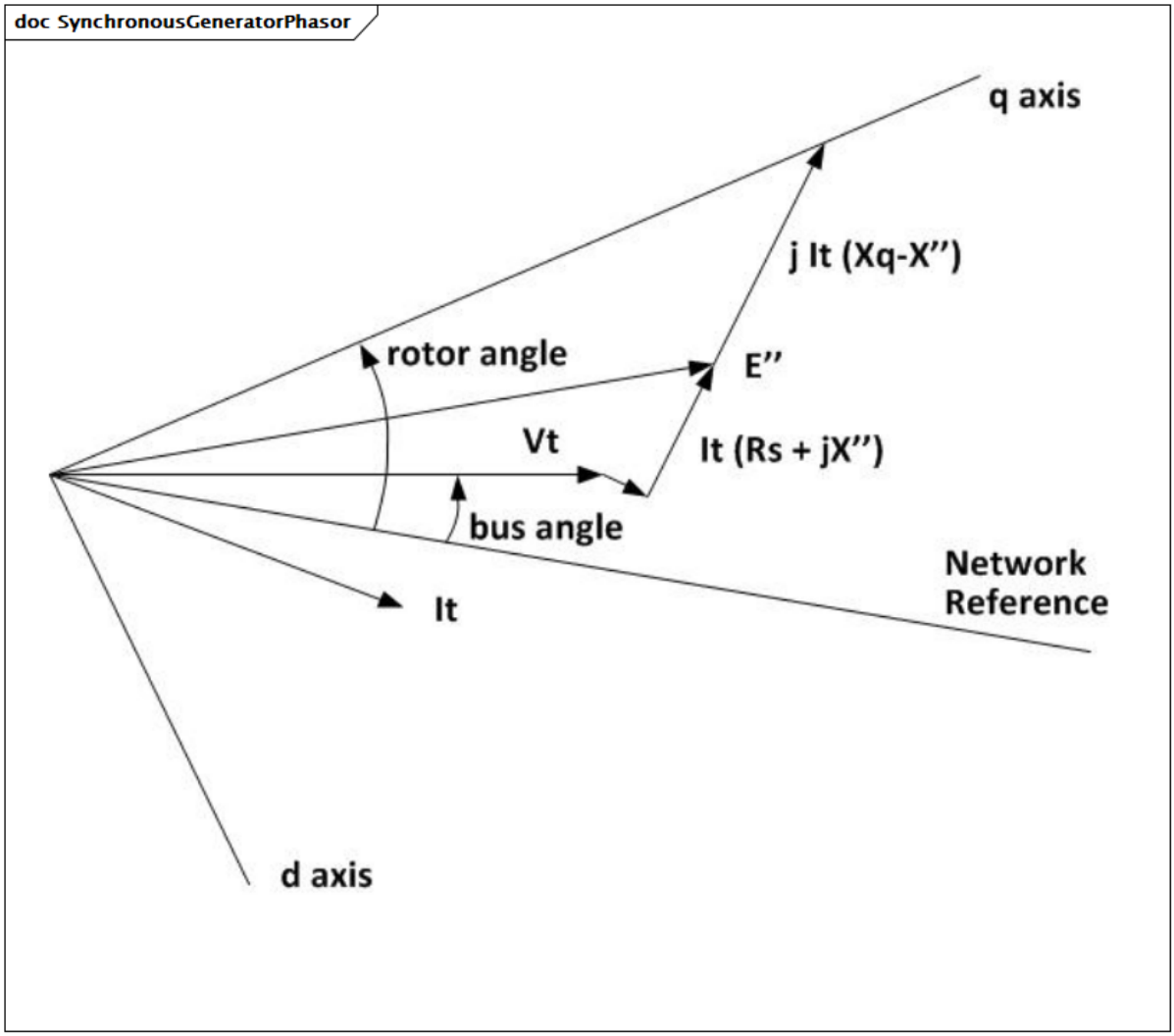

Figure 72 – SynchronousGeneratorPhasor

Figure 72: Diagram details:

1) The definition of d- and q-axis variables is based on the SynchronousGeneratorPhasor diagram (counter-clockwise rotation) for the case of an overexcited generator (generating reactive power).

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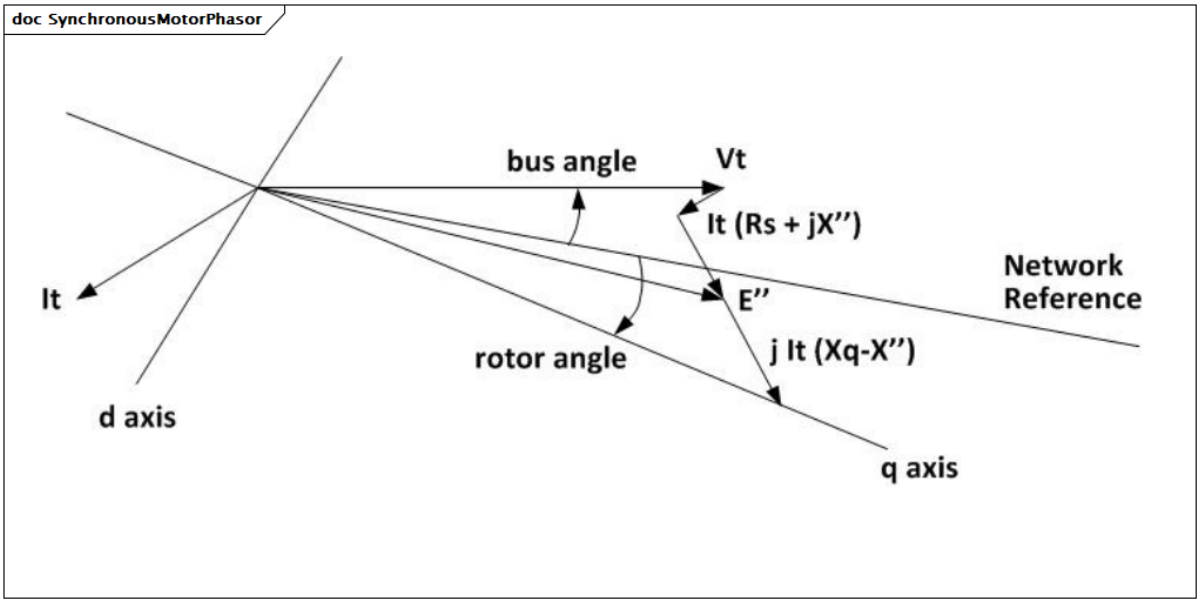

Figure 73 – SynchronousMotorPhasor

Figure 73: Diagram details:

1) The definition of d- and q-axis variables is based on the SynchronousMotorPhasor diagram (counter-clockwise rotation) for the case of a motor consuming active power and overexcited (generating reactive power).

14.3.3.4.2 SynchronousMachineSimplified

Inheritance path = SynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

The simplified model represents a synchronous generator as a constant internal voltage behind an impedance (Rs + jXp) as shown in the Simplified diagram.

Since internal voltage is held constant, there is no Efd input and any excitation system model will be ignored. There is also no Ifd output.

This model should not be used for representing a real generator except, perhaps, small generators whose response is insignificant.

The parameters used for the simplified model include:

– RotatingMachineDynamics.damping (D);
– RotatingMachineDynamics.inertia (H);
– RotatingMachineDynamics.statorLeakageReactance (used to exchange jXp for SynchronousMachineSimplified);
– RotatingMachineDynamics.statorResistance (Rs).

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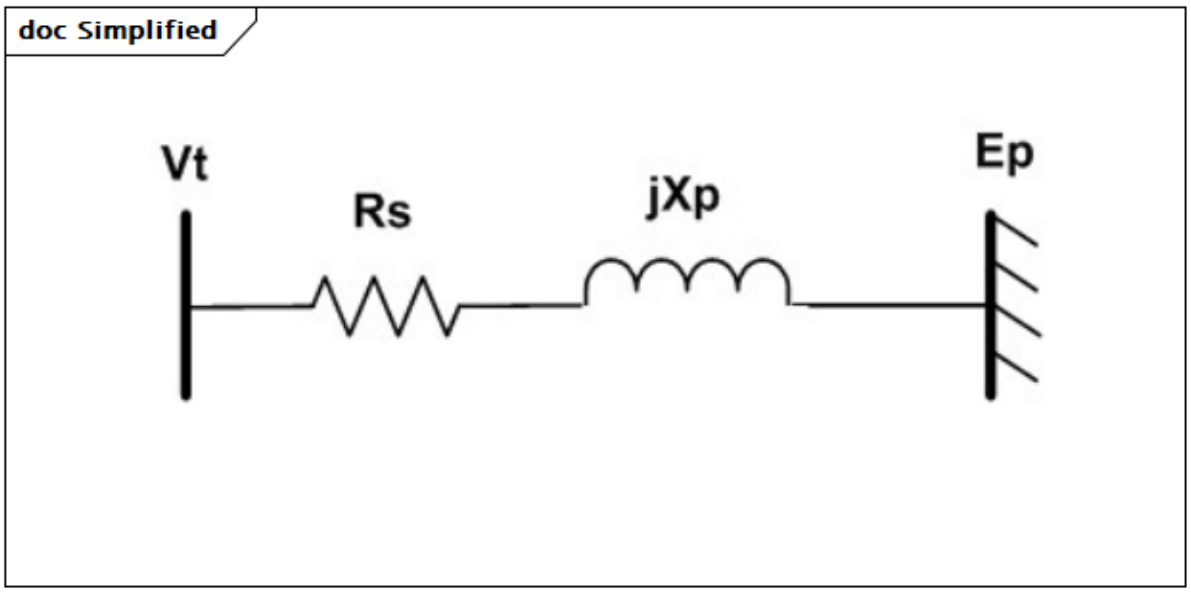

Figure 74 – Simplified

Figure 74: This diagram describes the simplified synchronous generator model.

Parameter details:

1) The model is often used for gross equivalents of parts of a system that are not represented in detail. In this case the MVA rating would be the combined rating of all generators in the equivalenced area.
2) Rs + jXp would be the short-circuit equivalent impedance at the location of the equivalent generator on the generator MVA rating base. Inertia constant (H) would be typical or average values for the generators in the equivalenced area.
3) The internal reactance (Xp) can be labelled in different ways (X?, X?, X?d, X?d) by different simulation tools.

Table 660 shows all attributes of SynchronousMachineSimplified.

Table 660 – Attributes of SynchronousMachineDynamics:: SynchronousMachineSimplified

name	mult	type	description
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 661 shows all association ends of SynchronousMachineSimplified with other classes.

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Table 661 – Association ends of SynchronousMachineDynamics:: SynchronousMachineSimplified with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	inherited from: SynchronousMachineDynamics

14.3.3.4.3 (abstract) SynchronousMachineDynamics

Inheritance path = RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

Synchronous machine whose behaviour is described by reference to a standard model expressed in one of the following forms:

– simplified (or classical), where a group of generators or motors is not modelled in detail;
– detailed, in equivalent circuit form;
– detailed, in time constant reactance form; or
– by definition of a user-defined model.

It is a common practice to represent small generators by a negative load rather than by a dynamic generator model when performing dynamics simulations. In this case, a SynchronousMachine in the static model is not represented by anything in the dynamics model, instead it is treated as an ordinary load.

Parameter details:

1) Synchronous machine parameters such as Xl, Xd, Xp etc. are actually used as inductances in the models, but are commonly referred to as reactances since, at nominal frequency, the PU values are the same. However, some references use the symbol L instead of X.

Table 662 shows all attributes of SynchronousMachineDynamics.

Table 662 – Attributes of SynchronousMachineDynamics:: SynchronousMachineDynamics

name	mult	type	description
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 663 shows all association ends of SynchronousMachineDynamics with other classes.

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Table 663 – Association ends of SynchronousMachineDynamics:: SynchronousMachineDynamics with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	Synchronous machine to which synchronous machine dynamics model applies.

14.3.3.4.4 (abstract) SynchronousMachineDetailed

Inheritance path = SynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

All synchronous machine detailed types use a subset of the same data parameters and input/output variables.

The several variations differ in the following ways:

– the number of equivalent windings that are included;

– he way in which saturation is incorporated into the model;

– whether or not “subtransient saliency” (X''q not = X''d) is represented.

It is not necessary for each simulation tool to have separate models for each of the model types. The same model can often be used for several types by alternative logic within the model. Also, differences in saturation representation might not result in significant model performance differences so model substitutions are often acceptable.

Table 664 shows all attributes of SynchronousMachineDetailed.

Table 664 – Attributes of SynchronousMachineDynamics::SynchronousMachineDetailed

name	mult	type	description
saturationFactorQAxis	0..1	Float	Quadrature-axis saturation factor at rated terminal voltage (S1q) (>= 0). Typical value = 0,02.
saturationFactor120QAxis	0..1	Float	Quadrature-axis saturation factor at 120% of rated terminal voltage (S12q) (>= SynchonousMachineDetailed.saturationFactorQAxis). Typical value = 0,12.
efdBaseRatio	1..1	Float	Ratio (exciter voltage/generator voltage) of Efd bases of exciter and generator models (> 0). Typical value = 1.
ifdBaseType	1..1	IfdBaseKind	Excitation base system mode. It should be equal to the value of WLMDV given by the user. WLMDV is the PU ratio between the field voltage and the excitation current: Efd = WLMDV x Ifd. Typical value = ifag.
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 665 shows all association ends of SynchronousMachineDetailed with other classes.

Table 665 – Association ends of SynchronousMachineDynamics:: SynchronousMachineDetailed with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	inherited from: SynchronousMachineDynamics

14.3.3.4.5 SynchronousMachineTimeConstantReactance

Inheritance path = SynchronousMachineDetailed: SynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

Synchronous machine detailed modelling types are defined by the combination of the attributes SynchronousMachineTimeConstantReactance.modelType and SynchronousMachineTimeConstantReactance.rotorType.

Parameter details:

1) The “p” in the time-related attribute names is a substitution for a “prime” in the usual parameter notation, e.g. tpdo refers to T'do.
2) The parameters used for models expressed in time constant reactance form include:
   – RotatingMachine.ratedS (MVAbase);
   – RotatingMachineDynamics.damping (D);
   – RotatingMachineDynamics.inertia (H);
   – RotatingMachineDynamics.saturationFactor (S1);
   – RotatingMachineDynamics.saturationFactor120 (S12);
   – RotatingMachineDynamics.statorLeakageReactance (Xl);
   – RotatingMachineDynamics.statorResistance (Rs);
   – SynchronousMachineTimeConstantReactance.ks (Ks);
   – SynchronousMachineDetailed.saturationFactorQAxis (S1q);
   – SynchronousMachineDetailed.saturationFactor120QAxis (S12q);
   – SynchronousMachineDetailed.efdBaseRatio;
   – SynchronousMachineDetailed.ifdBaseType;
   – .xDirectSync (Xd);
   – .xDirectTrans (X'd);
   – .xDirectSubtrans (X''d);
   – .xQuadSync (Xq);
   – .xQuadTrans (X'q);
   – .xQuadSubtrans (X''q);
   – .tpdo (T'do);
   – .tppdo (T''do);

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– .tpqo (T'qo);
– .tppqo (T''qo);
– .tc.

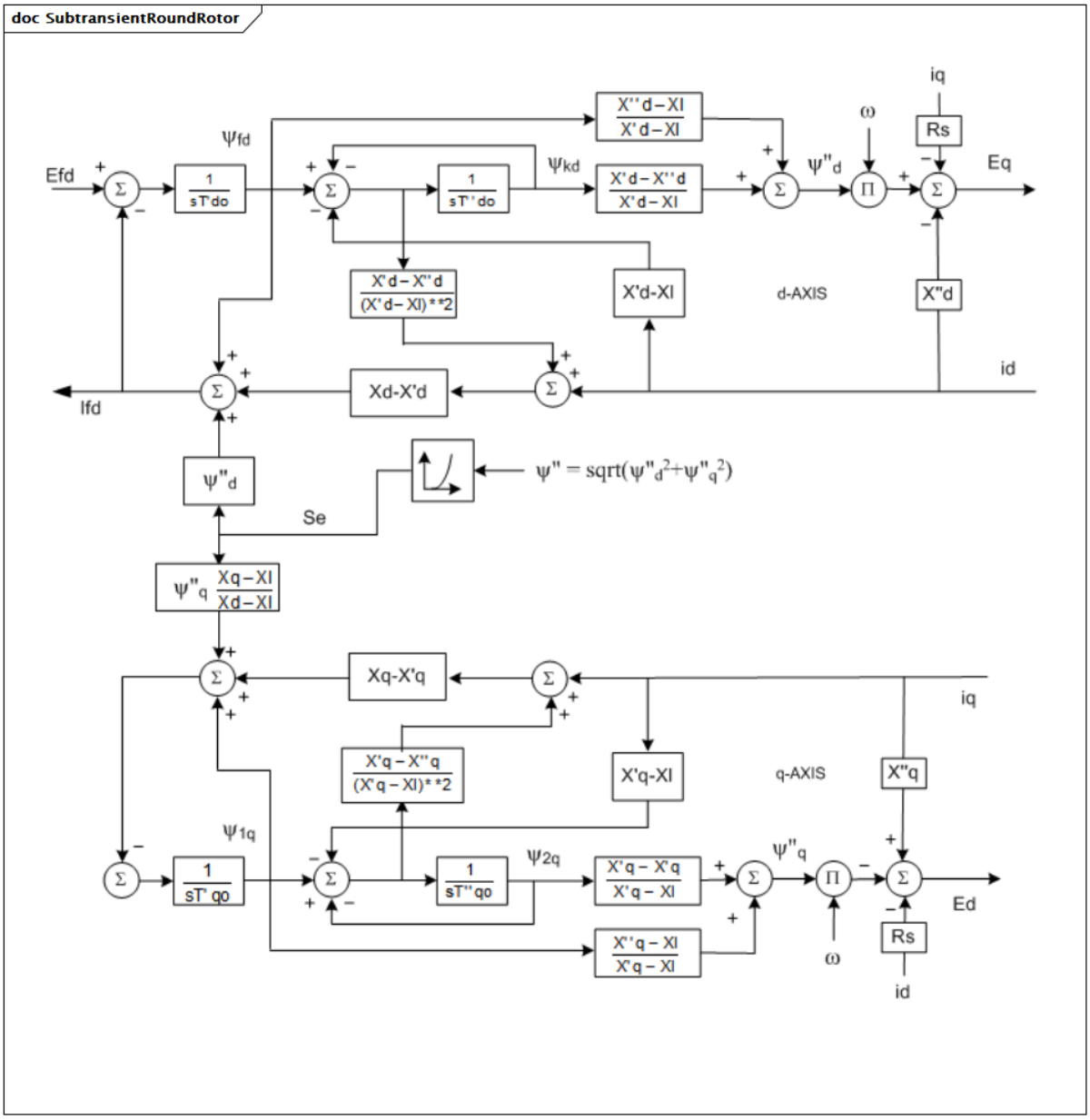

Figure 75 – SubtransientRoundRotor

Figure 75: The subtransient round rotor type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransient and .rotorType = roundRotor.

The complete equivalent circuit is used with two rotor windings in each axis.

Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in PU (IEEE 421.5:2005, Annex B). It is equal to Efd in steady-state.

Parameter details:

1) X''q can be assumed to be equal to X''d (no subtransient saliency).

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2) Saturation is modelled in both the direct- and quadrature- axes.
3) For generators, the following parameters can be omitted with defaults shown: X''q (= X''d), Ks (= 0), S1q (= S1), S12q (= S12), Tc (= 0).
4) For motors, the following parameter is not used: X''q.

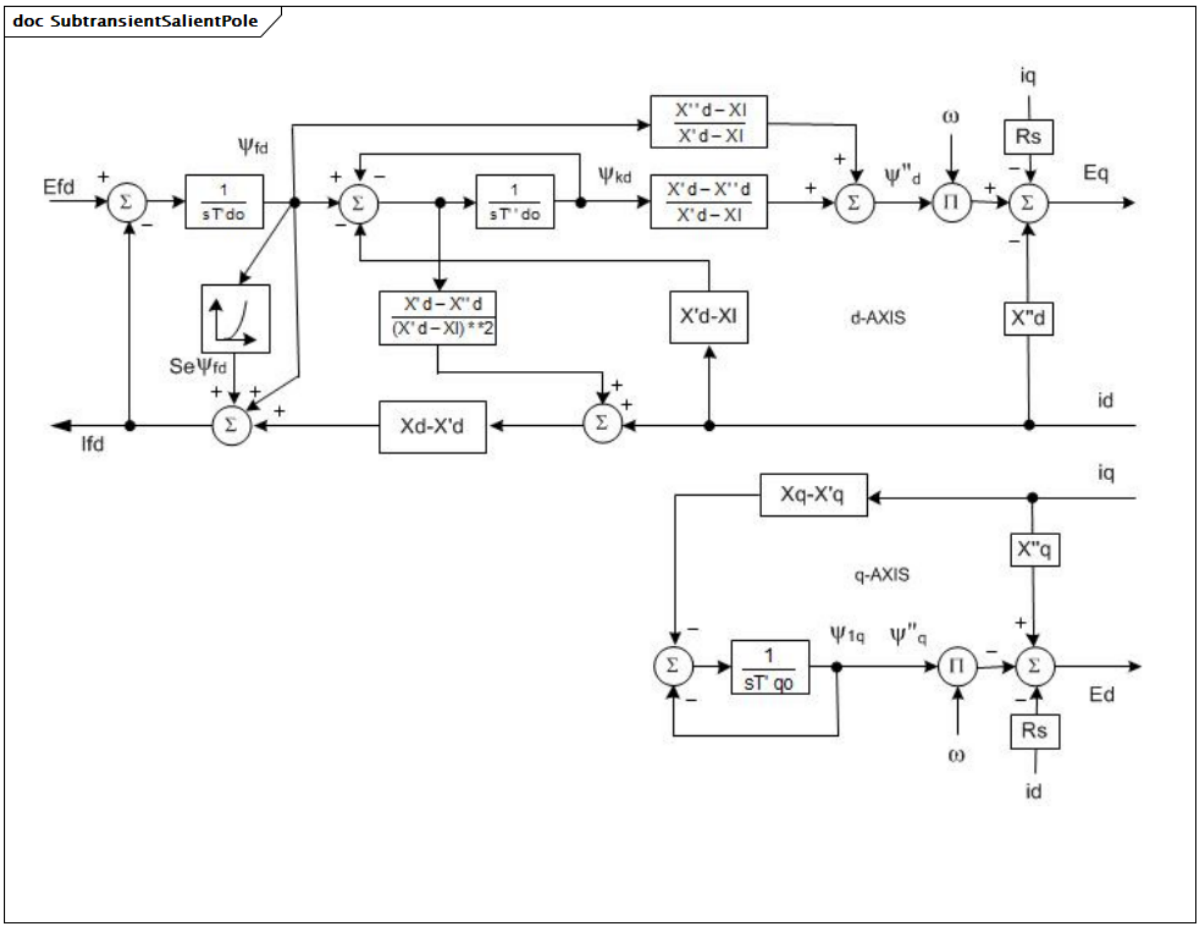

Figure 76 – SubtransientSalientPole

Figure 76: The subtransient salient pole type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransient and .rotorKind = salientPole.

The d-axis equivalent circuit is the same as for the subtransient round rotor type. The q-axis has only one equivalent rotor winding, which can be labelled as transient (X'q) or subtransient (X''q); X'q is used for the CIM description.

Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in PU (IEEE 421.5:2005, Annex B). It is equal to Efd in steady-state.

Parameter details:

1) X''q (=X'q) is assumed to be equal to X''d (no subtransient saliency).
2) Saturation is modelled in the d-axis only.
3) For generators, the following input parameters can be omitted with default shown: X'q (=X''d), X''q (=X''d), T''qo, Ks (=0), S1q (=0), S12q (=0), Tc (=0).
4) For motors, the following input parameters are not used: X'q, X''q, T''qo.

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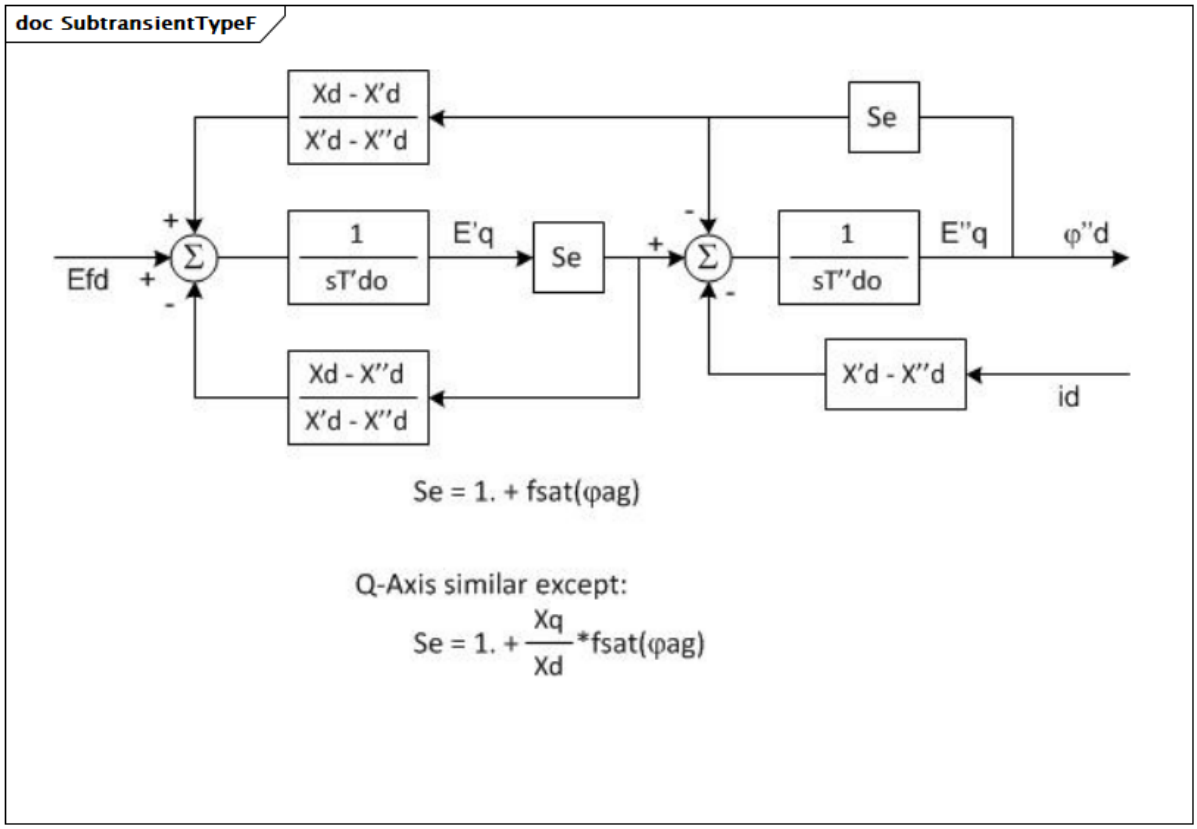

Figure 77 – SubtransientTypeF

Figure 77: The subtransient type F is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientTypeF.

This model is the WECC type F model with uniform inductance rotor modelling; shaft speed effects are neglected. It has a similar level of detail to the subtransient round rotor type but permits subtransient saliency (X''q not = X''d) and models saturation differently. The subtransient round rotor type can usually be substituted without significant loss of accuracy.

Saturation is modelled in both the direct- and quadrature- axes.

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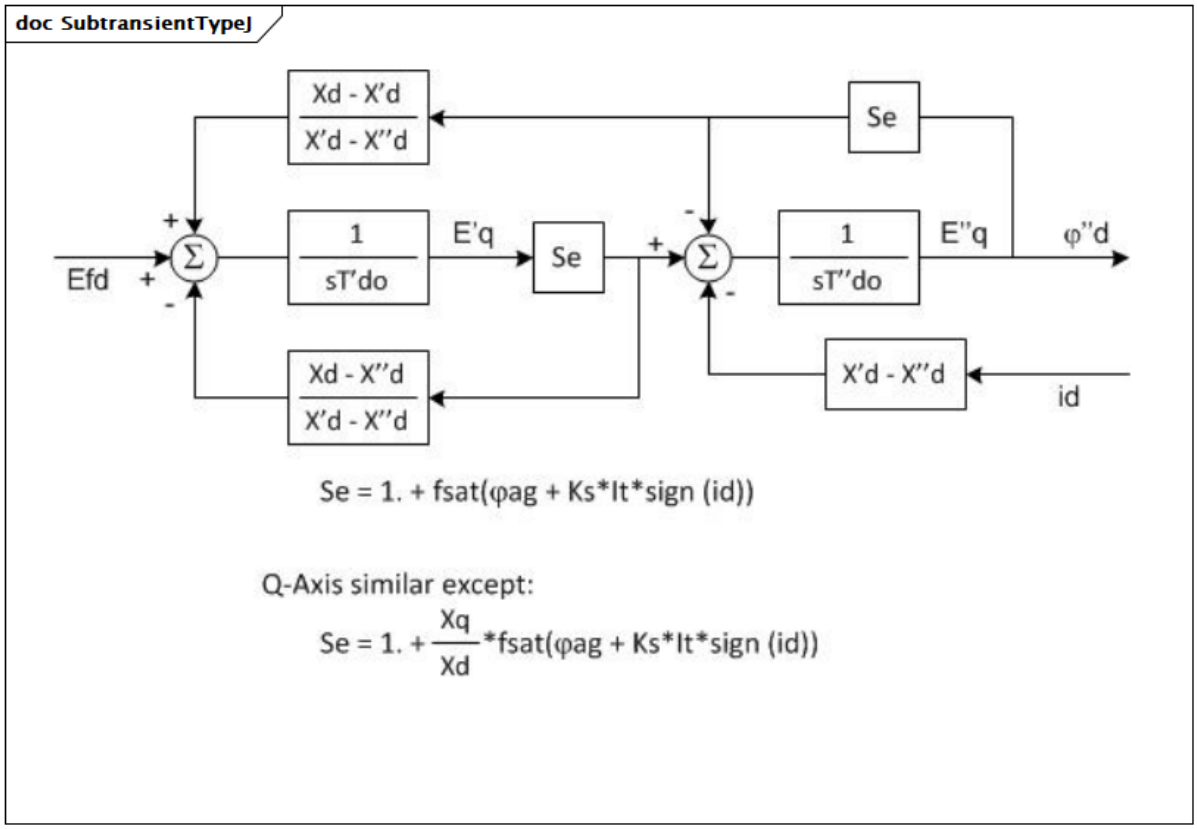

Figure 78 – SubtransientTypeJ

Figure 78: The subtransient type J is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientTypeJ.

This model is the WECC type J model with uniform inductance ratios rotor modelling with modified saturation model; shaft speed effects are neglected. This model is the same as subtransient type F but includes the effect of generator loading on saturation. With the usage of the saturation constant (Ks) parameter, it allows reproduction of the measured "V curves" over the whole range of reactive power, particularly on some hydro (salient pole) machines.

Saturation is modelled in both the direct- and quadrature- axes.

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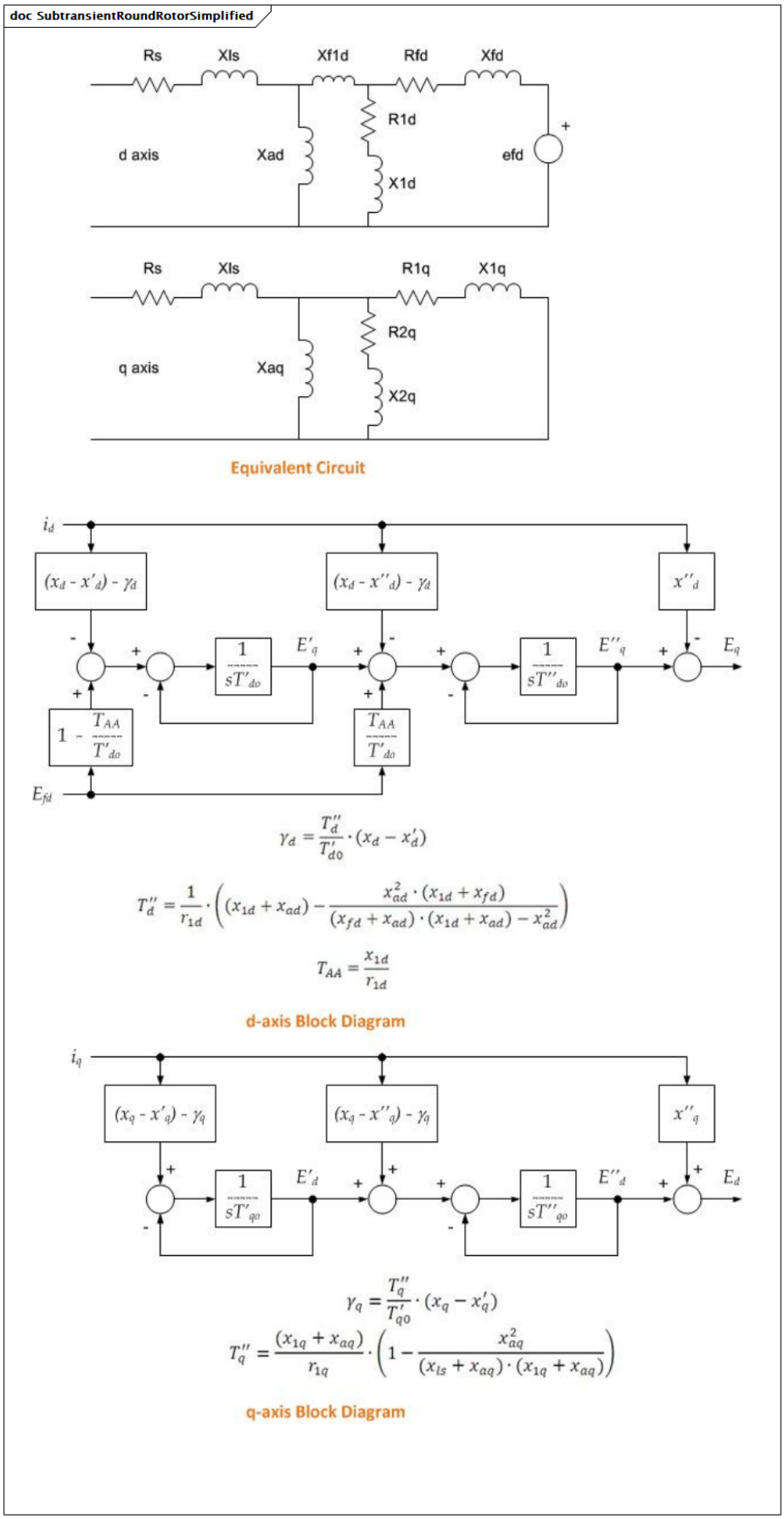

Figure 79 – SubtransientRoundRotorSimplified

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Figure 79: The subtransient round rotor simplified type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplified and .rotorKind = roundRotor.

It is a simplified version of subtransient round rotor type, magnetic coupling between the direct- and quadrature- axes is ignored.

Parameter details:

1) The effects of speed variation are neglected.
2) No armature resistance.
3) No saturation effects.
4) No subtransient saliency.
5) Direct-axis with field circuit and one equivalent damper circuit, X'd not = X''d.
6) Quadrature-axis with two equivalent damper circuit, X''q not = X'q.
7) For generators and motors the following input parameters are needed: Xd, X'd, X''d, T'do, T''do and Xq, X'q, X''q, T'qo, T''qo.

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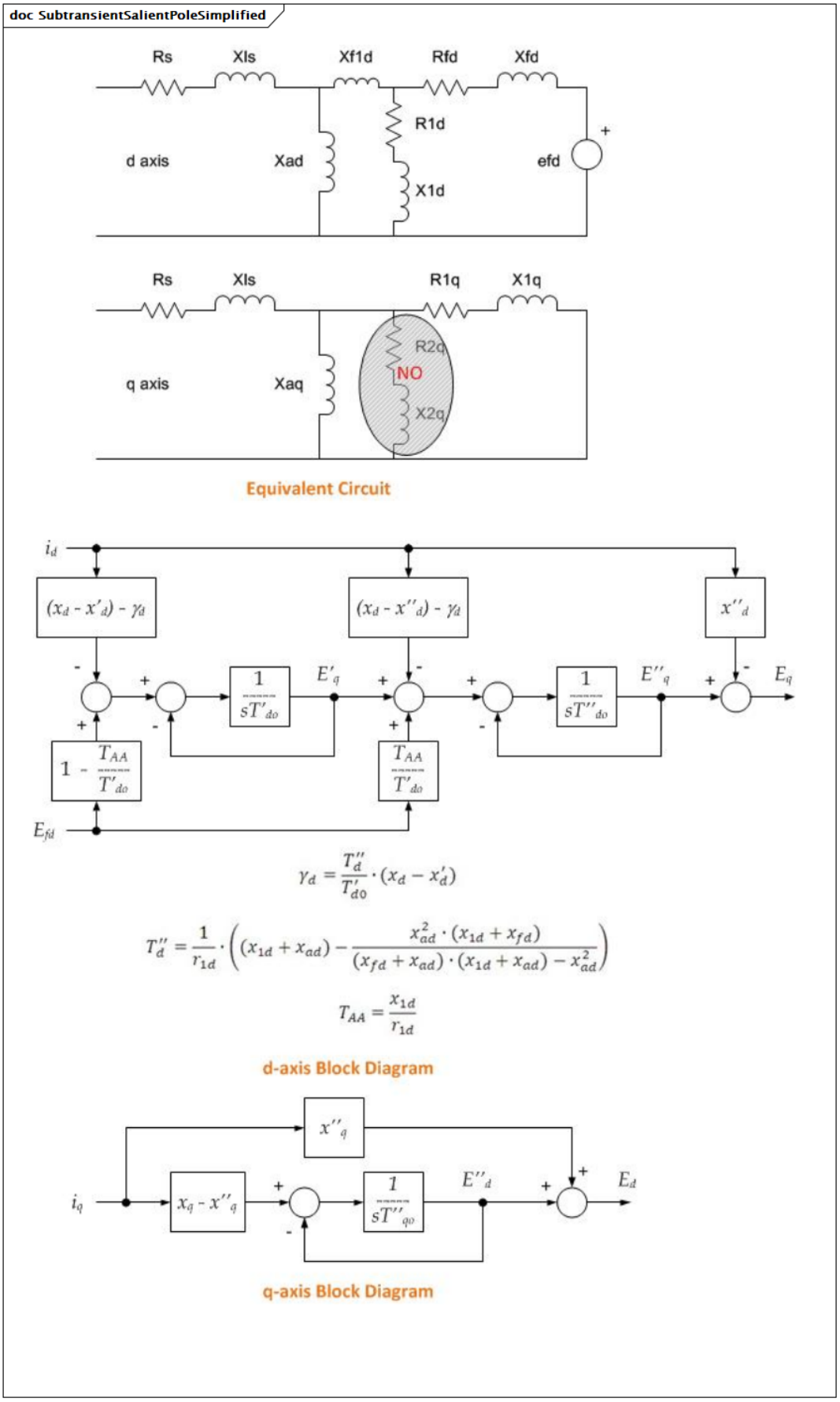

Figure 80 – SubtransientSalientPoleSimplified

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Figure 80: The subtransient salient pole simplified type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplified and .rotorKind = salientPole.

It is a simplified version of subtransient salient pole type, magnetic coupling between the direct- and quadrature- axes is ignored.

Parameter details:

1) The effects of speed variation are neglected.
2) No armature resistance.
3) No saturation effects.
4) No subtransient saliency.
5) Direct-axis with field circuit and one equivalent damper circuit, X'd not = X''d.
6) Quadrature-axis with only one equivalent damper circuit, which is labeled as subtransient (X''q), X''q =X'q.
7) For generators and motors the following input parameters are needed: Xd, X'd, X''d, T'do, T''do and Xq, X''q, T''qo.

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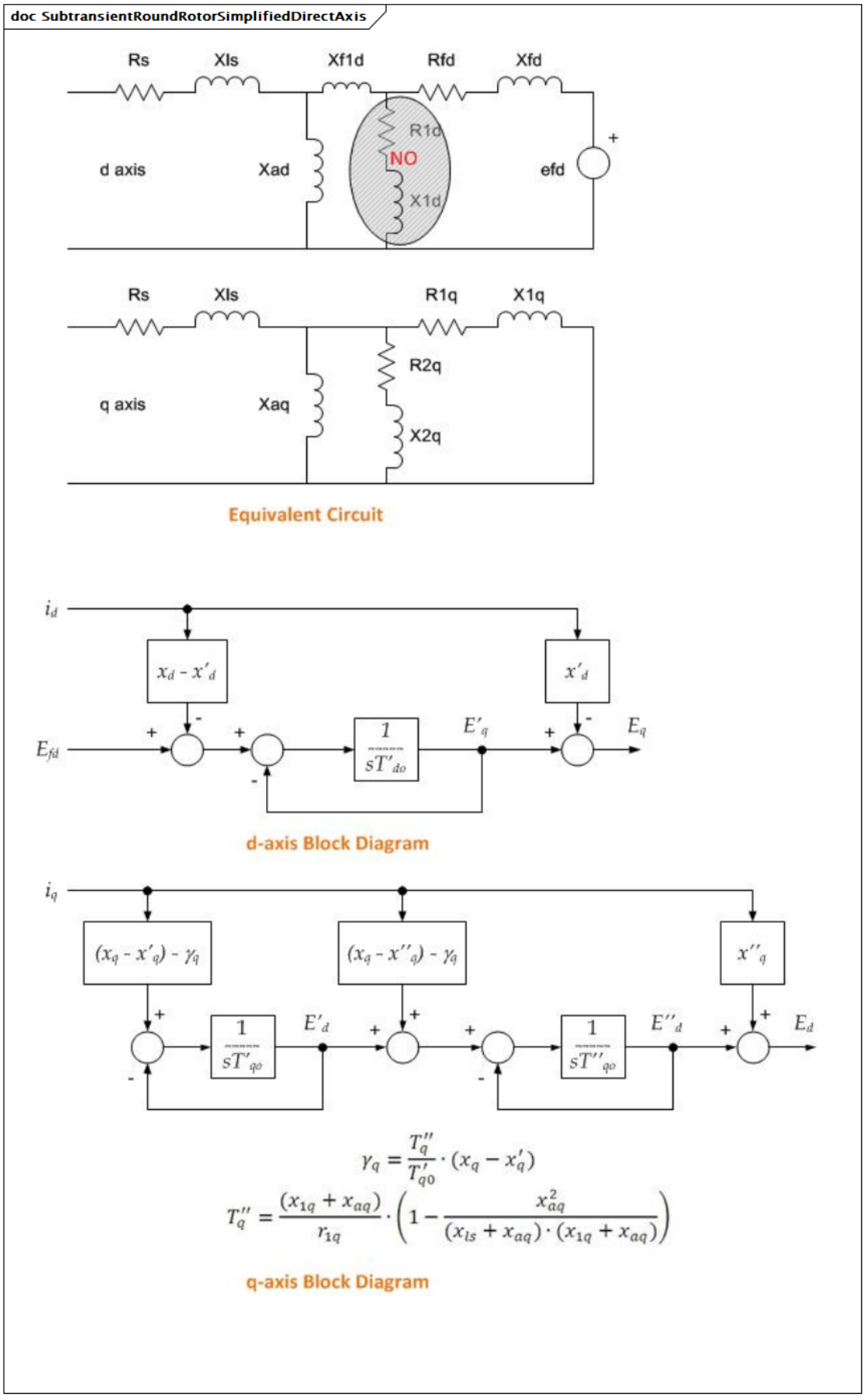

Figure 81 – SubtransientRoundRotorSimplifiedDirectAxis

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Figure 81: The subtransient round rotor simplified direct-axis type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplifiedDirectAxis and .rotorKind = roundRotor.

It is a simplified version of a subtransient round rotor type with no damper circuit on the d-axis. This simplification contributes to a reduction in computational effort by reducing the order of the differential equation system. The q-axis has two equivalent rotor windings. Magnetic coupling between the direct- and quadrature- axes is ignored.

Parameter details:

1) The effects of speed variation are neglected.
2) No armature resistance.
3) No saturation effects.
4) No subtransient saliency.
5) Direct-axis with field circuit only, no equivalent damper circuit. Approximation: X'd = X''d.
6) Quadrature-axis with two equivalent damper circuit, X''q not = X'q.
7) For generators and motors the following input parameters are needed: Xd, X'd, T'do and Xq, X'q, X''q, T'qo, T''qo.

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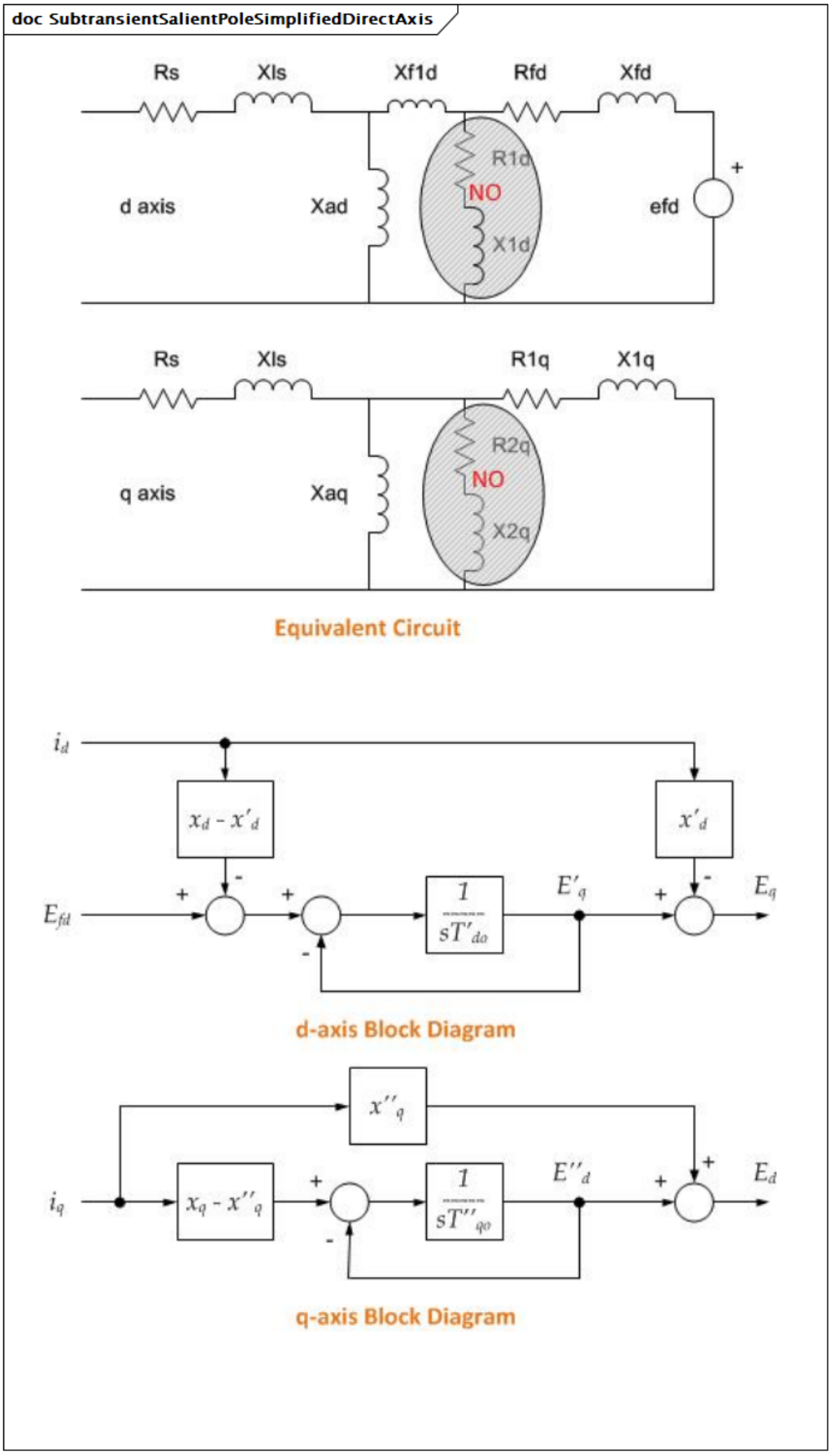

Figure 82 – SubtransientSalientPoleSimplifiedDirectAxis

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Figure 82: The subtransient salient pole simplified direct-axis type is modelled using one of the concrete child classes of the SynchronousMachineDetailed class. If the SynchronousMachineTimeConstantReactance class is used, .modelType = subtransientSimplifiedDirectAxis and .rotorKind = salientPole. It is a simplified version of a subtransient salient pole type with no damper circuit on the d-axis. This simplification contributes to a reduction in computational effort by reducing the order of the differential equation system. The q-axis has only one equivalent rotor winding. Magnetic coupling between the direct- and quadrature- axes is ignored.

Parameter details:

1) The effects of speed variation are neglected.
2) No armature resistance.
3) No saturation effects.
4) No subtransient saliency.
5) Direct-axis with field circuit only, no equivalent damper circuit. Approximation: X'd = X''d.
6) Quadrature-axis with only one equivalent damper circuit, which is labelled as subtransient (X''q), X''q =X'q.
7) For generators and motors the following input parameters are needed: Xd, X'd, T'do and Xq, X''q, T''qo.

Table 666 shows all attributes of SynchronousMachineTimeConstantReactance.

Table 666 – Attributes of SynchronousMachineDynamics:: SynchronousMachineTimeConstantReactance

name	mult	type	description
rotorType	0..1	RotorKind	Type of rotor on physical machine.
modelType	1..1	SynchronousMachineModelKind	Type of synchronous machine model used in dynamic simulation applications.
ks	1..1	Float	Saturation loading correction factor (Ks) (>= 0). Used only by type J model. Typical value = 0.
xDirectSync	1..1	PU	Direct-axis synchronous reactance (Xd) (>= SynchronousMachineTimeConstantReactance.xDirectTrans). The quotient of a sustained value of that AC component of armature voltage that is produced by the total direct-axis flux due to direct-axis armature current and the value of the AC component of this current, the machine running at rated speed. Typical value = 1,8.
xDirectTrans	1..1	PU	Direct-axis transient reactance (unsaturated) (X'd) (>= SynchronousMachineTimeConstantReactance.xDirectSubtrans). Typical value = 0,5.
xDirectSubtrans	1..1	PU	Direct-axis subtransient reactance (unsaturated) (X''d) (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
xQuadSync	1..1	PU	Quadrature-axis synchronous reactance (Xq) (>= SynchronousMachineTimeConstantReactance.xQuadTrans). The ratio of the component of reactive armature voltage, due to the quadrature-axis component of armature current, to this component of current, under steady state conditions and at rated frequency. Typical value = 1,6.
xQuadTrans	0..1	PU	Quadrature-axis transient reactance (X'q) (>= SynchronousMachineTimeConstantReactance.xQuadSubtrans). Typical value = 0,3.

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name	mult	type	description
xQuadSubtrans	1..1	PU	Quadrature-axis subtransient reactance (X''q) (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
tpdo	1..1	Seconds	Direct-axis transient rotor time constant (T'do) (> SynchronousMachineTimeConstantReactance.tppdo). Typical value = 5.
tppdo	1..1	Seconds	Direct-axis subtransient rotor time constant (T''do) (> 0). Typical value = 0,03.
tpqo	0..1	Seconds	Quadrature-axis transient rotor time constant (T'qo) (> SynchronousMachineTimeConstantReactance.tppqo). Typical value = 0,5.
tppqo	1..1	Seconds	Quadrature-axis subtransient rotor time constant (T''qo) (> 0). Typical value = 0,03.
tc	1..1	Seconds	Damping time constant for “Canay” reactance (>= 0). Typical value = 0.
saturationFactorQAxis	0..1	Float	inherited from: SynchronousMachineDetailed
saturationFactor120QAxis	0..1	Float	inherited from: SynchronousMachineDetailed
efdBaseRatio	1..1	Float	inherited from: SynchronousMachineDetailed
ifdBaseType	1..1	IfdBaseKind	inherited from: SynchronousMachineDetailed
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 667 shows all association ends of SynchronousMachineTimeConstantReactance with other classes.

Table 667 – Association ends of SynchronousMachineDynamics:: SynchronousMachineTimeConstantReactance with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	inherited from: SynchronousMachineDynamics

14.3.3.4.6 SynchronousMachineEquivalentCircuit

Inheritance path = SynchronousMachineDetailed: SynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

The electrical equations for all variations of the synchronous models are based on the SynchronousEquivalentCircuit diagram for the direct- and quadrature- axes.

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Equations for conversion between equivalent circuit and time constant reactance forms:

Xd = Xad + Xl

X’d = Xl + Xad x Xfd / (Xad + Xfd)

X”d = Xl + Xad x Xfd x X1d / (Xad x Xfd + Xad x X1d + Xfd x X1d)

Xq = Xaq + Xl

X’q = Xl + Xaq x X1q / (Xaq + X1q)

X”q = Xl + Xaq x X1q x X2q / (Xaq x X1q + Xaq x X2q + X1q x X2q)

T’do = (Xad + Xfd) / (omega0 x Rfd)

T”do = (Xad x Xfd + Xad x X1d + Xfd x X1d) / (omega0 x R1d x (Xad + Xfd)

T’qo = (Xaq + X1q) / (omega0 x R1q)

T”qo = (Xaq x X1q + Xaq x X2q + X1q x X2q) / (omega0 x R2q x (Xaq + X1q)

Same equations using CIM attributes from SynchronousMachineTimeConstantReactance class on left of "=" and SynchronousMachineEquivalentCircuit class on right (except as noted):

xDirectSync = xad + RotatingMachineDynamics.statorLeakageReactance

xDirectTrans = RotatingMachineDynamics.statorLeakageReactance + xad x xfd / (xad + xfd)

xDirectSubtrans = RotatingMachineDynamics.statorLeakageReactance + xad x xfd x x1d / (xad x xfd + xad x x1d + xfd x x1d)

xQuadSync = xaq + RotatingMachineDynamics.statorLeakageReactance

xQuadTrans = RotatingMachineDynamics.statorLeakageReactance + xaq x x1q / (xaq+ x1q)

xQuadSubtrans = RotatingMachineDynamics.statorLeakageReactance + xaq x x1q x x2q / (xaq x x1q + xaq x x2q + x1q x x2q)

tpdo = (xad + xfd) / (2 x pi x nominal frequency x rfd)

tppdo = (xad x xfd + xad x x1d + xfd x x1d) / (2 x pi x nominal frequency x r1d x (xad + xfd)

tpqo = (xaq + x1q) / (2 x pi x nominal frequency x r1q)

tppqo = (xaq x x1q + xaq x x2q + x1q x x2q) / (2 x pi x nominal frequency x r2q x (xaq + x1q)

These are only valid for a simplified model where "Canay" reactance is zero.

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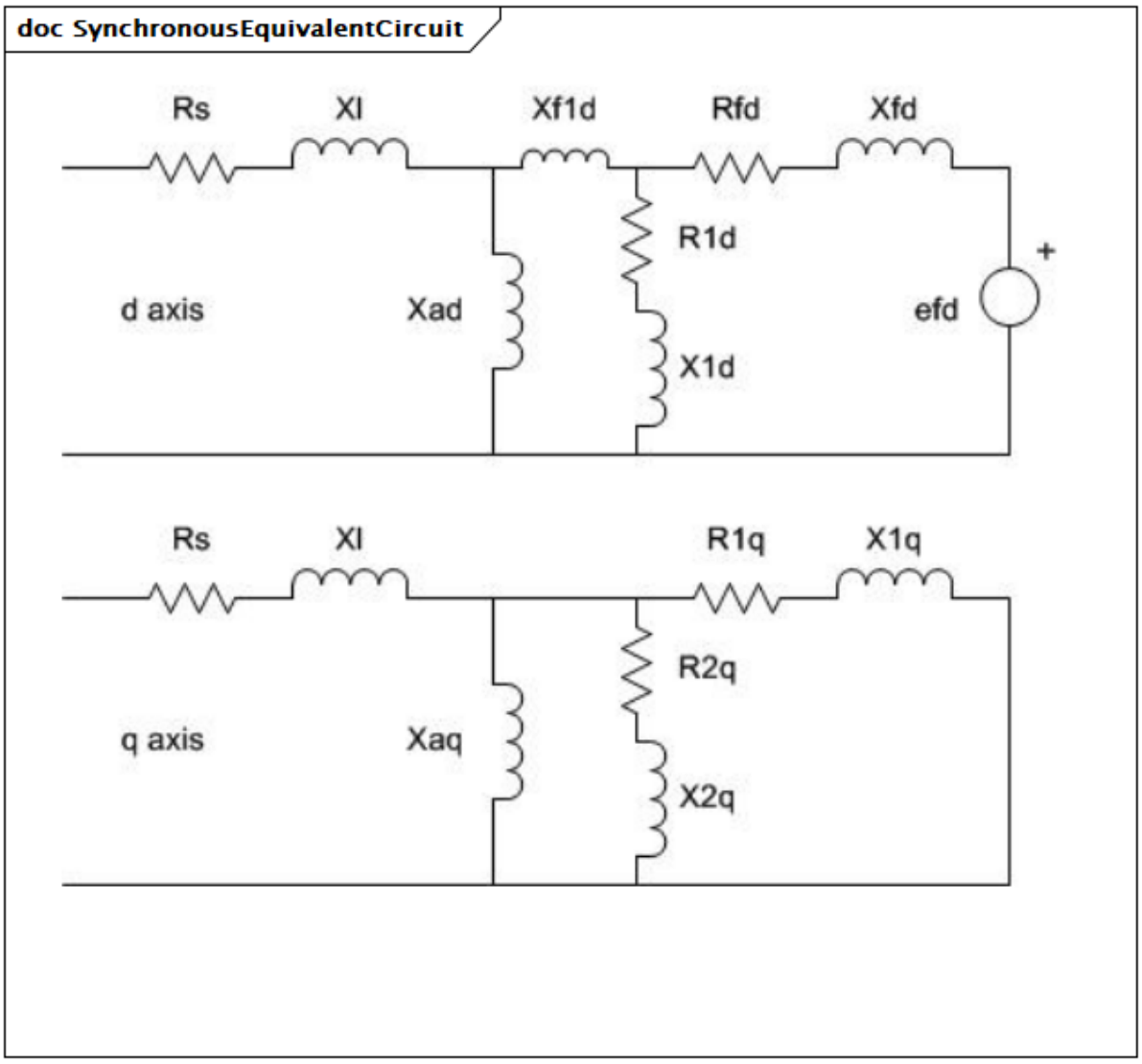

Figure 83 – SynchronousEquivalentCircuit

Figure 83: In each axis, the branches represent the stator leakage reactance (Xl) and resistance (Rs), the magnetizing reactance (Xad, Xaq), the physical field winding (Rfd, Xfd) on the rotor, and equivalent windings for eddy current flow in the rotor iron.

Table 668 shows all attributes of SynchronousMachineEquivalentCircuit.

Table 668 – Attributes of SynchronousMachineDynamics:: SynchronousMachineEquivalentCircuit

name	mult	type	description
xad	1..1	PU	Direct-axis mutual reactance.
rfd	1..1	PU	Field winding resistance.
xfd	1..1	PU	Field winding leakage reactance.
r1d	1..1	PU	Direct-axis damper 1 winding resistance.
x1d	1..1	PU	Direct-axis damper 1 winding leakage reactance.
xf1d	1..1	PU	Differential mutual (“Canay”) reactance.
xaq	1..1	PU	Quadrature-axis mutual reactance.
r1q	1..1	PU	Quadrature-axis damper 1 winding resistance.

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name	mult	type	description
x1q	1..1	PU	Quadrature-axis damper 1 winding leakage reactance.
r2q	1..1	PU	Quadrature-axis damper 2 winding resistance.
x2q	1..1	PU	Quadrature-axis damper 2 winding leakage reactance.
saturationFactorQAxis	0..1	Float	inherited from: SynchronousMachineDetailed
saturationFactor120QAxis	0..1	Float	inherited from: SynchronousMachineDetailed
efdBaseRatio	1..1	Float	inherited from: SynchronousMachineDetailed
ifdBaseType	1..1	IfdBaseKind	inherited from: SynchronousMachineDetailed
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 669 shows all association ends of SynchronousMachineEquivalentCircuit with other classes.

Table 669 – Association ends of SynchronousMachineDynamics:: SynchronousMachineEquivalentCircuit with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	inherited from: SynchronousMachineDynamics

14.3.3.5 Package AsynchronousMachineDynamics

14.3.3.5.1 General

An asynchronous machine model represents a (induction) generator or motor with no external connection to the rotor windings, e.g. a squirrel-cage induction machine.

The interconnection with the electrical network equations can differ among simulation tools. The program only needs to know the terminal to which this asynchronous machine is connected in order to establish the correct interconnection. The interconnection with the motor’s equipment could also differ due to input and output signals required by standard models.

The asynchronous machine model is used to model wind generators type 1 and type 2. For these, normal practice is to include the rotor flux transients and neglect the stator flux transients.

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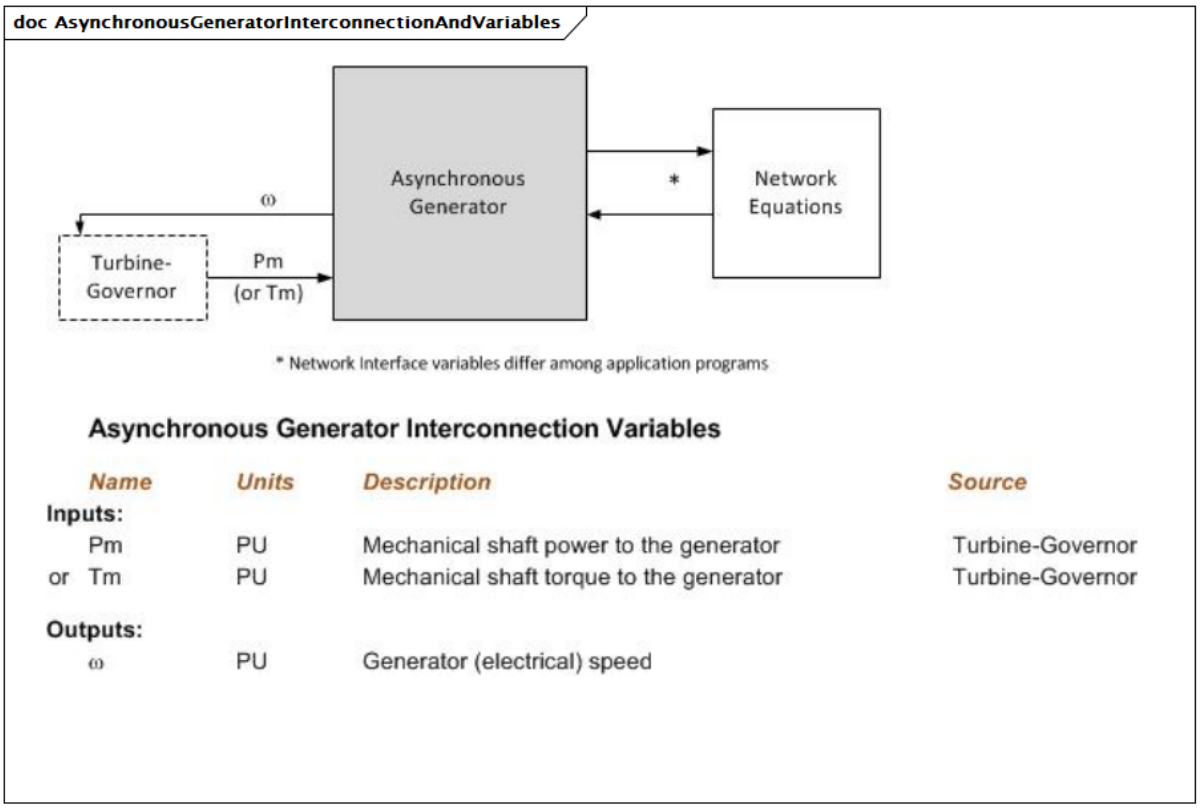

Figure 84 – AsynchronousGeneratorInterconnectionAndVariables

Figure 84: This diagram illustrates the inputs and outputs for an asynchronous generator.

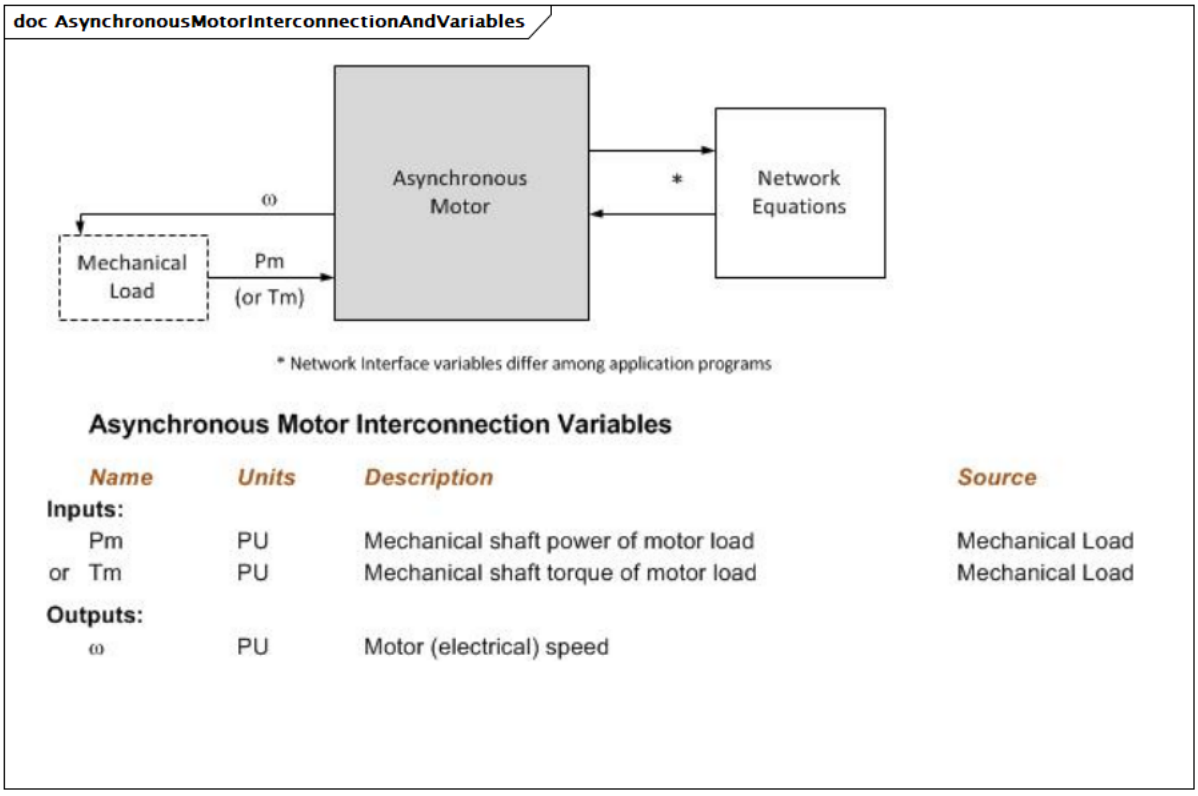

Figure 85 – AsynchronousMotorInterconnectionAndVariables

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Figure 85: This diagram illustrates the inputs and outputs for an asynchronous motor.

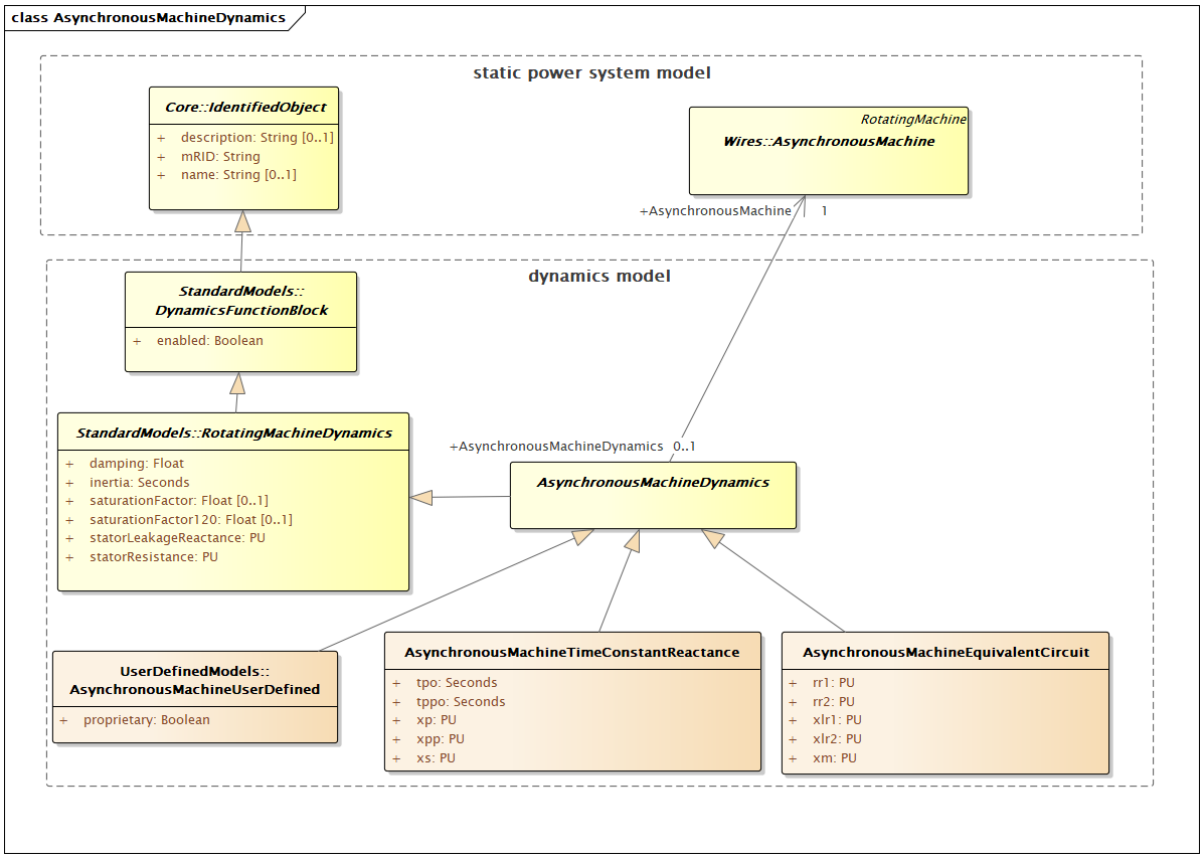

Figure 86 – Class diagram AsynchronousMachineDynamics:: AsynchronousMachineDynamics

Figure 86: This diagram shows static power system model classes and dynamics model classes related to the asynchronous machine dynamics model. An association between the AsynchronousMachineDynamics class and the AsynchronousMachine class provides a means for relating the dynamics model to the static model.

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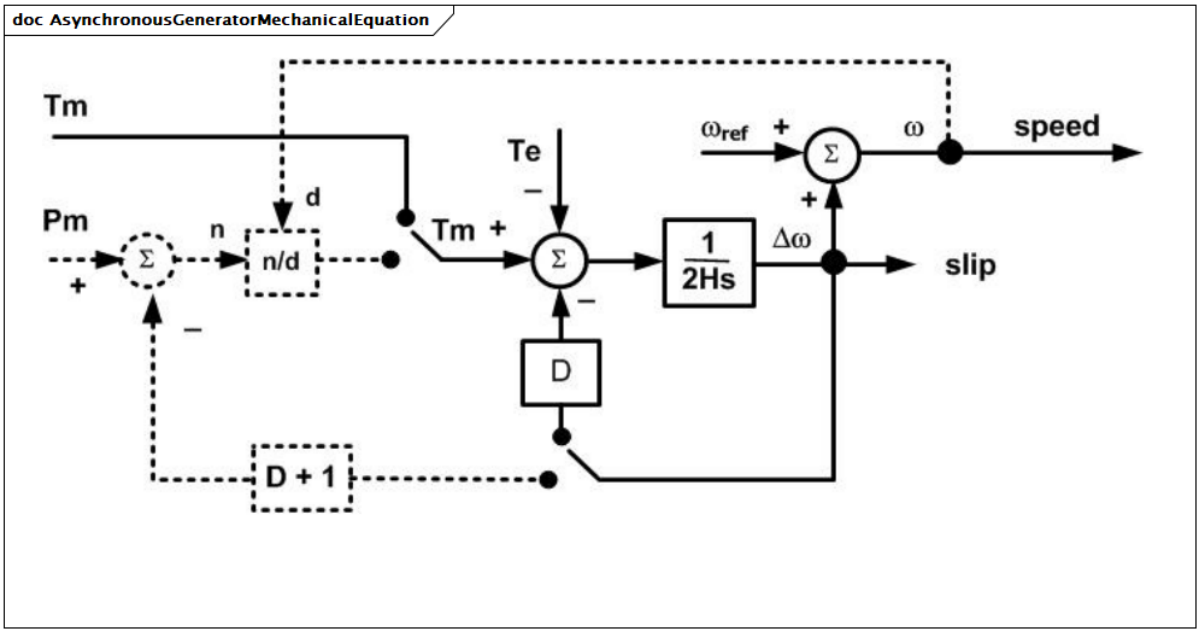

Figure 87 – AsynchronousGeneratorMechanicalEquation

Figure 87: The mechanical equations for all the variations of the asynchronous generator model are the same and can be represented by the AsynchronousGeneratorMechanicalEquation block diagram.

Parameter details:

1) All variables are PU on motor MVA base except angle, which is in radians.
2) omega0 is the system synchronous frequency in radians per s. omegaref is the PU system frequency (or isolated area frequency), which can be constant at the initial frequency or vary during a simulation depending on the application program.

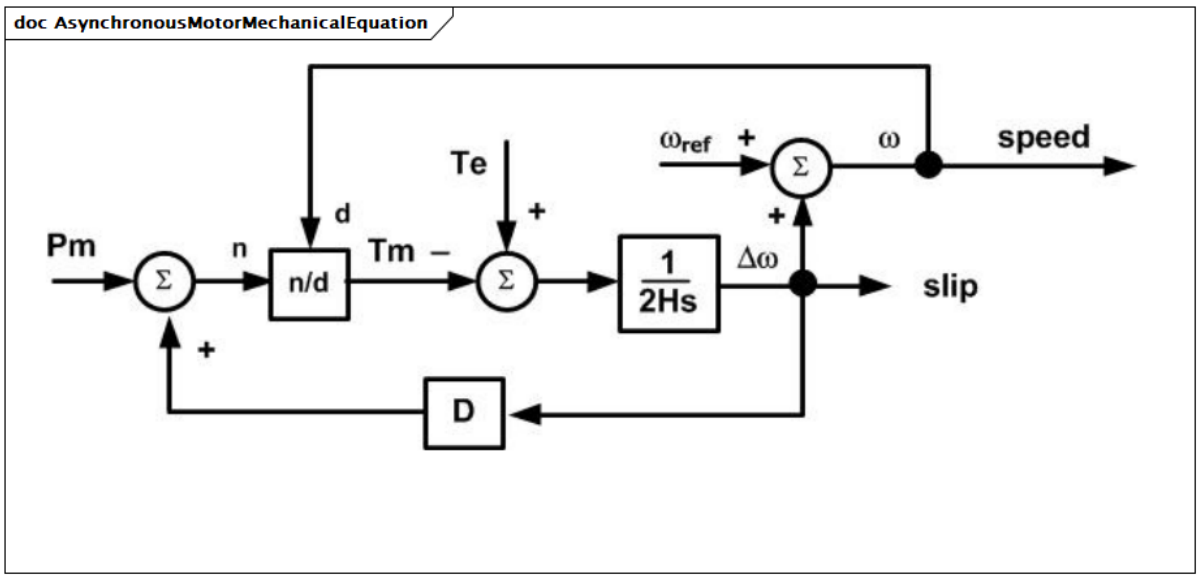

Figure 88 – AsynchronousMotorMechanicalEquation

Figure 88: The mechanical equations for all variations of the asynchronous motor model are the same and can be represented by the AsynchronousMotorMechanicalEquation diagram.

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Parameter details:

1) All variables are PU on motor MVA base except angle, which is in radians.
2) omega0 is the system synchronous frequency in radians per s. omegaref is the PU system frequency (or isolated area frequency), which can be constant at the initial frequency or vary during a simulation depending on the application program.

14.3.3.5.2 (abstract) AsynchronousMachineDynamics

Inheritance path = RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

Asynchronous machine whose behaviour is described by reference to a standard model expressed in either time constant reactance form or equivalent circuit form or by definition of a user-defined model.

Parameter details:

1) Asynchronous machine parameters such as Xl, Xs, etc. are actually used as inductances in the model, but are commonly referred to as reactances since, at nominal frequency, the PU values are the same. However, some references use the symbol L instead of X.

Table 670 shows all attributes of AsynchronousMachineDynamics.

Table 670 – Attributes of AsynchronousMachineDynamics:: AsynchronousMachineDynamics

name	mult	type	description
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 671 shows all association ends of AsynchronousMachineDynamics with other classes.

Table 671 – Association ends of AsynchronousMachineDynamics:: AsynchronousMachineDynamics with other classes

mult from	name	mult to	type	description
0..1	AsynchronousMachine	1..1	AsynchronousMachine	Asynchronous machine to which this asynchronous machine dynamics model applies.

14.3.3.5.3 AsynchronousMachineTimeConstantReactance

Inheritance path = AsynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

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Parameter details:

1) If X'' = X', a single cage (one equivalent rotor winding per axis) is modelled.
2) The “p” in the attribute names is a substitution for a “prime” in the usual parameter notation, e.g. tpo refers to T'o.

The parameters used for models expressed in time constant reactance form include:

– RotatingMachine.ratedS (MVAbase);
– RotatingMachineDynamics.damping (D);
– RotatingMachineDynamics.inertia (H);
– RotatingMachineDynamics.saturationFactor (S1);
– RotatingMachineDynamics.saturationFactor120 (S12);
– RotatingMachineDynamics.statorLeakageReactance (Xl);
– RotatingMachineDynamics.statorResistance (Rs);
– .xs (Xs);
– .xp (X');
– .xpp (X'');
– .tpo (T'o);
– .tppo (T''o).

Table 672 shows all attributes of AsynchronousMachineTimeConstantReactance.

Table 672 – Attributes of AsynchronousMachineDynamics:: AsynchronousMachineTimeConstantReactance

name	mult	type	description
xs	1..1	PU	Synchronous reactance (Xs) (>= AsynchronousMachineTimeConstantReactance.xp). Typical value = 1,8.
xp	1..1	PU	Transient reactance (unsaturated) (X') (>= AsynchronousMachineTimeConstantReactance.xpp). Typical value = 0,5.
xpp	1..1	PU	Subtransient reactance (unsaturated) (X'') (> RotatingMachineDynamics.statorLeakageReactance). Typical value = 0,2.
tpo	1..1	Seconds	Transient rotor time constant (T'o) (> AsynchronousMachineTimeConstantReactance.tppo). Typical value = 5.
tppo	1..1	Seconds	Subtransient rotor time constant (T''o) (> 0). Typical value = 0,03.
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 673 shows all association ends of AsynchronousMachineTimeConstantReactance with other classes.

Table 673 – Association ends of AsynchronousMachineDynamics:: AsynchronousMachineTimeConstantReactance with other classes

mult from	name	mult to	type	description
0..1	AsynchronousMachine	1..1	AsynchronousMachine	inherited from: AsynchronousMachineDynamics

14.3.3.5.4 AsynchronousMachineEquivalentCircuit

Inheritance path = AsynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

The electrical equations of all variations of the asynchronous model are based on the AsynchronousEquivalentCircuit diagram for the direct- and quadrature- axes, with two equivalent rotor windings in each axis.

Equations for conversion between equivalent circuit and time constant reactance forms:

Xs = Xm + Xl

X' = Xl + Xm x Xlr1 / (Xm + Xlr1)

X'' = Xl + Xm x Xlr1 x Xlr2 / (Xm x Xlr1 + Xm x Xlr2 + Xlr1 x Xlr2)

T'o = (Xm + Xlr1) / (omega0 x Rr1)

T''o = (Xm x Xlr1 + Xm x Xlr2 + Xlr1 x Xlr2) / (omega0 x Rr2 x (Xm + Xlr1)

Same equations using CIM attributes from AsynchronousMachineTimeConstantReactance class on left of "=" and AsynchronousMachineEquivalentCircuit class on right (except as noted):

xs = xm + RotatingMachineDynamics.statorLeakageReactance

xp = RotatingMachineDynamics.statorLeakageReactance + xm x xlr1 / (xm + xlr1)

xpp = RotatingMachineDynamics.statorLeakageReactance + xm x xlr1 x xlr2 / (xm x xlr1 + xm x xlr2 + xlr1 x xlr2)

tpo = (xm + xlr1) / (2 x pi x nominal frequency x rr1)

tppo = (xm x xlr1 + xm x xlr2 + xlr1 x xlr2) / (2 x pi x nominal frequency x rr2 x (xm + xlr1).

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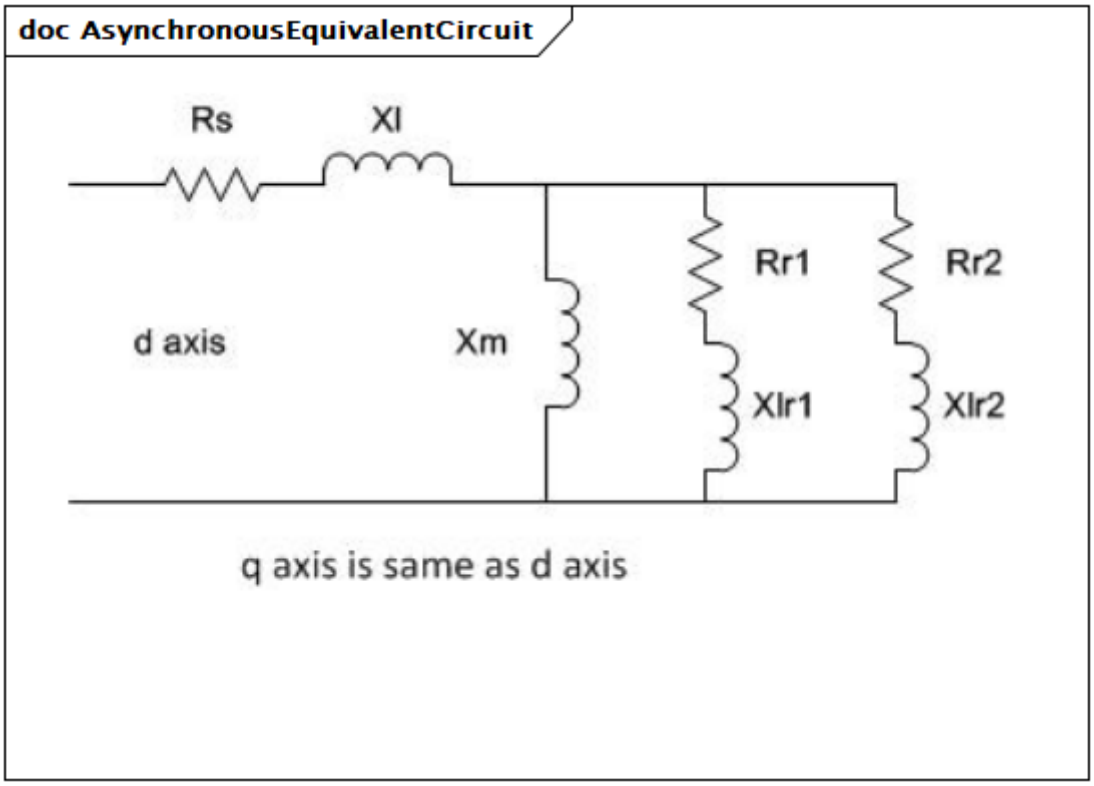

Figure 89 – AsynchronousEquivalentCircuit

Figure 89: In each axis, the branches represent the stator leakage reactance (Xl) and resistance (Rs), the magnetizing reactance (Xm), and the resistance and leakage reactance of equivalent windings (Rr1, Xlr1, etc.) on the rotor.

Table 674 shows all attributes of AsynchronousMachineEquivalentCircuit.

Table 674 – Attributes of AsynchronousMachineDynamics:: AsynchronousMachineEquivalentCircuit

name	mult	type	description
xm	1..1	PU	Magnetizing reactance.
rr1	1..1	PU	Damper 1 winding resistance.
xlr1	1..1	PU	Damper 1 winding leakage reactance.
rr2	1..1	PU	Damper 2 winding resistance.
xlr2	1..1	PU	Damper 2 winding leakage reactance.
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 675 shows all association ends of AsynchronousMachineEquivalentCircuit with other classes.

Table 675 – Association ends of AsynchronousMachineDynamics:: AsynchronousMachineEquivalentCircuit with other classes

mult from	name	mult to	type	description
0..1	AsynchronousMachine	1..1	AsynchronousMachine	inherited from: AsynchronousMachineDynamics

14.3.3.6 Package TurbineGovernorDynamics

14.3.3.6.1 General

The turbine-governor model is linked to one or two synchronous generators and determines the shaft mechanical power (Pm) or torque (Tm) for the generator model.

Unlike IEEE standard models for other function blocks, the three IEEE turbine-governor standard models (GovHydroIEEE0, GovHydroIEEE2, and GovSteamIEEE1) are documented in IEEE Transactions, not in IEEE standards. For that reason, diagrams are supplied for those models.

A 2012 IEEE report, Dynamic Models for Turbine-Governors in Power System Studies, provides updated information on a variety of models including IEEE, vendor and reliability authority models. Fully incorporating the results of that report into the CIM dynamics model is a future effort.

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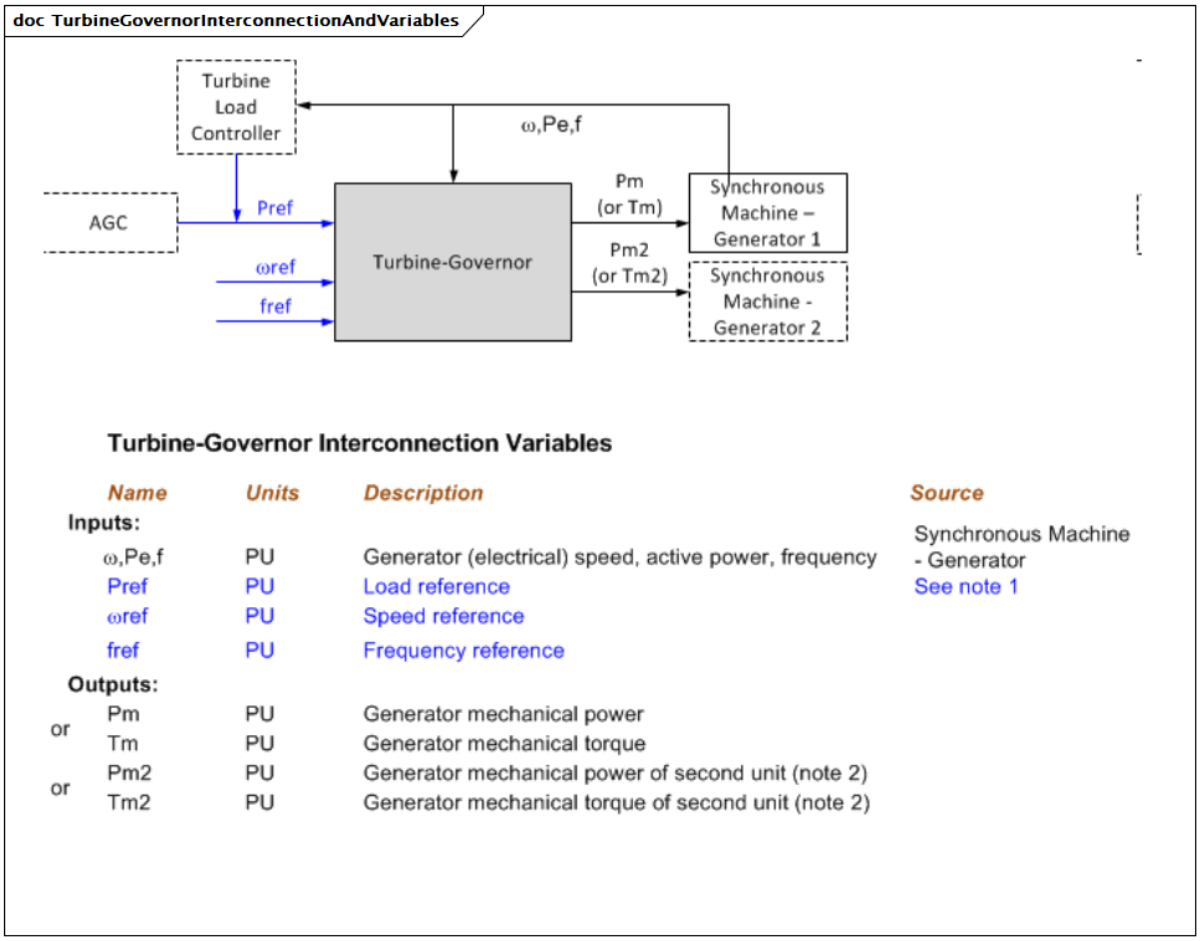

Turbine-Governor Interconnection Variables

	Name	*Units*	*Description*	*Source*
Inputs:				
	ω,Pe,f	PU	Generator (electrical) speed, active power, frequency	Synchronous Machine - Generator
	Pref	PU	Load reference	See note 1
	ωref	PU	Speed reference	
	fref	PU	Frequency reference	
Outputs:				
	Pm	PU	Generator mechanical power	
or	Tm	PU	Generator mechanical torque	
	Pm2	PU	Generator mechanical power of second unit (note 2)	
or	Tm2	PU	Generator mechanical torque of second unit (note 2)	

Figure 90 – TurbineGovernorInterconnectionAndVariables

Figure 90: This diagram illustrates the inputs and outputs for a turbine-governor. Some turbine-governor models are used in cross-compound interconnections, e.g. GovSteam1, where they are connected to two generators (synchronous machines).

Diagram details:

1) Pref is usually held constant for stability analysis, but can be determined by a user-defined model or, for long-term dynamics, an area-wide automatic generation control (AGC) model. For some models, Pref has units of PU speed instead of PU power.
2) The vast majority of turbine-governor models have MWbase as a parameter. This parameter corresponds to GeneratingUnit.ratedGrossMaxP from the Production package, but is specified explicitly in the Dynamics package to support use of a different value by transient analysis tools, if needed.

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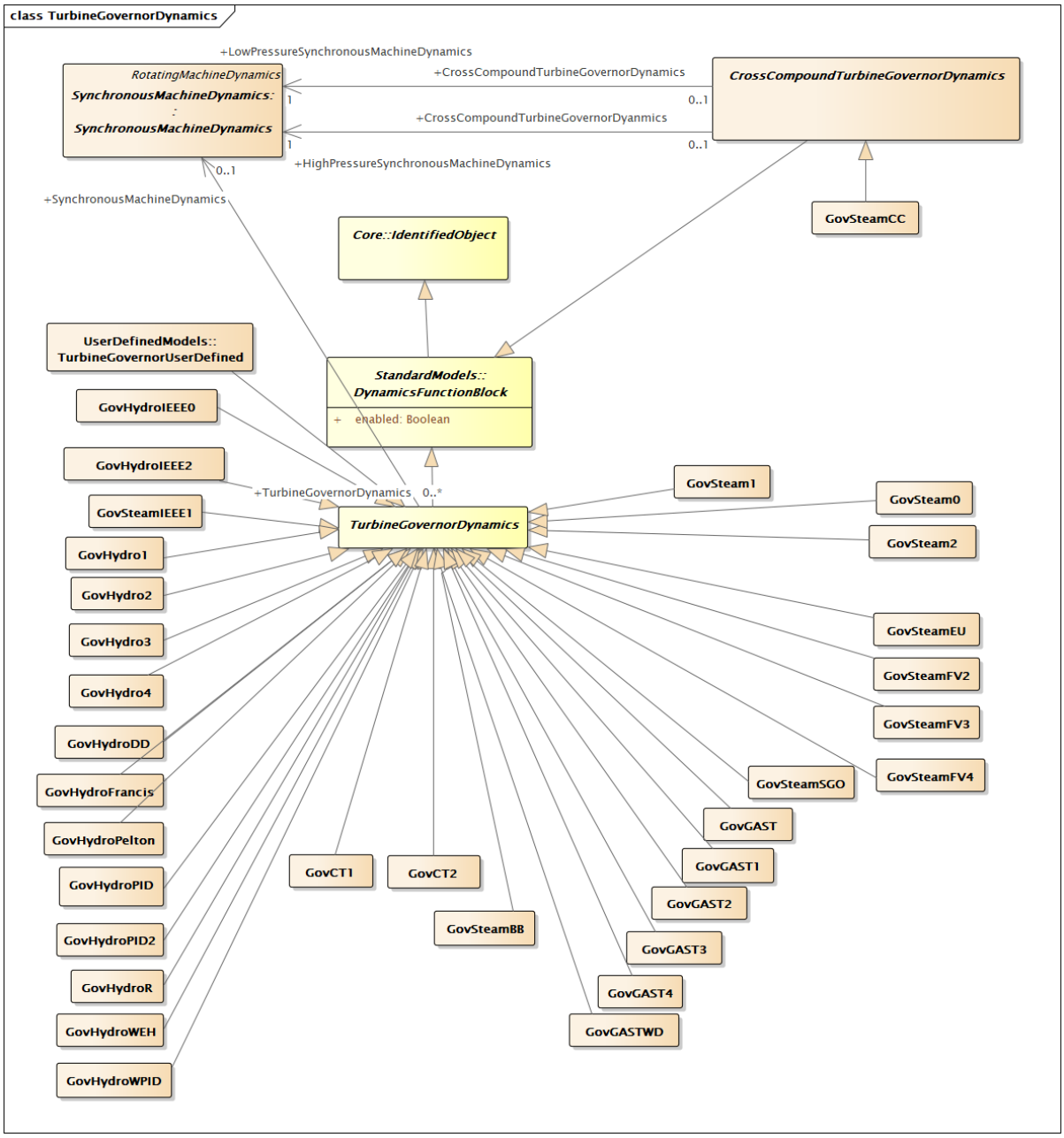

Figure 91 – Class diagram TurbineGovernorDynamics::TurbineGovernorDynamics

Figure 91: This diagram shows dynamic standard model turbine-governor classes and their inheritance from the TurbineGovernorDynamics class.

14.3.3.6.2 (abstract) CrossCompoundTurbineGovernorDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Turbine-governor cross-compound function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 676 shows all attributes of CrossCompoundTurbineGovernorDynamics.

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Table 676 – Attributes of TurbineGovernorDynamics:: CrossCompoundTurbineGovernorDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 677 shows all association ends of CrossCompoundTurbineGovernorDynamics with other classes.

Table 677 – Association ends of TurbineGovernorDynamics:: CrossCompoundTurbineGovernorDynamics with other classes

mult from	name	mult to	type	description
0..1	HighPressureSynchronousMachineDynamics	1..1	SynchronousMachineDynamics	High-pressure synchronous machine with which this cross-compound turbine governor is associated.
0..1	LowPressureSynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Low-pressure synchronous machine with which this cross-compound turbine governor is associated.

14.3.3.6.3 (abstract) TurbineGovernorDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Turbine-governor function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 678 shows all attributes of TurbineGovernorDynamics.

Table 678 – Attributes of TurbineGovernorDynamics::TurbineGovernorDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 679 shows all association ends of TurbineGovernorDynamics with other classes.

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Table 679 – Association ends of TurbineGovernorDynamics:: TurbineGovernorDynamics with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	Synchronous machine model with which this turbine-governor model is associated. TurbineGovernorDynamics shall have either an association to SynchronousMachineDynamics or to AsynchronousMachineDynamics.
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Asynchronous machine model with which this turbine-governor model is associated. TurbineGovernorDynamics shall have either an association to SynchronousMachineDynamics or to AsynchronousMachineDynamics.

14.3.3.6.4 GovHydroIEEE0

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE simplified hydro governor-turbine model. Used for mechanical-hydraulic and electro-hydraulic turbine governors, with or without steam feedback. Typical values given are for mechanical-hydraulic turbine-governor.

Reference: IEEE Transactions on Power Apparatus and Systems, November/December 1973, Volume PAS-92, Number 6, Dynamic Models for Steam and Hydro Turbines in Power System Studies, page 1904.

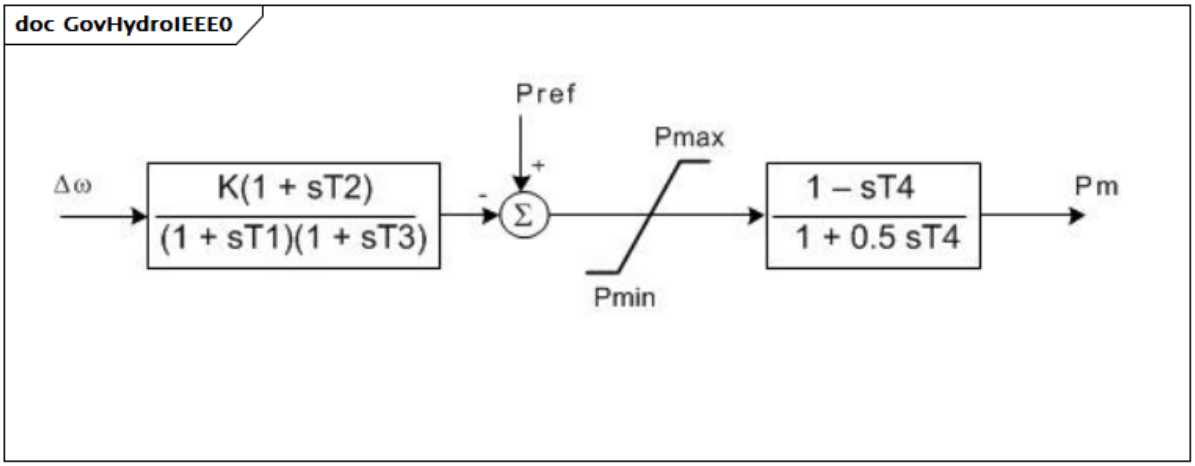

Figure 92 – GovHydroIEEE0

Figure 92: This diagram describes the CIM model interpretation of the IEEE simplified hydro governor-turbine model.

Table 680 shows all attributes of GovHydroIEEE0.

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Table 680 – Attributes of TurbineGovernorDynamics::GovHydroIEEE0

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
k	1..1	PU	Governor gain (K).
t1	1..1	Seconds	Governor lag time constant (T1) (>= 0). Typical value = 0,25.
t2	1..1	Seconds	Governor lead time constant (T2) (>= 0). Typical value = 0.
t3	1..1	Seconds	Gate actuator time constant (T3) (>= 0). Typical value = 0,1.
t4	1..1	Seconds	Water starting time (T4) (>= 0).
pmax	1..1	PU	Gate maximum (Pmax) (> GovHydroIEEE0.pmin).
pmin	1..1	PU	Gate minimum (Pmin) (< GovHydroIEEE.pmax).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 681 shows all association ends of GovHydroIEEE0 with other classes.

Table 681 – Association ends of TurbineGovernorDynamics:: GovHydroIEEE0 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.5 GovHydroIEEE2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE hydro turbine governor model represents plants with straightforward penstock configurations and hydraulic-dashpot governors.

Reference: IEEE Transactions on Power Apparatus and Systems, November/December 1973, Volume PAS-92, Number 6, Dynamic Models for Steam and Hydro Turbines in Power System Studies, page 1904.

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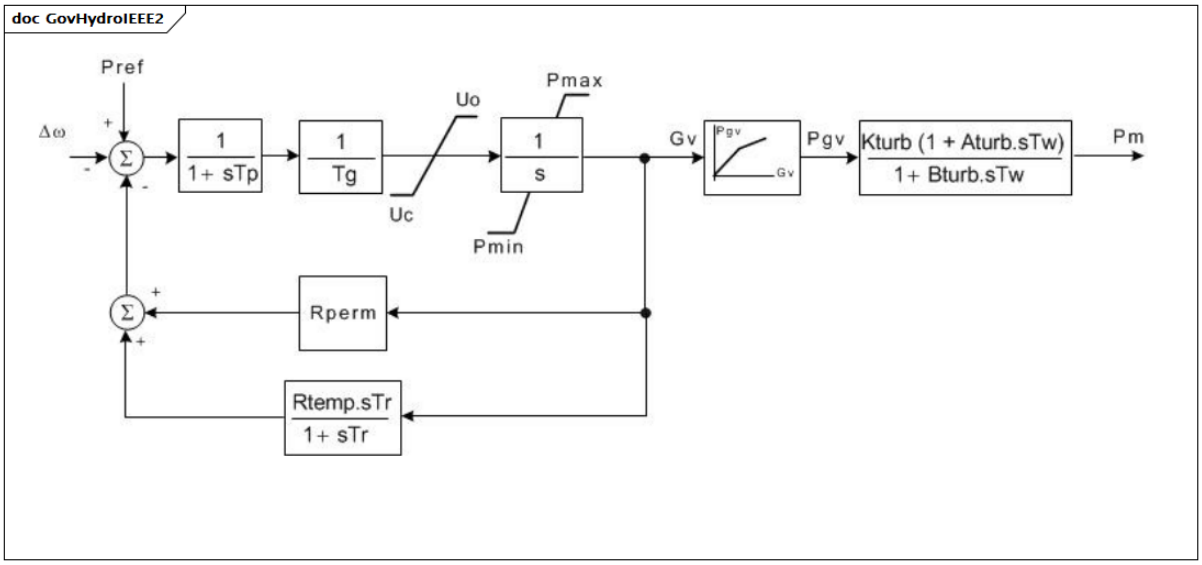

Figure 93 – GovHydroIEEE2

Figure 93: This diagram describes the CIM model interpretation of the IEEE hydro turbine governor model representing plants with straightforward penstock configurations and hydraulic-dashpot governors.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) This turbine governor model is a reasonable approximation to the performance of simple hydro plants for small motions of the gates about a given load, but it represents the variation of water inertia effect with load only if the values of the turbine parameters are properly adjusted by the user as load is varied. For idealized turbine model, set Kturb = 1, Aturb = -1, Bturb = 0,5.
3) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero values, the default hydro curve (between 0,0, 0,0 and 1,0, 1,0) will be used.

Table 682 shows all attributes of GovHydroIEEE2.

Table 682 – Attributes of TurbineGovernorDynamics::GovHydroIEEE2

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg) (>= 0). Typical value = 0,5.
tp	1..1	Seconds	Pilot servo valve time constant (Tp) (>= 0). Typical value = 0,03.
uo	1..1	Float	Maximum gate opening velocity (Uo). Unit = PU / s. Typical value = 0,1.
uc	1..1	Float	Maximum gate closing velocity (Uc) (<0). Typical value = -0,1.
pmax	1..1	PU	Maximum gate opening (Pmax) (> GovHydroIEEE2.pmin). Typical value = 1.
pmin	1..1	PU	Minimum gate opening (Pmin) (<GovHydroIEEE2.pmax). Typical value = 0.

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name	mult	type	description
rperm	1..1	PU	Permanent droop (Rperm). Typical value = 0,05.
rtemp	1..1	PU	Temporary droop (Rtemp). Typical value = 0,5.
tr	1..1	Seconds	Dashpot time constant (Tr) (>= 0). Typical value = 12.
tw	1..1	Seconds	Water inertia time constant (Tw) (>= 0). Typical value = 2.
kturb	1..1	PU	Turbine gain (Kturb). Typical value = 1.
aturb	1..1	PU	Turbine numerator multiplier (Aturb). Typical value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (> 0). Typical value = 0,5.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 683 shows all association ends of GovHydroIEEE2 with other classes.

Table 683 – Association ends of TurbineGovernorDynamics:: GovHydroIEEE2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

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14.3.3.6.6 GovSteamIEEE1

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE steam turbine governor model.

Reference: IEEE Transactions on Power Apparatus and Systems, November/December 1973, Volume PAS-92, Number 6, Dynamic Models for Steam and Hydro Turbines in Power System Studies, page 1904.

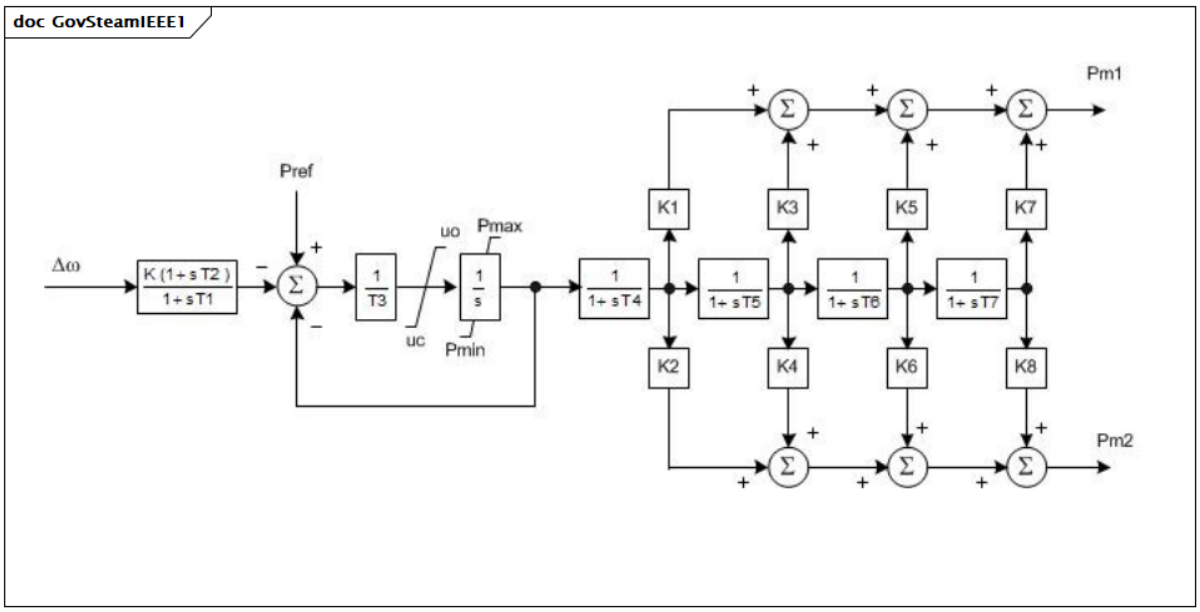

Figure 94 – GovSteamIEEE1

Figure 94: This diagram describes the CIM model interpretation of the IEEE steam turbine governor model.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) For a tandem-compound turbine the parameters K2, K4, K6, and K8 are ignored. For a cross-compound turbine, two generators are connected to this turbine-governor model.
3) Each generator shall be represented in the load flow by data on its own MVA base. The values of K1, K3, K5, and K7 shall be specified to describe the proportionate development of power on the first turbine shaft. K2, K4, K6, and K8 shall describe the second turbine shaft. Normally K1 + K3 + K5 + K7 = 1,0 and K2 + K4 + K6 + K8 = 1,0 (if second generator is present).
4) The division of power between the two shafts is in proportion to the values of MVA bases of the two generators. The initial condition load flow should, therefore, have the two generators loaded to the same fraction of each one's MVA base.

Table 684 shows all attributes of GovSteamIEEE1.

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Table 684 – Attributes of TurbineGovernorDynamics::GovSteamIEEE1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
k	1..1	PU	Governor gain (reciprocal of droop) (K) (> 0). Typical value = 25.
t1	1..1	Seconds	Governor lag time constant (T1) (>= 0). Typical value = 0.
t2	1..1	Seconds	Governor lead time constant (T2) (>= 0). Typical value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3) (> 0). Typical value = 0,1.
uo	1..1	Float	Maximum valve opening velocity (Uo) (> 0). Unit = PU / s. Typical value = 1.
uc	1..1	Float	Maximum valve closing velocity (Uc) (< 0). Unit = PU / s. Typical value = -10.
pmax	1..1	PU	Maximum valve opening (Pmax) (> GovSteamIEEE1.pmin). Typical value = 1.
pmin	1..1	PU	Minimum valve opening (Pmin) (>= 0 and < GovSteamIEEE1.pmax). Typical value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4) (>= 0). Typical value = 0,3.
k1	1..1	Float	Fraction of HP shaft power after first boiler pass (K1). Typical value = 0,2.
k2	1..1	Float	Fraction of LP shaft power after first boiler pass (K2). Typical value = 0.
t5	1..1	Seconds	Time constant of second boiler pass (T5) (>= 0). Typical value = 5.
k3	1..1	Float	Fraction of HP shaft power after second boiler pass (K3). Typical value = 0,3.
k4	1..1	Float	Fraction of LP shaft power after second boiler pass (K4). Typical value = 0.
t6	1..1	Seconds	Time constant of third boiler pass (T6) (>= 0). Typical value = 0,5.
k5	1..1	Float	Fraction of HP shaft power after third boiler pass (K5). Typical value = 0,5.
k6	1..1	Float	Fraction of LP shaft power after third boiler pass (K6). Typical value = 0.
t7	1..1	Seconds	Time constant of fourth boiler pass (T7) (>= 0). Typical value = 0.
k7	1..1	Float	Fraction of HP shaft power after fourth boiler pass (K7). Typical value = 0.
k8	1..1	Float	Fraction of LP shaft power after fourth boiler pass (K8). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 685 shows all association ends of GovSteamIEEE1 with other classes.

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Table 685 – Association ends of TurbineGovernorDynamics:: GovSteamIEEE1 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.7 GovCT1

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

General model for any prime mover with a PID governor, used primarily for combustion turbine and combined cycle units.

This model can be used to represent a variety of prime movers controlled by PID governors. It is suitable, for example, for the representation of:

– gas turbine and single shaft combined cycle turbines

– diesel engines with modern electronic or digital governors

– steam turbines where steam is supplied from a large boiler drum or a large header whose pressure is substantially constant over the period under study

– simple hydro turbines in dam configurations where the water column length is short and water inertia effects are minimal.

Additional information on this model is available in the 2012 IEEE report, Dynamic Models for Turbine-Governors in Power System Studies, 3.1.2.3 pages 3-4 (GGOV1).

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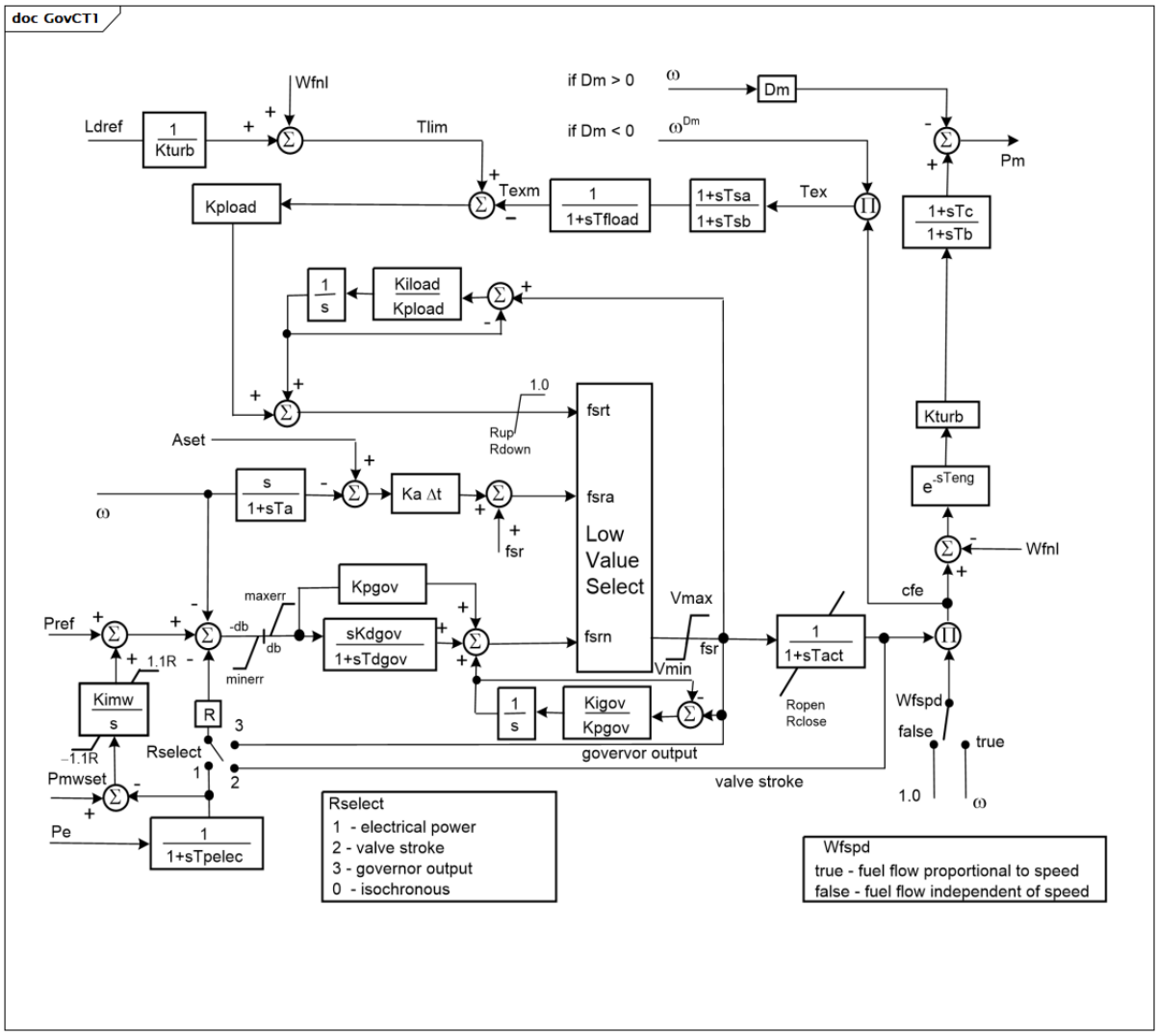

Figure 95 – GovCT1

Figure 95: This diagram describes the GovCT1 turbine governor model.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) The range of fuel valve travel and of fuel flow is unity. Thus the largest possible value of Vmax is 1,0 and the smallest possible value of Vmin is zero. Vmax can, however, be reduced below unity to represent a loading limit imposed by the operator or a supervisory control system. For gas turbines Vmin should normally be greater than zero and less than Wfnl to represent a minimum firing limit. The value of the fuel flow at maximum output shall be less than, or equal to unity, depending on the value of Kturb.
3) The load limiter module can be used to impose a maximum output limit such as an exhaust temperature limit. To do this the time constant Tfload should be set to represent the time constant in the measurement of temperature (or other signal), and the gains of the limiter, Kpload and Kiload, should be set to give prompt stable control when on limit. The load limit can be deactivated by setting the parameter Ldref to a high value.

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4) This model includes a simple representation of a supervisory load controller. This controller is active if the parameter Kimw is non-zero. The load controller is a slow acting reset loop that adjusts the speed/load reference of the turbine governor to hold the electrical power output of the unit at its initial condition value. This value is stored in the parameter Pmwset (which equals Pe, initial generated power) when the model is initialized, and can be changed thereafter. The load controller shall be adjusted to respond gently relative to the speed governor.
5) The parameters Aset, Ka, and Ta describe an acceleration limiter. Ta shall be non-zero, but the acceleration limiter can be disabled by setting Aset to a large value, such as 1.
6) The parameters Tsa and Tsb are provided to augment the exhaust gas temperature measurement subsystem in gas turbines. For example, they can be set to values such as 4,0 and 5,0, to represent the "radiation shield" element of large gas turbines.
7) The parameters Rup and Rdown specify the maximum rate of increase and decrease of the output of the load limit controller (Kpload / Kiload). These parameters should normally be set, or defaulted to 99 / -99, but can be given particular values to represent the temperature limit controls of some General ElectricTM heavy-duty engine controls. [Footnote: GE heavy-duty engine controls are an example of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.]
8) The fuel flow command fsr is determined by whichever is lowest of fsrt, fsra, and fsrn. Although not explicitly shown in the GovCT1 diagram, the signals that are not in control track fsr so that they do not "windup" beyond that value. This represents GE gas turbine control practice but might not be true for other controller designs.
9) As shown in the GovCT1 diagram, when Kpgov is non-zero, the governor PI control is implemented to "track" fsr to prevent windup when fsr is limited by another signal (fsrt, fsra) or Vmax/Vmin. If Kpgov is zero, the integral path is implemented directly. The same applies to the load limiter PI control with regard to Kpload.
10) The parameter delta t in the block diagram is the simulation time step. It is not exchanged with the dataset. In cases where applications are using variable time step the results of the simulations could be different. In such situations the application applies a default value which is 0,001.

Table 686 shows all attributes of GovCT1.

Table 686 – Attributes of TurbineGovernorDynamics::GovCT1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical value = 0,04.
rselect	1..1	DroopSignalFeedbackKind	Feedback signal for droop (Rselect). Typical value = electricalPower.
tpelec	1..1	Seconds	Electrical power transducer time constant (Tpelec) (> 0). Typical value = 1.
maxerr	1..1	PU	Maximum value for speed error signal (maxerr) (> GovCT1.minerr). Typical value = 0,05.
minerr	1..1	PU	Minimum value for speed error signal (minerr) (< GovCT1.maxerr). Typical value = -0,05.
kpgov	1..1	PU	Governor proportional gain (Kpgov). Typical value = 10.
kigov	1..1	PU	Governor integral gain (Kigov). Typical value = 2.
kdgov	1..1	PU	Governor derivative gain (Kdgov). Typical value = 0.
tdgov	1..1	Seconds	Governor derivative controller time constant (Tdgov) (>= 0). Typical value = 1.

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name	mult	type	description
vmax	1..1	PU	Maximum valve position limit (Vmax) (> GovCT1.vmin). Typical value = 1.
vmin	1..1	PU	Minimum valve position limit (Vmin) (< GovCT1.vmax). Typical value = 0,15.
tact	1..1	Seconds	Actuator time constant (Tact) (>= 0). Typical value = 0,5.
kturb	1..1	PU	Turbine gain (Kturb) (> 0). Typical value = 1,5.
wfnl	1..1	PU	No load fuel flow (Wfnl). Typical value = 0,2.
tb	1..1	Seconds	Turbine lag time constant (Tb) (> 0). Typical value = 0,5.
tc	1..1	Seconds	Turbine lead time constant (Tc) (>= 0). Typical value = 0.
wfspd	1..1	Boolean	Switch for fuel source characteristic to recognize that fuel flow, for a given fuel valve stroke, can be proportional to engine speed (Wfspd). true = fuel flow proportional to speed (for some gas turbines and diesel engines with positive displacement fuel injectors) false = fuel control system keeps fuel flow independent of engine speed. Typical value = true.
teng	1..1	Seconds	Transport time delay for diesel engine used in representing diesel engines where there is a small but measurable transport delay between a change in fuel flow setting and the development of torque (Teng) (>= 0). Teng should be zero in all but special cases where this transport delay is of particular concern. Typical value = 0.
tfload	1..1	Seconds	Load-limiter time constant (Tfload) (> 0). Typical value = 3.
kpload	1..1	PU	Load limiter proportional gain for PI controller (Kpload). Typical value = 2.
kiload	1..1	PU	Load limiter integral gain for PI controller (Kiload). Typical value = 0,67.
ldref	1..1	PU	Load limiter reference value (Ldref). Typical value = 1.
dm	1..1	PU	Speed sensitivity coefficient (Dm). Dm can represent either the variation of the engine power with the shaft speed or the variation of maximum power capability with shaft speed. If it is positive it describes the falling slope of the engine speed verses power characteristic as speed increases. A slightly falling characteristic is typical for reciprocating engines and some aero-derivative turbines. If it is negative the engine power is assumed to be unaffected by the shaft speed, but the maximum permissible fuel flow is taken to fall with falling shaft speed. This is characteristic of single-shaft industrial turbines due to exhaust temperature limits. Typical value = 0.
ropen	1..1	Float	Maximum valve opening rate (Ropen). Unit = PU / s. Typical value = 0,10.
rclose	1..1	Float	Minimum valve closing rate (Rclose). Unit = PU / s. Typical value = -0,1.
kimw	1..1	PU	Power controller (reset) gain (Kimw). The default value of 0,01 corresponds to a reset time of 100 s. A value of 0,001 corresponds to a relatively slow-acting load controller. Typical value = 0,01.
aset	1..1	Float	Acceleration limiter setpoint (Aset). Unit = PU / s. Typical value = 0,01.

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name	mult	type	description
ka	1..1	PU	Acceleration limiter gain (Ka). Typical value = 10.
ta	1..1	Seconds	Acceleration limiter time constant (Ta) (> 0). Typical value = 0,1.
db	1..1	PU	Speed governor deadband in PU speed (db). In the majority of applications, it is recommended that this value be set to zero. Typical value = 0.
tsa	1..1	Seconds	Temperature detection lead time constant (Tsa) (>= 0). Typical value = 4.
tsb	1..1	Seconds	Temperature detection lag time constant (Tsb) (>= 0). Typical value = 5.
rup	1..1	PU	Maximum rate of load limit increase (Rup). Typical value = 99.
rdown	1..1	PU	Maximum rate of load limit decrease (Rdown). Typical value = -99.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 687 shows all association ends of GovCT1 with other classes.

Table 687 – Association ends of TurbineGovernorDynamics::GovCT1 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.8 GovCT2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

General governor with frequency-dependent fuel flow limit. This model is a modification of the GovCT1 model in order to represent the frequency-dependent fuel flow limit of a specific gas turbine manufacturer.

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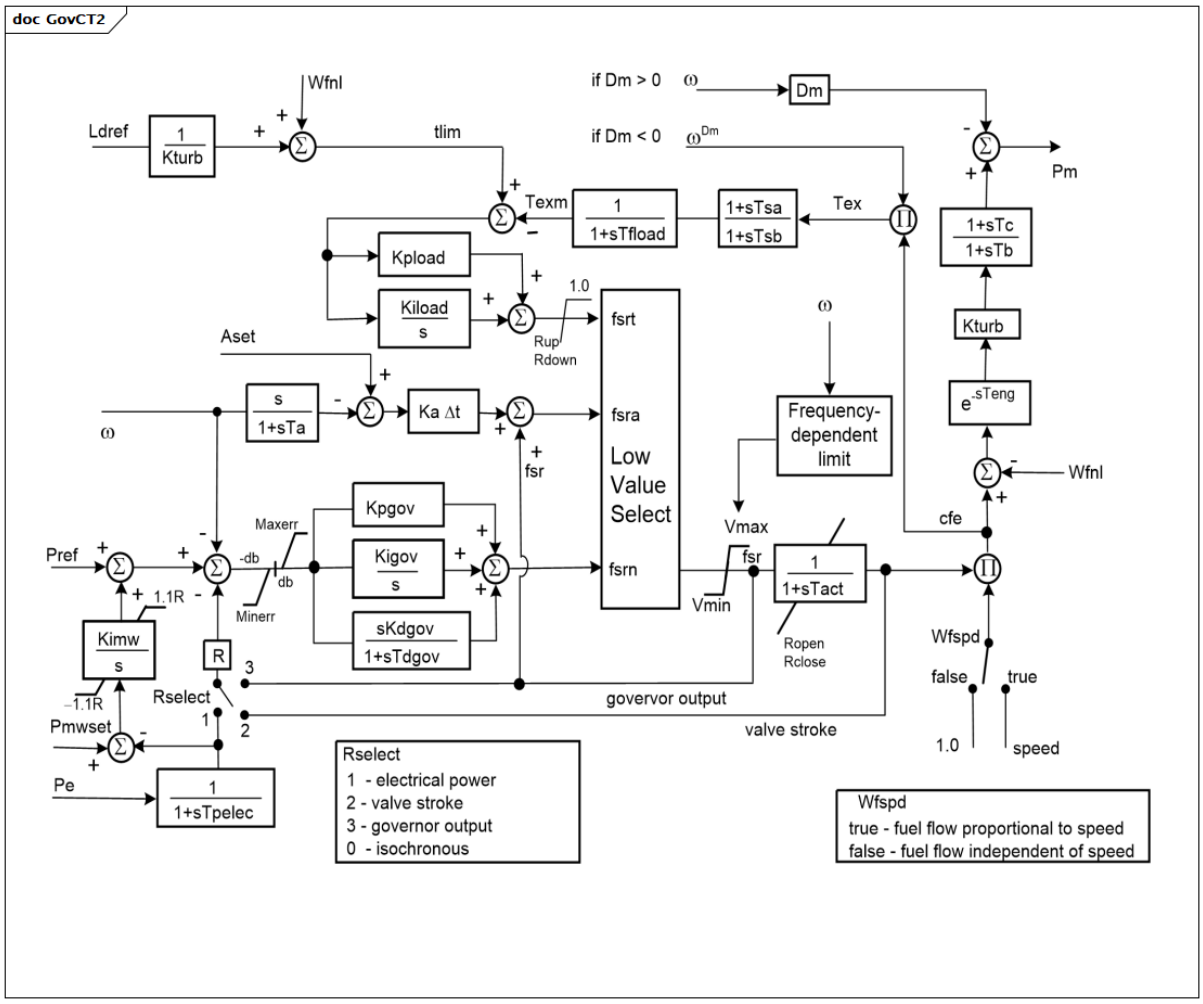

Figure 96 – GovCT2

Figure 96: This diagram describes the GovCT2 turbine governor model.

Parameter Details:

1) The frequency-dependent limit reduces the Vmax limit on fuel flow signal (fsr). For normal operation, the limiter performs no action. When frequency (generator speed) drops below Flim1, the highest frequency data point, the desired value for the power limit (Plim) is determined by linear interpolation between associated data pairs. The maximum value of the limiter (Vmax) will ramp to the fsr value corresponding to Plim (Plim / Kturb + Wfnl) at the Prate ramp rate. Plim will be updated as the frequency changes. If the frequency subsequently rises back above Flim1, the value of the limit will ramp up at the rate Prate back to the normal value of Vmax.
2) The frequency-dependent limit is defined by a set of up to 10 pairs of points relating frequency (generator speed in Hz) and power. The points shall be monotonically decreasing in both the magnitude of frequency and power. If fewer than 10 points are needed to define the relationship, values of zero shall be entered for the remaining frequencies and power limits. The value of the last non-zero data pair is used as a lower limit, that is, the last two values are not extrapolated to calculate lower power limits. If there is only one set of frequency and power limits, those values will be used as a single limit of power applied at that frequency and below.
3) Refer to the GovCT1 model for other parameter details. Aside from the frequency-dependent limit, GovCT2 is identical to GovCT1, except that the temperature fuel command fsrt does not track fsr when it is not in control. Instead, it stays at its upper limit (1).

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4) The Kpgov/Kigov and Kpload/Kiload controllers include tracking logic to ensure smooth transfer between active controllers. This logic is not shown on the GovCT2 diagram.
5) Pmwset equals Pe, initial generated power.
6) The parameter delta t in the block diagram is the simulation time step. It is not exchanged with the dataset. In cases where applications are using variable time step the results of the simulations could be different. In such situations the application applies a default value which is 0,001.

Table 688 shows all attributes of GovCT2.

Table 688 – Attributes of TurbineGovernorDynamics::GovCT2

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical value = 0,05.
rselect	1..1	DroopSignalFeedbackKind	Feedback signal for droop (Rselect). Typical value = electricalPower.
tpelec	1..1	Seconds	Electrical power transducer time constant (Tpelec) (>= 0). Typical value = 2,5.
maxerr	1..1	PU	Maximum value for speed error signal (Maxerr) (> GovCT2.minerr). Typical value = 1.
minerr	1..1	PU	Minimum value for speed error signal (Minerr) (< GovCT2.maxerr). Typical value = -1.
kpgov	1..1	PU	Governor proportional gain (Kpgov). Typical value = 4.
kigov	1..1	PU	Governor integral gain (Kigov). Typical value = 0,45.
kdgov	1..1	PU	Governor derivative gain (Kdgov). Typical value = 0.
tdgov	1..1	Seconds	Governor derivative controller time constant (Tdgov) (>= 0). Typical value = 1.
vmax	1..1	PU	Maximum valve position limit (Vmax) (> GovCT2.vmin). Typical value = 1.
vmin	1..1	PU	Minimum valve position limit (Vmin) (< GovCT2.vmax). Typical value = 0,175.
tact	1..1	Seconds	Actuator time constant (Tact) (>= 0). Typical value = 0,4.
kturb	1..1	PU	Turbine gain (Kturb). Typical value = 1,9168.
wfnl	1..1	PU	No load fuel flow (Wfnl). Typical value = 0,187.
tb	1..1	Seconds	Turbine lag time constant (Tb) (>= 0). Typical value = 0,1.
tc	1..1	Seconds	Turbine lead time constant (Tc) (>= 0). Typical value = 0.
wfspd	1..1	Boolean	Switch for fuel source characteristic to recognize that fuel flow, for a given fuel valve stroke, can be proportional to engine speed (Wfspd). true = fuel flow proportional to speed (for some gas turbines and diesel engines with positive displacement fuel injectors) false = fuel control system keeps fuel flow independent of engine speed. Typical value = false.

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name	mult	type	description
teng	1..1	Seconds	Transport time delay for diesel engine used in representing diesel engines where there is a small but measurable transport delay between a change in fuel flow setting and the development of torque (Teng) (>= 0). Teng should be zero in all but special cases where this transport delay is of particular concern. Typical value = 0.
tfload	1..1	Seconds	Load limiter time constant (Tfload) (>= 0). Typical value = 3.
kpload	1..1	PU	Load limiter proportional gain for PI controller (Kpload). Typical value = 1.
kiload	1..1	PU	Load limiter integral gain for PI controller (Kiload). Typical value = 1.
ldref	1..1	PU	Load limiter reference value (Ldref). Typical value = 1.
dm	1..1	PU	Speed sensitivity coefficient (Dm). Dm can represent either the variation of the engine power with the shaft speed or the variation of maximum power capability with shaft speed. If it is positive it describes the falling slope of the engine speed verses power characteristic as speed increases. A slightly falling characteristic is typical for reciprocating engines and some aero-derivative turbines. If it is negative the engine power is assumed to be unaffected by the shaft speed, but the maximum permissible fuel flow is taken to fall with falling shaft speed. This is characteristic of single-shaft industrial turbines due to exhaust temperature limits. Typical value = 0.
ropen	1..1	Float	Maximum valve opening rate (Ropen). Unit = PU / s. Typical value = 99.
rclose	1..1	Float	Minimum valve closing rate (Rclose). Unit = PU / s. Typical value = -99.
kimw	1..1	PU	Power controller (reset) gain (Kimw). The default value of 0,01 corresponds to a reset time of 100 seconds. A value of 0,001 corresponds to a relatively slow-acting load controller. Typical value = 0.
aset	1..1	Float	Acceleration limiter setpoint (Aset). Unit = PU / s. Typical value = 10.
ka	1..1	PU	Acceleration limiter gain (Ka). Typical value = 10.
ta	1..1	Seconds	Acceleration limiter time constant (Ta) (>= 0). Typical value = 1.
db	1..1	PU	Speed governor deadband in PU speed (db). In the majority of applications, it is recommended that this value be set to zero. Typical value = 0.
tsa	1..1	Seconds	Temperature detection lead time constant (Tsa) (>= 0). Typical value = 0.
tsb	1..1	Seconds	Temperature detection lag time constant (Tsb) (>= 0). Typical value = 50.
rup	1..1	PU	Maximum rate of load limit increase (Rup). Typical value = 99.
rdown	1..1	PU	Maximum rate of load limit decrease (Rdown). Typical value = -99.
prate	1..1	PU	Ramp rate for frequency-dependent power limit (Prate). Typical value = 0,017.
flim1	1..1	Frequency	Frequency threshold 1 (Flim1). Unit = Hz. Typical value = 59.
plim1	1..1	PU	Power limit 1 (Plim1). Typical value = 0,8325.

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name	mult	type	description
flim2	1..1	Frequency	Frequency threshold 2 (Flim2). Unit = Hz. Typical value = 0.
plim2	1..1	PU	Power limit 2 (Plim2). Typical value = 0.
flim3	1..1	Frequency	Frequency threshold 3 (Flim3). Unit = Hz. Typical value = 0.
plim3	1..1	PU	Power limit 3 (Plim3). Typical value = 0.
flim4	1..1	Frequency	Frequency threshold 4 (Flim4). Unit = Hz. Typical value = 0.
plim4	1..1	PU	Power limit 4 (Plim4). Typical value = 0.
flim5	1..1	Frequency	Frequency threshold 5 (Flim5). Unit = Hz. Typical value = 0.
plim5	1..1	PU	Power limit 5 (Plim5). Typical value = 0.
flim6	1..1	Frequency	Frequency threshold 6 (Flim6). Unit = Hz. Typical value = 0.
plim6	1..1	PU	Power limit 6 (Plim6). Typical value = 0.
flim7	1..1	Frequency	Frequency threshold 7 (Flim7). Unit = Hz. Typical value = 0.
plim7	1..1	PU	Power limit 7 (Plim7). Typical value = 0.
flim8	1..1	Frequency	Frequency threshold 8 (Flim8). Unit = Hz. Typical value = 0.
plim8	1..1	PU	Power limit 8 (Plim8). Typical value = 0.
flim9	1..1	Frequency	Frequency threshold 9 (Flim9). Unit = Hz. Typical value = 0.
plim9	1..1	PU	Power Limit 9 (Plim9). Typical value = 0.
flim10	1..1	Frequency	Frequency threshold 10 (Flim10). Unit = Hz. Typical value = 0.
plim10	1..1	PU	Power limit 10 (Plim10). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 689 shows all association ends of GovCT2 with other classes.

Table 689 – Association ends of TurbineGovernorDynamics::GovCT2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.9 GovGAST

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Single shaft gas turbine.

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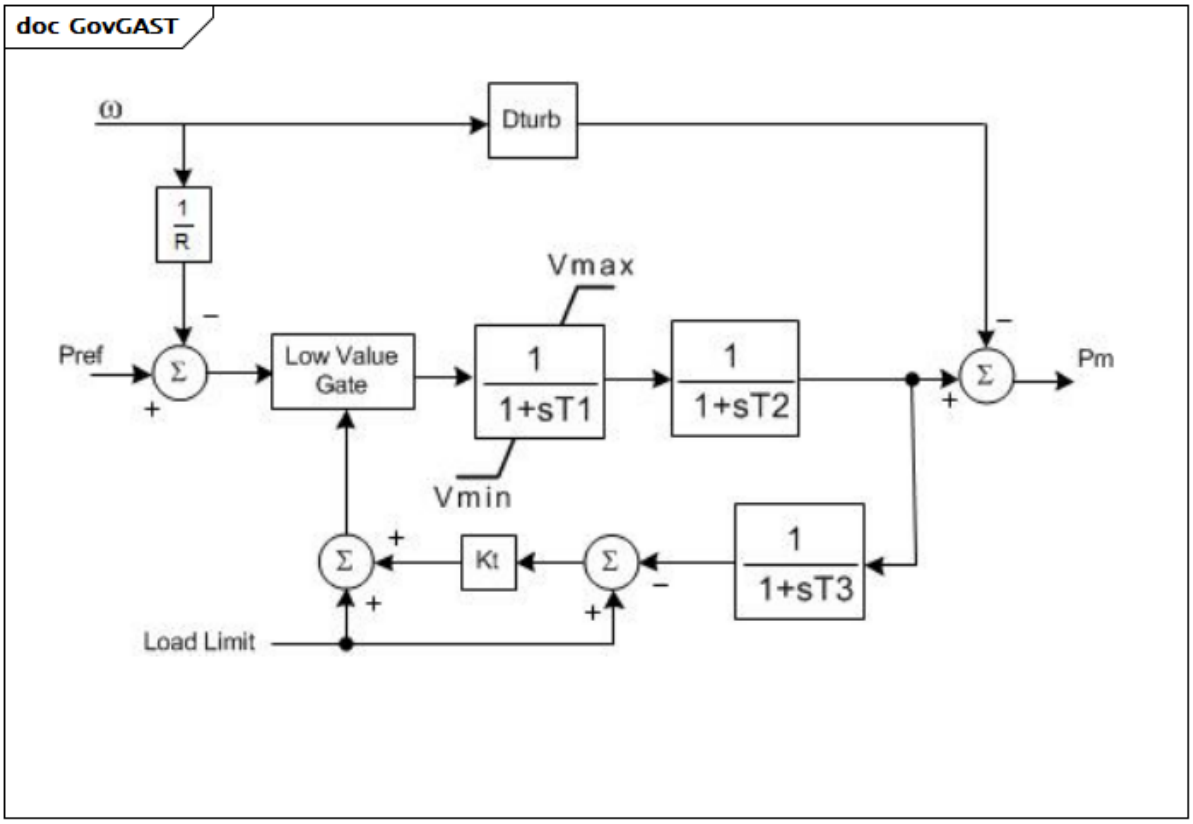

Figure 97 – GovGAST

Figure 97: This diagram describes the GovGAST turbine governor model.

Table 690 shows all attributes of GovGAST.

Table 690 – Attributes of TurbineGovernorDynamics::GovGAST

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R) (>0). Typical value = 0,04.
t1	1..1	Seconds	Governor mechanism time constant (T1) (>= 0). T1 represents the natural valve positioning time constant of the governor for small disturbances, as seen when rate limiting is not in effect. Typical value = 0,5.
t2	1..1	Seconds	Turbine power time constant (T2) (>= 0). T2 represents delay due to internal energy storage of the gas turbine engine. T2 can be used to give a rough approximation to the delay associated with acceleration of the compressor spool of a multi-shaft engine, or with the compressibility of gas in the plenum of a free power turbine of an aero-derivative unit, for example. Typical value = 0,5.
t3	1..1	Seconds	Turbine exhaust temperature time constant (T3) (>= 0). Typical value = 3.
at	1..1	PU	Ambient temperature load limit (Load Limit). Typical value = 1.
kt	1..1	PU	Temperature limiter gain (Kt). Typical value = 3.

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name	mult	type	description
vmax	1..1	PU	Maximum turbine power, PU of MWbase (Vmax) (> GovGAST.vmin). Typical value = 1.
vmin	1..1	PU	Minimum turbine power, PU of MWbase (Vmin) (< GovGAST.vmax). Typical value = 0.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical value = 0,18.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 691 shows all association ends of GovGAST with other classes.

Table 691 – Association ends of TurbineGovernorDynamics:: GovGAST with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.10 GovGAST1

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified single shaft gas turbine.

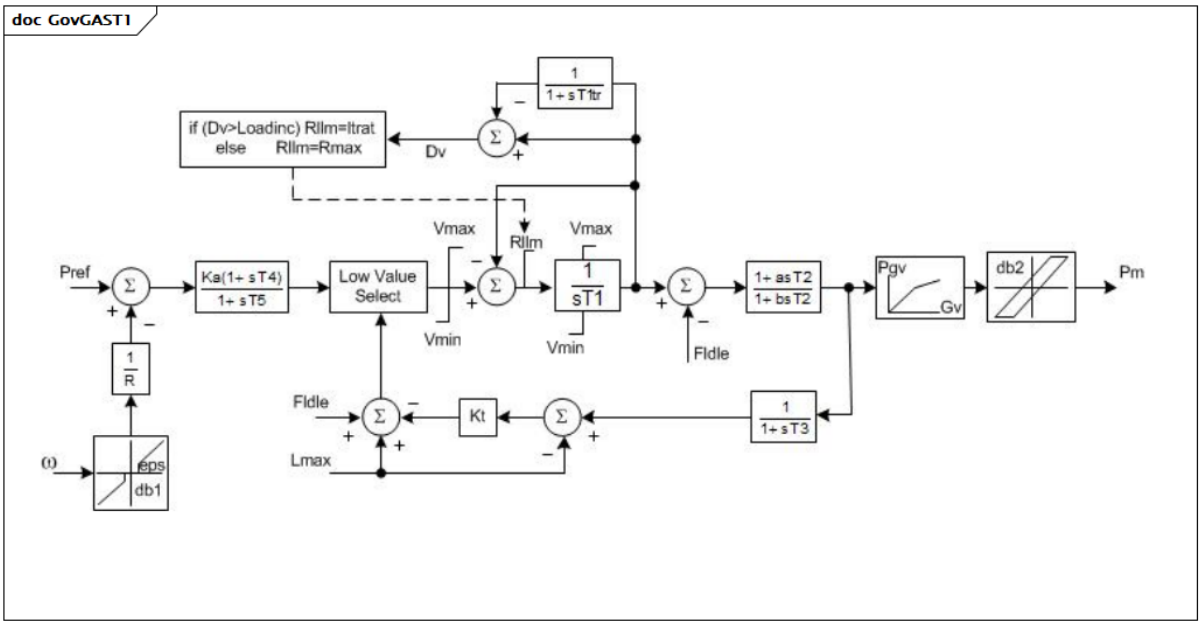

Figure 98 – GovGAST1

Figure 98: This diagram describes the GovGAST1 turbine governor model.

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Parameter details:

1) Rmax, Linc, Tltr, Ltrate represent a loading rate limit subsystem. For small excursions about a steady load, the fuel valve can open at a rate of Rmax PU / s. The time constant, Tltr, is used to obtain a long term average valve position. When a large output increase is called for, the fuel valve is allowed to open at the full rate only until the valve opening has moved ahead of the long term average position by Linc PU. After this the valve is permitted to open only at the rate, Ltrat, until the long term average position catches up.
2) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Vmax > 1,0, the input and output are scaled by Vmax. If Gv1 is a negative number or zero values, the default curve (between 0,0, 0,0 and 1,0, 1,0) will be used.

Table 692 shows all attributes of GovGAST1.

Table 692 – Attributes of TurbineGovernorDynamics::GovGAST1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R) (>0). Typical value = 0,04.
t1	1..1	Seconds	Governor mechanism time constant (T1) (>= 0). T1 represents the natural valve positioning time constant of the governor for small disturbances, as seen when rate limiting is not in effect. Typical value = 0,5.
t2	1..1	Seconds	Turbine power time constant (T2) (>= 0). T2 represents delay due to internal energy storage of the gas turbine engine. T2 can be used to give a rough approximation to the delay associated with acceleration of the compressor spool of a multi-shaft engine, or with the compressibility of gas in the plenum of the free power turbine of an aero-derivative unit, for example. Typical value = 0,5.
t3	1..1	Seconds	Turbine exhaust temperature time constant (T3) (>= 0). T3 represents delay in the exhaust temperature and load limiting system. Typical value = 3.
lmax	1..1	PU	Ambient temperature load limit (Lmax). Lmax is the turbine power output corresponding to the limiting exhaust gas temperature. Typical value = 1.
kt	1..1	PU	Temperature limiter gain (Kt). Typical value = 3.
vmax	1..1	PU	Maximum turbine power, PU of MWbase (Vmax) (> GovGAST1.vmin). Typical value = 1.
vmin	1..1	PU	Minimum turbine power, PU of MWbase (Vmin) (< GovGAST1.vmax). Typical value = 0.
fidle	1..1	PU	Fuel flow at zero power output (Fidle). Typical value = 0,18.
rmax	1..1	Float	Maximum fuel valve opening rate (Rmax). Unit = PU / s. Typical value = 1.
loadinc	1..1	PU	Valve position change allowed at fast rate (Loadinc). Typical value = 0,05.
tltr	1..1	Seconds	Valve position averaging time constant (Tltr) (>= 0). Typical value = 10.
ltrate	1..1	Float	Maximum long term fuel valve opening rate (Ltrate). Typical value = 0,02.
a	1..1	Float	Turbine power time constant numerator scale factor (a). Typical value = 0,8.

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name	mult	type	description
b	1..1	Float	Turbine power time constant denominator scale factor (b) (>0). Typical value = 1.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2,PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
ka	1..1	PU	Governor gain (Ka). Typical value = 0.
t4	1..1	Seconds	Governor lead time constant (T4) (>= 0). Typical value = 0.
t5	1..1	Seconds	Governor lag time constant (T5) (>= 0). If = 0, entire gain and lead-lag block is bypassed. Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 693 shows all association ends of GovGAST1 with other classes.

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Table 693 – Association ends of TurbineGovernorDynamics:: GovGAST1 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.11 GovGAST2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Gas turbine.

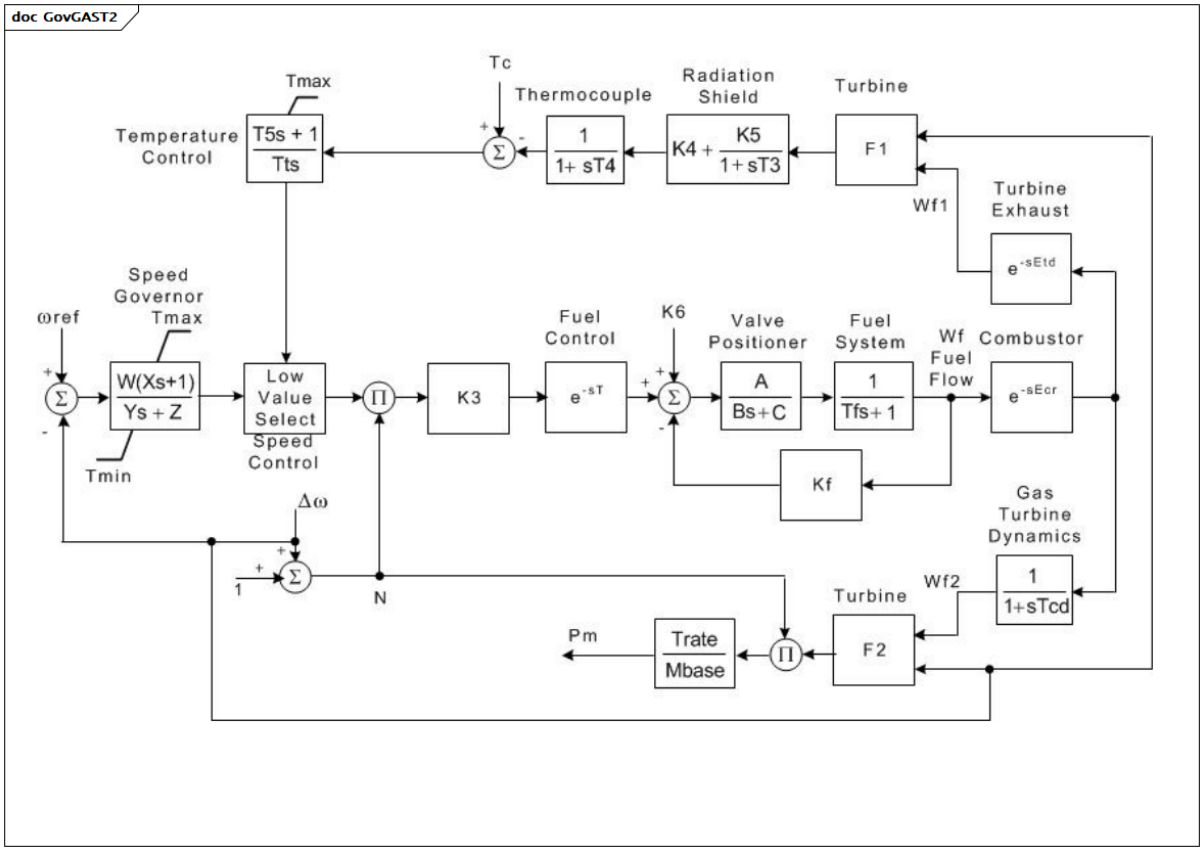

Figure 99 – GovGAST2

Figure 99: This diagram describes the GovGAST2 turbine governor model.

Parameter details:

1) Temperature control output is set to output of speed governor when temperature control input changes from positive to negative.
2) F1 = Tr – Af1(1 – Wf1) – Bf1(delta omega).
3) F2 = Af2 + Bf2(Wf2) – Cf2(delta omega).

Table 694 shows all attributes of GovGAST2.

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Table 694 – Attributes of TurbineGovernorDynamics::GovGAST2

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
w	1..1	PU	Governor gain (1/droop) on turbine rating (W).
x	1..1	Seconds	Governor lead time constant (X) (>= 0).
y	1..1	Seconds	Governor lag time constant (Y) (> 0).
z	1..1	Integer	Governor mode (Z). 1 = droop 0 = isochronous.
etd	1..1	Seconds	Turbine and exhaust delay (Etd) (>= 0).
tcd	1..1	Seconds	Compressor discharge time constant (Tcd) (>= 0).
trate	1..1	ActivePower	Turbine rating (Trate). Unit = MW.
t	1..1	Seconds	Fuel control time constant (T) (>= 0).
tmax	1..1	PU	Maximum turbine limit (Tmax) (> GovGAST2.tmin).
tmin	1..1	PU	Minimum turbine limit (Tmin) (< GovGAST2.tmax).
ecr	1..1	Seconds	Combustion reaction time delay (Ecr) (>= 0).
k3	1..1	PU	Ratio of fuel adjustment (K3).
a	1..1	Float	Valve positioner (A).
b	1..1	Float	Valve positioner (B).
c	1..1	Float	Valve positioner (C).
tf	1..1	Seconds	Fuel system time constant (Tf) (>= 0).
kf	1..1	PU	Fuel system feedback (Kf).
k5	1..1	PU	Gain of radiation shield (K5).
k4	1..1	PU	Gain of radiation shield (K4).
t3	1..1	Seconds	Radiation shield time constant (T3) (>= 0).
t4	1..1	Seconds	Thermocouple time constant (T4) (>= 0).
tt	1..1	Seconds	Temperature controller integration rate (Tt) (>= 0).
t5	1..1	Seconds	Temperature control time constant (T5) (>= 0).
af1	1..1	PU	Exhaust temperature parameter (Af1). Unit = PU temperature. Based on temperature in degrees C.
bf1	1..1	PU	(Bf1). Bf1 = E(1 – W) where E (speed sensitivity coefficient) is 0,55 to 0,65 x Tr. Unit = PU temperature. Based on temperature in degrees C.
af2	1..1	PU	Coefficient equal to 0,5(1-speed) (Af2).
bf2	1..1	PU	Turbine torque coefficient Khhv (depends on heating value of fuel stream in combustion chamber) (Bf2).
cf2	1..1	PU	Coefficient defining fuel flow where power output is 0 % (Cf2). Synchronous but no output. Typically 0,23 x Khhv (23 % fuel flow).
tr	1..1	Temperature	Rated temperature (Tr). Unit = °C depending on parameters Af1 and Bf1.
k6	1..1	PU	Minimum fuel flow (K6).

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name	mult	type	description
tc	1..1	Temperature	Temperature control (Tc). Unit = °F or °C depending on parameters Af1 and Bf1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 695 shows all association ends of GovGAST2 with other classes.

Table 695 – Association ends of TurbineGovernorDynamics:: GovGAST2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.12 GovGAST3

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Generic turbogas with acceleration and temperature controller.

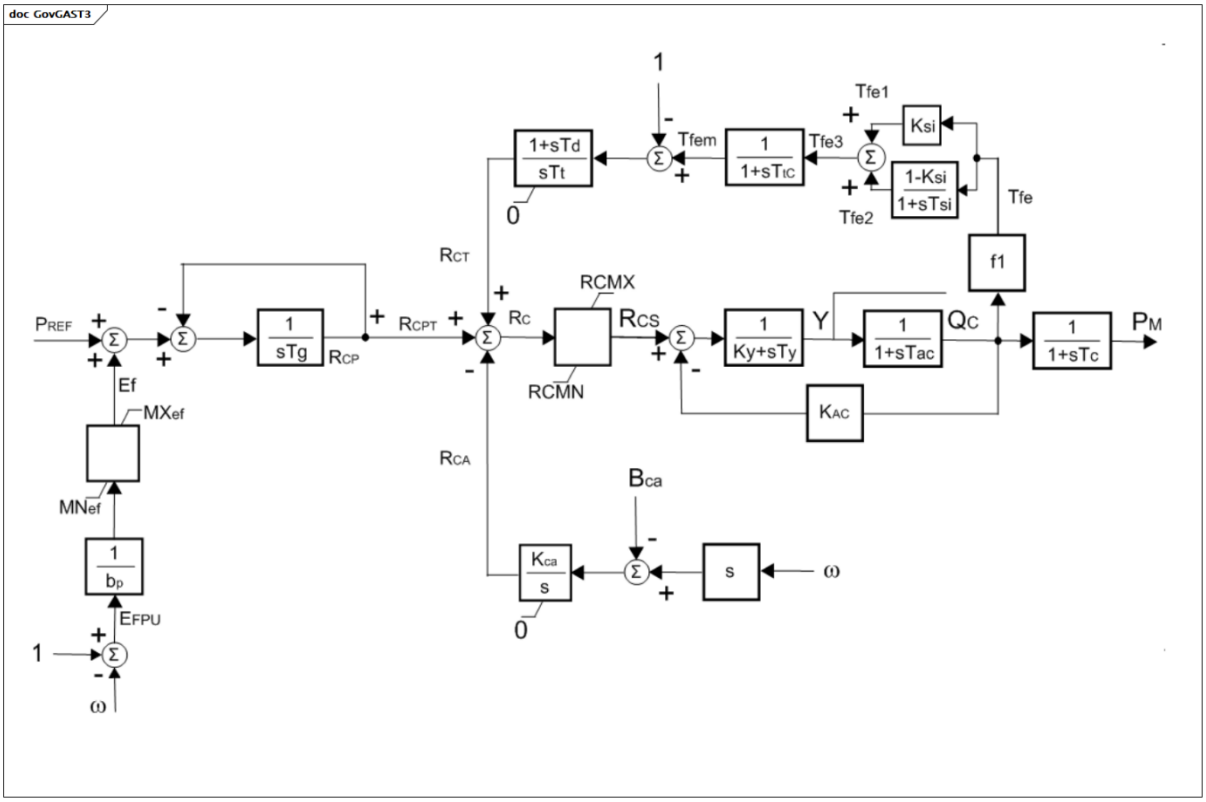

Figure 100 – GovGAST3

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Figure 100: This diagram describes the GovGAST3 turbine governor model.

Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) omega has units of PU speed.
3) Function for exhaust temperature calculation shown in GovGAST3ExhaustTemperature diagram.

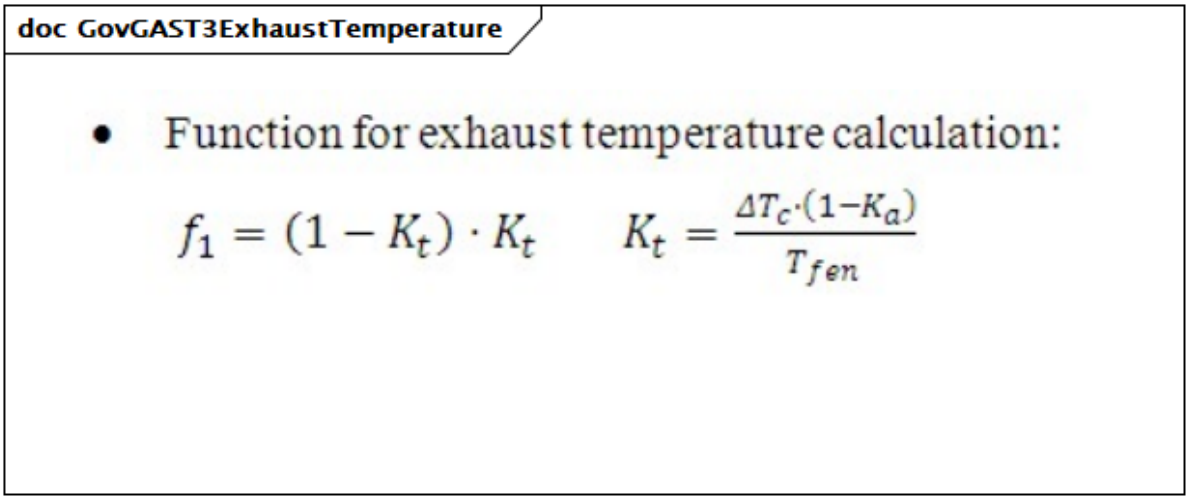

Figure 101 – GovGAST3ExhaustTemperature

Figure 101: This diagram illustrates the function for exhaust temperature calculation.

Table 696 shows all attributes of GovGAST3.

Table 696 – Attributes of TurbineGovernorDynamics::GovGAST3

name	mult	type	description
bp	1..1	PU	Droop (bp). Typical value = 0,05.
tg	1..1	Seconds	Time constant of speed governor (Tg) (>= 0). Typical value = 0,05.
rcmx	1..1	PU	Maximum fuel flow (RCMX). Typical value = 1.
rcmn	1..1	PU	Minimum fuel flow (RCMN). Typical value = -0,1.
ky	1..1	Float	Coefficient of transfer function of fuel valve positioner (Ky). Typical value = 1.
ty	1..1	Seconds	Time constant of fuel valve positioner (Ty) (>= 0). Typical value = 0,2.
tac	1..1	Seconds	Fuel control time constant (Tac) (>= 0). Typical value = 0,1.
kac	1..1	Float	Fuel system feedback (KAC). Typical value = 0.
tc	1..1	Seconds	Compressor discharge volume time constant (Tc) (>= 0). Typical value = 0,2.
bca	1..1	Float	Acceleration limit set-point (Bca). Unit = 1/s. Typical value = 0,01.
kca	1..1	Float	Acceleration control integral gain (Kca). Unit = 1/s. Typical value = 100.
dtc	1..1	Temperature	Exhaust temperature variation due to fuel flow increasing from 0 to 1 PU (deltaTc). Typical value = 390.

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name	mult	type	description
ka	1..1	PU	Minimum fuel flow (Ka). Typical value = 0,23.
tsi	1..1	Seconds	Time constant of radiation shield (Tsi) (>= 0). Typical value = 15.
ksi	1..1	Float	Gain of radiation shield (Ksi). Typical value = 0,8.
ttc	1..1	Seconds	Time constant of thermocouple (Ttc) (>= 0). Typical value = 2,5.
tfen	1..1	Temperature	Turbine rated exhaust temperature correspondent to Pm=1 PU (Tfen). Typical value = 540.
td	1..1	Seconds	Temperature controller derivative gain (Td) (>= 0). Typical value = 3,3.
tt	1..1	Temperature	Temperature controller integration rate (Tt). Typical value = 250.
mxef	1..1	PU	Fuel flow maximum positive error value (MXef). Typical value = 0,05.
mnef	1..1	PU	Fuel flow maximum negative error value (MNef). Typical value = -0,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 697 shows all association ends of GovGAST3 with other classes.

Table 697 – Association ends of TurbineGovernorDynamics:: GovGAST3 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.13 GovGAST4

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Generic turbogas.

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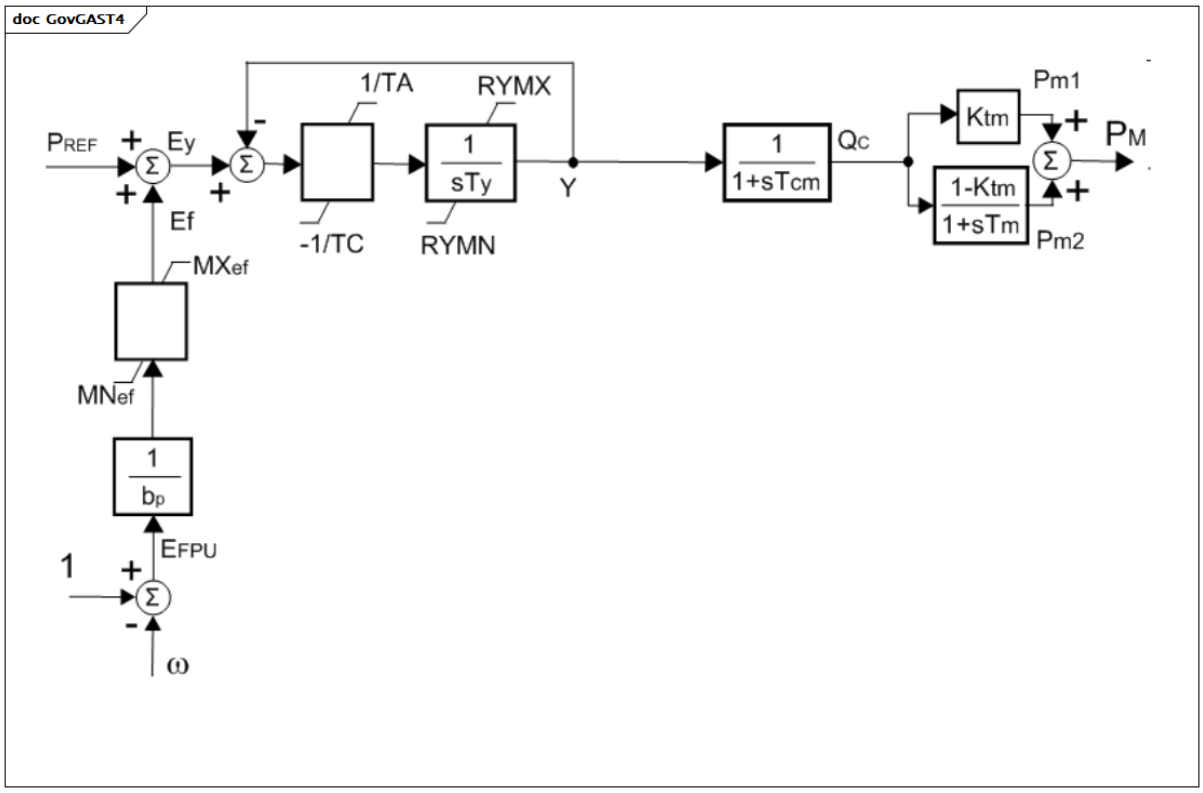

Figure 102 – GovGAST4

Figure 102: This diagram describes the GovGAST4 turbine governor model.

Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) omega has units of PU speed.

Table 698 shows all attributes of GovGAST4.

Table 698 – Attributes of TurbineGovernorDynamics::GovGAST4

name	mult	type	description
bp	1..1	PU	Droop (bp). Typical value = 0,05.
ty	1..1	Seconds	Time constant of fuel valve positioner (Ty) (>= 0). Typical value = 0,1.
ta	1..1	Seconds	Maximum gate opening velocity (TA) (>= 0). Typical value = 3.
tc	1..1	Seconds	Maximum gate closing velocity (TC) (>= 0). Typical value = 0,5.
tcm	1..1	Seconds	Fuel control time constant (Tcm) (>= 0). Typical value = 0,1.
ktm	1..1	PU	Compressor gain (Ktm). Typical value = 0.
tm	1..1	Seconds	Compressor discharge volume time constant (Tm) (>= 0). Typical value = 0,2.
rymx	1..1	PU	Maximum valve opening (RYMX). Typical value = 1,1.
rymn	1..1	PU	Minimum valve opening (RYMN). Typical value = 0.

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name	mult	type	description
mxef	1..1	PU	Fuel flow maximum positive error value (MXef). Typical value = 0,05.
mnef	1..1	PU	Fuel flow maximum negative error value (MNef). Typical value = -0,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 699 shows all association ends of GovGAST4 with other classes.

Table 699 – Association ends of TurbineGovernorDynamics:: GovGAST4 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.14 GovGASTWD

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Woodward™ gas turbine governor[1].

[1] Woodward™ gas turbines are an example of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

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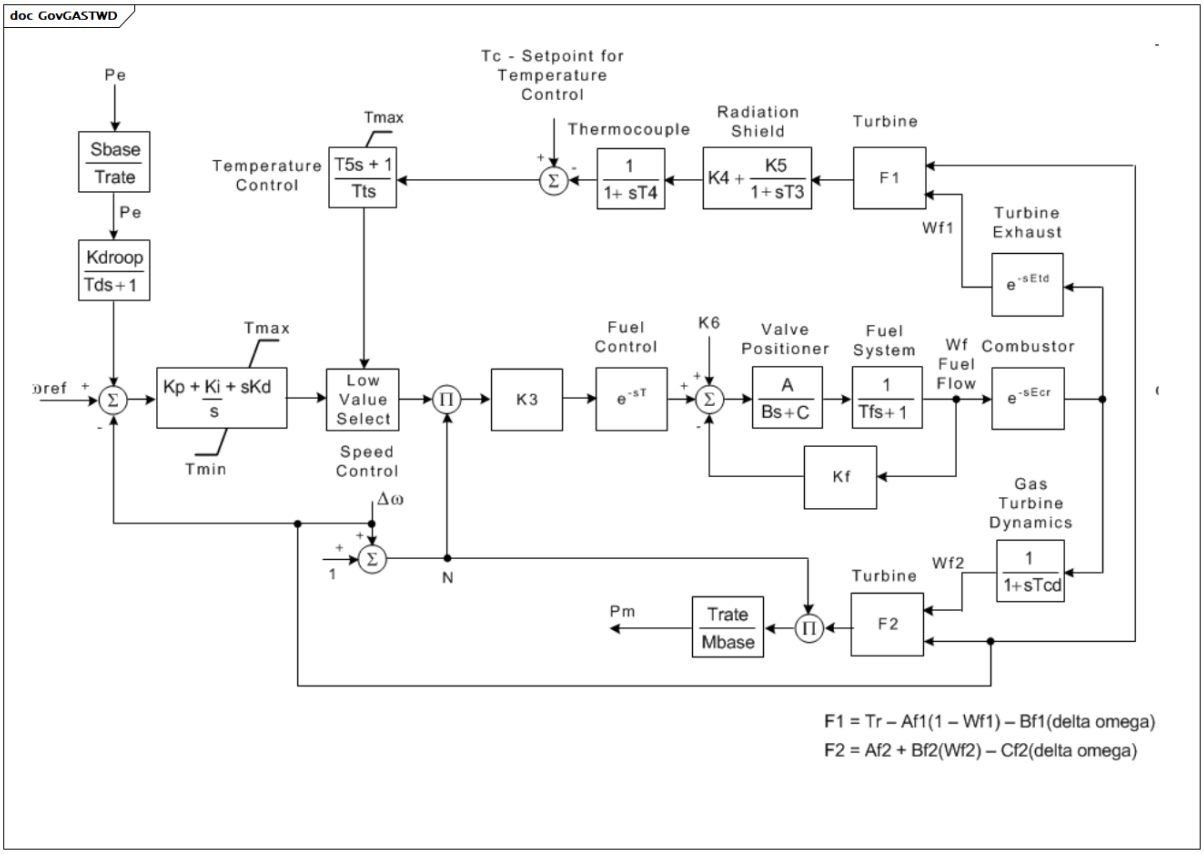

Figure 103 – GovGASTWD

Figure 103: This diagram describes the GovGASTWD turbine governor model.

Parameter details:

1) Temperature control output is set to output of speed governor when temperature control input changes from positive to negative.
2) F1 = Tr – Af1(1 – Wf1) – Bf1(delta omega)
3) F2 = Af2 + Bf2(Wf2) – Cf2(delta omega).

Table 700 shows all attributes of GovGASTWD.

Table 700 – Attributes of TurbineGovernorDynamics::GovGASTWD

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
kdroop	1..1	PU	(Kdroop) (>= 0).
kp	1..1	PU	PID proportional gain (Kp).
ki	1..1	PU	Isochronous Governor Gain (Ki).
kd	1..1	PU	Drop governor gain (Kd).
etd	1..1	Seconds	Turbine and exhaust delay (Etd) (>= 0).
tcd	1..1	Seconds	Compressor discharge time constant (Tcd) (>= 0).
trate	1..1	ActivePower	Turbine rating (Trate). Unit = MW.
t	1..1	Seconds	Fuel control time constant (T) (>= 0).

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name	mult	type	description
tmax	1..1	PU	Maximum Turbine limit (Tmax) (> GovGASTWD.tmin).
tmin	1..1	PU	Minimum turbine limit (Tmin) (< GovGASTWD.tmax).
ecr	1..1	Seconds	Combustion reaction time delay (Ecr) (>= 0).
k3	1..1	PU	Ratio of fuel adjustment (K3).
a	1..1	Float	Valve positioner (A).
b	1..1	Float	Valve positioner (B).
c	1..1	Float	Valve positioner (C).
tf	1..1	Seconds	Fuel system time constant (Tf) (>= 0).
kf	1..1	PU	Fuel system feedback (Kf).
k5	1..1	PU	Gain of radiation shield (K5).
k4	1..1	PU	Gain of radiation shield (K4).
t3	1..1	Seconds	Radiation shield time constant (T3) (>= 0).
t4	1..1	Seconds	Thermocouple time constant (T4) (>= 0).
tt	1..1	Seconds	Temperature controller integration rate (Tt) (>= 0).
t5	1..1	Seconds	Temperature control time constant (T5) (>= 0).
af1	1..1	PU	Exhaust temperature parameter (Af1).
bf1	1..1	PU	(Bf1). Bf1 = E(1-w) where E (speed sensitivity coefficient) is 0,55 to 0,65 x Tr.
af2	1..1	PU	Coefficient equal to 0,5(1-speed) (Af2).
bf2	1..1	PU	Turbine torque coefficient Khhv (depends on heating value of fuel stream in combustion chamber) (Bf2).
cf2	1..1	PU	Coefficient defining fuel flow where power output is 0 % (Cf2). Synchronous but no output. Typically 0,23 x Khhv (23 % fuel flow).
tr	1..1	Temperature	Rated temperature (Tr).
k6	1..1	PU	Minimum fuel flow (K6).
tc	1..1	Temperature	Temperature control (Tc).
td	1..1	Seconds	Power transducer time constant (Td) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 701 shows all association ends of GovGASTWD with other classes.

Table 701 – Association ends of TurbineGovernorDynamics:: GovGASTWD with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

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14.3.3.6.15 GovHydro1

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Basic hydro turbine governor.

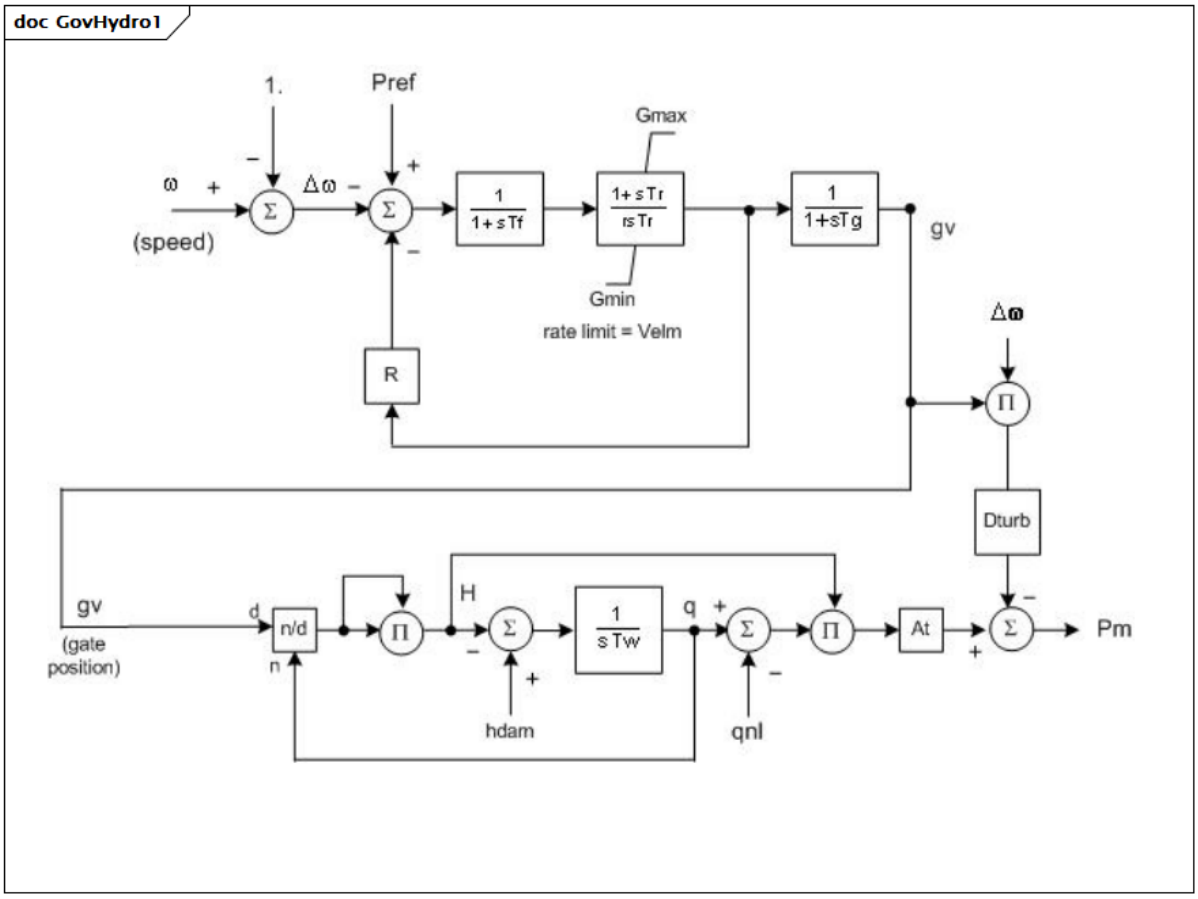

Figure 104 – GovHydro1

Figure 104: This diagram describes the GovHydro1 turbine governor model.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) The gates travel over a range of 1,0 PU from fully closed to fully opened. The gate position is normally greater than zero at zero power and normally less than 1,0 when power is 1,0 PU. Gmax and Gmin are operating limits.
3) Dturb has units of delta P (PU of mwcap) /delta speed (PU).

Table 702 shows all attributes of GovHydro1.

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Table 702 – Attributes of TurbineGovernorDynamics::GovHydro1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
rperm	1..1	PU	Permanent droop (R) (> 0). Typical value = 0,04.
rtemp	1..1	PU	Temporary droop (r) (> GovHydro1.rperm). Typical value = 0,3.
tr	1..1	Seconds	Washout time constant (Tr) (> 0). Typical value = 5.
tf	1..1	Seconds	Filter time constant (Tf) (> 0). Typical value = 0,05.
tg	1..1	Seconds	Gate servo time constant (Tg) (> 0). Typical value = 0,5.
velm	1..1	Float	Maximum gate velocity (Vlem) (> 0). Typical value = 0,2.
gmax	1..1	PU	Maximum gate opening (Gmax) (> 0 and > GovHydro.gmin). Typical value = 1.
gmin	1..1	PU	Minimum gate opening (Gmin) (>= 0 and < GovHydro1.gmax). Typical value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (> 0). Typical value = 1.
at	1..1	PU	Turbine gain (At) (> 0). Typical value = 1,2.
dturb	1..1	PU	Turbine damping factor (Dturb) (>= 0). Typical value = 0,5.
qnl	1..1	PU	No-load flow at nominal head (qnl) (>= 0). Typical value = 0,08.
hdam	1..1	PU	Turbine nominal head (hdam). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 703 shows all association ends of GovHydro1 with other classes.

Table 703 – Association ends of TurbineGovernorDynamics:: GovHydro1 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.16 GovHydro2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE hydro turbine governor with straightforward penstock configuration and hydraulic-dashpot governor.

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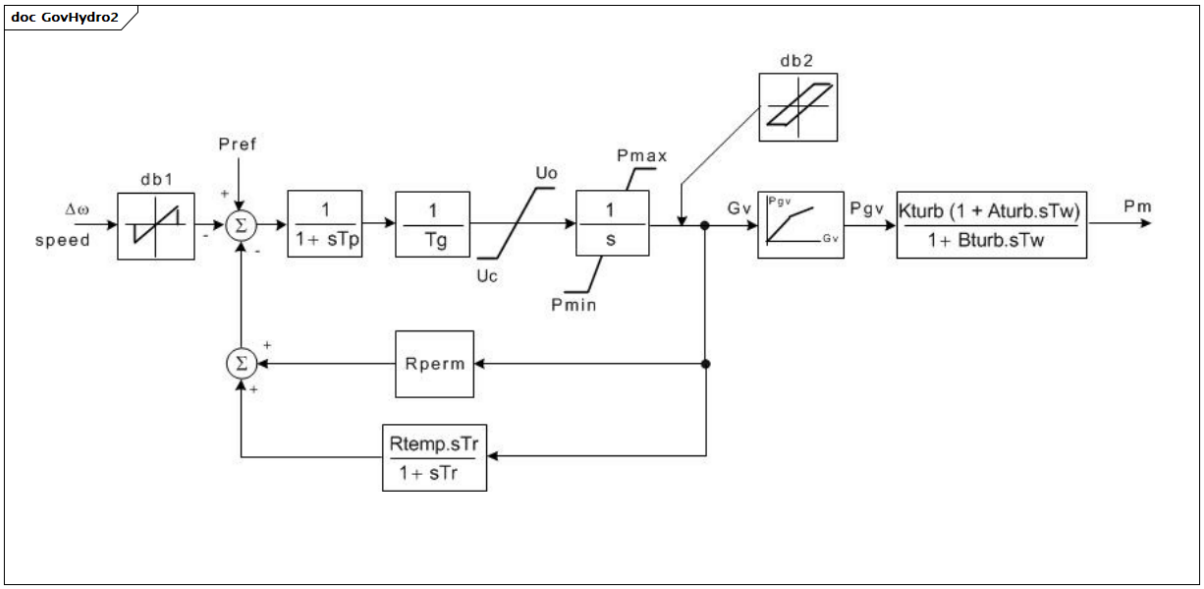

Figure 105 – GovHydro2

Figure 105: This diagram describes the GovHydro2 turbine governor model.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) This turbine governor model is a reasonable approximation to the performance of simple hydro plants for small motions of the gates about a given load, but it represents the variation of water inertia effect with load only if the values of the turbine parameters are properly adjusted by the user as load is varied. For idealized turbine model, set Kturb = 1,0, Aturb = -1,0, Bturb = 0,5.
3) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0.) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.
4) For compatibility with older versions of this model: Kturb = a23, Aturb = a11 – a13 x a21/a23, Bturb = a11.

Table 704 shows all attributes of GovHydro2.

Table 704 – Attributes of TurbineGovernorDynamics::GovHydro2

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg) (> 0). Typical value = 0,5.
tp	1..1	Seconds	Pilot servo valve time constant (Tp) (>= 0). Typical value = 0,03.
uo	1..1	Float	Maximum gate opening velocity (Uo). Unit = PU / s. Typical value = 0,1.
uc	1..1	Float	Maximum gate closing velocity (Uc) (< 0). Unit = PU / s. Typical value = -0,1.
pmax	1..1	PU	Maximum gate opening (Pmax) (> GovHydro2.pmin). Typical value = 1.

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name	mult	type	description
pmin	1..1	PU	Minimum gate opening (Pmin) (< GovHydro2.pmax). Typical value = 0.
rperm	1..1	PU	Permanent droop (Rperm). Typical value = 0,05.
rtemp	1..1	PU	Temporary droop (Rtemp). Typical value = 0,5.
tr	1..1	Seconds	Dashpot time constant (Tr) (>= 0). Typical value = 12.
tw	1..1	Seconds	Water inertia time constant (Tw) (>= 0). Typical value = 2.
kturb	1..1	PU	Turbine gain (Kturb). Typical value = 1.
aturb	1..1	PU	Turbine numerator multiplier (Aturb). Typical value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (> 0). Typical value = 0,5.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional deadband (db2). Unit = MW. Typical value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 705 shows all association ends of GovHydro2 with other classes.

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Table 705 – Association ends of TurbineGovernorDynamics:: GovHydro2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.17 GovHydro3

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE hydro governor-turbine. This model differs from that defined in the IEEE modelling guideline paper in that the limits on gate position and velocity do not permit "wind up" of the upstream signals.

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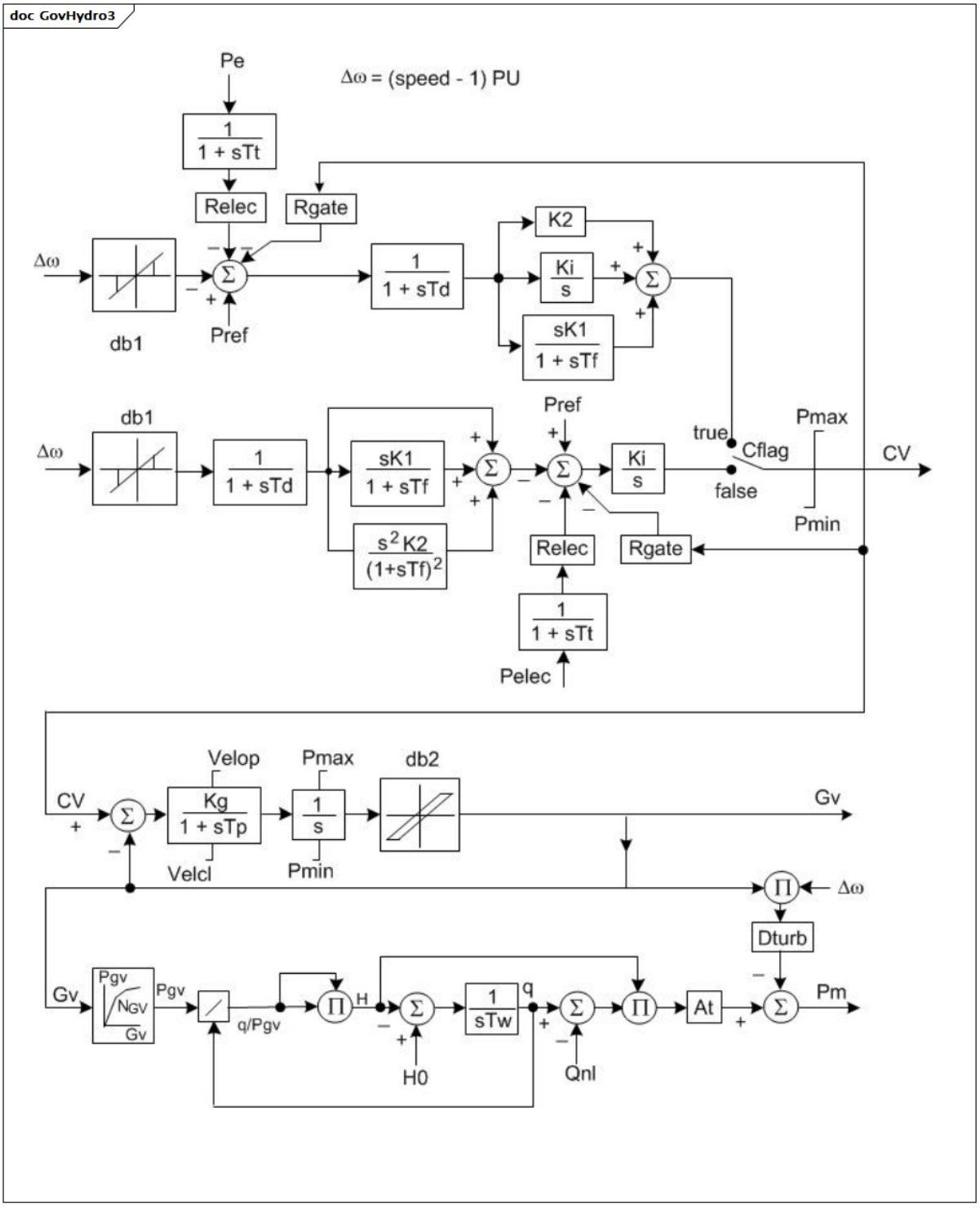

Figure 106 – GovHydro3

Figure 106: This diagram describes the GovHydro3 turbine governor model.

Parameter details:

1) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0 , 0,0) and (1,0 , 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default hydro curve (between (0,0 , 0,0) and (1,0 , 1,0)) will be used.

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Table 706 shows all attributes of GovHydro3.

Table 706 – Attributes of TurbineGovernorDynamics::GovHydro3

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax) (> GovHydro3.pmin). Typical value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin) (< GovHydro3.pmax). Typical value = 0.
governorControl	1..1	Boolean	Governor control flag (Cflag). true = PID control is active false = double derivative control is active. Typical value = true.
rgate	1..1	PU	Steady-state droop, PU, for governor output feedback (Rgate). Typical value = 0.
relec	1..1	PU	Steady-state droop, PU, for electrical power feedback (Relec). Typical value = 0,05.
td	1..1	Seconds	Input filter time constant (Td) (>= 0). Typical value = 0,05.
tf	1..1	Seconds	Washout time constant (Tf) (>= 0). Typical value = 0,1.
tp	1..1	Seconds	Gate servo time constant (Tp) (>= 0). Typical value = 0,05.
velop	1..1	Float	Maximum gate opening velocity (Velop). Unit = PU / s. Typical value = 0,2.
velcl	1..1	Float	Maximum gate closing velocity (Velcl). Unit = PU / s. Typical value = -0,2.
k1	1..1	PU	Derivative gain (K1). Typical value = 0,01.
k2	1..1	PU	Double derivative gain, if Cflag = -1 (K2). Typical value = 2,5.
ki	1..1	PU	Integral gain (Ki). Typical value = 0,5.
kg	1..1	PU	Gate servo gain (Kg). Typical value = 2.
tt	1..1	Seconds	Power feedback time constant (Tt) (>= 0). Typical value = 0,2.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (>= 0). If = 0, block is bypassed. Typical value = 1.
at	1..1	PU	Turbine gain (At) (>0). Typical value = 1,2.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical value = 0,2.
qnl	1..1	PU	No-load turbine flow at nominal head (Qnl). Typical value = 0,08.
h0	1..1	PU	Turbine nominal head (H0). Typical value = 1.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.

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name	mult	type	description
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 707 shows all association ends of GovHydro3 with other classes.

Table 707 – Association ends of TurbineGovernorDynamics:: GovHydro3 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.18 GovHydro4

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Hydro turbine and governor. Represents plants with straight-forward penstock configurations and hydraulic governors of the traditional 'dashpot' type. This model can be used to represent simple, Francis/Pelton or Kaplan turbines.

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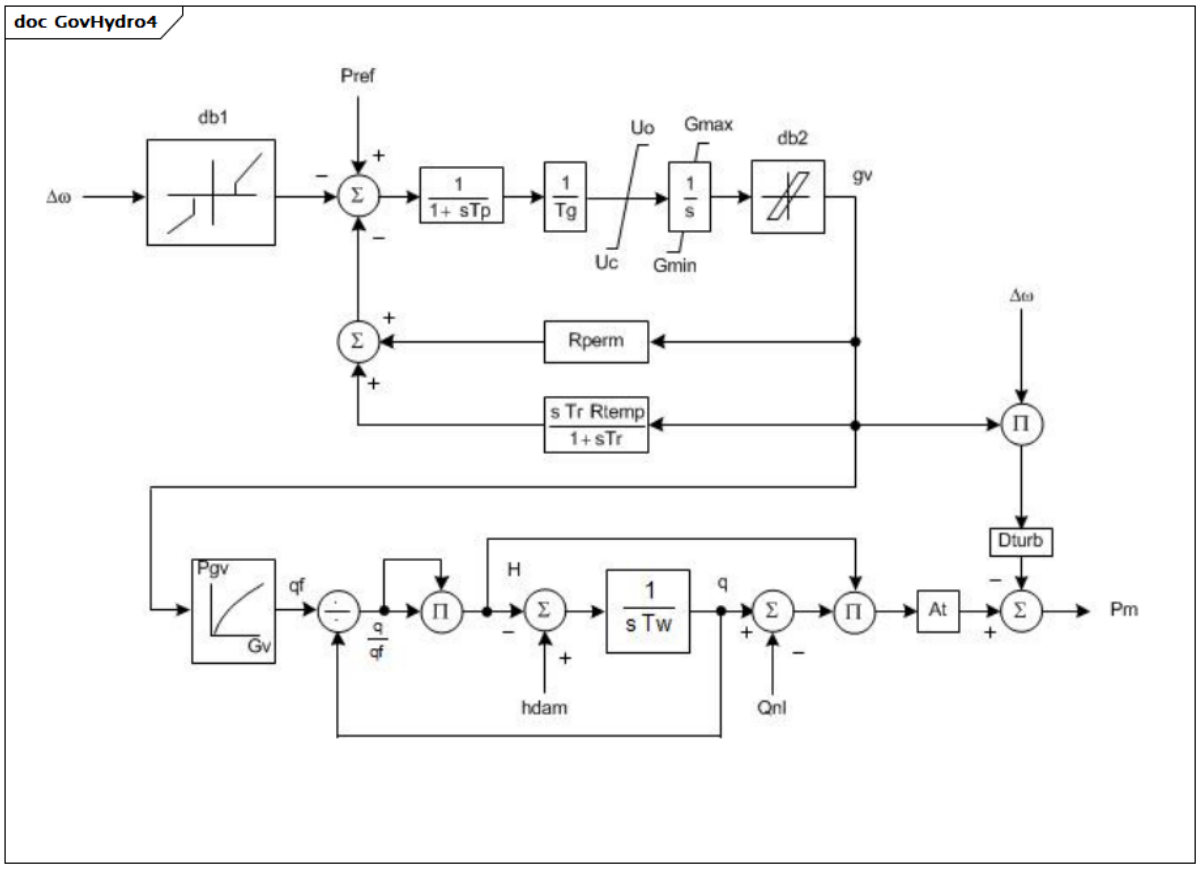

Figure 107 – GovHydro4

Figure 107: This diagram describes the GovHydro4 turbine governor model.

Parameter details:

1) The GovHydro4 model can be used to represent three different standard model variants: simple, Francis/Pelton and Kaplan.
2) The gates travel over a range of 1 to 0 PU from fully closed to fully opened. The gates are at a position greater than zero at zero power and, normally, less than 1 PU when power is 1 PU. Gmax and Gmin are operating limits.
3) The following parameter combinations are used by the variants to represent gate-head-flow power characteristics:
   - Simple: the line with slope At and intercept Qnl describes the gate opening-to-turbine power characteristic (as shown in the GovHydro4SimpleHydroTurbine diagram). This curve is proportional to turbine power, with the constant of proportionality being a function of available head. This constant is specified by the turbine gain, At. Parameters Gv0 to Gv5, Pgv0 to Pgv5, and Bgv0 to Bgv5 are not considered.
   - Francis/Pelton turbine model: (Gv0, Pgv0) thru (Gv5, Pgv5) are used to describe the non-linear gate opening-to-turbine power characteristic (as shown in the GovHydro4FrancisPeltonTurbine diagram). Bgv0 to Bgv5 are not considered.
   - Kaplan turbine model: (Gv0, Pgv0) thru (Gv5, Pgv5) and (Gv0, Bv0) thru (Gv5, Bgv5) are used to describe the non-linear characteristics of gate opening-to-turbine power and gate opening-to-blade angle (as shown in the GovHydro4KaplanTurbine diagram).

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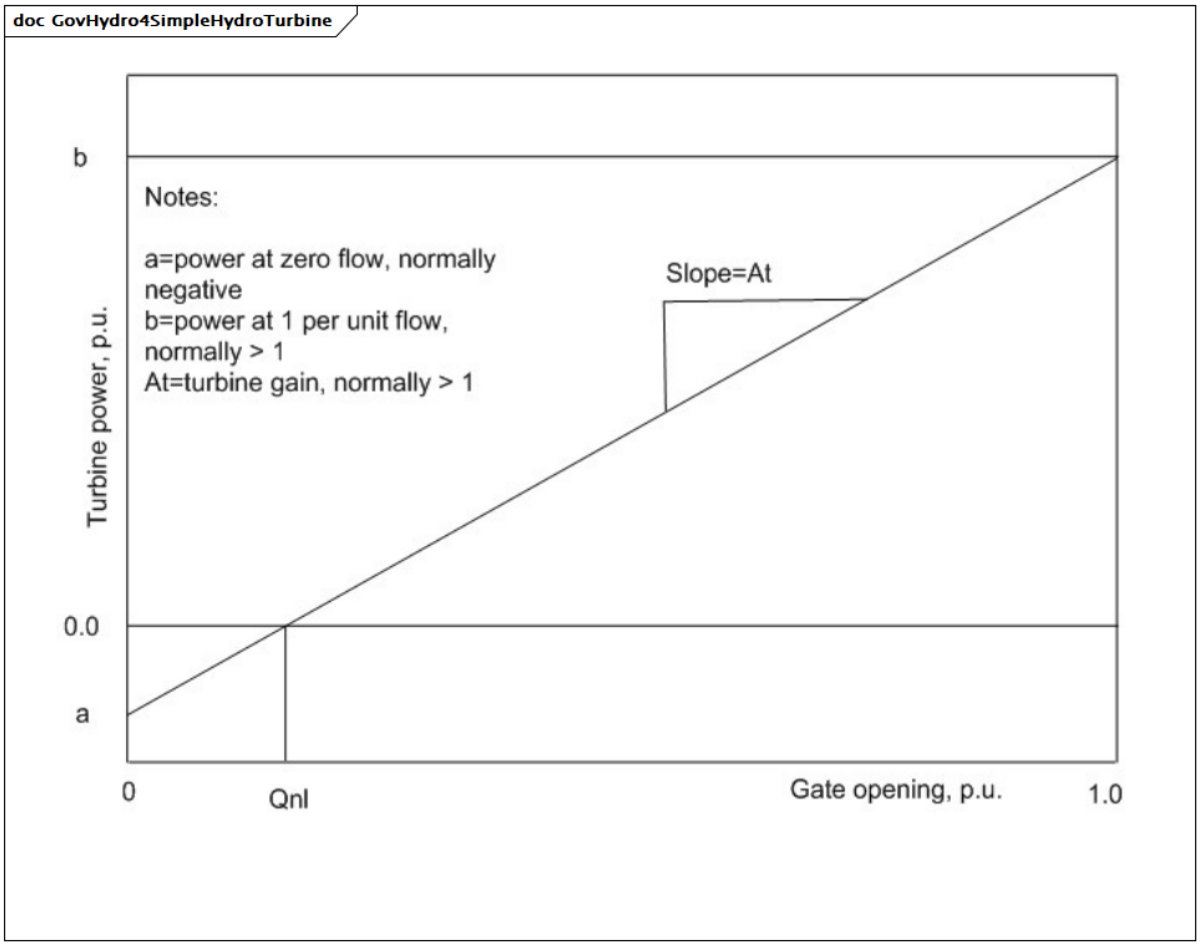

Figure 108 – GovHydro4SimpleHydroTurbine

Figure 108: This diagram illustrates the gate-head-flow-power characteristics for a simple general purpose model.

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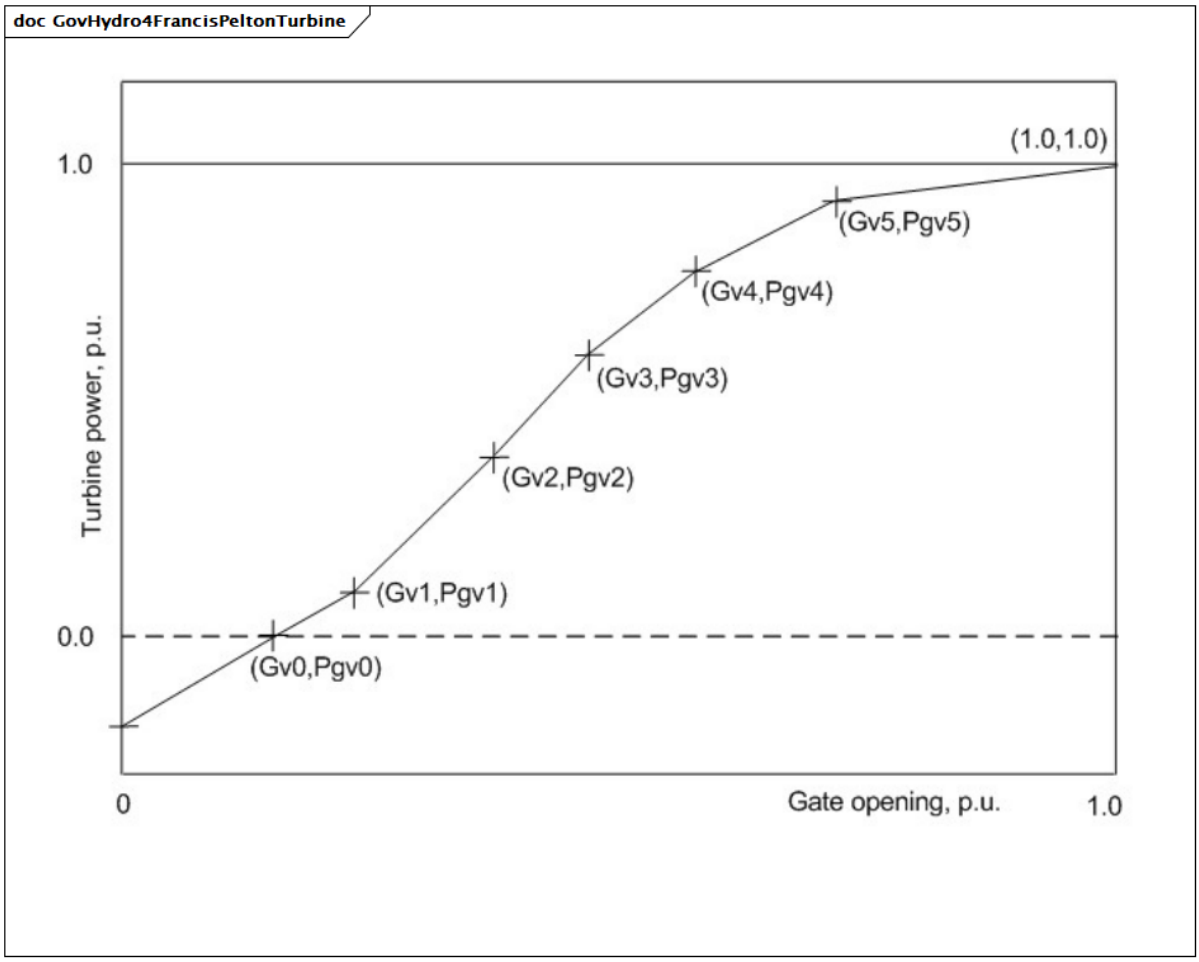

Figure 109 – GovHydro4FrancisPeltonTurbine

Figure 109: This diagram illustrates the gate-head-flow-power characteristics for a Francis/Pelton turbine model.

Parameter details:

1) The curve described by (Gv0, Pgv0) to (Gv5, Pgv5) describes the relationship of flow through the turbine to stroke of the gate actuating servomotor.
2) If Gv0 is greater than 0, the first two data points will be used to extrapolate downward for gate positions less than Gv0 If Gv0 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.

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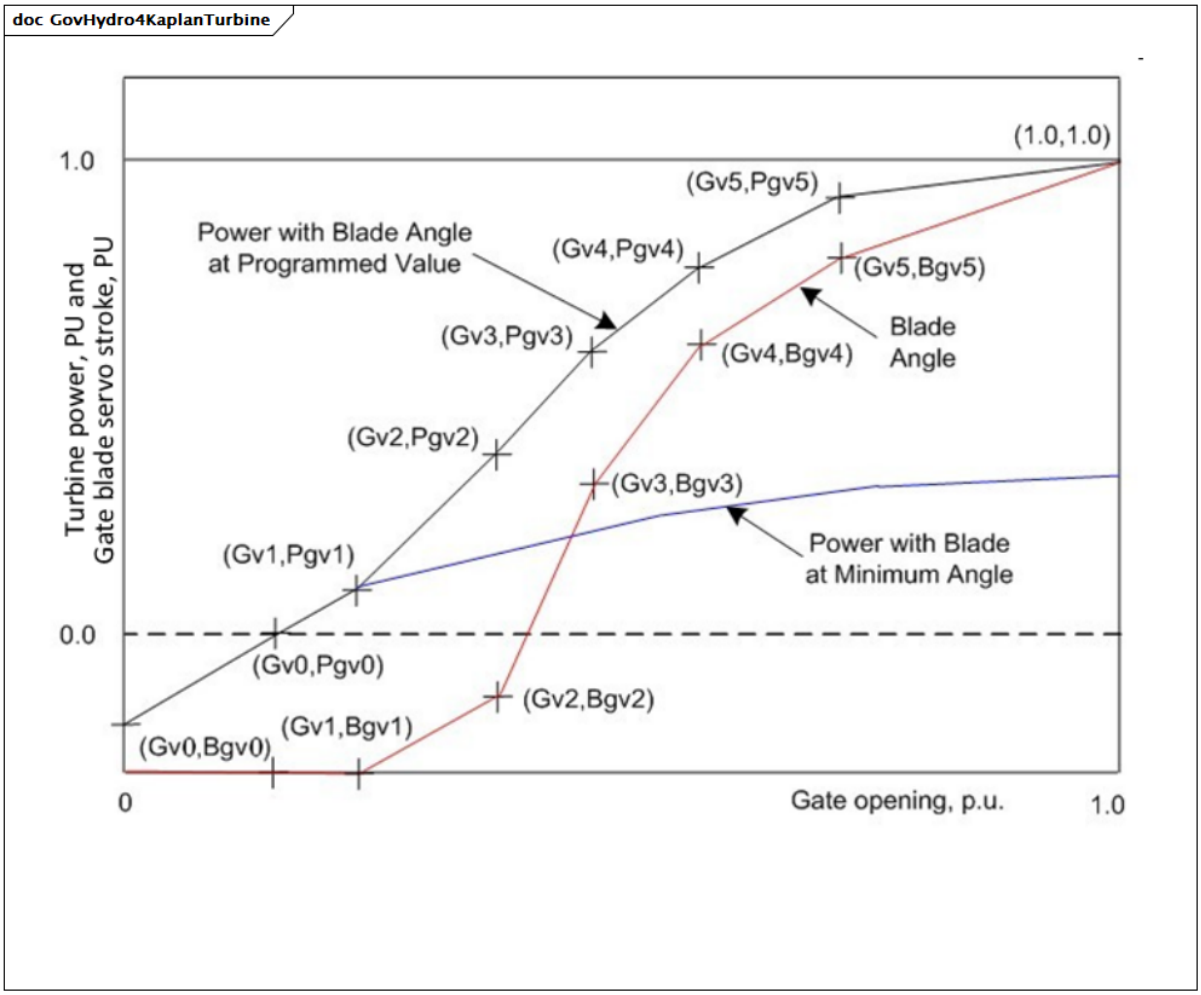

Figure 110 – GovHydro4KaplanTurbine

Figure 110: This diagram illustrates the non-linear characteristics of gate opening-to-turbine power and gate opening-to-blade angle for a Kaplan turbine model.

Parameter details:

1) The curve described by (Gv0, Pgv0) to (Gv5, Pgv5) describes the relationship of flow through the turbine to stroke of the gate actuating servomotor.
2) If Gv0 is greater than 0, the first two data points will be used to extrapolate downward for gate positions less than Gv0 If Gv0 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.

Table 708 shows all attributes of GovHydro4.

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Table 708 – Attributes of TurbineGovernorDynamics::GovHydro4

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tg	1..1	Seconds	Gate servo time constant (Tg) (> 0). Typical value = 0,5.
tp	1..1	Seconds	Pilot servo time constant (Tp) (>= 0). Typical value = 0,1.
uo	1..1	Float	Max gate opening velocity (Uo). Typical value = 0,2.
uc	1..1	Float	Max gate closing velocity (Uc). Typical value = 0,2.
gmax	1..1	PU	Maximum gate opening, PU of MWbase (Gmax) (> GovHydro4.gmin). Typical value = 1.
gmin	1..1	PU	Minimum gate opening, PU of MWbase (Gmin) (< GovHydro4.gmax). Typical value = 0.
rperm	1..1	Seconds	Permanent droop (Rperm) (>= 0). Typical value = 0,05.
rtemp	1..1	Seconds	Temporary droop (Rtemp) (>= 0). Typical value = 0,3.
tr	1..1	Seconds	Dashpot time constant (Tr) (>= 0). Typical value = 5.
tw	1..1	Seconds	Water inertia time constant (Tw) (> 0). Typical value = 1.
at	1..1	PU	Turbine gain (At). Typical value = 1,2.
dturb	1..1	PU	Turbine damping factor (Dturb). Unit = delta P (PU of MWbase) / delta speed (PU). Typical value for simple = 0,5, Francis/Pelton = 1,1, Kaplan = 1,1.
hdam	1..1	PU	Head available at dam (hdam). Typical value = 1.
qnl	1..1	PU	No-load flow at nominal head (Qnl). Typical value for simple = 0,08, Francis/Pelton = 0, Kaplan = 0.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
gv0	1..1	PU	Nonlinear gain point 0, PU gv (Gv0) (= 0 for simple). Typical for Francis/Pelton = 0,1, Kaplan = 0,1.
pgv0	1..1	PU	Nonlinear gain point 0, PU power (Pgv0) (= 0 for simple). Typical value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1) (= 0 for simple, > GovHydro4.gv0 for Francis/Pelton and Kaplan). Typical value for Francis/Pelton = 0,4, Kaplan = 0,4.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1) (= 0 for simple). Typical value for Francis/Pelton = 0,42, Kaplan = 0,35.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2) (= 0 for simple, > GovHydro4.gv1 for Francis/Pelton and Kaplan). Typical value for Francis/Pelton = 0,5, Kaplan = 0,5.

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name	mult	type	description
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2) (= 0 for simple). Typical value for Francis/Pelton = 0,56, Kaplan = 0,468.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3) (= 0 for simple, > GovHydro4.gv2 for Francis/Pelton and Kaplan). Typical value for Francis/Pelton = 0,7, Kaplan = 0,7.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3) (= 0 for simple). Typical value for Francis/Pelton = 0,8, Kaplan = 0,796.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4) (= 0 for simple, > GovHydro4.gv3 for Francis/Pelton and Kaplan). Typical value for Francis/Pelton = 0,8, Kaplan = 0,8.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4) (= 0 for simple). Typical value for Francis/Pelton = 0,9, Kaplan = 0,917.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5) (= 0 for simple, < 1 and > GovHydro4.gv4 for Francis/Pelton and Kaplan). Typical value for Francis/Pelton = 0,9, Kaplan = 0,9.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5) (= 0 for simple). Typical value for Francis/Pelton = 0,97, Kaplan = 0,99.
bgv0	1..1	PU	Kaplan blade servo point 0 (Bgv0) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 0.
bgv1	1..1	PU	Kaplan blade servo point 1 (Bgv1) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 0.
bgv2	1..1	PU	Kaplan blade servo point 2 (Bgv2) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 0,1.
bgv3	1..1	PU	Kaplan blade servo point 3 (Bgv3) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 0,667.
bgv4	1..1	PU	Kaplan blade servo point 4 (Bgv4) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 0,9.
bgv5	1..1	PU	Kaplan blade servo point 5 (Bgv5) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 1.
bmax	1..1	Float	Maximum blade adjustment factor (Bmax) (= 0 for simple, = 0 for Francis/Pelton). Typical value for Kaplan = 1,1276.
tblade	1..1	Seconds	Blade servo time constant (Tblade) (>= 0). Typical value = 100.
model	1..1	GovHydro4ModelKind	The kind of model being represented (simple, Francis/Pelton or Kaplan).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 709 shows all association ends of GovHydro4 with other classes.

Table 709 – Association ends of TurbineGovernorDynamics:: GovHydro4 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.19 GovHydroDD

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Double derivative hydro governor and turbine.

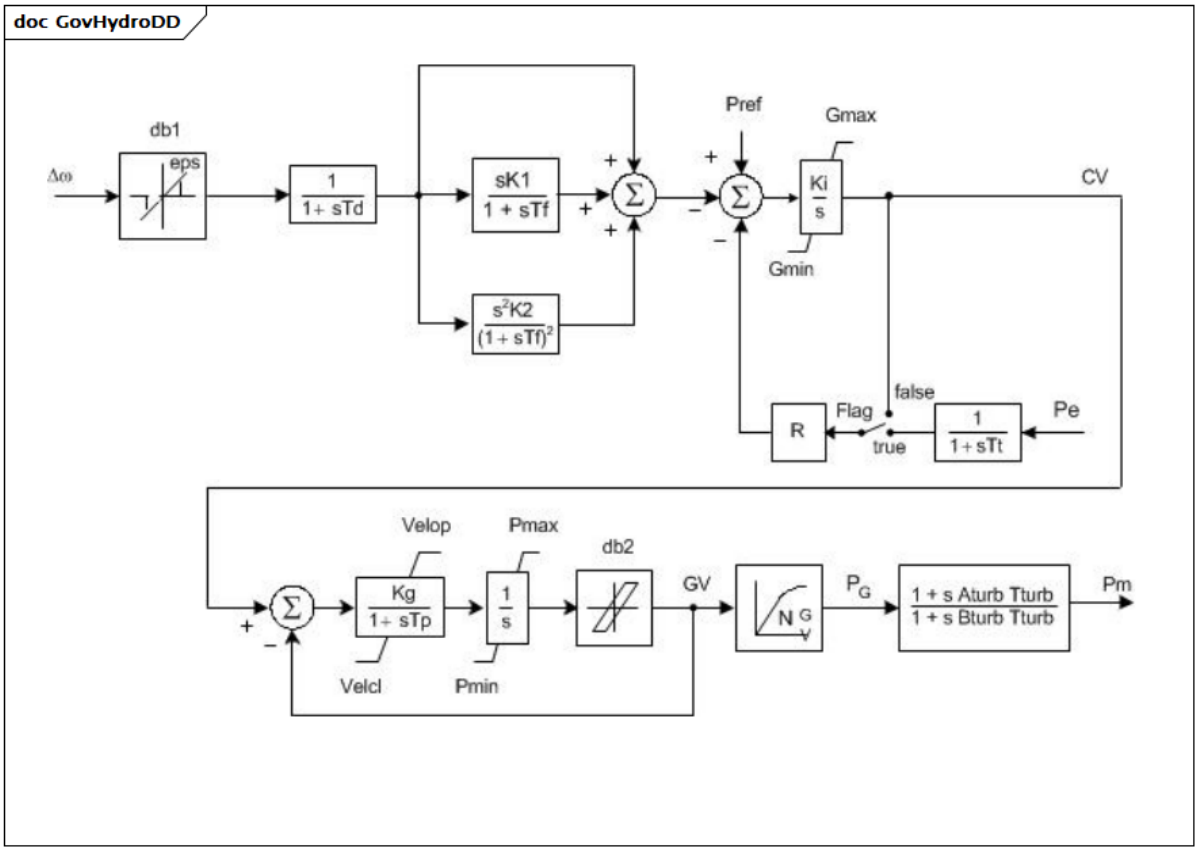

Figure 111 – GovHydroDD

Figure 111: This diagram describes the GovHydroDD turbine governor model.

Parameter details:

1) If Tf is input as zero, but K1 or K2 is not zero, Tf is set to a small value (4 times integration time step). If Bturb x Tturb is less than 4 times integration time step, Tturb is set to zero to bypass the block.

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2) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero values, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.
3) For a hydro turbine, set Tturb = Tw (water starting time), set Aturb = -1,0 and Bturb = 0,5. For a diesel engine or two-shaft gas turbine, set Aturb = 0,8 and Bturb = 1,0 and Tturb = 1 s. For a small steam turbine with a non-reheat steam supply or a single- shaft gas turbine, set Aturb = 0,0 and Bturb = 1,0 and Tturb = 0,2 s.

Table 710 shows all attributes of GovHydroDD.

Table 710 – Attributes of TurbineGovernorDynamics::GovHydroDD

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax) (> GovHydroDD.pmin). Typical value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin) (> GovHydroDD.pmax). Typical value = 0.
r	1..1	PU	Steady state droop (R). Typical value = 0,05.
td	1..1	Seconds	Input filter time constant (Td) (>= 0). If = 0, block is bypassed. Typical value = 0.
tf	1..1	Seconds	Washout time constant (Tf) (>= 0). Typical value = 0,1.
tp	1..1	Seconds	Gate servo time constant (Tp) (>= 0). If = 0, block is bypassed. Typical value = 0,35.
velop	1..1	Float	Maximum gate opening velocity (Velop). Unit = PU / s. Typical value = 0,09.
velcl	1..1	Float	Maximum gate closing velocity (Velcl). Unit = PU / s. Typical value = -0,14.
k1	1..1	PU	Single derivative gain (K1). Typical value = 3,6.
k2	1..1	PU	Double derivative gain (K2). Typical value = 0,2.
ki	1..1	PU	Integral gain (Ki). Typical value = 1.
kg	1..1	PU	Gate servo gain (Kg). Typical value = 3.
tturb	1..1	Seconds	Turbine time constant (Tturb) (>= 0). See parameter detail 3. Typical value = 0,8.
aturb	1..1	PU	Turbine numerator multiplier (Aturb) (see parameter detail 3). Typical value = -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (see parameter detail 3). Typical value = 0,5.
tt	1..1	Seconds	Power feedback time constant (Tt) (>= 0). If = 0, block is bypassed. Typical value = 0,02.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.

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name	mult	type	description
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
gmax	1..1	PU	Maximum gate opening (Gmax) (> GovHydroDD.gmin). Typical value = 0.
gmin	1..1	PU	Minimum gate opening (Gmin) (< GovHydroDD.gmax). Typical value = 0.
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tt is zero, Flag is set to false. If Tt is not zero, Flag is set to true. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 711 shows all association ends of GovHydroDD with other classes.

Table 711 – Association ends of TurbineGovernorDynamics:: GovHydroDD with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.20 GovHydroFrancis

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Detailed hydro unit – Francis model. This model can be used to represent three types of governors.

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A schematic of the hydraulic system of detailed hydro unit models, such as Francis and Pelton, is provided in the DetailedHydroModelHydraulicSystem diagram.

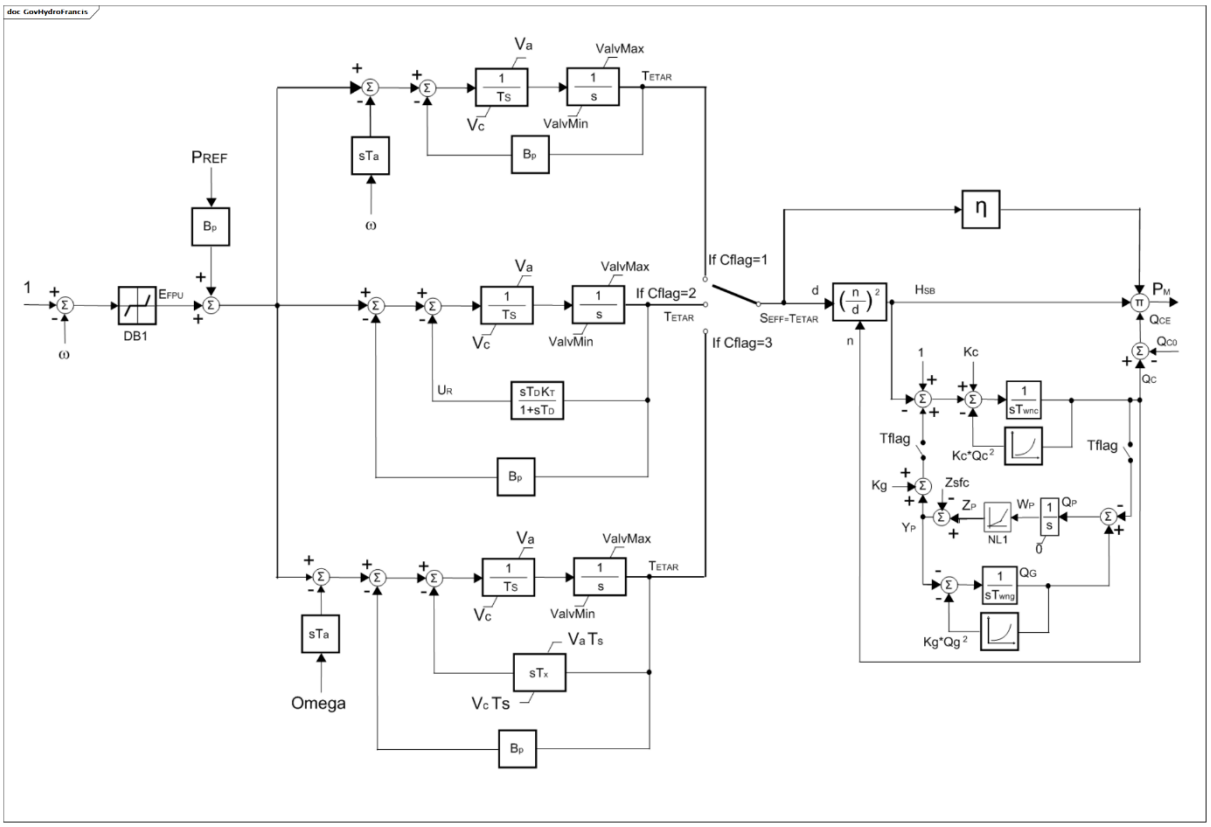

Figure 112 – GovHydroFrancis

Figure 112: This diagram describes the GovHydroFrancis turbine governor model.

Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) Rated hydraulic head, Hn in metres, and rated flow, Qn in cubic metres per s, are the design parameters of the hydraulic system capability.
3) Hydraulic system head attributes (Zsfc, H1, H2) are in metres but in the block diagram are in PU on base of Hn in metres.
4) Hydraulic area attributes (Av0, Av1) are in square metres but in the block diagram are in PU on base of Qn / Hn in square metres per s.
5) omega has units of PU speed.
6) Non-linear gain and efficiency equations are shown in GovHydroFrancisNonLinearGainAndEfficiency diagram.

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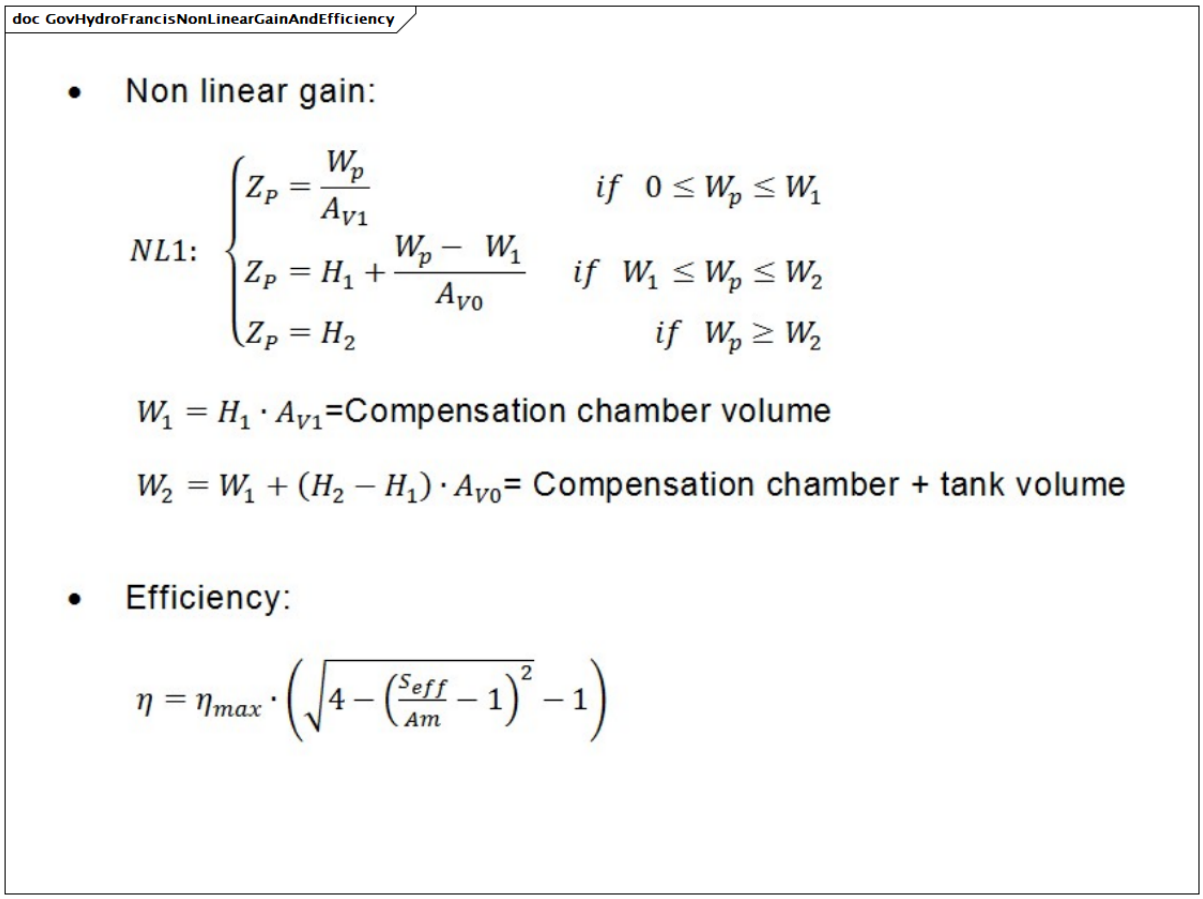

Figure 113 – GovHydroFrancisNonLinearGainAndEfficiency

Figure 113: This diagram illustrates the non-linear gain and efficiency equations.

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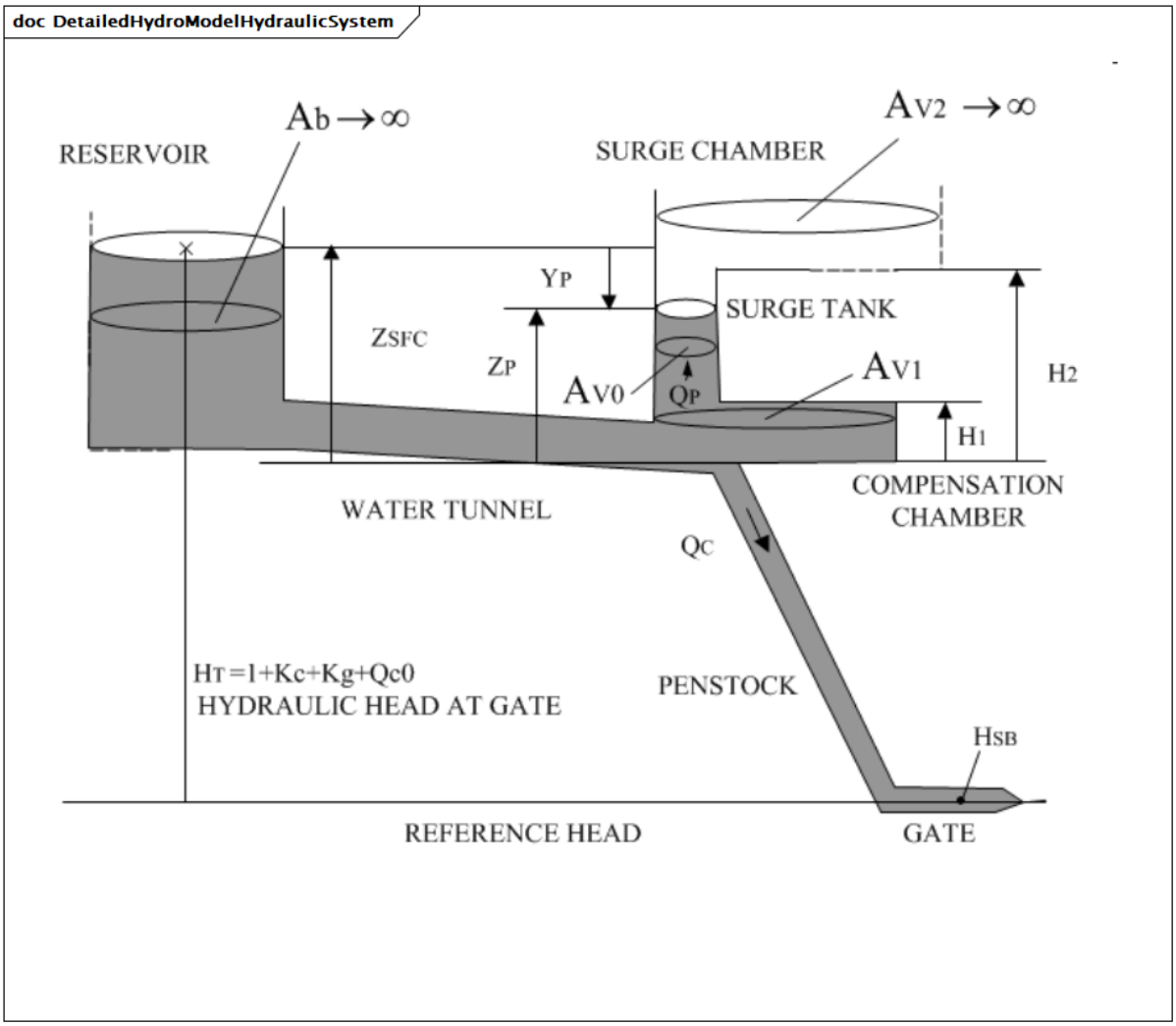

Figure 114 – DetailedHydroModelHydraulicSystem

Figure 114: This diagram illustrates the detailed hydro model of a hydraulic system.

Table 712 shows all attributes of GovHydroFrancis.

Table 712 – Attributes of TurbineGovernorDynamics::GovHydroFrancis

name	mult	type	description
am	1..1	PU	Opening section SEFF at the maximum efficiency (Am). Typical value = 0,7.
av0	1..1	Area	Area of the surge tank (AV0). Unit = m2. Typical value = 30.
av1	1..1	Area	Area of the compensation tank (AV1). Unit = m2. Typical value = 700.
bp	1..1	PU	Droop (Bp). Typical value = 0,05.
db1	1..1	Frequency	Intentional dead-band width (DB1). Unit = Hz. Typical value = 0.
etamax	1..1	PU	Maximum efficiency (EtaMax). Typical value = 1,05.
governorControl	1..1	FrancisGovernorControlKind	Governor control flag (Cflag). Typical value = mechanicHydrolicTachoAccelerator.

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name	mult	type	description
h1	1..1	Length	Head of compensation chamber water level with respect to the level of penstock (H1). Unit = km. Typical value = 0,004.
h2	1..1	Length	Head of surge tank water level with respect to the level of penstock (H2). Unit = km. Typical value = 0,040.
hn	1..1	Length	Rated hydraulic head (Hn). Unit = km. Typical value = 0,250.
kc	1..1	PU	Penstock loss coefficient (due to friction) (Kc). Typical value = 0,025.
kg	1..1	PU	Water tunnel and surge chamber loss coefficient (due to friction) (Kg). Typical value = 0,025.
kt	1..1	PU	Washout gain (Kt). Typical value = 0,25.
qc0	1..1	PU	No-load turbine flow at nominal head (Qc0). Typical value = 0,1.
qn	1..1	VolumeFlowRate	Rated flow (Qn). Unit = m3/s. Typical value = 250.
ta	1..1	Seconds	Derivative gain (Ta) (>= 0). Typical value = 3.
td	1..1	Seconds	Washout time constant (Td) (>= 0). Typical value = 6.
ts	1..1	Seconds	Gate servo time constant (Ts) (>= 0). Typical value = 0,5.
twnc	1..1	Seconds	Water inertia time constant (Twnc) (>= 0). Typical value = 1.
twng	1..1	Seconds	Water tunnel and surge chamber inertia time constant (Twng) (>= 0). Typical value = 3.
tx	1..1	Seconds	Derivative feedback gain (Tx) (>= 0). Typical value = 1.
va	1..1	Float	Maximum gate opening velocity (Va). Unit = PU / s. Typical value = 0,06.
valvmax	1..1	PU	Maximum gate opening (ValvMax) (> GovHydroFrancis.valvmin). Typical value = 1,1.
valvmin	1..1	PU	Minimum gate opening (ValvMin) (< GovHydroFrancis.valvmax). Typical value = 0.
vc	1..1	Float	Maximum gate closing velocity (Vc). Unit = PU / s. Typical value = -0,06.
waterTunnelSurgeChamberSimulation	1..1	Boolean	Water tunnel and surge chamber simulation (Tflag). true = enable of water tunnel and surge chamber simulation false = inhibit of water tunnel and surge chamber simulation. Typical value = false.
zsfc	1..1	Length	Head of upper water level with respect to the level of penstock (Zsfc). Unit = km. Typical value = 0,025.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 713 shows all association ends of GovHydroFrancis with other classes.

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Table 713 – Association ends of TurbineGovernorDynamics:: GovHydroFrancis with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.21 GovHydroPelton

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Detailed hydro unit – Pelton model. This model can be used to represent the dynamic related to water tunnel and surge chamber.

The DetailedHydroModelHydraulicSystem diagram, located under the GovHydroFrancis class, provides a schematic of the hydraulic system of detailed hydro unit models, such as Francis and Pelton.

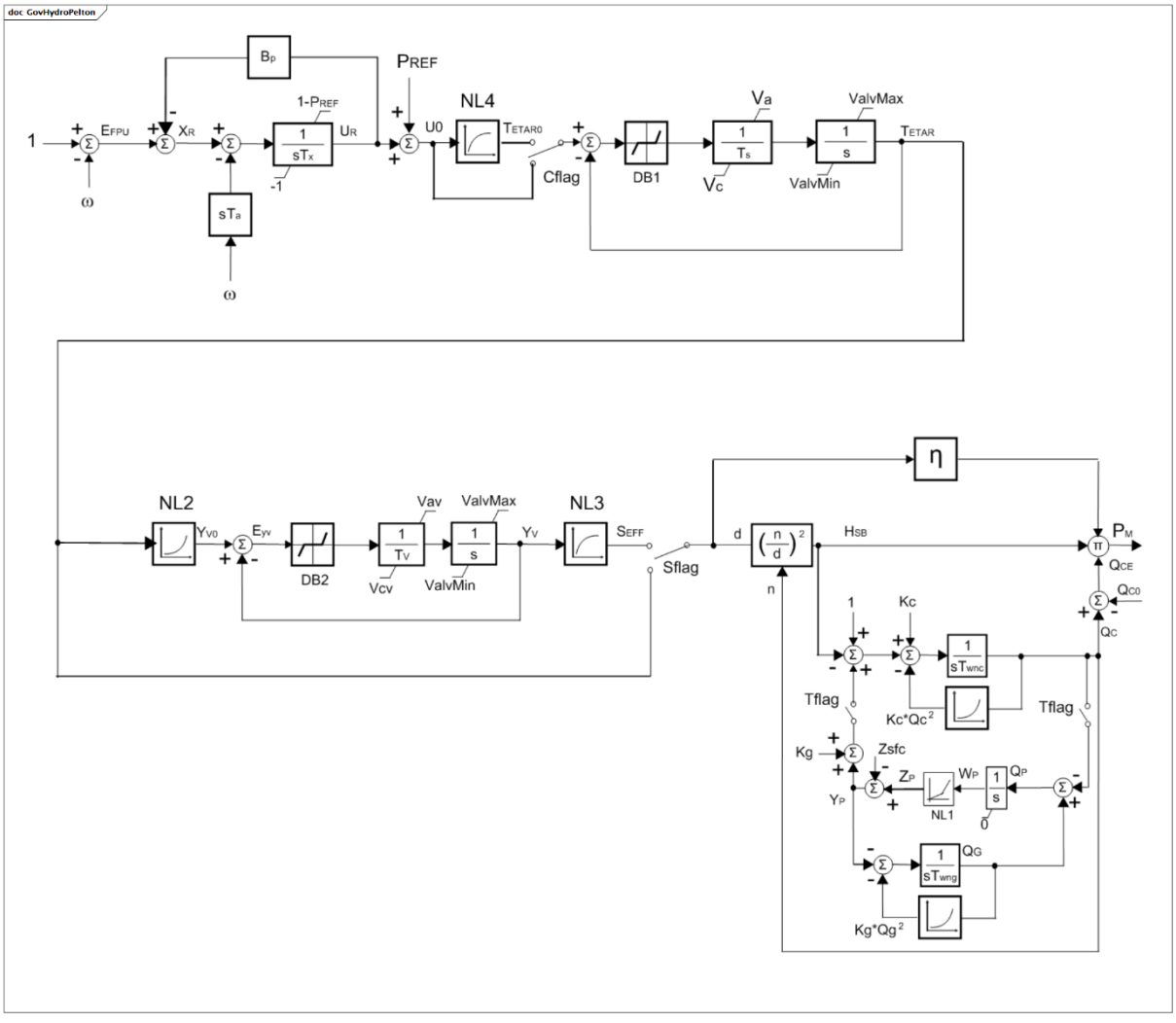

Figure 115 – GovHydroPelton

Figure 115: This diagram describes the GovHydroPelton turbine governor model.

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Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) Rated hydraulic head, Hn, in metres, and rated flow, Qn, in cubic metres per s, are the design parameters of the hydraulic system capability.
3) Hydraulic system head attributes (Zsfc, H1, H2) are in metres but in the block diagram are in PU on base of Hn in metres.
4) Hydraulic area attributes (Av0, Av1) are in square metres but in the block diagram are in PU on base of Qn / Hn in square metres per s.
5) omega has units of PU speed.
6) Non-linear gains and efficiency shown in GovHydroPeltonNonLinearGainAndEfficiency diagram.

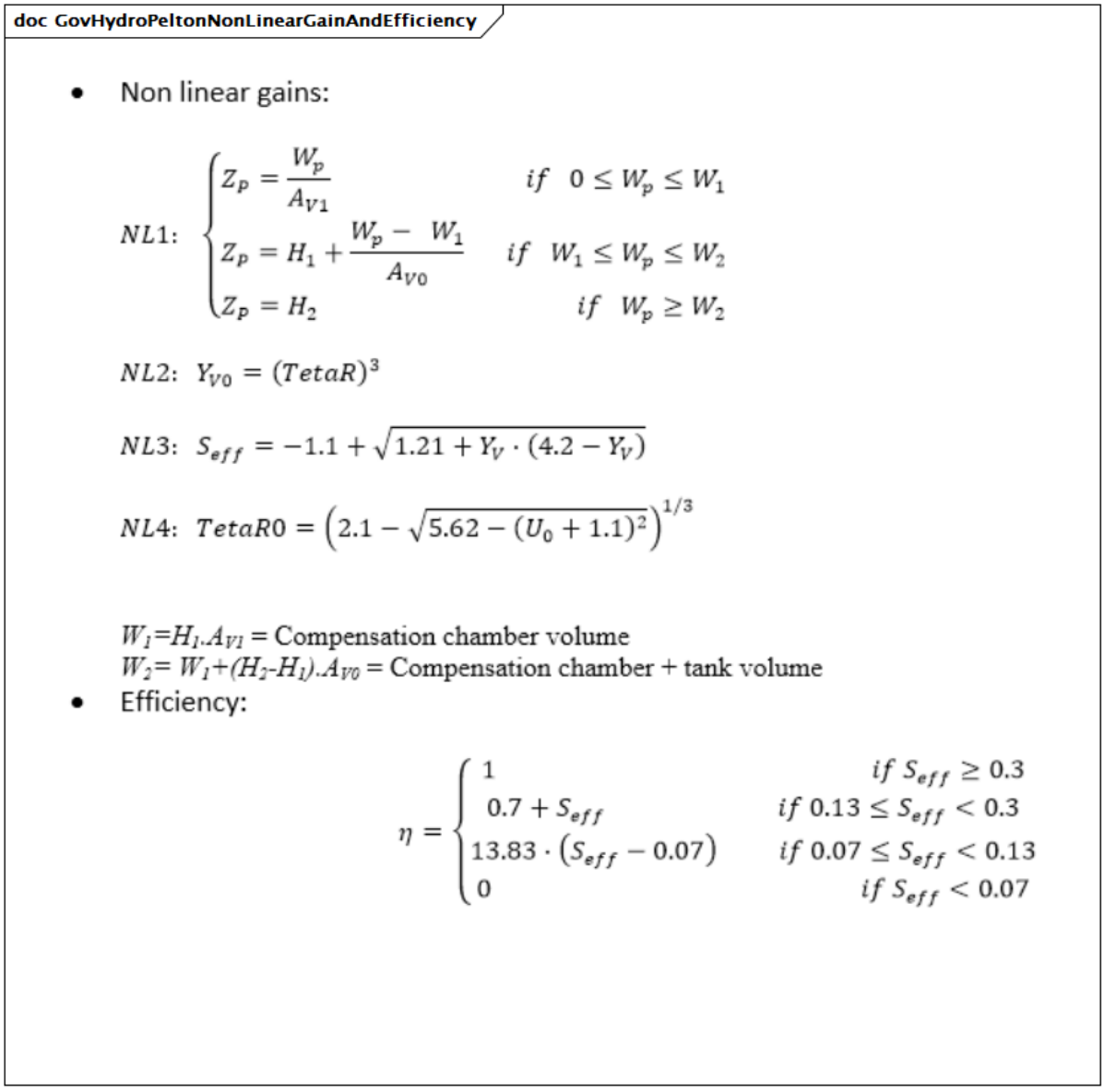

Figure 116 – GovHydroPeltonNonLinearGainAndEfficiency

Figure 116: This diagram illustrates the non-linear gain and efficiency equations.

Table 714 shows all attributes of GovHydroPelton.

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Table 714 – Attributes of TurbineGovernorDynamics::GovHydroPelton

name	mult	type	description
av0	1..1	Area	Area of the surge tank (AV0). Unit = m2. Typical value = 30.
av1	1..1	Area	Area of the compensation tank (AV1). Unit = m2. Typical value = 700.
bp	1..1	PU	Droop (bp). Typical value = 0,05.
db1	1..1	Frequency	Intentional dead-band width (DB1). Unit = Hz. Typical value = 0.
db2	1..1	Frequency	Intentional dead-band width of valve opening error (DB2). Unit = Hz. Typical value = 0,01.
h1	1..1	Length	Head of compensation chamber water level with respect to the level of penstock (H1). Unit = km. Typical value = 0,004.
h2	1..1	Length	Head of surge tank water level with respect to the level of penstock (H2). Unit = km. Typical value = 0,040.
hn	1..1	Length	Rated hydraulic head (Hn). Unit = km. Typical value = 0,250.
kc	1..1	PU	Penstock loss coefficient (due to friction) (Kc). Typical value = 0,025.
kg	1..1	PU	Water tunnel and surge chamber loss coefficient (due to friction) (Kg). Typical value = 0,025.
qc0	1..1	PU	No-load turbine flow at nominal head (Qc0). Typical value = 0,05.
qn	1..1	VolumeFlowRate	Rated flow (Qn). Unit = m3/s. Typical value = 250.
simplifiedPelton	1..1	Boolean	Simplified Pelton model simulation (Sflag). true = enable of simplified Pelton model simulation false = enable of complete Pelton model simulation (non-linear gain). Typical value = true.
staticCompensating	1..1	Boolean	Static compensating characteristic (Cflag). It should be true if simplifiedPelton = false. true = enable of static compensating characteristic false = inhibit of static compensating characteristic. Typical value = false.
ta	1..1	Seconds	Derivative gain (accelerometer time constant) (Ta) (>= 0). Typical value = 3.
ts	1..1	Seconds	Gate servo time constant (Ts) (>= 0). Typical value = 0,15.
tv	1..1	Seconds	Servomotor integrator time constant (Tv) (>= 0). Typical value = 0,3.
twnc	1..1	Seconds	Water inertia time constant (Twnc) (>= 0). Typical value = 1.
twng	1..1	Seconds	Water tunnel and surge chamber inertia time constant (Twng) (>= 0). Typical value = 3.
tx	1..1	Seconds	Electronic integrator time constant (Tx) (>= 0). Typical value = 0,5.
va	1..1	Float	Maximum gate opening velocity (Va). Unit = PU / s. Typical value = 0,06.
valvmax	1..1	PU	Maximum gate opening (ValvMax) (> GovHydroPelton.valvmin). Typical value = 1,1.

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name	mult	type	description
valvmin	1..1	PU	Minimum gate opening (ValvMin) (< GovHydroPelton.valvmax). Typical value = 0.
vav	1..1	PU	Maximum servomotor valve opening velocity (Vav). Typical value = 0,1.
vc	1..1	Float	Maximum gate closing velocity (Vc). Unit = PU / s. Typical value = -0,06.
vcv	1..1	PU	Maximum servomotor valve closing velocity (Vcv). Typical value = -0,1.
waterTunnelSurgeChamberSimulation	1..1	Boolean	Water tunnel and surge chamber simulation (Tflag). true = enable of water tunnel and surge chamber simulation false = inhibit of water tunnel and surge chamber simulation. Typical value = false.
zsfc	1..1	Length	Head of upper water level with respect to the level of penstock (Zsfc). Unit = km. Typical value = 0,025.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 715 shows all association ends of GovHydroPelton with other classes.

Table 715 – Association ends of TurbineGovernorDynamics:: GovHydroPelton with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.22 GovHydroPID

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

PID governor and turbine.

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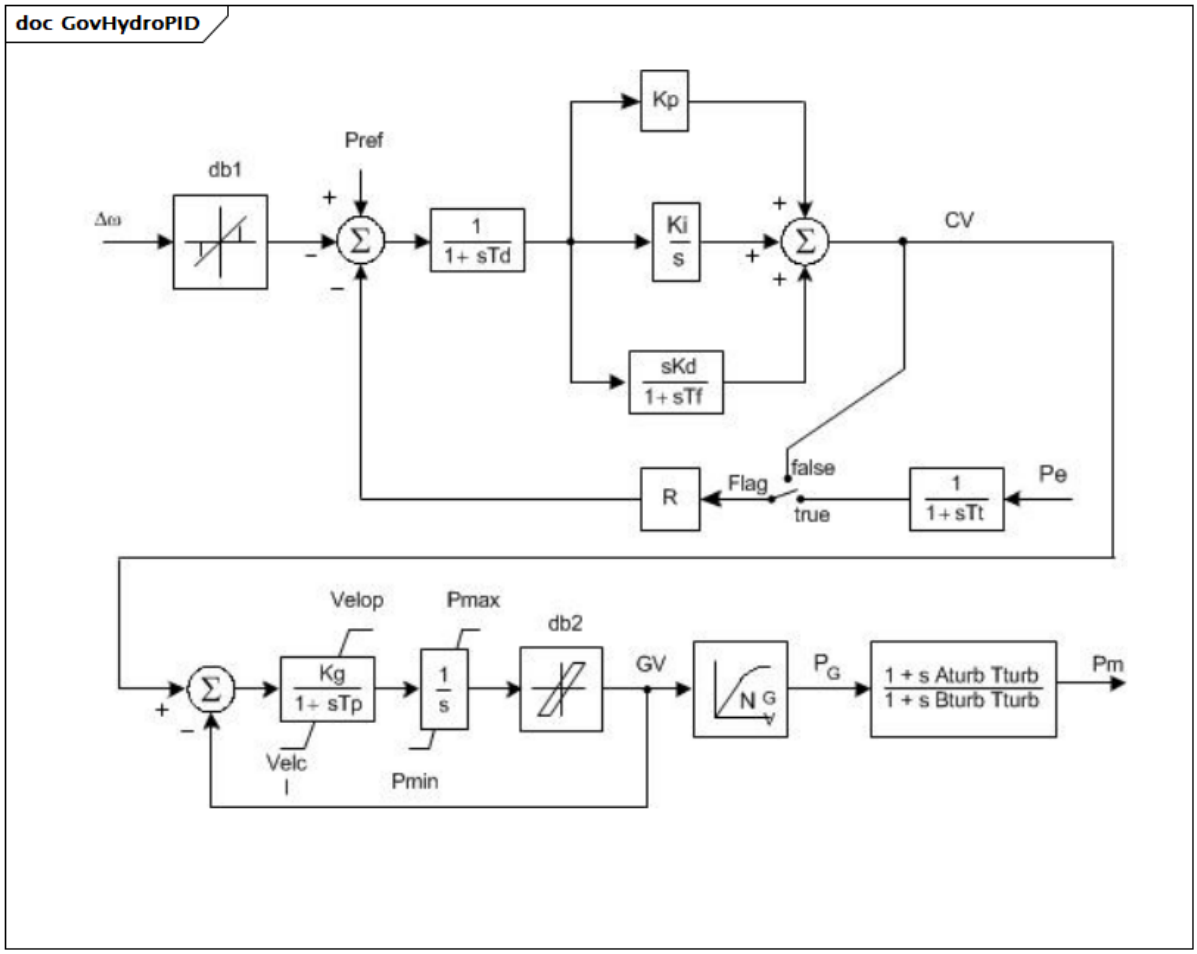

Figure 117 – GovHydroPID

Figure 117: This diagram describes the GovHydroPID turbine governor model.

Parameter details:

1) If Tf is zero, but Kd is not zero, Tf is set to a small value (4 times integration time step).
2) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.
3) For a hydro turbine, set Tturb = Tw (water starting time), set Aturb = -1 and Bturb = 0,5. For a diesel engine or two-shaft gas turbine, set Aturb = 0,8 and Bturb = 1 and Tturb = 1 s. For a small steam turbine with a non-reheat steam supply or a single- shaft gas turbine, set Aturb = 0 and Bturb = 1 and Tturb = 0 s.

Table 716 shows all attributes of GovHydroPID.

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Table 716 – Attributes of TurbineGovernorDynamics::GovHydroPID

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax) (> GovHydroPID.pmin). Typical value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin) (< GovHydroPID.pmax). Typical value = 0.
r	1..1	PU	Steady state droop (R). Typical value = 0,05.
td	1..1	Seconds	Input filter time constant (Td) (>= 0). If = 0, block is bypassed. Typical value = 0.
tf	1..1	Seconds	Washout time constant (Tf) (>= 0). Typical value = 0,1.
tp	1..1	Seconds	Gate servo time constant (Tp) (>= 0). If = 0, block is bypassed. Typical value = 0,35.
velop	1..1	Float	Maximum gate opening velocity (Velop). Unit = PU / s. Typical value = 0,09.
velcl	1..1	Float	Maximum gate closing velocity (Velcl). Unit = PU / s. Typical value = -0,14.
kd	1..1	PU	Derivative gain (Kd). Typical value = 1,11.
kp	1..1	PU	Proportional gain (Kp). Typical value = 0,1.
ki	1..1	PU	Integral gain (Ki). Typical value = 0,36.
kg	1..1	PU	Gate servo gain (Kg). Typical value = 2,5.
tturb	1..1	Seconds	Turbine time constant (Tturb) (>= 0). See Parameter detail 3. Typical value = 0,8.
aturb	1..1	PU	Turbine numerator multiplier (Aturb) (see parameter detail 3). Typical value -1.
bturb	1..1	PU	Turbine denominator multiplier (Bturb) (see parameter detail 3). Typical value = 0,5.
tt	1..1	Seconds	Power feedback time constant (Tt) (>= 0). If = 0, block is bypassed. Typical value = 0,02.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical value = 0.
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tt is zero, Flag is set to false. If Tt is not zero, Flag is set to true. Typical value = true.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.

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name	mult	type	description
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 717 shows all association ends of GovHydroPID with other classes.

Table 717 – Association ends of TurbineGovernorDynamics:: GovHydroPID with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.23 GovHydroPID2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Hydro turbine and governor. Represents plants with straightforward penstock configurations and "three term" electro-hydraulic governors (i.e. Woodward™ electronic)[2].

2 Woodward™ electronic governors are an example of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

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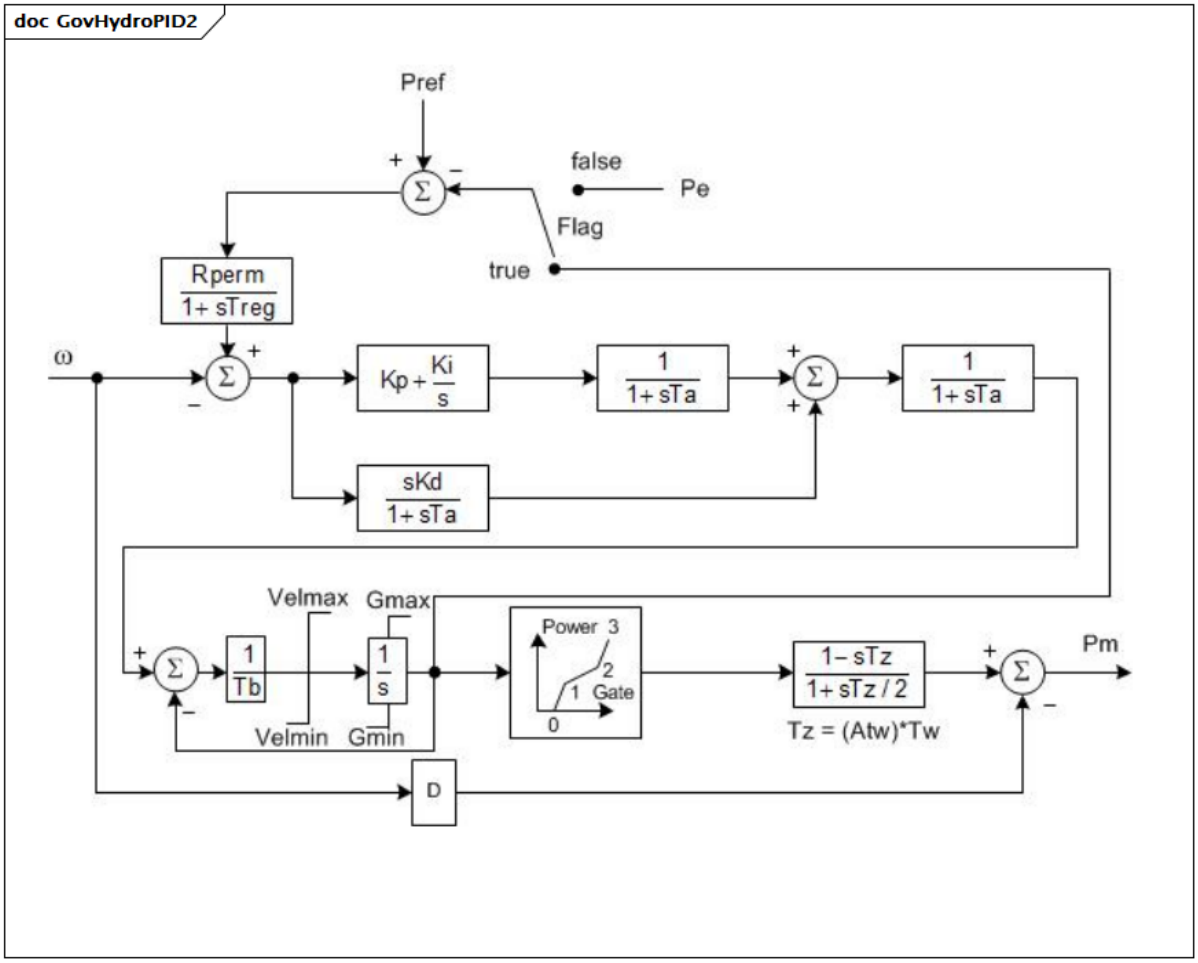

Figure 118 – GovHydroPID2

Figure 118: This diagram describes the GovHydroPID2 turbine governor model.

Parameter details:

1) The gates travel over a range of 1,0 PU from fully closed to fully opened. The gates are at a position greater than zero at zero power and, normally, less than 1,0 when power is 1,0 PU.
2) G0, G1, P1, G2, P2, and P3 specify the variation of power with gate G2 > G1 > G0.
3) This model uses a simplified turbine-penstock model that does not account for the variation of the water inertia effect with gate opening. To make this model correspond to other models using the "classical" turbine-penstock model, set Atw to unity. To approximate water inertia effects at part load, Atw can be set to the PU gate opening about which perturbations are centred.

Table 718 shows all attributes of GovHydroPID2.

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Table 718 – Attributes of TurbineGovernorDynamics::GovHydroPID2

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (>0). Unit = MW.
treg	1..1	Seconds	Speed detector time constant (Treg) (>= 0). Typical value = 0.
rperm	1..1	PU	Permanent drop (Rperm). Typical value = 0.
kp	1..1	PU	Proportional gain (Kp). Typical value = 0.
ki	1..1	Float	Reset gain (Ki). Unit = PU/s. Typical value = 0.
kd	1..1	PU	Derivative gain (Kd). Typical value = 0.
ta	1..1	Seconds	Controller time constant (Ta) (>= 0). Typical value = 0.
tb	1..1	Seconds	Gate servo time constant (Tb) (> 0).
velmax	1..1	Float	Maximum gate opening velocity (Velmax) (< GovHydroPID2.velmin). Unit = PU / s. Typical value = 0.
velmin	1..1	Float	Maximum gate closing velocity (Velmin) (> GovHydroPID2.velmax). Unit = PU / s. Typical value = 0.
gmax	1..1	PU	Maximum gate opening (Gmax) (> GovHydroPID2.gmin). Typical value = 0.
gmin	1..1	PU	Minimum gate opening (Gmin) (> GovHydroPID2.gmax). Typical value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (>= 0). Typical value = 0.
d	1..1	PU	Turbine damping factor (D). Unit = delta P / delta speed. Typical value = 0.
g0	1..1	PU	Gate opening at speed no load (G0). Typical value = 0.
g1	1..1	PU	Intermediate gate opening (G1). Typical value = 0.
p1	1..1	PU	Power at gate opening G1 (P1). Typical value = 0.
g2	1..1	PU	Intermediate gate opening (G2). Typical value = 0.
p2	1..1	PU	Power at gate opening G2 (P2). Typical value = 0.
p3	1..1	PU	Power at full opened gate (P3). Typical value = 0.
atw	1..1	PU	Factor multiplying Tw (Atw). Typical value = 0.
feedbackSignal	1..1	Boolean	Feedback signal type flag (Flag). true = use gate position feedback signal false = use Pe.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 719 shows all association ends of GovHydroPID2 with other classes.

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Table 719 – Association ends of TurbineGovernorDynamics:: GovHydroPID2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.24 GovHydroR

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Fourth order lead-lag governor and hydro turbine.

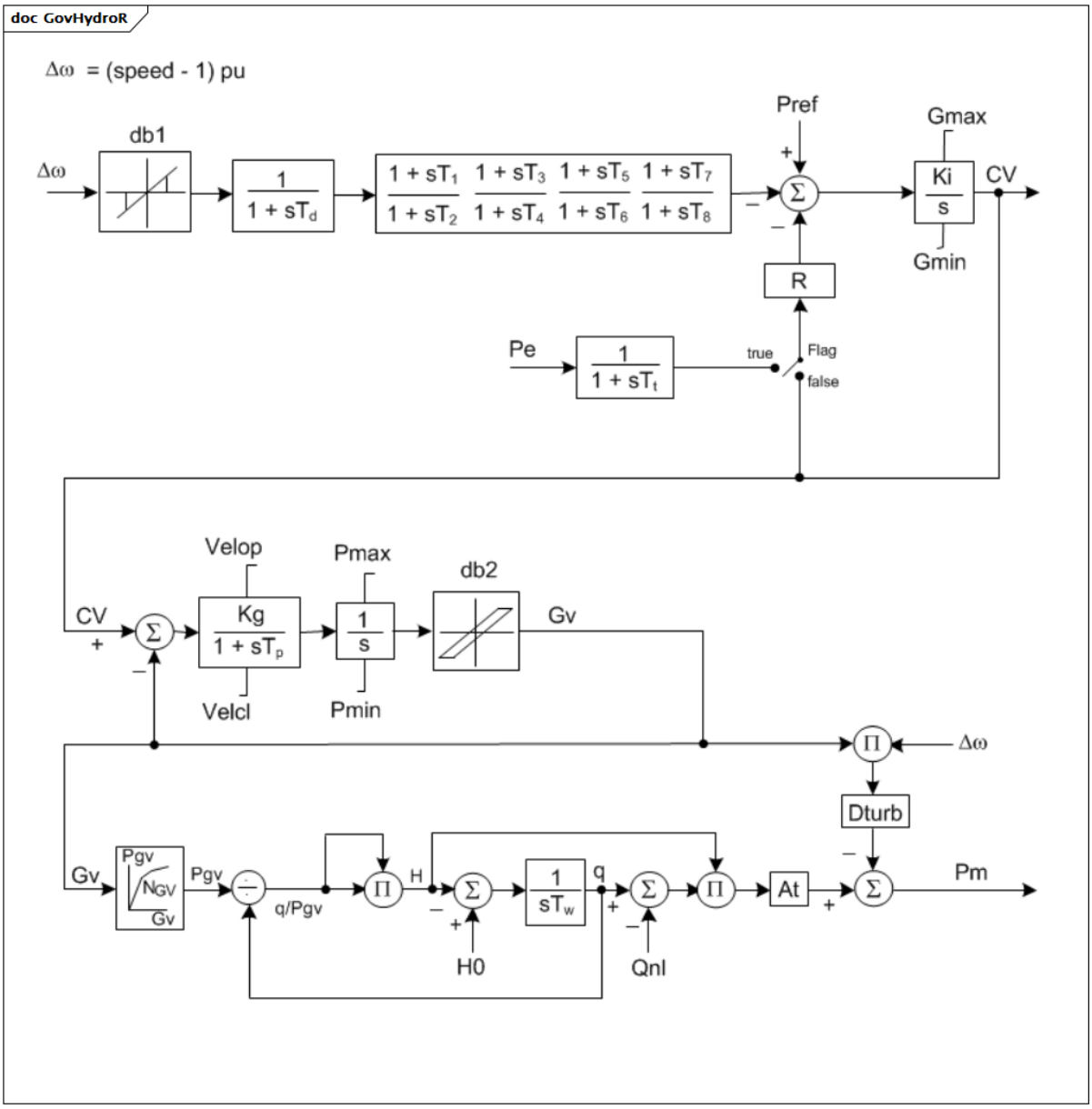

Figure 119 – GovHydroR

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Figure 119: This diagram describes the GovHydroR turbine governor model.

Parameter details:

1) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.

Table 720 shows all attributes of GovHydroR.

Table 720 – Attributes of TurbineGovernorDynamics::GovHydroR

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
pmax	1..1	PU	Maximum gate opening, PU of MWbase (Pmax) (> GovHydroR.pmin). Typical value = 1.
pmin	1..1	PU	Minimum gate opening, PU of MWbase (Pmin) (< GovHydroR.pmax). Typical value = 0.
r	1..1	PU	Steady-state droop (R). Typical value = 0,05.
td	1..1	Seconds	Input filter time constant (Td) (>= 0). Typical value = 0,05.
t1	1..1	Seconds	Lead time constant 1 (T1) (>= 0). Typical value = 1,5.
t2	1..1	Seconds	Lag time constant 1 (T2) (>= 0). Typical value = 0,1.
t3	1..1	Seconds	Lead time constant 2 (T3) (>= 0). Typical value = 1,5.
t4	1..1	Seconds	Lag time constant 2 (T4) (>= 0). Typical value = 0,1.
t5	1..1	Seconds	Lead time constant 3 (T5) (>= 0). Typical value = 0.
t6	1..1	Seconds	Lag time constant 3 (T6) (>= 0). Typical value = 0,05.
t7	1..1	Seconds	Lead time constant 4 (T7) (>= 0). Typical value = 0.
t8	1..1	Seconds	Lag time constant 4 (T8) (>= 0). Typical value = 0,05.
tp	1..1	Seconds	Gate servo time constant (Tp) (>= 0). Typical value = 0,05.
velop	1..1	Float	Maximum gate opening velocity (Velop). Unit = PU / s. Typical value = 0,2.
velcl	1..1	Float	Maximum gate closing velocity (Velcl). Unit = PU / s. Typical value = -0,2.
ki	1..1	PU	Integral gain (Ki). Typical value = 0,5.
kg	1..1	PU	Gate servo gain (Kg). Typical value = 2.
gmax	1..1	PU	Maximum governor output (Gmax) (> GovHydroR.gmin). Typical value = 1,05.
gmin	1..1	PU	Minimum governor output (Gmin) (< GovHydroR.gmax). Typical value = -0,05.
tt	1..1	Seconds	Power feedback time constant (Tt) (>= 0). Typical value = 0.

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name	mult	type	description
inputSignal	1..1	Boolean	Input signal switch (Flag). true = Pe input is used false = feedback is received from CV. Flag is normally dependent on Tt. If Tt is zero, Flag is set to false. If Tt is not zero, Flag is set to true. Typical value = true.
db1	1..1	Frequency	Intentional dead-band width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
db2	1..1	ActivePower	Unintentional dead-band (db2). Unit = MW. Typical value = 0.
tw	1..1	Seconds	Water inertia time constant (Tw) (> 0). Typical value = 1.
at	1..1	PU	Turbine gain (At). Typical value = 1,2.
dturb	1..1	PU	Turbine damping factor (Dturb). Typical value = 0,2.
qnl	1..1	PU	No-load turbine flow at nominal head (Qnl). Typical value = 0,08.
h0	1..1	PU	Turbine nominal head (H0). Typical value = 1.
gv1	1..1	PU	Nonlinear gain point 1, PU gv (Gv1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain point 1, PU power (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain point 2, PU gv (Gv2). Typical value = 0.
pgv2	1..1	PU	Nonlinear gain point 2, PU power (Pgv2). Typical value = 0.
gv3	1..1	PU	Nonlinear gain point 3, PU gv (Gv3). Typical value = 0.
pgv3	1..1	PU	Nonlinear gain point 3, PU power (Pgv3). Typical value = 0.
gv4	1..1	PU	Nonlinear gain point 4, PU gv (Gv4). Typical value = 0.
pgv4	1..1	PU	Nonlinear gain point 4, PU power (Pgv4). Typical value = 0.
gv5	1..1	PU	Nonlinear gain point 5, PU gv (Gv5). Typical value = 0.
pgv5	1..1	PU	Nonlinear gain point 5, PU power (Pgv5). Typical value = 0.
gv6	1..1	PU	Nonlinear gain point 6, PU gv (Gv6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain point 6, PU power (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 721 shows all association ends of GovHydroR with other classes.

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Table 721 – Association ends of TurbineGovernorDynamics:: GovHydroR with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.25 GovHydroWEH

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Woodward™ electric hydro governor[3].

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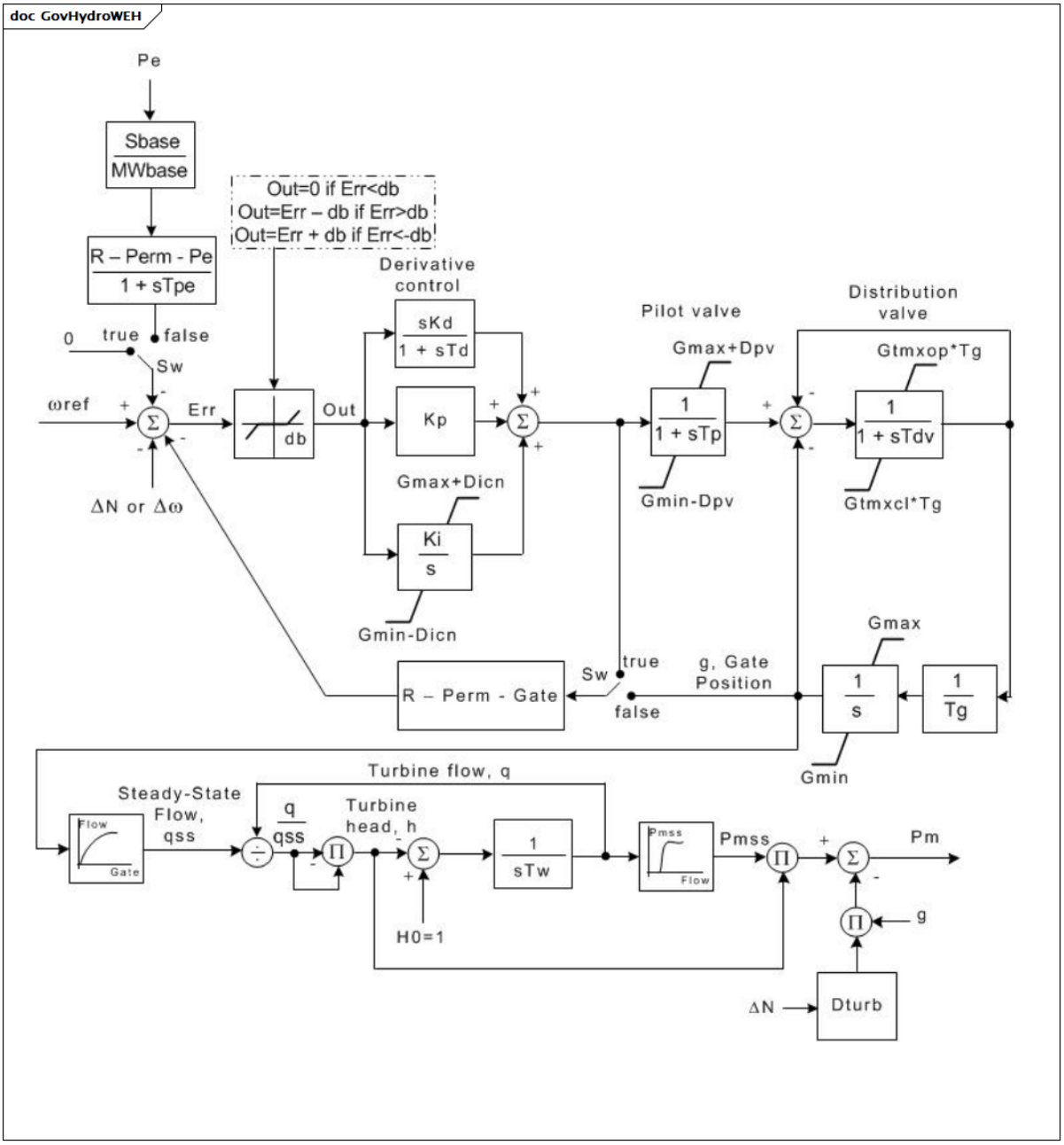

Figure 120 – GovHydroWEH

Figure 120: This diagram describes the GovHydroWEH turbine governor model.

Parameter details:

1) The nonlinear relationship between gate position and water flow can be input with up to 5 points specified by pairs of .gvn and .fln attributes.
2) The nonlinear relationship between turbine flow and mechanical power output (on machine MVA rating) can be input with up to 10 points specified by pairs of .fpn and .pmssn attributes.

Table 722 shows all attributes of GovHydroWEH.

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Table 722 – Attributes of TurbineGovernorDynamics::GovHydroWEH

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
rpg	1..1	Float	Permanent droop for governor output feedback (R-Perm-Gate).
rpp	1..1	Float	Permanent droop for electrical power feedback (R-Perm-Pe).
tpe	1..1	Seconds	Electrical power droop time constant (Tpe) (>= 0).
kp	1..1	PU	Derivative control gain (Kp).
ki	1..1	PU	Derivative controller Integral gain (Ki).
kd	1..1	PU	Derivative controller derivative gain (Kd).
td	1..1	Seconds	Derivative controller time constant (Td) (>= 0). Limits the derivative characteristic beyond a breakdown frequency to avoid amplification of high-frequency noise.
tp	1..1	Seconds	Pilot valve time lag time constant (Tp) (>= 0).
tdv	1..1	Seconds	Distributive valve time lag time constant (Tdv) (>= 0).
tg	1..1	Seconds	Value to allow the distribution valve controller to advance beyond the gate movement rate limit (Tg) (>= 0).
gtmxop	1..1	PU	Maximum gate opening rate (Gtmxop).
gtmxcl	1..1	PU	Maximum gate closing rate (Gtmxcl).
gmax	1..1	PU	Maximum gate position (Gmax) (> GovHydroWEH.gmin).
gmin	1..1	PU	Minimum gate position (Gmin) (< GovHydroWEH.gmax).
dturb	1..1	PU	Turbine damping factor (Dturb). Unit = delta P (PU of MWbase) / delta speed (PU).
tw	1..1	Seconds	Water inertia time constant (Tw) (> 0).
db	1..1	PU	Speed deadband (db).
dpv	1..1	PU	Value to allow the pilot valve controller to advance beyond the gate limits (Dpv).
dicn	1..1	PU	Value to allow the integral controller to advance beyond the gate limits (Dicn).
feedbackSignal	1..1	Boolean	Feedback signal selection (Sw). true = PID output (if R-Perm-Gate = droop and R-Perm-Pe = 0) false = electrical power (if R-Perm-Gate = 0 and R-Perm-Pe = droop) or false = gate position (if R-Perm-Gate = droop and R-Perm-Pe = 0). Typical value = false.
gv1	1..1	PU	Gate 1 (Gv1). Gate Position value for point 1 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv2	1..1	PU	Gate 2 (Gv2). Gate Position value for point 2 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.

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name	mult	type	description
gv3	1..1	PU	Gate 3 (Gv3). Gate Position value for point 3 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv4	1..1	PU	Gate 4 (Gv4). Gate Position value for point 4 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
gv5	1..1	PU	Gate 5 (Gv5). Gate Position value for point 5 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl1	1..1	PU	Flowgate 1 (Fl1). Flow value for gate position point 1 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl2	1..1	PU	Flowgate 2 (Fl2). Flow value for gate position point 2 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl3	1..1	PU	Flowgate 3 (Fl3). Flow value for gate position point 3 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl4	1..1	PU	Flowgate 4 (Fl4). Flow value for gate position point 4 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fl5	1..1	PU	Flowgate 5 (Fl5). Flow value for gate position point 5 for lookup table representing water flow through the turbine as a function of gate position to produce steady state flow.
fp1	1..1	PU	Flow P1 (Fp1). Turbine flow value for point 1 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp2	1..1	PU	Flow P2 (Fp2). Turbine flow value for point 2 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp3	1..1	PU	Flow P3 (Fp3). Turbine flow value for point 3 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp4	1..1	PU	Flow P4 (Fp4). Turbine flow value for point 4 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp5	1..1	PU	Flow P5 (Fp5). Turbine flow value for point 5 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp6	1..1	PU	Flow P6 (Fp6). Turbine flow value for point 6 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp7	1..1	PU	Flow P7 (Fp7). Turbine flow value for point 7 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp8	1..1	PU	Flow P8 (Fp8). Turbine flow value for point 8 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.

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name	mult	type	description
fp9	1..1	PU	Flow P9 (Fp9). Turbine flow value for point 9 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
fp10	1..1	PU	Flow P10 (Fp10). Turbine flow value for point 10 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss1	1..1	PU	Pmss flow P1 (Pmss1). Mechanical power output for turbine flow point 1 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss2	1..1	PU	Pmss flow P2 (Pmss2). Mechanical power output for turbine flow point 2 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss3	1..1	PU	Pmss flow P3 (Pmss3). Mechanical power output for turbine flow point 3 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss4	1..1	PU	Pmss flow P4 (Pmss4). Mechanical power output for turbine flow point 4 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss5	1..1	PU	Pmss flow P5 (Pmss5). Mechanical power output for turbine flow point 5 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss6	1..1	PU	Pmss flow P6 (Pmss6). Mechanical power output for turbine flow point 6 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss7	1..1	PU	Pmss flow P7 (Pmss7). Mechanical power output for turbine flow point 7 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss8	1..1	PU	Pmss flow P8 (Pmss8). Mechanical power output for turbine flow point 8 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss9	1..1	PU	Pmss flow P9 (Pmss9). Mechanical power output for turbine flow point 9 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
pmss10	1..1	PU	Pmss flow P10 (Pmss10). Mechanical power output for turbine flow point 10 for lookup table representing PU mechanical power on machine MVA rating as a function of turbine flow.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 723 shows all association ends of GovHydroWEH with other classes.

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Table 723 – Association ends of TurbineGovernorDynamics:: GovHydroWEH with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.26 GovHydroWPID

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Woodward™ PID hydro governor[4].

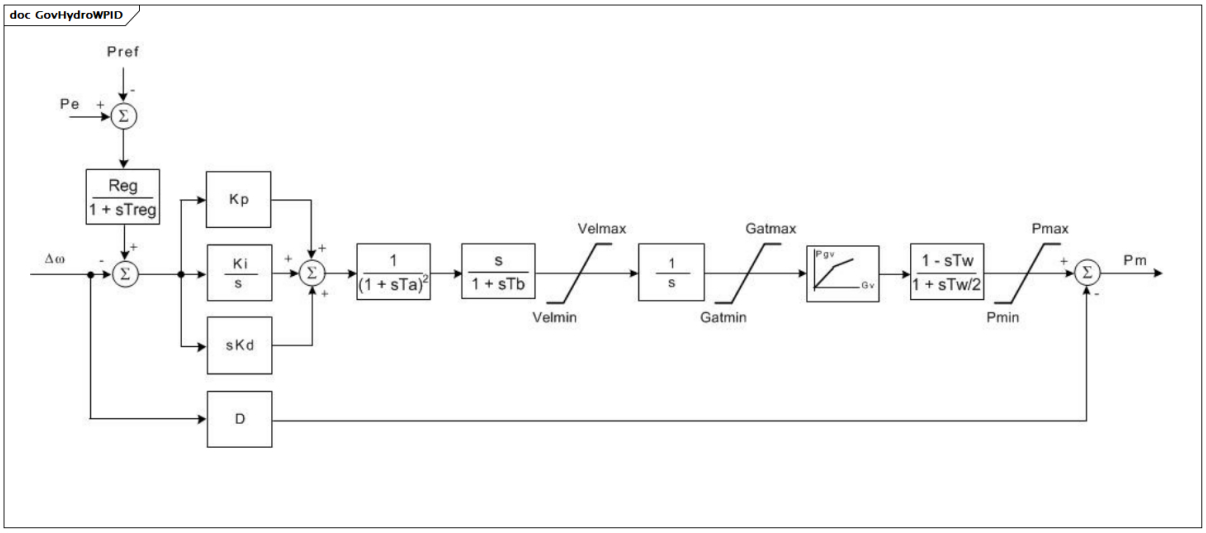

Figure 121 – GovHydroWPID

Figure 121: This diagram describes the GovHydroWPID turbine governor model.

Parameter details:

1) The nonlinear gain between gate position and power can be input with up to 3 points. The (0,0, 0,0) point can be assumed. In this model, Gv3 = 1,0 and the output is not allowed to go above 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default hydro curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.

Table 724 shows all attributes of GovHydroWPID.

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Table 724 – Attributes of TurbineGovernorDynamics::GovHydroWPID

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
treg	1..1	Seconds	Speed detector time constant (Treg) (>= 0).
reg	1..1	PU	Permanent drop (Reg).
kp	1..1	PU	Proportional gain (Kp). Typical value = 0,1.
ki	1..1	PU	Reset gain (Ki). Typical value = 0,36.
kd	1..1	PU	Derivative gain (Kd). Typical value = 1,11.
ta	1..1	Seconds	Controller time constant (Ta) (>= 0). Typical value = 0.
tb	1..1	Seconds	Gate servo time constant (Tb) (>= 0). Typical value = 0.
velmax	1..1	PU	Maximum gate opening velocity (Velmax) (> GovHydroWPID.velmin). Unit = PU / s. Typical value = 0.
velmin	1..1	PU	Maximum gate closing velocity (Velmin) (< GovHydroWPID.velmax). Unit = PU / s. Typical value = 0.
gatmax	1..1	PU	Gate opening limit maximum (Gatmax) (> GovHydroWPID.gatmin).
gatmin	1..1	PU	Gate opening limit minimum (Gatmin) (< GovHydroWPID.gatmax).
tw	1..1	Seconds	Water inertia time constant (Tw) (>= 0). Typical value = 0.
pmax	1..1	PU	Maximum power output (Pmax) (> GovHydroWPID.pmin).
pmin	1..1	PU	Minimum power output (Pmin) (< GovHydroWPID.pmax).
d	1..1	PU	Turbine damping factor (D). Unit = delta P / delta speed.
gv3	1..1	PU	Gate position 3 (Gv3) (= 1,0).
gv1	1..1	PU	Gate position 1 (Gv1).
pgv1	1..1	PU	Output at Gv1 PU of MWbase (Pgv1).
gv2	1..1	PU	Gate position 2 (Gv2).
pgv2	1..1	PU	Output at Gv2 PU of MWbase (Pgv2).
pgv3	1..1	PU	Output at Gv3 PU of MWbase (Pgv3).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 725 shows all association ends of GovHydroWPID with other classes.

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Table 725 – Association ends of TurbineGovernorDynamics:: GovHydroWPID with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.27 GovSteam0

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

A simplified steam turbine governor.

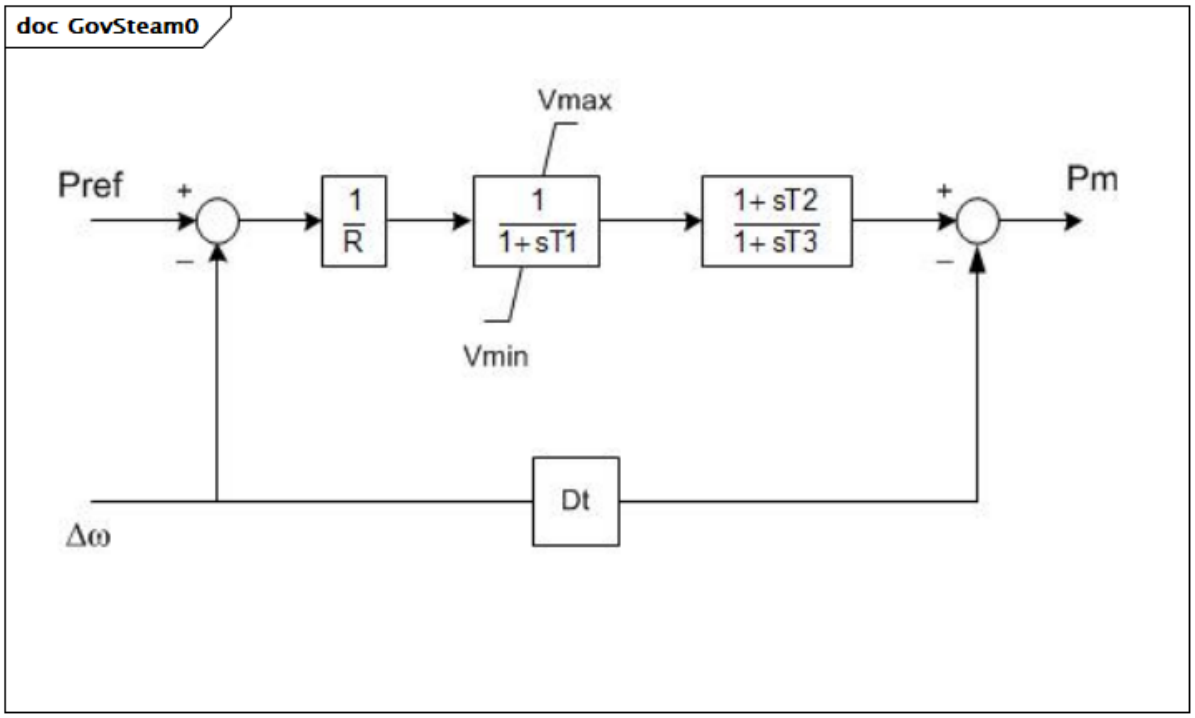

Figure 122 – GovSteam0

Figure 122: This diagram describes the GovSteam0 turbine governor model.

Parameter details:

1) To represent a non-reheat turbine set T2= 0 and T3 to about 0,5 s. For reheat turbine, set T2 = T3 x (HP turbine fraction).

Table 726 shows all attributes of GovSteam0.

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Table 726 – Attributes of TurbineGovernorDynamics::GovSteam0

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
r	1..1	PU	Permanent droop (R). Typical value = 0,05.
t1	1..1	Seconds	Steam bowl time constant (T1) (> 0). Typical value = 0,5.
vmax	1..1	PU	Maximum valve position, PU of mwcap (Vmax) (> GovSteam0.vmin). Typical value = 1.
vmin	1..1	PU	Minimum valve position, PU of mwcap (Vmin) (< GovSteam0.vmax). Typical value = 0.
t2	1..1	Seconds	Numerator time constant of T2/T3 block (T2) (>= 0). Typical value = 3.
t3	1..1	Seconds	Reheater time constant (T3) (> 0). Typical value = 10.
dt	1..1	PU	Turbine damping coefficient (Dt). Unit = delta P / delta speed. Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 727 shows all association ends of GovSteam0 with other classes.

Table 727 – Association ends of TurbineGovernorDynamics:: GovSteam0 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.28 GovSteam1

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Steam turbine governor, based on the GovSteamIEEE1 (with optional deadband and nonlinear valve gain added).

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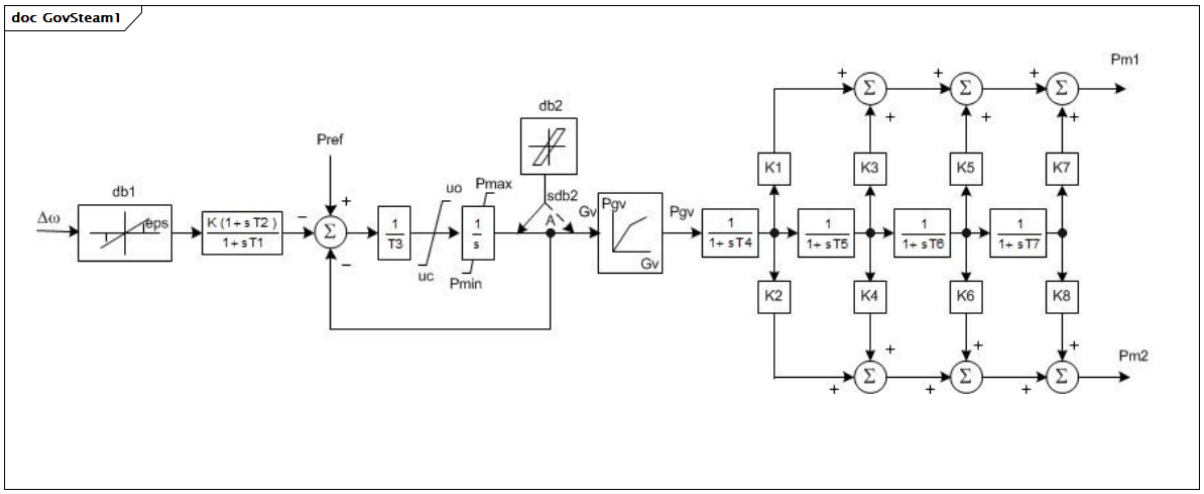

Figure 123 – GovSteam1

Figure 123: This diagram describes the GovSteam1 turbine governor model.

Parameter details:

1) PU parameters are on base of MWbase, which is normally the MW capability of the turbine.
2) For a tandem-compound turbine, the parameters K2, K4, K6, and K8 are ignored. For a cross-compound turbine, two generators are connected to this turbine-governor model.
3) Each generator shall be represented in the load flow by data on its own MVA base. The values of K1, K3, K5, and K7 shall be specified to describe the proportionate development of power on the first turbine shaft. K2, K4, K6, and K8 shall describe the second turbine shaft. Normally K1 + K3 + K5 + K7 = 1,0 and K2 + K4 + K6 + K8 = 1,0 (if second generator is present).
4) The division of power between the two shafts is in proportion to the values of MVA bases of the two generators. The initial condition load flow should, therefore, have the two generators loaded to the same fraction of each one's MVA base.
5) The intentional input speed deadband follows the relationship shown in the GovSteamIEEE1InputSpeedDeadband diagram.
6) The unintentional deadband can be used to represent backlash hysteresis in the control components by the relationship shown in the GovSteamIEEE1BacklashHysteresis diagram.
7) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero, the default curve (between (0,0, 0,0) and (1,0, 1,0)) will be used.

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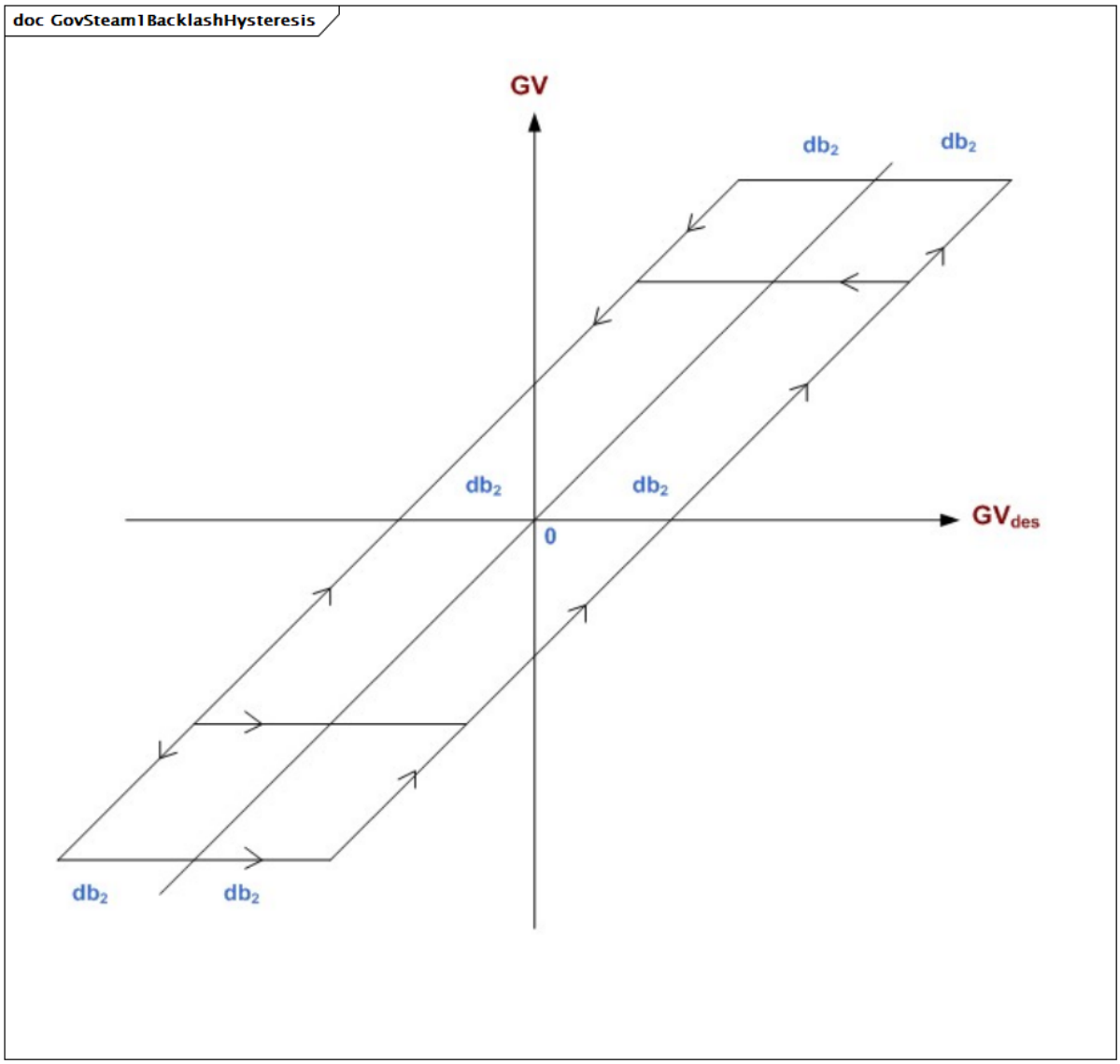

Figure 124 – GovSteam1BacklashHysteresis

Figure 124: This diagram illustrates the backlash hysteresis.

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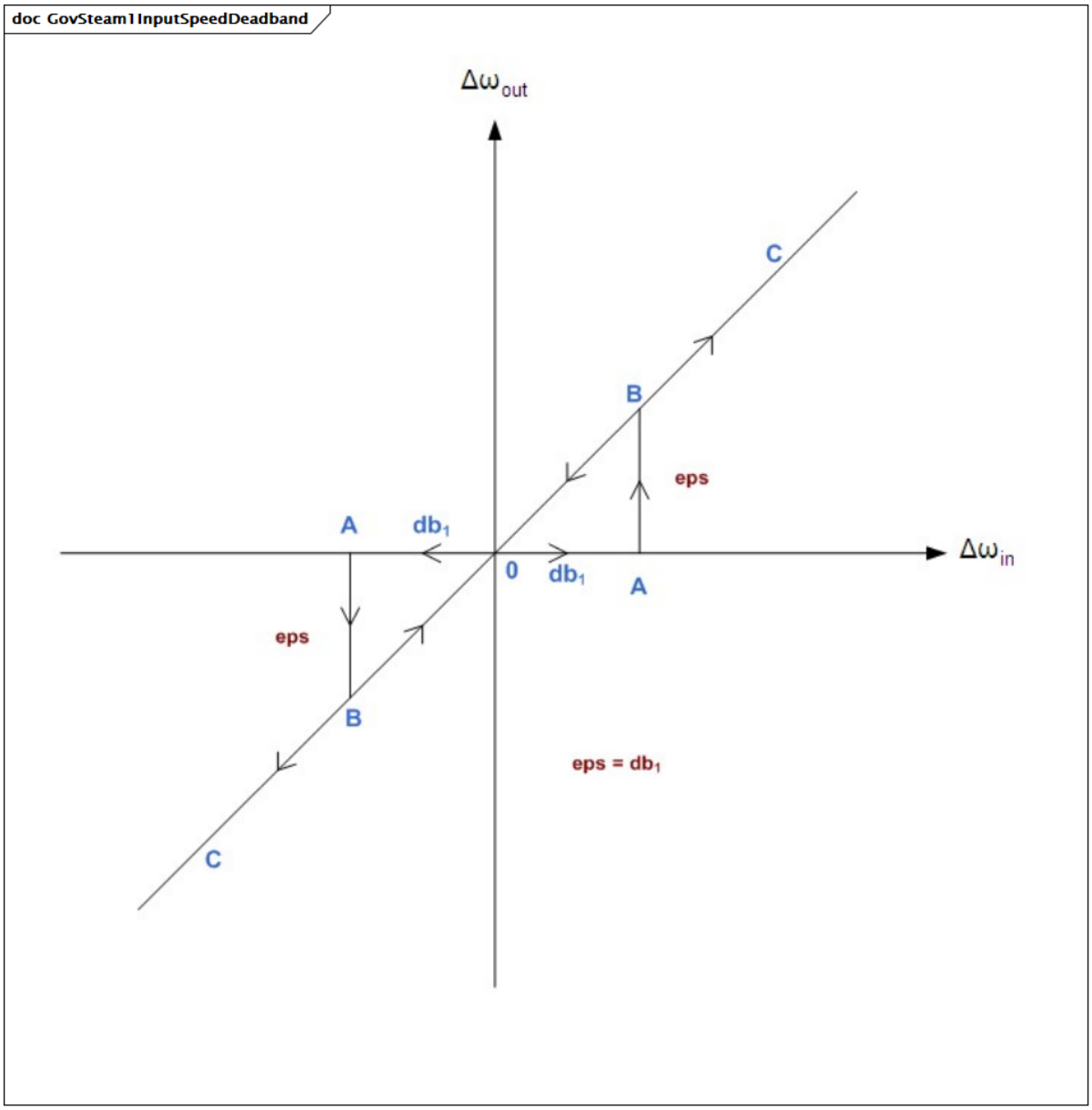

Figure 125 – GovSteam1InputSpeedDeadband

Figure 125: This diagram illustrates the input speed deadband.

Table 728 shows all attributes of GovSteam1.

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Table 728 – Attributes of TurbineGovernorDynamics::GovSteam1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
k	1..1	PU	Governor gain (reciprocal of droop) (K) (> 0). Typical value = 25.
t1	1..1	Seconds	Governor lag time constant (T1) (>= 0). Typical value = 0.
t2	1..1	Seconds	Governor lead time constant (T2) (>= 0). Typical value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3) (> 0). Typical value = 0,1.
uo	1..1	Float	Maximum valve opening velocity (Uo) (> 0). Unit = PU / s. Typical value = 1.
uc	1..1	Float	Maximum valve closing velocity (Uc) (< 0). Unit = PU / s. Typical value = -10.
pmax	1..1	PU	Maximum valve opening (Pmax) (> GovSteam1.pmin). Typical value = 1.
pmin	1..1	PU	Minimum valve opening (Pmin) (>= 0 and < GovSteam1.pmax). Typical value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4) (>= 0). Typical value = 0,3.
k1	1..1	Float	Fraction of HP shaft power after first boiler pass (K1). Typical value = 0,2.
k2	1..1	Float	Fraction of LP shaft power after first boiler pass (K2). Typical value = 0.
t5	1..1	Seconds	Time constant of second boiler pass (T5) (>= 0). Typical value = 5.
k3	1..1	Float	Fraction of HP shaft power after second boiler pass (K3). Typical value = 0,3.
k4	1..1	Float	Fraction of LP shaft power after second boiler pass (K4). Typical value = 0.
t6	1..1	Seconds	Time constant of third boiler pass (T6) (>= 0). Typical value = 0,5.
k5	1..1	Float	Fraction of HP shaft power after third boiler pass (K5). Typical value = 0,5.
k6	1..1	Float	Fraction of LP shaft power after third boiler pass (K6). Typical value = 0.
t7	1..1	Seconds	Time constant of fourth boiler pass (T7) (>= 0). Typical value = 0.
k7	1..1	Float	Fraction of HP shaft power after fourth boiler pass (K7). Typical value = 0.
k8	1..1	Float	Fraction of LP shaft power after fourth boiler pass (K8). Typical value = 0.
db1	1..1	Frequency	Intentional deadband width (db1). Unit = Hz. Typical value = 0.
eps	1..1	Frequency	Intentional db hysteresis (eps). Unit = Hz. Typical value = 0.
sdb1	1..1	Boolean	Intentional deadband indicator. true = intentional deadband is applied false = intentional deadband is not applied. Typical value = true.

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name	mult	type	description
sdb2	1..1	Boolean	Unintentional deadband location. true = intentional deadband is applied before point "A" false = intentional deadband is applied after point "A". Typical value = true.
db2	1..1	ActivePower	Unintentional deadband (db2). Unit = MW. Typical value = 0.
valve	1..1	Boolean	Nonlinear valve characteristic. true = nonlinear valve characteristic is used false = nonlinear valve characteristic is not used. Typical value = true.
gv1	1..1	PU	Nonlinear gain valve position point 1 (GV1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain power value point 1 (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain valve position point 2 (GV2). Typical value = 0,4.
pgv2	1..1	PU	Nonlinear gain power value point 2 (Pgv2). Typical value = 0,75.
gv3	1..1	PU	Nonlinear gain valve position point 3 (GV3). Typical value = 0,5.
pgv3	1..1	PU	Nonlinear gain power value point 3 (Pgv3). Typical value = 0,91.
gv4	1..1	PU	Nonlinear gain valve position point 4 (GV4). Typical value = 0,6.
pgv4	1..1	PU	Nonlinear gain power value point 4 (Pgv4). Typical value = 0,98.
gv5	1..1	PU	Nonlinear gain valve position point 5 (GV5). Typical value = 1.
pgv5	1..1	PU	Nonlinear gain power value point 5 (Pgv5). Typical value = 1.
gv6	1..1	PU	Nonlinear gain valve position point 6 (GV6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain power value point 6 (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 729 shows all association ends of GovSteam1 with other classes.

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Table 729 – Association ends of TurbineGovernorDynamics:: GovSteam1 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.29 GovSteam2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified governor.

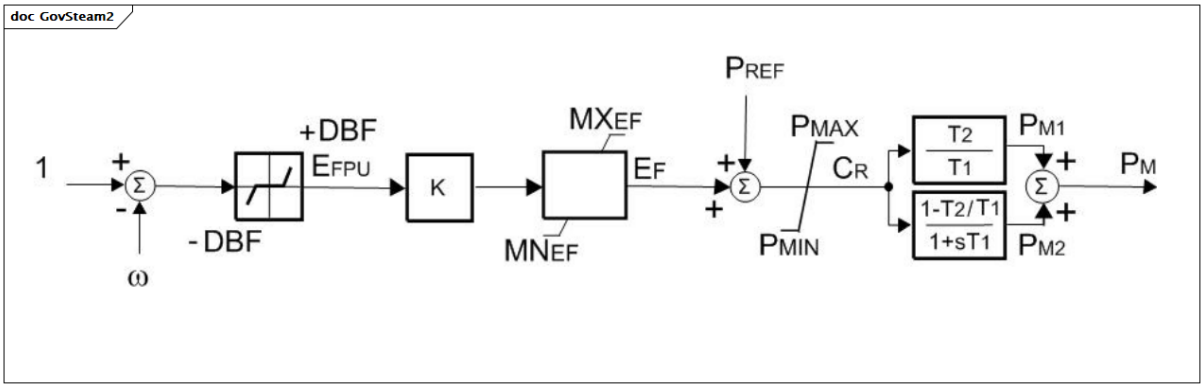

Figure 126 – GovSteam2

Figure 126: This diagram describes the GovSteam2 turbine governor model.

Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) This model can be used for both thermal and hydro power plants.
3) omega has units of per unit speed.

Table 730 shows all attributes of GovSteam2.

Table 730 – Attributes of TurbineGovernorDynamics::GovSteam2

name	mult	type	description
k	1..1	Float	Governor gain (reciprocal of droop) (K). Typical value = 20.
dbf	1..1	PU	Frequency deadband (DBF). Typical value = 0.
t1	1..1	Seconds	Governor lag time constant (T1) (> 0). Typical value = 0,45.
t2	1..1	Seconds	Governor lead time constant (T2) (>= 0). Typical value = 0.
pmax	1..1	PU	Maximum fuel flow (PMAX) (> GovSteam2.pmin). Typical value = 1.
pmin	1..1	PU	Minimum fuel flow (PMIN) (< GovSteam2.pmax). Typical value = 0.

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name	mult	type	description
mxef	1..1	PU	Fuel flow maximum positive error value (MXEF). Typical value = 1.
mnef	1..1	PU	Fuel flow maximum negative error value (MNEF). Typical value = -1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 731 shows all association ends of GovSteam2 with other classes.

Table 731 – Association ends of TurbineGovernorDynamics:: GovSteam2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.30 GovSteamBB

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

European governor model.

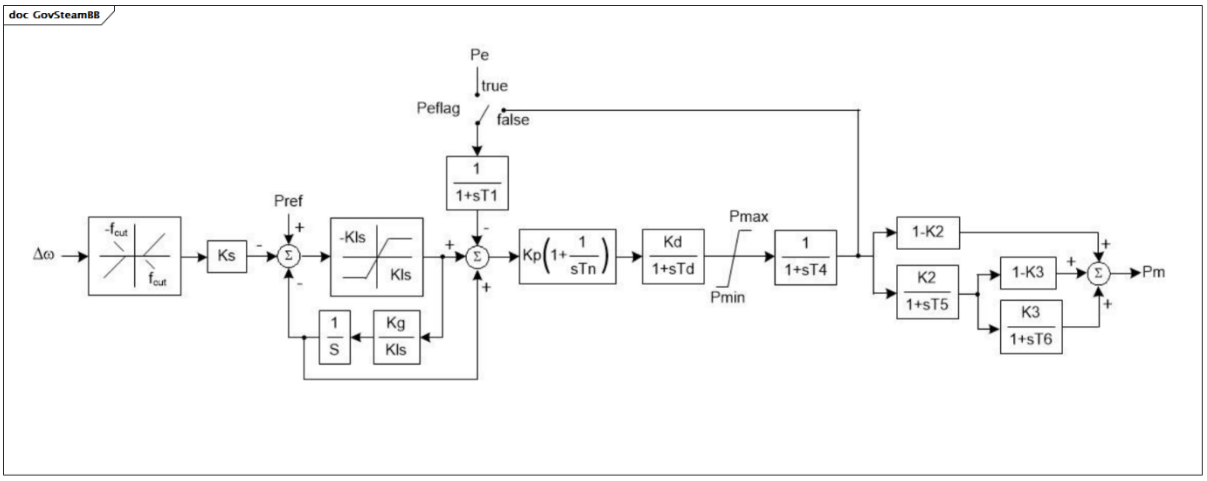

Figure 127 – GovSteamBB

Figure 127: This diagram describes the GovSteamBB turbine governor model.

Table 732 shows all attributes of GovSteamBB.

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Table 732 – Attributes of TurbineGovernorDynamics::GovSteamBB

name	mult	type	description
fcut	1..1	PU	Frequency deadband (fcut) (>= 0). Typical value = 0,002.
ks	1..1	PU	Gain (Ks). Typical value = 21,0.
kls	1..1	PU	Gain (Kls) (> 0). Typical value = 0,1.
kg	1..1	PU	Gain (Kg). Typical value = 1,0.
t1	1..1	Seconds	Time constant (T1). Typical value = 0,05.
kp	1..1	PU	Gain (Kp). Typical value = 1,0.
tn	1..1	Seconds	Time constant (Tn) (> 0). Typical value = 1,0.
kd	1..1	PU	Gain (Kd). Typical value = 1,0.
td	1..1	Seconds	Time constant (Td) (> 0). Typical value = 1,0.
pmax	1..1	PU	High power limit (Pmax) (> GovSteamBB.pmin). Typical value = 1,0.
pmin	1..1	PU	Low power limit (Pmin) (< GovSteamBB.pmax). Typical value = 0.
t4	1..1	Seconds	Time constant (T4). Typical value = 0,15.
k2	1..1	PU	Gain (K2). Typical value = 0,75.
t5	1..1	Seconds	Time constant (T5). Typical value = 12,0.
k3	1..1	PU	Gain (K3). Typical value = 0,5.
t6	1..1	Seconds	Time constant (T6). Typical value = 0,75.
peflag	1..1	Boolean	Electric power input selection (Peflag). true = electric power input false = feedback signal. Typical value = false.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 733 shows all association ends of GovSteamBB with other classes.

Table 733 – Association ends of TurbineGovernorDynamics:: GovSteamBB with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.31 GovSteamCC

Inheritance path = CrossCompoundTurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Cross compound turbine governor. Unlike tandem compound units, cross compound units are not on the same shaft.

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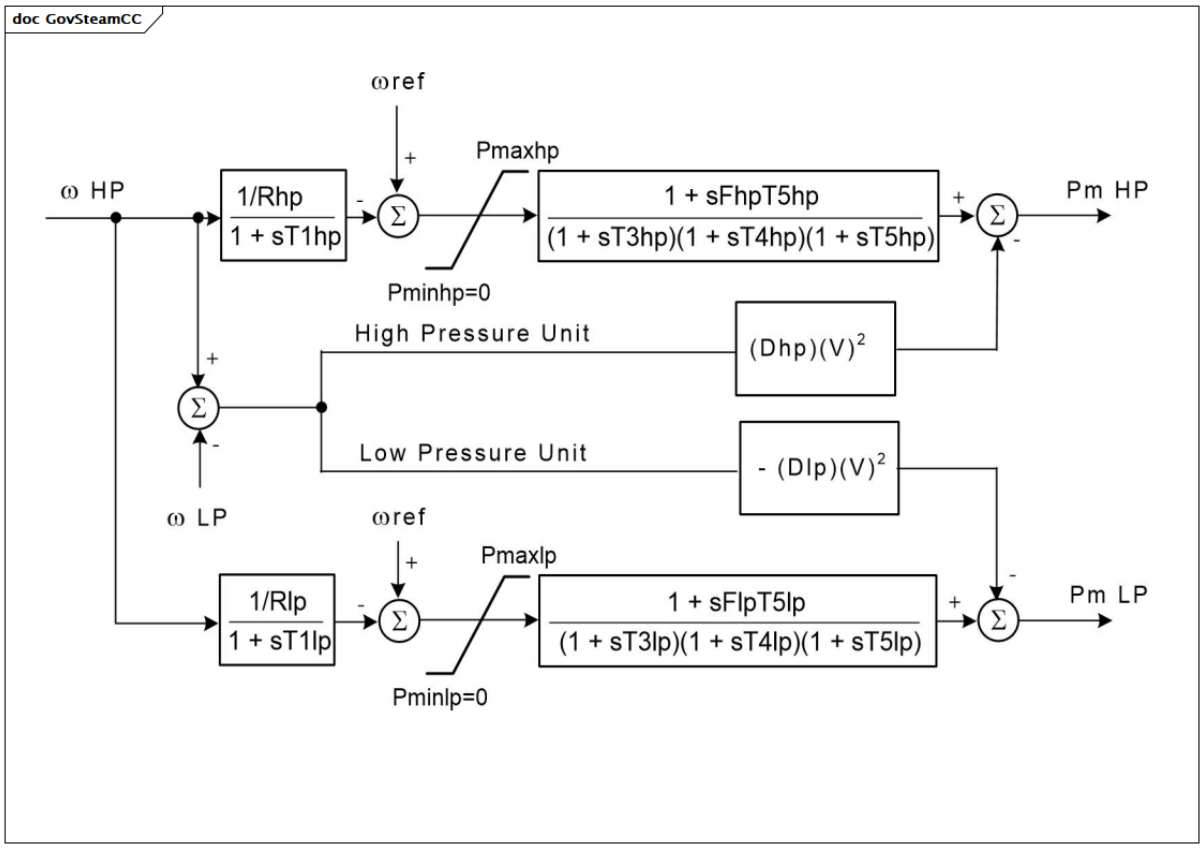

Figure 128 – GovSteamCC

Figure 128: This diagram describes the GovSteamCC turbine governor model.

Parameter details:

1) Both generators shall be present and in service.
2) Each generator shall be represented in the load flow by data stated on its own MVA base. The division of power between the two turbines is determined by the values of MWbase of the two generators, not by the values of Fhp and Flp.
3) The rotor speed reference values (omegaref) are the respective reference values of the units, each of which has its own separate shaft with each its own generator and exciter.
4) V is the terminal voltage.
5) Fhp and Flp can be used to represent the presence of dual reheat stages for a single turbine. If there is only one reheat stage, between the HP and LP turbines, Fhp and Flp should normally be unity and zero, respectively, and T5lp should be set to the reheater time constant.
6) Dhp and Dlp give a rough approximation of the damping of intermachine speed oscillations by electromagnetic effects. When the generators are represented by the round rotor model with proper values for subtransient time constants, this damping is represented accurately in the generator electrical torques and Dhp and Dlp should be set to zero.
7) If any of the time constants are zero, they are bypassed.

Table 734 shows all attributes of GovSteamCC.

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Table 734 – Attributes of TurbineGovernorDynamics::GovSteamCC

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
pmaxhp	1..1	PU	Maximum HP value position (Pmaxhp). Typical value = 1.
rhp	1..1	PU	HP governor droop (Rhp) (> 0). Typical value = 0,05.
t1hp	1..1	Seconds	HP governor time constant (T1hp) (>= 0). Typical value = 0,1.
t3hp	1..1	Seconds	HP turbine time constant (T3hp) (>= 0). Typical value = 0,1.
t4hp	1..1	Seconds	HP turbine time constant (T4hp) (>= 0). Typical value = 0,1.
t5hp	1..1	Seconds	HP reheater time constant (T5hp) (>= 0). Typical value = 10.
fhp	1..1	PU	Fraction of HP power ahead of reheater (Fhp). Typical value = 0,3.
dhp	1..1	PU	HP damping factor (Dhp). Typical value = 0.
pmaxlp	1..1	PU	Maximum LP value position (Pmaxlp). Typical value = 1.
rlp	1..1	PU	LP governor droop (Rlp) (> 0). Typical value = 0,05.
t1lp	1..1	Seconds	LP governor time constant (T1lp) (>= 0). Typical value = 0,1.
t3lp	1..1	Seconds	LP turbine time constant (T3lp) (>= 0). Typical value = 0,1.
t4lp	1..1	Seconds	LP turbine time constant (T4lp) (>= 0). Typical value = 0,1.
t5lp	1..1	Seconds	LP reheater time constant (T5lp) (>= 0). Typical value = 10.
flp	1..1	PU	Fraction of LP power ahead of reheater (Flp). Typical value = 0,7.
dlp	1..1	PU	LP damping factor (Dlp). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 735 shows all association ends of GovSteamCC with other classes.

Table 735 – Association ends of TurbineGovernorDynamics:: GovSteamCC with other classes

mult from	name	mult to	type	description
0..1	HighPressureSynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: CrossCompoundTurbineGovernorDynamics
0..1	LowPressureSynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: CrossCompoundTurbineGovernorDynamics

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14.3.3.6.32 GovSteamEU

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified boiler and steam turbine with PID governor.

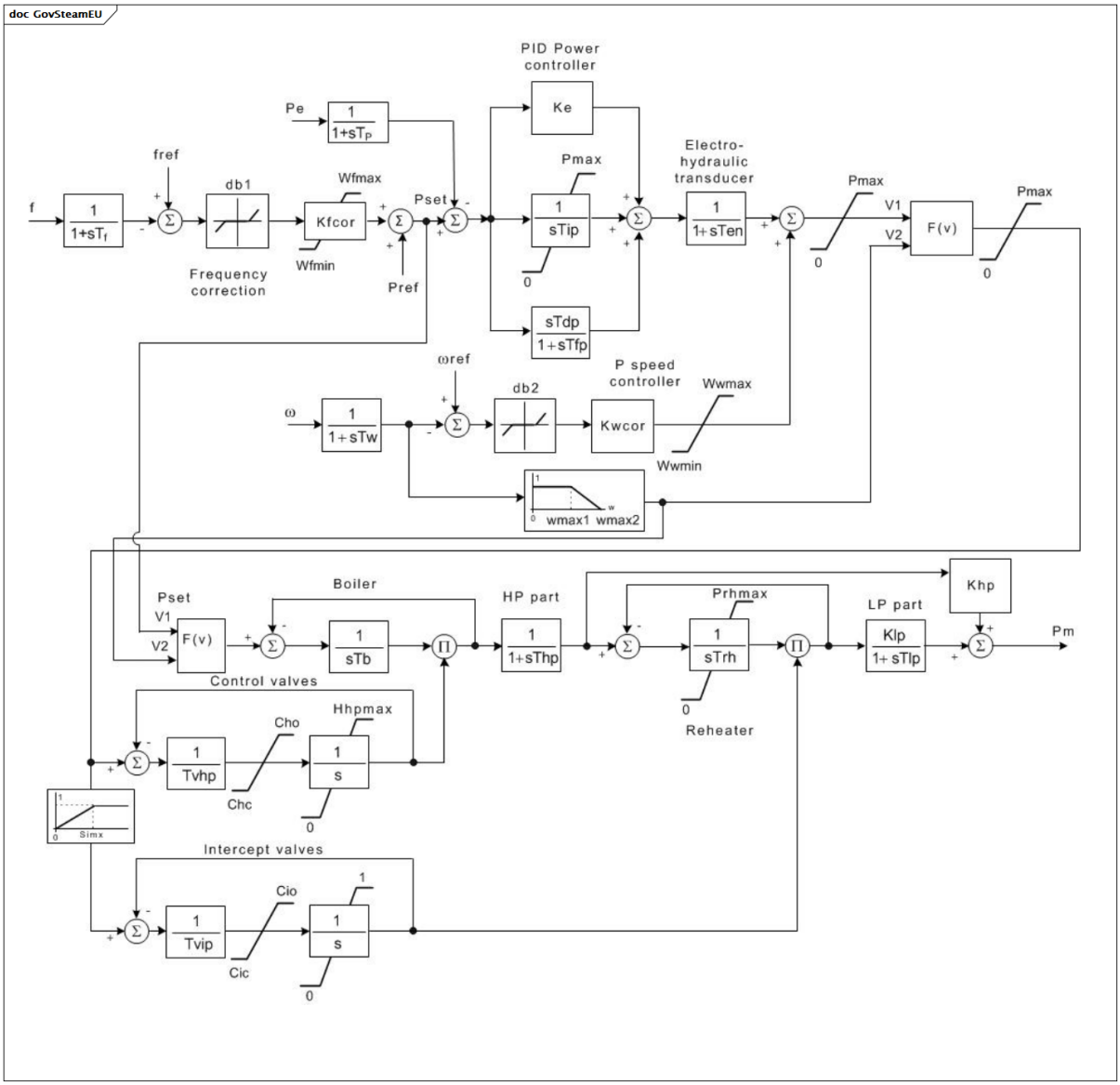

Figure 129 – GovSteamEU

Figure 129: This diagram describes the GovSteamEU turbine governor model.

Parameter details:

1) PU parameters are on base of turbine MW capability MWbase.
2) To represent a non-reheat turbine, set Trh = 0 and Tlp to about 0,5 s.
3) F(v) is minimum value selection.
4) Frequency (f) is local (nodal) frequency.

Table 736 shows all attributes of GovSteamEU.

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Table 736 – Attributes of TurbineGovernorDynamics::GovSteamEU

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
tp	1..1	Seconds	Power transducer time constant (Tp) (>= 0). Typical value = 0,07.
ke	1..1	PU	Gain of the power controller (Ke). Typical value = 0,65.
tip	1..1	Seconds	Integral time constant of the power controller (Tip) (>= 0). Typical value = 2.
tdp	1..1	Seconds	Derivative time constant of the power controller (Tdp) (>= 0). Typical value = 0.
tfp	1..1	Seconds	Time constant of the power controller (Tfp) (>= 0). Typical value = 0.
tf	1..1	Seconds	Frequency transducer time constant (Tf) (>= 0). Typical value = 0.
kfcor	1..1	PU	Gain of the frequency corrector (Kfcor). Typical value = 20.
db1	1..1	PU	Deadband of the frequency corrector (db1). Typical value = 0.
wfmax	1..1	PU	Upper limit for frequency correction (Wfmax) (> GovSteamEU.wfmin). Typical value = 0,05.
wfmin	1..1	PU	Lower limit for frequency correction (Wfmin) (< GovSteamEU.wfmax). Typical value = -0,05.
pmax	1..1	PU	Maximal active power of the turbine (Pmax). Typical value = 1.
ten	1..1	Seconds	Electro hydraulic transducer (Ten) (>= 0). Typical value = 0,1.
tw	1..1	Seconds	Speed transducer time constant (Tw) (>= 0). Typical value = 0,02.
komegacor	1..1	PU	Gain of the speed governor (Kwcor). Typical value = 20.
db2	1..1	PU	Deadband of the speed governor (db2). Typical value = 0,0004.
wwmax	1..1	PU	Upper limit for the speed governor (Wwmax) (> GovSteamEU.wwmin). Typical value = 0,1.
wwmin	1..1	PU	Lower limit for the speed governor frequency correction (Wwmin) (< GovSteamEU.wwmax). Typical value = -1.
wmax1	1..1	PU	Emergency speed control lower limit (wmax1). Typical value = 1,025.
wmax2	1..1	PU	Emergency speed control upper limit (wmax2). Typical value = 1,05.
tvhp	1..1	Seconds	Control valves servo time constant (Tvhp) (>= 0). Typical value = 0,1.
cho	1..1	Float	Control valves rate opening limit (Cho). Unit = PU / s. Typical value = 0,17.
chc	1..1	Float	Control valves rate closing limit (Chc). Unit = PU / s. Typical value = -3,3.
hhpmax	1..1	PU	Maximum control valve position (Hhpmax). Typical value = 1.
tvip	1..1	Seconds	Intercept valves servo time constant (Tvip) (>= 0). Typical value = 0,15.
cio	1..1	PU	Intercept valves rate opening limit (Cio). Typical value = 0,123.

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name	mult	type	description
cic	1..1	PU	Intercept valves rate closing limit (Cic). Typical value = -2,2.
simx	1..1	PU	Intercept valves transfer limit (Simx). Typical value = 0,425.
thp	1..1	Seconds	High pressure (HP) time constant of the turbine (Thp) (>= 0). Typical value = 0,31.
trh	1..1	Seconds	Reheater time constant of the turbine (Trh) (>= 0). Typical value = 8.
tlp	1..1	Seconds	Low pressure (LP) time constant of the turbine (Tlp) (>= 0). Typical value = 0,45.
prhmax	1..1	PU	Maximum low pressure limit (Prhmax). Typical value = 1,4.
khp	1..1	PU	Fraction of total turbine output generated by HP part (Khp). Typical value = 0,277.
klp	1..1	PU	Fraction of total turbine output generated by HP part (Klp). Typical value = 0,723.
tb	1..1	Seconds	Boiler time constant (Tb) (>= 0). Typical value = 100.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 737 shows all association ends of GovSteamEU with other classes.

Table 737 – Association ends of TurbineGovernorDynamics:: GovSteamEU with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.33 GovSteamFV2

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Steam turbine governor with reheat time constants and modelling of the effects of fast valve closing to reduce mechanical power.

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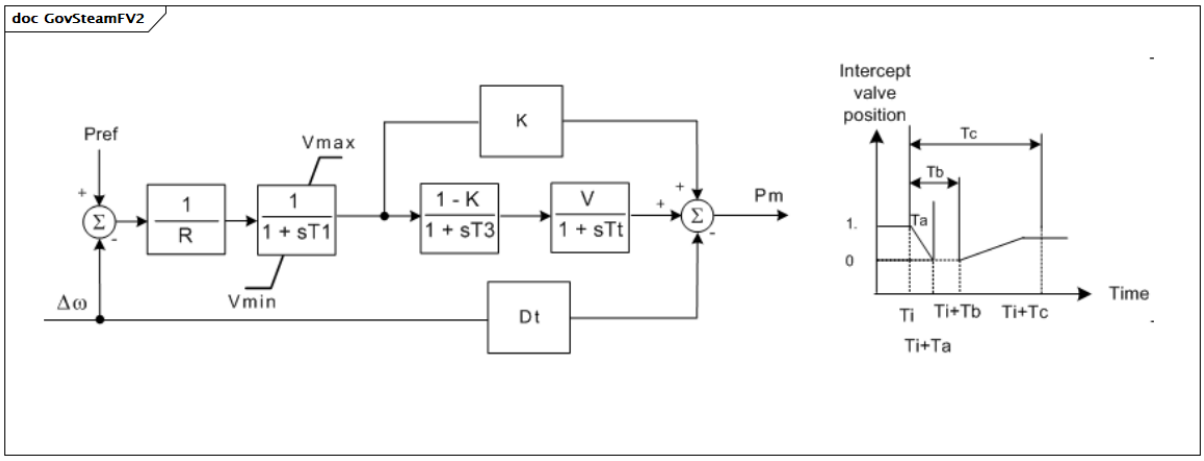

Figure 130 – GovSteamFV2

Figure 130: This diagram describes the GovSteamFV2 turbine governor model.

Parameter details:

1) Valve position is determined by the time curve. The valve position is either 1 or 0.
2) Ti (Initial time to begin fast valving) is a user-defined parameter at the time of the simulation and is not exchanged. Ti is assumed to be 0,1 s. It is used to review the effect of the fast valving.

Table 738 shows all attributes of GovSteamFV2.

Table 738 – Attributes of TurbineGovernorDynamics::GovSteamFV2

name	mult	type	description
mwbase	1..1	ActivePower	Alternate base used instead of machine base in equipment model if necessary (MWbase) (> 0). Unit = MW.
t1	1..1	Seconds	Governor time constant (T1) (>= 0).
vmax	1..1	PU	(Vmax) (> GovSteamFV2.vmin).
vmin	1..1	PU	(Vmin) (< GovSteamFV2.vmax).
k	1..1	PU	Fraction of the turbine power developed by turbine sections not involved in fast valving (K).
t3	1..1	Seconds	Reheater time constant (T3) (>= 0).
dt	1..1	PU	(Dt).
tt	1..1	Seconds	Time constant with which power falls off after intercept valve closure (Tt) (>= 0).
r	1..1	PU	(R).
ta	1..1	Seconds	Time after initial time for valve to close (Ta) (>= 0).
tb	1..1	Seconds	Time after initial time for valve to begin opening (Tb) (>= 0).
tc	1..1	Seconds	Time after initial time for valve to become fully open (Tc) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 739 shows all association ends of GovSteamFV2 with other classes.

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Table 739 – Association ends of TurbineGovernorDynamics:: GovSteamFV2 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.34 GovSteamFV3

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified GovSteamIEEE1 steam turbine governor with Prmax limit and fast valving.

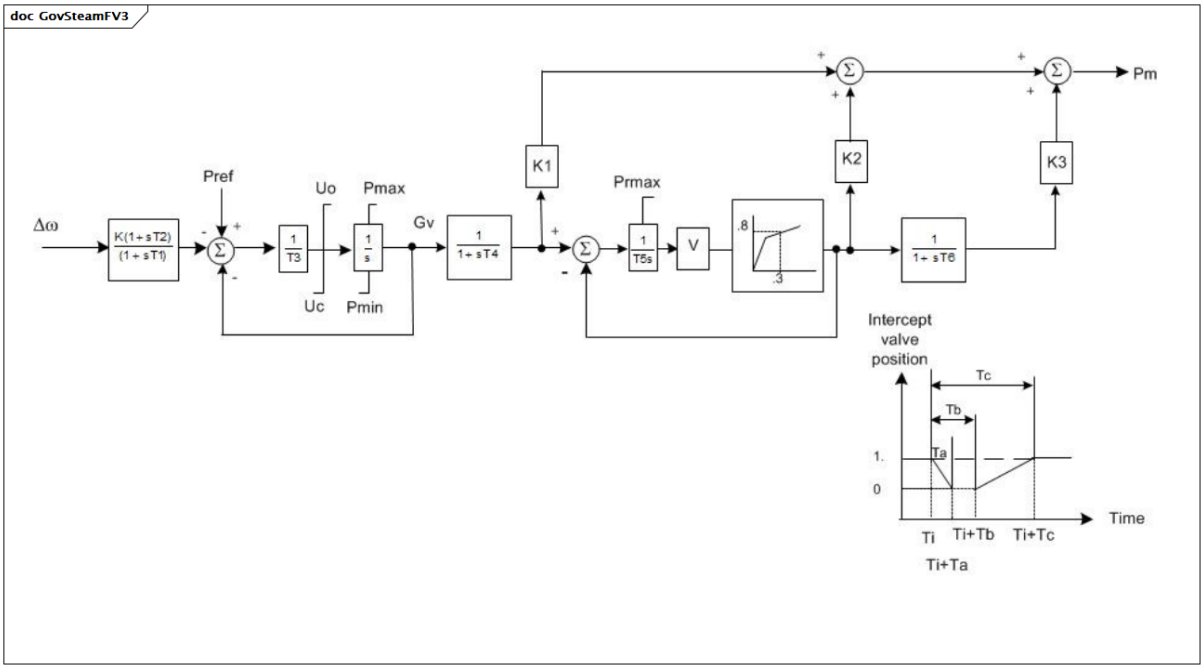

Figure 131 – GovSteamFV3

Figure 131: This diagram describes the GovSteamFV3 turbine governor model.

Parameter details:

1) The gains K1, K2, K3 describe the division of power output among turbine stages. Normally, K1 + K2 + K3 = 1.
2) The times Ta, Tb, and Tc are all with respect to the initiation time.
3) The value Prmax is a limit on the pressure in the reheater. Normally, this value should be about 1,10 PU.
4) The nonlinear gain between gate position and power can be input with up to 6 points. The (0,0, 0,0) and (1,0, 1,0) points can be assumed. The output is not allowed to go beyond 0,0 and 1,0. However, if Pmax > 1,0, the input and output are scaled by Pmax. If Gv1 is a negative number or zero values, the default curve (between 0,0, 0,0 and 1,0, 1,0) will be used.

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5) Ti (initial time to begin fast valving) is a user-defined parameter at the time of the simulation and it is not exchanged. Ti is assumed to be 0,1 s. It is used to review the effect of the fast valving.

Table 740 shows all attributes of GovSteamFV3.

Table 740 – Attributes of TurbineGovernorDynamics::GovSteamFV3

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
k	1..1	PU	Governor gain, (reciprocal of droop) (K). Typical value = 20.
t1	1..1	Seconds	Governor lead time constant (T1) (>= 0). Typical value = 0.
t2	1..1	Seconds	Governor lag time constant (T2) (>= 0). Typical value = 0.
t3	1..1	Seconds	Valve positioner time constant (T3) (> 0). Typical value = 0.
uo	1..1	Float	Maximum valve opening velocity (Uo). Unit = PU / s. Typical value = 0,1.
uc	1..1	Float	Maximum valve closing velocity (Uc). Unit = PU / s. Typical value = -1.
pmax	1..1	PU	Maximum valve opening, PU of MWbase (Pmax) (> GovSteamFV3.pmin). Typical value = 1.
pmin	1..1	PU	Minimum valve opening, PU of MWbase (Pmin) (< GovSteamFV3.pmax). Typical value = 0.
t4	1..1	Seconds	Inlet piping/steam bowl time constant (T4) (>= 0). Typical value = 0,2.
k1	1..1	PU	Fraction of turbine power developed after first boiler pass (K1). Typical value = 0,2.
t5	1..1	Seconds	Time constant of second boiler pass (i.e. reheater) (T5) (> 0 if fast valving is used, otherwise >= 0). Typical value = 0,5.
k2	1..1	PU	Fraction of turbine power developed after second boiler pass (K2). Typical value = 0,2.
t6	1..1	Seconds	Time constant of crossover or third boiler pass (T6) (>= 0). Typical value = 10.
k3	1..1	PU	Fraction of hp turbine power developed after crossover or third boiler pass (K3). Typical value = 0,6.
ta	1..1	Seconds	Time to close intercept valve (IV) (Ta) (>= 0). Typical value = 0,97.
tb	1..1	Seconds	Time until IV starts to reopen (Tb) (>= 0). Typical value = 0,98.
tc	1..1	Seconds	Time until IV is fully open (Tc) (>= 0). Typical value = 0,99.
prmax	1..1	PU	Max. pressure in reheater (Prmax). Typical value = 1.
gv1	1..1	PU	Nonlinear gain valve position point 1 (GV1). Typical value = 0.
pgv1	1..1	PU	Nonlinear gain power value point 1 (Pgv1). Typical value = 0.
gv2	1..1	PU	Nonlinear gain valve position point 2 (GV2). Typical value = 0,4.
pgv2	1..1	PU	Nonlinear gain power value point 2 (Pgv2). Typical value = 0,75.

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name	mult	type	description
gv3	1..1	PU	Nonlinear gain valve position point 3 (GV3). Typical value = 0,5.
pgv3	1..1	PU	Nonlinear gain power value point 3 (Pgv3). Typical value = 0,91.
gv4	1..1	PU	Nonlinear gain valve position point 4 (GV4). Typical value = 0,6.
pgv4	1..1	PU	Nonlinear gain power value point 4 (Pgv4). Typical value = 0,98.
gv5	1..1	PU	Nonlinear gain valve position point 5 (GV5). Typical value = 1.
pgv5	1..1	PU	Nonlinear gain power value point 5 (Pgv5). Typical value = 1.
gv6	1..1	PU	Nonlinear gain valve position point 6 (GV6). Typical value = 0.
pgv6	1..1	PU	Nonlinear gain power value point 6 (Pgv6). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 741 shows all association ends of GovSteamFV3 with other classes.

Table 741 – Association ends of TurbineGovernorDynamics:: GovSteamFV3 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.35 GovSteamFV4

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Detailed electro-hydraulic governor for steam unit.

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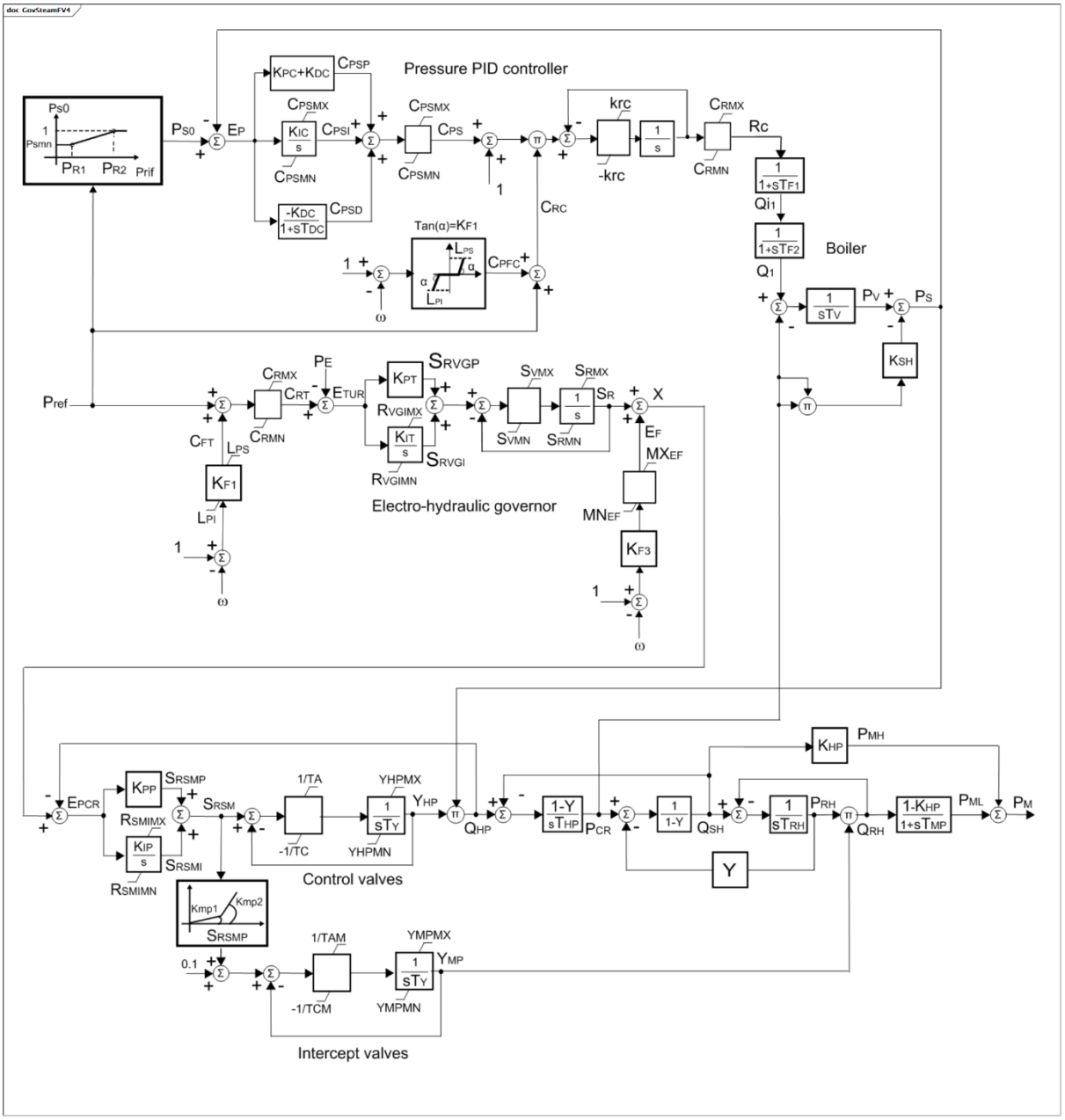

Figure 132 – GovSteamFV4

Figure 132: This diagram describes the GovSteamFV4 turbine governor model.

Parameter details:

1) PU parameters are on base of MW capability of the turbine.
2) This model can be used for steam units in coordinated control, not belonging to a combined-cycle power plant.
3) omega has units of per unit speed.

Table 742 shows all attributes of GovSteamFV4.

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Table 742 – Attributes of TurbineGovernorDynamics::GovSteamFV4

name	mult	type	description
kf1	1..1	PU	Frequency bias (reciprocal of droop) (Kf1). Typical value = 20.
kf3	1..1	PU	Frequency control (reciprocal of droop) (Kf3). Typical value = 20.
lps	1..1	PU	Maximum positive power error (Lps). Typical value = 0,03.
lpi	1..1	PU	Maximum negative power error (Lpi). Typical value = -0,15.
mxef	1..1	PU	Upper limit for frequency correction (MXEF). Typical value = 0,05.
mnef	1..1	PU	Lower limit for frequency correction (MNEF). Typical value = -0,05.
crmx	1..1	PU	Maximum value of regulator set-point (Crmx). Typical value = 1,2.
crmn	1..1	PU	Minimum value of regulator set-point (Crmn). Typical value = 0.
kpt	1..1	PU	Proportional gain of electro-hydraulic regulator (Kpt). Typical value = 0,3.
kit	1..1	PU	Integral gain of electro-hydraulic regulator (Kit). Typical value = 0,04.
rvgmx	1..1	PU	Maximum value of integral regulator (Rvgmx). Typical value = 1,2.
rvgmn	1..1	PU	Minimum value of integral regulator (Rvgmn). Typical value = 0.
svmx	1..1	Float	Maximum regulator gate opening velocity (Svmx). Typical value = 0,0333.
svmn	1..1	Float	Maximum regulator gate closing velocity (Svmn). Typical value = -0,0333.
srmx	1..1	PU	Maximum valve opening (Srmx). Typical value = 1,1.
srmn	1..1	PU	Minimum valve opening (Srmn). Typical value = 0.
kpp	1..1	PU	Proportional gain of pressure feedback regulator (Kpp). Typical value = 1.
kip	1..1	PU	Integral gain of pressure feedback regulator (Kip). Typical value = 0,5.
rsmimx	1..1	PU	Maximum value of integral regulator (Rsmimx). Typical value = 1,1.
rsmimn	1..1	PU	Minimum value of integral regulator (Rsmimn). Typical value = 0.
kmp1	1..1	PU	First gain coefficient of intercept valves characteristic (Kmp1). Typical value = 0,5.
kmp2	1..1	PU	Second gain coefficient of intercept valves characteristic (Kmp2). Typical value = 3,5.
srsmp	1..1	PU	Intercept valves characteristic discontinuity point (Srsmp). Typical value = 0,43.
ta	1..1	Seconds	Control valves rate opening time (Ta) (>= 0). Typical value = 0,8.
tc	1..1	Seconds	Control valves rate closing time (Tc) (>= 0). Typical value = 0,5.
ty	1..1	Seconds	Control valves servo time constant (Ty) (>= 0). Typical value = 0,1.
yhpmx	1..1	PU	Maximum control valve position (Yhpmx). Typical value = 1,1.

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name	mult	type	description
yhpmn	1..1	PU	Minimum control valve position (Yhpmn). Typical value = 0.
tam	1..1	Seconds	Intercept valves rate opening time (Tam) (>= 0). Typical value = 0,8.
tcm	1..1	Seconds	Intercept valves rate closing time (Tcm) (>= 0). Typical value = 0,5.
ympmx	1..1	PU	Maximum intercept valve position (Ympmx). Typical value = 1,1.
ympmn	1..1	PU	Minimum intercept valve position (Ympmn). Typical value = 0.
y	1..1	PU	Coefficient of linearized equations of turbine (Stodola formulation) (Y). Typical value = 0,13.
thp	1..1	Seconds	High pressure (HP) time constant of the turbine (Thp) (>= 0). Typical value = 0,15.
trh	1..1	Seconds	Reheater time constant of the turbine (Trh) (>= 0). Typical value = 10.
tmp	1..1	Seconds	Low pressure (LP) time constant of the turbine (Tmp) (>= 0). Typical value = 0,4.
khp	1..1	PU	Fraction of total turbine output generated by HP part (Khp). Typical value = 0,35.
pr1	1..1	PU	First value of pressure set point static characteristic (Pr1). Typical value = 0,2.
pr2	1..1	PU	Second value of pressure set point static characteristic, corresponding to Ps0 = 1,0 PU (Pr2). Typical value = 0,75.
psmn	1..1	PU	Minimum value of pressure set point static characteristic (Psmn). Typical value = 1.
kpc	1..1	PU	Proportional gain of pressure regulator (Kpc). Typical value = 0,5.
kic	1..1	PU	Integral gain of pressure regulator (Kic). Typical value = 0,0033.
kdc	1..1	PU	Derivative gain of pressure regulator (Kdc). Typical value = 1.
tdc	1..1	Seconds	Derivative time constant of pressure regulator (Tdc) (>= 0). Typical value = 90.
cpsmx	1..1	PU	Maximum value of pressure regulator output (Cpsmx). Typical value = 1.
cpsmn	1..1	PU	Minimum value of pressure regulator output (Cpsmn). Typical value = -1.
krc	1..1	PU	Maximum variation of fuel flow (Krc). Typical value = 0,05.
tf1	1..1	Seconds	Time constant of fuel regulation (Tf1) (>= 0). Typical value = 10.
tf2	1..1	Seconds	Time constant of steam chest (Tf2) (>= 0). Typical value = 10.
tv	1..1	Seconds	Boiler time constant (Tv) (>= 0). Typical value = 60.
ksh	1..1	PU	Pressure loss due to flow friction in the boiler tubes (Ksh). Typical value = 0,08.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 743 shows all association ends of GovSteamFV4 with other classes.

Table 743 – Association ends of TurbineGovernorDynamics:: GovSteamFV4 with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.6.36 GovSteamSGO

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified steam turbine governor.

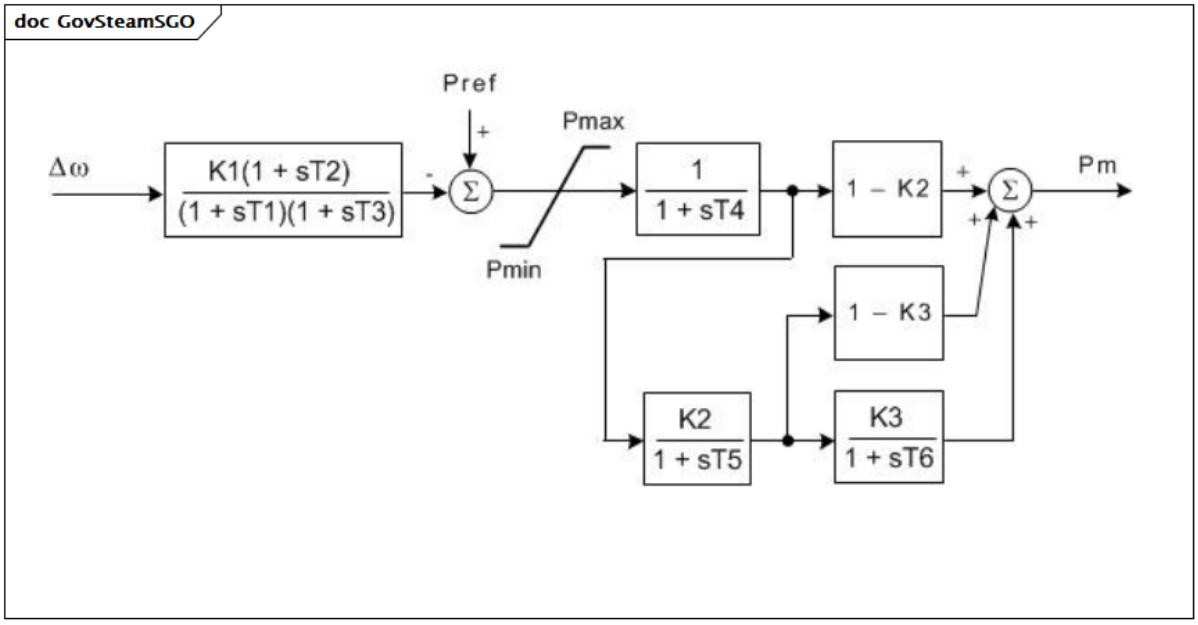

Figure 133 – GovSteamSGO

Figure 133: This diagram describes the GovSteamSGO turbine governor model.

Table 744 shows all attributes of GovSteamSGO.

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Table 744 – Attributes of TurbineGovernorDynamics::GovSteamSGO

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
t1	1..1	Seconds	Controller lag (T1) (>= 0).
t2	1..1	Seconds	Controller lead compensation (T2) (>= 0).
t3	1..1	Seconds	Governor lag (T3) (> 0).
t4	1..1	Seconds	Delay due to steam inlet volumes associated with steam chest and inlet piping (T4) (>= 0).
t5	1..1	Seconds	Reheater delay including hot and cold leads (T5) (>= 0).
t6	1..1	Seconds	Delay due to IP-LP turbine, crossover pipes and LP end hoods (T6) (>= 0).
k1	1..1	PU	One / PU regulation (K1).
k2	1..1	PU	Fraction (K2).
k3	1..1	PU	Fraction (K3).
pmax	1..1	PU	Upper power limit (Pmax) (> GovSteamSGO.pmin).
pmin	1..1	Seconds	Lower power limit (Pmin) (>= 0 and < GovSteamSGO.pmax).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 745 shows all association ends of GovSteamSGO with other classes.

Table 745 – Association ends of TurbineGovernorDynamics:: GovSteamSGO with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.3.7 Package TurbineLoadControllerDynamics

14.3.3.7.1 General

A turbine load controller acts to maintain turbine power at a set value by continuous adjustment of the turbine governor speed-load reference.

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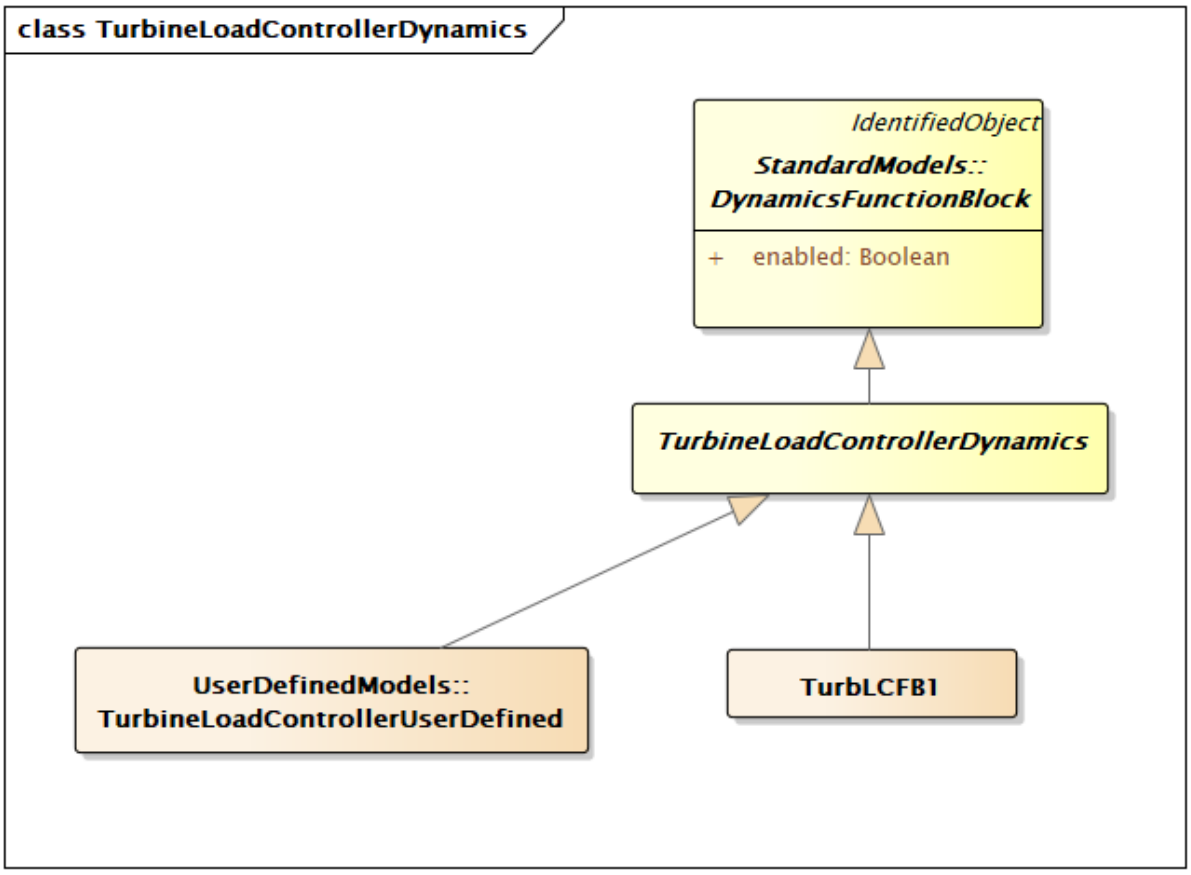

Figure 134 – Class diagram TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics

Figure 134: This diagram shows dynamic standard model turbine load controller classes and their inheritance from the TurbineLoadControllerDynamics class.

14.3.3.7.2 (abstract) TurbineLoadControllerDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Turbine load controller function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 746 shows all attributes of TurbineLoadControllerDynamics.

Table 746 – Attributes of TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 747 shows all association ends of TurbineLoadControllerDynamics with other classes.

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Table 747 – Association ends of TurbineLoadControllerDynamics:: TurbineLoadControllerDynamics with other classes

mult from	name	mult to	type	description
0..1	TurbineGovernorDynamics	1..1	TurbineGovernorDynamics	Turbine-governor controlled by this turbine load controller.

14.3.3.7.3 TurbLCFB1

Inheritance path = TurbineLoadControllerDynamics: DynamicsFunctionBlock: IdentifiedObject

Turbine load controller model developed by WECC. This model represents a supervisory turbine load controller that acts to maintain turbine power at a set value by continuous adjustment of the turbine governor speed-load reference. This model is intended to represent slow reset 'outer loop' controllers managing the action of the turbine governor.

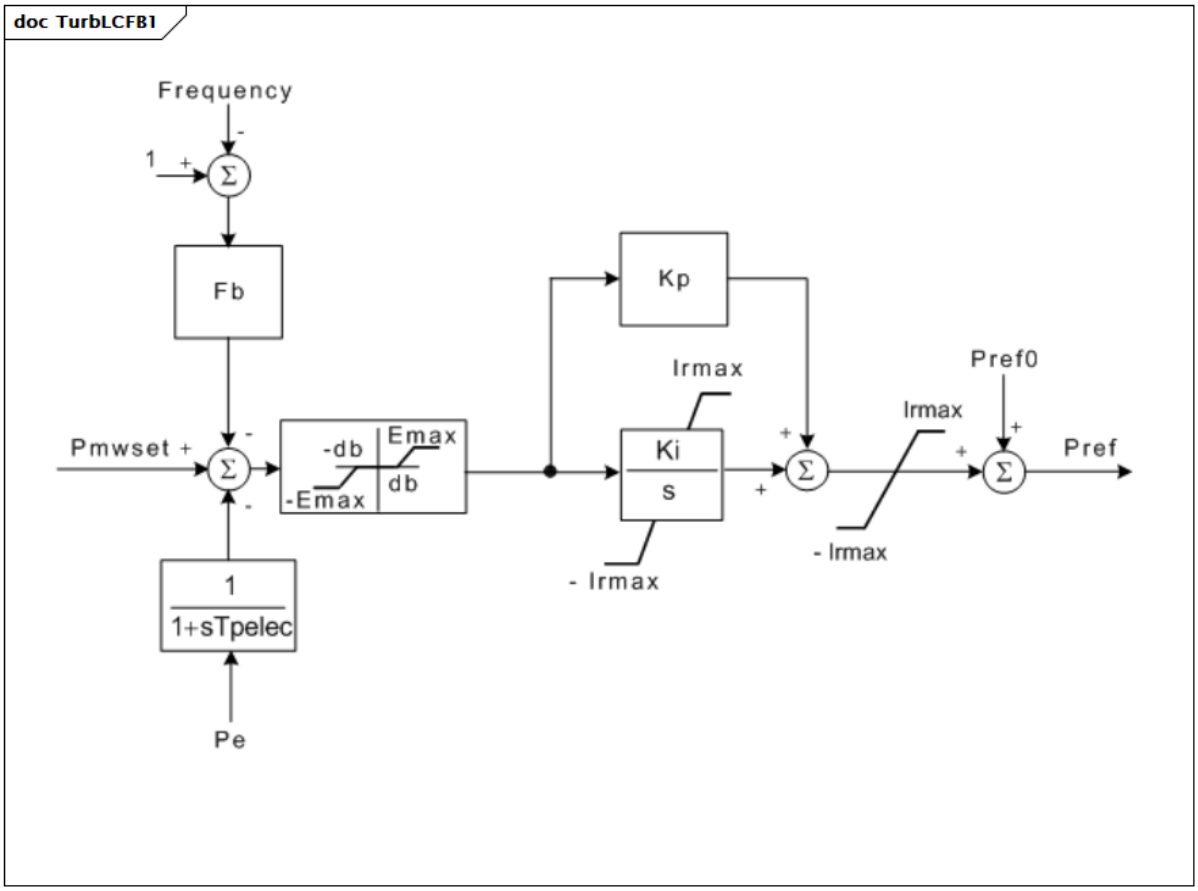

Figure 135 – TurbLCFB1

Figure 135: This diagram describes the TurbLCFB1 turbine load controller model.

Parameter details:

1) The load reference of this supervisory load control loop is accessible as the parameter, Pmwset. Pmwset is given a value automatically when the model is initialized. This value overwrites any value entered prior to initialization.
2) The controller acts by applying a bias to the turbine-governor speed load reference, Pref. This reference is initialized in the normal manner by the turbine-governor model.

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3) The output of the controller is limited to plus or minus Irmax. To prevent the controller from adjusting the turbine power excessively, Irmax should not exceed approximately one half of the governor droop or approximately 0,025 PU.

4) The error limit, Emax, is intended to set a maximum rate of change of turbine power. If limitation of load change rate is desired, Emax should be set such that Emax = (max rate, PU / s) x (droop of governor in PU) / Ki. Note that governor droop used in this calculation shall be in PU relative to the base power, MWbase, of the turbine. Note, also, that some governor models (such as GovSteamIEEE1) have their droop specified by the value of a gain whose value is the reciprocal of the droop. For example, in GovSteamIEEE1 the droop is specified by the gain, K, whose typical value is 1 / 0,04 = 25.

5) This model recognizes the two alternative ways of specifying the turbine-governor load reference. In models such as GovCT1 and GovHydro1, the reference is a speed setpoint. In other models such as GovSteamIEEE1, the reference is a PU load value. The references are generally as follows: 1) speed reference (1 + (initial power or valve position) x (droop)), 2) load reference (0 + (initial power or valve position)). The parameters of TurbLCFB1 shall always be set on the basis of a governor speed reference.

Table 748 shows all attributes of TurbLCFB1.

Table 748 – Attributes of TurbineLoadControllerDynamics::TurbLCFB1

name	mult	type	description
mwbase	1..1	ActivePower	Base for power values (MWbase) (> 0). Unit = MW.
speedReferenceGovernor	1..1	Boolean	Type of turbine governor reference (Type). true = speed reference governor false = load reference governor. Typical value = true.
db	1..1	PU	Controller deadband (db). Typical value = 0.
emax	1..1	PU	Maximum control error (Emax) (see parameter detail 4). Typical value = 0,02.
fb	1..1	PU	Frequency bias gain (Fb). Typical value = 0.
kp	1..1	PU	Proportional gain (Kp). Typical value = 0.
ki	1..1	PU	Integral gain (Ki). Typical value = 0.
fbf	1..1	Boolean	Frequency bias flag (Fbf). true = enable frequency bias false = disable frequency bias. Typical value = false.
pbf	1..1	Boolean	Power controller flag (Pbf). true = enable load controller false = disable load controller. Typical value = false.
tpelec	1..1	Seconds	Power transducer time constant (Tpelec) (>= 0). Typical value = 0.
irmax	1..1	PU	Maximum turbine speed/load reference bias (Irmax) (see parameter detail 3). Typical value = 0.
pmwset	1..1	ActivePower	Power controller setpoint (Pmwset) (see parameter detail 1). Unit = MW. Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	0..1	String	inherited from: IdentifiedObject

Table 749 shows all association ends of TurbLCFB1 with other classes.

Table 749 – Association ends of TurbineLoadControllerDynamics:: TurbLCFB1 with other classes

mult from	name	mult to	type	description
0..1	TurbineGovernorDynamics	1..1	TurbineGovernorDynamics	inherited from: TurbineLoadControllerDynamics

14.3.3.8 Package MechanicalLoadDynamics

14.3.3.8.1 General

A mechanical load represents the variation in a motor's shaft torque or power as a function of shaft speed.

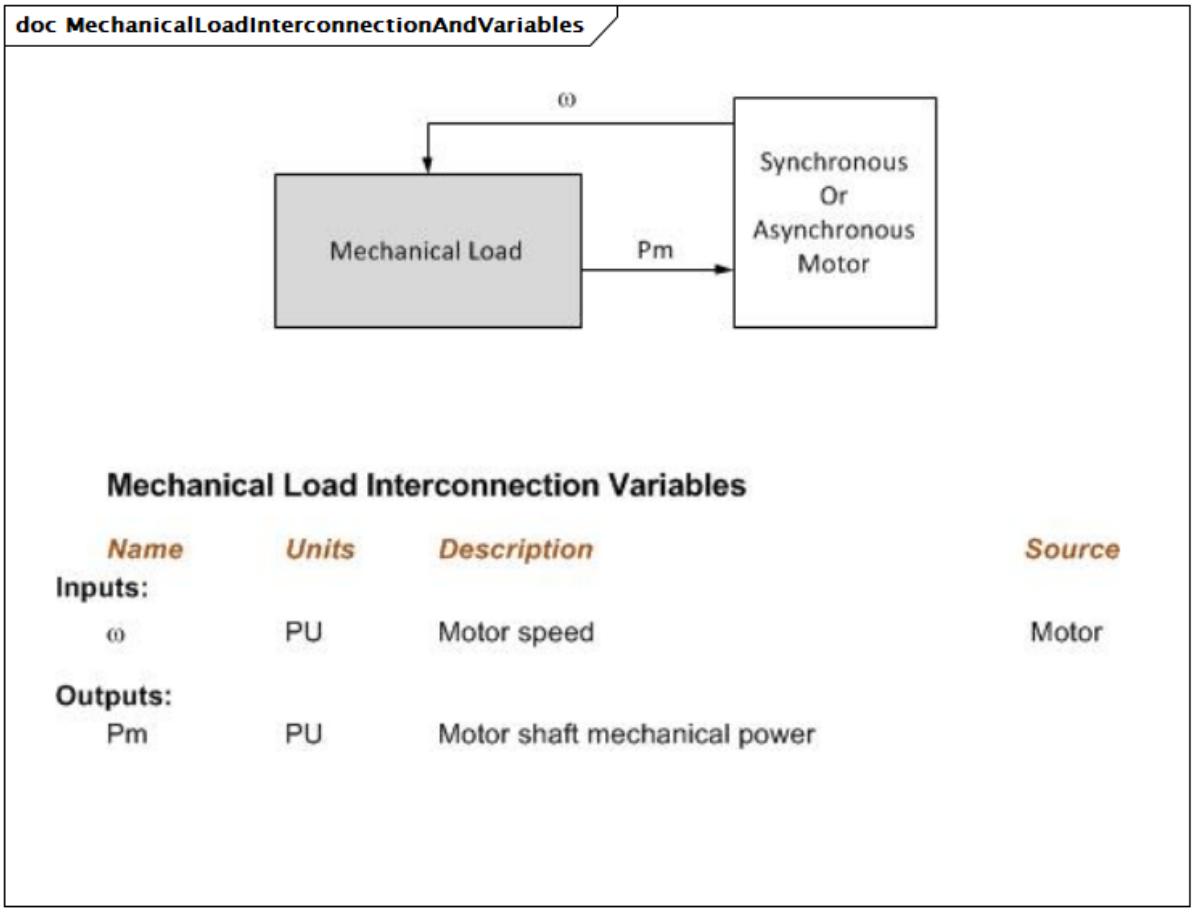

Figure 136 – MechanicalLoadInterconnectionAndVariables

Figure 136: This diagram illustrates the inputs and outputs for a mechanical load.

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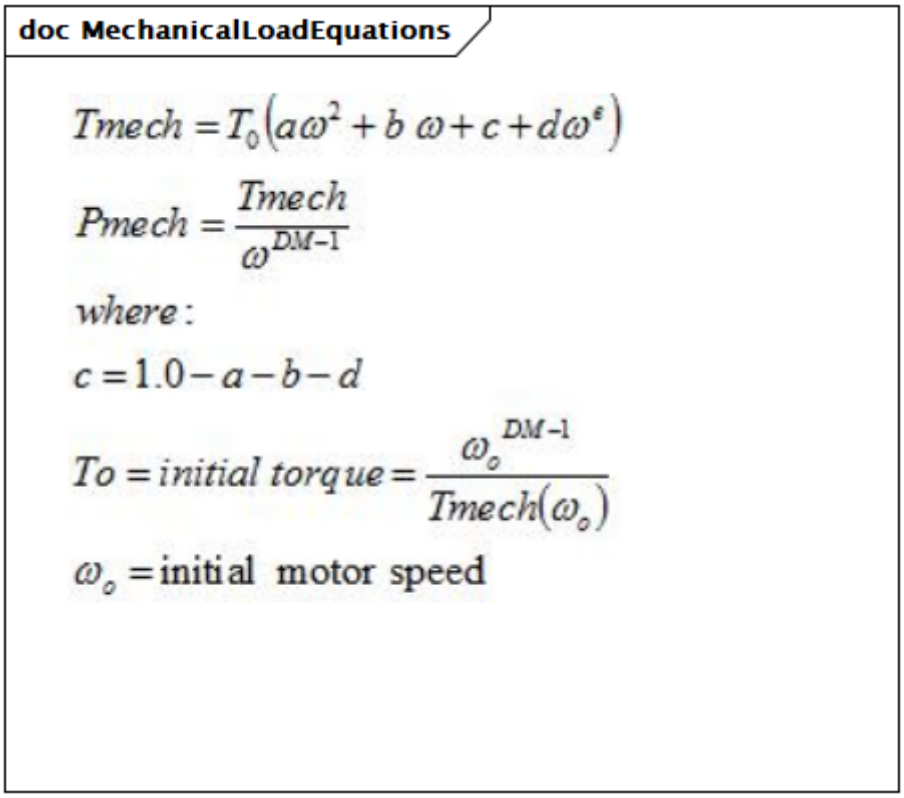

Figure 137 – MechanicalLoadEquations

Figure 137: This diagram illustrates the equations for a mechanical load.

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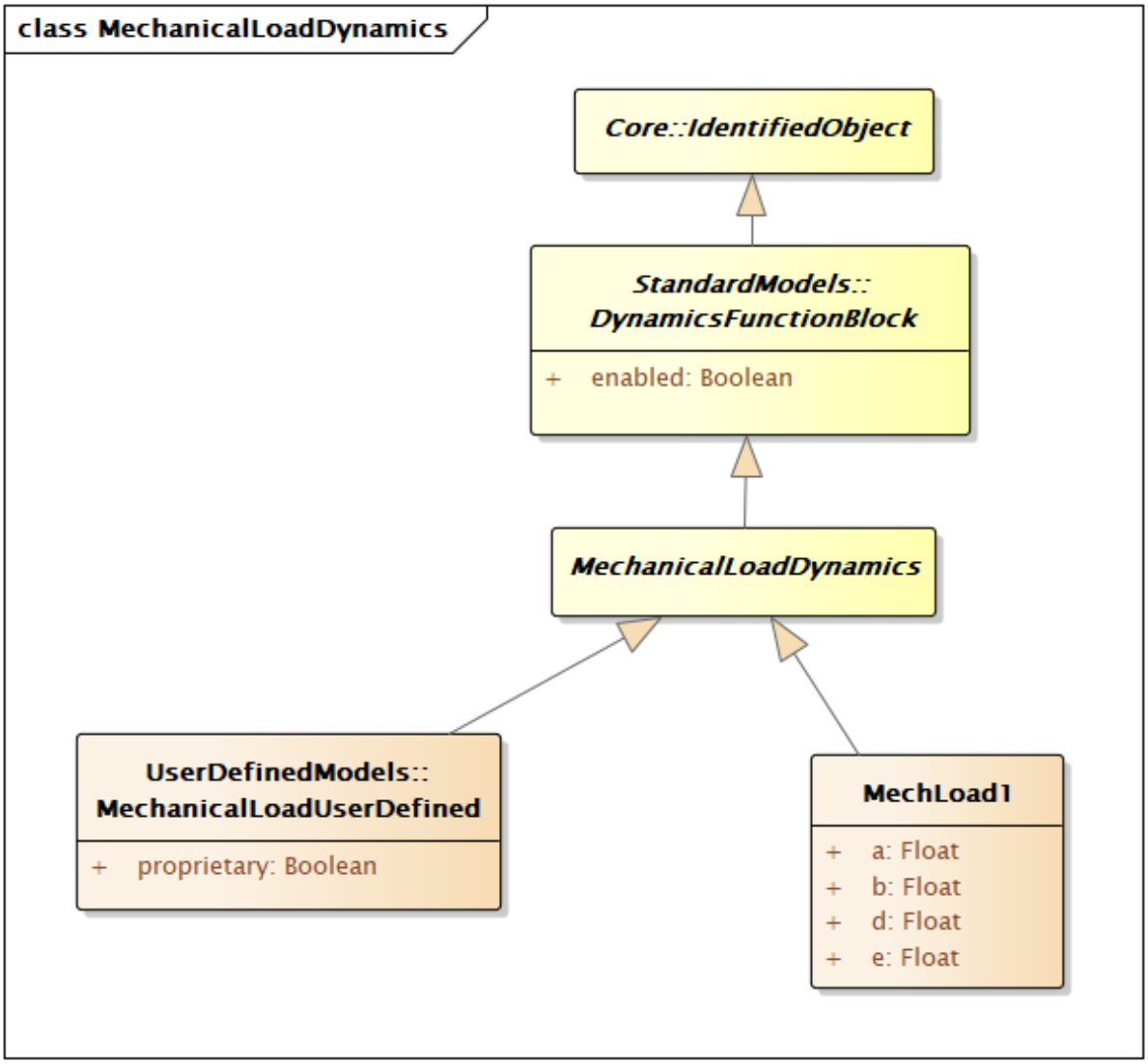

Figure 138 – Class diagram MechanicalLoadDynamics::MechanicalLoadDynamics

Figure 138: This diagram shows dynamic standard model mechanical load classes and their inheritance from the MechanicalLoadDynamics class.

14.3.3.8.2 (abstract) MechanicalLoadDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Mechanical load function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 750 shows all attributes of MechanicalLoadDynamics.

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Table 750 – Attributes of MechanicalLoadDynamics::MechanicalLoadDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 751 shows all association ends of MechanicalLoadDynamics with other classes.

Table 751 – Association ends of MechanicalLoadDynamics:: MechanicalLoadDynamics with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	Synchronous machine model with which this mechanical load model is associated. MechanicalLoadDynamics shall have either an association to SynchronousMachineDynamics or AsynchronousMachineDyanmics.
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	Asynchronous machine model with which this mechanical load model is associated. MechanicalLoadDynamics shall have either an association to SynchronousMachineDynamics or to AsynchronousMachineDynamics.

14.3.3.8.3 MechLoad1

Inheritance path = MechanicalLoadDynamics: DynamicsFunctionBlock: IdentifiedObject

Mechanical load model type 1.

Table 752 shows all attributes of MechLoad1.

Table 752 – Attributes of MechanicalLoadDynamics::MechLoad1

name	mult	type	description
a	1..1	Float	Speed squared coefficient (a).
b	1..1	Float	Speed coefficient (b).
d	1..1	Float	Speed to the exponent coefficient (d).
e	1..1	Float	Exponent (e).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 753 shows all association ends of MechLoad1 with other classes.

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Table 753 – Association ends of MechanicalLoadDynamics:: MechLoad1 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: MechanicalLoadDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: MechanicalLoadDynamics

14.3.3.9 Package ExcitationSystemDynamics

14.3.3.9.1 General

The excitation system model provides the field voltage (Efd) for a synchronous machine model. It is linked to a specific generator (synchronous machine).

The representation of all limits used by the models (not including IEEE standard models) shall comply with the representation defined in the Annex E of IEEE 421.5:2005, unless specified differently in the documentation of the model.

The parameters are different for each excitation system model; the same parameter name can have different meaning in different models.

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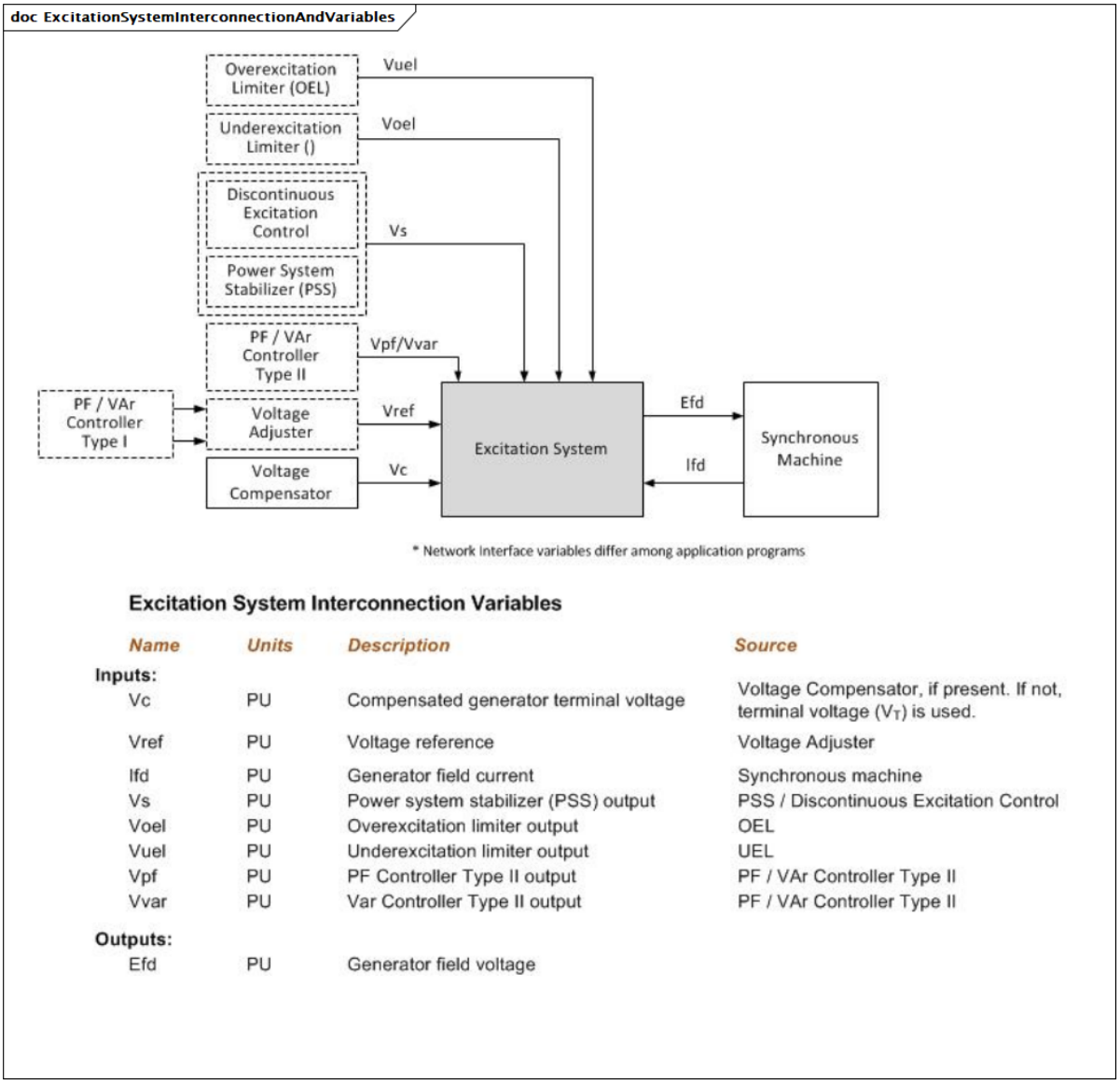

Excitation System Interconnection Variables

Name	*Units*	*Description*	*Source*
Inputs:			
Vc	PU	Compensated generator terminal voltage	Voltage Compensator, if present. If not, terminal voltage (V_T) is used.
Vref	PU	Voltage reference	Voltage Adjuster
Ifd	PU	Generator field current	Synchronous machine
Vs	PU	Power system stabilizer (PSS) output	PSS / Discontinuous Excitation Control
Voel	PU	Overexcitation limiter output	OEL
Vuel	PU	Underexcitation limiter output	UEL
Vpf	PU	PF Controller Type II output	PF / VAr Controller Type II
Vvar	PU	Var Controller Type II output	PF / VAr Controller Type II
Outputs:			
Efd	PU	Generator field voltage	

Figure 139 – ExcitationSystemInterconnectionAndVariables

Figure 139: This diagram illustrates the inputs and outputs for an excitation system model.

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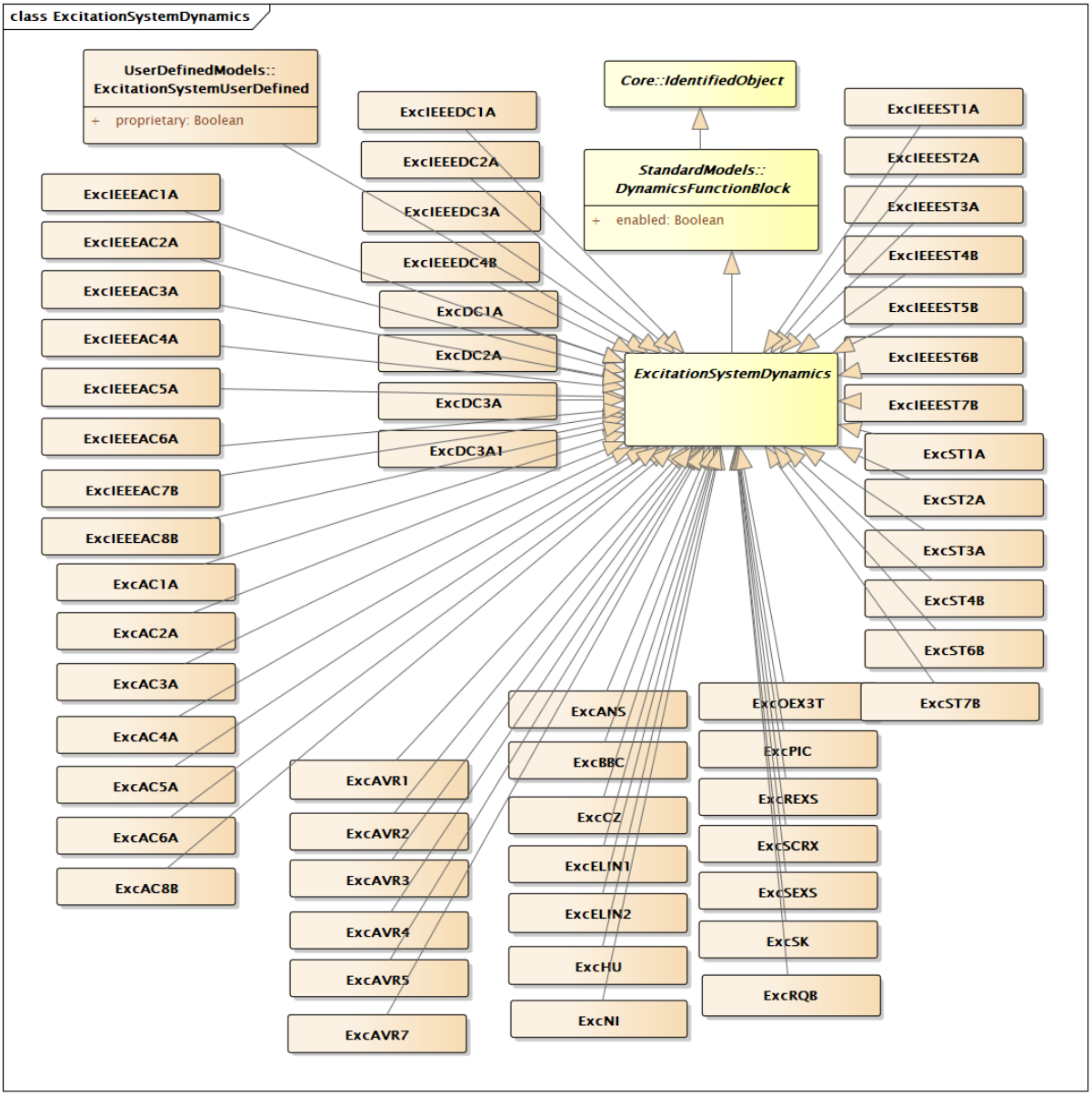

Figure 140 – Class diagram ExcitationSystemDynamics::ExcitationSystemDynamics

Figure 140: This diagram shows dynamic standard model excitation system classes and their inheritance from the ExcitationSystemDynamics class.

14.3.3.9.2 (abstract) ExcitationSystemDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Excitation system function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 754 shows all attributes of ExcitationSystemDynamics.

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Table 754 – Attributes of ExcitationSystemDynamics::ExcitationSystemDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 755 shows all association ends of ExcitationSystemDynamics with other classes.

Table 755 – Association ends of ExcitationSystemDynamics:: ExcitationSystemDynamics with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Synchronous machine model with which this excitation system model is associated.

14.3.3.9.3 ExcIEEEAC1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC1A model. The model represents the field-controlled alternator-rectifier excitation systems designated type AC1A. These excitation systems consist of an alternator main exciter with non-controlled rectifiers.

Reference: IEEE 421.5:2005, 6.1.

Table 756 shows all attributes of ExcIEEEAC1A.

Table 756 – Attributes of ExcitationSystemDynamics::ExcIEEEAC1A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 400.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,02.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 14,5.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -14,5.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,8.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0,03.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 1.

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name	mult	type	description
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,2.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 0,38.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 4,18.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 0,1.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 3,14.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 0,03.
vrmax	1..1	PU	Maximum voltage regulator outputs (VRMAX) (> 0). Typical value = 6,03.
vrmin	1..1	PU	Minimum voltage regulator outputs (VRMIN) (< 0). Typical value = -5,43.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 757 shows all association ends of ExcIEEEAC1A with other classes.

Table 757 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.4 ExcIEEEAC2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC2A model. The model represents a high initial response field-controlled alternator-rectifier excitation system. The alternator main exciter is used with non-controlled rectifiers. The type AC2A model is similar to that of type AC1A except for the inclusion of exciter time constant compensation and exciter field current limiting elements.

Reference: IEEE 421.5:2005, 6.2.

Table 758 shows all attributes of ExcIEEEAC2A.

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Table 758 – Attributes of ExcitationSystemDynamics::ExcIEEEAC2A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 400.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,02.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 8.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -8.
kb	1..1	PU	Second stage regulator gain (KB) (> 0). Typical value = 25.
vrmax	1..1	PU	Maximum voltage regulator outputs (VRMAX) (> 0). Typical value = 105.
vrmin	1..1	PU	Minimum voltage regulator outputs (VRMIN) (< 0). Typical value = -95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,6.
vfemax	1..1	PU	Exciter field current limit reference (VFEMAX) (> 0). Typical value = 4,4.
kh	1..1	PU	Exciter field current feedback gain (KH) (>= 0). Typical value = 1.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0,03.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,28.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 0,35.
ke	1..1	PU	Exciter constant related to self-excited field (KE) (>= 0). Typical value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 4,4.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 0,037.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 3,3.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 0,012.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 759 shows all association ends of ExcIEEEAC2A with other classes.

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Table 759 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.5 ExcIEEEAC3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC3A model. The model represents the field-controlled alternator-rectifier excitation systems designated type AC3A. These excitation systems include an alternator main exciter with non-controlled rectifiers. The exciter employs self-excitation, and the voltage regulator power is derived from the exciter output voltage. Therefore, this system has an additional nonlinearity, simulated by the use of a multiplier whose inputs are the voltage regulator command signal, Va, and the exciter output voltage, Efd, times KR. This model is applicable to excitation systems employing static voltage regulators.

Reference: IEEE 421.5:2005, 6.3.

Table 760 shows all attributes of ExcIEEEAC3A.

Table 760 – Attributes of ExcitationSystemDynamics::ExcIEEEAC3A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 45,62.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,013.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 1.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -0,95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 1,17.
vemin	1..1	PU	Minimum exciter voltage output (VEMIN) (<= 0). Typical value = 0.
kr	1..1	PU	Constant associated with regulator and alternator field power supply (KR) (> 0). Typical value = 3,77.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0,143.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 1.
kn	1..1	PU	Excitation control system stabilizer gain (KN) (>= 0). Typical value = 0,05.
efdn	1..1	PU	Value of Efd at which feedback gain changes (EFDN) (> 0). Typical value = 2,36.

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name	mult	type	description
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,104.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 0,499.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
vfemax	1..1	PU	Exciter field current limit reference (VFEMAX) (>= 0). Typical value = 16.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 6,24.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 1,143.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 4,68.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 761 shows all association ends of ExcIEEEAC3A with other classes.

Table 761 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.6 ExcIEEEAC4A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC4A model. The model represents type AC4A alternator-supplied controlled-rectifier excitation system which is quite different from the other types of AC systems. This high initial response excitation system utilizes a full thyristor bridge in the exciter output circuit. The voltage regulator controls the firing of the thyristor bridges. The exciter alternator uses an independent voltage regulator to control its output voltage to a constant value. These effects are not modelled; however, transient loading effects on the exciter alternator are included.

Reference: IEEE 421.5:2005, 6.4.

Table 762 shows all attributes of ExcIEEEAC4A.

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Table 762 – Attributes of ExcitationSystemDynamics::ExcIEEEAC4A

name	mult	type	description
vimax	1..1	PU	Maximum voltage regulator input limit (VIMAX) (> 0). Typical value = 10.
vimin	1..1	PU	Minimum voltage regulator input limit (VIMIN) (< 0). Typical value = -10.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 1.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 10.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 200.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,015.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 5,64.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -4,53.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 763 shows all association ends of ExcIEEEAC4A with other classes.

Table 763 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC4A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.7 ExcIEEEAC5A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC5A model. The model represents a simplified model for brushless excitation systems. The regulator is supplied from a source, such as a permanent magnet generator, which is not affected by system disturbances. Unlike other AC models, this model uses loaded rather than open circuit exciter saturation data in the same way as it is used for the DC models. Because the model has been widely implemented by the industry, it is sometimes used to represent other types of systems when either detailed data for them are not available or simplified models are required.

Reference: IEEE 421.5:2005, 6.5.

Table 764 shows all attributes of ExcIEEEAC5A.

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Table 764 – Attributes of ExcitationSystemDynamics::ExcIEEEAC5A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 400.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 7,3.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -7,3.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,8.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0,03.
tf1	1..1	Seconds	Excitation control system stabilizer time constant (TF1) (> 0). Typical value = 1.
tf2	1..1	Seconds	Excitation control system stabilizer time constant (TF2) (>= 0). Typical value = 1.
tf3	1..1	Seconds	Excitation control system stabilizer time constant (TF3) (>= 0). Typical value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD1) (> 0). Typical value = 5,6.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD1 (SE[EFD1]) (>= 0). Typical value = 0,86.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD2) (> 0). Typical value = 4,2.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD2 (SE[EFD2]) (>= 0). Typical value = 0,5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 765 shows all association ends of ExcIEEEAC5A with other classes.

Table 765 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC5A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.8 ExcIEEEAC6A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

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IEEE 421.5:2005 type AC6A model. The model represents field-controlled alternator-rectifier excitation systems with system-supplied electronic voltage regulators. The maximum output of the regulator, VR, is a function of terminal voltage, VT. The field current limiter included in the original model AC6A remains in the 2005 update.

Reference: IEEE 421.5:2005, 6.6.

Table 766 shows all attributes of ExcIEEEAC6A.

Table 766 – Attributes of ExcitationSystemDynamics::ExcIEEEAC6A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 536.
ta	1..1	Seconds	Voltage regulator time constant (TA) (>= 0). Typical value = 0,086.
tk	1..1	Seconds	Voltage regulator time constant (TK) (>= 0). Typical value = 0,18.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 9.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 3.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 75.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -75.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 44.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -36.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 1.
kh	1..1	PU	Exciter field current limiter gain (KH) (>= 0). Typical value = 92.
tj	1..1	Seconds	Exciter field current limiter time constant (TJ) (>= 0). Typical value = 0,02.
th	1..1	Seconds	Exciter field current limiter time constant (TH) (> 0). Typical value = 0,08.
vfelim	1..1	PU	Exciter field current limit reference (VFELIM) (> 0). Typical value = 19.
vhmax	1..1	PU	Maximum field current limiter signal reference (VHMAX) (> 0). Typical value = 75.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,173.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 1,91.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1,6.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 7,4.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 0,214.

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name	mult	type	description
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 5,55.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 0,044.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 767 shows all association ends of ExcIEEEAC6A with other classes.

Table 767 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC6A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.9 ExcIEEEAC7B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC7B model. The model represents excitation systems which consist of an AC alternator with either stationary or rotating rectifiers to produce the DC field requirements. It is an upgrade to earlier AC excitation systems, which replace only the controls but retain the AC alternator and diode rectifier bridge.

Reference: IEEE 421.5:2005, 6.7. Note, however, that in IEEE 421.5:2005, the [1 / sTE] block is shown as [1 / (1 + sTE)], which is incorrect.

Table 768 shows all attributes of ExcIEEEAC7B.

Table 768 – Attributes of ExcitationSystemDynamics::ExcIEEEAC7B

name	mult	type	description
kpr	1..1	PU	Voltage regulator proportional gain (KPR) (> 0 if ExcIEEEAC7B.kir = 0). Typical value = 4,24.
kir	1..1	PU	Voltage regulator integral gain (KIR) (>= 0). Typical value = 4,24.
kdr	1..1	PU	Voltage regulator derivative gain (KDR) (>= 0). Typical value = 0.
tdr	1..1	Seconds	Lag time constant (TDR) (>= 0). Typical value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 5,79.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -5,79.
kpa	1..1	PU	Voltage regulator proportional gain (KPA) (> 0 if ExcIEEEAC7B.kia = 0). Typical value = 65,36.

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name	mult	type	description
kia	1..1	PU	Voltage regulator integral gain (KIA) (>= 0). Typical value = 59,69.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 1.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -0,95.
kp	1..1	PU	Potential circuit gain coefficient (KP) (> 0). Typical value = 4,96.
kl	1..1	PU	Exciter field voltage lower limit parameter (KL). Typical value = 10.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 1,1.
vfemax	1..1	PU	Exciter field current limit reference (VFEMAX). Typical value = 6,9.
vemin	1..1	PU	Minimum exciter voltage output (VEMIN) (<= 0). Typical value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,18.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 0,02.
kf1	1..1	PU	Excitation control system stabilizer gain (KF1) (>= 0). Typical value = 0,212.
kf2	1..1	PU	Excitation control system stabilizer gain (KF2) (>= 0). Typical value = 0.
kf3	1..1	PU	Excitation control system stabilizer gain (KF3) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 6,3.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 0,44.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 3,02.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 0,075.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 769 shows all association ends of ExcIEEEAC7B with other classes.

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Table 769 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC7B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.10 ExcIEEEAC8B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type AC8B model. This model represents a PID voltage regulator with either a brushless exciter or DC exciter. The AVR in this model consists of PID control, with separate constants for the proportional (KPR), integral (KIR), and derivative (KDR) gains. The representation of the brushless exciter (TE, KE, SE, KC, KD) is similar to the model type AC2A. The type AC8B model can be used to represent static voltage regulators applied to brushless excitation systems. Digitally based voltage regulators feeding DC rotating main exciters can be represented with the AC type AC8B model with the parameters KC and KD set to 0. For thyristor power stages fed from the generator terminals, the limits VRMAX and VRMIN should be a function of terminal voltage: VT x VRMAX and VT x VRMIN.

Reference: IEEE 421.5:2005, 6.8.

Table 770 shows all attributes of ExcIEEEAC8B.

Table 770 – Attributes of ExcitationSystemDynamics::ExcIEEEAC8B

name	mult	type	description
kpr	1..1	PU	Voltage regulator proportional gain (KPR) (> 0 if ExcIEEEAC8B.kir = 0). Typical value = 80.
kir	1..1	PU	Voltage regulator integral gain (KIR) (>= 0). Typical value = 5.
kdr	1..1	PU	Voltage regulator derivative gain (KDR) (>= 0). Typical value = 10.
tdr	1..1	Seconds	Lag time constant (TDR) (> 0). Typical value = 0,1.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 35.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (<= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 1.
ta	1..1	Seconds	Voltage regulator time constant (TA) (>= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 1,2.
vfemax	1..1	PU	Exciter field current limit reference (VFEMAX). Typical value = 6.
vemin	1..1	PU	Minimum exciter voltage output (VEMIN) (<= 0). Typical value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,55.

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name	mult	type	description
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (KD) (>= 0). Typical value = 1,1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE1) (> 0). Typical value = 6,5.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE1, back of commutating reactance (SE[VE1]) (>= 0). Typical value = 0,3.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (VE2) (> 0). Typical value = 9.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, VE2, back of commutating reactance (SE[VE2]) (>= 0). Typical value = 3.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 771 shows all association ends of ExcIEEEAC8B with other classes.

Table 771 – Association ends of ExcitationSystemDynamics:: ExcIEEEAC8B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.11 ExcIEEEDC1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type DC1A model. This model represents field-controlled DC commutator exciters with continuously acting voltage regulators (especially the direct-acting rheostatic, rotating amplifier, and magnetic amplifier types). Because this model has been widely implemented by the industry, it is sometimes used to represent other types of systems when detailed data for them are not available or when a simplified model is required.

Reference: IEEE 421.5:2005, 5.1.

Table 772 shows all attributes of ExcIEEEDC1A.

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Table 772 – Attributes of ExcitationSystemDynamics::ExcIEEEDC1A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 46.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,06.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> ExcIEEEDC1A.vrmin). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0 and < ExcIEEEDC1A.vrmax). Typical value = -0,9.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,46.
kf	1..1	PU	Excitation control system stabilizer gain (KF) (>= 0). Typical value = 0.1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD1) (> 0). Typical value = 3,1.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD1 (SE[EFD1]) (>= 0). Typical value = 0.33.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD2) (> 0). Typical value = 2,3.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD2 (SE[EFD2]) (>= 0). Typical value = 0,1.
uelin	1..1	Boolean	UEL input (uelin). true = input is connected to the HV gate false = input connects to the error signal. Typical value = true.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 773 shows all association ends of ExcIEEEDC1A with other classes.

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Table 773 – Association ends of ExcitationSystemDynamics:: ExcIEEEDC1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.12 ExcIEEEDC2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type DC2A model. This model represents field-controlled DC commutator exciters with continuously acting voltage regulators having supplies obtained from the generator or auxiliary bus. It differs from the type DC1A model only in the voltage regulator output limits, which are now proportional to terminal voltage VT.

It is representative of solid-state replacements for various forms of older mechanical and rotating amplifier regulating equipment connected to DC commutator exciters.

Reference: IEEE 421.5:2005, 5.2.

Table 774 shows all attributes of ExcIEEEDC2A.

Table 774 – Attributes of ExcitationSystemDynamics::ExcIEEEDC2A

name	mult	type	description
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD1) (> 0). Typical value = 3,05.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD2) (> 0). Typical value = 2,29.
exclim	1..1	PU	(exclim). IEEE standard is ambiguous about lower limit on exciter output. Typical value = –999 which means that there is no limit applied.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 300.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
kf	1..1	PU	Excitation control system stabilizer gain (KF) (>= 0). Typical value = 0,1.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD1 (SE[EFD1]) (>= 0). Typical value = 0,279.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD2 (SE[EFD2]) (>= 0). Typical value = 0,117.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,01.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 1,33.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (> 0). Typical value = 0,675.

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name	mult	type	description
uelin	1..1	Boolean	UEL input (uelin). true = input is connected to the HV gate false = input connects to the error signal. Typical value = true.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX)(> ExcIEEEDC2A.vrmin). Typical value = 4,95.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0 and < ExcIEEEDC2A.vrmax). Typical value = -4,9.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 775 shows all association ends of ExcIEEEDC2A with other classes.

Table 775 – Association ends of ExcitationSystemDynamics:: ExcIEEEDC2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.13 ExcIEEEDC3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type DC3A model. This model represents older systems, in particular those DC commutator exciters with non-continuously acting regulators that were commonly used before the development of the continuously acting varieties. These systems respond at basically two different rates, depending upon the magnitude of voltage error. For small errors, adjustment is made periodically with a signal to a motor-operated rheostat. Larger errors cause resistors to be quickly shorted or inserted and a strong forcing signal applied to the exciter. Continuous motion of the motor-operated rheostat occurs for these larger error signals, even though it is bypassed by contactor action.

Reference: IEEE 421.5:2005, 5.3.

Table 776 shows all attributes of ExcIEEEDC3A.

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Table 776 – Attributes of ExcitationSystemDynamics::ExcIEEEDC3A

name	mult	type	description
trh	1..1	Seconds	Rheostat travel time (TRH) (> 0). Typical value = 20.
kv	1..1	PU	Fast raise/lower contact setting (KV) (> 0). Typical value = 0,05.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (<= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,5.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 0,05.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD1) (> 0). Typical value = 3,375.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD1 (SE[EFD1]) (>= 0). Typical value = 0,267.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD2) (> 0). Typical value = 3,15.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD2 (SE[EFD2]) (>= 0). Typical value = 0,068.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 777 shows all association ends of ExcIEEEDC3A with other classes.

Table 777 – Association ends of ExcitationSystemDynamics:: ExcIEEEDC3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.14 ExcIEEEDC4B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type DC4B model. These excitation systems utilize a field-controlled DC commutator exciter with a continuously acting voltage regulator having supplies obtained from the generator or auxiliary bus.

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Reference: IEEE 421.5:2005, 5.4.

Table 778 shows all attributes of ExcIEEEDC4B.

Table 778 – Attributes of ExcitationSystemDynamics::ExcIEEEDC4B

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 1.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,2.
kp	1..1	PU	Regulator proportional gain (KP) (>= 0). Typical value = 20.
ki	1..1	PU	Regulator integral gain (KI) (>= 0). Typical value = 20.
kd	1..1	PU	Regulator derivative gain (KD) (>= 0). Typical value = 20.
td	1..1	Seconds	Regulator derivative filter time constant (TD) (> 0 if ExcIEEEDC4B.kd > 0). Typical value = 0,01.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> ExcIEEEDC4B.vrmin). Typical value = 2,7.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (<= 0 and < ExcIEEEDC4B.vrmax). Typical value = -0,9.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,8.
kf	1..1	PU	Excitation control system stabilizer gain (KF) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (>= 0). Typical value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD1) (> 0). Typical value = 1,75.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD1 (SE[EFD1]) (>= 0). Typical value = 0,08.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (EFD2) (> 0). Typical value = 2,33.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, EFD2 (SE[EFD2]) (>= 0). Typical value = 0,27.
vemin	1..1	PU	Minimum exciter voltage output (VEMIN) (<= 0). Typical value = 0.
oelin	1..1	Boolean	OEL input (OELin). true = LV gate false = subtract from error signal. Typical value = true.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 779 shows all association ends of ExcIEEEDC4B with other classes.

Table 779 – Association ends of ExcitationSystemDynamics:: ExcIEEEDC4B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.15 ExcIEEEST1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST1A model. This model represents systems in which excitation power is supplied through a transformer from the generator terminals (or the unit's auxiliary bus) and is regulated by a controlled rectifier. The maximum exciter voltage available from such systems is directly related to the generator terminal voltage.

Reference: IEEE 421.5:2005, 7.1.

Table 780 shows all attributes of ExcIEEEST1A.

Table 780 – Attributes of ExcitationSystemDynamics::ExcIEEEST1A

name	mult	type	description
ilr	1..1	PU	Exciter output current limit reference (ILR). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 190.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,08.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0.
klr	1..1	PU	Exciter output current limiter gain (KLR). Typical value = 0.
pssin	1..1	Boolean	Selector of the Power System Stabilizer (PSS) input (PSSin). true = PSS input (Vs) added to error signal false = PSS input (Vs) added to voltage regulator output. Typical value = true.
ta	1..1	Seconds	Voltage regulator time constant (TA) (>= 0). Typical value = 0.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 10.
tb1	1..1	Seconds	Voltage regulator time constant (TB1) (>= 0). Typical value = 0.

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name	mult	type	description
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 1.
tc1	1..1	Seconds	Voltage regulator time constant (TC1) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (>= 0). Typical value = 1.
uelin	1..1	ExcIEEEST1AUELselectorKind	Selector of the connection of the UEL input (UELin). Typical value = ignoreUELsignal.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 14,5.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -14,5.
vimax	1..1	PU	Maximum voltage regulator input limit (VIMAX) (> 0). Typical value = 999.
vimin	1..1	PU	Minimum voltage regulator input limit (VIMIN) (< 0). Typical value = -999.
vrmax	1..1	PU	Maximum voltage regulator outputs (VRMAX) (> 0). Typical value = 7,8.
vrmin	1..1	PU	Minimum voltage regulator outputs (VRMIN) (< 0). Typical value = -6,7.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 781 shows all association ends of ExcIEEEST1A with other classes.

Table 781 – Association ends of ExcitationSystemDynamics:: ExcIEEEST1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.16 ExcIEEEST2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST2A model. Some static systems use both current and voltage sources (generator terminal quantities) to comprise the power source. The regulator controls the exciter output through controlled saturation of the power transformer components. These compound-source rectifier excitation systems are designated type ST2A and are represented by ExcIEEEST2A.

Reference: IEEE 421.5:2005, 7.2.

Table 782 shows all attributes of ExcIEEEST2A.

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Table 782 – Attributes of ExcitationSystemDynamics::ExcIEEEST2A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (KA) (> 0). Typical value = 120.
ta	1..1	Seconds	Voltage regulator time constant (TA) (> 0). Typical value = 0,15.
vrmax	1..1	PU	Maximum voltage regulator outputs (VRMAX) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator outputs (VRMIN) (<= 0). Typical value = 0.
ke	1..1	PU	Exciter constant related to self-excited field (KE). Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (TE) (> 0). Typical value = 0,5.
kf	1..1	PU	Excitation control system stabilizer gains (KF) (>= 0). Typical value = 0,05.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (>= 0). Typical value = 1.
kp	1..1	PU	Potential circuit gain coefficient (KP) (>= 0). Typical value = 4,88.
ki	1..1	PU	Potential circuit gain coefficient (KI) (>= 0). Typical value = 8.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 1,82.
efdmax	1..1	PU	Maximum field voltage (EFDMax) (>= 0). Typical value = 99.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 783 shows all association ends of ExcIEEEST2A with other classes.

Table 783 – Association ends of ExcitationSystemDynamics:: ExcIEEEST2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

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14.3.3.9.17 ExcIEEEST3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST3A model. Some static systems utilize a field voltage control loop to linearize the exciter control characteristic. This also makes the output independent of supply source variations until supply limitations are reached. These systems utilize a variety of controlled-rectifier designs: full thyristor complements or hybrid bridges in either series or shunt configurations. The power source can consist of only a potential source, either fed from the machine terminals or from internal windings. Some designs can have compound power sources utilizing both machine potential and current. These power sources are represented as phasor combinations of machine terminal current and voltage and are accommodated by suitable parameters in model type ST3A which is represented by ExcIEEEST3A.

Reference: IEEE 421.5:2005, 7.3.

Table 784 shows all attributes of ExcIEEEST3A.

Table 784 – Attributes of ExcitationSystemDynamics::ExcIEEEST3A

name	mult	type	description
vimax	1..1	PU	Maximum voltage regulator input limit (VIMAX) (> 0). Typical value = 0,2.
vimin	1..1	PU	Minimum voltage regulator input limit (VIMIN) (< 0). Typical value = -0,2.
ka	1..1	PU	Voltage regulator gain (KA) (> 0). This is parameter K in the IEEE standard. Typical value = 200.
ta	1..1	Seconds	Voltage regulator time constant (TA) (>= 0). Typical value = 0.
tb	1..1	Seconds	Voltage regulator time constant (TB) (>= 0). Typical value = 10.
tc	1..1	Seconds	Voltage regulator time constant (TC) (>= 0). Typical value = 1.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 10.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -10.
km	1..1	PU	Forward gain constant of the inner loop field regulator (KM) (> 0). Typical value = 7,93.
tm	1..1	Seconds	Forward time constant of inner loop field regulator (TM) (> 0). Typical value = 0,4.
vmmax	1..1	PU	Maximum inner loop output (VMMax) (> 0). Typical value = 1.
vmmin	1..1	PU	Minimum inner loop output (VMMin) (<= 0). Typical value = 0.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (KG) (>= 0). Typical value = 1.
kp	1..1	PU	Potential circuit gain coefficient (KP) (> 0). Typical value = 6,15.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical value = 0.
ki	1..1	PU	Potential circuit gain coefficient (KI) (>= 0). Typical value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,2.

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name	mult	type	description
xl	1..1	PU	Reactance associated with potential source (XL) (>= 0). Typical value = 0,081.
vbmax	1..1	PU	Maximum excitation voltage (VBMax) (> 0). Typical value = 6,9.
vgmax	1..1	PU	Maximum inner loop feedback voltage (VGMax) (>= 0). Typical value = 5,8.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 785 shows all association ends of ExcIEEEST3A with other classes.

Table 785 – Association ends of ExcitationSystemDynamics:: ExcIEEEST3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.18 ExcIEEEST4B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST4B model. This model is a variation of the type ST3A model, with a proportional plus integral (PI) regulator block replacing the lag-lead regulator characteristic that is in the ST3A model. Both potential and compound source rectifier excitation systems are modelled. The PI regulator blocks have non-windup limits that are represented. The voltage regulator of this model is typically implemented digitally.

Reference: IEEE 421.5:2005, 7.4.

Table 786 shows all attributes of ExcIEEEST4B.

Table 786 – Attributes of ExcitationSystemDynamics::ExcIEEEST4B

name	mult	type	description
kpr	1..1	PU	Voltage regulator proportional gain (KPR). Typical value = 10,75.
kir	1..1	PU	Voltage regulator integral gain (KIR). Typical value = 10,75.
ta	1..1	Seconds	Voltage regulator time constant (TA) (>= 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -0,87.
kpm	1..1	PU	Voltage regulator proportional gain output (KPM). Typical value = 1.
kim	1..1	PU	Voltage regulator integral gain output (KIM). Typical value = 0.

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name	mult	type	description
vmmax	1..1	PU	Maximum inner loop output (VMMax) (> ExcIEEEST4B.vmmin). Typical value = 99.
vmmin	1..1	PU	Minimum inner loop output (VMMin) (< ExcIEEEST4B.vmmax). Typical value = -99.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (KG) (>= 0). Typical value = 0.
kp	1..1	PU	Potential circuit gain coefficient (KP) (> 0). Typical value = 9,3.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical value = 0.
ki	1..1	PU	Potential circuit gain coefficient (KI) (>= 0). Typical value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (KC) (>= 0). Typical value = 0,113.
xl	1..1	PU	Reactance associated with potential source (XL) (>= 0). Typical value = 0,124.
vbmax	1..1	PU	Maximum excitation voltage (VBMax) (> 0). Typical value = 11,63.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 787 shows all association ends of ExcIEEEST4B with other classes.

Table 787 – Association ends of ExcitationSystemDynamics:: ExcIEEEST4B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.19 ExcIEEEST5B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST5B model. The type ST5B excitation system is a variation of the type ST1A model, with alternative overexcitation and underexcitation inputs and additional limits.

The block diagram in the IEEE 421.5 standard has input signal Vc and does not indicate the summation point with Vref. The implementation of the ExcIEEEST5B shall consider summation point with Vref.

Reference: IEEE 421.5:2005, 7.5.

Table 788 shows all attributes of ExcIEEEST5B.

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Table 788 – Attributes of ExcitationSystemDynamics::ExcIEEEST5B

name	mult	type	description
kr	1..1	PU	Regulator gain (KR) (> 0). Typical value = 200.
t1	1..1	Seconds	Firing circuit time constant (T1) (>= 0). Typical value = 0,004.
kc	1..1	PU	Rectifier regulation factor (KC) (>= 0). Typical value = 0,004.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -4.
tc1	1..1	Seconds	Regulator lead time constant (TC1) (>= 0). Typical value = 0,8.
tb1	1..1	Seconds	Regulator lag time constant (TB1) (>= 0). Typical value = 6.
tc2	1..1	Seconds	Regulator lead time constant (TC2) (>= 0). Typical value = 0,08.
tb2	1..1	Seconds	Regulator lag time constant (TB2) (>= 0). Typical value = 0,01.
toc1	1..1	Seconds	OEL lead time constant (TOC1) (>= 0). Typical value = 0,1.
tob1	1..1	Seconds	OEL lag time constant (TOB1) (>= 0). Typical value = 2.
toc2	1..1	Seconds	OEL lead time constant (TOC2) (>= 0). Typical value = 0,08.
tob2	1..1	Seconds	OEL lag time constant (TOB2) (>= 0). Typical value = 0,08.
tuc1	1..1	Seconds	UEL lead time constant (TUC1) (>= 0). Typical value = 2.
tub1	1..1	Seconds	UEL lag time constant (TUB1) (>= 0). Typical value = 10.
tuc2	1..1	Seconds	UEL lead time constant (TUC2) (>= 0). Typical value = 0,1.
tub2	1..1	Seconds	UEL lag time constant (TUB2) (>= 0). Typical value = 0,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 789 shows all association ends of ExcIEEEST5B with other classes.

Table 789 – Association ends of ExcitationSystemDynamics:: ExcIEEEST5B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.20 ExcIEEEST6B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

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IEEE 421.5:2005 type ST6B model. This model consists of a PI voltage regulator with an inner loop field voltage regulator and pre-control. The field voltage regulator implements a proportional control. The pre-control and the delay in the feedback circuit increase the dynamic response.

Reference: IEEE 421.5:2005, 7.6.

Table 790 shows all attributes of ExcIEEEST6B.

Table 790 – Attributes of ExcitationSystemDynamics::ExcIEEEST6B

name	mult	type	description
ilr	1..1	PU	Exciter output current limit reference (ILR) (> 0). Typical value = 4,164.
kci	1..1	PU	Exciter output current limit adjustment (KCI) (> 0). Typical value = 1,0577.
kff	1..1	PU	Pre-control gain constant of the inner loop field regulator (KFF). Typical value = 1.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (KG) (>= 0). Typical value = 1.
kia	1..1	PU	Voltage regulator integral gain (KIA) (> 0). Typical value = 45,094.
klr	1..1	PU	Exciter output current limiter gain (KLR) (> 0). Typical value = 17,33.
km	1..1	PU	Forward gain constant of the inner loop field regulator (KM). Typical value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (KPA) (> 0). Typical value = 18,038.
oelin	1..1	ExcST6BOELselectorKind	OEL input selector (OELin). Typical value = noOELinput.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (TG) (>= 0). Typical value = 0,02.
vamax	1..1	PU	Maximum voltage regulator output (VAMAX) (> 0). Typical value = 4,81.
vamin	1..1	PU	Minimum voltage regulator output (VAMIN) (< 0). Typical value = -3,85.
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 4,81.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -3,85.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 791 shows all association ends of ExcIEEEST6B with other classes.

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Table 791 – Association ends of ExcitationSystemDynamics:: ExcIEEEST6B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.21 ExcIEEEST7B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type ST7B model. This model is representative of static potential-source excitation systems. In this system, the AVR consists of a PI voltage regulator. A phase lead-lag filter in series allows the introduction of a derivative function, typically used with brushless excitation systems. In that case, the regulator is of the PID type. In addition, the terminal voltage channel includes a phase lead-lag filter. The AVR includes the appropriate inputs on its reference for overexcitation limiter (OEL1), underexcitation limiter (UEL), stator current limiter (SCL), and current compensator (DROOP). All these limitations, when they work at voltage reference level, keep the PSS (VS signal from PSS) in operation. However, the UEL limitation can also be transferred to the high value (HV) gate acting on the output signal. In addition, the output signal passes through a low value (LV) gate for a ceiling overexcitation limiter (OEL2).

Reference: IEEE 421.5:2005, 7.7.

Table 792 shows all attributes of ExcIEEEST7B.

Table 792 – Attributes of ExcitationSystemDynamics::ExcIEEEST7B

name	mult	type	description
kh	1..1	PU	High-value gate feedback gain (KH) (>= 0). Typical value = 1.
kia	1..1	PU	Voltage regulator integral gain (KIA) (>= 0). Typical value = 1.
kl	1..1	PU	Low-value gate feedback gain (KL) (>= 0). Typical value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (KPA) (> 0). Typical value = 40.
oelin	1..1	ExcST7BOELselectorKind	OEL input selector (OELin). Typical value = noOELinput.
tb	1..1	Seconds	Regulator lag time constant (TB) (>= 0). Typical value = 1.
tc	1..1	Seconds	Regulator lead time constant (TC) (>= 0). Typical value = 1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (TF) (>= 0). Typical value = 1.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (TG) (>= 0). Typical value = 1.
tia	1..1	Seconds	Feedback time constant (TIA) (>= 0). Typical value = 3.
uelin	1..1	ExcST7BUELselectorKind	UEL input selector (UELin). Typical value = noUELinput.
vmax	1..1	PU	Maximum voltage reference signal (VMAX) (> 0 and > ExcIEEEST7B.vmin). Typical value = 1,1.
vmin	1..1	PU	Minimum voltage reference signal (VMIN) (> 0 and < ExcIEEEST7B.vmax). Typical value = 0,9.

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name	mult	type	description
vrmax	1..1	PU	Maximum voltage regulator output (VRMAX) (> 0). Typical value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (VRMIN) (< 0). Typical value = -4,5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 793 shows all association ends of ExcIEEEST7B with other classes.

Table 793 – Association ends of ExcitationSystemDynamics:: ExcIEEEST7B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.22 ExcAC1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC1A alternator-supplied rectifier excitation system with different rate feedback source.

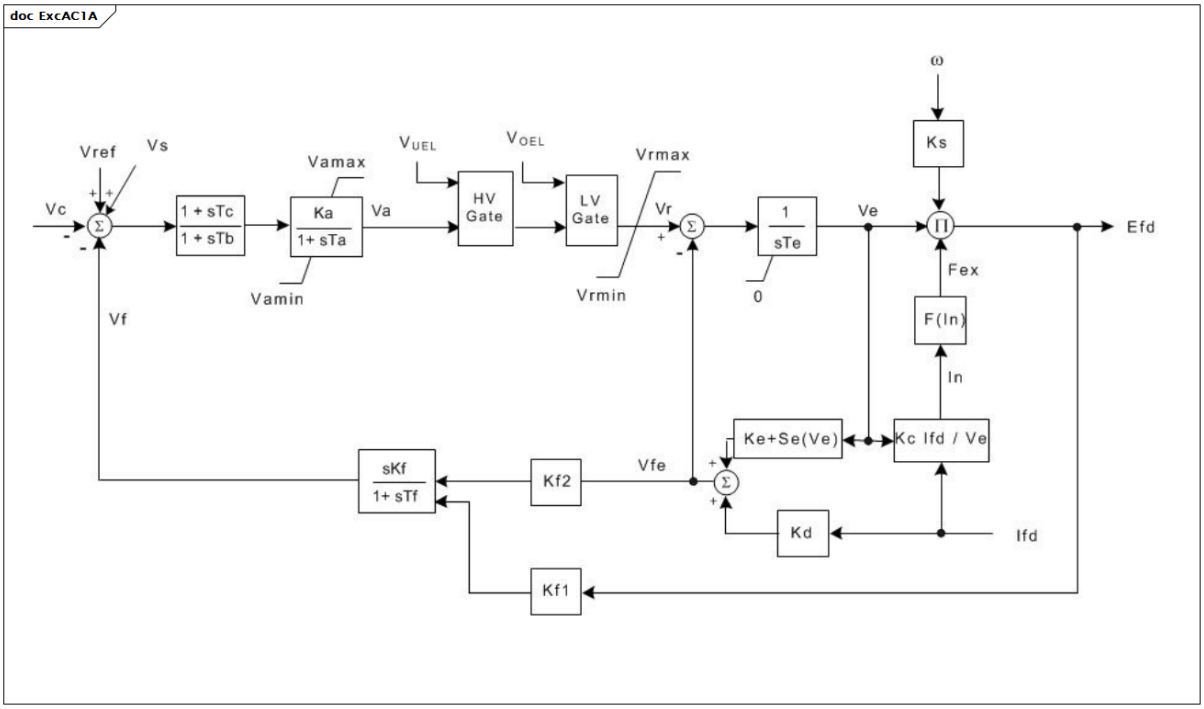

Figure 141 – ExcAC1A

Figure 141: This diagram describes the ExcAC1A excitation system model.

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Parameter details:

1) If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Table 794 shows all attributes of ExcAC1A.

Table 794 – Attributes of ExcitationSystemDynamics::ExcAC1A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 400.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,02.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 14,5.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -14,5.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 0,8.
kf	1..1	PU	Excitation control system stabilizer gains (Kf) (>= 0). Typical value = 0,03.
kf1	1..1	PU	Coefficient to allow different usage of the model (Kf1) (>= 0). Typical value = 0.
kf2	1..1	PU	Coefficient to allow different usage of the model (Kf2) (>= 0). Typical value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (> 0). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,2.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd) (>= 0). Typical value = 0,38.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 4,18.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 0,1.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 3,14.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 0,03.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax) (> 0). Typical value = 6,03.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin) (< 0). Typical value = -5,43.

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name	mult	type	description
hvlvgates	1..1	Boolean	Indicates if both HV gate and LV gate are active (HVLVgates). true = gates are used false = gates are not used. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 795 shows all association ends of ExcAC1A with other classes.

Table 795 – Association ends of ExcitationSystemDynamics:: ExcAC1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.23 ExcAC2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC2A alternator-supplied rectifier excitation system with different field current limit.

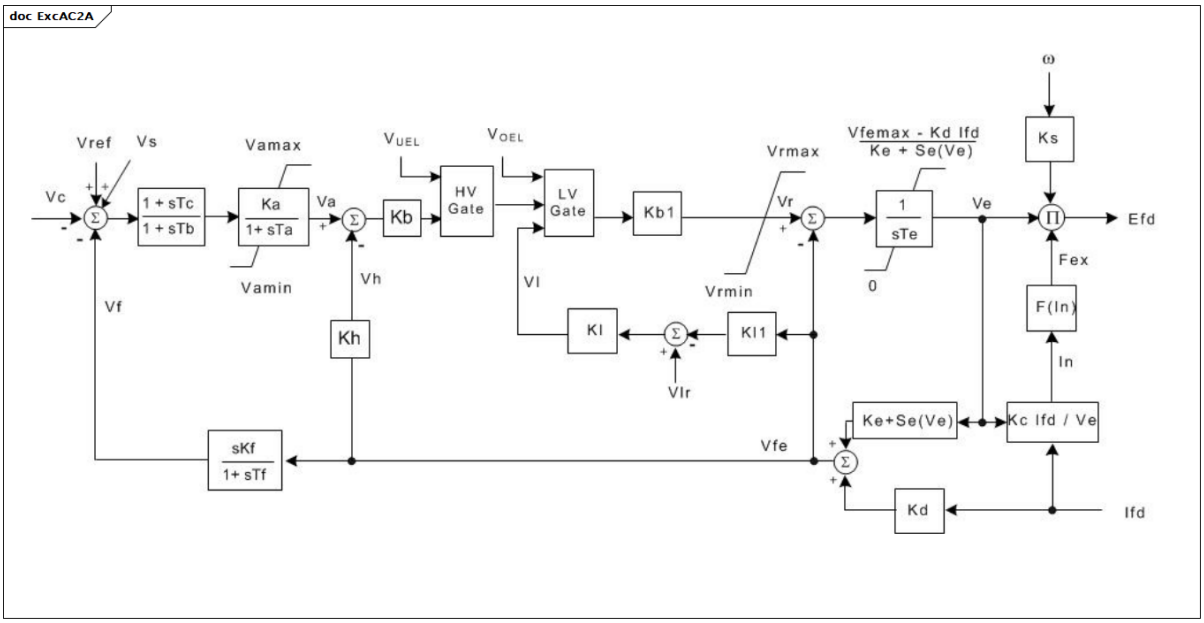

Figure 142 – ExcAC2A

Figure 142: This diagram describes the ExcAC2A excitation system model.

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Parameter details:

1) If Ks is set to 1 the output (Efd) is multiplied by generator speed;
2) The upper limit on Ve represents the effect of the field current limiter. If Vfemax is zero, this limit will not be enforced. In the real system, the limiter is implemented by a low value gate just before Kb. The input to this LV gate is Kl x (Vlr – Vfe). If the values of Kl and Vlr are given, Vfemax can be calculated as Vlr x Kl x Kb / (1 + Kl x Kb).

Table 796 shows all attributes of ExcAC2A.

Table 796 – Attributes of ExcitationSystemDynamics::ExcAC2A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 400.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,02.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 8.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -8.
kb	1..1	PU	Second stage regulator gain (Kb) (> 0). Exciter field current controller gain. Typical value = 25.
kb1	1..1	PU	Second stage regulator gain (Kb1). It is exciter field current controller gain used as alternative to Kb to represent a variant of the ExcAC2A model. Typical value = 25.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax) (> 0). Typical value = 105.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin) (< 0). Typical value = -95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 0,6.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax) (>= 0). Typical value = 4,4.
kh	1..1	PU	Exciter field current feedback gain (Kh) (>= 0). Typical value = 1.
kf	1..1	PU	Excitation control system stabilizer gains (Kf) (>= 0). Typical value = 0,03.
kl	1..1	PU	Exciter field current limiter gain (Kl). Typical value = 10.
vlr	1..1	PU	Maximum exciter field current (Vlr) (> 0). Typical value = 4,4.
kl1	1..1	PU	Coefficient to allow different usage of the model (Kl1). Typical value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (> 0). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,28.

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name	mult	type	description
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd) (>= 0). Typical value = 0,35.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 4,4.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 0,037.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 3,3.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 0,012.
hvgate	1..1	Boolean	Indicates if HV gate is active (HVgate). true = gate is used false = gate is not used. Typical value = true.
lvgate	1..1	Boolean	Indicates if LV gate is active (LVgate). true = gate is used false = gate is not used. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 797 shows all association ends of ExcAC2A with other classes.

Table 797 – Association ends of ExcitationSystemDynamics:: ExcAC2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.24 ExcAC3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC3A alternator-supplied rectifier excitation system with different field current limit.

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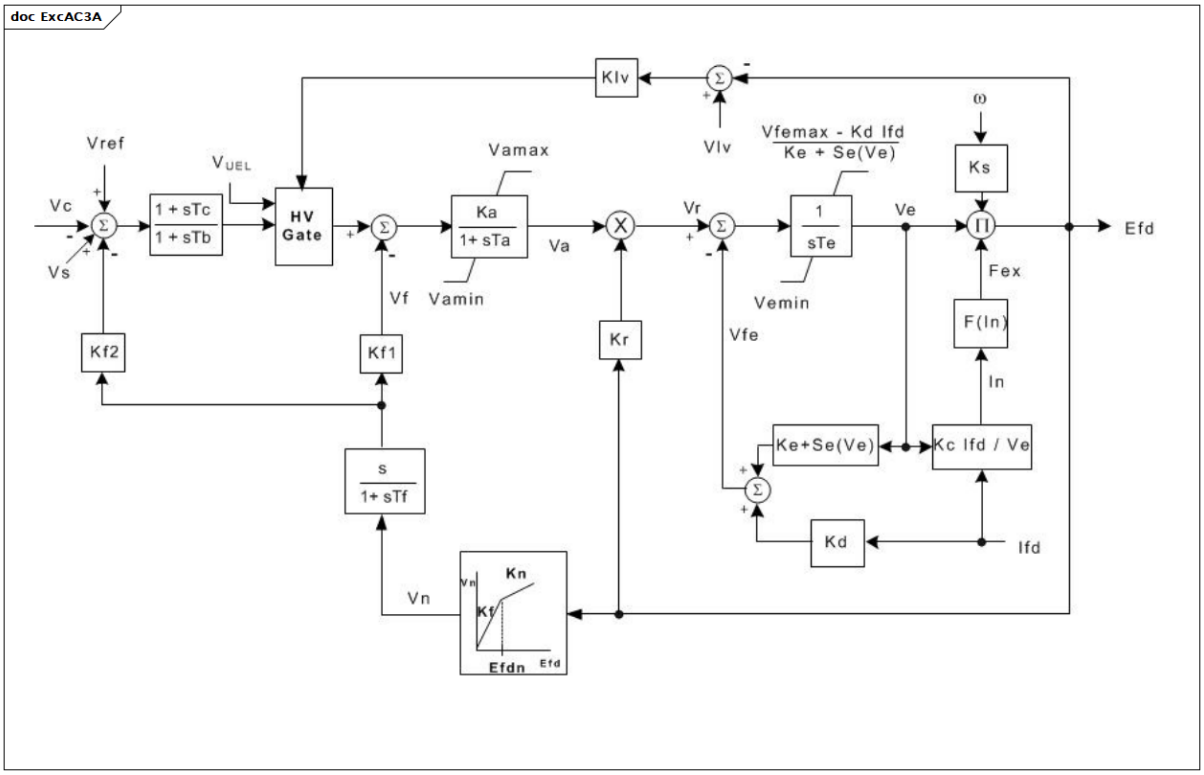

Figure 143 – ExcAC3A

Figure 143: This diagram describes the ExcAC3A excitation system model.

Parameter details:

1) If Ks is set to 1, the output (Efd) is multiplied by generator speed.
2) The upper limit on Ve is an approximate representation of the effect of the maximum field current limiter. If Vfemax is zero, this limit will not be enforced. The lower limit (Vemin) on Ve is an approximate representation of the minimum field voltage limiter.

Table 798 shows all attributes of ExcAC3A.

Table 798 – Attributes of ExcitationSystemDynamics::ExcAC3A

name	mult	type	description
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
ka	1..1	Seconds	Voltage regulator gain (Ka) (> 0). Typical value = 45,62.
ta	1..1	PU	Voltage regulator time constant (Ta) (> 0). Typical value = 0,013.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 1.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -0,95.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1,17.

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name	mult	type	description
vemin	1..1	PU	Minimum exciter voltage output (Vemin) (<= 0). Typical value = 0.
kr	1..1	PU	Constant associated with regulator and alternator field power supply (Kr) (> 0). Typical value =3,77.
kf	1..1	PU	Excitation control system stabilizer gains (Kf) (>= 0). Typical value = 0,143.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (> 0). Typical value = 1.
kn	1..1	PU	Excitation control system stabilizer gain (Kn) (>= 0). Typical value =0,05.
efdn	1..1	PU	Value of Efd at which feedback gain changes (Efdn) (> 0). Typical value = 2,36.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,104.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd) (>= 0). Typical value = 0,499.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
klv	1..1	PU	Gain used in the minimum field voltage limiter loop (Klv). Typical value = 0,194.
kf1	1..1	PU	Coefficient to allow different usage of the model (Kf1). Typical value = 1.
kf2	1..1	PU	Coefficient to allow different usage of the model (Kf2). Typical value = 0.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax) (>= 0). Typical value = 16.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 6.24.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 1,143.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 4,68.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 0,1.
vlv	1..1	PU	Field voltage used in the minimum field voltage limiter loop (Vlv). Typical value = 0,79.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 799 shows all association ends of ExcAC3A with other classes.

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Table 799 – Association ends of ExcitationSystemDynamics:: ExcAC3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.25 ExcAC4A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC4A alternator-supplied rectifier excitation system with different minimum controller output.

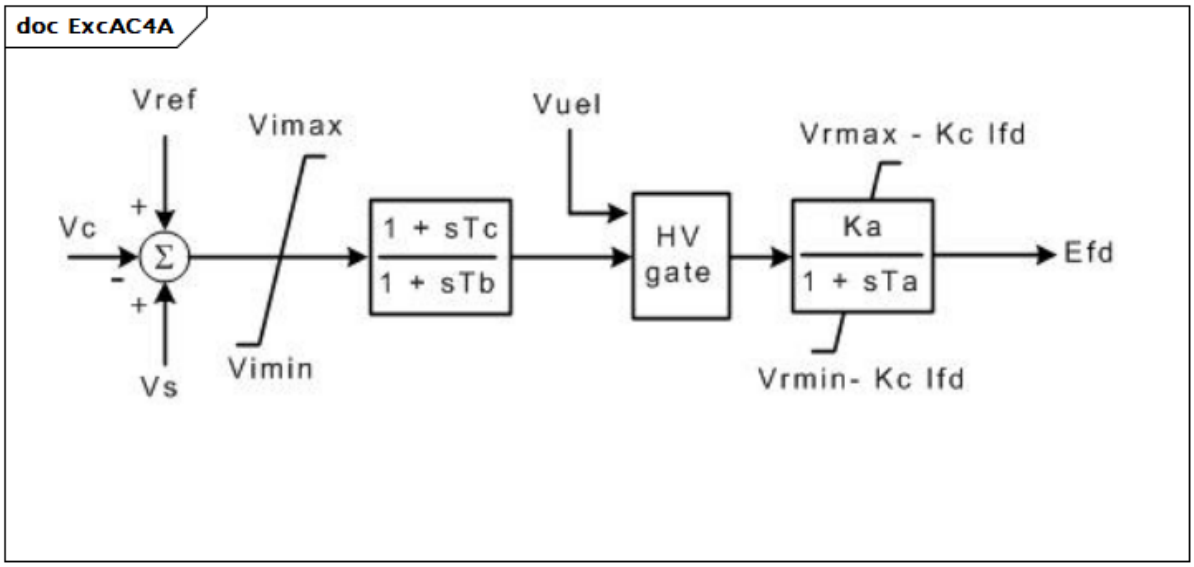

Figure 144 – ExcAC4A

Figure 144: This diagram describes the ExcAC4A excitation system model.

Table 800 shows all attributes of ExcAC4A.

Table 800 – Attributes of ExcitationSystemDynamics::ExcAC4A

name	mult	type	description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax) (> 0). Typical value = 10.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin) (< 0). Typical value = -10.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 1.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 10.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 200.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,015.

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name	mult	type	description
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 5,64.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -4,53.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 801 shows all association ends of ExcAC4A with other classes.

Table 801 – Association ends of ExcitationSystemDynamics:: ExcAC4A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.26 ExcAC5A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC5A alternator-supplied rectifier excitation system with different minimum controller output.

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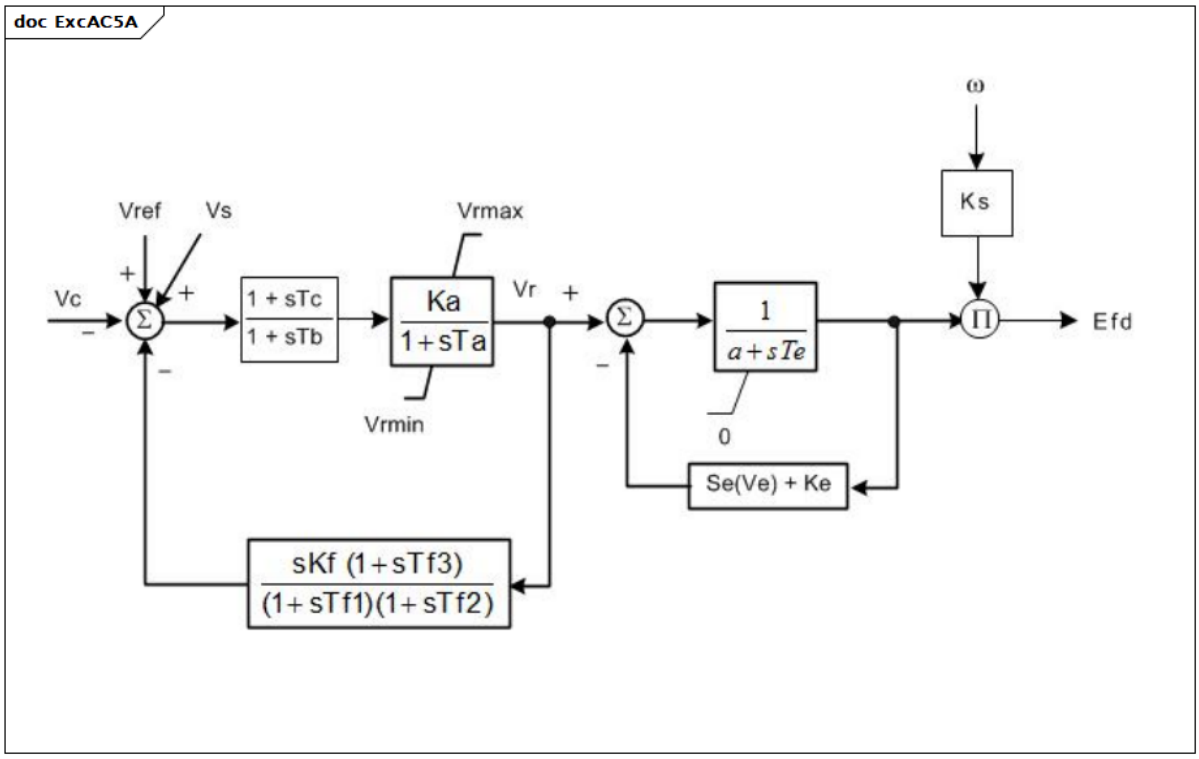

Figure 145 – ExcAC5A

Figure 145: This diagram describes the ExcAC5A excitation system model.

Parameter details:

1) If Ks is set to 1, the output (Efd) is multiplied by generator speed.

Table 802 shows all attributes of ExcAC5A.

Table 802 – Attributes of ExcitationSystemDynamics::ExcAC5A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 400.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 7,3.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value =-7,3.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 0,8.

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name	mult	type	description
kf	1..1	PU	Excitation control system stabilizer gains (Kf) (>= 0). Typical value = 0,03.
tf1	1..1	Seconds	Excitation control system stabilizer time constant (Tf1) (> 0). Typical value = 1.
tf2	1..1	Seconds	Excitation control system stabilizer time constant (Tf2) (>= 0). Typical value = 0,8.
tf3	1..1	Seconds	Excitation control system stabilizer time constant (Tf3) (>= 0). Typical value = 0.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1) (> 0). Typical value = 5,6.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Efd1]) (>= 0). Typical value = 0,86.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2) (> 0). Typical value = 4,2.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Efd2]) (>= 0). Typical value = 0,5.
a	1..1	Float	Coefficient to allow different usage of the model (a). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 803 shows all association ends of ExcAC5A with other classes.

Table 803 – Association ends of ExcitationSystemDynamics:: ExcAC5A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.27 ExcAC6A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC6A alternator-supplied rectifier excitation system with speed input.

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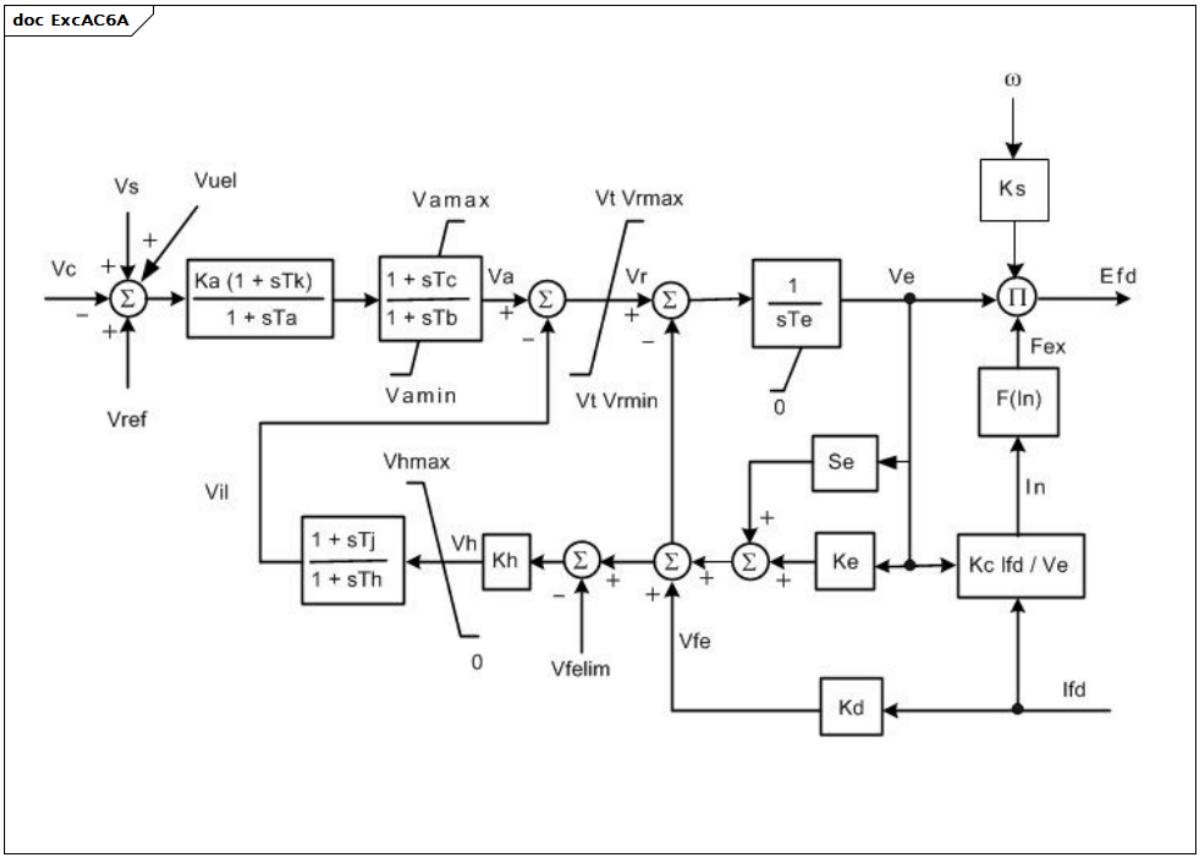

Figure 146 – ExcAC6A

Figure 146: This diagram describes the ExcAC6A excitation system model.

Parameter details:

1) If Ks is set to 1 the output (Efd) is multiplied by generator speed.

Table 804 shows all attributes of ExcAC6A.

Table 804 – Attributes of ExcitationSystemDynamics::ExcAC6A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 536.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (>= 0). Typical value = 0,086.
tk	1..1	Seconds	Voltage regulator time constant (Tk) (>= 0). Typical value = 0,18.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 9.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 3.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 75.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -75.

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name	mult	type	description
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 44.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -36.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1.
kh	1..1	PU	Exciter field current limiter gain (Kh) (>= 0). Typical value = 92.
tj	1..1	Seconds	Exciter field current limiter time constant (Tj) (>= 0). Typical value = 0,02.
th	1..1	Seconds	Exciter field current limiter time constant (Th) (> 0). Typical value = 0,08.
vfelim	1..1	PU	Exciter field current limit reference (Vfelim) (> 0). Typical value = 19.
vhmax	1..1	PU	Maximum field current limiter signal reference (Vhmax) (> 0). Typical value = 75.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,173.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd) (>= 0). Typical value = 1,91.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1,6.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 7,4.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 0,214.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 5,55.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 0,044.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 805 shows all association ends of ExcAC6A with other classes.

Table 805 – Association ends of ExcitationSystemDynamics:: ExcAC6A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

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14.3.3.9.28 ExcAC8B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE AC8B alternator-supplied rectifier excitation system with speed input and input limiter.

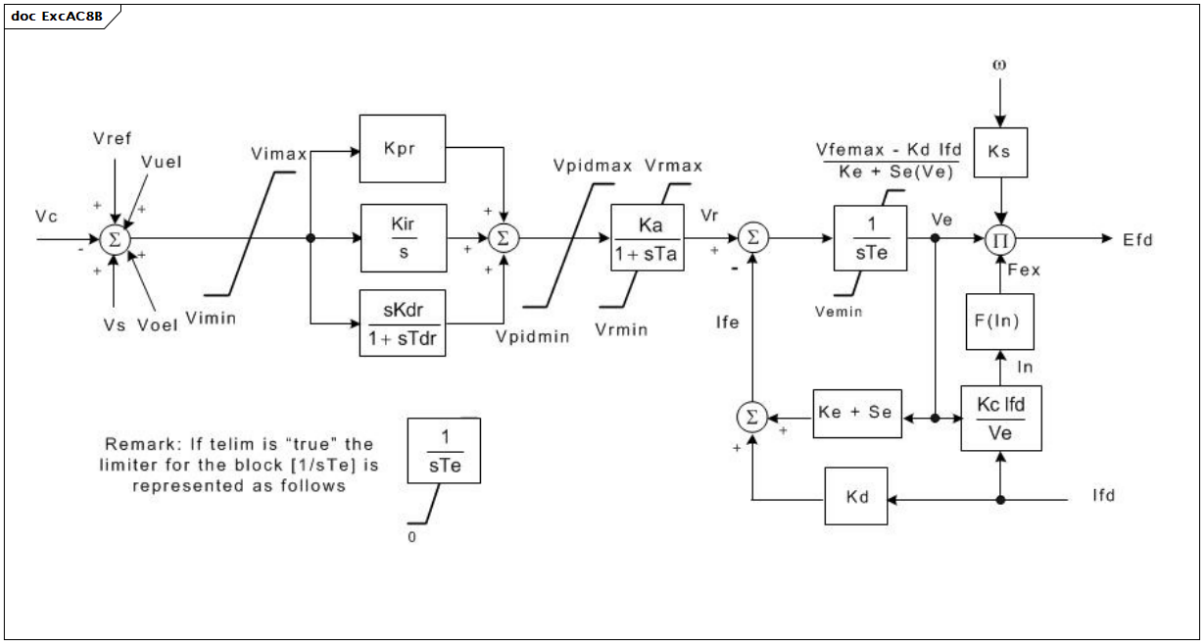

Figure 147 – ExcAC8B

Figure 147: This diagram describes the ExcAC8A excitation system model.

Parameter details:

1) The upper limit on Ve represents the effect of the field current limiter. If Vfemax is zero, this limit will not be enforced.
2) The non-windup limits are defined as in Annex E of the IEEE 421.5:2005.
3) If Ks is set to 1, the output (Efd) is multiplied by generator speed.

Table 806 shows all attributes of ExcAC8B.

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Table 806 – Attributes of ExcitationSystemDynamics::ExcAC8B

name	mult	type	description
inlim	1..1	Boolean	Input limiter indicator. true = input limiter Vimax and Vimin is considered false = input limiter Vimax and Vimin is not considered. Typical value = true.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 1.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,55.
kd	1..1	PU	Demagnetizing factor, a function of exciter alternator reactances (Kd) (>= 0). Typical value = 1,1.
kdr	1..1	PU	Voltage regulator derivative gain (Kdr) (>= 0). Typical value = 10.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
kir	1..1	PU	Voltage regulator integral gain (Kir) (>= 0). Typical value = 5.
kpr	1..1	PU	Voltage regulator proportional gain (Kpr) (> 0 if ExcAC8B.kir = 0). Typical value = 80.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
pidlim	1..1	Boolean	PID limiter indicator. true = input limiter Vpidmax and Vpidmin is considered false = input limiter Vpidmax and Vpidmin is not considered. Typical value = true.
seve1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 0,3.
seve2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 3.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (>= 0). Typical value = 0.
tdr	1..1	Seconds	Lag time constant (Tdr) (> 0 if ExcAC8B.kdr > 0). Typical value = 0,1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1,2.
telim	1..1	Boolean	Selector for the limiter on the block (1/sTe). See diagram for meaning of true and false. Typical value = false.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 6,5.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 9.

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name	mult	type	description
vemin	1..1	PU	Minimum exciter voltage output (Vemin) (<= 0). Typical value = 0.
vfemax	1..1	PU	Exciter field current limit reference (Vfemax). Typical value = 6.
vimax	1..1	PU	Input signal maximum (Vimax) (> ExcAC8B.vimin). Typical value = 35.
vimin	1..1	PU	Input signal minimum (Vimin) (< ExcAC8B.vimax). Typical value = -10.
vpidmax	1..1	PU	PID maximum controller output (Vpidmax) (> ExcAC8B.vpidmin). Typical value = 35.
vpidmin	1..1	PU	PID minimum controller output (Vpidmin) (< ExcAC8B.vpidmax). Typical value = -10.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 35.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = 0.
vtmult	1..1	Boolean	Multiply by generator's terminal voltage indicator. true =the limits Vrmax and Vrmin are multiplied by the generator’s terminal voltage to represent a thyristor power stage fed from the generator terminals false = limits are not multiplied by generator's terminal voltage. Typical value = false.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 807 shows all association ends of ExcAC8B with other classes.

Table 807 – Association ends of ExcitationSystemDynamics:: ExcAC8B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.29 ExcANS

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian excitation system. It represents static field voltage or excitation current feedback excitation system.

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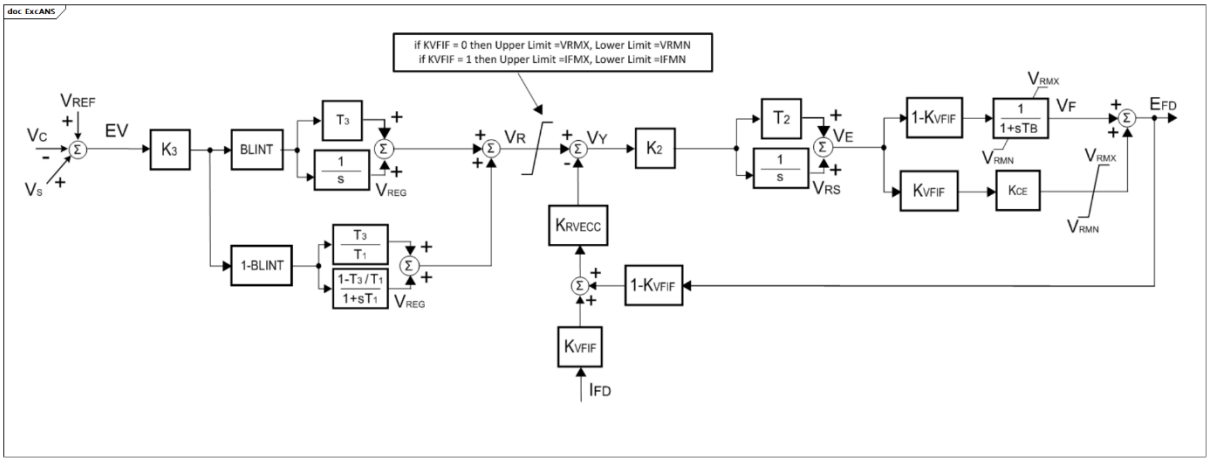

Figure 148 – ExcANS

Figure 148: This diagram describes the ExcANS excitation system model.

Parameter details:

1) Limits on VR signal are modelled as follows:
   – If KVFIF = 0, then upper limit = VRMX and lower limit = VRMN;
   – If KVFIF = 1, then upper limit = IFMX and lower limit = IFMN.

Table 808 shows all attributes of ExcANS.

Table 808 – Attributes of ExcitationSystemDynamics::ExcANS

name	mult	type	description
k3	1..1	Float	AVR gain (K3). Typical value = 1000.
k2	1..1	Float	Exciter gain (K2). Typical value = 20.
kce	1..1	Float	Ceiling factor (KCE). Typical value = 1.
t3	1..1	Seconds	Time constant (T3) (>= 0). Typical value = 1,6.
t2	1..1	Seconds	Time constant (T2) (>= 0). Typical value = 0,05.
t1	1..1	Seconds	Time constant (T1) (>= 0). Typical value = 20.
blint	1..1	Integer	Governor control flag (BLINT). 0 = lead-lag regulator 1 = proportional integral regulator. Typical value = 0.
kvfif	1..1	Integer	Rate feedback signal flag (KVFIF). 0 = output voltage of the exciter 1 = exciter field current. Typical value = 0.
ifmn	1..1	PU	Minimum exciter current (IFMN). Typical value = -5,2.
ifmx	1..1	PU	Maximum exciter current (IFMX). Typical value = 6,5.
vrmn	1..1	PU	Minimum AVR output (VRMN). Typical value = -5,2.
vrmx	1..1	PU	Maximum AVR output (VRMX). Typical value = 6,5.

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name	mult	type	description
krvecc	1..1	Integer	Feedback enabling (KRVECC). 0 = open loop control 1 = closed loop control. Typical value = 1.
tb	1..1	Seconds	Exciter time constant (TB) (>= 0). Typical value = 0,04.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 809 shows all association ends of ExcANS with other classes.

Table 809 – Association ends of ExcitationSystemDynamics::ExcANS with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.30 ExcAVR1

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian excitation system corresponding to IEEE (1968) type 1 model. It represents an exciter dynamo and electromechanical regulator.

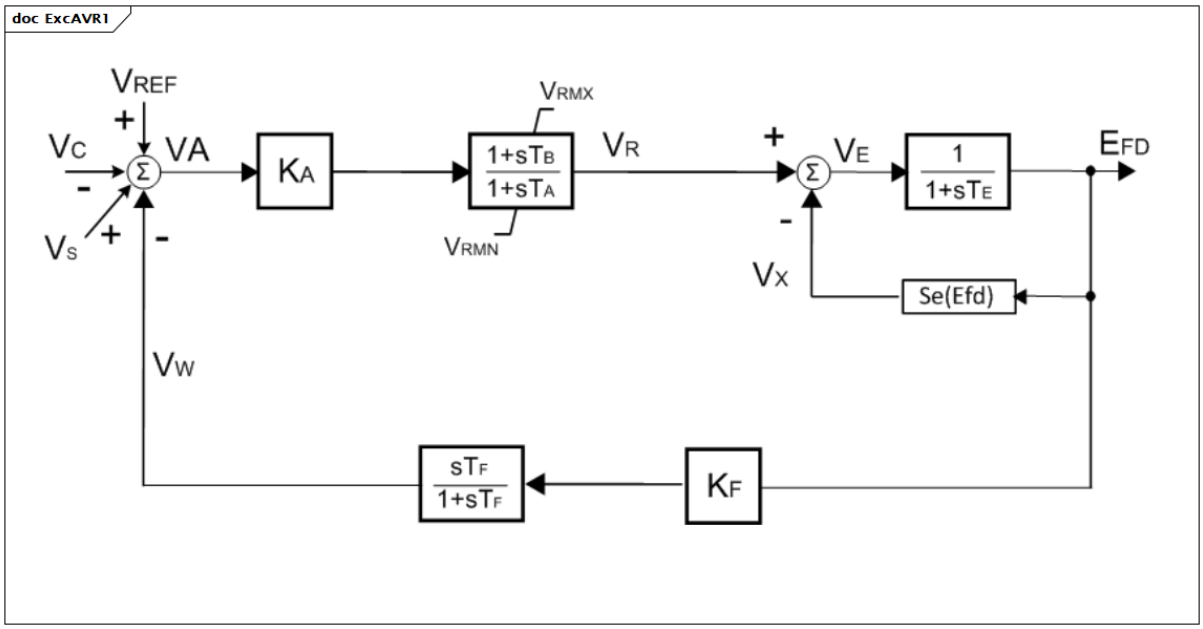

Figure 149 – ExcAVR1

Figure 149: This diagram describes the ExcAVR1 excitation system model.

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Table 810 shows all attributes of ExcAVR1.

Table 810 – Attributes of ExcitationSystemDynamics::ExcAVR1

name	mult	type	description
ka	1..1	Float	AVR gain (KA). Typical value = 500.
vrmn	1..1	PU	Minimum AVR output (VRMN). Typical value = -6.
vrmx	1..1	PU	Maximum AVR output (VRMX). Typical value = 7.
ta	1..1	Seconds	AVR time constant (TA) (>= 0). Typical value = 0,2.
tb	1..1	Seconds	AVR time constant (TB) (>= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant (TE) (>= 0). Typical value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical value = 4.18.
se1	1..1	Float	Saturation factor at E1 (S[E1]). Typical value = 0,1.
e2	1..1	PU	Field voltage value 2 (E2). Typical value = 3,14.
se2	1..1	Float	Saturation factor at E2 (S[E2]). Typical value = 0,03.
kf	1..1	Float	Rate feedback gain (KF). Typical value = 0,12.
tf	1..1	Seconds	Rate feedback time constant (TF) (>= 0). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 811 shows all association ends of ExcAVR1 with other classes.

Table 811 – Association ends of ExcitationSystemDynamics:: ExcAVR1 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.31 ExcAVR2

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian excitation system corresponding to IEEE (1968) type 2 model. It represents an alternator and rotating diodes and electromechanic voltage regulators.

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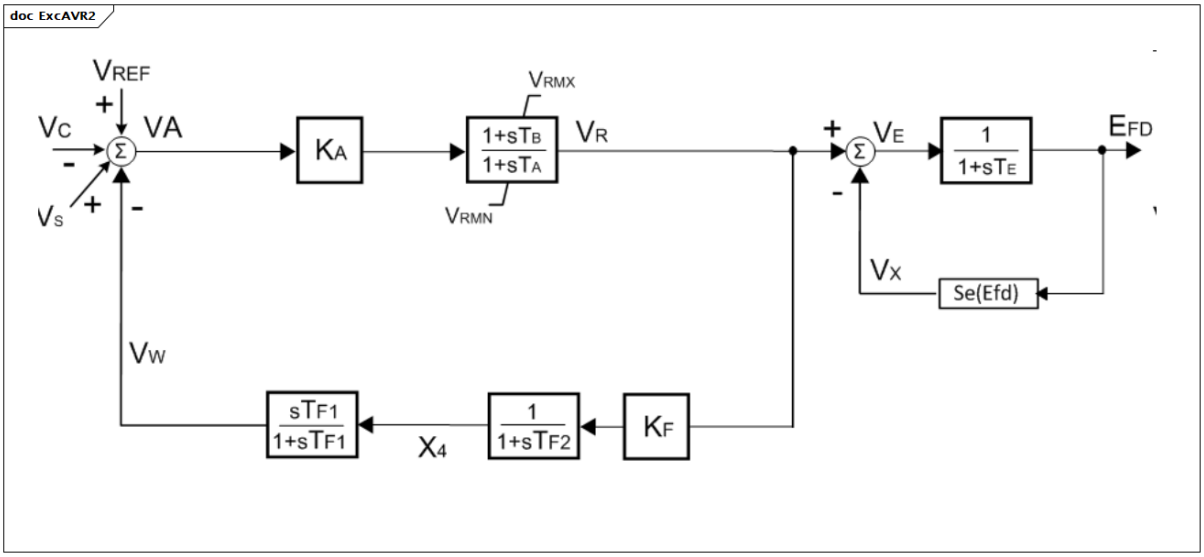

Figure 150 – ExcAVR2

Figure 150: This diagram describes the ExcAVR2 excitation system model.

Table 812 shows all attributes of ExcAVR2.

Table 812 – Attributes of ExcitationSystemDynamics::ExcAVR2

name	mult	type	description
ka	1..1	Float	AVR gain (KA). Typical value = 500.
vrmn	1..1	PU	Minimum AVR output (VRMN). Typical value = -6.
vrmx	1..1	PU	Maximum AVR output (VRMX). Typical value = 7.
ta	1..1	Seconds	AVR time constant (TA) (>= 0). Typical value = 0,02.
tb	1..1	Seconds	AVR time constant (TB) (>= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant (TE) (>= 0). Typical value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical value = 4,18.
se1	1..1	Float	Saturation factor at E1 (S[E1]). Typical value = 0.1.
e2	1..1	PU	Field voltage value 2 (E2). Typical value = 3,14.
se2	1..1	Float	Saturation factor at E2 (S[E2]). Typical value = 0,03.
kf	1..1	Float	Rate feedback gain (KF). Typical value = 0,12.
tf1	1..1	Seconds	Rate feedback time constant (TF1) (>= 0). Typical value = 1.
tf2	1..1	Seconds	Rate feedback time constant (TF2) (>= 0). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 813 shows all association ends of ExcAVR2 with other classes.

Table 813 – Association ends of ExcitationSystemDynamics:: ExcAVR2 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.32 ExcAVR3

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian excitation system. It represents an exciter dynamo and electric regulator.

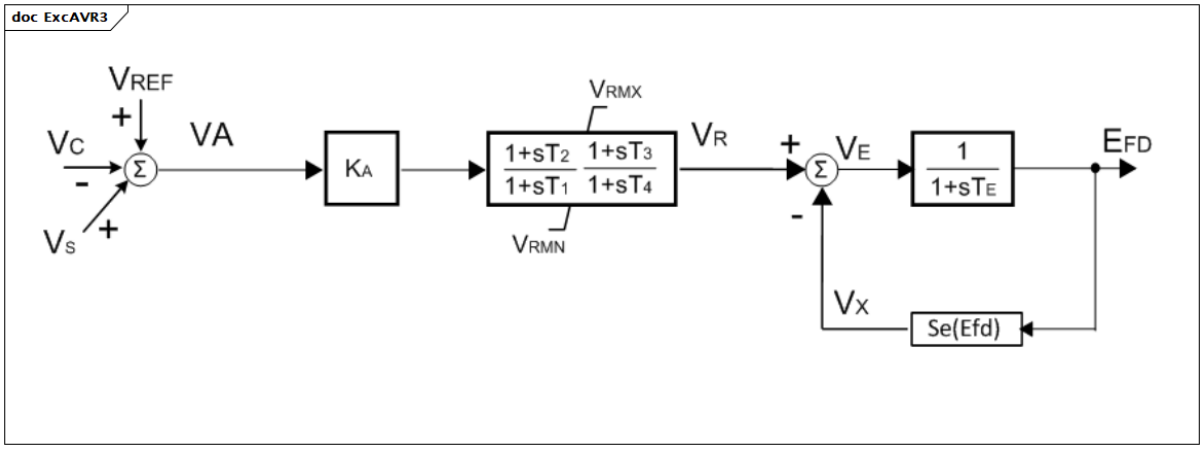

Figure 151 – ExcAVR3

Figure 151: This diagram describes the ExcAVR3 excitation system model.

Parameter details:

1) The non-windup limit is defined as in Annex E of the IEEE 421.5:2005.

Table 814 shows all attributes of ExcAVR3.

Table 814 – Attributes of ExcitationSystemDynamics::ExcAVR3

name	mult	type	description
ka	1..1	Float	AVR gain (KA). Typical value = 100.
vrmn	1..1	PU	Minimum AVR output (VRMN). Typical value = -7,5.
vrmx	1..1	PU	Maximum AVR output (VRMX). Typical value = 7,5.
t1	1..1	Seconds	AVR time constant (T1) (>= 0). Typical value = 20.
t2	1..1	Seconds	AVR time constant (T2) (>= 0). Typical value = 1,6.
t3	1..1	Seconds	AVR time constant (T3) (>= 0). Typical value = 0,66.

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name	mult	type	description
t4	1..1	Seconds	AVR time constant (T4) (>= 0). Typical value = 0,07.
te	1..1	Seconds	Exciter time constant (TE) (>= 0). Typical value = 1.
e1	1..1	PU	Field voltage value 1 (E1). Typical value = 4,18.
se1	1..1	Float	Saturation factor at E1 (S[E1]). Typical value = 0,1.
e2	1..1	PU	Field voltage value 2 (E2). Typical value = 3,14.
se2	1..1	Float	Saturation factor at E2 (S[E2]). Typical value = 0,03.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 815 shows all association ends of ExcAVR3 with other classes.

Table 815 – Association ends of ExcitationSystemDynamics:: ExcAVR3 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.33 ExcAVR4

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian excitation system. It represents a static exciter and electric voltage regulator.

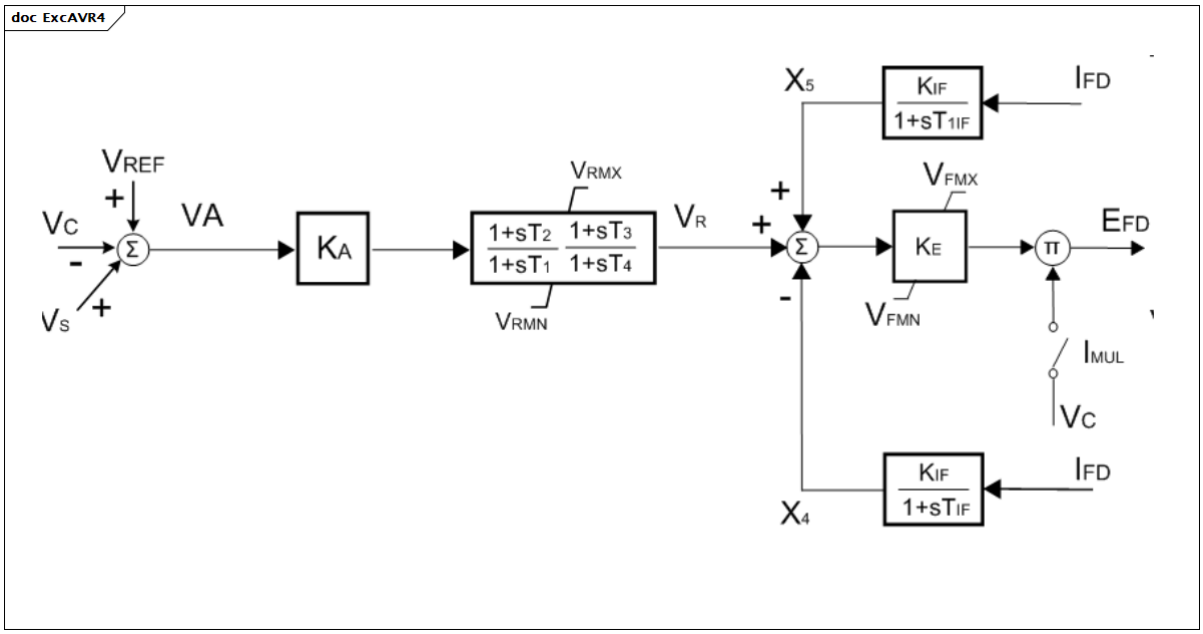

Figure 152 – ExcAVR4

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Figure 152: This diagram describes the ExcAVR4 excitation system model.

Table 816 shows all attributes of ExcAVR4.

Table 816 – Attributes of ExcitationSystemDynamics::ExcAVR4

name	mult	type	description
ka	1..1	Float	AVR gain (KA). Typical value = 300.
vrmn	1..1	PU	Minimum AVR output (VRMN). Typical value = 0.
vrmx	1..1	PU	Maximum AVR output (VRMX). Typical value = 5.
t1	1..1	Seconds	AVR time constant (T1) (>= 0). Typical value = 4,8.
t2	1..1	Seconds	AVR time constant (T2) (>= 0). Typical value = 1,5.
t3	1..1	Seconds	AVR time constant (T3) (>= 0). Typical value = 0.
t4	1..1	Seconds	AVR time constant (T4) (>= 0). Typical value = 0.
ke	1..1	Float	Exciter gain (KE). Typical value = 1.
vfmx	1..1	PU	Maximum exciter output (VFMX). Typical value = 5.
vfmn	1..1	PU	Minimum exciter output (VFMN). Typical value = 0.
kif	1..1	Float	Exciter internal reactance (KIF). Typical value = 0.
tif	1..1	Seconds	Exciter current feedback time constant (TIF) (>= 0). Typical value = 0.
t1if	1..1	Seconds	Exciter current feedback time constant (T1IF) (>= 0). Typical value = 60.
imul	1..1	Boolean	AVR output voltage dependency selector (IMUL). true = selector is connected false = selector is not connected. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 817 shows all association ends of ExcAVR4 with other classes.

Table 817 – Association ends of ExcitationSystemDynamics:: ExcAVR4 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

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14.3.3.9.34 ExcAVR5

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Manual excitation control with field circuit resistance. This model can be used as a very simple representation of manual voltage control.

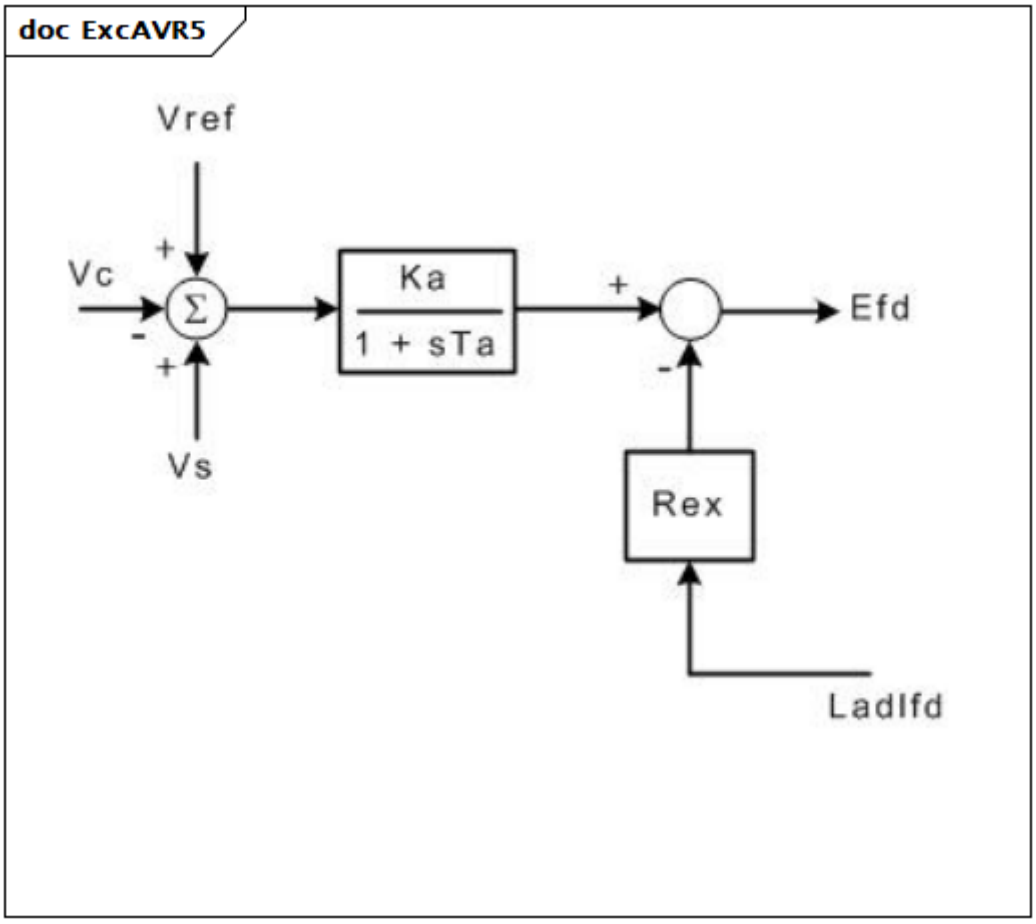

Figure 153 – ExcAVR5

Figure 153: This diagram describes the ExcAVR5 excitation system model.

Table 818 shows all attributes of ExcAVR5.

Table 818 – Attributes of ExcitationSystemDynamics::ExcAVR5

name	mult	type	description
ka	1..1	PU	Gain (Ka).
ta	1..1	Seconds	Time constant (Ta) (>= 0).
rex	1..1	PU	Effective output resistance (Rex). Rex represents the effective output resistance seen by the excitation system.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 819 shows all association ends of ExcAVR5 with other classes.

Table 819 – Association ends of ExcitationSystemDynamics:: ExcAVR5 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.35 ExcAVR7

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

IVO excitation system.

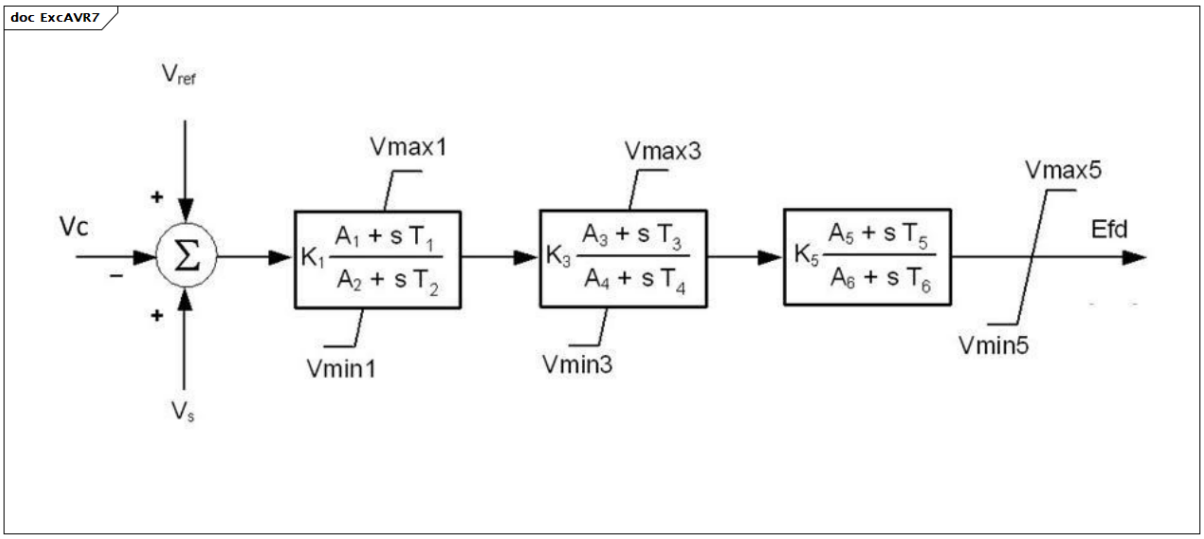

Figure 154 – ExcAVR7

Figure 154: This diagram describes the ExcAVR7 excitation system model.

Table 820 shows all attributes of ExcAVR7.

Table 820 – Attributes of ExcitationSystemDynamics::ExcAVR7

name	mult	type	description
k1	1..1	PU	Gain (K1). Typical value = 1.
a1	1..1	PU	Lead coefficient (A1). Typical value = 0,5.
a2	1..1	PU	Lag coefficient (A2). Typical value = 0,5.
t1	1..1	Seconds	Lead time constant (T1) (>= 0). Typical value = 0,05.
t2	1..1	Seconds	Lag time constant (T2) (>= 0). Typical value = 0,1.
vmax1	1..1	PU	Lead-lag maximum limit (Vmax1) (> ExcAVR7.vmin1). Typical value = 5.
vmin1	1..1	PU	Lead-lag minimum limit (Vmin1) (< ExcAVR7.vmax1). Typical value = -5.
k3	1..1	PU	Gain (K3). Typical value = 3.

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name	mult	type	description
a3	1..1	PU	Lead coefficient (A3). Typical value = 0,5.
a4	1..1	PU	Lag coefficient (A4). Typical value = 0,5.
t3	1..1	Seconds	Lead time constant (T3) (>= 0). Typical value = 0,1.
t4	1..1	Seconds	Lag time constant (T4) (>= 0). Typical value = 0,1.
vmax3	1..1	PU	Lead-lag maximum limit (Vmax3) (> ExcAVR7.vmin3). Typical value = 5.
vmin3	1..1	PU	Lead-lag minimum limit (Vmin3) (< ExcAVR7.vmax3). Typical value = -5.
k5	1..1	PU	Gain (K5). Typical value = 1.
a5	1..1	PU	Lead coefficient (A5). Typical value = 0,5.
a6	1..1	PU	Lag coefficient (A6). Typical value = 0,5.
t5	1..1	Seconds	Lead time constant (T5) (>= 0). Typical value = 0,1.
t6	1..1	Seconds	Lag time constant (T6) (>= 0). Typical value = 0,1.
vmax5	1..1	PU	Lead-lag maximum limit (Vmax5) (> ExcAVR7.vmin5). Typical value = 5.
vmin5	1..1	PU	Lead-lag minimum limit (Vmin5) (< ExcAVR7.vmax5). Typical value = -2.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 821 shows all association ends of ExcAVR7 with other classes.

Table 821 – Association ends of ExcitationSystemDynamics:: ExcAVR7 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.36 ExcBBC

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Transformer fed static excitation system (static with ABB regulator). This model represents a static excitation system in which a gated thyristor bridge fed by a transformer at the main generator terminals feeds the main generator directly.

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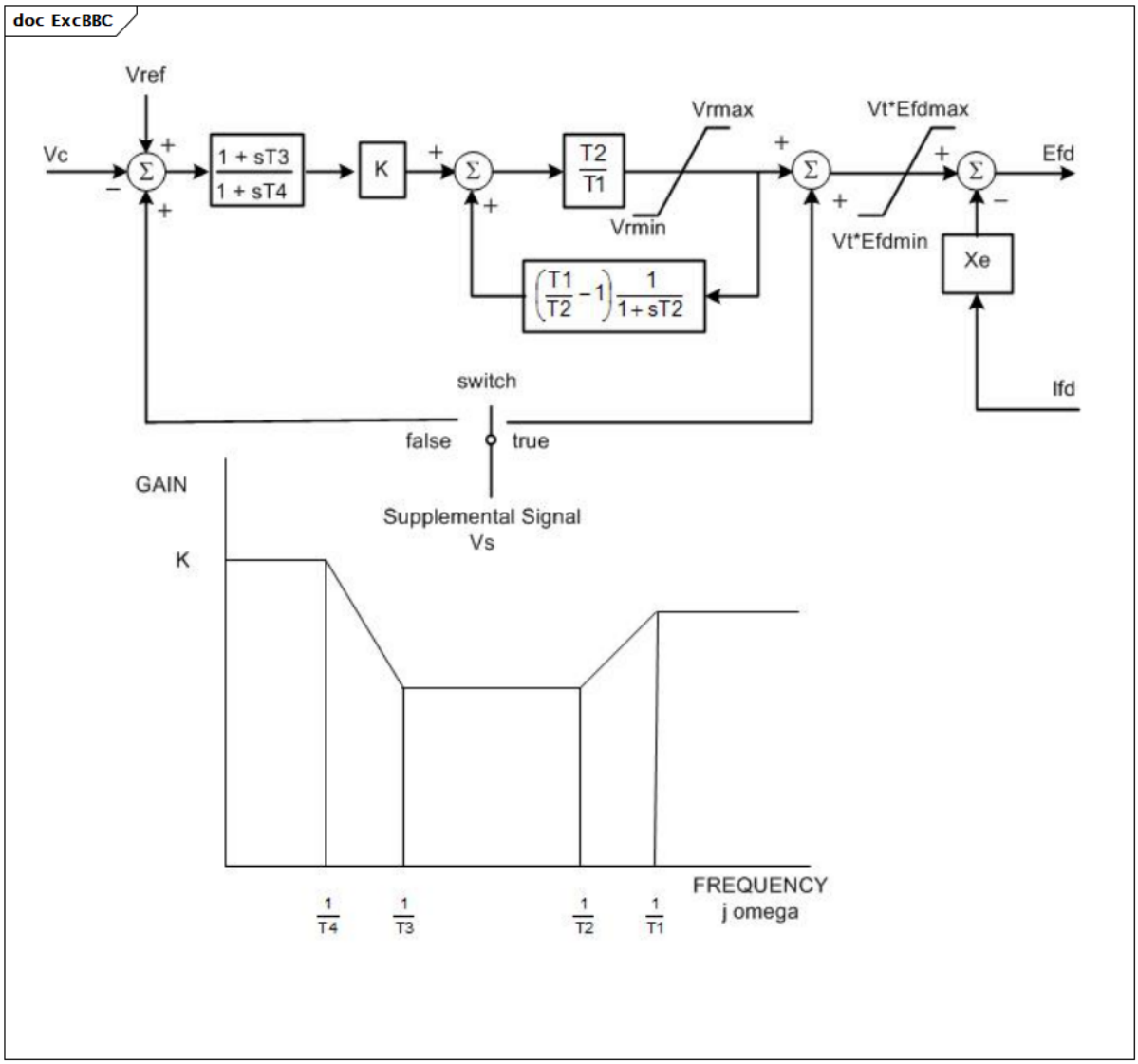

Figure 155 – ExcBBC

Figure 155: This diagram describes the ExcBBC excitation system model.

Table 822 shows all attributes of ExcBBC.

Table 822 – Attributes of ExcitationSystemDynamics::ExcBBC

name	mult	type	description
t1	1..1	Seconds	Controller time constant (T1) (>= 0). Typical value = 6.
t2	1..1	Seconds	Controller time constant (T2) (>= 0). Typical value = 1.
t3	1..1	Seconds	Lead/lag time constant (T3) (>= 0). If = 0, block is bypassed. Typical value = 0,05.
t4	1..1	Seconds	Lead/lag time constant (T4) (>= 0). If = 0, block is bypassed. Typical value = 0,01.
k	1..1	PU	Steady state gain (K) (not = 0). Typical value = 300.
vrmin	1..1	PU	Minimum control element output (Vrmin) (< ExcBBC.vrmax). Typical value = -5.

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name	mult	type	description
vrmax	1..1	PU	Maximum control element output (Vrmax) (> ExcBBC.vrmin). Typical value = 5.
efdmin	1..1	PU	Minimum open circuit exciter voltage (Efdmin) (< ExcBBC.efdmax). Typical value = -5.
efdmax	1..1	PU	Maximum open circuit exciter voltage (Efdmax) (> ExcBBC.efdmin). Typical value = 5.
xe	1..1	PU	Effective excitation transformer reactance (Xe) (>= 0). Xe models the regulation of the transformer/rectifier unit. Typical value = 0,05.
switch	1..1	Boolean	Supplementary signal routing selector (switch). true = Vs connected to 3rd summing point false = Vs connected to 1st summing point (see diagram). Typical value = false.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 823 shows all association ends of ExcBBC with other classes.

Table 823 – Association ends of ExcitationSystemDynamics:: ExcBBC with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.37 ExcCZ

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Czech proportion/integral exciter.

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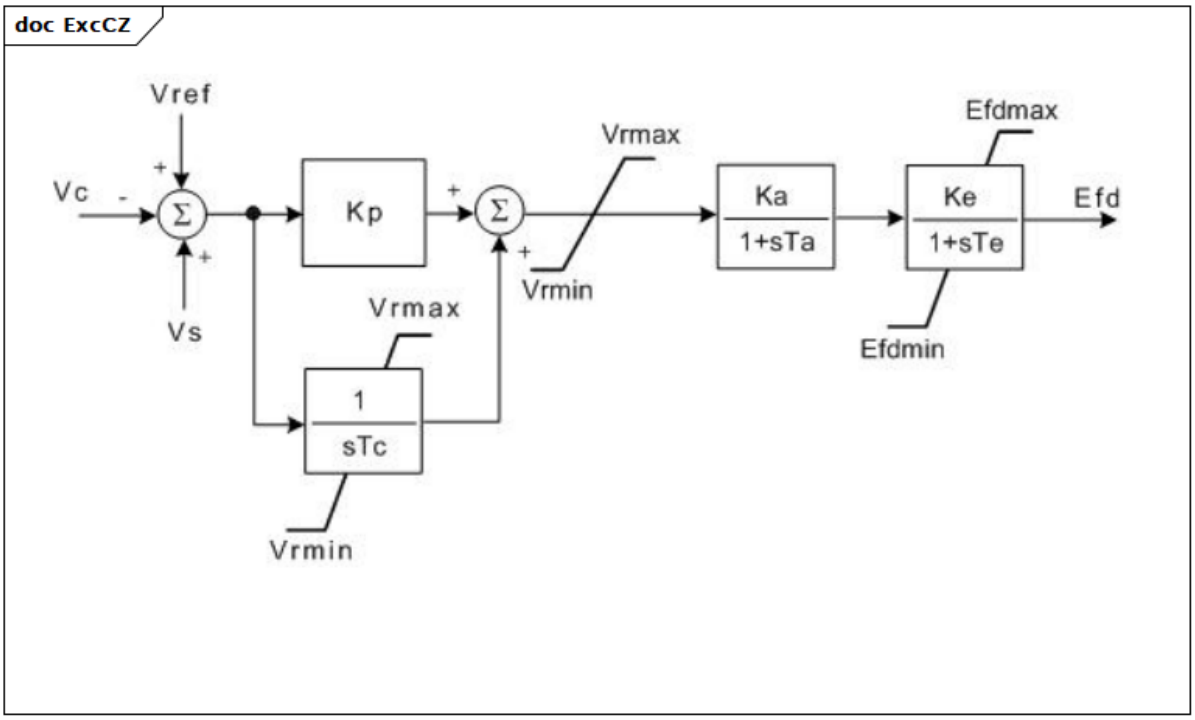

Figure 156 – ExcCZ

Figure 156: This diagram describes the ExcCZ excitation system model.

Table 824 shows all attributes of ExcCZ.

Table 824 – Attributes of ExcitationSystemDynamics::ExcCZ

name	mult	type	description
kp	1..1	PU	Regulator proportional gain (Kp).
tc	1..1	Seconds	Regulator integral time constant (Tc) (>= 0).
vrmax	1..1	PU	Voltage regulator maximum limit (Vrmax) (> ExcCZ.vrmin).
vrmin	1..1	PU	Voltage regulator minimum limit (Vrmin) (< ExcCZ.vrmax).
ka	1..1	PU	Regulator gain (Ka).
ta	1..1	Seconds	Regulator time constant (Ta) (>= 0).
ke	1..1	PU	Exciter constant related to self-excited field (Ke).
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (>= 0).
efdmax	1..1	PU	Exciter output maximum limit (Efdmax) (> ExcCZ.efdmin).
efdmin	1..1	PU	Exciter output minimum limit (Efdmin) (< ExcCZ.efdmax).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 825 shows all association ends of ExcCZ with other classes.

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Table 825 – Association ends of ExcitationSystemDynamics::ExcCZ with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.38 ExcDC1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE DC1A direct current commutator exciter with speed input and without underexcitation limiters (UEL) inputs.

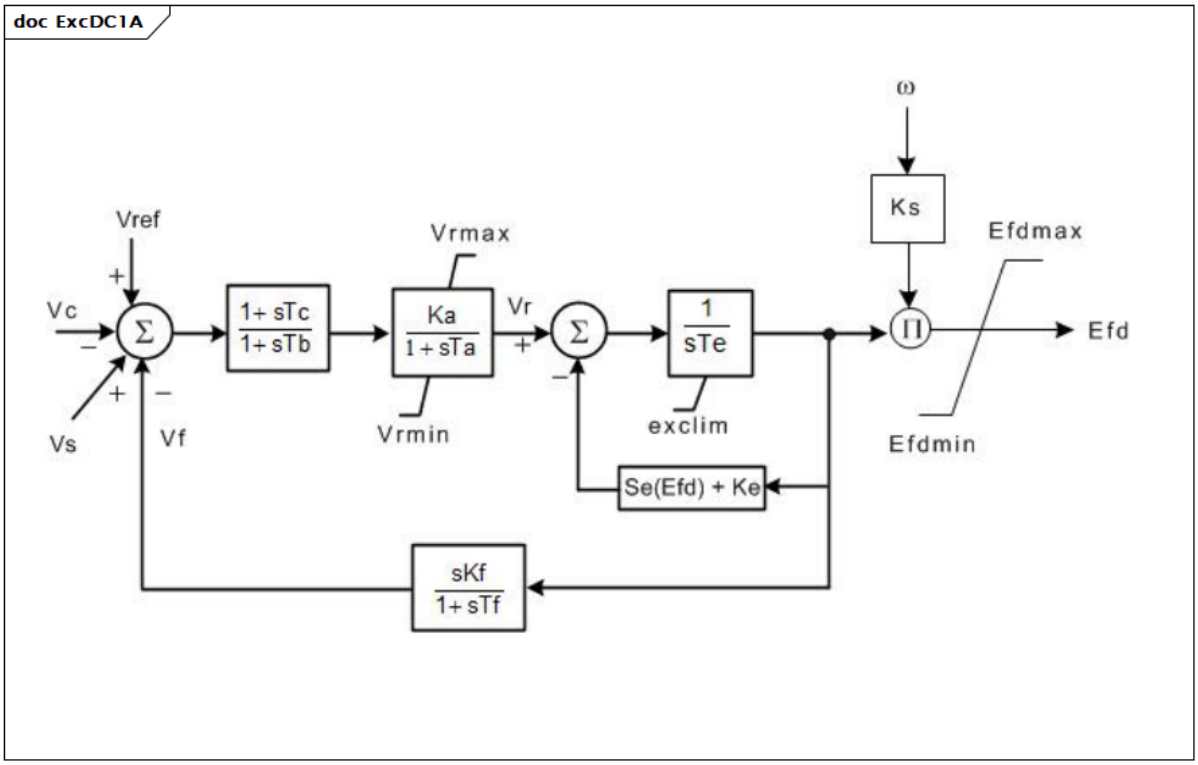

Figure 157 – ExcDC1A

Figure 157: This diagram describes the ExcDC1A excitation system model.

Parameter details:

1) If Ks is set to 1, the output (Efd) is multiplied by generator speed.

Table 826 shows all attributes of ExcDC1A.

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Table 826 – Attributes of ExcitationSystemDynamics::ExcDC1A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 46.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,06.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> ExcDC1A.vrmin). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0 and < ExcDC1A.vrmax). Typical value = -0,9.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 0,46.
kf	1..1	PU	Excitation control system stabilizer gain (Kf) (>= 0). Typical value = 0,1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (> 0). Typical value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1) (> 0). Typical value = 3,1.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Eefd1]) (>= 0). Typical value = 0,33.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2) (> 0). Typical value = 2,3.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Eefd2]) (>= 0). Typical value = 0,1.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical value = true.
efdmin	1..1	PU	Minimum voltage exciter output limiter (Efdmin) (< ExcDC1A.edfmax). Typical value = -99.
efdmax	1..1	PU	Maximum voltage exciter output limiter (Efdmax) (> ExcDC1A.efdmin). Typical value = 99.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 827 shows all association ends of ExcDC1A with other classes.

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Table 827 – Association ends of ExcitationSystemDynamics:: ExcDC1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.39 ExcDC2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE DC2A direct current commutator exciter with speed input, one more leg block in feedback loop and without underexcitation limiters (UEL) inputs. DC type 2 excitation system model with added speed multiplier, added lead-lag, and voltage-dependent limits.

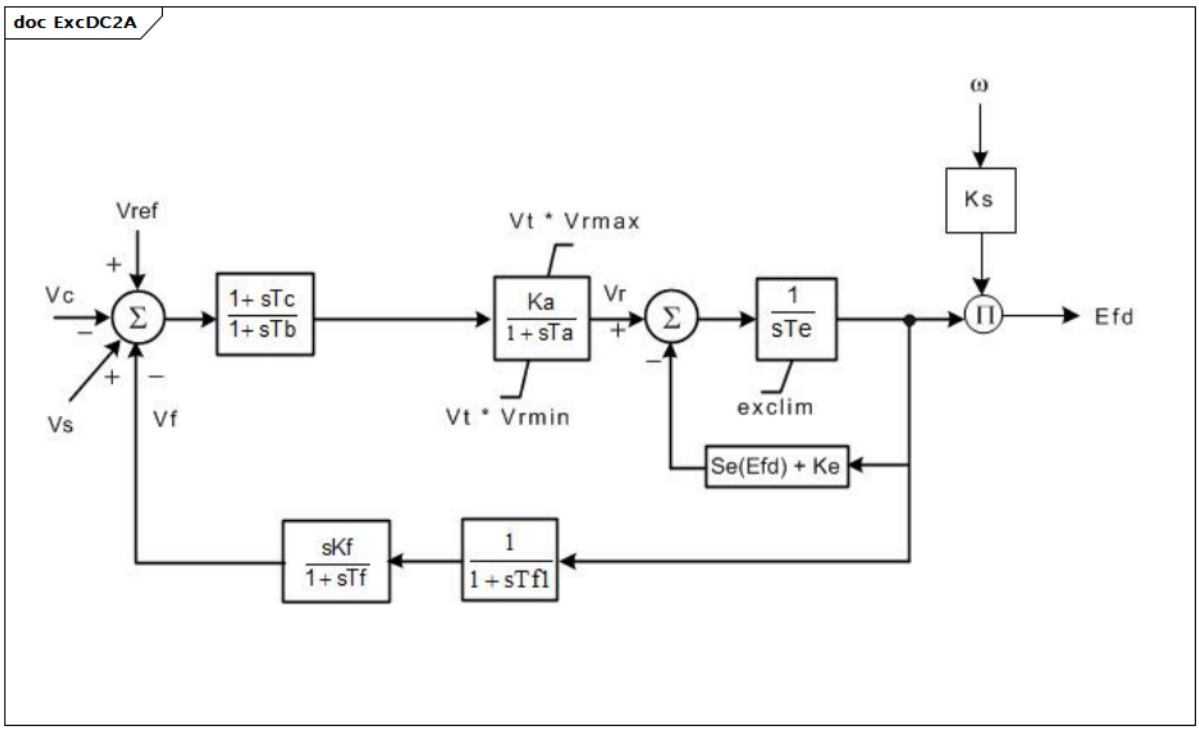

Figure 158 – ExcDC2A

Figure 158: This diagram describes the ExcDC2A excitation system model.

Parameter details:

1) If Ks is set to 1, the output (Efd) is multiplied by generator speed.

Table 828 shows all attributes of ExcDC2A.

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Table 828 – Attributes of ExcitationSystemDynamics::ExcDC2A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 300.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,01.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> ExcDC2A.vrmin). Typical value = 4,95.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0 and < ExcDC2A.vrmax). Typical value = -4,9.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). If Ke is entered as zero, the model calculates an effective value of Ke such that the initial condition value of Vr is zero. The zero value of Ke is not changed. If Ke is entered as non-zero, its value is used directly, without change. Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1,33.
kf	1..1	PU	Excitation control system stabilizer gain (Kf) (>= 0). Typical value = 0,1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (> 0). Typical value = 0,675.
tf1	1..1	Seconds	Excitation control system stabilizer time constant (Tf1) (>= 0). Typical value = 0.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1) (> 0). Typical value = 3,05.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Efd1]) (>= 0). Typical value = 0,279.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2) (> 0). Typical value = 2,29.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Efd2]) (>= 0). Typical value = 0,117.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero is not applied to integrator output. Typical value = true.
vtlim	1..1	Boolean	(Vtlim). true = limiter at the block (Ka / [1 + sTa]) is dependent on Vt false = limiter at the block is not dependent on Vt. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 829 shows all association ends of ExcDC2A with other classes.

Table 829 – Association ends of ExcitationSystemDynamics:: ExcDC2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.40 ExcDC3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE DC3A direct current commutator exciter with speed input, and deadband. DC old type 4.

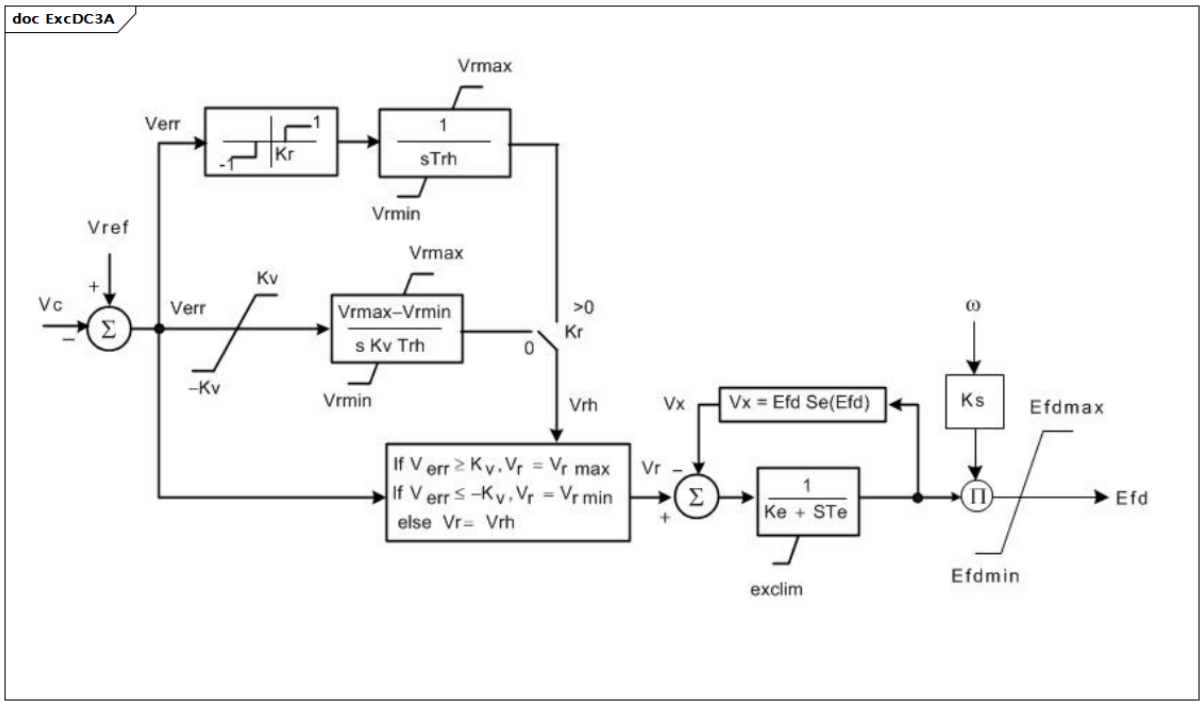

Figure 159 – ExcDC3A

Figure 159: This diagram describes the ExcDC3A excitation system model.

Parameter details:

1) If Ks is set to 1, the output (Efd) is multiplied by generator speed.
2) If Kr is not zero, the voltage regulator input changes at a constant rate if Verr > Kr or Verr < -Kr as per the IEEE (1968) type 4 model. If Kr is zero, the error signal drives the voltage regulator continuously as per the IEEE (1980) DC3 and IEEE (1992, 2005) DC3A models.

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Table 830 shows all attributes of ExcDC3A.

Table 830 – Attributes of ExcitationSystemDynamics::ExcDC3A

name	mult	type	description
trh	1..1	Seconds	Rheostat travel time (Trh) (> 0). Typical value = 20.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
kr	1..1	PU	Deadband (Kr). Typical value = 0.
kv	1..1	PU	Fast raise/lower contact setting (Kv) (> 0). Typical value = 0,05.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (<= 0). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1,83.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
efd1	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd1) (> 0). Typical value = 2,6.
seefd1	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd1 (Se[Efd1]) (>= 0). Typical value = 0,1.
efd2	1..1	PU	Exciter voltage at which exciter saturation is defined (Efd2) (> 0). Typical value = 3,45.
seefd2	1..1	Float	Exciter saturation function value at the corresponding exciter voltage, Efd2 (Se[Efd2]) (>= 0). Typical value = 0,35.
exclim	1..1	Boolean	(exclim). IEEE standard is ambiguous about lower limit on exciter output. true = a lower limit of zero is applied to integrator output false = a lower limit of zero not applied to integrator output. Typical value = true.
efdmax	1..1	PU	Maximum voltage exciter output limiter (Efdmax) (> ExcDC3A.efdmin). Typical value = 99.
efdmin	1..1	PU	Minimum voltage exciter output limiter (Efdmin) (< ExcDC3A.efdmax). Typical value = -99.
efdlim	1..1	Boolean	(Efdlim). true = exciter output limiter is active false = exciter output limiter not active. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 831 shows all association ends of ExcDC3A with other classes.

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Table 831 – Association ends of ExcitationSystemDynamics:: ExcDC3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.41 ExcDC3A1

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified old IEEE type 3 excitation system.

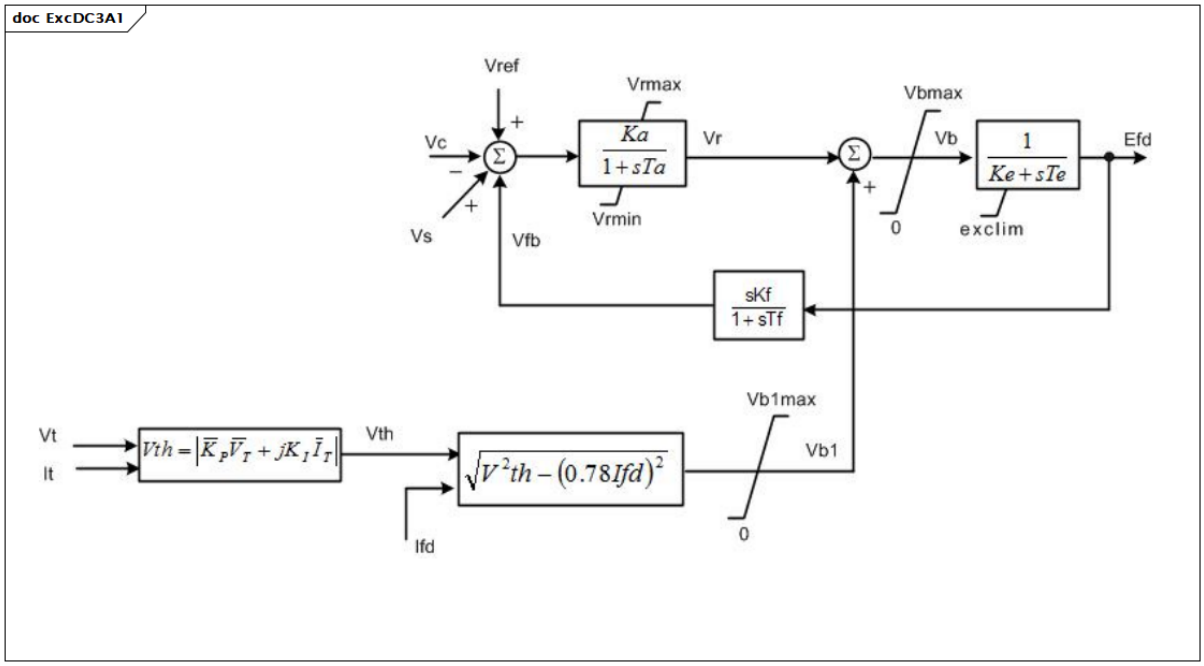

Figure 160 – ExcDC3A1

Figure 160: This diagram describes the ExcDC3A1 excitation system model.

Parameter details:

1) Ifd represents LadIfd (XadIfd) and is a voltage proportional to field current in PU (IEEE 421.5:2005, Annex B). It is equal to Efd in steady-state.

Table 832 shows all attributes of ExcDC3A1.

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Table 832 – Attributes of ExcitationSystemDynamics::ExcDC3A1

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 300.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,01.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> ExcDC3A1.vrmin). Typical value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0 and < ExcDC3A1.vrmax). Typical value = 0.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 1,83.
kf	1..1	PU	Excitation control system stabilizer gain (Kf) (>= 0). Typical value = 0,1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (>= 0). Typical value = 0,675.
kp	1..1	PU	Potential circuit gain coefficient (Kp) (>= 0). Typical value = 4,37.
ki	1..1	PU	Potential circuit gain coefficient (Ki) (>= 0). Typical value = 4,83.
vbmax	1..1	PU	Available exciter voltage limiter (Vbmax) (> 0). Typical value = 11,63.
exclim	1..1	Boolean	(exclim). true = lower limit of zero is applied to integrator output false = lower limit of zero not applied to integrator output. Typical value = true.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
vb1max	1..1	PU	Available exciter voltage limiter (Vb1max) (> 0). Typical value = 11,63.
vblim	1..1	Boolean	Vb limiter indicator. true = exciter Vbmax limiter is active false = Vb1max is active. Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 833 shows all association ends of ExcDC3A1 with other classes.

Table 833 – Association ends of ExcitationSystemDynamics:: ExcDC3A1 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

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14.3.3.9.42 ExcELIN1

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Static PI transformer fed excitation system ELIN (VATECH) – simplified model. This model represents an all-static excitation system. A PI voltage controller establishes a desired field current set point for a proportional current controller. The integrator of the PI controller has a follow-up input to match its signal to the present field current. A power system stabilizer with power input is included in the model.

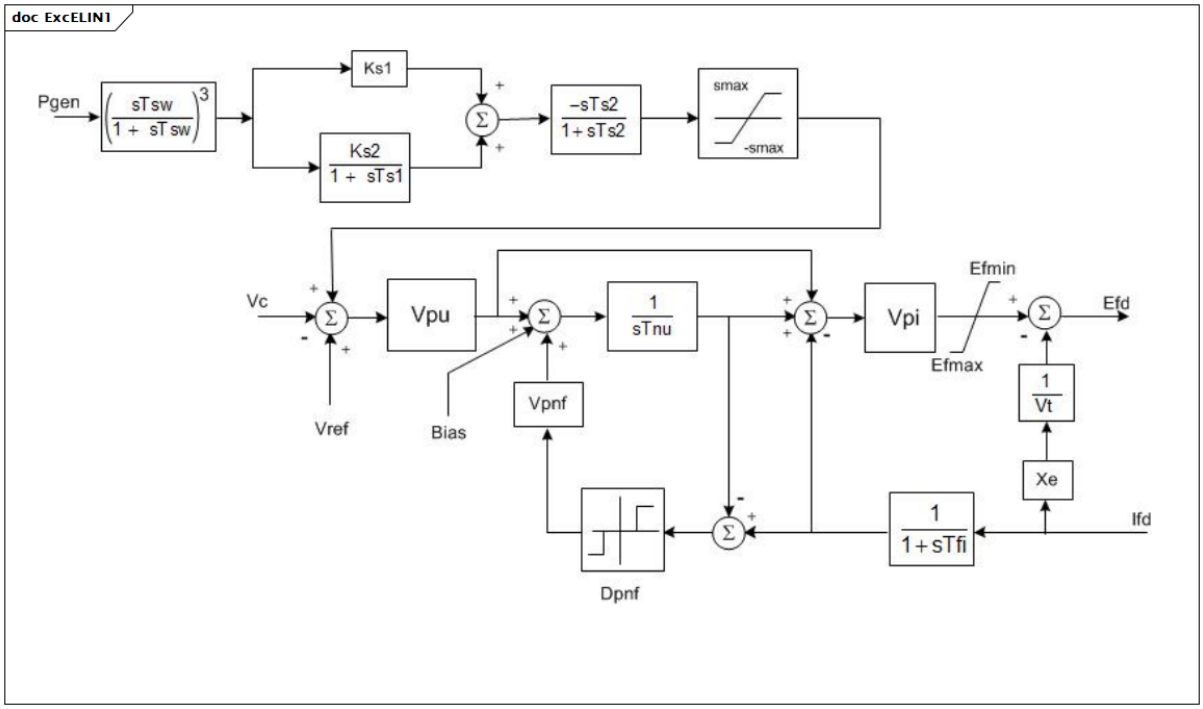

Figure 161 – ExcELIN1

Figure 161: This diagram describes the ExcELIN1 excitation system model.

Table 834 shows all attributes of ExcELIN1.

Table 834 – Attributes of ExcitationSystemDynamics::ExcELIN1

name	mult	type	description
tfi	1..1	Seconds	Current transducer time constant (Tfi) (>= 0). Typical value = 0.
tnu	1..1	Seconds	Controller reset time constant (Tnu) (>= 0). Typical value = 2.
vpu	1..1	PU	Voltage controller proportional gain (Vpu). Typical value = 34,5.
vpi	1..1	PU	Current controller gain (Vpi). Typical value = 12,45.
vpnf	1..1	PU	Controller follow up gain (Vpnf). Typical value = 2.
dpnf	1..1	PU	Controller follow up deadband (Dpnf). Typical value = 0.
tsw	1..1	Seconds	Stabilizer parameters (Tsw) (>= 0). Typical value = 3.

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name	mult	type	description
efmin	1..1	PU	Minimum open circuit excitation voltage (Efmin) (< ExcELIN1.efmax). Typical value = -5.
efmax	1..1	PU	Maximum open circuit excitation voltage (Efmax) (> ExcELIN1.efmin). Typical value = 5.
xe	1..1	PU	Excitation transformer effective reactance (Xe) (>= 0). Xe represents the regulation of the transformer/rectifier unit. Typical value = 0,06.
ks1	1..1	PU	Stabilizer gain 1 (Ks1). Typical value = 0.
ks2	1..1	PU	Stabilizer gain 2 (Ks2). Typical value = 0.
ts1	1..1	Seconds	Stabilizer phase lag time constant (Ts1) (>= 0). Typical value = 1.
ts2	1..1	Seconds	Stabilizer filter time constant (Ts2) (>= 0). Typical value = 1.
smax	1..1	PU	Stabilizer limit output (smax). Typical value = 0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 835 shows all association ends of ExcELIN1 with other classes.

Table 835 – Association ends of ExcitationSystemDynamics:: ExcELIN1 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.43 ExcELIN2

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Detailed excitation system ELIN (VATECH). This model represents an all-static excitation system. A PI voltage controller establishes a desired field current set point for a proportional current controller. The integrator of the PI controller has a follow-up input to match its signal to the present field current. Power system stabilizer models used in conjunction with this excitation system model: PssELIN2, PssIEEE2B, Pss2B.

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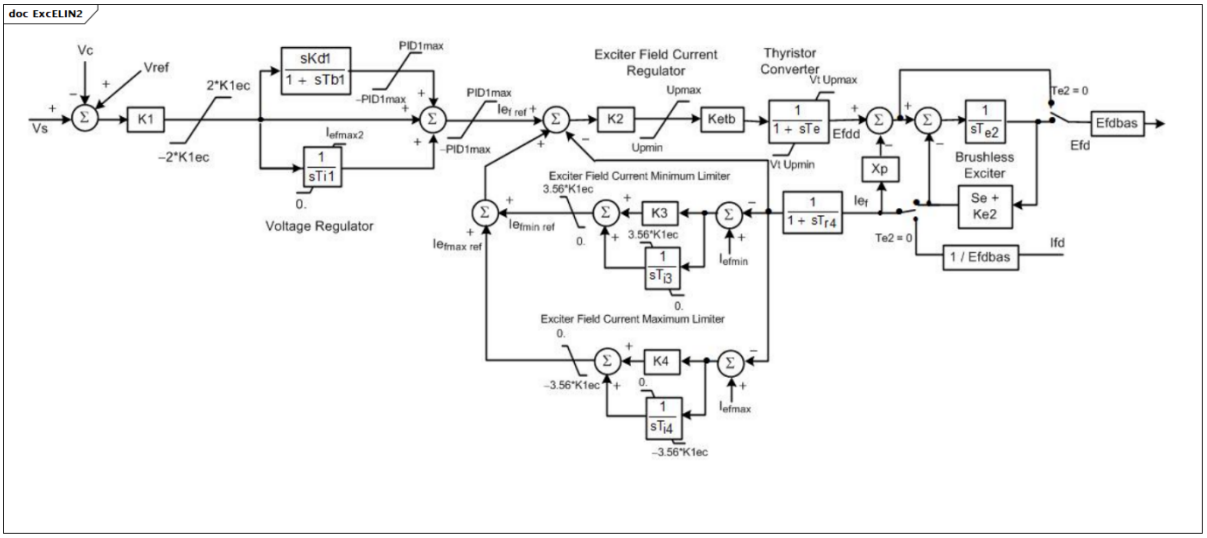

Figure 162 – ExcELIN2

Figure 162: This diagram describes the ExcELIN2 excitation system model.

Table 836 shows all attributes of ExcELIN2.

Table 836 – Attributes of ExcitationSystemDynamics::ExcELIN2

name	mult	type	description
k1	1..1	PU	Voltage regulator input gain (K1). Typical value = 0.
k1ec	1..1	PU	Voltage regulator input limit (K1ec). Typical value = 2.
kd1	1..1	PU	Voltage controller derivative gain (Kd1). Typical value = 34,5.
tb1	1..1	Seconds	Voltage controller derivative washout time constant (Tb1) (>= 0). Typical value = 12,45.
pid1max	1..1	PU	Controller follow up gain (PID1max). Typical value = 2.
ti1	1..1	PU	Controller follow up deadband (Ti1). Typical value = 0.
iefmax2	1..1	PU	Minimum open circuit excitation voltage (Iefmax2). Typical value = -5.
k2	1..1	PU	Gain (K2). Typical value = 5.
ketb	1..1	PU	Gain (Ketb). Typical value = 0,06.
upmax	1..1	PU	Limiter (Upmax) (> ExcELIN2.upmin). Typical value = 3.
upmin	1..1	PU	Limiter (Upmin) (< ExcELIN2.upmax). Typical value = 0.
te	1..1	Seconds	Time constant (Te) (>= 0). Typical value = 0.
xp	1..1	PU	Excitation transformer effective reactance (Xp). Typical value = 1.
te2	1..1	Seconds	Time Constant (Te2) (>= 0). Typical value = 1.
ke2	1..1	PU	Gain (Ke2). Typical value = 0,1.
ve1	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve1) (> 0). Typical value = 3.

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name	mult	type	description
seve1	1..1	PU	Exciter saturation function value at the corresponding exciter voltage, Ve1, back of commutating reactance (Se[Ve1]) (>= 0). Typical value = 0.
ve2	1..1	PU	Exciter alternator output voltages back of commutating reactance at which saturation is defined (Ve2) (> 0). Typical value = 0.
seve2	1..1	PU	Exciter saturation function value at the corresponding exciter voltage, Ve2, back of commutating reactance (Se[Ve2]) (>= 0). Typical value = 1.
tr4	1..1	Seconds	Time constant (Tr4) (>= 0). Typical value = 1.
k3	1..1	PU	Gain (K3). Typical value = 0,1.
ti3	1..1	Seconds	Time constant (Ti3) (>= 0). Typical value = 3.
k4	1..1	PU	Gain (K4). Typical value = 0.
ti4	1..1	Seconds	Time constant (Ti4) (>= 0). Typical value = 0.
iefmax	1..1	PU	Limiter (Iefmax) (> ExcELIN2.iefmin). Typical value = 1.
iefmin	1..1	PU	Limiter (Iefmin) (< ExcELIN2.iefmax). Typical value = 1.
efdbas	1..1	PU	Gain (Efdbas). Typical value = 0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 837 shows all association ends of ExcELIN2 with other classes.

Table 837 – Association ends of ExcitationSystemDynamics:: ExcELIN2 with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.44 ExcHU

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Hungarian excitation system, with built-in voltage transducer.

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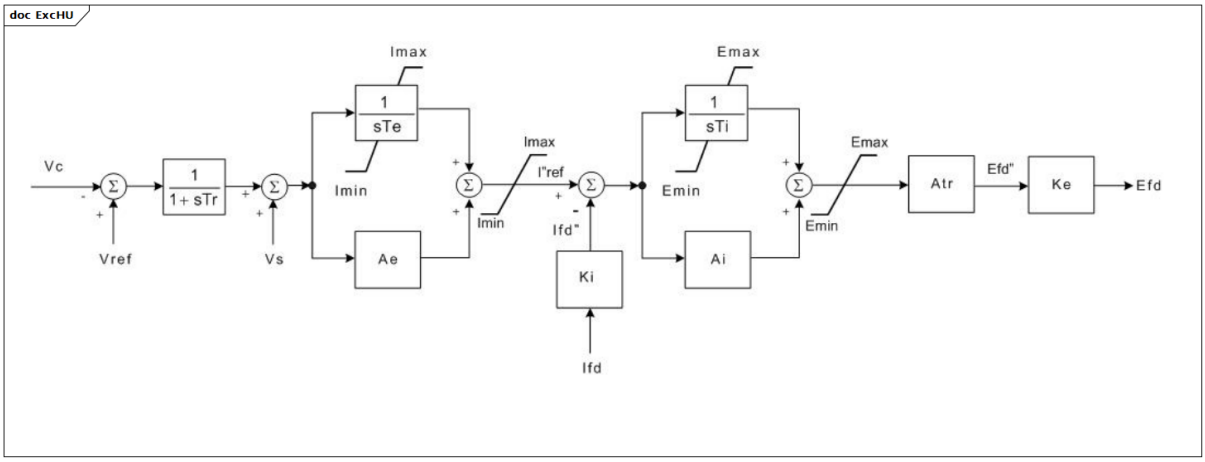

Figure 163 – ExcHU

Figure 163: This diagram describes the ExcHU excitation system model.

Table 838 shows all attributes of ExcHU.

Table 838 – Attributes of ExcitationSystemDynamics::ExcHU

name	mult	type	description
tr	1..1	Seconds	Filter time constant (Tr) (>= 0). If a voltage compensator is used in conjunction with this excitation system model, Tr should be set to 0. Typical value = 0,01.
te	1..1	Seconds	Major loop PI tag integration time constant (Te) (>= 0). Typical value = 0,154.
imin	1..1	PU	Major loop PI tag output signal lower limit (Imin) (< ExcHU.imax). Typical value = 0,1.
imax	1..1	PU	Major loop PI tag output signal upper limit (Imax) (> ExcHU.imin). Typical value = 2,19.
ae	1..1	PU	Major loop PI tag gain factor (Ae). Typical value = 3.
emin	1..1	PU	Field voltage control signal lower limit on AVR base (Emin) (< ExcHU.emax). Typical value = -0,866.
emax	1..1	PU	Field voltage control signal upper limit on AVR base (Emax) (> ExcHU.emin). Typical value = 0,996.
ki	1..1	Float	Current base conversion constant (Ki). Typical value = 0,21428.
ai	1..1	PU	Minor loop PI tag gain factor (Ai). Typical value = 22.
ti	1..1	Seconds	Minor loop PI control tag integration time constant (Ti) (>= 0). Typical value = 0,01333.
atr	1..1	PU	AVR constant (Atr). Typical value = 2,19.
ke	1..1	Float	Voltage base conversion constant (Ke). Typical value = 4,666.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 839 shows all association ends of ExcHU with other classes.

Table 839 – Association ends of ExcitationSystemDynamics::ExcHU with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.45 ExcNI

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Bus or solid fed SCR (silicon-controlled rectifier) bridge excitation system model type NI (NVE).

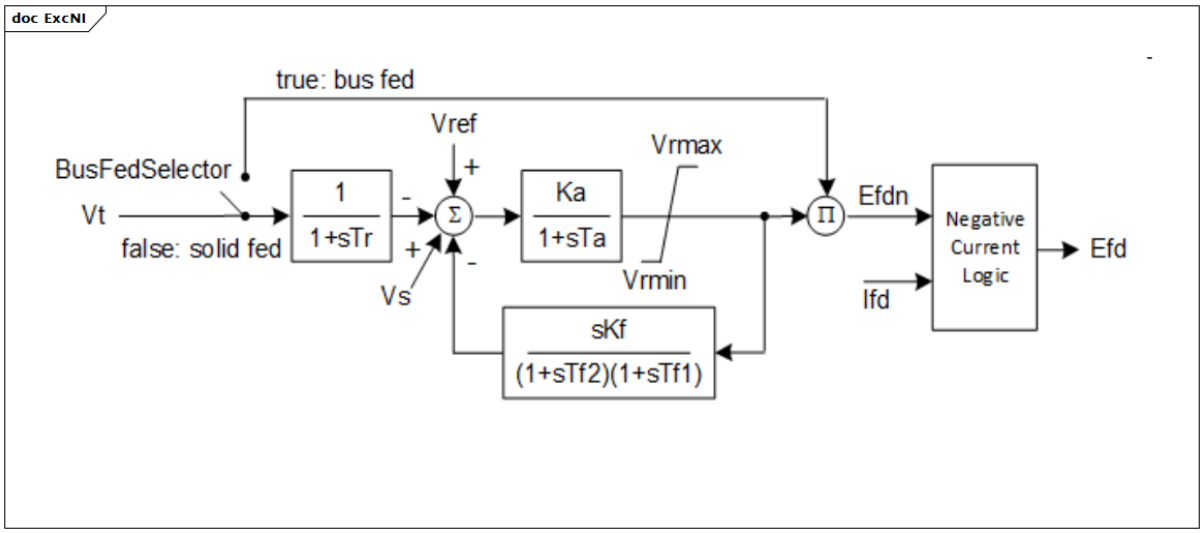

Figure 164 – ExcNI

Figure 164: This diagram describes the ExcNI excitation system model.

Parameter details:

1) R shall either be = 0 (meaning the exciter has negative current capability) or > 0 (meaning the exciter has no negative current capability).
2) The 'Negative Current Logic' block logic uses the R parameter and the Efdn and Ifd inputs. The output Efd is set equal to the input Efdn unless R x Ifd > 0 and R x Ifd > Efdn, in which case Efd is set equal to R x Ifd. This means that the negative current logic is active (R and Ifd are used to calculate Efd) only when the current is positive and the exciter does not have current capability (i.e. when ExcNi.r is greater than 0).

Table 840 shows all attributes of ExcNI.

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Table 840 – Attributes of ExcitationSystemDynamics::ExcNI

name	mult	type	description
busFedSelector	1..1	Boolean	Fed by selector (BusFedSelector). true = bus fed (switch is closed) false = solid fed (switch is open). Typical value = true.
tr	1..1	Seconds	Time constant (Tr) (>= 0). Typical value = 0,02.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 210.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator ouput (Vrmax) (> ExcNI.vrmin). Typical value = 5,0.
vrmin	1..1	PU	Minimum voltage regulator ouput (Vrmin) (< ExcNI.vrmax). Typical value = -2,0.
kf	1..1	PU	Excitation control system stabilizer gain (Kf) (> 0). Typical value 0,01.
tf2	1..1	Seconds	Excitation control system stabilizer time constant (Tf2) (> 0). Typical value = 0,1.
tf1	1..1	Seconds	Excitation control system stabilizer time constant (Tf1) (> 0). Typical value = 1,0.
r	1..1	PU	rc / rfd (R) (>= 0). 0 means exciter has negative current capability > 0 means exciter does not have negative current capability. Typical value = 5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 841 shows all association ends of ExcNI with other classes.

Table 841 – Association ends of ExcitationSystemDynamics::ExcNI with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.46 ExcOEX3T

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE type ST1 excitation system with semi-continuous and acting terminal voltage limiter.

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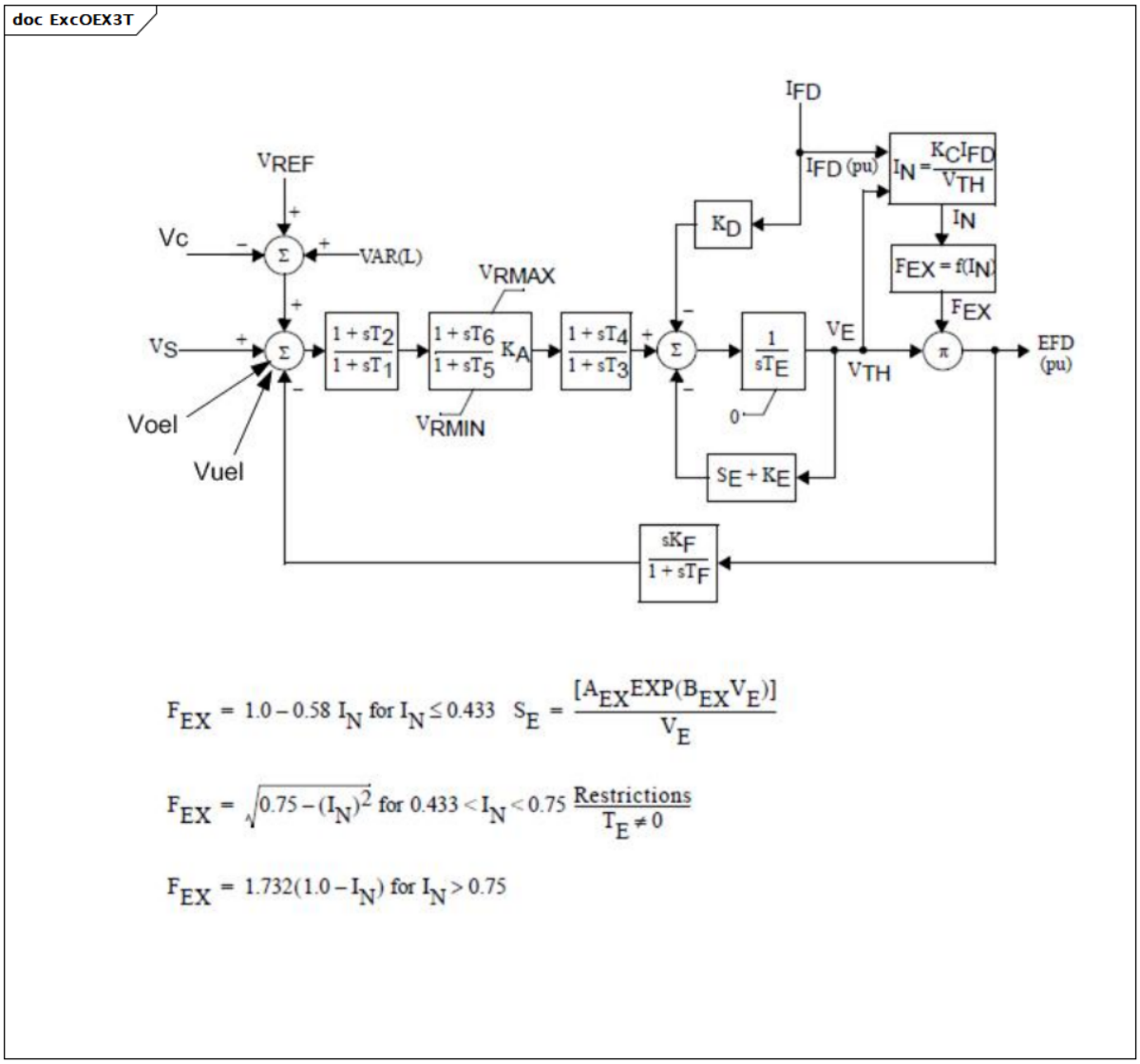

Figure 165 – ExcOEX3T

Figure 165: This diagram describes the ExcOEX3T excitation system model.

Table 842 shows all attributes of ExcOEX3T.

Table 842 – Attributes of ExcitationSystemDynamics::ExcOEX3T

name	mult	type	description
t1	1..1	Seconds	Time constant (T1) (>= 0).
t2	1..1	Seconds	Time constant (T2) (>= 0).
t3	1..1	Seconds	Time constant (T3) (>= 0).
t4	1..1	Seconds	Time constant (T4) (>= 0).
ka	1..1	PU	Gain (KA).
t5	1..1	Seconds	Time constant (T5) (>= 0).
t6	1..1	Seconds	Time constant (T6) (>= 0).
vrmax	1..1	PU	Limiter (VRMAX) (> ExcOEX3T.vrmin).
vrmin	1..1	PU	Limiter (VRMIN) (< ExcOEX3T.vrmax).
te	1..1	Seconds	Time constant (TE) (>= 0).

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name	mult	type	description
kf	1..1	PU	Gain (KF).
tf	1..1	Seconds	Time constant (TF) (>= 0).
kc	1..1	PU	Gain (KC).
kd	1..1	PU	Gain (KD).
ke	1..1	PU	Gain (KE).
e1	1..1	PU	Saturation parameter (E1).
see1	1..1	PU	Saturation parameter (SE[E1]).
e2	1..1	PU	Saturation parameter (E2).
see2	1..1	PU	Saturation parameter (SE[E2]).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 843 shows all association ends of ExcOEX3T with other classes.

Table 843 – Association ends of ExcitationSystemDynamics:: ExcOEX3T with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.47 ExcPIC

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Proportional/integral regulator excitation system. This model can be used to represent excitation systems with a proportional-integral (PI) voltage regulator controller.

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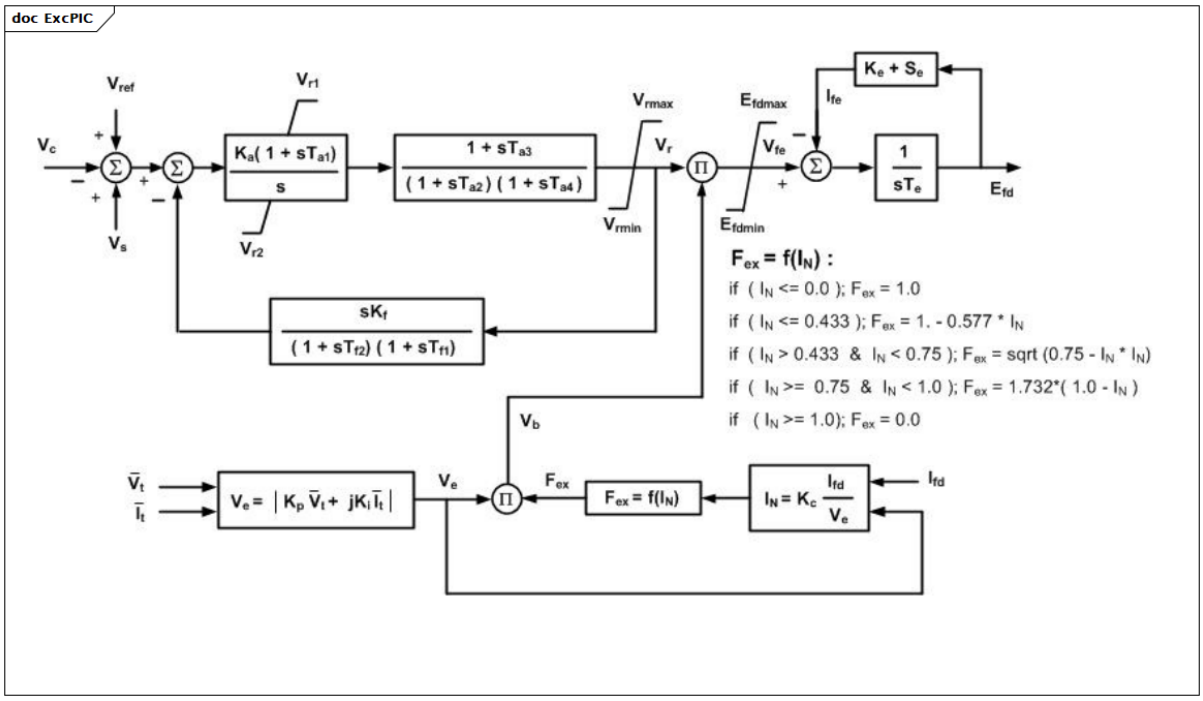

Figure 166 – ExcPIC

Figure 166: This diagram describes the ExcPIC excitation system model.

Table 844 shows all attributes of ExcPIC.

Table 844 – Attributes of ExcitationSystemDynamics::ExcPIC

name	mult	type	description
ka	1..1	PU	PI controller gain (Ka). Typical value = 3,15.
ta1	1..1	Seconds	PI controller time constant (Ta1) (>= 0). Typical value = 1.
vr1	1..1	PU	PI maximum limit (Vr1). Typical value = 1.
vr2	1..1	PU	PI minimum limit (Vr2). Typical value = -0,87.
ta2	1..1	Seconds	Voltage regulator time constant (Ta2) (>= 0). Typical value = 0,01.
ta3	1..1	Seconds	Lead time constant (Ta3) (>= 0). Typical value = 0.
ta4	1..1	Seconds	Lag time constant (Ta4) (>= 0). Typical value = 0.
vrmax	1..1	PU	Voltage regulator maximum limit (Vrmax) (> ExcPIC.vrmin). Typical value = 1.
vrmin	1..1	PU	Voltage regulator minimum limit (Vrmin) (< ExcPIC.vrmax). Typical value = -0,87.
kf	1..1	PU	Rate feedback gain (Kf). Typical value = 0.
tf1	1..1	Seconds	Rate feedback time constant (Tf1) (>= 0). Typical value = 0.
tf2	1..1	Seconds	Rate feedback lag time constant (Tf2) (>= 0). Typical value = 0.
efdmax	1..1	PU	Exciter maximum limit (Efdmax) (> ExcPIC.efdmin). Typical value = 8.
efdmin	1..1	PU	Exciter minimum limit (Efdmin) (< ExcPIC.efdmax). Typical value = -0,87.

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name	mult	type	description
ke	1..1	PU	Exciter constant (Ke). Typical value = 0.
te	1..1	Seconds	Exciter time constant (Te) (>= 0). Typical value = 0.
e1	1..1	PU	Field voltage value 1 (E1). Typical value = 0.
se1	1..1	PU	Saturation factor at E1 (Se1). Typical value = 0.
e2	1..1	PU	Field voltage value 2 (E2). Typical value = 0.
se2	1..1	PU	Saturation factor at E2 (Se2). Typical value = 0.
kp	1..1	PU	Potential source gain (Kp). Typical value = 6,5.
ki	1..1	PU	Current source gain (Ki). Typical value = 0.
kc	1..1	PU	Exciter regulation factor (Kc). Typical value = 0,08.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 845 shows all association ends of ExcPIC with other classes.

Table 845 – Association ends of ExcitationSystemDynamics::ExcPIC with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.48 ExcREXS

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

General purpose rotating excitation system. This model can be used to represent a wide range of excitation systems whose DC power source is an AC or DC generator. It encompasses IEEE type AC1, AC2, DC1, and DC2 excitation system models.

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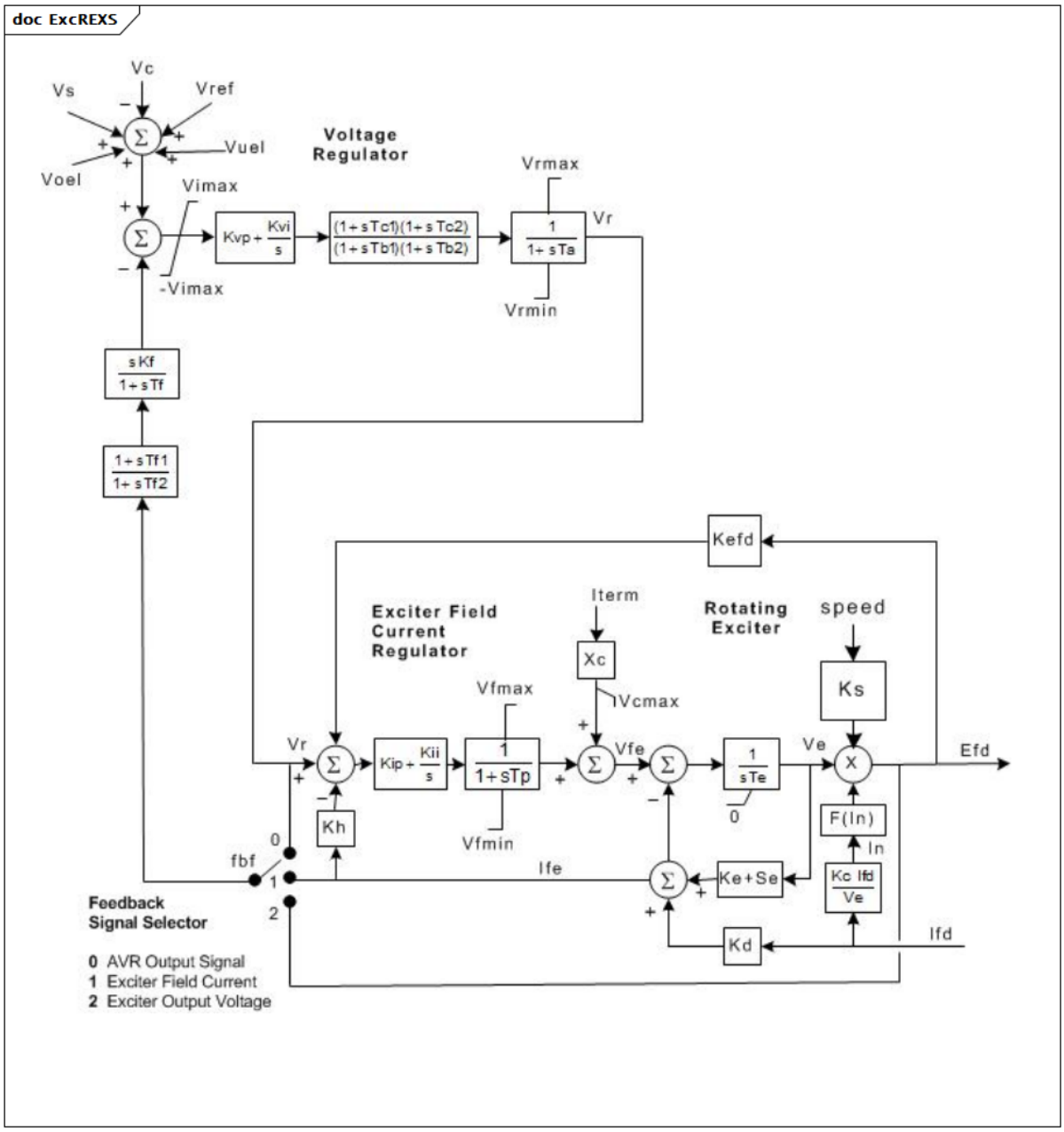

Figure 167 – ExcREXS

Figure 167: This diagram describes the ExcREXS excitation system model.

Parameter details:

1) The rotating exciter can be of either DC or brushless AC type. To model a DC exciter, set Kc = Kd = 0.
2) An AC exciter can be modelled either by setting Kc and Kd to non- zero values to represent the armature reaction of the exciter and the regulation of the rotating rectifier, respectively, or by specifying the saturation factors, Se1 and Se2, to represent the excitation-output-voltage curve of the exciter as loaded by the field resistance of the main generator. It is preferable to use the first approach (Kd = 0) since the later approach does not represent the effect of armature reaction in the exciter on its field current and hence invalidates the model when field current is sensed by the voltage regulator.

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3) When this model is used to represent a separately-excited exciter, the value of Ke should normally be close to unity. A shunt-excited DC exciter can be modelled by setting Ke to the appropriate small negative value needed to represent the relationship between the exciter's air gap line and field resistance line. Note, though, that this model does not determine the value of Ke automatically when a shunt field DC exciter is being represented
4) The voltage regulator can be of proportional or proportional-plus-integral form. Either Kvp or Kvi, but not both, can be zero.
5) The voltage regulator output can be either the field voltage applied directly to the exciter or the input to an exciter field current regulator by choice of values of Kip, Kii, and Kh. Either or both of Kip and Kii shall be non-zero. If Kii is non-zero, Kh should normally be non-zero.
6) The voltage regulator reference is modified by a volts-per-hertz limiter when shaft speed falls below Nvphz. When the shaft speed is below Nvphz the regulator reference is reduced as follows: effective Vref = Vref (1,0 – Kvphz (speed – Nvphz)) The volts-per-hertz limiter can be disabled by setting Nvphz = 0 or Kvphz = 0.
7) Saturation parameters are given by the IEEE saturation factor definition using the open circuit magnetization of the exciter. Either point [E1, S(E1) or E2, S(E2)] can be the higher value and the other the lower.
8) If Ks is set to 1, the output (Efd) is multiplied by generator speed.
9) The voltage regulator output limits, Vrmax and Vrmin, shall be stated as multiples of the value of exciter field current needed to maintain the exciter output at its base value. The field current limit, Vlr, shall be stated as a multiple of the exciter field current needed to maintain the exciter output at its base value. That is, Vrmax and Vrmin shall be stated in terms of the PU exciter output voltage to which they correspond in the steady state.
10) The regulator output limit flag indicates whether the control power supply for the voltage regulator is a transformer at the generator terminals or an independent source such as a permanent magnet generator. Set Flimf as follows:
    – Flimf = 0 the limits on regulator output are Vrmax, Vrmin, Vfmax and Vfmin;
    – Flimf = 1 the limits on regulator output are (Vrmax x Vterm), (Vrmin x Vterm), (Vfmax x Vterm), and (Vfmin x Vterm).
12) Feedback signal selector uses ExcREXSFeedbackSignalKind, where:
    – 0 (AVR output signal) = ExcREXSFeedbackSignalKind.fieldVoltage;
    – 1 (exciter field current) = ExcREXSFeedbackSignalKind.fieldCurent;
    – 2 (exciter output voltage) = ExcREXSFeedbackSignalKind.outputVoltage.

Table 846 shows all attributes of ExcREXS.

Table 846 – Attributes of ExcitationSystemDynamics::ExcREXS

name	mult	type	description
e1	1..1	PU	Field voltage value 1 (E1). Typical value = 3.
e2	1..1	PU	Field voltage value 2 (E2). Typical value = 4.
fbf	1..1	ExcREXSFeedbackSignalKind	Rate feedback signal flag (fbf). Typical value = fieldCurrent.
flimf	1..1	PU	Limit type flag (Flimf). Typical value = 0.
kc	1..1	PU	Rectifier regulation factor (Kc). Typical value = 0,05.
kd	1..1	PU	Exciter regulation factor (Kd). Typical value = 2.
ke	1..1	PU	Exciter field proportional constant (Ke). Typical value = 1.
kefd	1..1	PU	Field voltage feedback gain (Kefd). Typical value = 0.
kf	1..1	Seconds	Rate feedback gain (Kf) (>= 0). Typical value = 0,05.

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name	mult	type	description
kh	1..1	PU	Field voltage controller feedback gain (Kh). Typical value = 0.
kii	1..1	PU	Field current regulator integral gain (Kii). Typical value = 0.
kip	1..1	PU	Field current regulator proportional gain (Kip). Typical value = 1.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
kvi	1..1	PU	Voltage regulator integral gain (Kvi). Typical value = 0.
kvp	1..1	PU	Voltage regulator proportional gain (Kvp). Typical value = 2800.
kvphz	1..1	PU	V/Hz limiter gain (Kvphz). Typical value = 0.
nvphz	1..1	PU	Pickup speed of V/Hz limiter (Nvphz). Typical value = 0.
se1	1..1	PU	Saturation factor at E1 (Se1). Typical value = 0,0001.
se2	1..1	PU	Saturation factor at E2 (Se2). Typical value = 0,001.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (>= 0). If = 0, block is bypassed. Typical value = 0,01.
tb1	1..1	Seconds	Lag time constant (Tb1) (>= 0). If = 0, block is bypassed. Typical value = 0.
tb2	1..1	Seconds	Lag time constant (Tb2) (>= 0). If = 0, block is bypassed. Typical value = 0.
tc1	1..1	Seconds	Lead time constant (Tc1) (>= 0). Typical value = 0.
tc2	1..1	Seconds	Lead time constant (Tc2) (>= 0). Typical value = 0.
te	1..1	Seconds	Exciter field time constant (Te) (> 0). Typical value = 1,2.
tf	1..1	Seconds	Rate feedback time constant (Tf) (>= 0). If = 0, the feedback path is not used. Typical value = 1.
tf1	1..1	Seconds	Feedback lead time constant (Tf1) (>= 0). Typical value = 0.
tf2	1..1	Seconds	Feedback lag time constant (Tf2) (>= 0). If = 0, block is bypassed. Typical value = 0.
tp	1..1	Seconds	Field current bridge time constant (Tp) (>= 0). Typical value = 0.
vcmax	1..1	PU	Maximum compounding voltage (Vcmax). Typical value = 0.
vfmax	1..1	PU	Maximum exciter field current (Vfmax) (> ExcREXS.vfmin). Typical value = 47.
vfmin	1..1	PU	Minimum exciter field current (Vfmin) (< ExcREXS.vfmax). Typical value = -20.
vimax	1..1	PU	Voltage regulator input limit (Vimax). Typical value = 0,1.
vrmax	1..1	PU	Maximum controller output (Vrmax) (> ExcREXS.vrmin). Typical value = 47.
vrmin	1..1	PU	Minimum controller output (Vrmin) (< ExcREXS.vrmax). Typical value = -20.
xc	1..1	PU	Exciter compounding reactance (Xc). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 847 shows all association ends of ExcREXS with other classes.

Table 847 – Association ends of ExcitationSystemDynamics:: ExcREXS with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.49 ExcRQB

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Excitation system type RQB (four-loop regulator, r?gulateur quatre boucles, developed in France) primarily used in nuclear or thermal generating units. This excitation system shall be always used together with power system stabilizer type PssRQB.

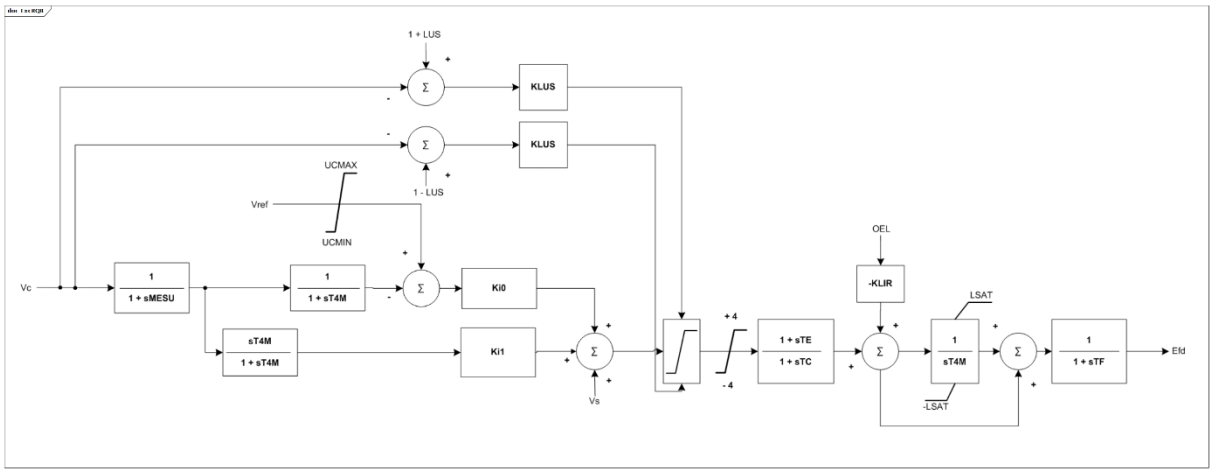

Figure 168 – ExcRQB

Figure 168: This diagram describes the ExcRQB excitation system model.

Table 848 shows all attributes of ExcRQB.

Table 848 – Attributes of ExcitationSystemDynamics::ExcRQB

name	mult	type	description
ki0	1..1	Float	Voltage reference input gain (Ki0). Typical value = 12,7.
ki1	1..1	Float	Voltage input gain (Ki1). Typical value = -16,8.
te	1..1	Seconds	Lead lag time constant (TE) (>= 0). Typical value = 0,22.
tc	1..1	Seconds	Lead lag time constant (TC) (>= 0). Typical value = 0,02.

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name	mult	type	description
klir	1..1	Float	OEL input gain (KLIR). Typical value = 12,13.
ucmin	1..1	PU	Minimum voltage reference limit (UCMIN) (< ExcRQB.ucmax). Typical value = 0,9.
ucmax	1..1	PU	Maximum voltage reference limit (UCMAX) (> ExcRQB.ucmin). Typical value = 1,1.
lus	1..1	PU	Setpoint (LUS). Typical value = 0,12.
klus	1..1	Float	Limiter gain (KLUS). Typical value = 50.
mesu	1..1	Seconds	Voltage input time constant (MESU) (>= 0). Typical value = 0,02.
t4m	1..1	Seconds	Input time constant (T4M) (>= 0). Typical value = 5.
lsat	1..1	PU	Integrator limiter (LSAT). Typical value = 5,73.
tf	1..1	Seconds	Exciter time constant (TF) (>= 0). Typical value = 0,01.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 849 shows all association ends of ExcRQB with other classes.

Table 849 – Association ends of ExcitationSystemDynamics:: ExcRQB with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.50 ExcSCRX

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Simple excitation system with generic characteristics typical of many excitation systems; intended for use where negative field current could be a problem.

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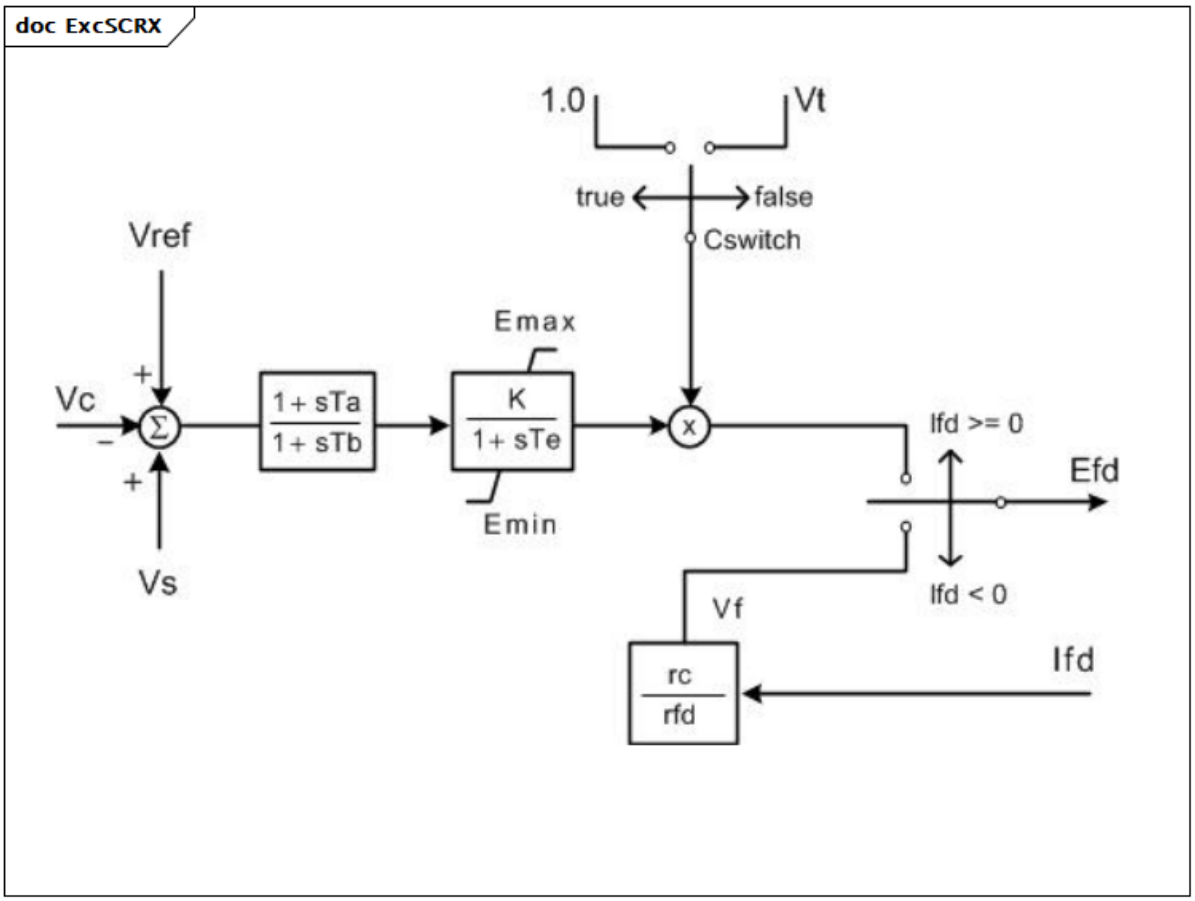

Figure 169 – ExcSCRX

Figure 169: This diagram describes the ExcSCRX excitation system model.

Table 850 shows all attributes of ExcSCRX.

Table 850 – Attributes of ExcitationSystemDynamics::ExcSCRX

name	mult	type	description
tatb	1..1	Float	Gain reduction ratio of lag-lead element ([Ta / Tb]). The parameter Ta is not defined explicitly. Typical value = 0.1.
tb	1..1	Seconds	Denominator time constant of lag-lead block (Tb) (>= 0). Typical value = 10.
k	1..1	PU	Gain (K) (> 0). Typical value = 200.
te	1..1	Seconds	Time constant of gain block (Te) (> 0). Typical value = 0,02.
emin	1..1	PU	Minimum field voltage output (Emin) (< ExcSCRX.emax). Typical value = 0.
emax	1..1	PU	Maximum field voltage output (Emax) (> ExcSCRX.emin). Typical value = 5.
cswitch	1..1	Boolean	Power source switch (Cswitch). true = fixed voltage of 1.0 PU false = generator terminal voltage.
rcrfd	1..1	Float	Ratio of field discharge resistance to field winding resistance ([rc / rfd]). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock

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name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 851 shows all association ends of ExcSCRX with other classes.

Table 851 – Association ends of ExcitationSystemDynamics:: ExcSCRX with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.51 ExcSEXS

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified excitation system.

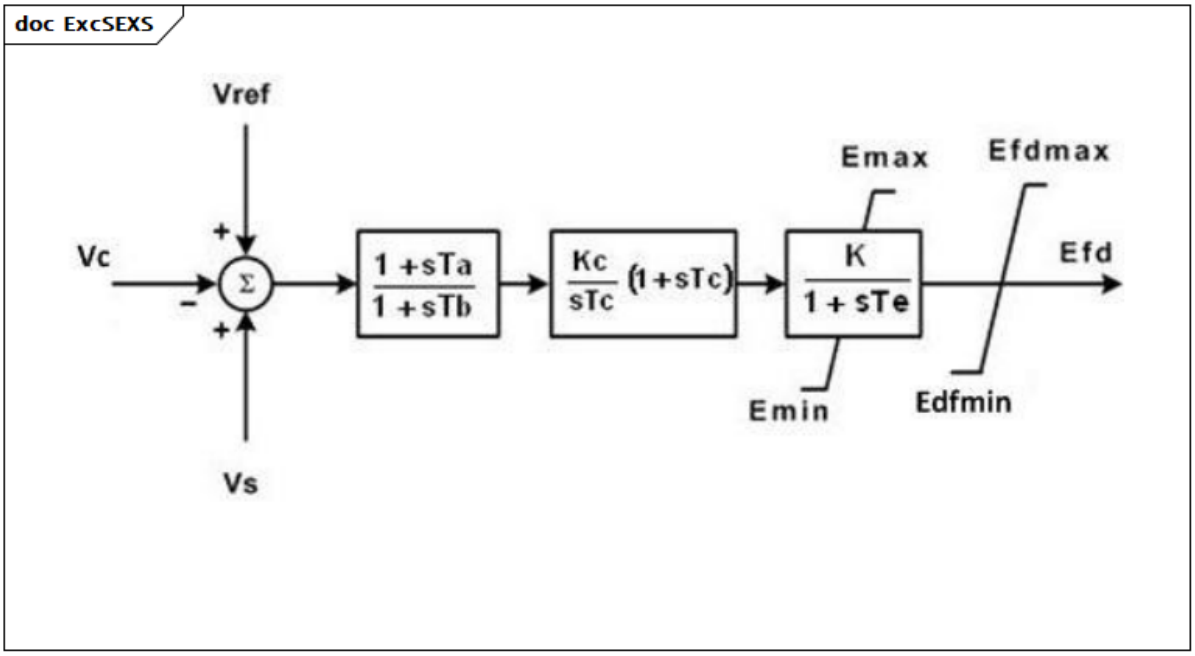

Figure 170 – ExcSEXS

Figure 170: This diagram describes the ExcSEXS excitation system model.

Table 852 shows all attributes of ExcSEXS.

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Table 852 – Attributes of ExcitationSystemDynamics::ExcSEXS

name	mult	type	description
tatb	1..1	Float	Gain reduction ratio of lag-lead element ([Ta / Tb]). Typical value = 0,1.
tb	1..1	Seconds	Denominator time constant of lag-lead block (Tb) (>= 0). Typical value = 10.
k	1..1	PU	Gain (K) (> 0). Typical value = 100.
te	1..1	Seconds	Time constant of gain block (Te) (> 0). Typical value = 0,05.
emin	1..1	PU	Minimum field voltage output (Emin) (< ExcSEXS.emax). Typical value = -5.
emax	1..1	PU	Maximum field voltage output (Emax) (> ExcSEXS.emin). Typical value = 5.
kc	1..1	PU	PI controller gain (Kc) (> 0 if ExcSEXS.tc > 0). Typical value = 0,08.
tc	1..1	Seconds	PI controller phase lead time constant (Tc) (>= 0). Typical value = 0.
efdmin	1..1	PU	Field voltage clipping minimum limit (Efdmin) (< ExcSEXS.efdmax). Typical value = -5.
efdmax	1..1	PU	Field voltage clipping maximum limit (Efdmax) (> ExcSEXS.efdmin). Typical value = 5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 853 shows all association ends of ExcSEXS with other classes.

Table 853 – Association ends of ExcitationSystemDynamics:: ExcSEXS with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.52 ExcSK

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Slovakian excitation system. UEL and secondary voltage control are included in this model. When this model is used, there cannot be a separate underexcitation limiter or VAr controller model.

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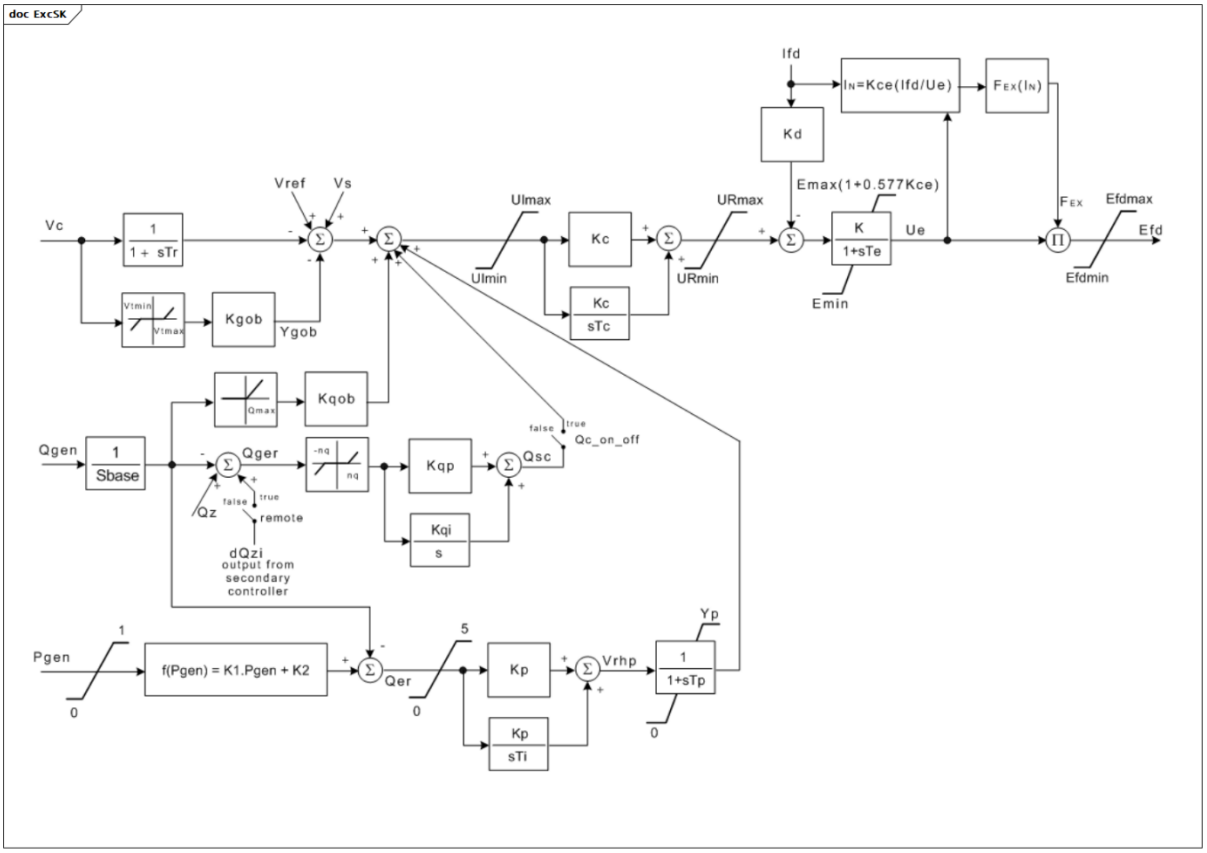

Figure 171 – ExcSK

Figure 171: This diagram describes the ExcSK excitation system model.

Parameter details:

1) Qmax is calculated using this equation: Qmax = K1 x Pgen + K2.
2) Pgen and Qgen are in PU.
3) The part of the model which includes Qmax and Kqob represents back-coupling from the reactive power with statics. It is used for parallel co-operation with other generators to share the reactive power demand. This can be neglected if the configuration is a single generating unit.
4) The block Fex (In) is modelled as described in IEEE 421.5 – part on rectifier regulation equations.

Table 854 shows all attributes of ExcSK.

Table 854 – Attributes of ExcitationSystemDynamics::ExcSK

name	mult	type	description
efdmax	1..1	PU	Field voltage clipping upper level limit (Efdmax) (> ExcSK.efdmin).
efdmin	1..1	PU	Field voltage clipping lower level limit (Efdmin) (< ExcSK.efdmax).
emax	1..1	PU	Maximum field voltage output (Emax) (> ExcSK.emin). Typical value = 20.
emin	1..1	PU	Minimum field voltage output (Emin) (< ExcSK.emax). Typical value = -20.
k	1..1	PU	Gain (K). Typical value = 1.

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name	mult	type	description
k1	1..1	PU	Parameter of underexcitation limit (K1). Typical value = 0,1364.
k2	1..1	PU	Parameter of underexcitation limit (K2). Typical value = -0,3861.
kc	1..1	PU	PI controller gain (Kc). Typical value = 70.
kce	1..1	PU	Rectifier regulation factor (Kce). Typical value = 0.
kd	1..1	PU	Exciter internal reactance (Kd). Typical value = 0.
kgob	1..1	PU	P controller gain (Kgob). Typical value = 10.
kp	1..1	PU	PI controller gain (Kp). Typical value = 1.
kqi	1..1	PU	PI controller gain of integral component (Kqi). Typical value = 0.
kqob	1..1	PU	Rate of rise of the reactive power (Kqob).
kqp	1..1	PU	PI controller gain (Kqp). Typical value = 0.
nq	1..1	PU	Deadband of reactive power (nq). Determines the range of sensitivity. Typical value = 0,001.
qconoff	1..1	Boolean	Secondary voltage control state (Qc_on_off). true = secondary voltage control is on false = secondary voltage control is off. Typical value = false.
qz	1..1	PU	Desired value (setpoint) of reactive power, manual setting (Qz).
remote	1..1	Boolean	Selector to apply automatic calculation in secondary controller model (remote). true = automatic calculation is activated false = manual set is active; the use of desired value of reactive power (Qz) is required. Typical value = true.
sbase	1..1	ApparentPower	Apparent power of the unit (Sbase) (> 0). Unit = MVA. Typical value = 259.
tc	1..1	Seconds	PI controller phase lead time constant (Tc) (>= 0). Typical value = 8.
te	1..1	Seconds	Time constant of gain block (Te) (>= 0). Typical value = 0,1.
ti	1..1	Seconds	PI controller phase lead time constant (Ti) (>= 0). Typical value = 2.
tp	1..1	Seconds	Time constant (Tp) (>= 0). Typical value = 0,1.
tr	1..1	Seconds	Voltage transducer time constant (Tr) (>= 0). Typical value = 0,01.
uimax	1..1	PU	Maximum error (UImax) (> ExcSK.uimin). Typical value = 10.
uimin	1..1	PU	Minimum error (UImin) (< ExcSK.uimax). Typical value = -10.
urmax	1..1	PU	Maximum controller output (URmax) (> ExcSK.urmin). Typical value = 10.
urmin	1..1	PU	Minimum controller output (URmin) (< ExcSK.urmax). Typical value = -10.
vtmax	1..1	PU	Maximum terminal voltage input (Vtmax) (> ExcSK.vtmin). Determines the range of voltage deadband. Typical value = 1,05.

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name	mult	type	description
vtmin	1..1	PU	Minimum terminal voltage input (Vtmin) (< ExcSK.vtmax). Determines the range of voltage deadband. Typical value = 0,95.
yp	1..1	PU	Maximum output (Yp). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 855 shows all association ends of ExcSK with other classes.

Table 855 – Association ends of ExcitationSystemDynamics::ExcSK with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.53 ExcST1A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modification of an old IEEE ST1A static excitation system without overexcitation limiter (OEL) and underexcitation limiter (UEL).

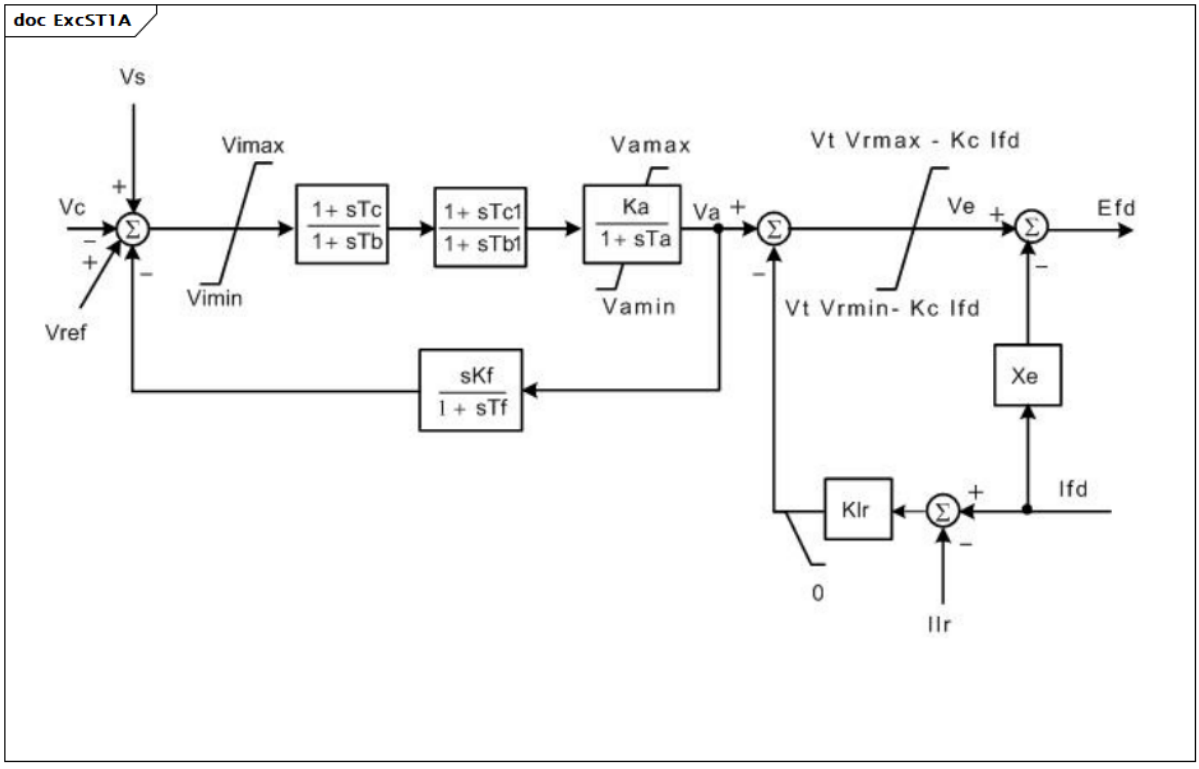

Figure 172 – ExcST1A

Figure 172: This diagram describes the ExcST1A excitation system model.

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Table 856 shows all attributes of ExcST1A.

Table 856 – Attributes of ExcitationSystemDynamics::ExcST1A

name	mult	type	description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax) (> 0). Typical value = 999.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin) (< 0). Typical value = -999.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 1.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 10.
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 190.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (>= 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax) (> 0) . Typical value = 7,8.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin) (< 0). Typical value = -6,7.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,05.
kf	1..1	PU	Excitation control system stabilizer gains (Kf) (>= 0). Typical value = 0.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (>= 0). Typical value = 1.
tc1	1..1	Seconds	Voltage regulator time constant (Tc1) (>= 0). Typical value = 0.
tb1	1..1	Seconds	Voltage regulator time constant (Tb1) (>= 0). Typical value = 0.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 999.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -999.
ilr	1..1	PU	Exciter output current limit reference (Ilr). Typical value = 0.
klr	1..1	PU	Exciter output current limiter gain (Klr). Typical value = 0.
xe	1..1	PU	Excitation xfmr effective reactance (Xe). Typical value = 0,04.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 857 shows all association ends of ExcST1A with other classes.

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Table 857 – Association ends of ExcitationSystemDynamics:: ExcST1A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.54 ExcST2A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE ST2A static excitation system with another lead-lag block added to match the model defined by WECC.

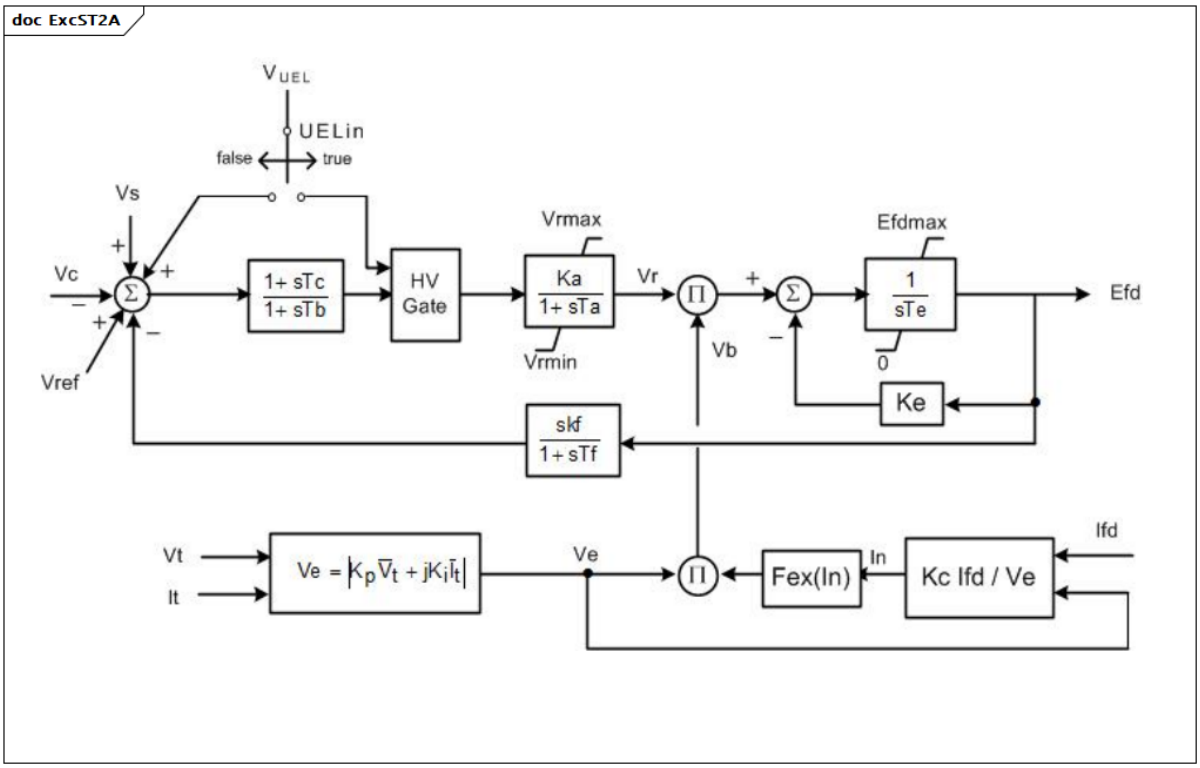

Figure 173 – ExcST2A

Figure 173: This diagram describes the ExcST2A excitation system model.

Table 858 shows all attributes of ExcST2A.

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Table 858 – Attributes of ExcitationSystemDynamics::ExcST2A

name	mult	type	description
ka	1..1	PU	Voltage regulator gain (Ka) (> 0). Typical value = 120.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (> 0). Typical value = 0,15.
vrmax	1..1	PU	Maximum voltage regulator outputs (Vrmax) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator outputs (Vrmin) (< 0). Typical value = -1.
ke	1..1	PU	Exciter constant related to self-excited field (Ke). Typical value = 1.
te	1..1	Seconds	Exciter time constant, integration rate associated with exciter control (Te) (> 0). Typical value = 0,5.
kf	1..1	PU	Excitation control system stabilizer gains (kf) (>= 0). Typical value = 0,05.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (>= 0). Typical value = 0,7.
kp	1..1	PU	Potential circuit gain coefficient (Kp) (>= 0). Typical value = 4,88.
ki	1..1	PU	Potential circuit gain coefficient (Ki) (>= 0). Typical value = 8.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 1,82.
efdmax	1..1	PU	Maximum field voltage (Efdmax) (>= 0). Typical value = 99.
uelin	1..1	Boolean	UEL input (UELin). true = HV gate false = add to error signal. Typical value = false.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 0.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 859 shows all association ends of ExcST2A with other classes.

Table 859 – Association ends of ExcitationSystemDynamics:: ExcST2A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

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14.3.3.9.55 ExcST3A

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE ST3A static excitation system with added speed multiplier.

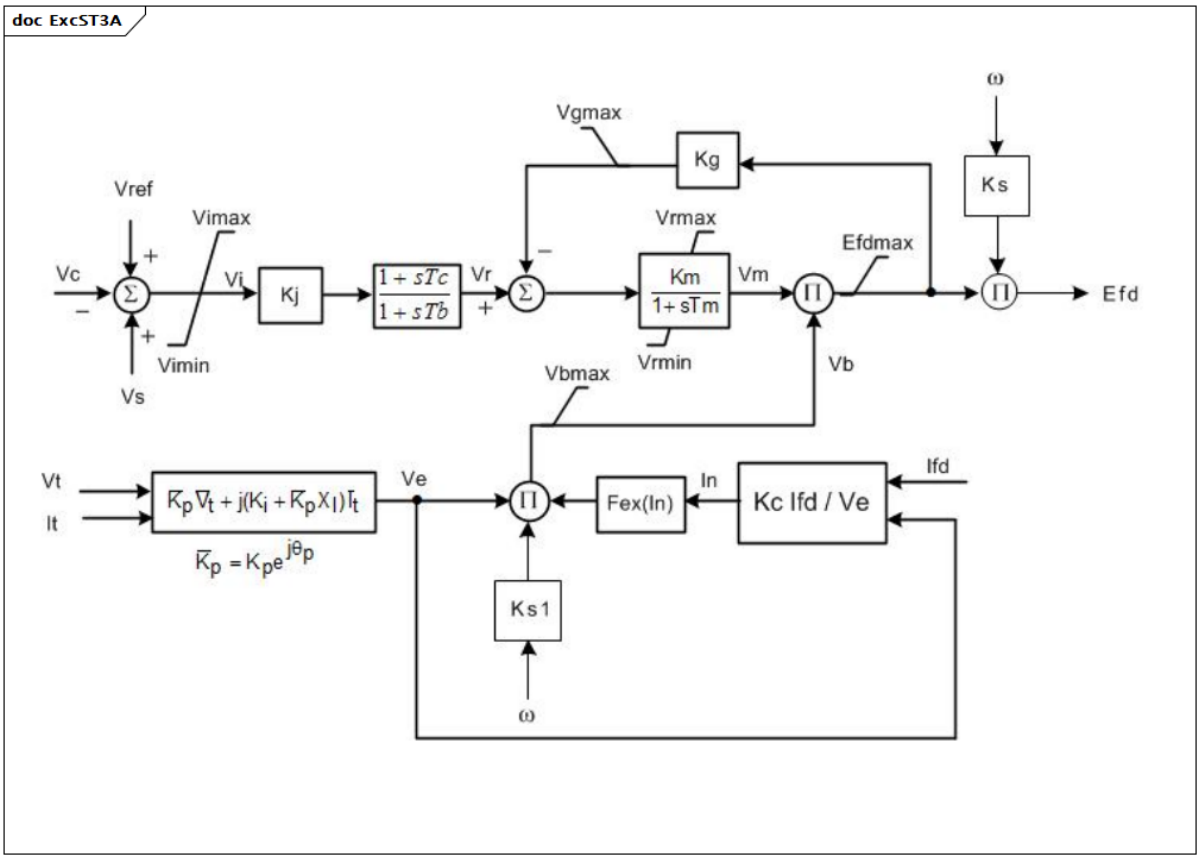

Figure 174 – ExcST3A

Figure 174: This diagram describes the ExcST3A excitation system model.

Parameter details:

1) If Ks or Ks1 is set to 1, the output (Efd) is multiplied by generator speed.

Table 860 shows all attributes of ExcST3A.

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Table 860 – Attributes of ExcitationSystemDynamics::ExcST3A

name	mult	type	description
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax) (> 0). Typical value = 0,2.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin) (< 0). Typical value = -0,2.
kj	1..1	PU	AVR gain (Kj) (> 0). Typical value = 200.
tb	1..1	Seconds	Voltage regulator time constant (Tb) (>= 0). Typical value = 6,67.
tc	1..1	Seconds	Voltage regulator time constant (Tc) (>= 0). Typical value = 1.
efdmax	1..1	PU	Maximum AVR output (Efdmax) (>= 0). Typical value = 6,9.
km	1..1	PU	Forward gain constant of the inner loop field regulator (Km) (> 0). Typical value = 7,04.
tm	1..1	Seconds	Forward time constant of inner loop field regulator (Tm) (> 0). Typical value = 1.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -1.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg) (>= 0). Typical value = 1.
kp	1..1	PU	Potential source gain (Kp) (> 0). Typical value = 4,37.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical value = 20.
ki	1..1	PU	Potential circuit gain coefficient (Ki) (>= 0). Typical value = 4,83.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 1,1.
xl	1..1	PU	Reactance associated with potential source (Xl) (>= 0). Typical value = 0,09.
vbmax	1..1	PU	Maximum excitation voltage (Vbmax) (> 0). Typical value = 8,63.
vgmax	1..1	PU	Maximum inner loop feedback voltage (Vgmax) (>= 0). Typical value = 6,53.
ks	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks). Typical value = 0.
ks1	1..1	PU	Coefficient to allow different usage of the model-speed coefficient (Ks1). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 861 shows all association ends of ExcST3A with other classes.

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Table 861 – Association ends of ExcitationSystemDynamics:: ExcST3A with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.56 ExcST4B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE ST4B static excitation system with maximum inner loop feedback gain Vgmax.

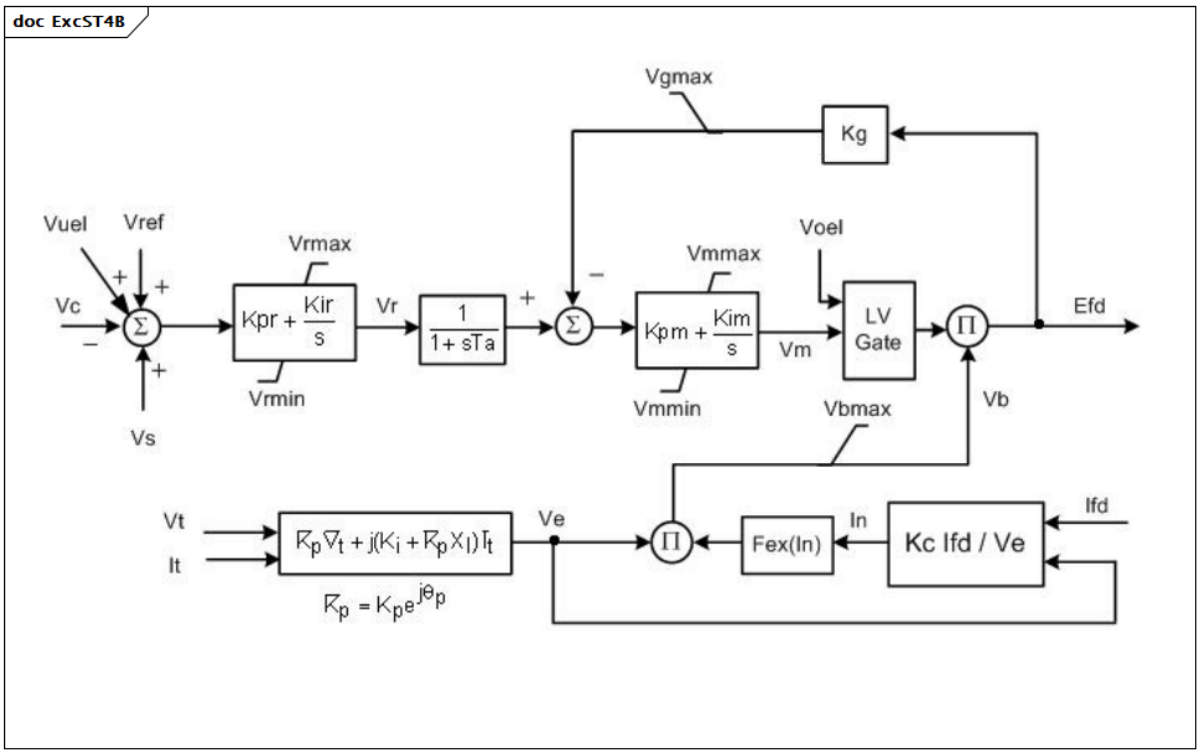

Figure 175 – ExcST4B

Figure 175: This diagram describes the ExcST4B excitation system model.

Table 862 shows all attributes of ExcST4B.

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Table 862 – Attributes of ExcitationSystemDynamics::ExcST4B

name	mult	type	description
kpr	1..1	PU	Voltage regulator proportional gain (Kpr). Typical value = 10,75.
kir	1..1	PU	Voltage regulator integral gain (Kir). Typical value = 10,75.
ta	1..1	Seconds	Voltage regulator time constant (Ta) (>= 0). Typical value = 0,02.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 1.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -0,87.
kpm	1..1	PU	Voltage regulator proportional gain output (Kpm). Typical value = 1.
kim	1..1	PU	Voltage regulator integral gain output (Kim). Typical value = 0.
vmmax	1..1	PU	Maximum inner loop output (Vmmax) (> ExcST4B.vmmin). Typical value = 99.
vmmin	1..1	PU	Minimum inner loop output (Vmmin) (< ExcST4B.vmmax). Typical value = -99.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg) (>= 0). Typical value = 0.
kp	1..1	PU	Potential circuit gain coefficient (Kp) (> 0). Typical value = 9,3.
thetap	1..1	AngleDegrees	Potential circuit phase angle (thetap). Typical value = 0.
ki	1..1	PU	Potential circuit gain coefficient (Ki) (>= 0). Typical value = 0.
kc	1..1	PU	Rectifier loading factor proportional to commutating reactance (Kc) (>= 0). Typical value = 0,113.
xl	1..1	PU	Reactance associated with potential source (Xl) (>= 0). Typical value = 0,124.
vbmax	1..1	PU	Maximum excitation voltage (Vbmax) (> 0). Typical value = 11,63.
vgmax	1..1	PU	Maximum inner loop feedback voltage (Vgmax) (>= 0). Typical value = 5,8.
uel	1..1	Boolean	Selector (UEL). true = UEL is part of block diagram false = UEL is not part of block diagram. Typical value = false.
lvgate	1..1	Boolean	Selector (LVGate). true = LVGate is part of the block diagram false = LVGate is not part of the block diagram. Typical value = false.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 863 shows all association ends of ExcST4B with other classes.

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Table 863 – Association ends of ExcitationSystemDynamics:: ExcST4B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.57 ExcST6B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE ST6B static excitation system with PID controller and optional inner feedback loop.

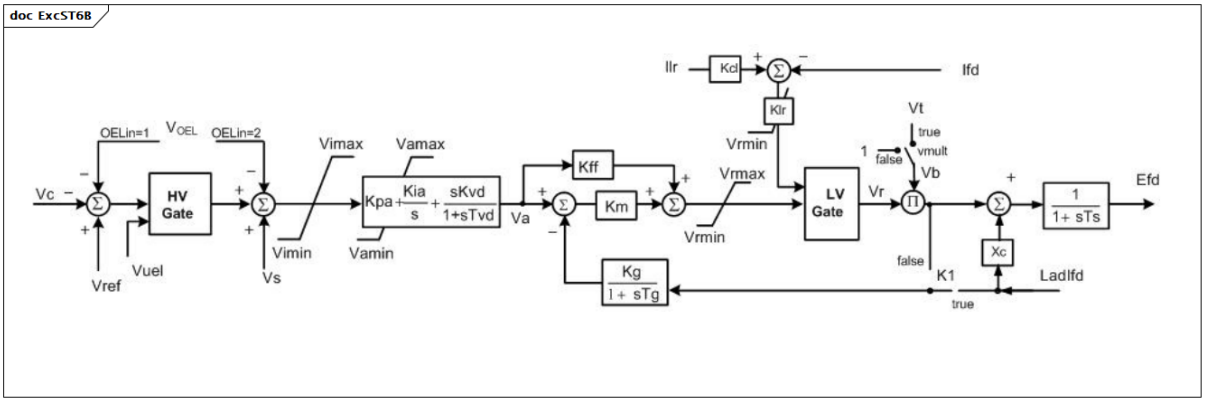

Figure 176 – ExcST6B

Figure 176: This diagram describes the ExcST6B excitation system model.

Table 864 shows all attributes of ExcST6B.

Table 864 – Attributes of ExcitationSystemDynamics::ExcST6B

name	mult	type	description
ilr	1..1	PU	Exciter output current limit reference (Ilr) (> 0). Typical value = 4,164.
k1	1..1	Boolean	Selector (K1). true = feedback is from Ifd false = feedback is not from Ifd. Typical value = true.
kcl	1..1	PU	Exciter output current limit adjustment (Kcl) (> 0). Typical value = 1,0577.
kff	1..1	PU	Pre-control gain constant of the inner loop field regulator (Kff). Typical value = 1.
kg	1..1	PU	Feedback gain constant of the inner loop field regulator (Kg) (>= 0). Typical value = 1.
kia	1..1	PU	Voltage regulator integral gain (Kia) (> 0). Typical value = 45,094.
klr	1..1	PU	Exciter output current limit adjustment (Kcl) (> 0). Typical value = 17,33.

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name	mult	type	description
km	1..1	PU	Forward gain constant of the inner loop field regulator (Km). Typical value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (Kpa) (> 0). Typical value = 18,038.
kvd	1..1	PU	Voltage regulator derivative gain (Kvd). Typical value = 0.
oelin	1..1	ExcST6BOELselectorKind	OEL input selector (OELin). Typical value = noOELinput (corresponds to OELin = 0 on diagram).
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (Tg) (>= 0). Typical value = 0,02.
ts	1..1	Seconds	Rectifier firing time constant (Ts) (>= 0). Typical value = 0.
tvd	1..1	Seconds	Voltage regulator derivative gain (Tvd) (>= 0). Typical value = 0.
vamax	1..1	PU	Maximum voltage regulator output (Vamax) (> 0). Typical value = 4,81.
vamin	1..1	PU	Minimum voltage regulator output (Vamin) (< 0). Typical value = -3,85.
vilim	1..1	Boolean	Selector (Vilim). true = Vimin-Vimax limiter is active false = Vimin-Vimax limiter is not active. Typical value = true.
vimax	1..1	PU	Maximum voltage regulator input limit (Vimax) (> ExcST6B.vimin). Typical value = 10.
vimin	1..1	PU	Minimum voltage regulator input limit (Vimin) (< ExcST6B.vimax). Typical value = -10.
vmult	1..1	Boolean	Selector (vmult). true = multiply regulator output by terminal voltage false = do not multiply regulator output by terminal voltage. Typical value = true.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 4,81.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -3,85.
xc	1..1	PU	Excitation source reactance (Xc). Typical value = 0,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 865 shows all association ends of ExcST6B with other classes.

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Table 865 – Association ends of ExcitationSystemDynamics:: ExcST6B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.9.58 ExcST7B

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE ST7B static excitation system without stator current limiter (SCL) and current compensator (DROOP) inputs.

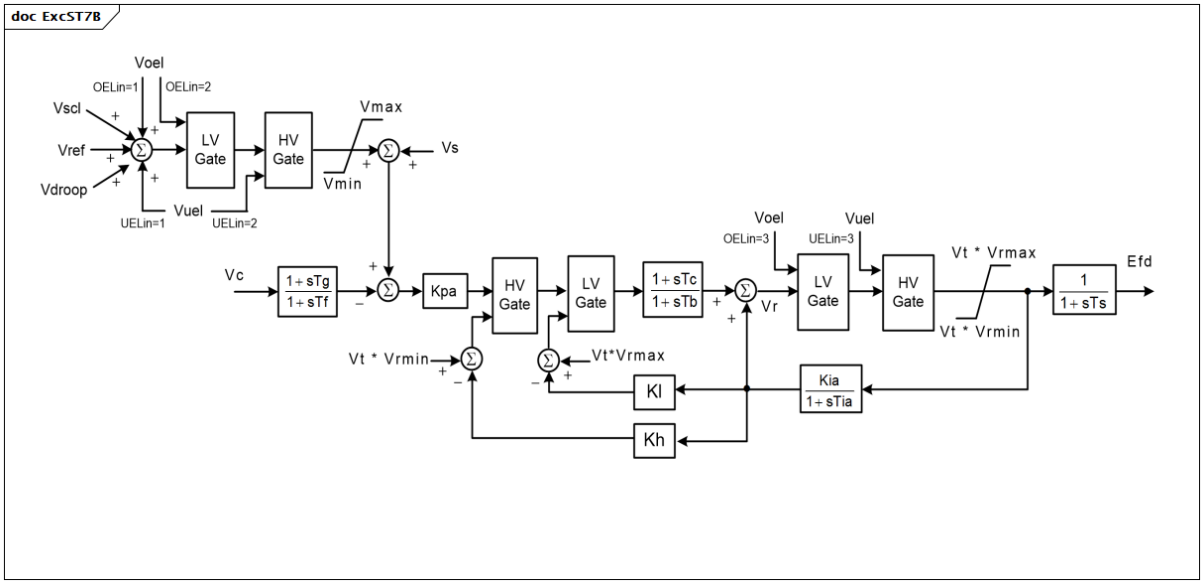

Figure 177 – ExcST7B

Figure 177: This diagram describes the ExcST7B excitation system model.

Table 866 shows all attributes of ExcST7B.

Table 866 – Attributes of ExcitationSystemDynamics::ExcST7B

name	mult	type	description
kh	1..1	PU	High-value gate feedback gain (Kh) (>= 0). Typical value = 1.
kia	1..1	PU	Voltage regulator integral gain (Kia) (>= 0). Typical value = 1.
kl	1..1	PU	Low-value gate feedback gain (Kl) (>= 0). Typical value = 1.
kpa	1..1	PU	Voltage regulator proportional gain (Kpa) (> 0). Typical value = 40.
oelin	1..1	ExcST7BOELselectorKind	OEL input selector (OELin). Typical value = noOELinput.
tb	1..1	Seconds	Regulator lag time constant (Tb) (>= 0). Typical value = 1.

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name	mult	type	description
tc	1..1	Seconds	Regulator lead time constant (Tc) (>= 0). Typical value = 1.
tf	1..1	Seconds	Excitation control system stabilizer time constant (Tf) (>= 0). Typical value = 1.
tg	1..1	Seconds	Feedback time constant of inner loop field voltage regulator (Tg) (>= 0). Typical value = 1.
tia	1..1	Seconds	Feedback time constant (Tia) (>= 0). Typical value = 3.
ts	1..1	Seconds	Rectifier firing time constant (Ts) (>= 0). Typical value = 0.
uelin	1..1	ExcST7BUELselectorKind	UEL input selector (UELin). Typical value = noUELinput.
vmax	1..1	PU	Maximum voltage reference signal (Vmax) (> 0 and > ExcST7B.vmin)). Typical value = 1,1.
vmin	1..1	PU	Minimum voltage reference signal (Vmin) (> 0 and < ExcST7B.vmax). Typical value = 0,9.
vrmax	1..1	PU	Maximum voltage regulator output (Vrmax) (> 0). Typical value = 5.
vrmin	1..1	PU	Minimum voltage regulator output (Vrmin) (< 0). Typical value = -4,5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 867 shows all association ends of ExcST7B with other classes.

Table 867 – Association ends of ExcitationSystemDynamics:: ExcST7B with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.3.10 Package OverexcitationLimiterDynamics

14.3.3.10.1 General

Overexcitation limiters (OELs) are also referred to as maximum excitation limiters and field current limiters. The possibility of voltage collapse in stressed power systems increases the importance of modelling these limiters in studies of system conditions that cause machines to operate at high levels of excitation for a sustained period, such as voltage collapse or system-islanding. Such events typically occur over a long time frame compared with transient or small-signal stability simulations.

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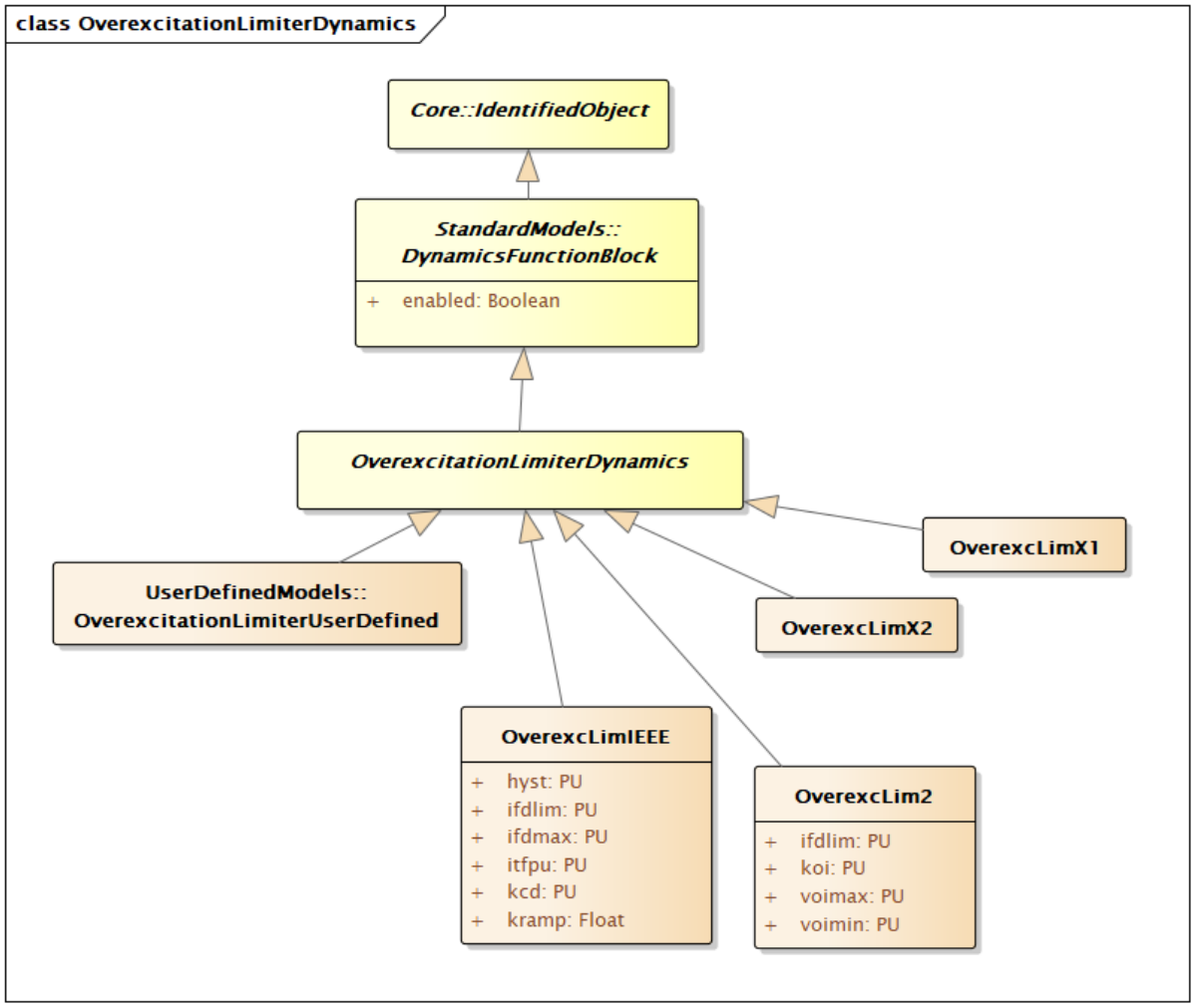

**Figure 178 – Class diagram OverexcitationLimiterDynamics::
OverexcitationLimiterDynamics**

Figure 178: This diagram shows dynamic standard model overexcitation limiter classes and their inheritance from the OverexcitationLimiterDynamics class.

14.3.3.10.2 (abstract) OverexcitationLimiterDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Overexcitation limiter function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 868 shows all attributes of OverexcitationLimiterDynamics.

**Table 868 – Attributes of OverexcitationLimiterDynamics::
OverexcitationLimiterDynamics**

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 869 shows all association ends of OverexcitationLimiterDynamics with other classes.

Table 869 – Association ends of OverexcitationLimiterDynamics:: OverexcitationLimiterDynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this overexcitation limiter model is associated.

14.3.3.10.3 OverexcLimIEEE

Inheritance path = OverexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

The over excitation limiter model is intended to represent the significant features of OELs necessary for some large-scale system studies. It is the result of a pragmatic approach to obtain a model that can be widely applied with attainable data from generator owners. An attempt to include all variations in the functionality of OELs and duplicate how they interact with the rest of the excitation systems would likely result in a level of application insufficient for the studies for which they are intended.

Reference: IEEE OEL 421.5:2005, 9.

Table 870 shows all attributes of OverexcLimIEEE.

Table 870 – Attributes of OverexcitationLimiterDynamics::OverexcLimIEEE

name	mult	type	description
itfpu	1..1	PU	OEL timed field current limiter pickup level (ITFPU). Typical value = 1,05.
ifdmax	1..1	PU	OEL instantaneous field current limit (IFDMAX). Typical value = 1,5.
ifdlim	1..1	PU	OEL timed field current limit (IFDLIM). Typical value = 1,05.
hyst	1..1	PU	OEL pickup/drop-out hysteresis (HYST). Typical value = 0,03.
kcd	1..1	PU	OEL cooldown gain (KCD). Typical value = 1.
kramp	1..1	Float	OEL ramped limit rate (KRAMP). Unit = PU / s. Typical value = 10.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 871 shows all association ends of OverexcLimIEEE with other classes.

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Table 871 – Association ends of OverexcitationLimiterDynamics:: OverexcLimIEEE with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: OverexcitationLimiterDynamics

14.3.3.10.4 OverexcLim2

Inheritance path = OverexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Different from LimIEEEOEL, LimOEL2 has a fixed pickup threshold and reduces the excitation set-point by means of a non-windup integral regulator.

Irated is the rated machine excitation current (calculated from nameplate conditions: Vnom, Pnom, CosPhinom).

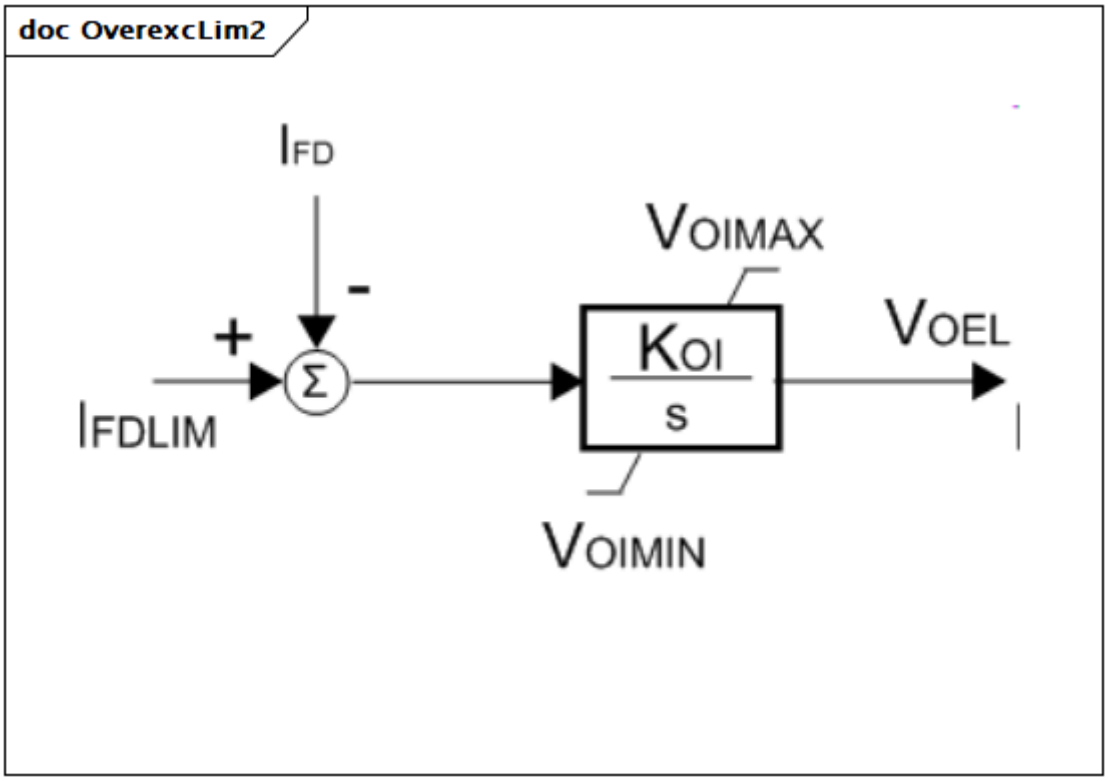

Figure 179 – OverexcLim2

Figure 179: This diagram describes the OverexcLim2 overexcitation limiter model.

Table 872 shows all attributes of OverexcLim2.

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Table 872 – Attributes of OverexcitationLimiterDynamics::OverexcLim2

name	mult	type	description
koi	1..1	PU	Gain Over excitation limiter (KOI). Typical value = 0,1.
voimax	1..1	PU	Maximum error signal (VOIMAX) (> OverexcLim2.voimin). Typical value = 0.
voimin	1..1	PU	Minimum error signal (VOIMIN) (< OverexcLim2.voimax). Typical value = -9999.
ifdlim	1..1	PU	Limit value of rated field current (IFDLIM). Typical value = 1,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 873 shows all association ends of OverexcLim2 with other classes.

Table 873 – Association ends of OverexcitationLimiterDynamics:: OverexcLim2 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: OverexcitationLimiterDynamics

14.3.3.10.5 OverexcLimX1

Inheritance path = OverexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Field voltage over excitation limiter.

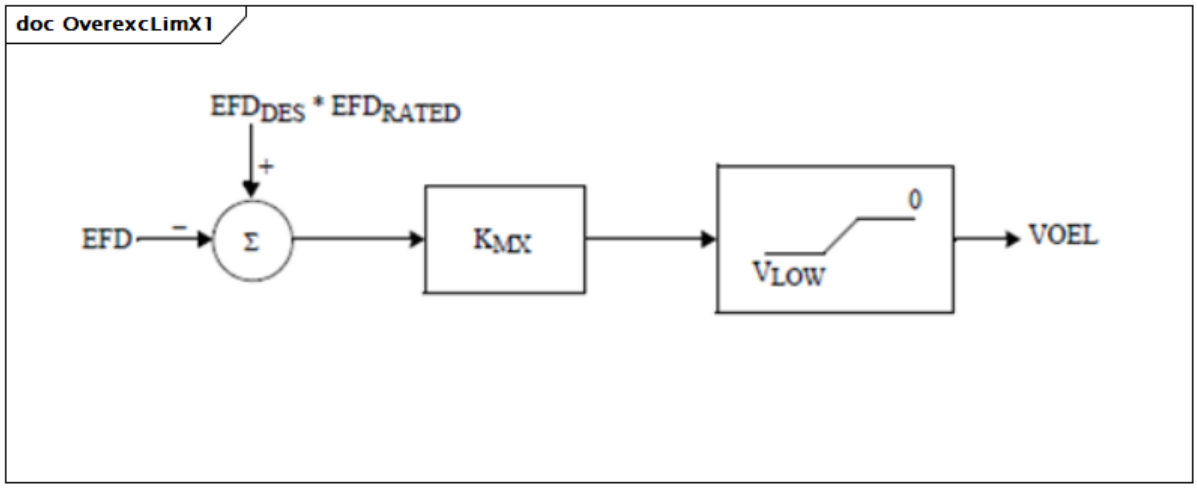

Figure 180 – OverexcLimX1

Figure 180: This diagram describes the OverexcLimX1 overexcitation limiter model.

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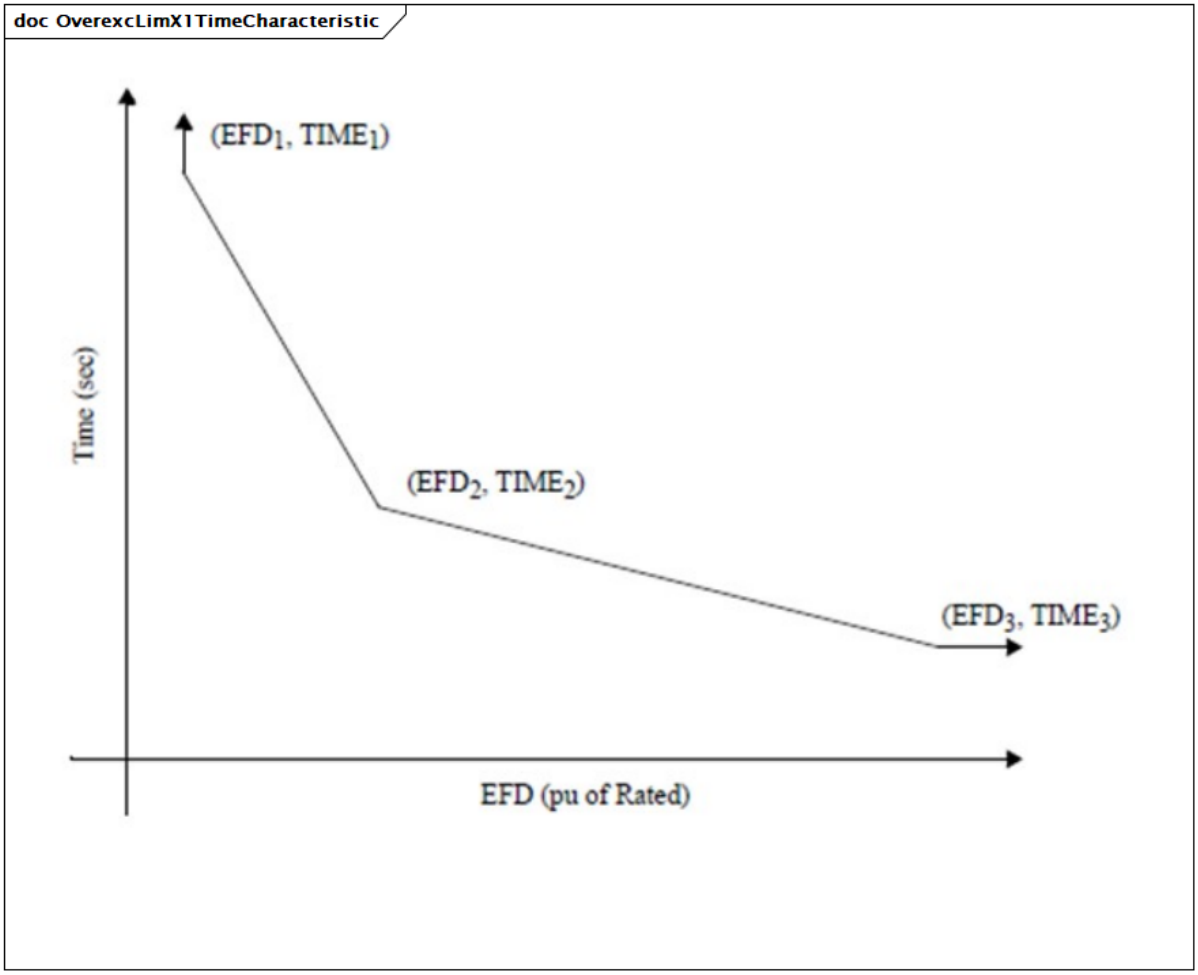

Figure 181 – OverexcLimX1TimeCharacteristic

Figure 181: This model assumes an inverse time characteristic, where operating time is a function of the magnitude of the overvoltage.

Table 874 shows all attributes of OverexcLimX1.

Table 874 – Attributes of OverexcitationLimiterDynamics::OverexcLimX1

name	mult	type	description
efdrated	1..1	PU	Rated field voltage (EFDRATED). Typical value = 1,05.
efd1	1..1	PU	Low voltage point on the inverse time characteristic (EFD1). Typical value = 1,1.
t1	1..1	Seconds	Time to trip the exciter at the low voltage point on the inverse time characteristic (TIME1) (>= 0). Typical value = 120.
efd2	1..1	PU	Mid voltage point on the inverse time characteristic (EFD2). Typical value = 1,2.
t2	1..1	Seconds	Time to trip the exciter at the mid voltage point on the inverse time characteristic (TIME2) (>= 0). Typical value = 40.
efd3	1..1	PU	High voltage point on the inverse time characteristic (EFD3). Typical value = 1,5.

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name	mult	type	description
t3	1..1	Seconds	Time to trip the exciter at the high voltage point on the inverse time characteristic (TIME3) (>= 0). Typical value = 15.
efddes	1..1	PU	Desired field voltage (EFDDES). Typical value = 0,9.
kmx	1..1	PU	Gain (KMX). Typical value = 0,01.
vlow	1..1	PU	Low voltage limit (VLOW) (> 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 875 shows all association ends of OverexcLimX1 with other classes.

Table 875 – Association ends of OverexcitationLimiterDynamics:: OverexcLimX1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: OverexcitationLimiterDynamics

14.3.3.10.6 OverexcLimX2

Inheritance path = OverexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Field voltage or current overexcitation limiter designed to protect the generator field of an AC machine with automatic excitation control from overheating due to prolonged overexcitation.

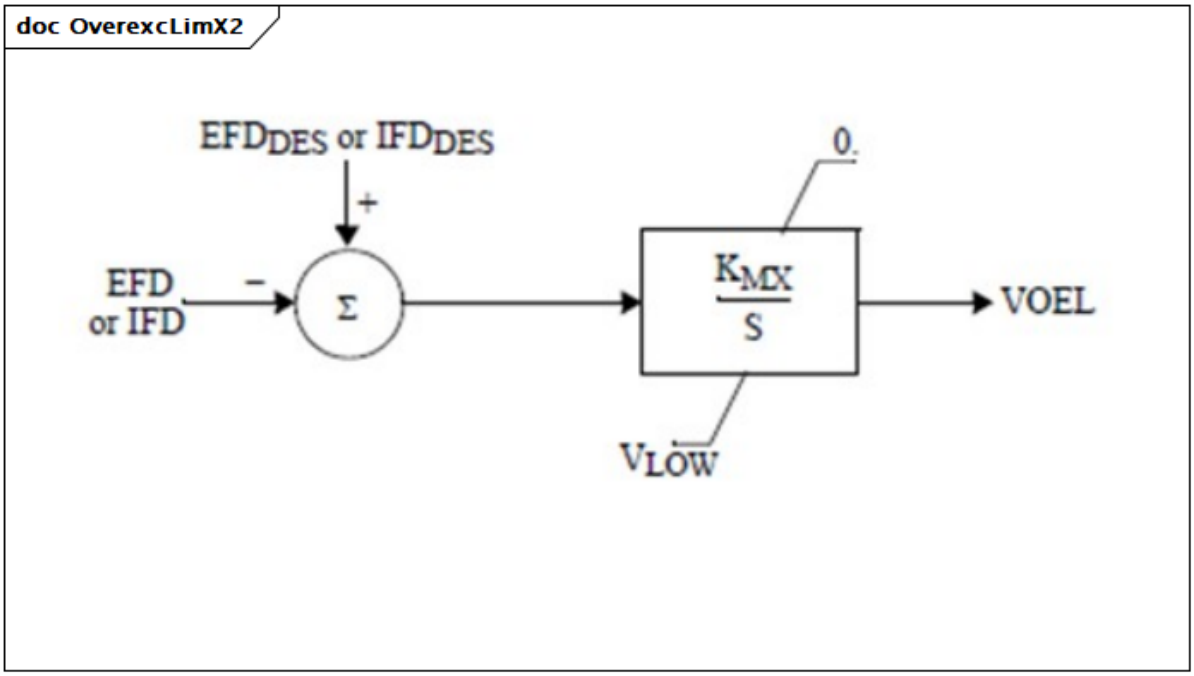

Figure 182 – OverexcLimX2

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Figure 182: This diagram describes the OverexcLimX2 overexcitation limiter model.

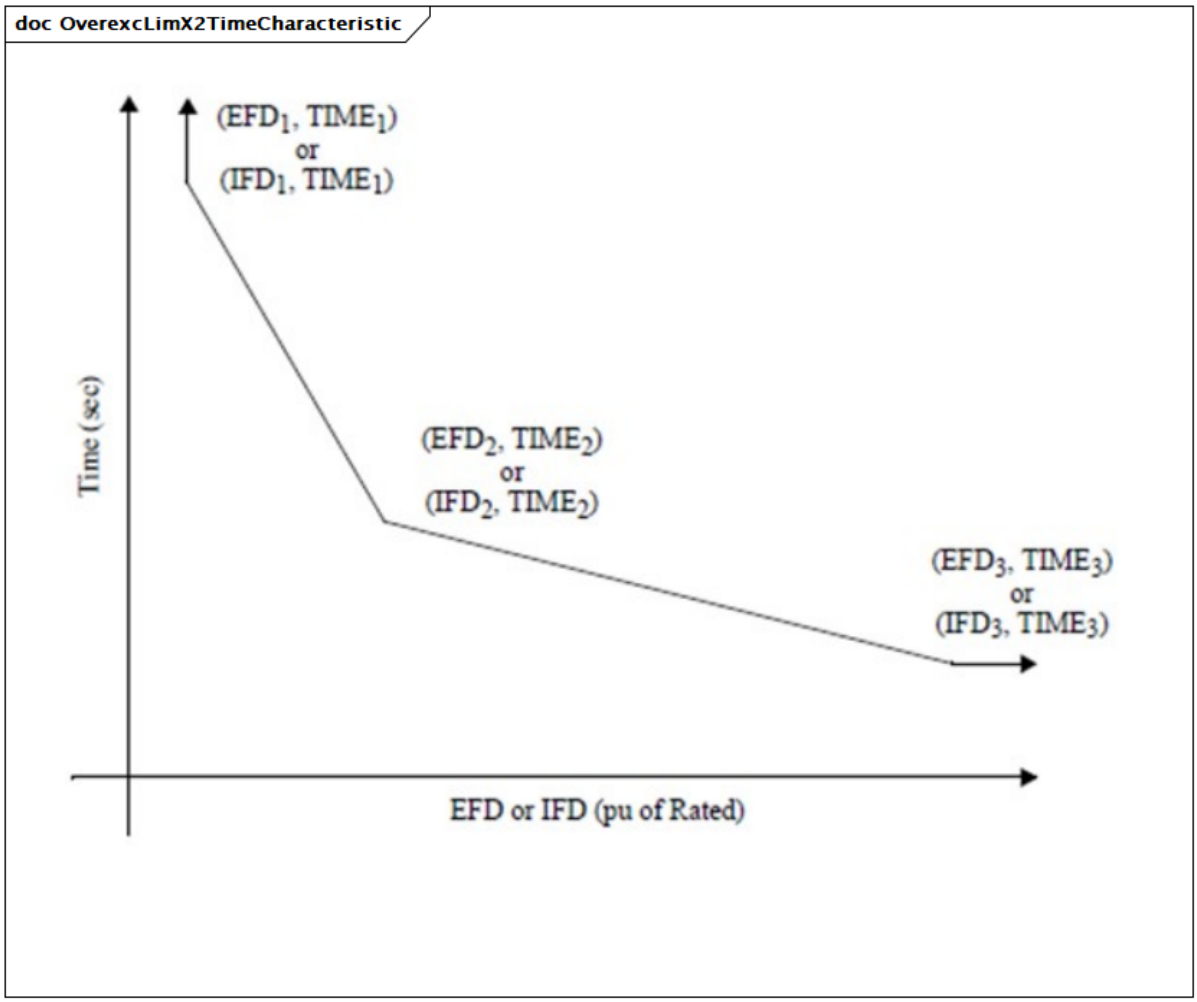

Figure 183 – OverexcLimX2TimeCharacteristic

Figure 183: This model assumes an inverse time characteristic, where operating time is a function of the magnitude of the overvoltage.

Table 876 shows all attributes of OverexcLimX2.

Table 876 – Attributes of OverexcitationLimiterDynamics::OverexcLimX2

name	mult	type	description
m	1..1	Boolean	(m). true = IFD limiting false = EFD limiting.
efdrated	1..1	PU	Rated field voltage if m = false or rated field current if m = true (EFDRATED). Typical value = 1,05.
efd1	1..1	PU	Low voltage or current point on the inverse time characteristic (EFD1). Typical value = 1,1.
t1	1..1	Seconds	Time to trip the exciter at the low voltage or current point on the inverse time characteristic (TIME1) (>= 0). Typical value = 120.
efd2	1..1	PU	Mid voltage or current point on the inverse time characteristic (EFD2). Typical value = 1,2.

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name	mult	type	description
t2	1..1	Seconds	Time to trip the exciter at the mid voltage or current point on the inverse time characteristic (TIME2) (>= 0). Typical value = 40.
efd3	1..1	PU	High voltage or current point on the inverse time characteristic (EFD3). Typical value = 1,5.
t3	1..1	Seconds	Time to trip the exciter at the high voltage or current point on the inverse time characteristic (TIME3) (>= 0). Typical value = 15.
efddes	1..1	PU	Desired field voltage if m = false or desired field current if m = true (EFDDES). Typical value = 1.
kmx	1..1	PU	Gain (KMX). Typical value = 0,002.
vlow	1..1	PU	Low voltage limit (VLOW) (> 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 877 shows all association ends of OverexcLimX2 with other classes.

Table 877 – Association ends of OverexcitationLimiterDynamics:: OverexcLimX2 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: OverexcitationLimiterDynamics

14.3.3.11 Package UnderexcitationLimiterDynamics

14.3.3.11.1 General

Underexcitation limiters (UELs) act to boost excitation. The UEL typically senses either a combination of voltage and current of the synchronous machine or a combination of real and reactive power. Some UELs utilize a temperature or pressure recalibration feature, in which the UEL characteristic is shifted depending upon the generator cooling gas temperature or pressure.

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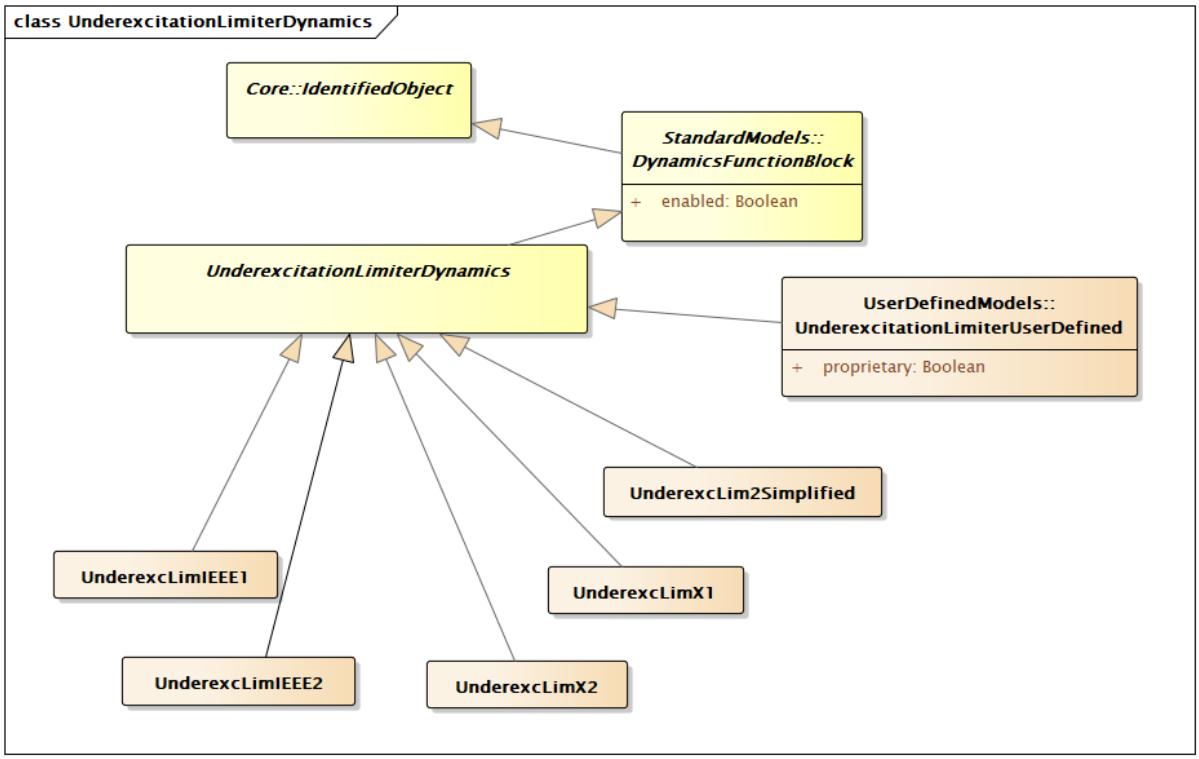

Figure 184 – Class diagram UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics

Figure 184: This diagram shows dynamic standard model underexcitation limiter classes and their inheritance from the UnderexcitationLimiterDynamics class.

14.3.3.11.2 (abstract) UnderexcitationLimiterDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Underexcitation limiter function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 878 shows all attributes of UnderexcitationLimiterDynamics.

Table 878 – Attributes of UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 879 shows all association ends of UnderexcitationLimiterDynamics with other classes.

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Table 879 – Association ends of UnderexcitationLimiterDynamics:: UnderexcitationLimiterDynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this underexcitation limiter model is associated.

14.3.3.11.3 UnderexcLimIEEE1

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Type UEL1 model which has a circular limit boundary when plotted in terms of machine reactive power vs. real power output.

Reference: IEEE UEL1 421.5:2005, 10.1.

Table 880 shows all attributes of UnderexcLimIEEE1.

Table 880 – Attributes of UnderexcitationLimiterDynamics::UnderexcLimIEEE1

name	mult	type	description
kur	1..1	PU	UEL radius setting (KUR). Typical value = 1,95.
kuc	1..1	PU	UEL centre setting (KUC). Typical value = 1,38.
kuf	1..1	PU	UEL excitation system stabilizer gain (KUF). Typical value = 3,3.
vurmax	1..1	PU	UEL maximum limit for radius phasor magnitude (VURMAX). Typical value = 5,8.
vucmax	1..1	PU	UEL maximum limit for operating point phasor magnitude (VUCMAX). Typical value = 5,8.
kui	1..1	PU	UEL integral gain (KUI). Typical value = 0.
kul	1..1	PU	UEL proportional gain (KUL). Typical value = 100.
vuimax	1..1	PU	UEL integrator output maximum limit (VUIMAX) (> UnderexcLimIEEE1.vuimin).
vuimin	1..1	PU	UEL integrator output minimum limit (VUIMIN) (< UnderexcLimIEEE1.vuimax).
tu1	1..1	Seconds	UEL lead time constant (TU1) (>= 0). Typical value = 0.
tu2	1..1	Seconds	UEL lag time constant (TU2) (>= 0). Typical value = 0,05.
tu3	1..1	Seconds	UEL lead time constant (TU3) (>= 0). Typical value = 0.
tu4	1..1	Seconds	UEL lag time constant (TU4) (>= 0). Typical value = 0.
vulmax	1..1	PU	UEL output maximum limit (VULMAX) (> UnderexcLimIEEE1.vulmin). Typical value = 18.
vulmin	1..1	PU	UEL output minimum limit (VULMIN) (< UnderexcLimIEEE1.vulmax). Typical value = -18.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 881 shows all association ends of UnderexcLimIEEE1 with other classes.

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Table 881 – Association ends of UnderexcitationLimiterDynamics:: UnderexcLimIEEE1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.3.11.4 UnderexcLimIEEE2

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Type UEL2 underexcitation limiter which has either a straight-line or multi-segment characteristic when plotted in terms of machine reactive power output vs. real power output.

Reference: IEEE UEL2 421.5:2005, 10.2 (limit characteristic lookup table shown in Figure 10.4 (p 32)).

Table 882 shows all attributes of UnderexcLimIEEE2.

Table 882 – Attributes of UnderexcitationLimiterDynamics::UnderexcLimIEEE2

name	mult	type	description
tuv	1..1	Seconds	Voltage filter time constant (TUV) (>= 0). Typical value = 5.
tup	1..1	Seconds	Real power filter time constant (TUP) (>= 0). Typical value = 5.
tuq	1..1	Seconds	Reactive power filter time constant (TUQ) (>= 0). Typical value = 0.
kui	1..1	PU	UEL integral gain (KUI). Typical value = 0,5.
kul	1..1	PU	UEL proportional gain (KUL). Typical value = 0,8.
vuimax	1..1	PU	UEL integrator output maximum limit (VUIMAX) (> UnderexcLimIEEE2.vuimin). Typical value = 0,25.
vuimin	1..1	PU	UEL integrator output minimum limit (VUIMIN) (< UnderexcLimIEEE2.vuimax). Typical value = 0.
kuf	1..1	PU	UEL excitation system stabilizer gain (KUF). Typical value = 0.
kfb	1..1	PU	Gain associated with optional integrator feedback input signal to UEL (KFB). Typical value = 0.
tul	1..1	Seconds	Time constant associated with optional integrator feedback input signal to UEL (TUL) (>= 0). Typical value = 0.
tu1	1..1	Seconds	UEL lead time constant (TU1) (>= 0). Typical value = 0.
tu2	1..1	Seconds	UEL lag time constant (TU2) (>= 0). Typical value = 0.
tu3	1..1	Seconds	UEL lead time constant (TU3) (>= 0). Typical value = 0.
tu4	1..1	Seconds	UEL lag time constant (TU4) (>= 0). Typical value = 0.
vulmax	1..1	PU	UEL output maximum limit (VULMAX) (> UnderexcLimIEEE2.vulmin). Typical value = 0,25.

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name	mult	type	description
vulmin	1..1	PU	UEL output minimum limit (VULMIN) (< UnderexcLimIEEE2.vulmax). Typical value = 0.
p0	1..1	PU	Real power values for endpoints (P0). Typical value = 0.
q0	1..1	PU	Reactive power values for endpoints (Q0). Typical value = -0,31.
p1	1..1	PU	Real power values for endpoints (P1). Typical value = 0,3.
q1	1..1	PU	Reactive power values for endpoints (Q1). Typical value = -0,31.
p2	1..1	PU	Real power values for endpoints (P2). Typical value = 0,6.
q2	1..1	PU	Reactive power values for endpoints (Q2). Typical value = -0,28.
p3	1..1	PU	Real power values for endpoints (P3). Typical value = 0,9.
q3	1..1	PU	Reactive power values for endpoints (Q3). Typical value = -0,21.
p4	1..1	PU	Real power values for endpoints (P4). Typical value = 1,02.
q4	1..1	PU	Reactive power values for endpoints (Q4). Typical value = 0.
p5	1..1	PU	Real power values for endpoints (P5).
q5	1..1	PU	Reactive power values for endpoints (Q5).
p6	1..1	PU	Real power values for endpoints (P6).
q6	1..1	PU	Reactive power values for endpoints (Q6).
p7	1..1	PU	Real power values for endpoints (P7).
q7	1..1	PU	Reactive power values for endpoints (Q7).
p8	1..1	PU	Real power values for endpoints (P8).
q8	1..1	PU	Reactive power values for endpoints (Q8).
p9	1..1	PU	Real power values for endpoints (P9).
q9	1..1	PU	Reactive power values for endpoints (Q9).
p10	1..1	PU	Real power values for endpoints (P10).
q10	1..1	PU	Reactive power values for endpoints (Q10).
k1	1..1	Float	UEL terminal voltage exponent applied to real power input to UEL limit look-up table (k1). Typical value = 2.
k2	1..1	Float	UEL terminal voltage exponent applied to reactive power output from UEL limit look-up table (k2). Typical value = 2.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 883 shows all association ends of UnderexcLimIEEE2 with other classes.

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Table 883 – Association ends of UnderexcitationLimiterDynamics:: UnderexcLimIEEE2 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.3.11.5 UnderexcLim2Simplified

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified type UEL2 underexcitation limiter. This model can be derived from UnderexcLimIEEE2. The limit characteristic (look –up table) is a single straight-line, the same as UnderexcLimIEEE2 (see Figure 10.4 (p 32), IEEE 421.5:2005, 10.2).

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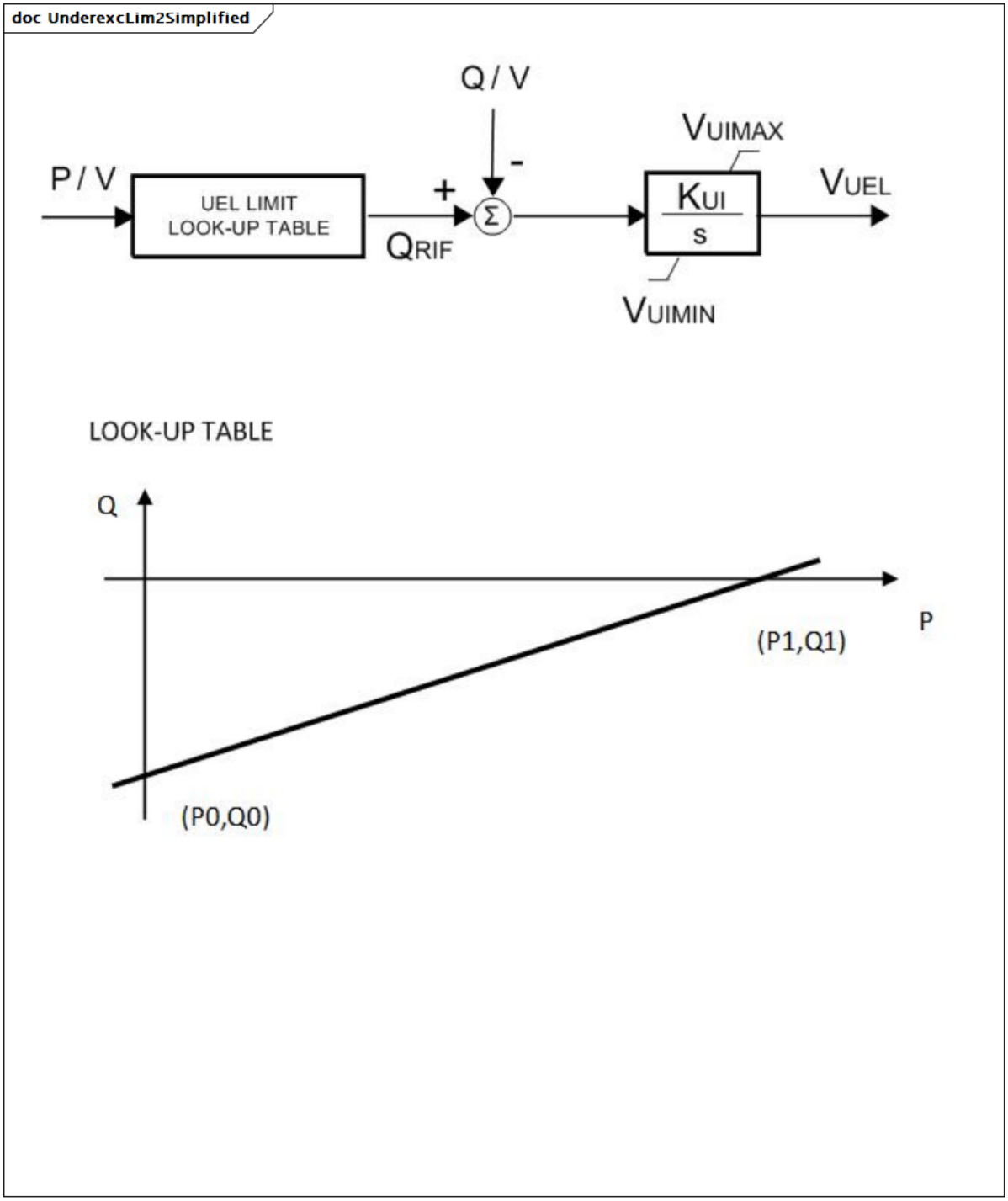

Figure 185 – UnderexcLim2Simplified

Figure 185: This diagram describes the UnderexcLim2Simplified underexcitation limiter model.

Table 884 shows all attributes of UnderexcLim2Simplified.

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Table 884 – Attributes of UnderexcitationLimiterDynamics::UnderexcLim2Simplified

name	mult	type	description
q0	1..1	PU	Segment Q initial point (Q0). Typical value = -0,31.
q1	1..1	PU	Segment Q end point (Q1). Typical value = -0,1.
p0	1..1	PU	Segment P initial point (P0). Typical value = 0.
p1	1..1	PU	Segment P end point (P1). Typical value = 1.
kui	1..1	PU	Gain Under excitation limiter (KUI). Typical value = 0,1.
vuimin	1..1	PU	Minimum error signal (VUIMIN) (< UnderexcLim2Simplified.vuimax). Typical value = 0.
vuimax	1..1	PU	Maximum error signal (VUIMAX) (> UnderexcLim2Simplified.vuimin). Typical value = 1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 885 shows all association ends of UnderexcLim2Simplified with other classes.

Table 885 – Association ends of UnderexcitationLimiterDynamics:: UnderexcLim2Simplified with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.3.11.6 UnderexcLimX1

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Allis-Chalmers minimum excitation limiter.

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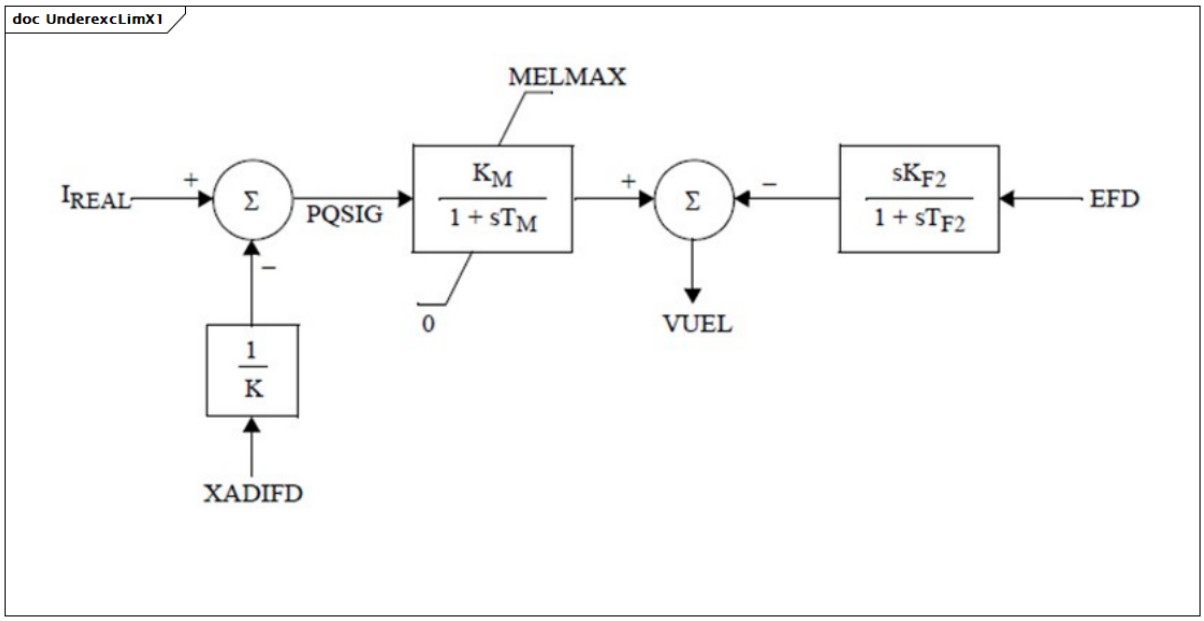

Figure 186 – UnderexcLimX1

Figure 186: This diagram describes the UnderexcLimX1 underexcitation limiter model.

Parameter details:

1) IREAL is the real part of stator current.
2) XADIFD is the signal Ifd as shown in the diagram of the SubtransientRoundRotor model, i.e. the XADIFD = E'q + [(E'q – psid) x (Xd – X'd) x (X'd – X"d) + Id x (Xd – X'd) x (X"d – Xl) x (X'd – Xl)] / (X'd – Xl) ** 2, where the psi in psid is the Greek symbol psi.

Table 886 shows all attributes of UnderexcLimX1.

Table 886 – Attributes of UnderexcitationLimiterDynamics::UnderexcLimX1

name	mult	type	description
kf2	1..1	PU	Differential gain (KF2).
tf2	1..1	Seconds	Differential time constant (TF2) (>= 0).
km	1..1	PU	Minimum excitation limit gain (KM).
tm	1..1	Seconds	Minimum excitation limit time constant (TM) (>= 0).
melmax	1..1	PU	Minimum excitation limit value (MELMAX).
k	1..1	PU	Minimum excitation limit slope (K) (> 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 887 shows all association ends of UnderexcLimX1 with other classes.

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Table 887 – Association ends of UnderexcitationLimiterDynamics:: UnderexcLimX1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.3.11.7 UnderexcLimX2

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Westinghouse minimum excitation limiter.

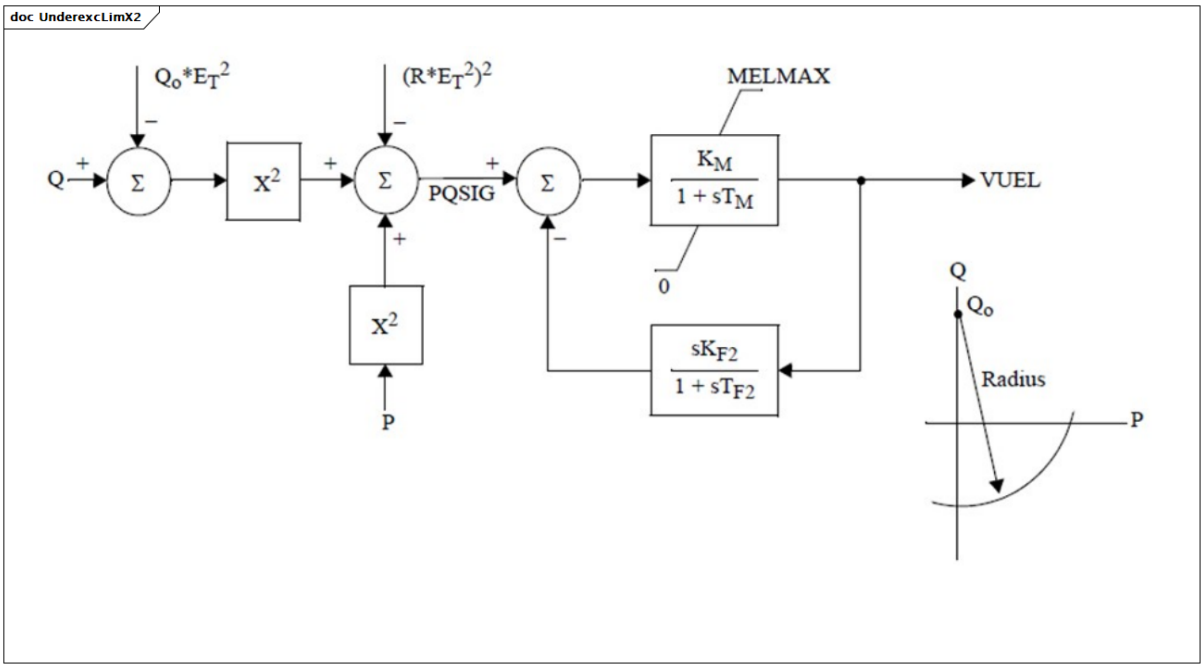

Figure 187 – UnderexcLimX2

Figure 187: This diagram describes the UnderexcLimX2 underexcitation limiter model.

Table 888 shows all attributes of UnderexcLimX2.

Table 888 – Attributes of UnderexcitationLimiterDynamics::UnderexcLimX2

name	mult	type	description
kf2	1..1	PU	Differential gain (KF2).
tf2	1..1	Seconds	Differential time constant (TF2) (>= 0).
km	1..1	PU	Minimum excitation limit gain (KM).
tm	1..1	Seconds	Minimum excitation limit time constant (TM) (>= 0).
melmax	1..1	PU	Minimum excitation limit value (MELMAX).
qo	1..1	PU	Excitation centre setting (QO).
r	1..1	PU	Excitation radius (R).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock

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name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 889 shows all association ends of UnderexcLimX2 with other classes.

Table 889 – Association ends of UnderexcitationLimiterDynamics:: UnderexcLimX2 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.3.12 Package PowerSystemStabilizerDynamics

14.3.3.12.1 General

The power system stabilizer (PSS) model provides an input (Vs) to the excitation system model to improve damping of system oscillations. A variety of input signals can be used depending on the particular design.

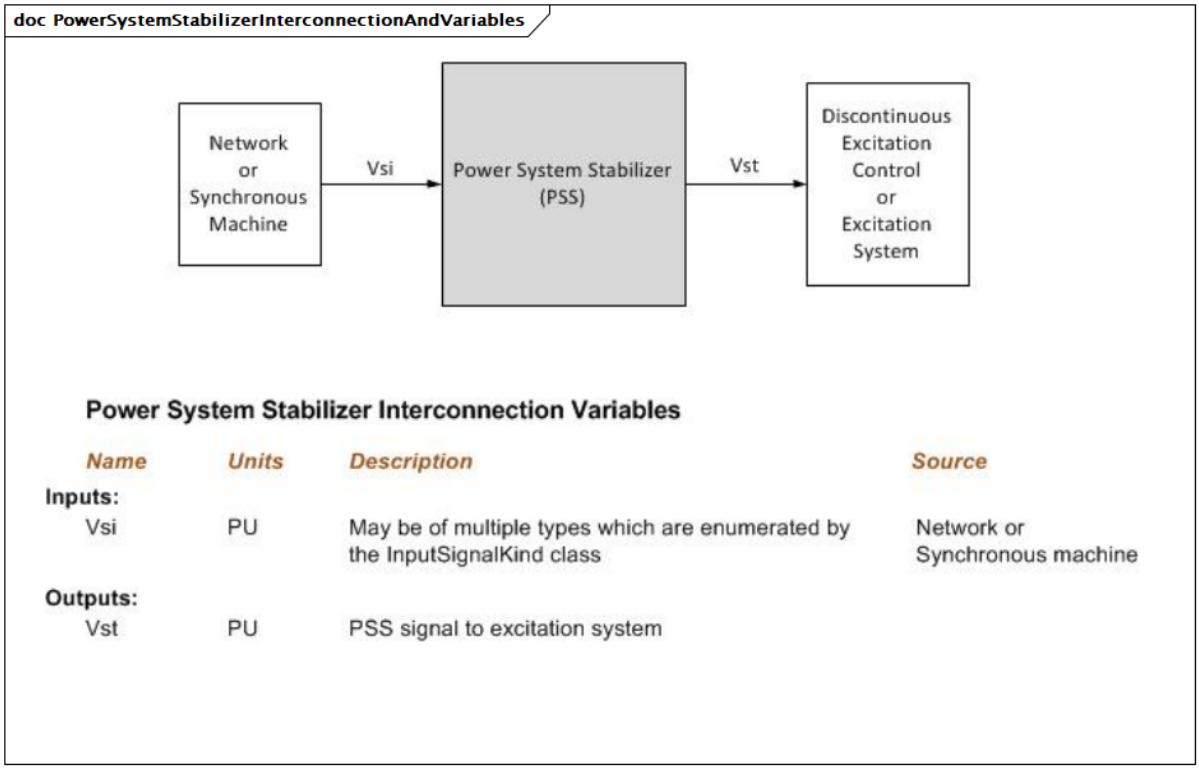

Figure 188 – PowerSystemStabilizerInterconnectionAndVariables

Figure 188: This diagram illustrates the inputs and outputs for a power system stabilizer.

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Diagram details:

1) If there is no discontinuous excitation control function block present, then output is Vs, not Vst
2) Vs is always initialized to zero by the excitation system model.
3) If bus voltage frequency is not available from network, the model can calculate it as derivative of the bus voltage angle.

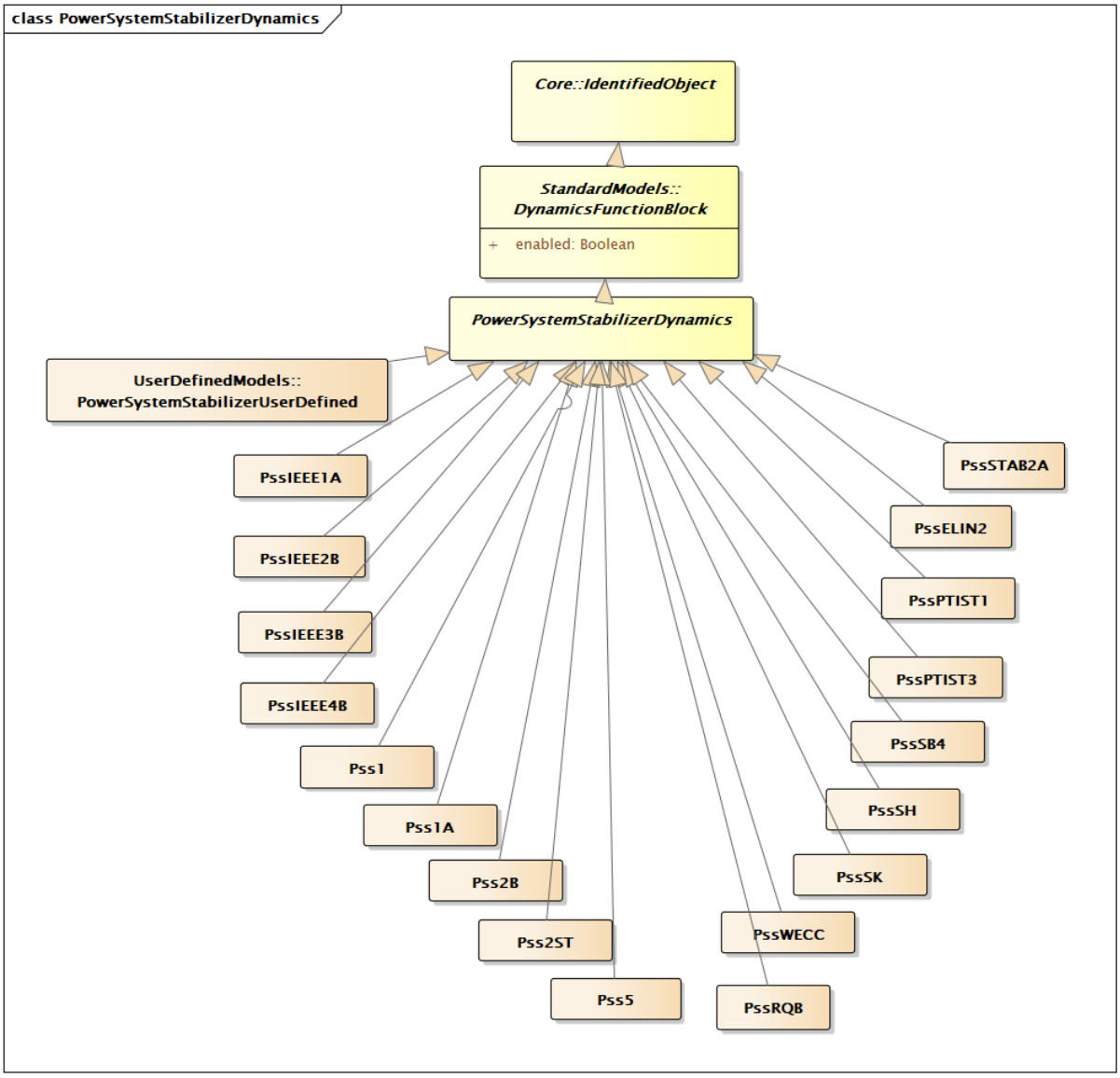

Figure 189 – Class diagram PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics

Figure 189: This diagram shows dynamic standard model power system stabilizer classes and their inheritance from the PowerSystemStabilizerDynamics class.

14.3.3.12.2 (abstract) PowerSystemStabilizerDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Power system stabilizer function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

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Table 890 shows all attributes of PowerSystemStabilizerDynamics.

Table 890 – Attributes of PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 891 shows all association ends of PowerSystemStabilizerDynamics with other classes.

Table 891 – Association ends of PowerSystemStabilizerDynamics:: PowerSystemStabilizerDynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this power system stabilizer model is associated.

14.3.3.12.3 PssIEEE1A

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type PSS1A power system stabilizer model. PSS1A is the generalized form of a PSS with a single input signal.

Reference: IEEE 1A 421.5:2005, 8.1.

Table 892 shows all attributes of PssIEEE1A.

Table 892 – Attributes of PowerSystemStabilizerDynamics::PssIEEE1A

name	mult	type	description
inputSignalType	1..1	InputSignalKind	Type of input signal (rotorAngularFrequencyDeviation, generatorElectricalPower, or busFrequencyDeviation). Typical value = rotorAngularFrequencyDeviation.
a1	1..1	PU	PSS signal conditioning frequency filter constant (A1). Typical value = 0,061.
a2	1..1	PU	PSS signal conditioning frequency filter constant (A2). Typical value = 0,0017.
t1	1..1	Seconds	Lead/lag time constant (T1) (>= 0). Typical value = 0,3.
t2	1..1	Seconds	Lead/lag time constant (T2) (>= 0). Typical value = 0,03.
t3	1..1	Seconds	Lead/lag time constant (T3) (>= 0). Typical value = 0,3.
t4	1..1	Seconds	Lead/lag time constant (T4) (>= 0). Typical value = 0,03.
t5	1..1	Seconds	Washout time constant (T5) (>= 0). Typical value = 10.

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name	mult	type	description
t6	1..1	Seconds	Transducer time constant (T6) (>= 0). Typical value = 0,01.
ks	1..1	PU	Stabilizer gain (Ks). Typical value = 5.
vrmax	1..1	PU	Maximum stabilizer output (Vrmax) (> PssIEEE1A.vrmin). Typical value = 0,05.
vrmin	1..1	PU	Minimum stabilizer output (Vrmin) (< PssIEEE1A.vrmax). Typical value = -0,05.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 893 shows all association ends of PssIEEE1A with other classes.

Table 893 – Association ends of PowerSystemStabilizerDynamics:: PssIEEE1A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.4 PssIEEE2B

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type PSS2B power system stabilizer model. This stabilizer model is designed to represent a variety of dual-input stabilizers, which normally use combinations of power and speed or frequency to derive the stabilizing signal.

Reference: IEEE 2B 421.5:2005, 8.2.

Table 894 shows all attributes of PssIEEE2B.

Table 894 – Attributes of PowerSystemStabilizerDynamics::PssIEEE2B

name	mult	type	description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1 (rotorAngularFrequencyDeviation, busFrequencyDeviation). Typical value = rotorAngularFrequencyDeviation.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2 (generatorElectricalPower). Typical value = generatorElectricalPower.
vsi1max	1..1	PU	Input signal #1 maximum limit (Vsi1max) (> PssIEEE2B.vsi1min). Typical value = 2.
vsi1min	1..1	PU	Input signal #1 minimum limit (Vsi1min) (< PssIEEE2B.vsi1max). Typical value = -2.
tw1	1..1	Seconds	First washout on signal #1 (Tw1) (>= 0). Typical value = 2.
tw2	1..1	Seconds	Second washout on signal #1 (Tw2) (>= 0). Typical value = 2.

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name	mult	type	description
vsi2max	1..1	PU	Input signal #2 maximum limit (Vsi2max) (> PssIEEE2B.vsi2min). Typical value = 2.
vsi2min	1..1	PU	Input signal #2 minimum limit (Vsi2min) (< PssIEEE2B.vsi2max). Typical value = -2.
tw3	1..1	Seconds	First washout on signal #2 (Tw3) (>= 0). Typical value = 2.
tw4	1..1	Seconds	Second washout on signal #2 (Tw4) (>= 0). Typical value = 0.
t1	1..1	Seconds	Lead/lag time constant (T1) (>= 0). Typical value = 0,12.
t2	1..1	Seconds	Lead/lag time constant (T2) (>= 0). Typical value = 0,02.
t3	1..1	Seconds	Lead/lag time constant (T3) (>= 0). Typical value = 0,3.
t4	1..1	Seconds	Lead/lag time constant (T4) (>= 0). Typical value = 0,02.
t6	1..1	Seconds	Time constant on signal #1 (T6) (>= 0). Typical value = 0.
t7	1..1	Seconds	Time constant on signal #2 (T7) (>= 0). Typical value = 2.
t8	1..1	Seconds	Lead of ramp tracking filter (T8) (>= 0). Typical value = 0,2.
t9	1..1	Seconds	Lag of ramp tracking filter (T9) (>= 0). Typical value = 0,1.
t10	1..1	Seconds	Lead/lag time constant (T10) (>= 0). Typical value = 0.
t11	1..1	Seconds	Lead/lag time constant (T11) (>= 0). Typical value = 0.
ks1	1..1	PU	Stabilizer gain (Ks1). Typical value = 12.
ks2	1..1	PU	Gain on signal #2 (Ks2). Typical value = 0,2.
ks3	1..1	PU	Gain on signal #2 input before ramp-tracking filter (Ks3). Typical value = 1.
n	1..1	Integer	Order of ramp tracking filter (N). Typical value = 1.
m	1..1	Integer	Denominator order of ramp tracking filter (M). Typical value = 5.
vstmax	1..1	PU	Stabilizer output maximum limit (Vstmax) (> PssIEEE2B.vstmin). Typical value = 0,1.
vstmin	1..1	PU	Stabilizer output minimum limit (Vstmin) (< PssIEEE2B.vstmax). Typical value = -0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 895 shows all association ends of PssIEEE2B with other classes.

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Table 895 – Association ends of PowerSystemStabilizerDynamics:: PssIEEE2B with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.5 PssIEEE3B

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type PSS3B power system stabilizer model. The PSS model PSS3B has dual inputs of electrical power and rotor angular frequency deviation. The signals are used to derive an equivalent mechanical power signal.

This model has 2 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): the first one is of rotorAngleFrequencyDeviation type and the second one is of generatorElectricalPower type.

Reference: IEEE 3B 421.5:2005, 8.3.

Table 896 shows all attributes of PssIEEE3B.

Table 896 – Attributes of PowerSystemStabilizerDynamics::PssIEEE3B

name	mult	type	description
t1	1..1	Seconds	Transducer time constant (T1) (>= 0). Typical value = 0,012.
t2	1..1	Seconds	Transducer time constant (T2) (>= 0). Typical value = 0,012.
tw1	1..1	Seconds	Washout time constant (Tw1) (>= 0). Typical value = 0,3.
tw2	1..1	Seconds	Washout time constant (Tw2) (>= 0). Typical value = 0,3.
tw3	1..1	Seconds	Washout time constant (Tw3) (>= 0). Typical value = 0,6.
ks1	1..1	PU	Gain on signal # 1 (Ks1). Typical value = -0,602.
ks2	1..1	PU	Gain on signal # 2 (Ks2). Typical value = 30,12.
a1	1..1	PU	Notch filter parameter (A1). Typical value = 0,359.
a2	1..1	PU	Notch filter parameter (A2). Typical value = 0,586.
a3	1..1	PU	Notch filter parameter (A3). Typical value = 0,429.
a4	1..1	PU	Notch filter parameter (A4). Typical value = 0,564.
a5	1..1	PU	Notch filter parameter (A5). Typical value = 0,001.
a6	1..1	PU	Notch filter parameter (A6). Typical value = 0.
a7	1..1	PU	Notch filter parameter (A7). Typical value = 0,031.
a8	1..1	PU	Notch filter parameter (A8). Typical value = 0.
vstmax	1..1	PU	Stabilizer output maximum limit (Vstmax) (> PssIEEE3B.vstmin). Typical value = 0,1.

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name	mult	type	description
vstmin	1..1	PU	Stabilizer output minimum limit (Vstmin) (< PssIEEE3B.vstmax). Typical value = -0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 897 shows all association ends of PssIEEE3B with other classes.

Table 897 – Association ends of PowerSystemStabilizerDynamics:: PssIEEE3B with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.6 PssIEEE4B

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE 421.5:2005 type PSS4B power system stabilizer. The PSS4B model represents a structure based on multiple working frequency bands. Three separate bands, respectively dedicated to the low-, intermediate- and high-frequency modes of oscillations, are used in this delta omega (speed input) PSS.

There is an error in the in IEEE 421.5:2005 PSS4B model: the Pe input should read –Pe. This implies that the input Pe needs to be multiplied by -1.

Reference: IEEE 4B 421.5:2005, 8.4.

Parameter details:

1) This model has 2 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): the first one is of rotorAngleFrequencyDeviation type and the second one is of generatorElectricalPower type.

Table 898 shows all attributes of PssIEEE4B.

Table 898 – Attributes of PowerSystemStabilizerDynamics::PssIEEE4B

name	mult	type	description
bwh1	1..1	Float	Notch filter 1 (high-frequency band): three dB bandwidth (Bwi).
bwh2	1..1	Float	Notch filter 2 (high-frequency band): three dB bandwidth (Bwi).
bwl1	1..1	Float	Notch filter 1 (low-frequency band): three dB bandwidth (Bwi).
bwl2	1..1	Float	Notch filter 2 (low-frequency band): three dB bandwidth (Bwi).
kh	1..1	PU	High band gain (KH). Typical value = 120.
kh1	1..1	PU	High band differential filter gain (KH1). Typical value = 66.

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name	mult	type	description
kh11	1..1	PU	High band first lead-lag blocks coefficient (KH11). Typical value = 1.
kh17	1..1	PU	High band first lead-lag blocks coefficient (KH17). Typical value = 1.
kh2	1..1	PU	High band differential filter gain (KH2). Typical value = 66.
ki	1..1	PU	Intermediate band gain (KI). Typical value = 30.
ki1	1..1	PU	Intermediate band differential filter gain (KI1). Typical value = 66.
ki11	1..1	PU	Intermediate band first lead-lag blocks coefficient (KI11). Typical value = 1.
ki17	1..1	PU	Intermediate band first lead-lag blocks coefficient (KI17). Typical value = 1.
ki2	1..1	PU	Intermediate band differential filter gain (KI2). Typical value = 66.
kl	1..1	PU	Low band gain (KL). Typical value = 7.5.
kl1	1..1	PU	Low band differential filter gain (KL1). Typical value = 66.
kl11	1..1	PU	Low band first lead-lag blocks coefficient (KL11). Typical value = 1.
kl17	1..1	PU	Low band first lead-lag blocks coefficient (KL17). Typical value = 1.
kl2	1..1	PU	Low band differential filter gain (KL2). Typical value = 66.
omeganh1	1..1	Float	Notch filter 1 (high-frequency band): filter frequency (omegani).
omeganh2	1..1	Float	Notch filter 2 (high-frequency band): filter frequency (omegani).
omeganl1	1..1	Float	Notch filter 1 (low-frequency band): filter frequency (omegani).
omeganl2	1..1	Float	Notch filter 2 (low-frequency band): filter frequency (omegani).
th1	1..1	Seconds	High band time constant (TH1) (>= 0). Typical value = 0,01513.
th10	1..1	Seconds	High band time constant (TH10) (>= 0). Typical value = 0.
th11	1..1	Seconds	High band time constant (TH11) (>= 0). Typical value = 0.
th12	1..1	Seconds	High band time constant (TH12) (>= 0). Typical value = 0.
th2	1..1	Seconds	High band time constant (TH2) (>= 0). Typical value = 0,01816.
th3	1..1	Seconds	High band time constant (TH3) (>= 0). Typical value = 0.
th4	1..1	Seconds	High band time constant (TH4) (>= 0). Typical value = 0.
th5	1..1	Seconds	High band time constant (TH5) (>= 0). Typical value = 0.
th6	1..1	Seconds	High band time constant (TH6) (>= 0). Typical value = 0.
th7	1..1	Seconds	High band time constant (TH7) (>= 0). Typical value = 0,01816.
th8	1..1	Seconds	High band time constant (TH8) (>= 0). Typical value = 0,02179.

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name	mult	type	description
th9	1..1	Seconds	High band time constant (TH9) (>= 0). Typical value = 0.
ti1	1..1	Seconds	Intermediate band time constant (TI1) (>= 0). Typical value = 0,173.
ti10	1..1	Seconds	Intermediate band time constant (TI10) (>= 0). Typical value = 0.
ti11	1..1	Seconds	Intermediate band time constant (TI11) (>= 0). Typical value = 0.
ti12	1..1	Seconds	Intermediate band time constant (TI12) (>= 0). Typical value = 0.
ti2	1..1	Seconds	Intermediate band time constant (TI2) (>= 0). Typical value = 0,2075.
ti3	1..1	Seconds	Intermediate band time constant (TI3) (>= 0). Typical value = 0.
ti4	1..1	Seconds	Intermediate band time constant (TI4) (>= 0). Typical value = 0.
ti5	1..1	Seconds	Intermediate band time constant (TI5) (>= 0). Typical value = 0.
ti6	1..1	Seconds	Intermediate band time constant (TI6) (>= 0). Typical value = 0.
ti7	1..1	Seconds	Intermediate band time constant (TI7) (>= 0). Typical value = 0,2075.
ti8	1..1	Seconds	Intermediate band time constant (TI8) (>= 0). Typical value = 0,2491.
ti9	1..1	Seconds	Intermediate band time constant (TI9) (>= 0). Typical value = 0.
tl1	1..1	Seconds	Low band time constant (TL1) (>= 0). Typical value = 1,73.
tl10	1..1	Seconds	Low band time constant (TL10) (>= 0). Typical value = 0.
tl11	1..1	Seconds	Low band time constant (TL11) (>= 0). Typical value = 0.
tl12	1..1	Seconds	Low band time constant (TL12) (>= 0). Typical value = 0.
tl2	1..1	Seconds	Low band time constant (TL2) (>= 0). Typical value = 2,075.
tl3	1..1	Seconds	Low band time constant (TL3) (>= 0). Typical value = 0.
tl4	1..1	Seconds	Low band time constant (TL4) (>= 0). Typical value = 0.
tl5	1..1	Seconds	Low band time constant (TL5) (>= 0). Typical value = 0.
tl6	1..1	Seconds	Low band time constant (TL6) (>= 0). Typical value = 0.
tl7	1..1	Seconds	Low band time constant (TL7) (>= 0). Typical value = 2,075.
tl8	1..1	Seconds	Low band time constant (TL8) (>= 0). Typical value = 2,491.
tl9	1..1	Seconds	Low band time constant (TL9) (>= 0). Typical value = 0.
vhmax	1..1	PU	High band output maximum limit (VHmax) (> PssIEEE4B.vhmin). Typical value = 0,6.
vhmin	1..1	PU	High band output minimum limit (VHmin) (< PssIEEE4V.vhmax). Typical value = -0,6.
vimax	1..1	PU	Intermediate band output maximum limit (VImax) (> PssIEEE4B.vimin). Typical value = 0,6.

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name	mult	type	description
vimin	1..1	PU	Intermediate band output minimum limit (VImin) (< PssIEEE4B.vimax). Typical value = -0,6.
vlmax	1..1	PU	Low band output maximum limit (VLmax) (> PssIEEE4B.vlmin). Typical value = 0,075.
vlmin	1..1	PU	Low band output minimum limit (VLmin) (< PssIEEE4B.vlmax). Typical value = -0,075.
vstmax	1..1	PU	PSS output maximum limit (VSTmax) (> PssIEEE4B.vstmin). Typical value = 0,15.
vstmin	1..1	PU	PSS output minimum limit (VSTmin) (< PssIEEE4B.vstmax). Typical value = -0,15.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 899 shows all association ends of PssIEEE4B with other classes.

Table 899 – Association ends of PowerSystemStabilizerDynamics:: PssIEEE4B with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.7 Pss1

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Italian PSS with three inputs (speed, frequency, power).

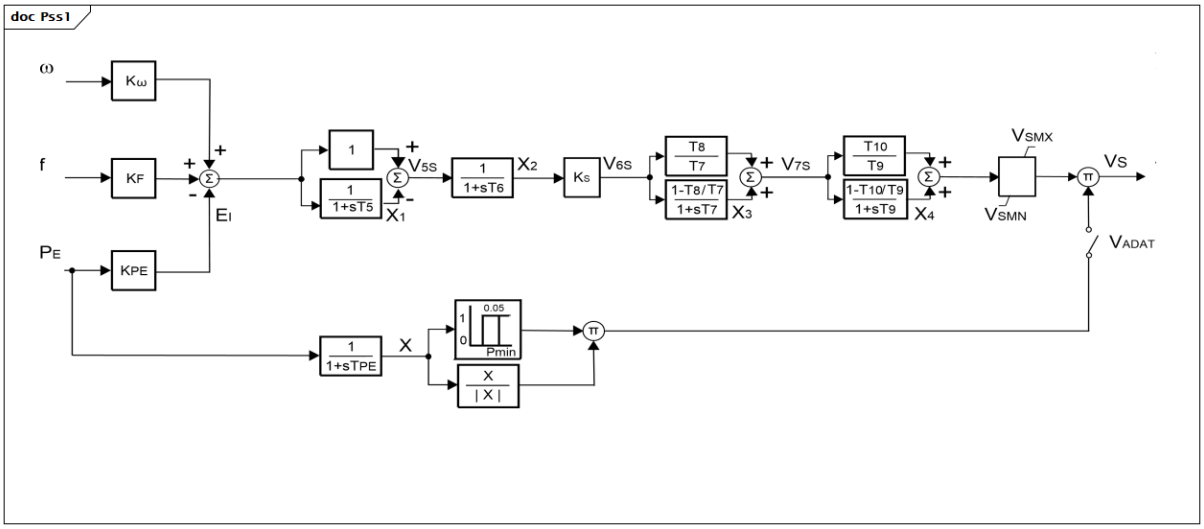

Figure 190 – Pss1

Figure 190: This diagram describes the Pss1 power system stabilizer model.

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Parameter details:

1) This model has 3 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): omega is of rotorSpeed type, f is of busFrequency type, and PE is of generatorElectricalPower type.
2) X is not a parameter, but rather an input signal to the block [X /|X|]. The block reflects the sign of Pe. When Pe is negative (for example, for a unit in pumping mode), the output of the block [X /|X|] is negative. When Pe is positive, the output of the block is positive.

Table 900 shows all attributes of Pss1.

Table 900 – Attributes of PowerSystemStabilizerDynamics::Pss1

name	mult	type	description
komega	1..1	Float	Shaft speed power input gain (Komega). Typical value = 0.
kf	1..1	Float	Frequency power input gain (KF). Typical value = 5.
kpe	1..1	Float	Electric power input gain (KPE). Typical value = 0,3.
pmin	1..1	PU	Minimum power PSS enabling (Pmin). Typical value = 0,25.
ks	1..1	Float	PSS gain (Ks). Typical value = 1.
vsmn	1..1	PU	Stabilizer output maximum limit (VSMN). Typical value = -0,06.
vsmx	1..1	PU	Stabilizer output minimum limit (VSMX). Typical value = 0,06.
tpe	1..1	Seconds	Electric power filter time constant (TPE) (>= 0). Typical value = 0,05.
t5	1..1	Seconds	Washout (T5) (>= 0). Typical value = 3,5.
t6	1..1	Seconds	Filter time constant (T6) (>= 0). Typical value = 0.
t7	1..1	Seconds	Lead/lag time constant (T7) (>= 0). If = 0, both blocks are bypassed. Typical value = 0.
t8	1..1	Seconds	Lead/lag time constant (T8) (>= 0). Typical value = 0.
t9	1..1	Seconds	Lead/lag time constant (T9) (>= 0). If = 0, both blocks are bypassed. Typical value = 0.
t10	1..1	Seconds	Lead/lag time constant (T10) (>= 0). Typical value = 0.
vadat	1..1	Boolean	Signal selector (VADAT). true = closed (generator power is greater than Pmin) false = open (Pe is smaller than Pmin). Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 901 shows all association ends of Pss1 with other classes.

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Table 901 – Association ends of PowerSystemStabilizerDynamics:: Pss1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.8 Pss1A

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Single input power system stabilizer. It is a modified version in order to allow representation of various vendors' implementations on PSS type 1A.

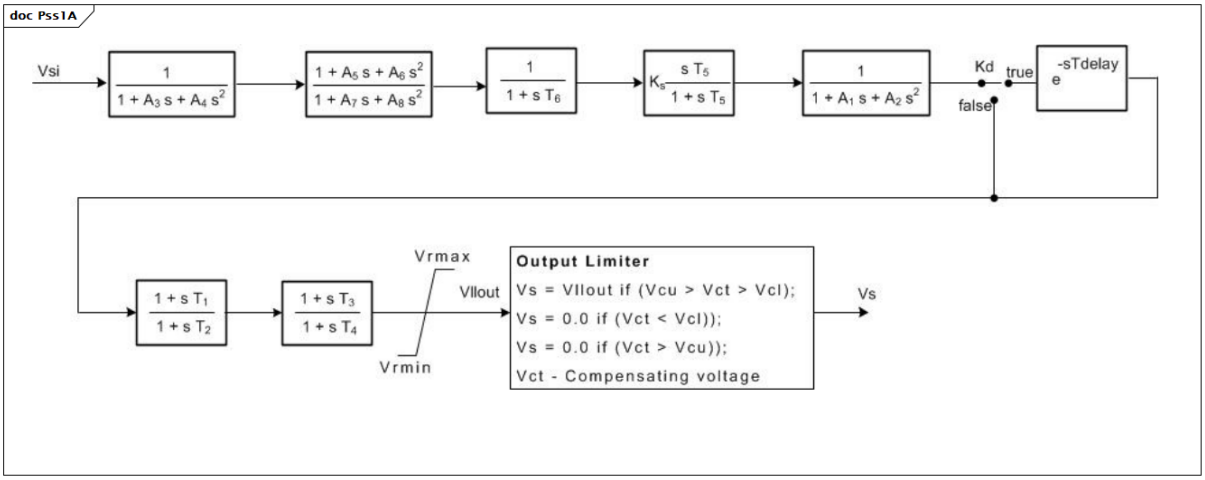

Figure 191 – Pss1A

Figure 191: This diagram describes the Pss1A power system stabilizer model.

Table 902 shows all attributes of Pss1A.

Table 902 – Attributes of PowerSystemStabilizerDynamics::Pss1A

name	mult	type	description
inputSignalType	1..1	InputSignalKind	Type of input signal (rotorAngularFrequencyDeviation, busFrequencyDeviation, generatorElectricalPower, generatorAcceleratingPower, busVoltage, or busVoltageDerivative).
a1	1..1	PU	Notch filter parameter (A1).
a2	1..1	PU	Notch filter parameter (A2).
t1	1..1	Seconds	Lead/lag time constant (T1) (>= 0).
t2	1..1	Seconds	Lead/lag time constant (T2) (>= 0).
t3	1..1	Seconds	Lead/lag time constant (T3) (>= 0).
t4	1..1	Seconds	Lead/lag time constant (T4) (>= 0).
t5	1..1	Seconds	Washout time constant (T5) (>= 0).
t6	1..1	Seconds	Transducer time constant (T6) (>= 0).
ks	1..1	PU	Stabilizer gain (Ks).

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name	mult	type	description
vrmax	1..1	PU	Maximum stabilizer output (Vrmax) (> Pss1A.vrmin).
vrmin	1..1	PU	Minimum stabilizer output (Vrmin) (< Pss1A.vrmax).
vcu	1..1	PU	Stabilizer input cutoff threshold (Vcu).
vcl	1..1	PU	Stabilizer input cutoff threshold (Vcl).
a3	1..1	PU	Notch filter parameter (A3).
a4	1..1	PU	Notch filter parameter (A4).
a5	1..1	PU	Notch filter parameter (A5).
a6	1..1	PU	Notch filter parameter (A6).
a7	1..1	PU	Notch filter parameter (A7).
a8	1..1	PU	Notch filter parameter (A8).
kd	1..1	Boolean	Selector (Kd). true = e-sTdelay used false = e-sTdelay not used.
tdelay	1..1	Seconds	Time constant (Tdelay) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 903 shows all association ends of Pss1A with other classes.

Table 903 – Association ends of PowerSystemStabilizerDynamics:: Pss1A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.9 Pss2B

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Modified IEEE PSS2B. Extra lead/lag (or rate) block added at end (up to 4 lead/lags total).

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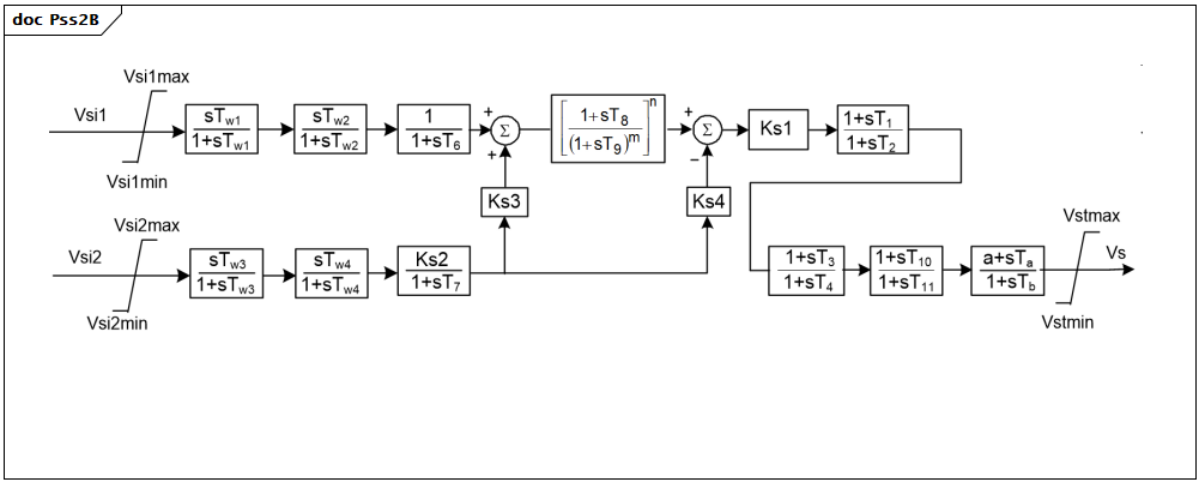

Figure 192 – Pss2B

Figure 192: This diagram describes the Pss2B power system stabilizer model.

Parameter details:

1) This model has 2 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): Vsi1 is of rotorSpeed type and Vsi2 is of generatorElectricalPower type.

Table 904 shows all attributes of Pss2B.

Table 904 – Attributes of PowerSystemStabilizerDynamics::Pss2B

name	mult	type	description
vsi1max	1..1	PU	Input signal #1 maximum limit (Vsi1max) (> Pss2B.vsi1min). Typical value = 2.
vsi1min	1..1	PU	Input signal #1 minimum limit (Vsi1min) (< Pss2B.vsi1max). Typical value = -2.
tw1	1..1	Seconds	First washout on signal #1 (Tw1) (>= 0). Typical value = 2.
tw2	1..1	Seconds	Second washout on signal #1 (Tw2) (>= 0). Typical value = 2.
vsi2max	1..1	PU	Input signal #2 maximum limit (Vsi2max) (> Pss2B.vsi2min). Typical value = 2.
vsi2min	1..1	PU	Input signal #2 minimum limit (Vsi2min) (< Pss2B.vsi2max). Typical value = -2.
tw3	1..1	Seconds	First washout on signal #2 (Tw3) (>= 0). Typical value = 2.
tw4	1..1	Seconds	Second washout on signal #2 (Tw4) (>= 0). Typical value = 0.
t1	1..1	Seconds	Lead/lag time constant (T1) (>= 0). Typical value = 0,12.
t2	1..1	Seconds	Lead/lag time constant (T2) (>= 0). Typical value = 0,02.
t3	1..1	Seconds	Lead/lag time constant (T3) (>= 0). Typical value = 0,3.
t4	1..1	Seconds	Lead/lag time constant (T4) (>= 0). Typical value = 0,02.
t6	1..1	Seconds	Time constant on signal #1 (T6) (>= 0). Typical value = 0.

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name	mult	type	description
t7	1..1	Seconds	Time constant on signal #2 (T7) (>= 0). Typical value = 2.
t8	1..1	Seconds	Lead of ramp tracking filter (T8) (>= 0). Typical value = 0,2.
t9	1..1	Seconds	Lag of ramp tracking filter (T9) (>= 0). Typical value = 0,1.
t10	1..1	Seconds	Lead/lag time constant (T10) (>= 0). Typical value = 0.
t11	1..1	Seconds	Lead/lag time constant (T11) (>= 0). Typical value = 0.
ks1	1..1	PU	Stabilizer gain (Ks1). Typical value = 12.
ks2	1..1	PU	Gain on signal #2 (Ks2). Typical value = 0,2.
ks3	1..1	PU	Gain on signal #2 input before ramp-tracking filter (Ks3). Typical value = 1.
ks4	1..1	PU	Gain on signal #2 input after ramp-tracking filter (Ks4). Typical value = 1.
n	1..1	Integer	Order of ramp tracking filter (n). Typical value = 1.
m	1..1	Integer	Denominator order of ramp tracking filter (m). Typical value = 5.
vstmax	1..1	PU	Stabilizer output maximum limit (Vstmax) (> Pss2B.vstmin). Typical value = 0,1.
vstmin	1..1	PU	Stabilizer output minimum limit (Vstmin) (< Pss2B.vstmax). Typical value = -0,1.
a	1..1	Float	Numerator constant (a). Typical value = 1.
ta	1..1	Seconds	Lead constant (Ta) (>= 0). Typical value = 0.
tb	1..1	Seconds	Lag time constant (Tb) (>= 0). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 905 shows all association ends of Pss2B with other classes.

Table 905 – Association ends of PowerSystemStabilizerDynamics:: Pss2B with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.10 Pss2ST

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

PTI microprocessor-based stabilizer type 1.

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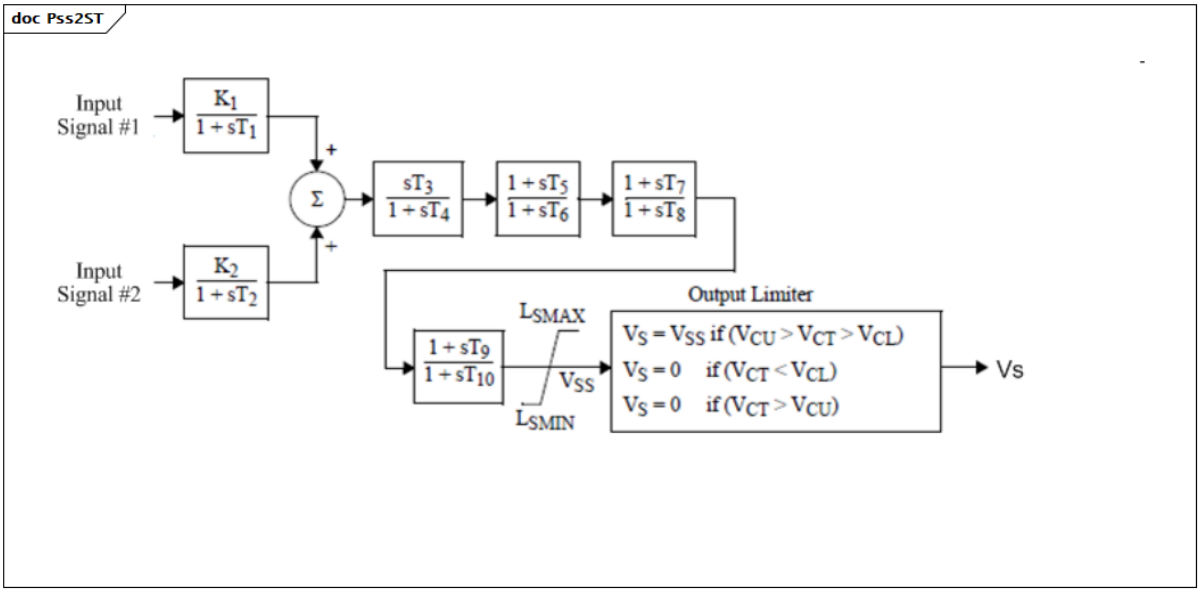

Figure 193 – Pss2ST

Figure 193: This diagram describes the Pss2ST power system stabilizer model.

Table 906 shows all attributes of Pss2ST.

Table 906 – Attributes of PowerSystemStabilizerDynamics::Pss2ST

name	mult	type	description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1 (rotorAngularFrequencyDeviation, busFrequencyDeviation, generatorElectricalPower, generatorAcceleratingPower, busVoltage, or busVoltageDerivative – shall be different than Pss2ST.inputSignal2Type). Typical value = rotorAngularFrequencyDeviation.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2 (rotorAngularFrequencyDeviation, busFrequencyDeviation, generatorElectricalPower, generatorAcceleratingPower, busVoltage, or busVoltageDerivative – shall be different than Pss2ST.inputSignal1Type). Typical value = busVoltageDerivative.
k1	1..1	PU	Gain (K1).
k2	1..1	PU	Gain (K2).
t1	1..1	Seconds	Time constant (T1) (>= 0).
t2	1..1	Seconds	Time constant (T2) (>= 0).
t3	1..1	Seconds	Time constant (T3) (>= 0).
t4	1..1	Seconds	Time constant (T4) (>= 0).
t5	1..1	Seconds	Time constant (T5) (>= 0).
t6	1..1	Seconds	Time constant (T6) (>= 0).
t7	1..1	Seconds	Time constant (T7) (>= 0).
t8	1..1	Seconds	Time constant (T8) (>= 0).
t9	1..1	Seconds	Time constant (T9) (>= 0).
t10	1..1	Seconds	Time constant (T10) (>= 0).

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name	mult	type	description
lsmax	1..1	PU	Limiter (LSMAX) (> Pss2ST.lsmin).
lsmin	1..1	PU	Limiter (LSMIN) (< Pss2ST.lsmax).
vcu	1..1	PU	Cutoff limiter (VCU).
vcl	1..1	PU	Cutoff limiter (VCL).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 907 shows all association ends of Pss2ST with other classes.

Table 907 – Association ends of PowerSystemStabilizerDynamics:: Pss2ST with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.11 Pss5

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Detailed Italian PSS.

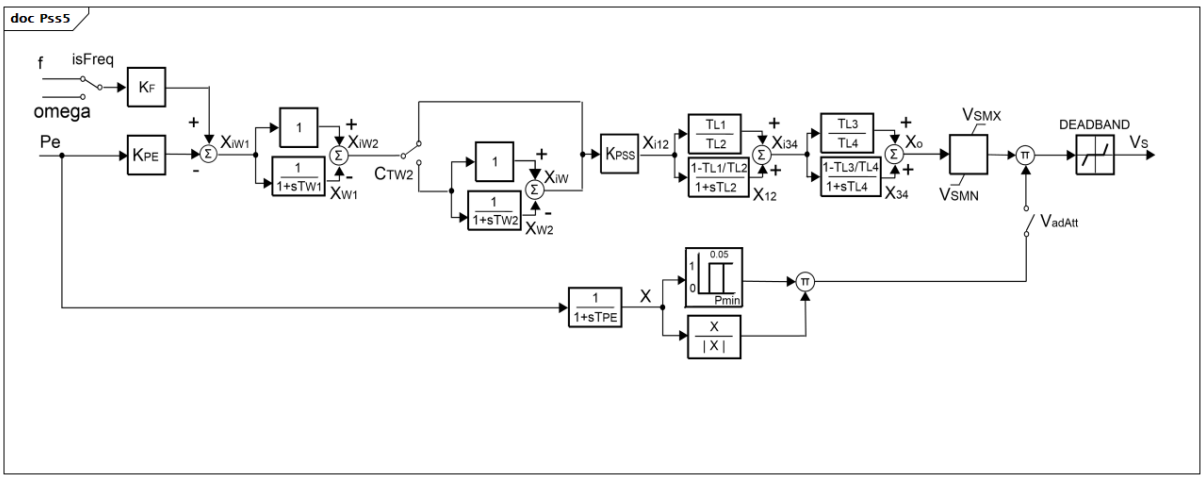

Figure 194 – Pss5

Figure 194: This diagram describes the Pss5 power system stabilizer model.

Parameter Details:

1) This model has 2 input signals. The type of first input signal is controlled by the isFreq switch (allowing the input to be either f with a type of busFrequency or omega with a type of rotorSpeed). The type of the second one, Pe, is generatorElectricalPower. (Types are described here as they are defined by InputSignalKind values).

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2) X is not a parameter, but rather an input signal to the block [X /|X|]. The block reflects the sign of Pe. When Pe is negative (for example, for a unit in pumping mode), the output of the block [X /|X|] is negative. When Pe is positive, the output of the block is positive.

Table 908 shows all attributes of Pss5.

Table 908 – Attributes of PowerSystemStabilizerDynamics::Pss5

name	mult	type	description
kpe	1..1	Float	Electric power input gain (KPE). Typical value = 0,3.
kf	1..1	Float	Frequency/shaft speed input gain (KF). Typical value = 5.
isfreq	1..1	Boolean	Selector for frequency/shaft speed input (isFreq). true = speed (same meaning as InputSignaKind.rotorSpeed) false = frequency (same meaning as InputSignalKind.busFrequency). Typical value = true (same meaning as InputSignalKind.rotorSpeed).
kpss	1..1	Float	PSS gain (KPSS). Typical value = 1.
ctw2	1..1	Boolean	Selector for second washout enabling (CTW2). true = second washout filter is bypassed false = second washout filter in use. Typical value = true.
tw1	1..1	Seconds	First washout (TW1) (>= 0). Typical value = 3,5.
tw2	1..1	Seconds	Second washout (TW2) (>= 0). Typical value = 0.
tl1	1..1	Seconds	Lead/lag time constant (TL1) (>= 0). Typical value = 0.
tl2	1..1	Seconds	Lead/lag time constant (TL2) (>= 0). If = 0, both blocks are bypassed. Typical value = 0.
tl3	1..1	Seconds	Lead/lag time constant (TL3) (>= 0). Typical value = 0.
tl4	1..1	Seconds	Lead/lag time constant (TL4) (>= 0). If = 0, both blocks are bypassed. Typical value = 0.
vsmn	1..1	PU	Stabilizer output maximum limit (VSMN). Typical value = -0,1.
vsmx	1..1	PU	Stabilizer output minimum limit (VSMX). Typical value = 0,1.
tpe	1..1	Seconds	Electric power filter time constant (TPE) (>= 0). Typical value = 0,05.
pmin	1..1	PU	Minimum power PSS enabling (Pmin). Typical value = 0,25.
deadband	1..1	PU	Stabilizer output deadband (DEADBAND). Typical value = 0.
vadat	1..1	Boolean	Signal selector (VadAtt). true = closed (generator power is greater than Pmin) false = open (Pe is smaller than Pmin). Typical value = true.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	0..1	String	inherited from: IdentifiedObject

Table 909 shows all association ends of Pss5 with other classes.

Table 909 – Association ends of PowerSystemStabilizerDynamics:: Pss5 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.12 PssELIN2

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Power system stabilizer typically associated with ExcELIN2 (though PssIEEE2B or Pss2B can also be used).

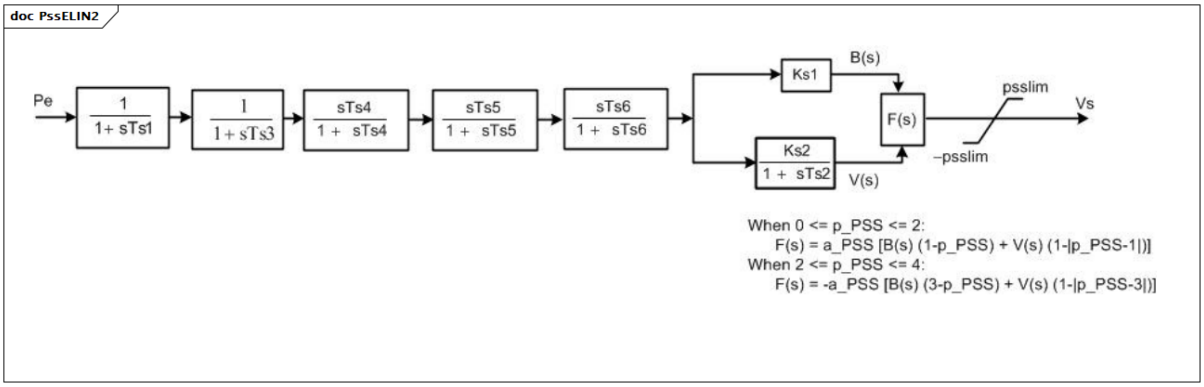

Figure 195 – PssELIN2

Figure 195: This diagram describes the PssELIN2 power system stabilizer model.

Parameter details:

1) The input Pe has the fixed type (expressed in terms of inputSignalKind value) of generatorElectricalPower.

Table 910 shows all attributes of PssELIN2.

Table 910 – Attributes of PowerSystemStabilizerDynamics::PssELIN2

name	mult	type	description
ts1	1..1	Seconds	Time constant (Ts1) (>= 0). Typical value = 0.
ts2	1..1	Seconds	Time constant (Ts2) (>= 0). Typical value = 1.
ts3	1..1	Seconds	Time constant (Ts3) (>= 0). Typical value = 1.
ts4	1..1	Seconds	Time constant (Ts4) (>= 0). Typical value = 0,1.
ts5	1..1	Seconds	Time constant (Ts5) (>= 0). Typical value = 0.
ts6	1..1	Seconds	Time constant (Ts6) (>= 0). Typical value = 1.

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name	mult	type	description
ks1	1..1	PU	Gain (Ks1). Typical value = 1.
ks2	1..1	PU	Gain (Ks2). Typical value = 0,1.
ppss	1..1	PU	Coefficient (p_PSS) (>= 0 and <= 4). Typical value = 0,1.
apss	1..1	PU	Coefficient (a_PSS). Typical value = 0,1.
psslim	1..1	PU	PSS limiter (psslim). Typical value = 0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 911 shows all association ends of PssELIN2 with other classes.

Table 911 – Association ends of PowerSystemStabilizerDynamics:: PssELIN2 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.13 PssPTIST1

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

PTI microprocessor-based stabilizer type 1.

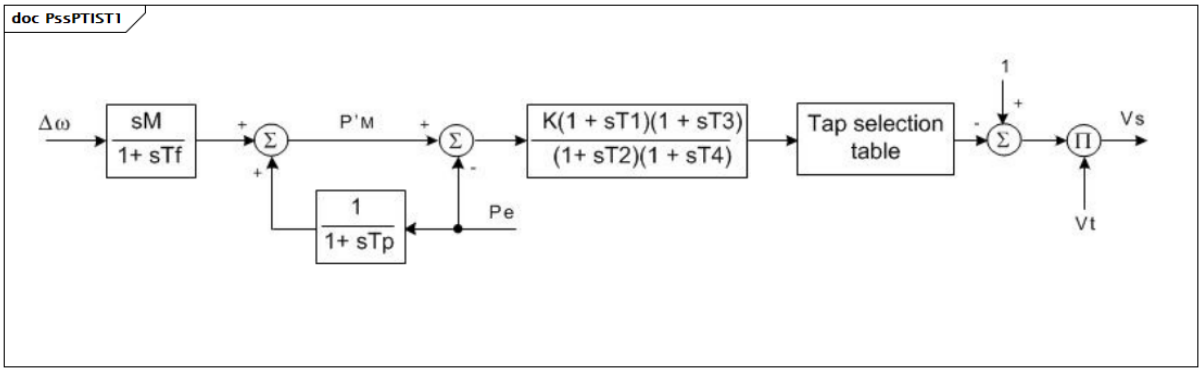

Figure 196 – PssPTIST1

Figure 196: This diagram describes the PssPTIST1 power system stabilizer model.

Parameter details:

1) Tap selection table enables the output control signal to be used to select an autotransformer tap position, which, in turn, modulates the feedback terminal voltage (digital output option).
2) This model has 2 input signals. The type (expressed in terms of InputSignalKind values) of the first signal (delta omega) is rotorAngularFrequency and the type of the second one, Pe, is generatorElectricalPower.

Table 912 shows all attributes of PssPTIST1.

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Table 912 – Attributes of PowerSystemStabilizerDynamics::PssPTIST1

name	mult	type	description
m	1..1	PU	(M). M = 2 x H. Typical value = 5.
tf	1..1	Seconds	Time constant (Tf) (>= 0). Typical value = 0,2.
tp	1..1	Seconds	Time constant (Tp) (>= 0). Typical value = 0,2.
t1	1..1	Seconds	Time constant (T1) (>= 0). Typical value = 0,3.
t2	1..1	Seconds	Time constant (T2) (>= 0). Typical value = 1.
t3	1..1	Seconds	Time constant (T3) (>= 0). Typical value = 0,2.
t4	1..1	Seconds	Time constant (T4) (>= 0). Typical value = 0,05.
k	1..1	PU	Gain (K). Typical value = 9.
dtf	1..1	Seconds	Time step frequency calculation (deltatf) (>= 0). Typical value = 0,025.
dtc	1..1	Seconds	Time step related to activation of controls (deltatc) (>= 0). Typical value = 0,025.
dtp	1..1	Seconds	Time step active power calculation (deltatp) (>= 0). Typical value = 0,0125.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 913 shows all association ends of PssPTIST1 with other classes.

Table 913 – Association ends of PowerSystemStabilizerDynamics:: PssPTIST1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.14 PssPTIST3

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

PTI microprocessor-based stabilizer type 3.

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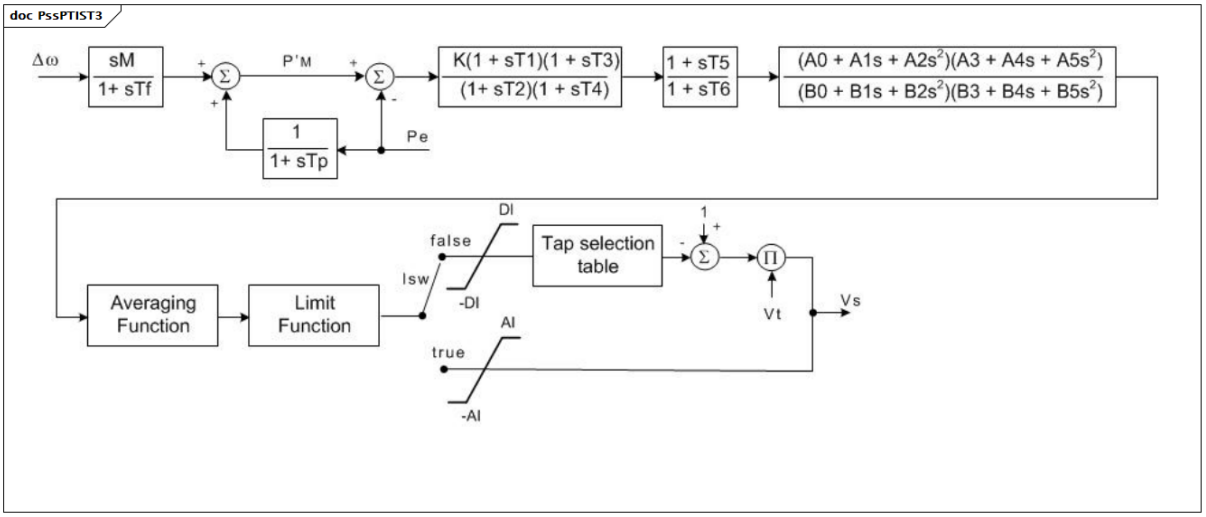

Figure 197 – PssPTIST3

Figure 197: This diagram describes the PssPTIST3 power system stabilizer model.

Parameter details:

1) This model has 2 input signals. The type (expressed in terms of InputSignalKind values) of the first signal (delta omega) is rotorAngularFrequency and the type of the second one, Pe, is generatorElectricalPower.
2) 'Tap selection table' enables the output control signal to be used to select an autotransformer tap position, which, in turn, modulates the feedback terminal voltage (digital output option).
3) 'Averaging Function' keeps a running average of the last NAV control outputs (clicks) up to a maximum of 16. This feature can be disabled by setting NAV equal to 1 (i.e., average 1 output). The last NAV outputs are stored in the output averaging table, VAR(1) to VAR(16). The sum of the last outputs is stored in the output accumulator. The averaging feature was added to reduce tap chattering under steady-state conditions. If the control output at any given click exceeds the threshold specified in Athres, the 'Averaging Function' will be automatically bypassed and that output signal will be passed directly to the 'Limit Function'. This feature has been added such that under transient conditions the stabilizer reverts to normal operation. When the absolute value of output signal is beyond the threshold, the output averaging table entries and accumulator are zeroed out. When the signal falls below the threshold, the table is rebuilt starting with the first signal within the threshold. If the output averaging is enabled, then the threshold value would probably be specified in the range from 0,002 to 0,02.
4) 'Limit Function' counts the number of control outputs that the output signal continuously exceeds a user-defined threshold as set in Lthres and, after the output signal has continuously exceeded the threshold for a number of user-defined counts as specified by NCL, the function ramps the tap position to nominal or the analogue output to zero over a user-specified number of counts as defined in NCR.

Table 914 shows all attributes of PssPTIST3.

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Table 914 – Attributes of PowerSystemStabilizerDynamics::PssPTIST3

name	mult	type	description
m	1..1	PU	(M). M = 2 x H. Typical value = 5.
tf	1..1	Seconds	Time constant (Tf) (>= 0). Typical value = 0,2.
tp	1..1	Seconds	Time constant (Tp) (>= 0). Typical value = 0,2.
t1	1..1	Seconds	Time constant (T1) (>= 0). Typical value = 0,3.
t2	1..1	Seconds	Time constant (T2) (>= 0). Typical value = 1.
t3	1..1	Seconds	Time constant (T3) (>= 0). Typical value = 0,2.
t4	1..1	Seconds	Time constant (T4) (>= 0). Typical value = 0,05.
k	1..1	PU	Gain (K). Typical value = 9.
dtf	1..1	Seconds	Time step frequency calculation (deltatf) (>= 0). Typical value = 0,025 (0,03 for 50 Hz).
dtc	1..1	Seconds	Time step related to activation of controls (deltatc) (>= 0). Typical value = 0,025 (0,03 for 50 Hz).
dtp	1..1	Seconds	Time step active power calculation (deltatp) (>= 0). Typical value = 0,0125 (0,015 for 50 Hz).
t5	1..1	Seconds	Time constant (T5) (>= 0).
t6	1..1	Seconds	Time constant (T6) (>= 0).
a0	1..1	PU	Filter coefficient (A0).
a1	1..1	PU	Limiter (Al).
a2	1..1	PU	Filter coefficient (A2).
b0	1..1	PU	Filter coefficient (B0).
b1	1..1	PU	Filter coefficient (B1).
b2	1..1	PU	Filter coefficient (B2).
a3	1..1	PU	Filter coefficient (A3).
a4	1..1	PU	Filter coefficient (A4).
a5	1..1	PU	Filter coefficient (A5).
b3	1..1	PU	Filter coefficient (B3).
b4	1..1	PU	Filter coefficient (B4).
b5	1..1	PU	Filter coefficient (B5).
athres	1..1	PU	Threshold value above which output averaging will be bypassed (Athres). Typical value = 0,005.
dl	1..1	PU	Limiter (Dl).
al	1..1	PU	Limiter (Al).
lthres	1..1	PU	Threshold value (Lthres).
pmin	1..1	PU	(Pmin).
isw	1..1	Boolean	Digital/analogue output switch (Isw). true = produce analogue output false = convert to digital output, using tap selection table.
nav	1..1	Float	Number of control outputs to average (NAV) (1 <= NAV <= 16). Typical value = 4.
ncl	1..1	Float	Number of counts at limit to active limit function (NCL) (> 0).
ncr	1..1	Float	Number of counts until reset after limit function is triggered (NCR).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock

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name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 915 shows all association ends of PssPTIST3 with other classes.

Table 915 – Association ends of PowerSystemStabilizerDynamics:: PssPTIST3 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.15 PssRQB

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Power system stabilizer type RQB. This power system stabilizer is intended to be used together with excitation system type ExcRQB, which is primarily used in nuclear or thermal generating units.

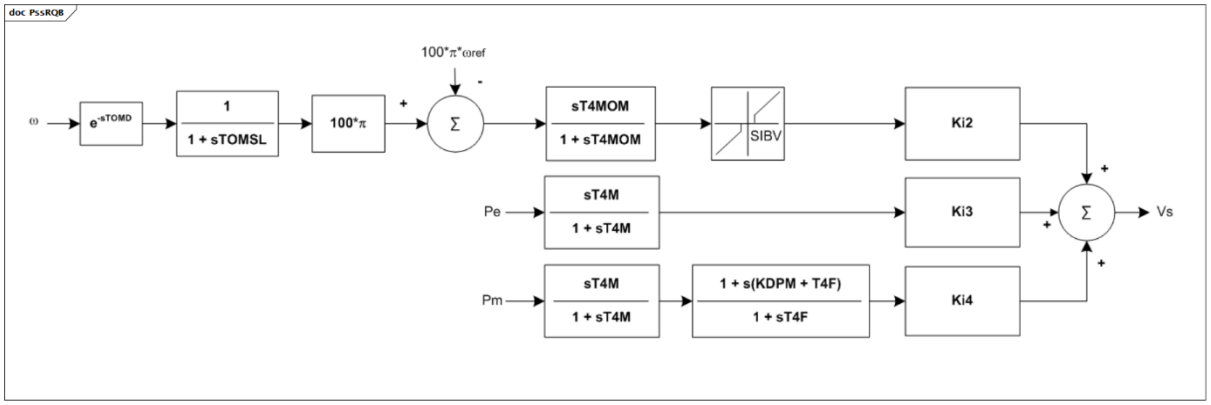

Figure 198 – PssRQB

Figure 198: This diagram describes the PssRQB power system stabilizer model.

Parameter details:

1) This model has 3 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): omega is of rotorSpeed type, Pe is of generatorElectricalPower type, and Pm is of generatorMechanicalPower type.
2) The block containing SIBV describes the following behaviour:
   – if input signal value < -SIBV, output signal = input + SIBV;
   – if input signal value is between -SIBV and SIBV, output signal = 0;
   – if input signal value > SIBV, output signal = input – SIBV.

Table 916 shows all attributes of PssRQB.

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Table 916 – Attributes of PowerSystemStabilizerDynamics::PssRQB

name	mult	type	description
ki2	1..1	Float	Speed input gain (Ki2). Typical value = 3,43.
ki3	1..1	Float	Electrical power input gain (Ki3). Typical value = -11,45.
ki4	1..1	Float	Mechanical power input gain (Ki4). Typical value = 11,86.
t4m	1..1	Seconds	Input time constant (T4M) (>= 0). Typical value = 5.
tomd	1..1	Seconds	Speed delay (TOMD) (>= 0). Typical value = 0,02.
tomsl	1..1	Seconds	Speed time constant (TOMSL) (>= 0). Typical value = 0,04.
t4mom	1..1	Seconds	Speed time constant (T4MOM) (>= 0). Typical value = 1,27.
sibv	1..1	PU	Speed deadband (SIBV). Typical value = 0,006.
kdpm	1..1	Float	Lead lag gain (KDPM). Typical value = 0,185.
t4f	1..1	Seconds	Lead lag time constant (T4F) (>= 0). Typical value = 0,045.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 917 shows all association ends of PssRQB with other classes.

Table 917 – Association ends of PowerSystemStabilizerDynamics:: PssRQB with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.16 PssSB4

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Power sensitive stabilizer model.

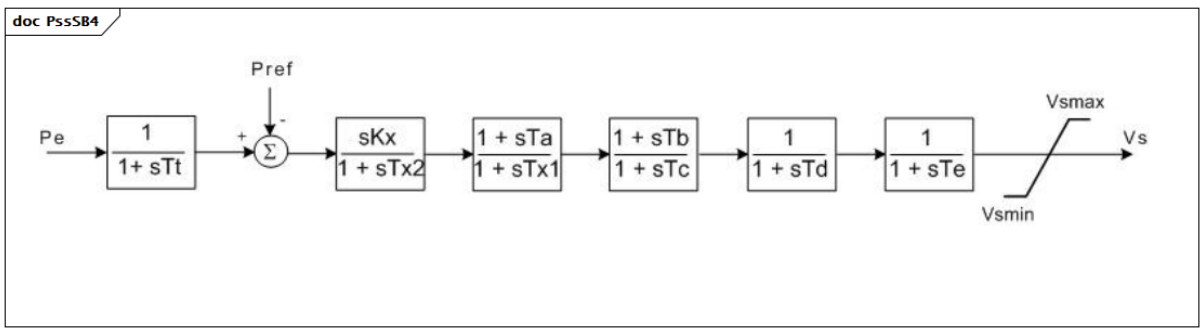

Figure 199 – PssSB4

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Figure 199: This diagram describes the PssSB4 power system stabilizer model.

Parameter details:

1) The input Pe has the fixed type (expressed in terms of inputSignalKind value) of generatorElectricalPower.

Table 918 shows all attributes of PssSB4.

Table 918 – Attributes of PowerSystemStabilizerDynamics::PssSB4

name	mult	type	description
tt	1..1	Seconds	Time constant (Tt) (>= 0). Typical value = 0,18.
kx	1..1	PU	Gain (Kx). Typical value = 2,7.
tx2	1..1	Seconds	Time constant (Tx2) (>= 0). Typical value = 5,0.
ta	1..1	Seconds	Time constant (Ta) (>= 0). Typical value = 0,37.
tx1	1..1	Seconds	Reset time constant (Tx1) (>= 0). Typical value = 0,035.
tb	1..1	Seconds	Time constant (Tb) (>= 0). Typical value = 0,37.
tc	1..1	Seconds	Time constant (Tc) (>= 0). Typical value = 0,035.
td	1..1	Seconds	Time constant (Td) (>= 0). Typical value = 0,0.
te	1..1	Seconds	Time constant (Te) (>= 0). Typical value = 0,0169.
vsmax	1..1	PU	Limiter (Vsmax) (> PssSB4.vsmin). Typical value = 0,062.
vsmin	1..1	PU	Limiter (Vsmin) (< PssSB4.vsmax). Typical value = -0,062.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 919 shows all association ends of PssSB4 with other classes.

Table 919 – Association ends of PowerSystemStabilizerDynamics:: PssSB4 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.17 PssSH

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Siemens™ “H infinity” power system stabilizer with generator electrical power input[5].

5 Siemens™ "H infinity" power system stabilizers are an example of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

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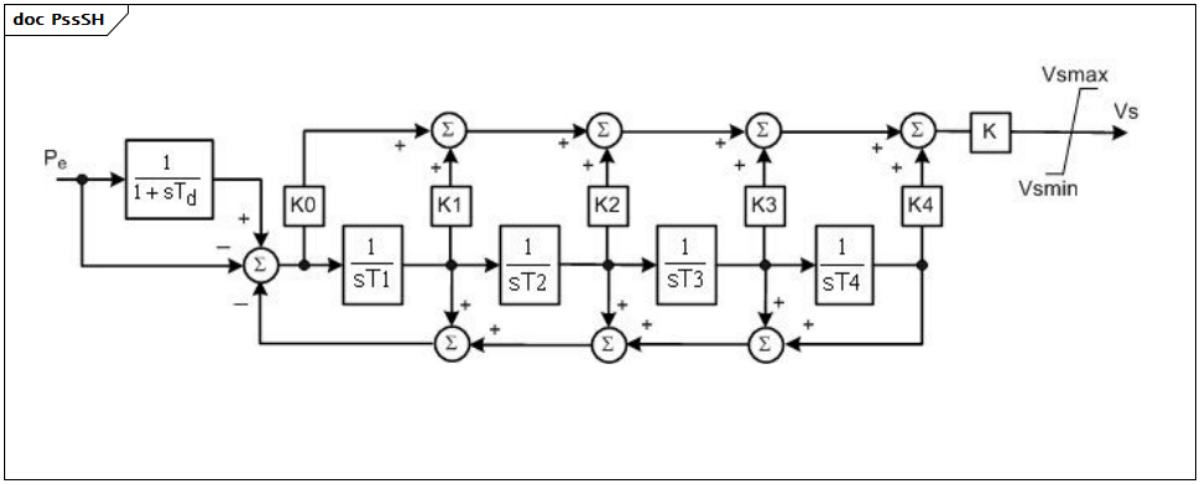

Figure 200 – PssSH

Figure 200: This diagram describes the PssSH power system stabilizer model.

Parameter details:

1) The input Pe has the fixed type (expressed in terms of inputSignalKind value) of generatorElectricalPower.

Table 920 shows all attributes of PssSH.

Table 920 – Attributes of PowerSystemStabilizerDynamics::PssSH

name	mult	type	description
k	1..1	PU	Main gain (K). Typical value = 1.
k0	1..1	PU	Gain 0 (K0). Typical value = 0,012.
k1	1..1	PU	Gain 1 (K1). Typical value = 0,488.
k2	1..1	PU	Gain 2 (K2). Typical value = 0,064.
k3	1..1	PU	Gain 3 (K3). Typical value = 0,224.
k4	1..1	PU	Gain 4 (K4). Typical value = 0,1.
td	1..1	Seconds	Input time constant (Td) (>= 0). Typical value = 10.
t1	1..1	Seconds	Time constant 1 (T1) (> 0). Typical value = 0,076.
t2	1..1	Seconds	Time constant 2 (T2) (> 0). Typical value = 0,086.
t3	1..1	Seconds	Time constant 3 (T3) (> 0). Typical value = 1,068.
t4	1..1	Seconds	Time constant 4 (T4) (> 0). Typical value = 1,913.
vsmax	1..1	PU	Output maximum limit (Vsmax) (> PssSH.vsmin). Typical value = 0,1.
vsmin	1..1	PU	Output minimum limit (Vsmin) (< PssSH.vsmax). Typical value = -0,1.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 921 shows all association ends of PssSH with other classes.

Table 921 – Association ends of PowerSystemStabilizerDynamics:: PssSH with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.18 PssSK

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Slovakian PSS with three inputs.

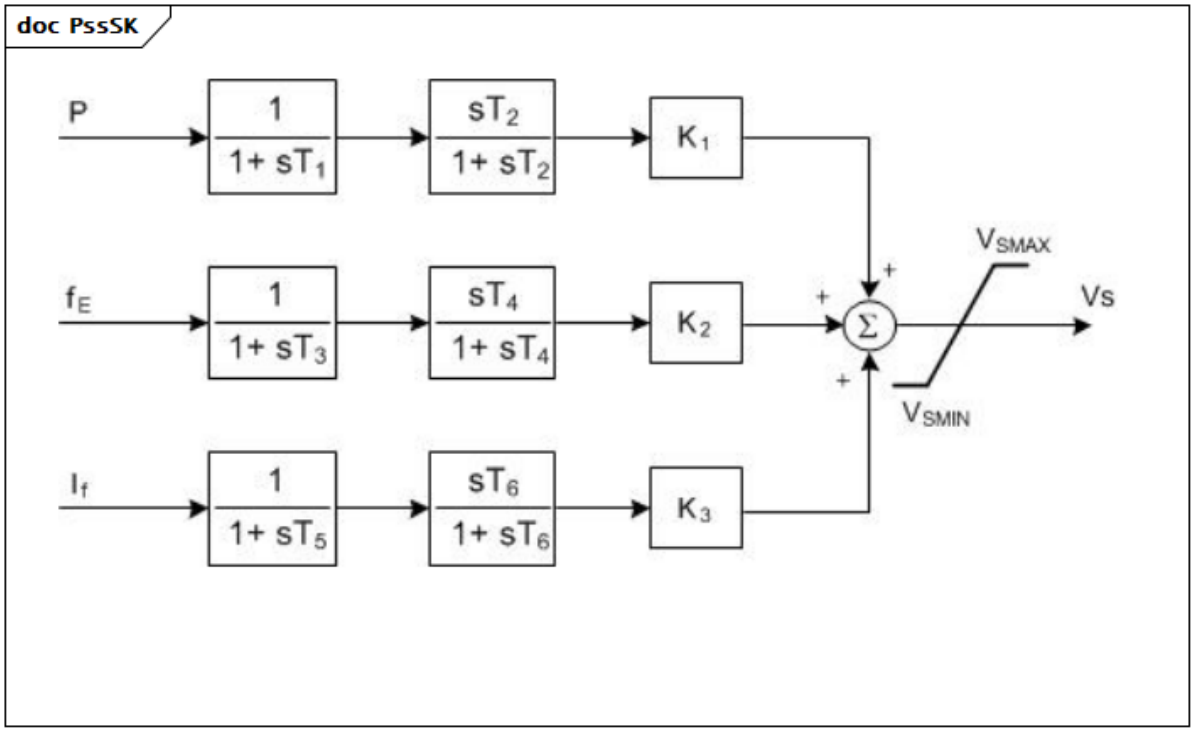

Figure 201 – PssSK

Figure 201: This diagram describes the PssSK power system stabilizer model.

Parameter details:

1) This model has 3 input signals. They have the following fixed types (expressed in terms of InputSignalKind values): P is of generatorElectricalPower type, fE is of busFrequency type, and If is of fieldCurrent type.

Table 922 shows all attributes of PssSK.

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Table 922 – Attributes of PowerSystemStabilizerDynamics::PssSK

name	mult	type	description
k1	1..1	PU	Gain P (K1). Typical value = -0,3.
k2	1..1	PU	Gain fE (K2). Typical value = -0,15.
k3	1..1	PU	Gain If (K3). Typical value = 10.
t1	1..1	Seconds	Denominator time constant (T1) (> 0,005). Typical value = 0,3.
t2	1..1	Seconds	Filter time constant (T2) (> 0,005). Typical value = 0,35.
t3	1..1	Seconds	Denominator time constant (T3) (> 0,005). Typical value = 0,22.
t4	1..1	Seconds	Filter time constant (T4) (> 0,005). Typical value = 0,02.
t5	1..1	Seconds	Denominator time constant (T5) (> 0,005). Typical value = 0,02.
t6	1..1	Seconds	Filter time constant (T6) (> 0,005). Typical value = 0,02.
vsmax	1..1	PU	Stabilizer output maximum limit (VSMAX) (> PssSK.vsmin). Typical value = 0,4.
vsmin	1..1	PU	Stabilizer output minimum limit (VSMIN) (< PssSK.vsmax). Typical value = -0.4.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 923 shows all association ends of PssSK with other classes.

Table 923 – Association ends of PowerSystemStabilizerDynamics:: PssSK with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.12.19 PssSTAB2A

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Power system stabilizer part of an ABB excitation system[6].

6 ABB excitation systems are an example of suitable products available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of these products.

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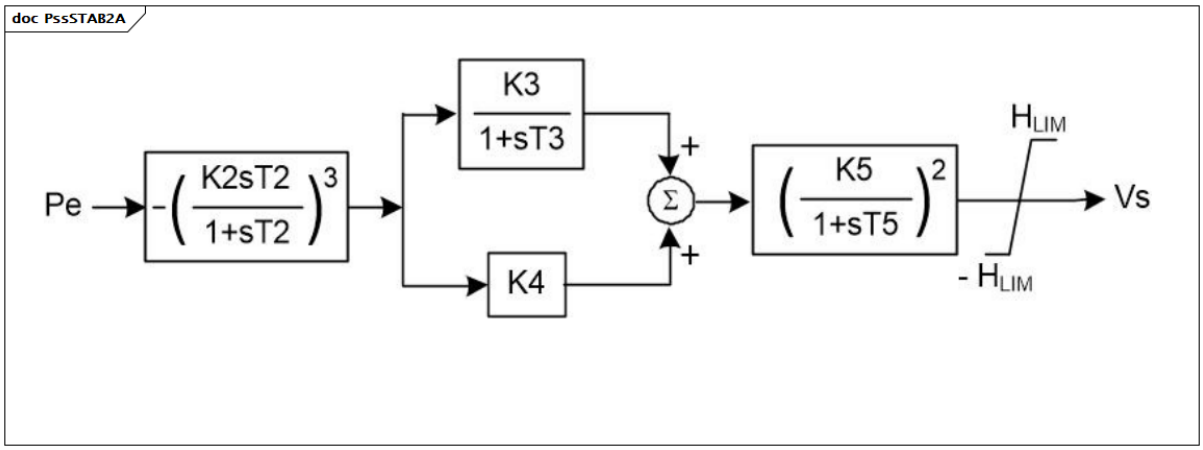

Figure 202 – PssSTAB2A

Figure 202: This diagram describes the PssSTAB2A power system stabilizer model.

Table 924 shows all attributes of PssSTAB2A.

Table 924 – Attributes of PowerSystemStabilizerDynamics::PssSTAB2A

name	mult	type	description
k2	1..1	PU	Gain (K2). Typical value = 1,0.
k3	1..1	PU	Gain (K3). Typical value = 0,25.
k4	1..1	PU	Gain (K4). Typical value = 0,075.
k5	1..1	PU	Gain (K5). Typical value = 2,5.
t2	1..1	Seconds	Time constant (T2). Typical value = 4,0.
t3	1..1	Seconds	Time constant (T3). Typical value = 2,0.
t5	1..1	Seconds	Time constant (T5). Typical value = 4,5.
hlim	1..1	PU	Stabilizer output limiter (HLIM). Typical value = 0,5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 925 shows all association ends of PssSTAB2A with other classes.

Table 925 – Association ends of PowerSystemStabilizerDynamics:: PssSTAB2A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

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14.3.3.12.20 PssWECC

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Dual input power system stabilizer, based on IEEE type 2, with modified output limiter defined by WECC (Western Electricity Coordinating Council, USA).

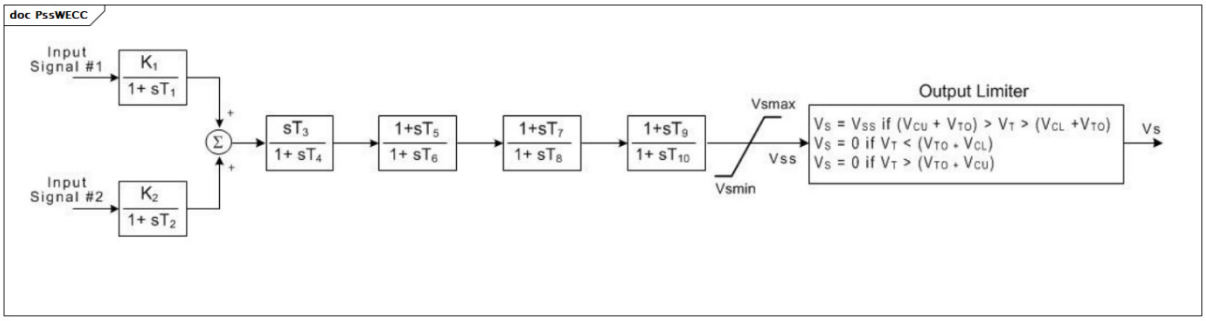

Figure 203 – PssWECC

Figure 203: This diagram describes the PssWECC power system stabilizer model.

Parameter details:

1) VT is the terminal voltage.
2) VTO is the initial terminal voltage.

Table 926 shows all attributes of PssWECC.

Table 926 – Attributes of PowerSystemStabilizerDynamics::PssWECC

name	mult	type	description
inputSignal1Type	1..1	InputSignalKind	Type of input signal #1 (rotorAngularFrequencyDeviation, busFrequencyDeviation, generatorElectricalPower, generatorAcceleratingPower, busVoltage, or busVoltageDerivative – shall be different than PssWECC.inputSignal2Type). Typical value = rotorAngularFrequencyDeviation.
inputSignal2Type	1..1	InputSignalKind	Type of input signal #2 (rotorAngularFrequencyDeviation, busFrequencyDeviation, generatorElectricalPower, generatorAcceleratingPower, busVoltage, busVoltageDerivative – shall be different than PssWECC.inputSignal1Type). Typical value = busVoltageDerivative.
k1	1..1	PU	Input signal 1 gain (K1). Typical value = 1,13.
t1	1..1	Seconds	Input signal 1 transducer time constant (T1) (>= 0). Typical value = 0,037.
k2	1..1	PU	Input signal 2 gain (K2). Typical value = 0,0.
t2	1..1	Seconds	Input signal 2 transducer time constant (T2) (>= 0). Typical value = 0,0.
t3	1..1	Seconds	Stabilizer washout time constant (T3) (>= 0). Typical value = 9,5.
t4	1..1	Seconds	Stabilizer washout time lag constant (T4) (>= 0). Typical value = 9,5.

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name	mult	type	description
t5	1..1	Seconds	Lead time constant (T5) (>= 0). Typical value = 1,7.
t6	1..1	Seconds	Lag time constant (T6) (>= 0). Typical value = 1,5.
t7	1..1	Seconds	Lead time constant (T7) (>= 0). Typical value = 1,7.
t8	1..1	Seconds	Lag time constant (T8) (>= 0). Typical value = 1,5.
t10	1..1	Seconds	Lag time constant (T10) (>= 0). Typical value = 0.
t9	1..1	Seconds	Lead time constant (T9) (>= 0). Typical value = 0.
vsmax	1..1	PU	Maximum output signal (Vsmax) (> PssWECC.vsmin). Typical value = 0,05.
vsmin	1..1	PU	Minimum output signal (Vsmin) (< PssWECC.vsmax). Typical value = -0,05.
vcu	1..1	PU	Maximum value for voltage compensator output (VCU). Typical value = 0.
vcl	1..1	PU	Minimum value for voltage compensator output (VCL). Typical value = 0.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 927 shows all association ends of PssWECC with other classes.

Table 927 – Association ends of PowerSystemStabilizerDynamics:: PssWECC with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.3.13 Package DiscontinuousExcitationControlDynamics

14.3.3.13.1 General

In certain system configurations, continuous excitation control with terminal voltage and power system stabilizing regulator input signals does not ensure that the potential of the excitation system for improving system stability is fully exploited. For these situations, discontinuous excitation control signals can be employed to enhance stability following large transient disturbances.

For additional information please refer to IEEE 421.5:2005, 12.

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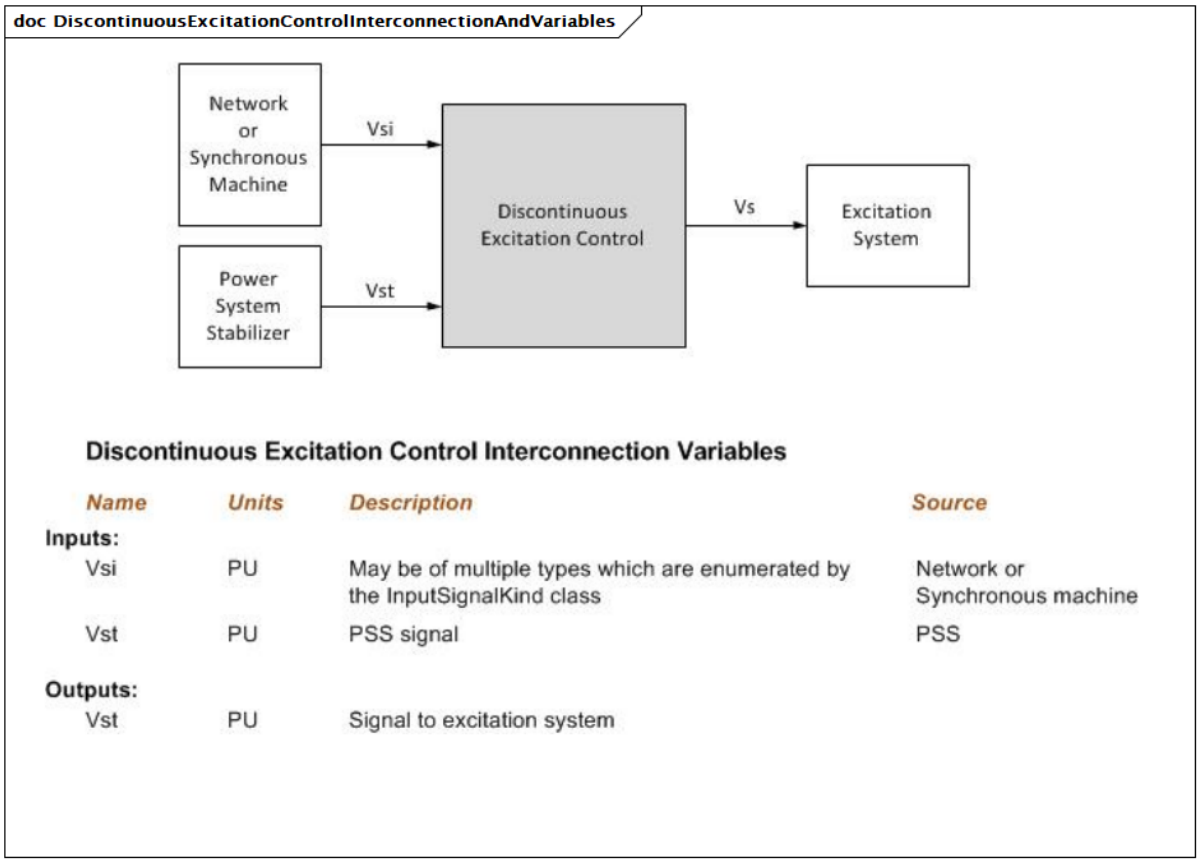

Discontinuous Excitation Control Interconnection Variables

Name	*Units*	*Description*	*Source*
Inputs:			
Vsi	PU	May be of multiple types which are enumerated by the InputSignalKind class	Network or Synchronous machine
Vst	PU	PSS signal	PSS
Outputs:			
Vst	PU	Signal to excitation system	

Figure 204 – DiscontinuousExcitationControlInterconnectionAndVariables

Figure 204: This diagram illustrates the inputs and outputs for a discontinuous excitation control.

Diagram details:

1) Vs is always initialized to zero by the excitation system model.
2) If bus voltage frequency is not available from network, the model can calculate it as derivative of the bus voltage angle.

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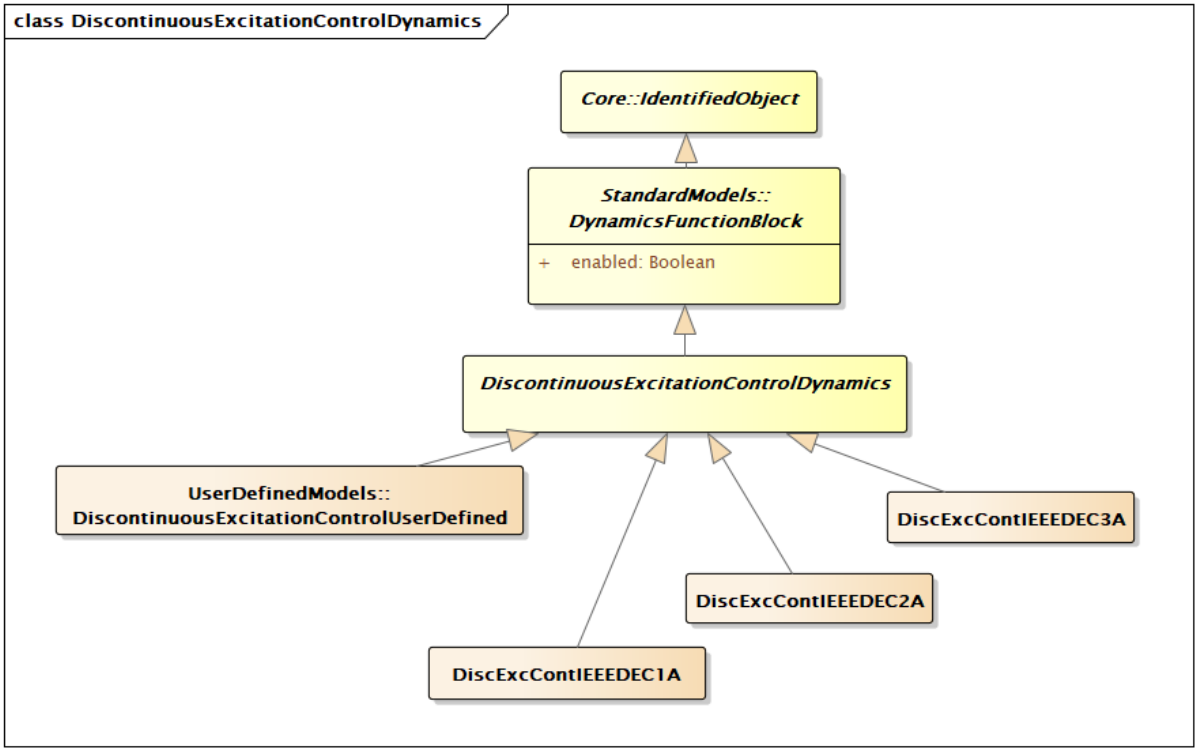

Figure 205 – Class diagram DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics

Figure 205: This diagram shows dynamic standard model discontinuous excitation control classes and their inheritance from the DiscontinuousExcitationControlDynamics class.

14.3.3.13.2 (abstract) DiscontinuousExcitationControlDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Discontinuous excitation control function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 928 shows all attributes of DiscontinuousExcitationControlDynamics.

Table 928 – Attributes of DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 929 shows all association ends of DiscontinuousExcitationControlDynamics with other classes.

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Table 929 – Association ends of DiscontinuousExcitationControlDynamics:: DiscontinuousExcitationControlDynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this discontinuous excitation control model is associated.

14.3.3.13.3 DiscExcContIEEEDEC1A

Inheritance path = DiscontinuousExcitationControlDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE type DEC1A discontinuous excitation control model that boosts generator excitation to a level higher than that demanded by the voltage regulator and stabilizer immediately following a system fault.

Reference: IEEE 421.5:2005, 12.2.

Table 930 shows all attributes of DiscExcContIEEEDEC1A.

Table 930 – Attributes of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC1A

name	mult	type	description
vtlmt	1..1	PU	Voltage reference (VTLMT). Typical value = 1,1.
vomax	1..1	PU	Limiter (VOMAX) (> DiscExcContIEEEDEC1A.vomin). Typical value = 0,3.
vomin	1..1	PU	Limiter (VOMIN) (< DiscExcContIEEEDEC1A.vomax). Typical value = 0,1.
ketl	1..1	PU	Terminal voltage limiter gain (KETL). Typical value = 47.
vtc	1..1	PU	Terminal voltage level reference (VTC). Typical value = 0,95.
val	1..1	PU	Regulator voltage reference (VAL). Typical value = 5,5.
esc	1..1	PU	Speed change reference (ESC). Typical value = 0,0015.
kan	1..1	PU	Discontinuous controller gain (KAN). Typical value = 400.
tan	1..1	Seconds	Discontinuous controller time constant (TAN) (>= 0). Typical value = 0,08.
tw5	1..1	Seconds	DEC washout time constant (TW5) (>= 0). Typical value = 5.
vsmax	1..1	PU	Limiter (VSMAX)(> DiscExcContIEEEDEC1A.vsmin). Typical value = 0,2.
vsmin	1..1	PU	Limiter (VSMIN) (< DiscExcContIEEEDEC1A.vsmax). Typical value = -0,066.
td	1..1	Seconds	Time constant (TD) (>= 0). Typical value = 0,03.
tl1	1..1	Seconds	Time constant (TL1) (>= 0). Typical value = 0,025.
tl2	1..1	Seconds	Time constant (TL2) (>= 0). Typical value = 1,25.

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name	mult	type	description
vtm	1..1	PU	Voltage limits (VTM). Typical value = 1,13.
vtn	1..1	PU	Voltage limits (VTN). Typical value = 1,12.
vanmax	1..1	PU	Limiter for Van (VANMAX).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 931 shows all association ends of DiscExcContIEEEDEC1A with other classes.

Table 931 – Association ends of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC1A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: DiscontinuousExcitationControlDynamics

14.3.3.13.4 DiscExcContIEEEDEC2A

Inheritance path = DiscontinuousExcitationControlDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE type DEC2A model for discontinuous excitation control. This system provides transient excitation boosting via an open-loop control as initiated by a trigger signal generated remotely.

Reference: IEEE 421.5:2005 12.3.

Table 932 shows all attributes of DiscExcContIEEEDEC2A.

Table 932 – Attributes of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC2A

name	mult	type	description
vk	1..1	PU	Discontinuous controller input reference (VK).
td1	1..1	Seconds	Discontinuous controller time constant (TD1) (>= 0).
td2	1..1	Seconds	Discontinuous controller washout time constant (TD2) (>= 0).
vdmin	1..1	PU	Limiter (VDMIN) (< DiscExcContIEEEDEC2A.vdmax).
vdmax	1..1	PU	Limiter (VDMAX) (> DiscExcContIEEEDEC2A.vdmin).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 933 shows all association ends of DiscExcContIEEEDEC2A with other classes.

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Table 933 – Association ends of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC2A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: DiscontinuousExcitationControlDynamics

14.3.3.13.5 DiscExcContIEEEDEC3A

Inheritance path = DiscontinuousExcitationControlDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE type DEC3A model. In some systems, the stabilizer output is disconnected from the regulator immediately following a severe fault to prevent the stabilizer from competing with action of voltage regulator during the first swing.

Reference: IEEE 421.5:2005 12.4.

Table 934 shows all attributes of DiscExcContIEEEDEC3A.

Table 934 – Attributes of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC3A

name	mult	type	description
vtmin	1..1	PU	Terminal undervoltage comparison level (VTMIN).
tdr	1..1	Seconds	Reset time delay (TDR) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 935 shows all association ends of DiscExcContIEEEDEC3A with other classes.

Table 935 – Association ends of DiscontinuousExcitationControlDynamics:: DiscExcContIEEEDEC3A with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: DiscontinuousExcitationControlDynamics

14.3.3.14 Package PFVArControllerType1Dynamics

14.3.3.14.1 General

Excitation systems for synchronous machines are sometimes supplied with an optional means of automatically adjusting generator output reactive power (VAr) or power factor (PF) to a user-specified value. This can be accomplished with either a reactive power or power factor controller or regulator. A reactive power or power factor controller is defined as a PF/VAr controller in IEEE 421.1 as “a control function that acts through the reference adjuster to modify the voltage regulator set point to maintain the synchronous machine steady-state power factor or reactive power at a predetermined value.”

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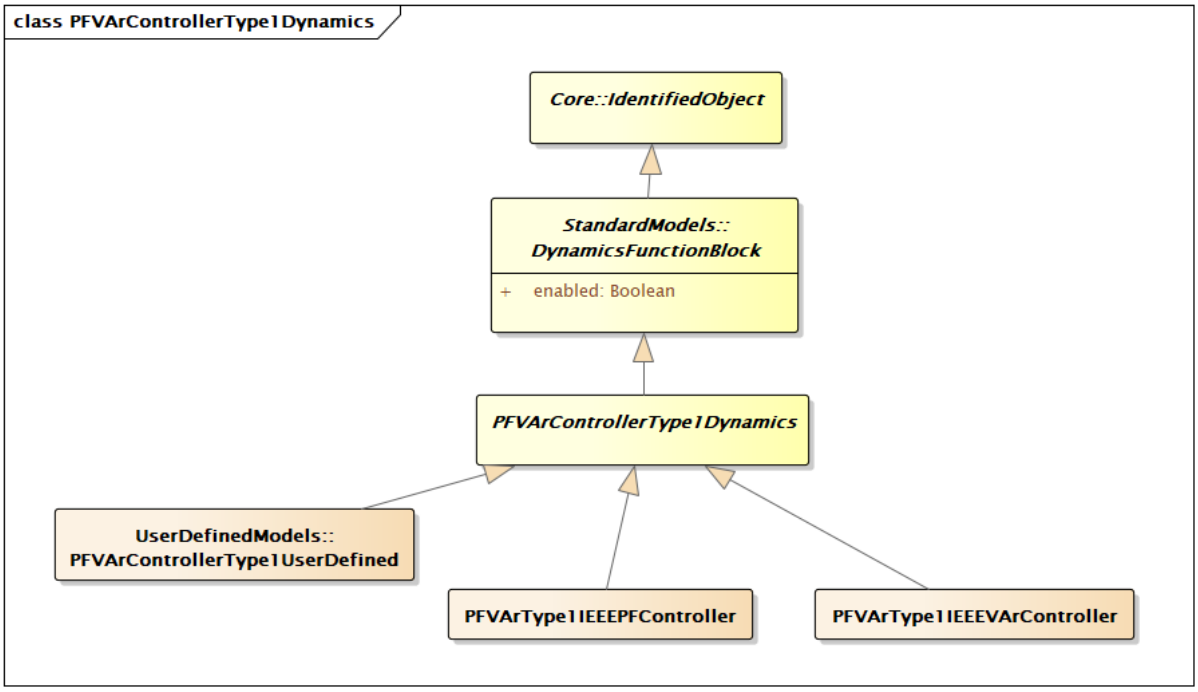

**Figure 206 – Class diagram PFVArControllerType1Dynamics::
PFVArControllerType1Dynamics**

Figure 206: This diagram shows dynamic standard model power factor or VAr controller type 1 classes and their inheritance from the PFVArControllerType1Dynamics class.

14.3.3.14.2 (abstract) PFVArControllerType1Dynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Power factor or VAr controller type 1 function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 936 shows all attributes of PFVArControllerType1Dynamics.

**Table 936 – Attributes of PFVArControllerType1Dynamics::
PFVArControllerType1Dynamics**

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 937 shows all association ends of PFVArControllerType1Dynamics with other classes.

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Table 937 – Association ends of PFVArControllerType1Dynamics:: PFVArControllerType1Dynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this power actor or VAr controller type 1 model is associated.

14.3.3.14.3 PFVArType1IEEEPFController

Inheritance path = PFVArControllerType1Dynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE PF controller type 1 which operates by moving the voltage reference directly.

Reference: IEEE 421.5:2005, 11.2.

Table 938 shows all attributes of PFVArType1IEEEPFController.

Table 938 – Attributes of PFVArControllerType1Dynamics:: PFVArType1IEEEPFController

name	mult	type	description
ovex	1..1	Boolean	Overexcitation Flag (OVEX) true = overexcited false = underexcited.
tpfc	1..1	Seconds	PF controller time delay (TPFC) (>= 0). Typical value = 5.
vitmin	1..1	PU	Minimum machine terminal current needed to enable pf/var controller (VITMIN).
vpf	1..1	PU	Synchronous machine power factor (VPF).
vpfcbw	1..1	Float	PF controller deadband (VPFC_BW). Typical value = 0,05.
vpfref	1..1	PU	PF controller reference (VPFREF).
vvtmax	1..1	PU	Maximum machine terminal voltage needed for pf/var controller to be enabled (VVTMAX) (> PFVArType1IEEEPFController.vvtmin).
vvtmin	1..1	PU	Minimum machine terminal voltage needed to enable pf/var controller (VVTMIN) (< PFVArType1IEEEPFController.vvtmax).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 939 shows all association ends of PFVArType1IEEEPFController with other classes.

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Table 939 – Association ends of PFVArControllerType1Dynamics:: PFVArType1IEEEPFController with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType1Dynamics

14.3.3.14.4 PFVArType1IEEEVArController

Inheritance path = PFVArControllerType1Dynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE VAR controller type 1 which operates by moving the voltage reference directly.

Reference: IEEE 421.5:2005, 11.3.

Table 940 shows all attributes of PFVArType1IEEEVArController.

Table 940 – Attributes of PFVArControllerType1Dynamics:: PFVArType1IEEEVArController

name	mult	type	description
tvarc	1..1	Seconds	Var controller time delay (TVARC) (>= 0). Typical value = 5.
vvar	1..1	PU	Synchronous machine power factor (VVAR).
vvarcbw	1..1	Float	Var controller deadband (VVARC_BW). Typical value = 0,02.
vvarref	1..1	PU	Var controller reference (VVARREF).
vvtmax	1..1	PU	Maximum machine terminal voltage needed for pf/VAr controller to be enabled (VVTMAX) (> PVFArType1IEEEVArController.vvtmin).
vvtmin	1..1	PU	Minimum machine terminal voltage needed to enable pf/var controller (VVTMIN) (< PVFArType1IEEEVArController.vvtmax).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 941 shows all association ends of PFVArType1IEEEVArController with other classes.

Table 941 – Association ends of PFVArControllerType1Dynamics:: PFVArType1IEEEVArController with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType1Dynamics

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14.3.3.15 Package VoltageAdjusterDynamics

14.3.3.15.1 General

A voltage adjuster is a reference adjuster that uses inputs from a reactive power or power factor controller to modify the voltage regulator set point to maintain the synchronous machine steady-state power factor or reactive power at a predetermined value.

For additional information please refer to IEEE 421.5:2005, 11.

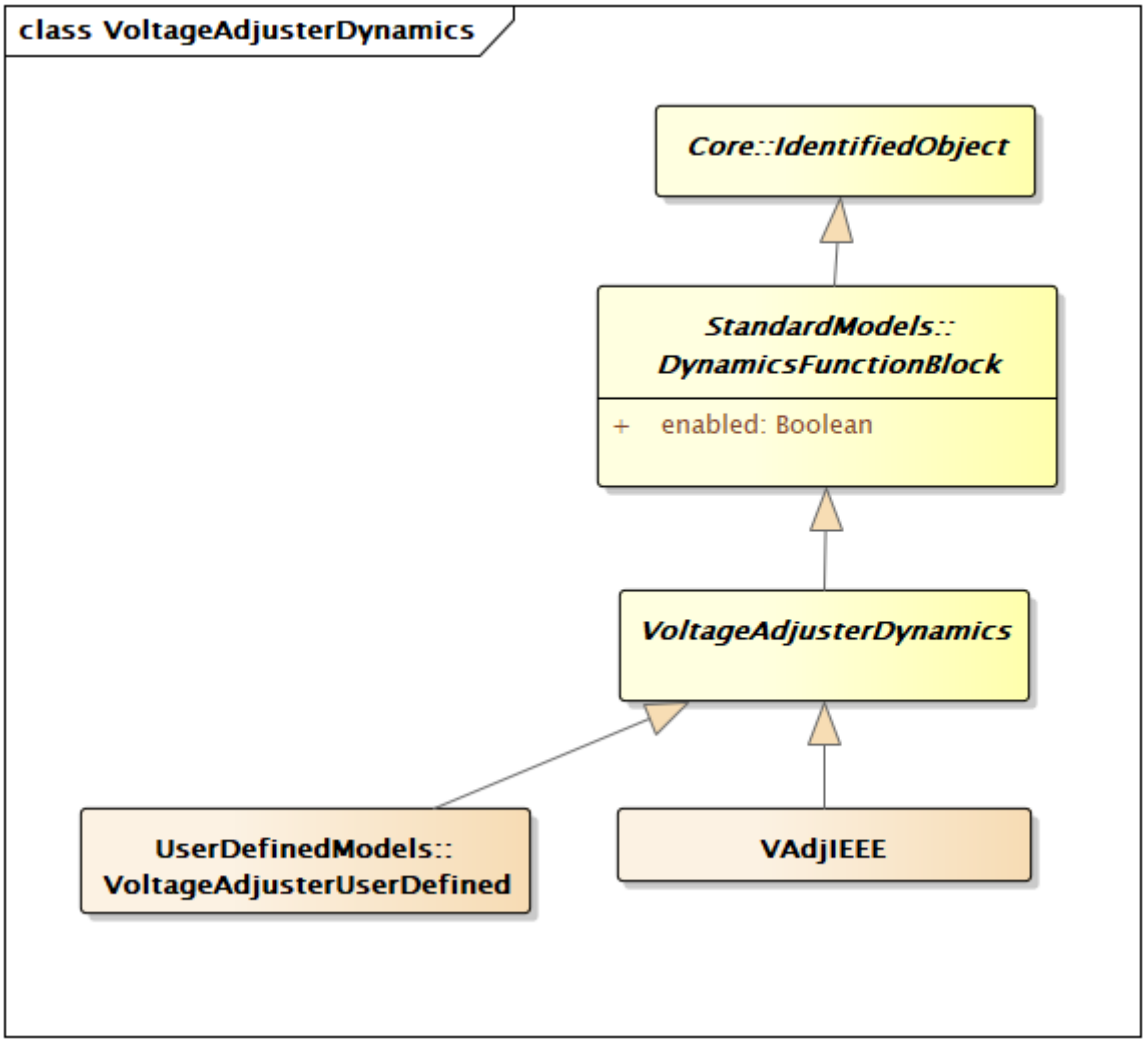

Figure 207 – Class diagram VoltageAdjusterDynamics::VoltageAdjusterDynamics

Figure 207: This diagram shows dynamic standard model voltage adjuster classes and their inheritance from the VoltageAdjusterDynamics class.

14.3.3.15.2 (abstract) VoltageAdjusterDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Voltage adjuster function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 942 shows all attributes of VoltageAdjusterDynamics.

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Table 942 – Attributes of VoltageAdjusterDynamics::VoltageAdjusterDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 943 shows all association ends of VoltageAdjusterDynamics with other classes.

Table 943 – Association ends of VoltageAdjusterDynamics:: VoltageAdjusterDynamics with other classes

mult from	name	mult to	type	description
0..1	PFVArControllerType1Dynamics	1..1	PFVArControllerType1Dynamics	Power factor or VAr controller type 1 model with which this voltage adjuster is associated.

14.3.3.15.3 VAdjIEEE

Inheritance path = VoltageAdjusterDynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE voltage adjuster which is used to represent the voltage adjuster in either a power factor or VAr control system.

Reference: IEEE 421.5:2005, 11.1.

Table 944 shows all attributes of VAdjIEEE.

Table 944 – Attributes of VoltageAdjusterDynamics::VAdjIEEE

name	mult	type	description
vadjf	1..1	Float	Set high to provide a continuous raise or lower (VADJF).
adjslew	1..1	Float	Rate at which output of adjuster changes (ADJ_SLEW). Unit = s / PU. Typical value = 300.
vadjmax	1..1	PU	Maximum output of the adjuster (VADJMAX) (> VAdjIEEE.vadjmin). Typical value = 1,1.
vadjmin	1..1	PU	Minimum output of the adjuster (VADJMIN) (< VAdjIEEE.vadjmax). Typical value = 0,9.
taon	1..1	Seconds	Time that adjuster pulses are on (TAON) (>= 0). Typical value = 0,1.
taoff	1..1	Seconds	Time that adjuster pulses are off (TAOFF) (>= 0). Typical value = 0,5.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 945 shows all association ends of VAdjIEEE with other classes.

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Table 945 – Association ends of VoltageAdjusterDynamics:: VAdjIEEE with other classes

mult from	name	mult to	type	description
0..1	PFVArControllerType1Dynamics	1..1	PFVArControllerType1Dynamics	inherited from: VoltageAdjusterDynamics

14.3.3.16 Package PFVArControllerType2Dynamics

14.3.3.16.1 General

A var/pf regulator is defined as “a synchronous machine regulator that functions to maintain the power factor or reactive component of power at a predetermined value.”

For additional information please refer to IEEE 421.5:2005, 11.

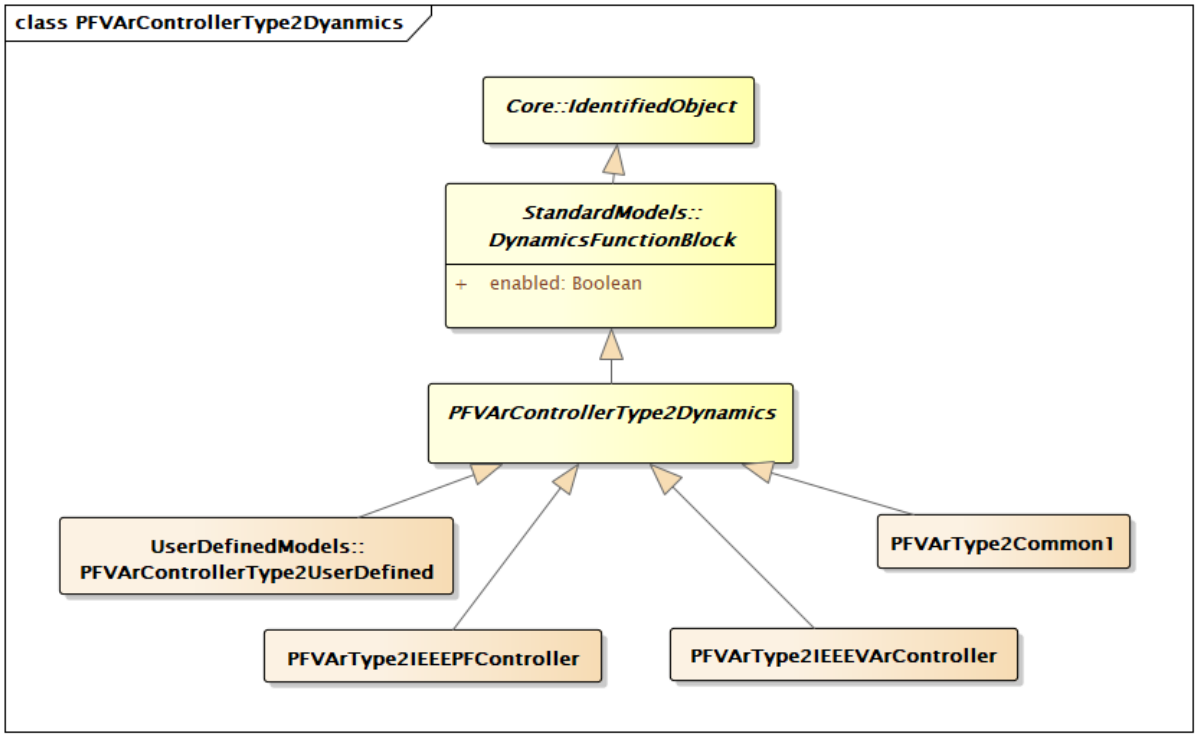

Figure 208 – Class diagram PFVArControllerType2Dynamics:: PFVArControllerType2Dyanmics

Figure 208: This diagram shows dynamic standard model power factor or VAr controller type 2 classes and their inheritance from the PFVArControllerType2Dynamics class.

14.3.3.16.2 (abstract) PFVArControllerType2Dynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Power factor or VAr controller type 2 function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 946 shows all attributes of PFVArControllerType2Dynamics.

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Table 946 – Attributes of PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 947 shows all association ends of PFVArControllerType2Dynamics with other classes.

Table 947 – Association ends of PFVArControllerType2Dynamics:: PFVArControllerType2Dynamics with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this power factor or VAr controller type 2 is associated.

14.3.3.16.3 PFVArType2IEEEPFController

Inheritance path = PFVArControllerType2Dynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE PF controller type 2 which is a summing point type controller making up the outside loop of a two-loop system. This controller is implemented as a slow PI type controller. The voltage regulator forms the inner loop and is implemented as a fast controller.

Reference: IEEE 421.5:2005, 11.4.

Table 948 shows all attributes of PFVArType2IEEEPFController.

Table 948 – Attributes of PFVArControllerType2Dynamics:: PFVArType2IEEEPFController

name	mult	type	description
pfref	1..1	PU	Power factor reference (PFREF).
vref	1..1	PU	Voltage regulator reference (VREF).
vclmt	1..1	PU	Maximum output of the pf controller (VCLMT). Typical value = 0,1.
kp	1..1	PU	Proportional gain of the pf controller (KP). Typical value = 1.
ki	1..1	PU	Integral gain of the pf controller (KI). Typical value = 1.
vs	1..1	Float	Generator sensing voltage (VS).
exlon	1..1	Boolean	Overexcitation or under excitation flag (EXLON) true = 1 (not in the overexcitation or underexcitation state, integral action is active) false = 0 (in the overexcitation or underexcitation state, so integral action is disabled to allow the limiter to play its role).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 949 shows all association ends of PFVArType2IEEEPFController with other classes.

Table 949 – Association ends of PFVArControllerType2Dynamics:: PFVArType2IEEEPFController with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType2Dynamics

14.3.3.16.4 PFVArType2IEEEVArController

Inheritance path = PFVArControllerType2Dynamics: DynamicsFunctionBlock: IdentifiedObject

IEEE VAR controller type 2 which is a summing point type controller. It makes up the outside loop of a two-loop system. This controller is implemented as a slow PI type controller, and the voltage regulator forms the inner loop and is implemented as a fast controller.

Reference: IEEE 421.5:2005, 11.5.

Table 950 shows all attributes of PFVArType2IEEEVArController.

Table 950 – Attributes of PFVArControllerType2Dynamics:: PFVArType2IEEEVArController

name	mult	type	description
qref	1..1	PU	Reactive power reference (QREF).
vref	1..1	PU	Voltage regulator reference (VREF).
vclmt	1..1	PU	Maximum output of the pf controller (VCLMT).
kp	1..1	PU	Proportional gain of the pf controller (KP).
ki	1..1	PU	Integral gain of the pf controller (KI).
vs	1..1	Float	Generator sensing voltage (VS).
exlon	1..1	Boolean	Overexcitation or under excitation flag (EXLON) true = 1 (not in the overexcitation or underexcitation state, integral action is active) false = 0 (in the overexcitation or underexcitation state, so integral action is disabled to allow the limiter to play its role).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 951 shows all association ends of PFVArType2IEEEVArController with other classes.

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Table 951 – Association ends of PFVArControllerType2Dynamics:: PFVArType2IEEEVArController with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType2Dynamics

14.3.3.16.5 PFVArType2Common1

Inheritance path = PFVArControllerType2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Power factor / reactive power regulator. This model represents the power factor or reactive power controller such as the Basler SCP-250[7]. The controller measures power factor or reactive power (PU on generator rated power) and compares it with the operator's set point.

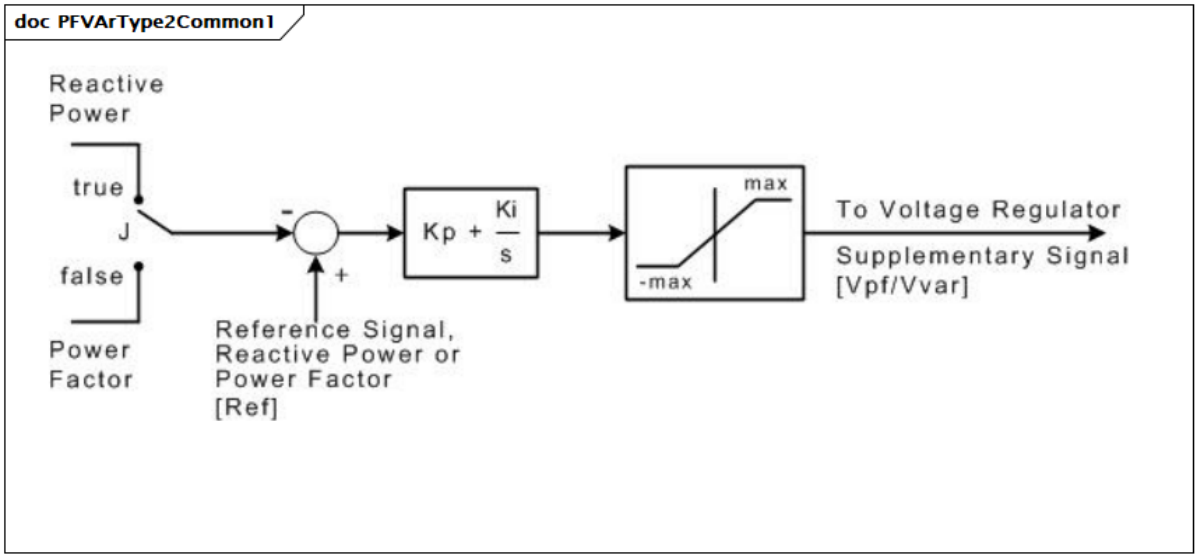

Figure 209 – PFVArType2Common1

Figure 209: This diagram describes the PFVArType2Common1 power factor / reactive power regulator.

Table 952 shows all attributes of PFVArType2Common1.

Table 952 – Attributes of PFVArControllerType2Dynamics::PFVArType2Common1

name	mult	type	description
j	1..1	Boolean	Selector (J). true = control mode for reactive power false = control mode for power factor.
kp	1..1	PU	Proportional gain (Kp).
ki	1..1	PU	Reset gain (Ki).

[7] Basler SCP-250 is an example of a suitable product available commercially. This information is given for the convenience of users of this document and does not constitute an endorsement by IEC of this product.

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name	mult	type	description
max	1..1	PU	Output limit (max).
ref	1..1	PU	Reference value of reactive power or power factor (Ref). The reference value is initialised by this model. This initialisation can override the value exchanged by this attribute to represent a plant operator's change of the reference setting.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 953 shows all association ends of PFVArType2Common1 with other classes.

Table 953 – Association ends of PFVArControllerType2Dynamics:: PFVArType2Common1 with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType2Dynamics

14.3.3.17 Package VoltageCompensatorDynamics

14.3.3.17.1 General

Synchronous machine terminal voltage transducer and current compensator models adjust the terminal voltage feedback to the excitation system by adding a quantity that is proportional to the terminal current of the generator. It is linked to a specific generator (synchronous machine).

Several types of compensation are available on most excitation systems. Synchronous machine active and reactive current compensation are the most common. Either reactive droop compensation and/or line-drop compensation can be used, simulating an impedance drop and effectively regulating at some point other than the terminals of the machine. The impedance or range of adjustment and type of compensation should be specified for different types.

Care shall be taken to ensure that a consistent PU system is utilized for the compensator parameters and the synchronous machine current base.

For further information see IEEE 421.5:2005, 4.

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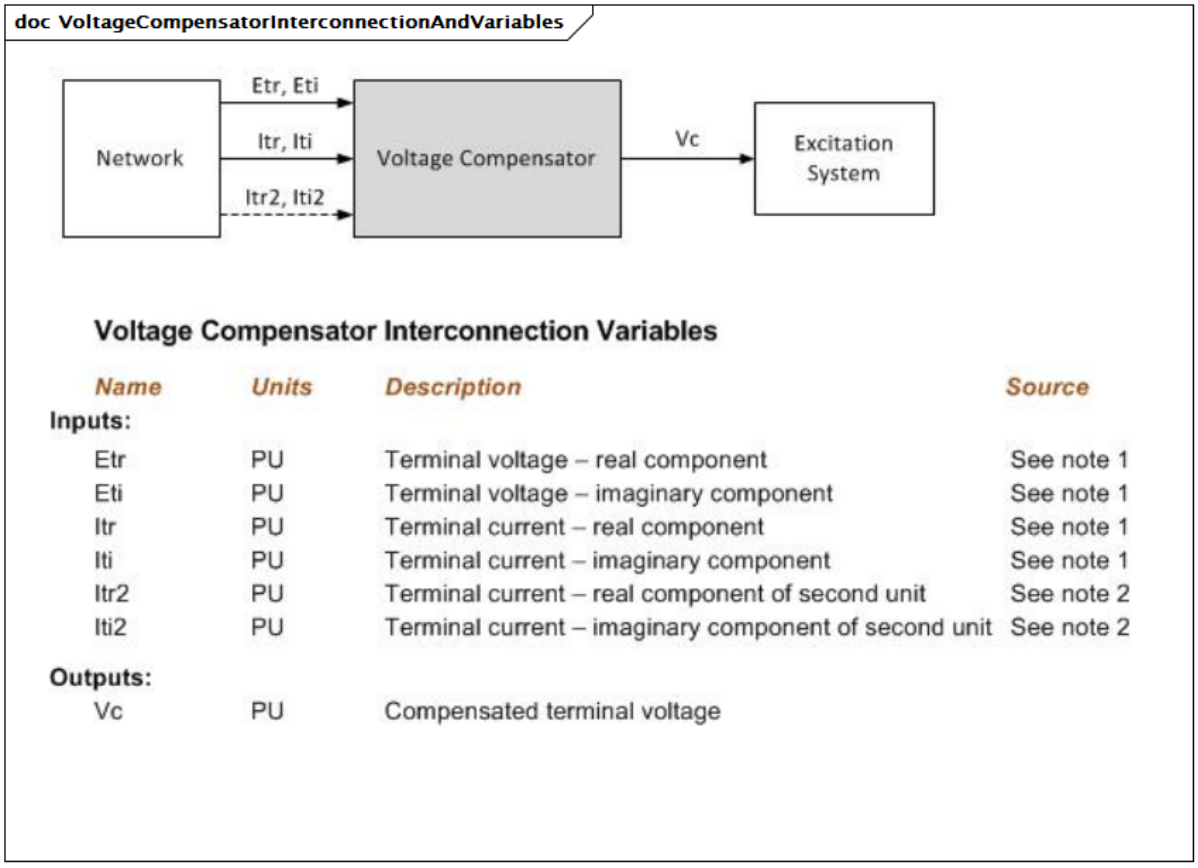

Voltage Compensator Interconnection Variables

Name	*Units*	*Description*	*Source*
Inputs:			
Etr	PU	Terminal voltage – real component	See note 1
Eti	PU	Terminal voltage – imaginary component	See note 1
Itr	PU	Terminal current – real component	See note 1
Iti	PU	Terminal current – imaginary component	See note 1
Itr2	PU	Terminal current – real component of second unit	See note 2
Iti2	PU	Terminal current – imaginary component of second unit	See note 2
Outputs:			
Vc	PU	Compensated terminal voltage	

Figure 210 – VoltageCompensatorInterconnectionAndVariables

Figure 210: This diagram illustrates the inputs and outputs for a voltage compensator.

Diagram details:

1) The source of the generator complex voltage components is application-dependent. It can come from the generator or from the network equations.
2) Itr2 and Iti2 relate to a second generator connected to the same terminal bus, usually the other unit of a cross-compound pair. These inputs are not used by all voltage compensation models.

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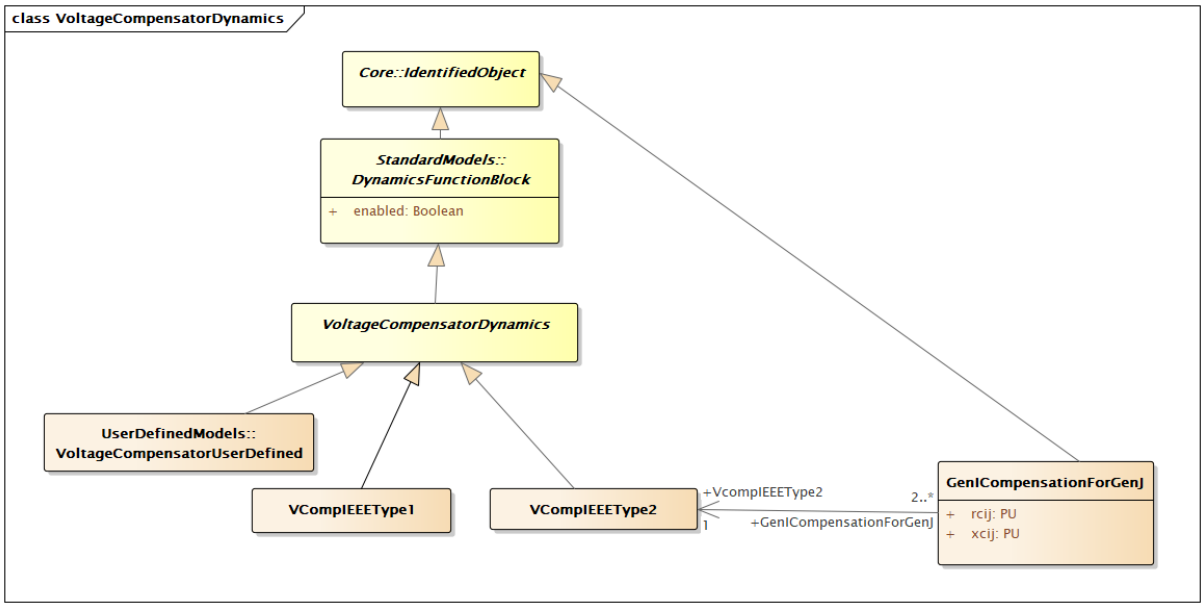

Figure 211 – Class diagram VoltageCompensatorDynamics::VoltageCompensatorDynamics

Figure 211: This diagram shows dynamic standard model voltage compensator classes and their inheritance from the VoltageCompensatorDynamics class.

14.3.3.17.2 (abstract) VoltageCompensatorDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Voltage compensator function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 954 shows all attributes of VoltageCompensatorDynamics.

Table 954 – Attributes of VoltageCompensatorDynamics::VoltageCompensatorDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 955 shows all association ends of VoltageCompensatorDynamics with other classes.

Table 955 – Association ends of VoltageCompensatorDynamics:: VoltageCompensatorDynamics with other classes

mult from	name	mult to	type	description
1..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	Excitation system model with which this voltage compensator is associated.

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14.3.3.17.3 VCompIEEEType1

Inheritance path = VoltageCompensatorDynamics: DynamicsFunctionBlock: IdentifiedObject

Terminal voltage transducer and load compensator as defined in IEEE 421.5:2005, 4. This model is common to all excitation system models described in the IEEE Standard.

Parameter details:

1) If Rc and Xc are set to zero, the load compensation is not employed and the behaviour is as a simple sensing circuit.
2) If all parameters (Rc, Xc and Tr) are set to zero, the standard model VCompIEEEType1 is bypassed.

Reference: IEEE 421.5:2005 4.

Table 956 shows all attributes of VCompIEEEType1.

Table 956 – Attributes of VoltageCompensatorDynamics::VCompIEEEType1

name	mult	type	description
rc	1..1	PU	Resistive component of compensation of a generator (Rc) (>= 0).
xc	1..1	PU	Reactive component of compensation of a generator (Xc) (>= 0).
tr	1..1	Seconds	Time constant which is used for the combined voltage sensing and compensation signal (Tr) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 957 shows all association ends of VCompIEEEType1 with other classes.

Table 957 – Association ends of VoltageCompensatorDynamics:: VCompIEEEType1 with other classes

mult from	name	mult to	type	description
1..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: VoltageCompensatorDynamics

14.3.3.17.4 VCompIEEEType2

Inheritance path = VoltageCompensatorDynamics: DynamicsFunctionBlock: IdentifiedObject

Terminal voltage transducer and load compensator as defined in IEEE 421.5:2005, 4. This model is designed to cover the following types of compensation:

– reactive droop;
– transformer-drop or line-drop compensation;
– reactive differential compensation known also as cross-current compensation.

Reference: IEEE 421.5:2005, 4.

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Table 958 shows all attributes of VCompIEEEType2.

Table 958 – Attributes of VoltageCompensatorDynamics::VCompIEEEType2

name	mult	type	description
tr	1..1	Seconds	Time constant which is used for the combined voltage sensing and compensation signal (Tr) (>= 0).
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 959 shows all association ends of VCompIEEEType2 with other classes.

Table 959 – Association ends of VoltageCompensatorDynamics:: VCompIEEEType2 with other classes

mult from	name	mult to	type	description
1..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: VoltageCompensatorDynamics

14.3.3.17.5 GenICompensationForGenJ

Inheritance path = IdentifiedObject

Resistive and reactive components of compensation for generator associated with IEEE type 2 voltage compensator for current flow out of another generator in the interconnection.

Table 960 shows all attributes of GenICompensationForGenJ.

Table 960 – Attributes of VoltageCompensatorDynamics::GenICompensationForGenJ

name	mult	type	description
rcij	1..1	PU	Resistive component of compensation of generator associated with this IEEE type 2 voltage compensator for current flow out of another generator (Rcij).
xcij	1..1	PU	Reactive component of compensation of generator associated with this IEEE type 2 voltage compensator for current flow out of another generator (Xcij).
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 961 shows all association ends of GenICompensationForGenJ with other classes.

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Table 961 – Association ends of VoltageCompensatorDynamics:: GenICompensationForGenJ with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	Standard synchronous machine out of which current flow is being compensated for.
2..*	VcomplEEEType2	1..1	VCompIEEEType2	The standard IEEE type 2 voltage compensator of this compensation.

14.3.3.18 Package WindDynamics

14.3.3.18.1 General

Wind turbines are generally divided into four types, which are currently significant in power systems. The four types have the following characteristics:

– type 1: wind turbine with directly grid connected asynchronous generator with fixed rotor resistance (typically squirrel cage);

– type 2: wind turbine with directly grid connected asynchronous generator with variable rotor resistance;

– type 3: wind turbines with doubly-fed asynchronous generators (directly connected stator and rotor connected through power converter);

– type 4: wind turbines connected to the grid through a full size power converter.

Models included in this package are according to IEC 61400-27-1:2015.

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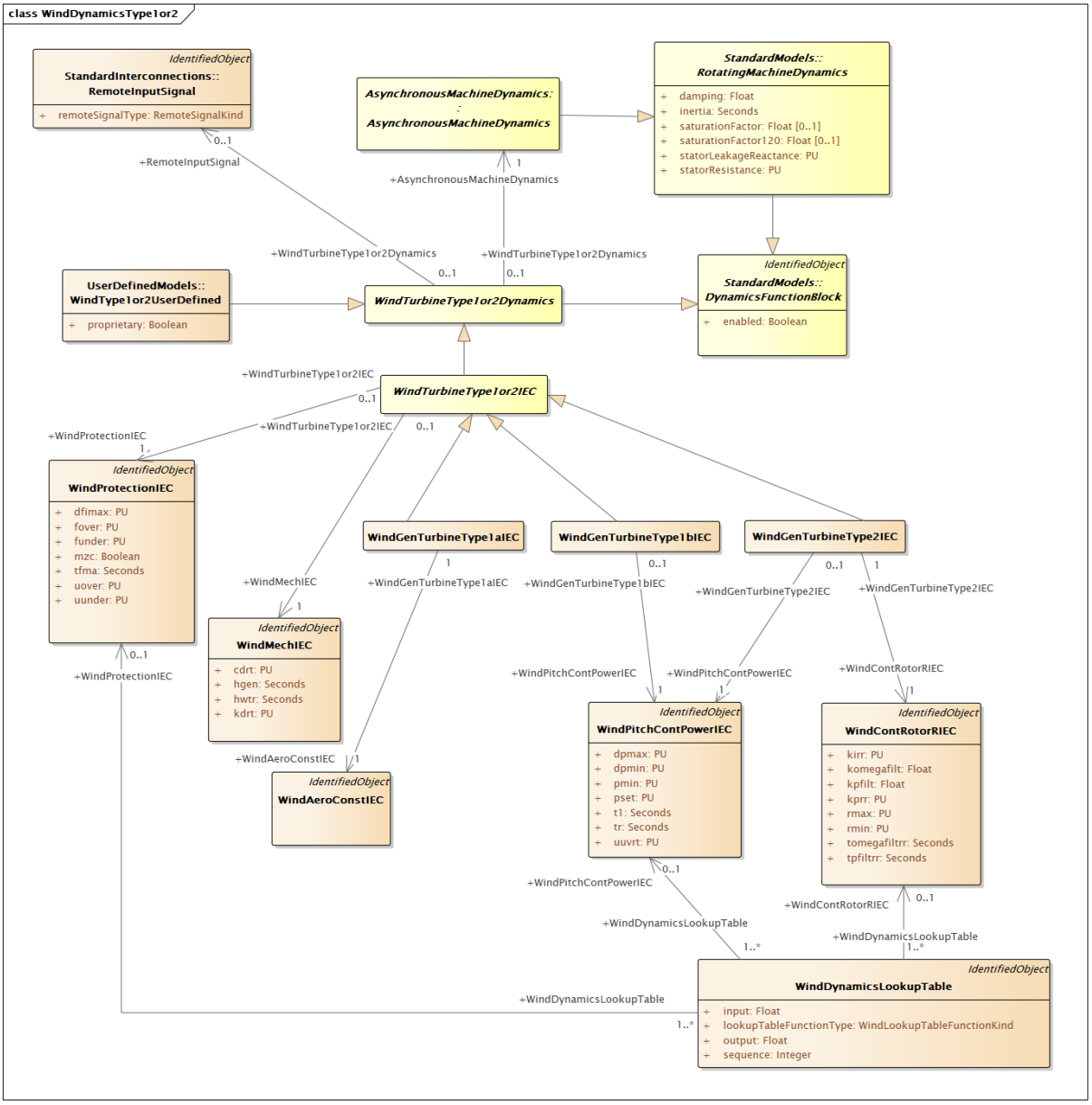

Figure 212 – Class diagram WindDynamics::WindDynamicsType1or2

Figure 212: This diagram shows dynamic standard model wind classes (for wind turbines type 1 and type 2) and their relationships. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes, not abstract function block classes, are specified in the wind interconnection models.

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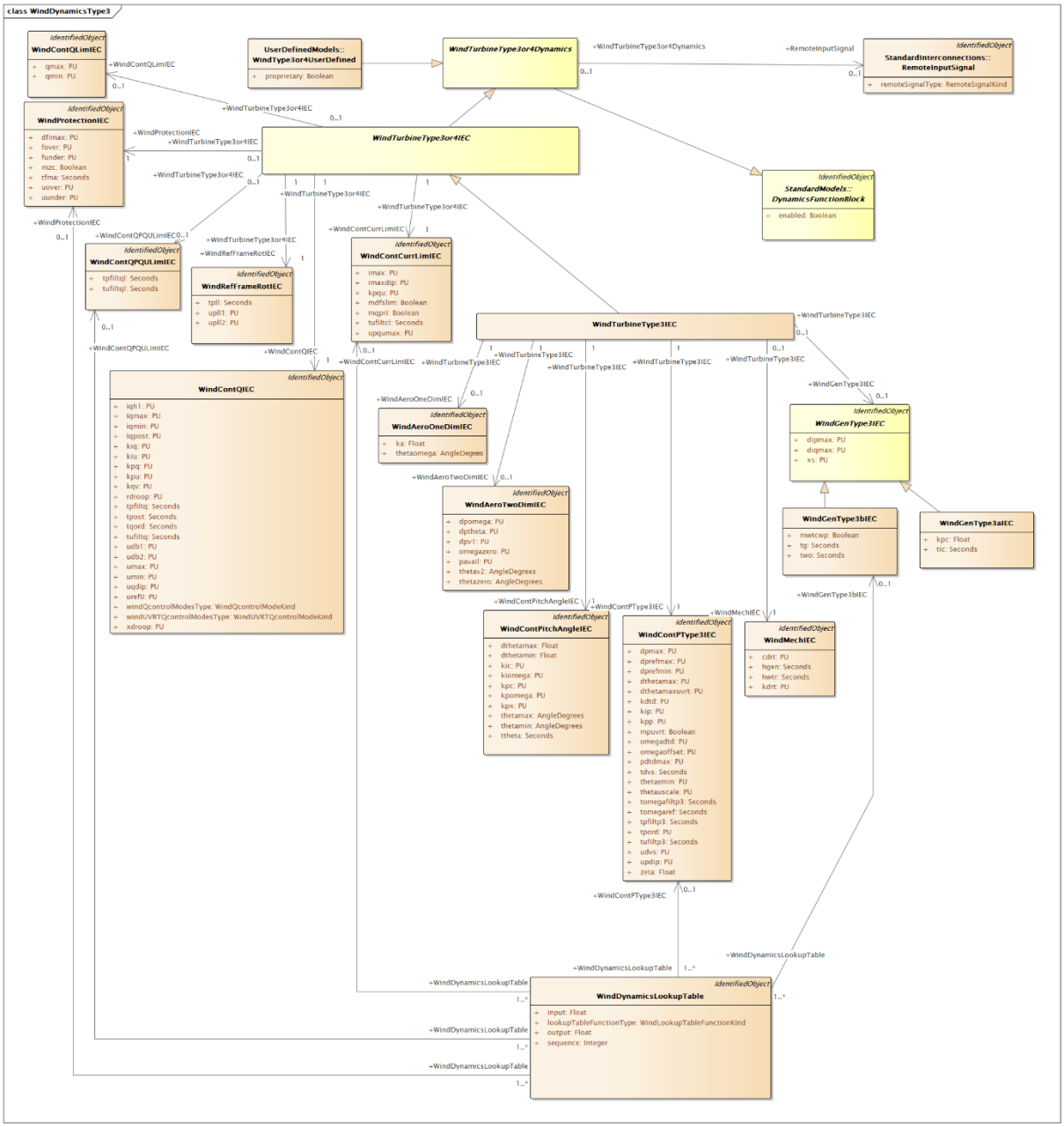

Figure 213 – Class diagram WindDynamics::WindDynamicsType3

Figure 213: This diagram shows dynamic standard model wind classes for wind turbines type 3 and their relationships. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes, not abstract function block classes, are specified in the wind interconnection models.

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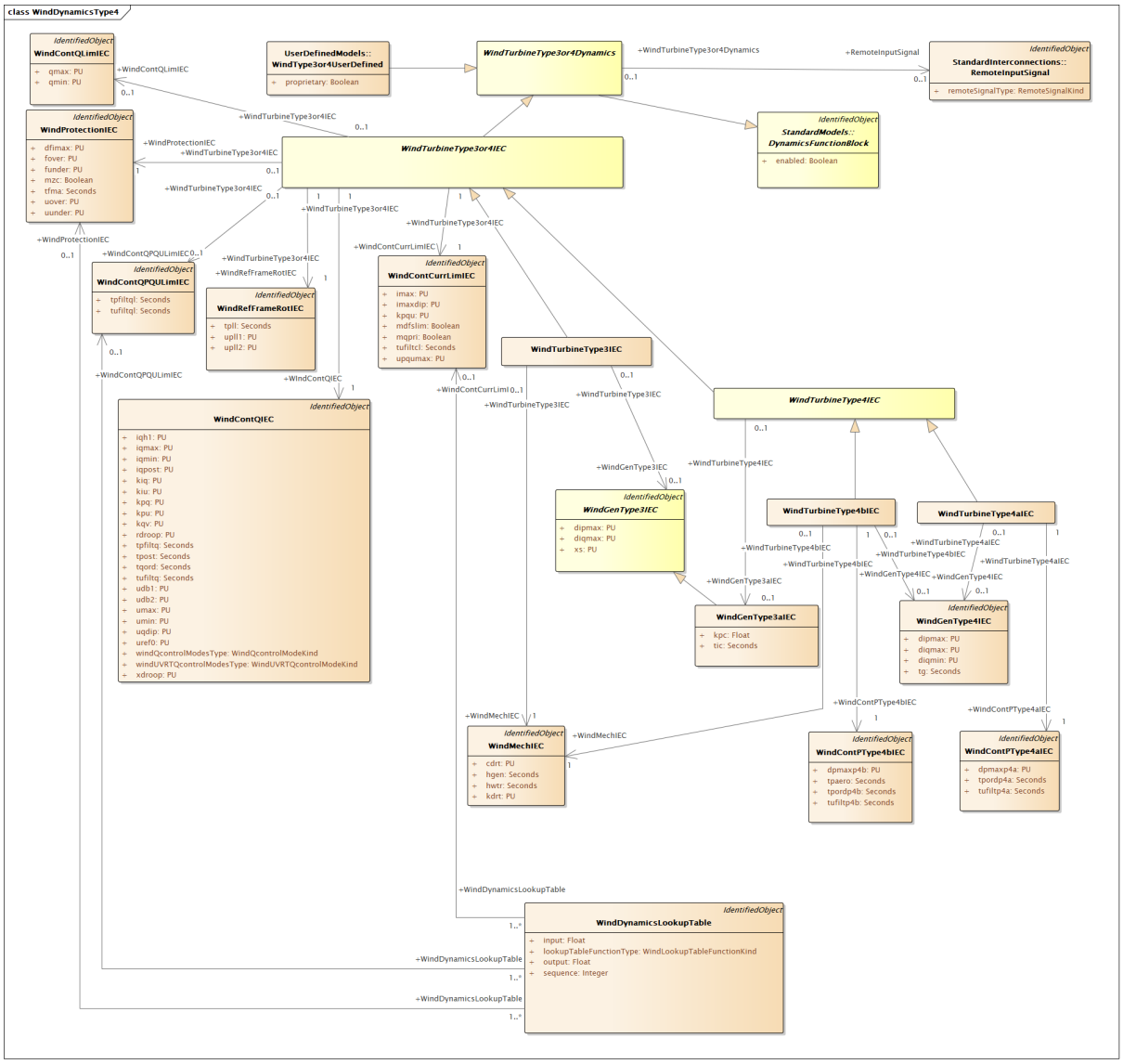

Figure 214 – Class diagram WindDynamics::WindDynamicsType4

Figure 214: This diagram shows dynamic standard model wind classes for wind turbines type 4 and their relationships. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes, not abstract function block classes, are specified in the wind interconnection models.

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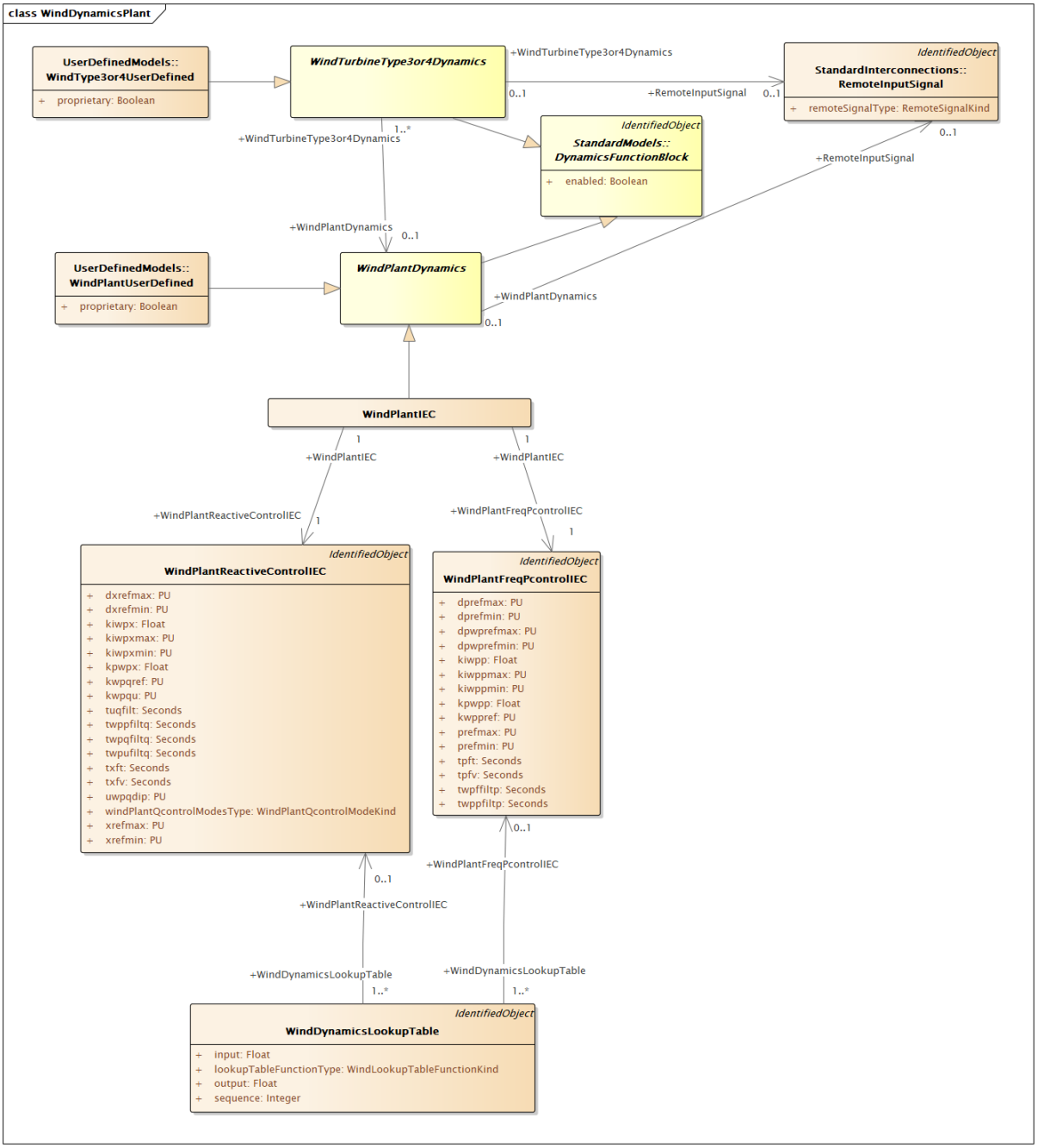

Figure 215 – Class diagram WindDynamics::WindDynamicsPlant

Figure 215: This diagram shows dynamic standard model wind classes for wind plant. Note that classes in the Wind package are not grouped by function block classes. This is because specific standard model classes, not abstract function block classes, are specified in the wind interconnection models.

14.3.3.18.2 WindAeroConstIEC

Inheritance path = IdentifiedObject

Constant aerodynamic torque model which assumes that the aerodynamic torque is constant.

Reference: IEC 61400-27-1:2015, 5.6.1.1.

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Table 962 shows all attributes of WindAeroConstIEC.

Table 962 – Attributes of WindDynamics::WindAeroConstIEC

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.3 WindAeroOneDimIEC

Inheritance path = IdentifiedObject

One-dimensional aerodynamic model.

Reference: IEC 61400-27-1:2015, 5.6.1.2.

Table 963 shows all attributes of WindAeroOneDimIEC.

Table 963 – Attributes of WindDynamics::WindAeroOneDimIEC

name	mult	type	description
ka	1..1	Float	Aerodynamic gain (ka). It is a type-dependent parameter.
thetaomega	1..1	AngleDegrees	Initial pitch angle (thetaomega0). It is a case-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.4 WindAeroTwoDimIEC

Inheritance path = IdentifiedObject

Two-dimensional aerodynamic model.

Reference: IEC 61400-27-1:2015, 5.6.1.3.

Table 964 shows all attributes of WindAeroTwoDimIEC.

Table 964 – Attributes of WindDynamics::WindAeroTwoDimIEC

name	mult	type	description
dpomega	1..1	PU	Partial derivative of aerodynamic power with respect to changes in WTR speed (dpomega). It is a type-dependent parameter.
dptheta	1..1	PU	Partial derivative of aerodynamic power with respect to changes in pitch angle (dptheta). It is a type-dependent parameter.
dpv1	1..1	PU	Partial derivative (dpv1). It is a type-dependent parameter.
omegazero	1..1	PU	Rotor speed if the wind turbine is not derated (omega0). It is a type-dependent parameter.

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name	mult	type	description
pavail	1..1	PU	Available aerodynamic power (pavail). It is a case-dependent parameter.
thetazero	1..1	AngleDegrees	Pitch angle if the wind turbine is not derated (theta0). It is a case-dependent parameter.
thetav2	1..1	AngleDegrees	Blade angle at twice rated wind speed (thetav2). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.5 WindContCurrLimIEC

Inheritance path = IdentifiedObject

Current limitation model. The current limitation model combines the physical limits and the control limits.

Reference: IEC 61400-27-1:2015, 5.6.5.8.

Table 965 shows all attributes of WindContCurrLimIEC.

Table 965 – Attributes of WindDynamics::WindContCurrLimIEC

name	mult	type	description
imax	1..1	PU	Maximum continuous current at the wind turbine terminals (imax). It is a type-dependent parameter.
imaxdip	1..1	PU	Maximum current during voltage dip at the wind turbine terminals (imaxdip). It is a project-dependent parameter.
kpqu	1..1	PU	Partial derivative of reactive current limit (Kpqu) versus voltage. It is a type-dependent parameter.
mdfslim	1..1	Boolean	Limitation of type 3 stator current (MDFSLim). MDFSLim = 1 for wind turbines type 4. It is a type-dependent parameter. false= total current limitation (0 in the IEC model) true=stator current limitation (1 in the IEC model).
mqpri	1..1	Boolean	Prioritisation of Q control during UVRT (Mqpri). It is a project-dependent parameter. true = reactive power priority (1 in the IEC model) false = active power priority (0 in the IEC model).
tufiltcl	1..1	Seconds	Voltage measurement filter time constant (Tufiltcl) (>= 0). It is a type-dependent parameter.
upqumax	1..1	PU	Wind turbine voltage in the operation point where zero reactive current can be delivered (upqumax). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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14.3.3.18.6 WindContPitchAngleIEC

Inheritance path = IdentifiedObject

Pitch angle control model.

Reference: IEC 61400-27-1:2015, 5.6.5.2.

Table 966 shows all attributes of WindContPitchAngleIEC.

Table 966 – Attributes of WindDynamics::WindContPitchAngleIEC

name	mult	type	description
dthetamax	1..1	Float	Maximum pitch positive ramp rate (dthetamax) (> WindContPitchAngleIEC.dthetamin). It is a type-dependent parameter. Unit = degrees / s.
dthetamin	1..1	Float	Maximum pitch negative ramp rate (dthetamin) (< WindContPitchAngleIEC.dthetamax). It is a type-dependent parameter. Unit = degrees / s.
kic	1..1	PU	Power PI controller integration gain (KIc). It is a type-dependent parameter.
kiomega	1..1	PU	Speed PI controller integration gain (KIomega). It is a type-dependent parameter.
kpc	1..1	PU	Power PI controller proportional gain (KPc). It is a type-dependent parameter.
kpomega	1..1	PU	Speed PI controller proportional gain (KPomega). It is a type-dependent parameter.
kpx	1..1	PU	Pitch cross coupling gain (KPX). It is a type-dependent parameter.
thetamax	1..1	AngleDegrees	Maximum pitch angle (thetamax) (> WindContPitchAngleIEC.thetamin). It is a type-dependent parameter.
thetamin	1..1	AngleDegrees	Minimum pitch angle (thetamin) (< WindContPitchAngleIEC.thetamax). It is a type-dependent parameter.
ttheta	1..1	Seconds	Pitch time constant (ttheta) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.7 WindContPType3IEC

Inheritance path = IdentifiedObject

P control model type 3.

Reference: IEC 61400-27-1:2015, 5.6.5.4.

Table 967 shows all attributes of WindContPType3IEC.

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Table 967 – Attributes of WindDynamics::WindContPType3IEC

name	mult	type	description
dpmax	1..1	PU	Maximum wind turbine power ramp rate (dpmax). It is a type-dependent parameter.
dprefmax	1..1	PU	Maximum ramp rate of wind turbine reference power (dprefmax). It is a project-dependent parameter.
dprefmin	1..1	PU	Minimum ramp rate of wind turbine reference power (dprefmin). It is a project-dependent parameter.
dthetamax	1..1	PU	Ramp limitation of torque, required in some grid codes (dtmax). It is a project-dependent parameter.
dthetamaxuvrt	1..1	PU	Limitation of torque rise rate during UVRT (dthetamaxUVRT). It is a project-dependent parameter.
kdtd	1..1	PU	Gain for active drive train damping (KDTD). It is a type-dependent parameter.
kip	1..1	PU	PI controller integration parameter (KIp). It is a type-dependent parameter.
kpp	1..1	PU	PI controller proportional gain (KPp). It is a type-dependent parameter.
mpuvrt	1..1	Boolean	Enable UVRT power control mode (MpUVRT). It is a project-dependent parameter. true = voltage control (1 in the IEC model) false = reactive power control (0 in the IEC model).
omegaoffset	1..1	PU	Offset to reference value that limits controller action during rotor speed changes (omegaoffset). It is a case-dependent parameter.
pdtdmax	1..1	PU	Maximum active drive train damping power (pDTDmax). It is a type-dependent parameter.
tdvs	1..1	Seconds	Time delay after deep voltage sags (TDVS) (>= 0). It is a project-dependent parameter.
thetaemin	1..1	PU	Minimum electrical generator torque (temin). It is a type-dependent parameter.
thetauscale	1..1	PU	Voltage scaling factor of reset-torque (tuscale). It is a project-dependent parameter.
tomegafiltp3	1..1	Seconds	Filter time constant for generator speed measurement (Tomegafiltp3) (>= 0). It is a type-dependent parameter.
tpfiltp3	1..1	Seconds	Filter time constant for power measurement (Tpfiltp3) (>= 0). It is a type-dependent parameter.
tpord	1..1	PU	Time constant in power order lag (Tpord). It is a type-dependent parameter.
tufiltp3	1..1	Seconds	Filter time constant for voltage measurement (Tufiltp3) (>= 0). It is a type-dependent parameter.
tomegaref	1..1	Seconds	Time constant in speed reference filter (Tomega,ref) (>= 0). It is a type-dependent parameter.
udvs	1..1	PU	Voltage limit for hold UVRT status after deep voltage sags (uDVS). It is a project-dependent parameter.
updip	1..1	PU	Voltage dip threshold for P-control (uPdip). Part of turbine control, often different (e.g 0.8) from converter thresholds. It is a project-dependent parameter.

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name	mult	type	description
omegadtd	1..1	PU	Active drive train damping frequency (omegaDTD). It can be calculated from two mass model parameters. It is a type-dependent parameter.
zeta	1..1	Float	Coefficient for active drive train damping (zeta). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.8 WindContPType4aIEC

Inheritance path = IdentifiedObject

P control model type 4A.

Reference: IEC 61400-27-1:2015, 5.6.5.5.

Table 968 shows all attributes of WindContPType4aIEC.

Table 968 – Attributes of WindDynamics::WindContPType4aIEC

name	mult	type	description
dpmaxp4a	1..1	PU	Maximum wind turbine power ramp rate (dpmaxp4A). It is a project-dependent parameter.
tpordp4a	1..1	Seconds	Time constant in power order lag (Tpordp4A) (>= 0). It is a type-dependent parameter.
tufiltp4a	1..1	Seconds	Voltage measurement filter time constant (Tufiltp4A) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.9 WindContPType4bIEC

Inheritance path = IdentifiedObject

P control model type 4B.

Reference: IEC 61400-27-1:2015, 5.6.5.6.

Table 969 shows all attributes of WindContPType4bIEC.

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Table 969 – Attributes of WindDynamics::WindContPType4bIEC

name	mult	type	description
dpmaxp4b	1..1	PU	Maximum wind turbine power ramp rate (dpmaxp4B). It is a project-dependent parameter.
tpaero	1..1	Seconds	Time constant in aerodynamic power response (Tpaero) (>= 0). It is a type-dependent parameter.
tpordp4b	1..1	Seconds	Time constant in power order lag (Tpordp4B) (>= 0). It is a type-dependent parameter.
tufiltp4b	1..1	Seconds	Voltage measurement filter time constant (Tufiltp4B) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.10 WindContQIEC

Inheritance path = IdentifiedObject

Q control model.

Reference: IEC 61400-27-1:2015, 5.6.5.7.

Table 970 shows all attributes of WindContQIEC.

Table 970 – Attributes of WindDynamics::WindContQIEC

name	mult	type	description
iqh1	1..1	PU	Maximum reactive current injection during dip (iqh1). It is a type-dependent parameter.
iqmax	1..1	PU	Maximum reactive current injection (iqmax) (> WindContQIEC.iqmin). It is a type-dependent parameter.
iqmin	1..1	PU	Minimum reactive current injection (iqmin) (< WindContQIEC.iqmax). It is a type-dependent parameter.
iqpost	1..1	PU	Post fault reactive current injection (iqpost). It is a project-dependent parameter.
kiq	1..1	PU	Reactive power PI controller integration gain (KI,q). It is a type-dependent parameter.
kiu	1..1	PU	Voltage PI controller integration gain (KI,u). It is a type-dependent parameter.
kpq	1..1	PU	Reactive power PI controller proportional gain (KP,q). It is a type-dependent parameter.
kpu	1..1	PU	Voltage PI controller proportional gain (KP,u). It is a type-dependent parameter.
kqv	1..1	PU	Voltage scaling factor for UVRT current (Kqv). It is a project-dependent parameter.
tpfiltq	1..1	Seconds	Power measurement filter time constant (Tpfiltq) (>= 0). It is a type-dependent parameter.
rdroop	1..1	PU	Resistive component of voltage drop impedance (rdroop) (>= 0). It is a project-dependent parameter.

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name	mult	type	description
tufiltq	1..1	Seconds	Voltage measurement filter time constant (Tufiltq) (>= 0). It is a type-dependent parameter.
tpost	1..1	Seconds	Length of time period where post fault reactive power is injected (Tpost) (>= 0). It is a project-dependent parameter.
tqord	1..1	Seconds	Time constant in reactive power order lag (Tqord) (>= 0). It is a type-dependent parameter.
udb1	1..1	PU	Voltage deadband lower limit (udb1). It is a type-dependent parameter.
udb2	1..1	PU	Voltage deadband upper limit (udb2). It is a type-dependent parameter.
umax	1..1	PU	Maximum voltage in voltage PI controller integral term (umax) (> WindContQIEC.umin). It is a type-dependent parameter.
umin	1..1	PU	Minimum voltage in voltage PI controller integral term (umin) (< WindContQIEC.umax). It is a type-dependent parameter.
uqdip	1..1	PU	Voltage threshold for UVRT detection in Q control (uqdip). It is a type-dependent parameter.
uref0	1..1	PU	User-defined bias in voltage reference (uref0). It is a case-dependent parameter.
windQcontrolModesType	1..1	WindQcontrolModeKind	Types of general wind turbine Q control modes (MqG). It is a project-dependent parameter.
windUVRTQcontrolModesType	1..1	WindUVRTQcontrolModeKind	Types of UVRT Q control modes (MqUVRT). It is a project-dependent parameter.
xdroop	1..1	PU	Inductive component of voltage drop impedance (xdroop) (>= 0). It is a project-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.11 WindContQLimIEC

Inheritance path = IdentifiedObject

Constant Q limitation model.

Reference: IEC 61400-27-1:2015, 5.6.5.9.

Table 971 shows all attributes of WindContQLimIEC.

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Table 971 – Attributes of WindDynamics::WindContQLimIEC

name	mult	type	description
qmax	1..1	PU	Maximum reactive power (qmax) (> WindContQLimIEC.qmin). It is a type-dependent parameter.
qmin	1..1	PU	Minimum reactive power (qmin) (< WindContQLimIEC.qmax). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.12 WindContQPQULimIEC

Inheritance path = IdentifiedObject

QP and QU limitation model.

Reference: IEC 61400-27-1:2015, 5.6.5.10.

Table 972 shows all attributes of WindContQPQULimIEC.

Table 972 – Attributes of WindDynamics::WindContQPQULimIEC

name	mult	type	description
tpfiltql	1..1	Seconds	Power measurement filter time constant for Q capacity (Tpfiltql) (>= 0). It is a type-dependent parameter.
tufiltql	1..1	Seconds	Voltage measurement filter time constant for Q capacity (Tufiltql) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.13 WindContRotorRIEC

Inheritance path = IdentifiedObject

Rotor resistance control model.

Reference: IEC 61400-27-1:2015, 5.6.5.3.

Table 973 shows all attributes of WindContRotorRIEC.

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Table 973 – Attributes of WindDynamics::WindContRotorRIEC

name	mult	type	description
kirr	1..1	PU	Integral gain in rotor resistance PI controller (KIrr). It is a type-dependent parameter.
komegafilt	1..1	Float	Filter gain for generator speed measurement (Komegafilt). It is a type-dependent parameter.
kpfilt	1..1	Float	Filter gain for power measurement (Kpfilt). It is a type-dependent parameter.
kprr	1..1	PU	Proportional gain in rotor resistance PI controller (KPrr). It is a type-dependent parameter.
rmax	1..1	PU	Maximum rotor resistance (rmax) (> WindContRotorRIEC.rmin). It is a type-dependent parameter.
rmin	1..1	PU	Minimum rotor resistance (rmin) (< WindContRotorRIEC.rmax). It is a type-dependent parameter.
tomegafiltrr	1..1	Seconds	Filter time constant for generator speed measurement (Tomegafiltrr) (>= 0). It is a type-dependent parameter.
tpfiltrr	1..1	Seconds	Filter time constant for power measurement (Tpfiltrr) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.14 WindDynamicsLookupTable

Inheritance path = IdentifiedObject

Look up table for the purpose of wind standard models.

Table 974 shows all attributes of WindDynamicsLookupTable.

Table 974 – Attributes of WindDynamics::WindDynamicsLookupTable

name	mult	type	description
input	1..1	Float	Input value (x) for the lookup table function.
lookupTableFunctionType	1..1	WindLookupTableFunctionKind	Type of the lookup table function.
output	1..1	Float	Output value (y) for the lookup table function.
sequence	1..1	Integer	Sequence numbers of the pairs of the input (x) and the output (y) of the lookup table function.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 975 shows all association ends of WindDynamicsLookupTable with other classes.

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Table 975 – Association ends of WindDynamics:: WindDynamicsLookupTable with other classes

mult from	name	mult to	type	description
1..*	WindContRotorRIEC	0..1	WindContRotorRIEC	The rotor resistance control model with which this wind dynamics lookup table is associated.
1..*	WindPitchContPowerIEC	0..1	WindPitchContPowerIEC	The pitch control power model with which this wind dynamics lookup table is associated.
1..*	WindContCurrLimIEC	0..1	WindContCurrLimIEC	The current control limitation model with which this wind dynamics lookup table is associated.
1..*	WindContPType3IEC	0..1	WindContPType3IEC	The P control type 3 model with which this wind dynamics lookup table is associated.
1..*	WindContQPQULimIEC	0..1	WindContQPQULimIEC	The QP and QU limitation model with which this wind dynamics lookup table is associated.
1..*	WindPlantFreqPcontrolIEC	0..1	WindPlantFreqPcontrolIEC	The frequency and active power wind plant control model with which this wind dynamics lookup table is associated.
1..*	WindProtectionIEC	0..1	WindProtectionIEC	The grid protection model with which this wind dynamics lookup table is associated.
1..*	WindPlantReactiveControlIEC	0..1	WindPlantReactiveControlIEC	The voltage and reactive power wind plant control model with which this wind dynamics lookup table is associated.
1..*	WindGenType3bIEC	0..1	WindGenType3bIEC	The generator type 3B model with which this wind dynamics lookup table is associated.

14.3.3.18.15 WindGenTurbineType1aIEC

Inheritance path = WindTurbineType1or2IEC: WindTurbineType1or2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind turbine IEC type 1A.

Reference: IEC 61400-27-1:2015, 5.5.2.2.

Table 976 shows all attributes of WindGenTurbineType1aIEC.

Table 976 – Attributes of WindDynamics::WindGenTurbineType1aIEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 977 shows all association ends of WindGenTurbineType1aIEC with other classes.

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Table 977 – Association ends of WindDynamics:: WindGenTurbineType1aIEC with other classes

mult from	name	mult to	type	description
1..1	WindAeroConstIEC	1..1	WindAeroConstIEC	Wind aerodynamic model associated with this wind turbine type 1A model.
0..1	WindMechIEC	1..1	WindMechIEC	inherited from: WindTurbineType1or2IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType1or2IEC
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType1or2Dynamics
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	inherited from: WindTurbineType1or2Dynamics

14.3.3.18.16 WindGenTurbineType1bIEC

Inheritance path = WindTurbineType1or2IEC: WindTurbineType1or2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind turbine IEC type 1B.

Reference: IEC 61400-27-1:2015, 5.5.2.3.

Table 978 shows all attributes of WindGenTurbineType1bIEC.

Table 978 – Attributes of WindDynamics::WindGenTurbineType1bIEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 979 shows all association ends of WindGenTurbineType1bIEC with other classes.

Table 979 – Association ends of WindDynamics:: WindGenTurbineType1bIEC with other classes

mult from	name	mult to	type	description
0..1	WindPitchContPowerIEC	1..1	WindPitchContPowerIEC	Pitch control power model associated with this wind turbine type 1B model.
0..1	WindMechIEC	1..1	WindMechIEC	inherited from: WindTurbineType1or2IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType1or2IEC
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType1or2Dynamics
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	inherited from: WindTurbineType1or2Dynamics

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14.3.3.18.17 WindGenTurbineType2IEC

Inheritance path = WindTurbineType1or2IEC: WindTurbineType1or2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind turbine IEC type 2.

Reference: IEC 61400-27-1:2015, 5.5.3.

Table 980 shows all attributes of WindGenTurbineType2IEC.

Table 980 – Attributes of WindDynamics::WindGenTurbineType2IEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 981 shows all association ends of WindGenTurbineType2IEC with other classes.

Table 981 – Association ends of WindDynamics:: WindGenTurbineType2IEC with other classes

mult from	name	mult to	type	description
1..1	WindContRotorRIEC	1..1	WindContRotorRIEC	Wind control rotor resistance model associated with wind turbine type 2 model.
0..1	WindPitchContPowerIEC	1..1	WindPitchContPowerIEC	Pitch control power model associated with this wind turbine type 2 model.
0..1	WindMechIEC	1..1	WindMechIEC	inherited from: WindTurbineType1or2IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType1or2IEC
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType1or2Dynamics
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	inherited from: WindTurbineType1or2Dynamics

14.3.3.18.18 WindGenType3aIEC

Inheritance path = WindGenType3IEC: IdentifiedObject

IEC type 3A generator set model.

Reference: IEC 61400-27-1:2015, 5.6.3.2.

Table 982 shows all attributes of WindGenType3aIEC.

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Table 982 – Attributes of WindDynamics::WindGenType3aIEC

name	mult	type	description
kpc	1..1	Float	Current PI controller proportional gain (KPc). It is a type-dependent parameter.
tic	1..1	Seconds	Current PI controller integration time constant (TIc) (>= 0). It is a type-dependent parameter.
dipmax	1..1	PU	inherited from: WindGenType3IEC
diqmax	1..1	PU	inherited from: WindGenType3IEC
xs	1..1	PU	inherited from: WindGenType3IEC
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.19 WindGenType3bIEC

Inheritance path = WindGenType3IEC: IdentifiedObject

IEC type 3B generator set model.

Reference: IEC 61400-27-1:2015, 5.6.3.3.

Table 983 shows all attributes of WindGenType3bIEC.

Table 983 – Attributes of WindDynamics::WindGenType3bIEC

name	mult	type	description
mwtcwp	1..1	Boolean	Crowbar control mode (MWTcwp). It is a case-dependent parameter. true = 1 in the IEC model false = 0 in the IEC model.
tg	1..1	Seconds	Current generation time constant (Tg) (>= 0). It is a type-dependent parameter.
two	1..1	Seconds	Time constant for crowbar washout filter (Two) (>= 0). It is a case-dependent parameter.
dipmax	1..1	PU	inherited from: WindGenType3IEC
diqmax	1..1	PU	inherited from: WindGenType3IEC
xs	1..1	PU	inherited from: WindGenType3IEC
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.20 (abstract) WindGenType3IEC

Inheritance path = IdentifiedObject

Parent class supporting relationships to IEC wind turbines type 3 generator models of IEC type 3A and 3B.

Table 984 shows all attributes of WindGenType3IEC.

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Table 984 – Attributes of WindDynamics::WindGenType3IEC

name	mult	type	description
dipmax	1..1	PU	Maximum active current ramp rate (dipmax). It is a project-dependent parameter.
diqmax	1..1	PU	Maximum reactive current ramp rate (diqmax). It is a project-dependent parameter.
xs	1..1	PU	Electromagnetic transient reactance (xS). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.21 WindGenType4IEC

Inheritance path = IdentifiedObject

IEC type 4 generator set model.

Reference: IEC 61400-27-1:2015, 5.6.3.4.

Table 985 shows all attributes of WindGenType4IEC.

Table 985 – Attributes of WindDynamics::WindGenType4IEC

name	mult	type	description
dipmax	1..1	PU	Maximum active current ramp rate (dipmax). It is a project-dependent parameter.
diqmin	1..1	PU	Minimum reactive current ramp rate (diqmin). It is a project-dependent parameter.
diqmax	1..1	PU	Maximum reactive current ramp rate (diqmax). It is a project-dependent parameter.
tg	1..1	Seconds	Time constant (Tg) (>= 0). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.22 WindMechIEC

Inheritance path = IdentifiedObject

Two mass model.

Reference: IEC 61400-27-1:2015, 5.6.2.1.

Table 986 shows all attributes of WindMechIEC.

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Table 986 – Attributes of WindDynamics::WindMechIEC

name	mult	type	description
cdrt	1..1	PU	Drive train damping (cdrt). It is a type-dependent parameter.
hgen	1..1	Seconds	Inertia constant of generator (Hgen) (>= 0). It is a type-dependent parameter.
hwtr	1..1	Seconds	Inertia constant of wind turbine rotor (HWTR) (>= 0). It is a type-dependent parameter.
kdrt	1..1	PU	Drive train stiffness (kdrt). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.23 WindPitchContPowerIEC

Inheritance path = IdentifiedObject

Pitch control power model.

Reference: IEC 61400-27-1:2015, 5.6.5.1.

Table 987 shows all attributes of WindPitchContPowerIEC.

Table 987 – Attributes of WindDynamics::WindPitchContPowerIEC

name	mult	type	description
dpmax	1..1	PU	Rate limit for increasing power (dpmax) (> WindPitchContPowerIEC.dpmin). It is a type-dependent parameter.
dpmin	1..1	PU	Rate limit for decreasing power (dpmin) (< WindPitchContPowerIEC.dpmax). It is a type-dependent parameter.
pmin	1..1	PU	Minimum power setting (pmin). It is a type-dependent parameter.
pset	1..1	PU	If pinit < pset then power will be ramped down to pmin. It is (pset) in the IEC 61400-27-1:2015. It is a type-dependent parameter.
t1	1..1	Seconds	Lag time constant (T1) (>= 0). It is a type-dependent parameter.
tr	1..1	Seconds	Voltage measurement time constant (Tr) (>= 0). It is a type-dependent parameter.
uuvrt	1..1	PU	Dip detection threshold (uUVRT). It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.24 (abstract) WindPlantDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

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Parent class supporting relationships to wind turbines type 3 and type 4 and wind plant IEC and user-defined wind plants including their control models.

Table 988 shows all attributes of WindPlantDynamics.

Table 988 – Attributes of WindDynamics::WindPlantDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 989 shows all association ends of WindPlantDynamics with other classes.

Table 989 – Association ends of WindDynamics::WindPlantDynamics with other classes

mult from	name	mult to	type	description
0..1	RemoteInputSignal	0..1	RemoteInputSignal	The remote signal with which this power plant is associated.

14.3.3.18.25 WindPlantFreqPcontrolIEC

Inheritance path = IdentifiedObject

Frequency and active power controller model.

Reference: IEC 61400-27-1:2015, Annex D.

Table 990 shows all attributes of WindPlantFreqPcontrolIEC.

Table 990 – Attributes of WindDynamics::WindPlantFreqPcontrolIEC

name	mult	type	description
dprefmax	1..1	PU	Maximum ramp rate of pWTref request from the plant controller to the wind turbines (dprefmax) (> WindPlantFreqPcontrolIEC.dprefmin). It is a case-dependent parameter.
dprefmin	1..1	PU	Minimum (negative) ramp rate of pWTref request from the plant controller to the wind turbines (dprefmin) (< WindPlantFreqPcontrolIEC.dprefmax). It is a project-dependent parameter.
dpwprefmax	1..1	PU	Maximum positive ramp rate for wind plant power reference (dpWPrefmax) (> WindPlantFreqPcontrolIEC.dpwprefmin). It is a project-dependent parameter.
dpwprefmin	1..1	PU	Maximum negative ramp rate for wind plant power reference (dpWPrefmin) (< WindPlantFreqPcontrolIEC.dpwprefmax). It is a project-dependent parameter.
prefmax	1..1	PU	Maximum pWTref request from the plant controller to the wind turbines (prefmax) (> WindPlantFreqPcontrolIEC.prefmin). It is a project-dependent parameter.

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name	mult	type	description
prefmin	1..1	PU	Minimum pWTref request from the plant controller to the wind turbines (prefmin) (< WindPlantFreqPcontrolIEC.prefmax). It is a project-dependent parameter.
kiwpp	1..1	Float	Plant P controller integral gain (KIWPp). It is a project-dependent parameter.
kiwppmax	1..1	PU	Maximum PI integrator term (KIWPpmax) (> WindPlantFreqPcontrolIEC.kiwppmin). It is a project-dependent parameter.
kiwppmin	1..1	PU	Minimum PI integrator term (KIWPpmin) (< WindPlantFreqPcontrolIEC.kiwppmax). It is a project-dependent parameter.
kpwpp	1..1	Float	Plant P controller proportional gain (KPWPp). It is a project-dependent parameter.
kwppref	1..1	PU	Power reference gain (KWPpref). It is a project-dependent parameter.
tpft	1..1	Seconds	Lead time constant in reference value transfer function (Tpft) (>= 0). It is a project-dependent parameter.
tpfv	1..1	Seconds	Lag time constant in reference value transfer function (Tpfv) (>= 0). It is a project-dependent parameter.
twpffiltp	1..1	Seconds	Filter time constant for frequency measurement (TWPffiltp) (>= 0). It is a project-dependent parameter.
twppfiltp	1..1	Seconds	Filter time constant for active power measurement (TWPpfiltp) (>= 0). It is a project-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.26 WindPlantIEC

Inheritance path = WindPlantDynamics: DynamicsFunctionBlock: IdentifiedObject

Simplified IEC type plant level model.

Reference: IEC 61400-27-1:2015, Annex D.

Table 991 shows all attributes of WindPlantIEC.

Table 991 – Attributes of WindDynamics::WindPlantIEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 992 shows all association ends of WindPlantIEC with other classes.

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Table 992 – Association ends of WindDynamics::WindPlantIEC with other classes

mult from	name	mult to	type	description
1..1	WindPlantFreqPcontrolIEC	1..1	WindPlantFreqPcontrolIEC	Wind plant frequency and active power control model associated with this wind plant.
1..1	WindPlantReactiveControlIEC	1..1	WindPlantReactiveControlIEC	Wind plant model with which this wind reactive control is associated.
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindPlantDynamics

14.3.3.18.27 WindPlantReactiveControlIEC

Inheritance path = IdentifiedObject

Simplified plant voltage and reactive power control model for use with type 3 and type 4 wind turbine models.

Reference: IEC 61400-27-1:2015, Annex D.

Table 993 shows all attributes of WindPlantReactiveControlIEC.

Table 993 – Attributes of WindDynamics::WindPlantReactiveControlIEC

name	mult	type	description
dxrefmax	1..1	PU	Maximum positive ramp rate for wind turbine reactive power/voltage reference (dxrefmax) (> WindPlantReactiveControlIEC.dxrefmin). It is a project-dependent parameter.
dxrefmin	1..1	PU	Maximum negative ramp rate for wind turbine reactive power/voltage reference (dxrefmin) (< WindPlantReactiveControlIEC.dxrefmax). It is a project-dependent parameter.
kiwpx	1..1	Float	Plant Q controller integral gain (KIWPx). It is a project-dependent parameter.
kiwpxmax	1..1	PU	Maximum reactive power/voltage reference from integration (KIWPxmax) (> WindPlantReactiveControlIEC.kiwpxmin). It is a project-dependent parameter.
kiwpxmin	1..1	PU	Minimum reactive power/voltage reference from integration (KIWPxmin) (< WindPlantReactiveControlIEC.kiwpxmax). It is a project-dependent parameter.
kpwpx	1..1	Float	Plant Q controller proportional gain (KPWPx). It is a project-dependent parameter.
kwpqref	1..1	PU	Reactive power reference gain (KWPqref). It is a project-dependent parameter.
kwpqu	1..1	PU	Plant voltage control droop (KWPqu). It is a project-dependent parameter.
tuqfilt	1..1	Seconds	Filter time constant for voltage-dependent reactive power (Tuqfilt) (>= 0). It is a project-dependent parameter.
twppfiltq	1..1	Seconds	Filter time constant for active power measurement (TWPpfiltq) (>= 0). It is a project-dependent parameter.
twpqfiltq	1..1	Seconds	Filter time constant for reactive power measurement (TWPqfiltq) (>= 0). It is a project-dependent parameter.

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name	mult	type	description
twpufiltq	1..1	Seconds	Filter time constant for voltage measurement (TWPufiltq) (>= 0). It is a project-dependent parameter.
txft	1..1	Seconds	Lead time constant in reference value transfer function (Txft) (>= 0). It is a project-dependent parameter.
txfv	1..1	Seconds	Lag time constant in reference value transfer function (Txfv) (>= 0). It is a project-dependent parameter.
uwpqdip	1..1	PU	Voltage threshold for UVRT detection in Q control (uWPqdip). It is a project-dependent parameter.
windPlantQcontrolModesType	1..1	WindPlantQcontrolModeKind	Reactive power/voltage controller mode (MWPqmode). It is a case-dependent parameter.
xrefmax	1..1	PU	Maximum xWTref (qWTref or delta uWTref) request from the plant controller (xrefmax) (> WindPlantReactiveControlIEC.xrefmin). It is a case-dependent parameter.
xrefmin	1..1	PU	Minimum xWTref (qWTref or delta uWTref) request from the plant controller (xrefmin) (< WindPlantReactiveControlIEC.xrefmax). It is a project-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.28 WindProtectionIEC

Inheritance path = IdentifiedObject

The grid protection model includes protection against over- and under-voltage, and against over- and under-frequency.

Reference: IEC 61400-27-1:2015, 5.6.6.

Table 994 shows all attributes of WindProtectionIEC.

Table 994 – Attributes of WindDynamics::WindProtectionIEC

name	mult	type	description
dfimax	1..1	PU	Maximum rate of change of frequency (dFmax). It is a type-dependent parameter.
fover	1..1	PU	Wind turbine over frequency protection activation threshold (fover). It is a project-dependent parameter.
funder	1..1	PU	Wind turbine under frequency protection activation threshold (funder). It is a project-dependent parameter.
mzc	1..1	Boolean	Zero crossing measurement mode (Mzc). It is a type-dependent parameter. true = WT protection system uses zero crossings to detect frequency (1 in the IEC model) false = WT protection system does not use zero crossings to detect frequency (0 in the IEC model).

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name	mult	type	description
tfma	1..1	Seconds	Time interval of moving average window (TfMA) (>= 0). It is a type-dependent parameter.
uover	1..1	PU	Wind turbine over voltage protection activation threshold (uover). It is a project-dependent parameter.
uunder	1..1	PU	Wind turbine under voltage protection activation threshold (uunder). It is a project-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.29 WindRefFrameRotIEC

Inheritance path = IdentifiedObject

Reference frame rotation model.

Reference: IEC 61400-27-1:2015, 5.6.3.5.

Table 995 shows all attributes of WindRefFrameRotIEC.

Table 995 – Attributes of WindDynamics::WindRefFrameRotIEC

name	mult	type	description
tpll	1..1	Seconds	Time constant for PLL first order filter model (TPLL) (>= 0). It is a type-dependent parameter.
upll1	1..1	PU	Voltage below which the angle of the voltage is filtered and possibly also frozen (uPLL1). It is a type-dependent parameter.
upll2	1..1	PU	Voltage (uPLL2) below which the angle of the voltage is frozen if uPLL2 is smaller or equal to uPLL1 . It is a type-dependent parameter.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.18.30 (abstract) WindTurbineType1or2Dynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to wind turbines type 1 and type 2 and their control models. Generator model for wind turbine of type 1 or type 2 is a standard asynchronous generator model.

Table 996 shows all attributes of WindTurbineType1or2Dynamics.

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Table 996 – Attributes of WindDynamics::WindTurbineType1or2Dynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 997 shows all association ends of WindTurbineType1or2Dynamics with other classes.

Table 997 – Association ends of WindDynamics:: WindTurbineType1or2Dynamics with other classes

mult from	name	mult to	type	description
0..1	RemoteInputSignal	0..1	RemoteInputSignal	Remote input signal used by this wind generator type 1 or type 2 model.
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	Asynchronous machine model with which this wind generator type 1 or type 2 model is associated.

14.3.3.18.31 (abstract) WindTurbineType1or2IEC

Inheritance path = WindTurbineType1or2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to IEC wind turbines type 1 and type 2 including their control models.

Generator model for wind turbine of IEC type 1 or type 2 is a standard asynchronous generator model.

Reference: IEC 61400-27-1:2015, 5.5.2 and 5.5.3.

Table 998 shows all attributes of WindTurbineType1or2IEC.

Table 998 – Attributes of WindDynamics::WindTurbineType1or2IEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 999 shows all association ends of WindTurbineType1or2IEC with other classes.

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Table 999 – Association ends of WindDynamics:: WindTurbineType1or2IEC with other classes

mult from	name	mult to	type	description
0..1	WindMechIEC	1..1	WindMechIEC	Wind mechanical model associated with this wind generator type 1 or type 2 model.
0..1	WindProtectionIEC	1..1	WindProtectionIEC	Wind turbune protection model associated with this wind generator type 1 or type 2 model.
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType1or2Dynamics
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	inherited from: WindTurbineType1or2Dynamics

14.3.3.18.32 WindTurbineType3IEC

Inheritance path = WindTurbineType3or4IEC: WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to IEC wind turbines type 3 including their control models.

Table 1000 shows all attributes of WindTurbineType3IEC.

Table 1000 – Attributes of WindDynamics::WindTurbineType3IEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1001 shows all association ends of WindTurbineType3IEC with other classes.

Table 1001 – Association ends of WindDynamics:: WindTurbineType3IEC with other classes

mult from	name	mult to	type	description
1..1	WindAeroOneDimIEC	0..1	WindAeroOneDimIEC	Wind aerodynamic model associated with this wind generator type 3 model.
1..1	WindAeroTwoDimIEC	0..1	WindAeroTwoDimIEC	Wind aerodynamic model associated with this wind turbine type 3 model.
1..1	WindContPitchAngleIEC	1..1	WindContPitchAngleIEC	Wind control pitch angle model associated with this wind turbine type 3.
1..1	WindContPType3IEC	1..1	WindContPType3IEC	Wind control P type 3 model associated with this wind turbine type 3 model.
0..1	WindMechIEC	1..1	WindMechIEC	Wind mechanical model associated with this wind turbine type 3 model.
0..1	WindGenType3IEC	0..1	WindGenType3IEC	Wind generator type 3 model associated with this wind turbine type 3 model.
1..1	WindContCurrLimIEC	1..1	WindContCurrLimIEC	inherited from: WindTurbineType3or4IEC
1..1	WIndContQIEC	1..1	WindContQIEC	inherited from: WindTurbineType3or4IEC

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mult from	name	mult to	type	description
0..1	WindContQLimIEC	0..1	WindContQLimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQPQULimIEC	0..1	WindContQPQULimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType3or4IEC
1..1	WindRefFrameRotIEC	1..1	WindRefFrameRotIEC	inherited from: WindTurbineType3or4IEC
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.3.18.33 (abstract) WindTurbineType3or4Dynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to wind turbines type 3 and type 4 and wind plant including their control models.

Table 1002 shows all attributes of WindTurbineType3or4Dynamics.

Table 1002 – Attributes of WindDynamics::WindTurbineType3or4Dynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1003 shows all association ends of WindTurbineType3or4Dynamics with other classes.

Table 1003 – Association ends of WindDynamics:: WindTurbineType3or4Dynamics with other classes

mult from	name	mult to	type	description
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	The power electronics connection associated with this wind turbine type 3 or type 4 dynamics model.
0..1	RemoteInputSignal	0..1	RemoteInputSignal	Remote input signal used by these wind turbine type 3 or type 4 models.
1..*	WindPlantDynamics	0..1	WindPlantDynamics	The wind plant with which the wind turbines type 3 or type 4 are associated.

14.3.3.18.34 (abstract) WindTurbineType3or4IEC

Inheritance path = WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to IEC wind turbines type 3 and type 4 including their control models.

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Table 1004 shows all attributes of WindTurbineType3or4IEC.

Table 1004 – Attributes of WindDynamics::WindTurbineType3or4IEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1005 shows all association ends of WindTurbineType3or4IEC with other classes.

Table 1005 – Association ends of WindDynamics:: WindTurbineType3or4IEC with other classes

mult from	name	mult to	type	description
1..1	WindContCurrLimIEC	1..1	WindContCurrLimIEC	Wind control current limitation model associated with this wind turbine type 3 or type 4 model.
1..1	WIndContQIEC	1..1	WindContQIEC	Wind control Q model associated with this wind turbine type 3 or type 4 model.
0..1	WindContQLimIEC	0..1	WindContQLimIEC	Constant Q limitation model associated with this wind generator type 3 or type 4 model.
0..1	WindContQPQULimIEC	0..1	WindContQPQULimIEC	QP and QU limitation model associated with this wind generator type 3 or type 4 model.
0..1	WindProtectionIEC	1..1	WindProtectionIEC	Wind turbune protection model associated with this wind generator type 3 or type 4 model.
1..1	WindRefFrameRotIEC	1..1	WindRefFrameRotIEC	Reference frame rotation model associated with this wind turbine type 3 or type 4 model.
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.3.18.35 WindTurbineType4aIEC

Inheritance path = WindTurbineType4IEC: WindTurbineType3or4IEC: WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind turbine IEC type 4A.

Reference: IEC 61400-27-1:2015, 5.5.5.2.

Table 1006 shows all attributes of WindTurbineType4aIEC.

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Table 1006 – Attributes of WindDynamics::WindTurbineType4aIEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1007 shows all association ends of WindTurbineType4aIEC with other classes.

Table 1007 – Association ends of WindDynamics:: WindTurbineType4aIEC with other classes

mult from	name	mult to	type	description
1..1	WindContPType4aIEC	1..1	WindContPType4aIEC	Wind control P type 4A model associated with this wind turbine type 4A model.
0..1	WindGenType4IEC	0..1	WindGenType4IEC	Wind generator type 4 model associated with this wind turbine type 4A model.
0..1	WindGenType3aIEC	0..1	WindGenType3aIEC	inherited from: WindTurbineType4IEC
1..1	WindContCurrLimIEC	1..1	WindContCurrLimIEC	inherited from: WindTurbineType3or4IEC
1..1	WIndContQIEC	1..1	WindContQIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQLimIEC	0..1	WindContQLimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQPQULimIEC	0..1	WindContQPQULimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType3or4IEC
1..1	WindRefFrameRotIEC	1..1	WindRefFrameRotIEC	inherited from: WindTurbineType3or4IEC
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.3.18.36 WindTurbineType4bIEC

Inheritance path = WindTurbineType4IEC: WindTurbineType3or4IEC: WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind turbine IEC type 4B.

Reference: IEC 61400-27-1:2015, 5.5.5.3.

Table 1008 shows all attributes of WindTurbineType4bIEC.

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Table 1008 – Attributes of WindDynamics::WindTurbineType4bIEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1009 shows all association ends of WindTurbineType4bIEC with other classes.

Table 1009 – Association ends of WindDynamics:: WindTurbineType4bIEC with other classes

mult from	name	mult to	type	description
1..1	WindContPType4bIEC	1..1	WindContPType4bIEC	Wind control P type 4B model associated with this wind turbine type 4B model.
0..1	WindGenType4IEC	0..1	WindGenType4IEC	Wind generator type 4 model associated with this wind turbine type 4B model.
0..1	WindMechIEC	1..1	WindMechIEC	Wind mechanical model associated with this wind turbine type 4B model.
0..1	WindGenType3aIEC	0..1	WindGenType3aIEC	inherited from: WindTurbineType4IEC
1..1	WindContCurrLimIEC	1..1	WindContCurrLimIEC	inherited from: WindTurbineType3or4IEC
1..1	WIndContQIEC	1..1	WindContQIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQLimIEC	0..1	WindContQLimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQPQULimIEC	0..1	WindContQPQULimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType3or4IEC
1..1	WindRefFrameRotIEC	1..1	WindRefFrameRotIEC	inherited from: WindTurbineType3or4IEC
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.3.18.37 (abstract) WindTurbineType4IEC

Inheritance path = WindTurbineType3or4IEC: WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Parent class supporting relationships to IEC wind turbines type 4 including their control models.

Table 1010 shows all attributes of WindTurbineType4IEC.

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Table 1010 – Attributes of WindDynamics::WindTurbineType4IEC

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1011 shows all association ends of WindTurbineType4IEC with other classes.

Table 1011 – Association ends of WindDynamics:: WindTurbineType4IEC with other classes

mult from	name	mult to	type	description
0..1	WindGenType3aIEC	0..1	WindGenType3aIEC	Wind generator type 3A model associated with this wind turbine type 4 model.
1..1	WindContCurrLimIEC	1..1	WindContCurrLimIEC	inherited from: WindTurbineType3or4IEC
1..1	WIndContQIEC	1..1	WindContQIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQLimIEC	0..1	WindContQLimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindContQPQULimIEC	0..1	WindContQPQULimIEC	inherited from: WindTurbineType3or4IEC
0..1	WindProtectionIEC	1..1	WindProtectionIEC	inherited from: WindTurbineType3or4IEC
1..1	WindRefFrameRotIEC	1..1	WindRefFrameRotIEC	inherited from: WindTurbineType3or4IEC
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.3.19 Package LoadDynamics

14.3.3.19.1 General

Dynamic load models are used to represent the dynamic real and reactive load behaviour of a load from the static power flow model.

Dynamic load models can be defined as applying either to a single load (energy consumer) or to a group of energy consumers.

Large industrial motors or groups of similar motors can be represented by a synchronous machine model (SynchronousMachineDynamics) or an asynchronous machine model (AsynchronousMachineDynamics), which are usually represented as generators with negative active power output in the static (power flow) data.

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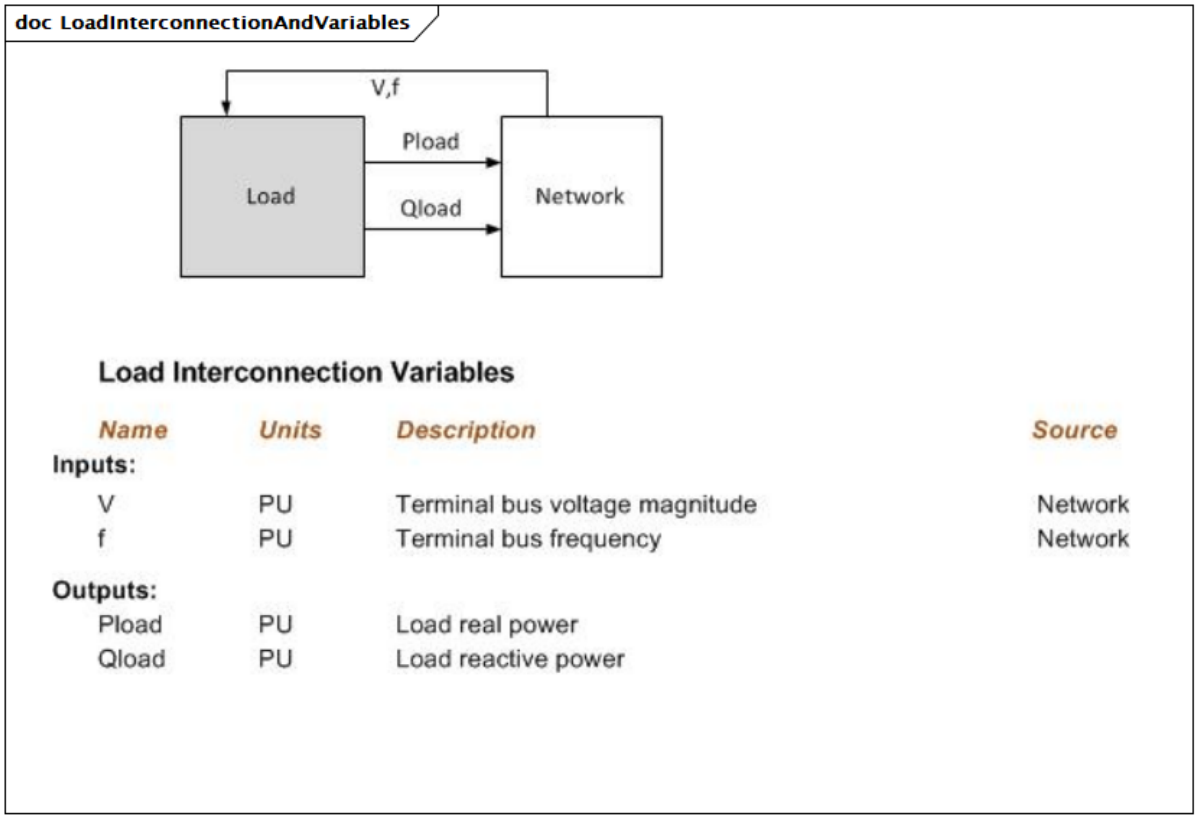

Load Interconnection Variables

Name	Units	Description	Source
Inputs:			
V	PU	Terminal bus voltage magnitude	Network
f	PU	Terminal bus frequency	Network
Outputs:			
Pload	PU	Load real power	
Qload	PU	Load reactive power	

Figure 216 – LoadInterconnectionAndVariables

Figure 216: This diagram illustrates the inputs and outputs for a load.

Diagram details:

1) The application program converts the P and Q of the load into a current injection at the bus.

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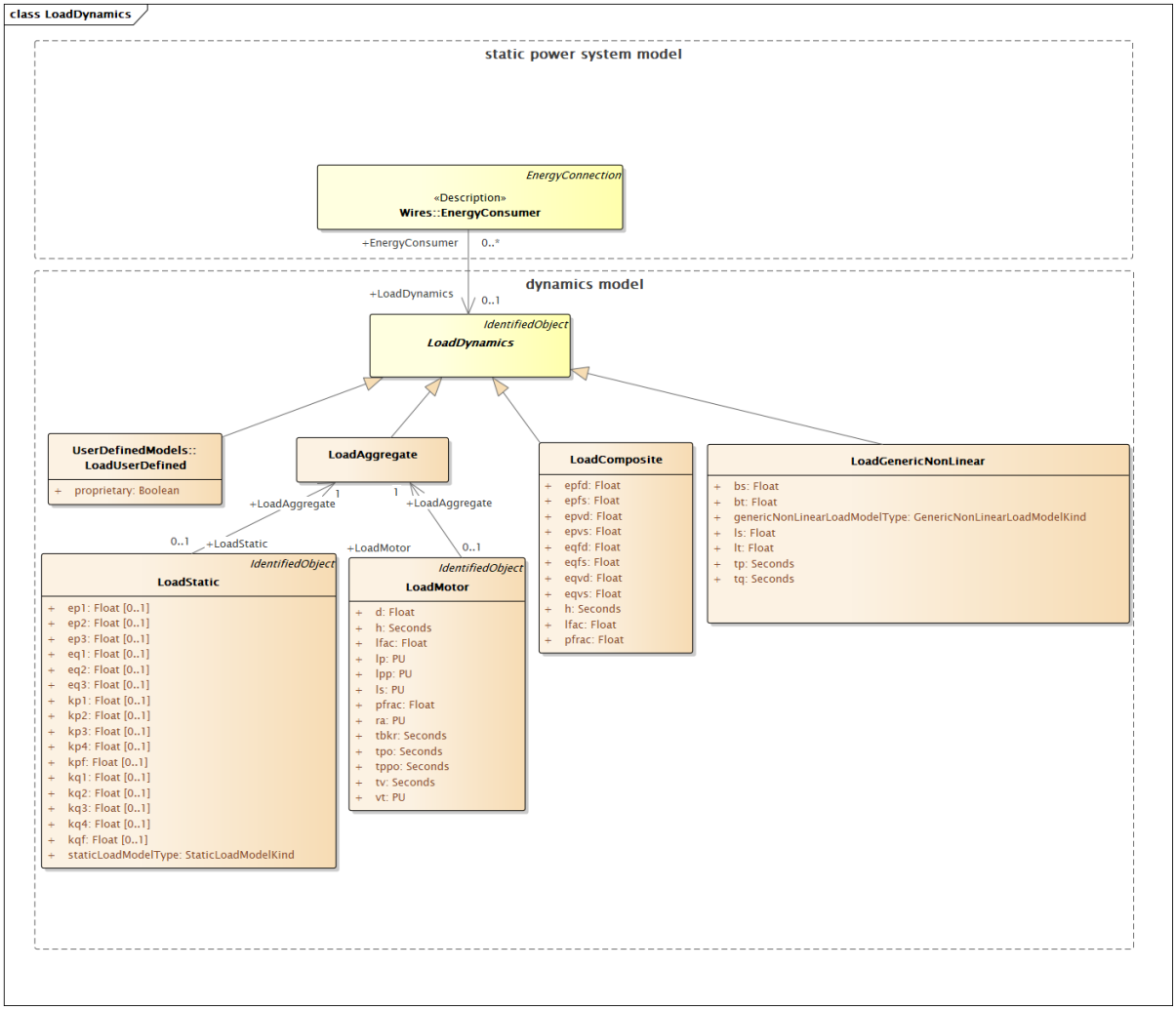

Figure 217 – Class diagram LoadDynamics::LoadDynamics

Figure 217: This diagram shows static power system model classes and dynamics model classes related to modelling of load for dynamics. An association between the EnergyConsumerDynamics class and the EnergyConsumer class provides a means for relating the dynamics model to the static model.

14.3.3.19.2 LoadComposite

Inheritance path = LoadDynamics: IdentifiedObject

Combined static load and induction motor load effects.

The dynamics of the motor are simplified by linearizing the induction machine equations.

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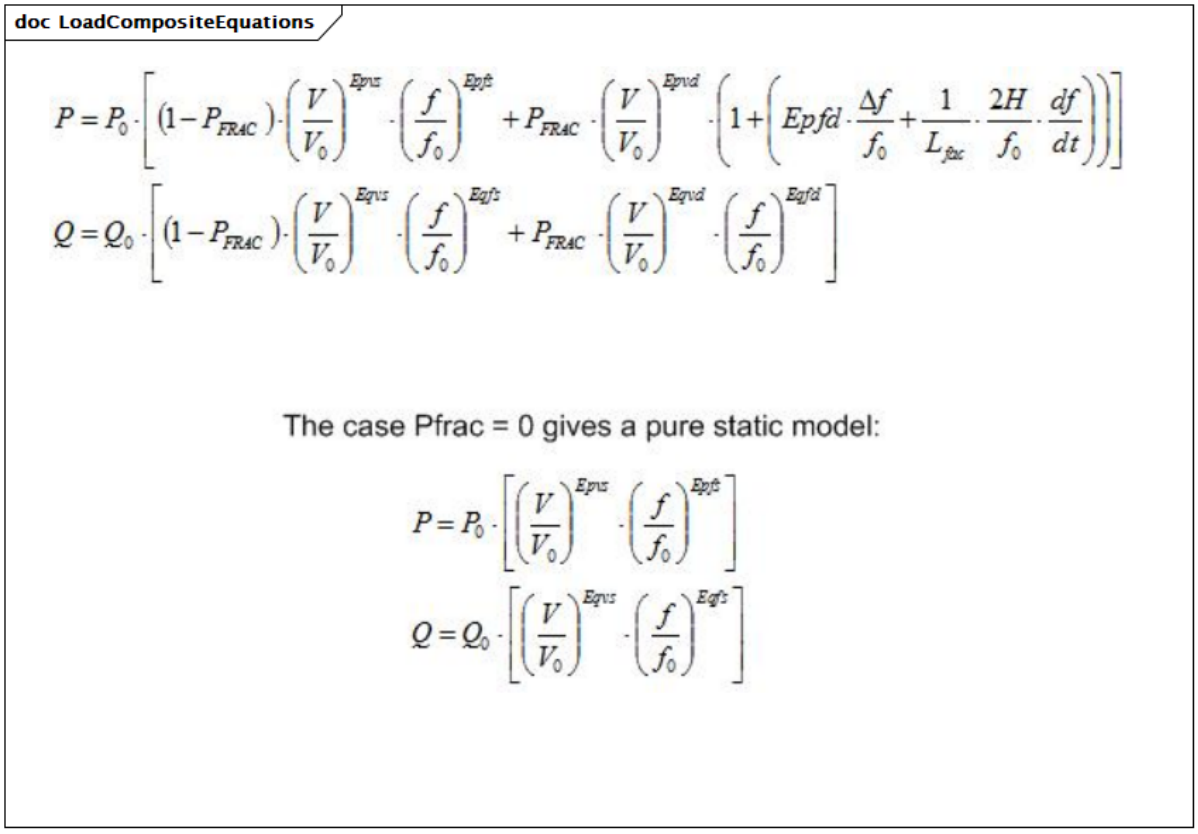

Figure 218 – LoadCompositeEquations

Figure 218: This diagram illustrates the equations of the model which combines static load and induction motor load effects.

Table 1012 shows all attributes of LoadComposite.

Table 1012 – Attributes of LoadDynamics::LoadComposite

name	mult	type	description
epvs	1..1	Float	Active load-voltage dependence index (static) (Epvs). Typical value = 0,7.
epfs	1..1	Float	Active load-frequency dependence index (static) (Epfs). Typical value = 1,5.
eqvs	1..1	Float	Reactive load-voltage dependence index (static) (Eqvs). Typical value = 2.
eqfs	1..1	Float	Reactive load-frequency dependence index (static) (Eqfs). Typical value = 0.
epvd	1..1	Float	Active load-voltage dependence index (dynamic) (Epvd). Typical value = 0,7.
epfd	1..1	Float	Active load-frequency dependence index (dynamic) (Epfd). Typical value = 1,5.
eqvd	1..1	Float	Reactive load-voltage dependence index (dynamic) (Eqvd). Typical value = 2.
eqfd	1..1	Float	Reactive load-frequency dependence index (dynamic) (Eqfd). Typical value = 0.
lfac	1..1	Float	Loading factor (Lfac). The ratio of initial P to motor MVA base. Typical value = 0,8.
h	1..1	Seconds	Inertia constant (H) (>= 0). Typical value = 2,5.

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name	mult	type	description
pfrac	1..1	Float	Fraction of constant-power load to be represented by this motor model (PFRAC) (>= 0,0 and <= 1,0). Typical value = 0,5.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.19.3 LoadGenericNonLinear

Inheritance path = LoadDynamics: IdentifiedObject

Generic non-linear dynamic (GNLD) load. This model can be used in mid-term and long-term voltage stability simulations (i.e., to study voltage collapse), as it can replace a more detailed representation of aggregate load, including induction motors, thermostatically controlled and static loads.

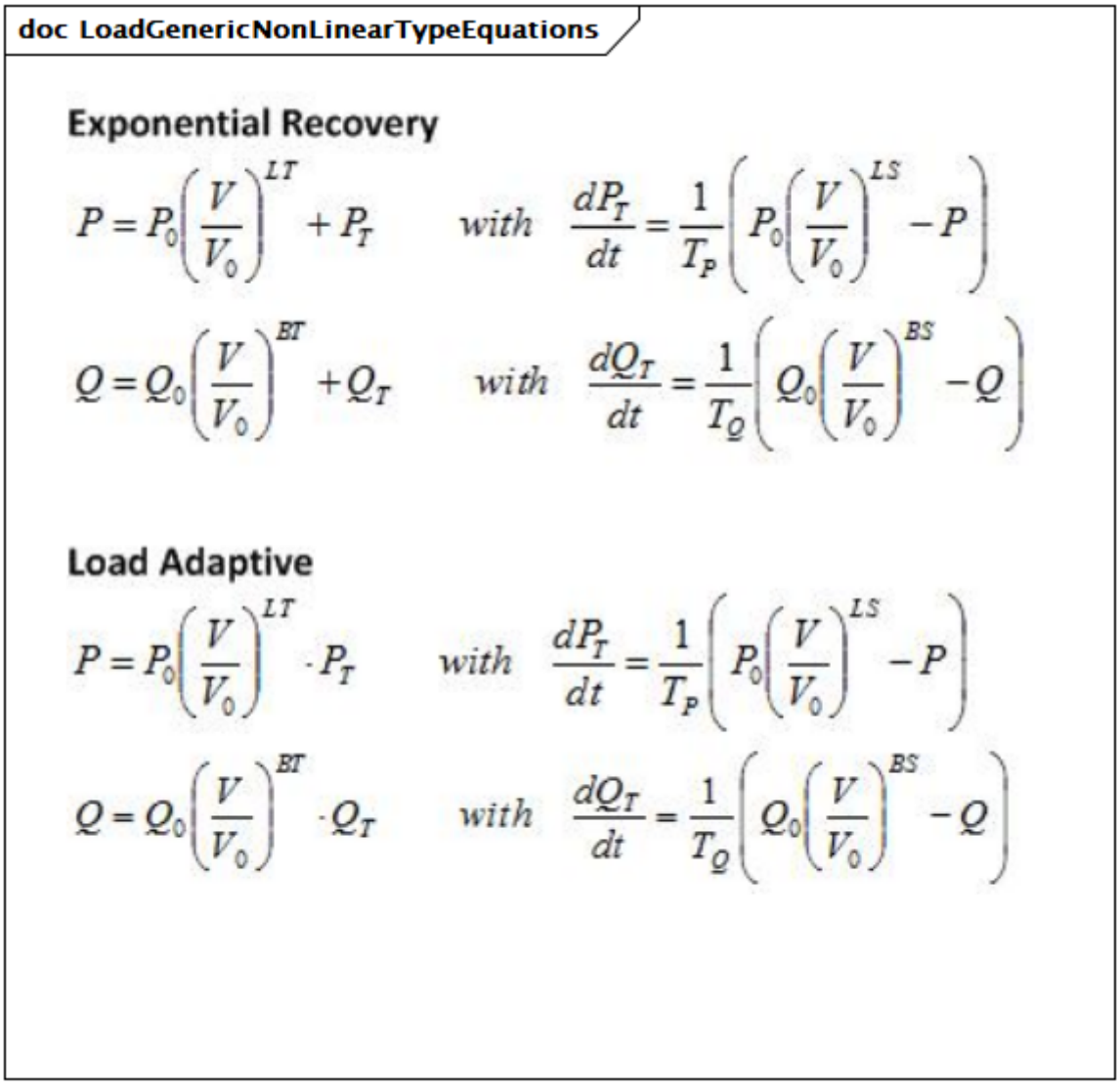

Figure 219 – LoadGenericNonLinearTypeEquations

Figure 219: This diagram illustrates the generic non-linear type equations.

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Diagram details:

1) P0 and Q0 are initial active and reactive power from the steady state solution (SvPowerFlow.p and SvPowerFlow.q).
2) V0 is the initial voltage from the steady state solution (SvPowerFlow.v).
3) V is the implied voltage signal from the bus to which the load is attached.

Table 1013 shows all attributes of LoadGenericNonLinear.

Table 1013 – Attributes of LoadDynamics::LoadGenericNonLinear

name	mult	type	description
genericNonLinearLoadModelType	1..1	GenericNonLinearLoadModelKind	Type of generic non-linear load model.
tp	1..1	Seconds	Time constant of lag function of active power (TP) (> 0).
tq	1..1	Seconds	Time constant of lag function of reactive power (TQ) (> 0).
ls	1..1	Float	Steady state voltage index for active power (LS).
lt	1..1	Float	Transient voltage index for active power (LT).
bs	1..1	Float	Steady state voltage index for reactive power (BS).
bt	1..1	Float	Transient voltage index for reactive power (BT).
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.19.4 (abstract) LoadDynamics

Inheritance path = IdentifiedObject

Load whose behaviour is described by reference to a standard model or by definition of a user-defined model.

A standard feature of dynamic load behaviour modelling is the ability to associate the same behaviour to multiple energy consumers by means of a single load definition. The load model is always applied to individual bus loads (energy consumers).

Table 1014 shows all attributes of LoadDynamics.

Table 1014 – Attributes of LoadDynamics::LoadDynamics

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.19.5 LoadAggregate

Inheritance path = LoadDynamics: IdentifiedObject

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Aggregate loads are used to represent all or part of the real and reactive load from one or more loads in the static (power flow) data. This load is usually the aggregation of many individual load devices and the load model is an approximate representation of the aggregate response of the load devices to system disturbances.

Standard aggregate load model comprised of static and/or dynamic components. A static load model represents the sensitivity of the real and reactive power consumed by the load to the amplitude and frequency of the bus voltage. A dynamic load model can be used to represent the aggregate response of the motor components of the load.

Table 1015 shows all attributes of LoadAggregate.

Table 1015 – Attributes of LoadDynamics::LoadAggregate

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.19.6 LoadStatic

Inheritance path = IdentifiedObject

General static load. This model represents the sensitivity of the real and reactive power consumed by the load to the amplitude and frequency of the bus voltage.

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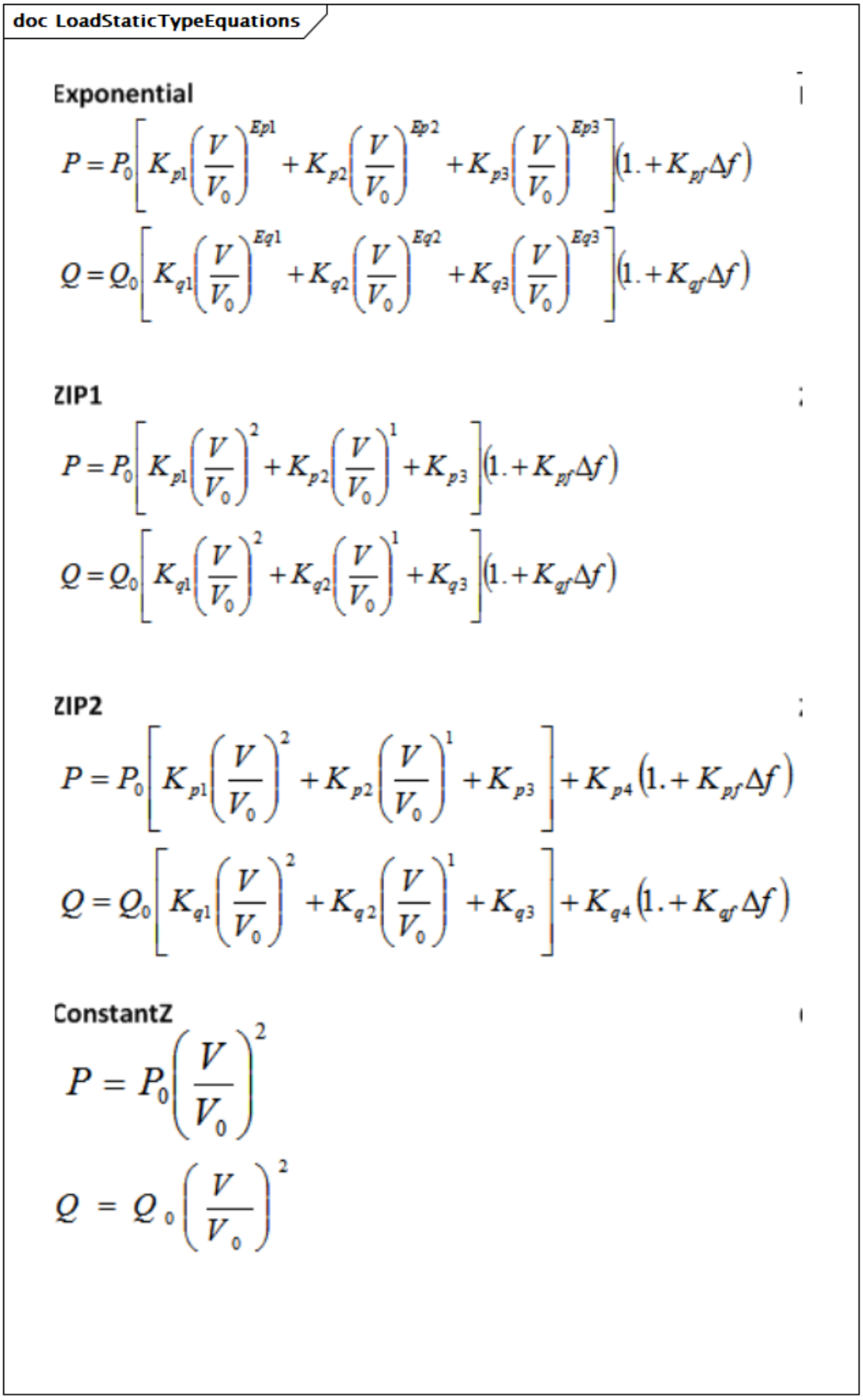

Figure 220 – LoadStaticTypeEquations

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Figure 220: This diagram illustrates the variations of the static load model which can be specified by the staticLoadModelType attribute.

Table 1016 shows all attributes of LoadStatic.

Table 1016 – Attributes of LoadDynamics::LoadStatic

name	mult	type	description
staticLoadModelType	1..1	StaticLoadModelKind	Type of static load model. Typical value = constantZ.
kp1	0..1	Float	First term voltage coefficient for active power (Kp1). Not used when .staticLoadModelType = constantZ.
kp2	0..1	Float	Second term voltage coefficient for active power (Kp2). Not used when .staticLoadModelType = constantZ.
kp3	0..1	Float	Third term voltage coefficient for active power (Kp3). Not used when .staticLoadModelType = constantZ.
kp4	0..1	Float	Frequency coefficient for active power (Kp4) (not = 0 if .staticLoadModelType = zIP2). Used only when .staticLoadModelType = zIP2.
ep1	0..1	Float	First term voltage exponent for active power (Ep1). Used only when .staticLoadModelType = exponential.
ep2	0..1	Float	Second term voltage exponent for active power (Ep2). Used only when .staticLoadModelType = exponential.
ep3	0..1	Float	Third term voltage exponent for active power (Ep3). Used only when .staticLoadModelType = exponential.
kpf	0..1	Float	Frequency deviation coefficient for active power (Kpf). Not used when .staticLoadModelType = constantZ.
kq1	0..1	Float	First term voltage coefficient for reactive power (Kq1). Not used when .staticLoadModelType = constantZ.
kq2	0..1	Float	Second term voltage coefficient for reactive power (Kq2). Not used when .staticLoadModelType = constantZ.
kq3	0..1	Float	Third term voltage coefficient for reactive power (Kq3). Not used when .staticLoadModelType = constantZ.
kq4	0..1	Float	Frequency coefficient for reactive power (Kq4) (not = 0 when .staticLoadModelType = zIP2). Used only when .staticLoadModelType – zIP2.
eq1	0..1	Float	First term voltage exponent for reactive power (Eq1). Used only when .staticLoadModelType = exponential.
eq2	0..1	Float	Second term voltage exponent for reactive power (Eq2). Used only when .staticLoadModelType = exponential.
eq3	0..1	Float	Third term voltage exponent for reactive power (Eq3). Used only when .staticLoadModelType = exponential.
kqf	0..1	Float	Frequency deviation coefficient for reactive power (Kqf). Not used when .staticLoadModelType = constantZ.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
name	0..1	String	inherited from: IdentifiedObject

Table 1017 shows all association ends of LoadStatic with other classes.

Table 1017 – Association ends of LoadDynamics::LoadStatic with other classes

mult from	name	mult to	type	description
0..1	LoadAggregate	1..1	LoadAggregate	Aggregate load to which this aggregate static load belongs.

14.3.3.19.7 LoadMotor

Inheritance path = IdentifiedObject

Aggregate induction motor load. This model is used to represent a fraction of an ordinary load as "induction motor load". It allows a load that is treated as an ordinary constant power in power flow analysis to be represented by an induction motor in dynamic simulation. This model is intended for representation of aggregations of many motors dispersed through a load represented at a high voltage bus but where there is no information on the characteristics of individual motors.

Either a "one-cage" or "two-cage" model of the induction machine can be modelled. Magnetic saturation is not modelled.

This model treats a fraction of the constant power part of a load as a motor. During initialisation, the initial power drawn by the motor is set equal to Pfrac times the constant P part of the static load. The remainder of the load is left as a static load.

The reactive power demand of the motor is calculated during initialisation as a function of voltage at the load bus. This reactive power demand can be less than or greater than the constant Q component of the load. If the motor's reactive demand is greater than the constant Q component of the load, the model inserts a shunt capacitor at the terminal of the motor to bring its reactive demand down to equal the constant Q reactive load.

If an induction motor load model and a static load model are both present for a load, the motor Pfrac is assumed to be subtracted from the power flow constant P load before the static load model is applied. The remainder of the load, if any, is then represented by the static load model.

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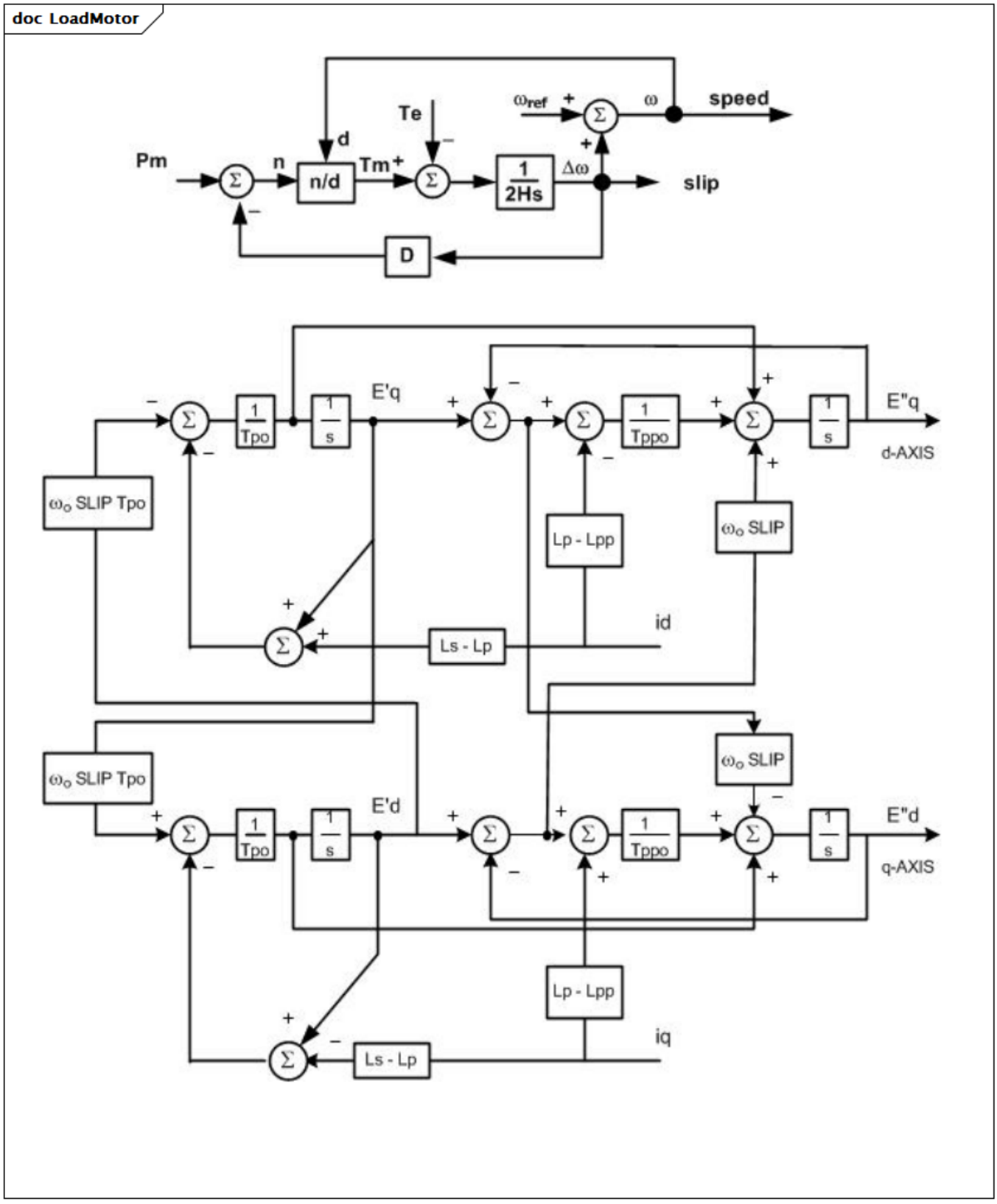

Figure 221 – LoadMotor

Figure 221: These diagrams describe the aggregate induction motor load model. The upper diagram represents the mechanical load model. The lower is the 3-phase induction motor electrical model.

Parameter details:

1) Ls, Lp, Lpp shall all be specified.
2) If Lpp = 0 or Lpp = Lp, or Tppo = 0, only one cage is represented.

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3) The initial value of Pm or the value coming from an separate mechanical load model, if used, is multiplied by speed raised to the power D to determine the effective mechanical load seen by the motor.
4) The parameters Vt and Tv define the under-voltage tripping logic. A timer is started if the terminal voltage falls below Vt PU and continues to run until the voltage rises above Vt. If voltage has not risen above Vt when the timer reaches Tv, the motor is tripped after a delay of circuit breaker operating time (Tbkr).
5) PU parameters are on the motor MVA base. The MVA base is calculated during initialisation as the initial motor P in MW divided by the loading factor (LoadMotor.lfac).

Table 1018 shows all attributes of LoadMotor.

Table 1018 – Attributes of LoadDynamics::LoadMotor

name	mult	type	description
pfrac	1..1	Float	Fraction of constant-power load to be represented by this motor model (Pfrac) (>= 0,0 and <= 1,0). Typical value = 0,3.
lfac	1..1	Float	Loading factor (Lfac). The ratio of initial P to motor MVA base. Typical value = 0,8.
ls	1..1	PU	Synchronous reactance (Ls). Typical value = 3,2.
lp	1..1	PU	Transient reactance (Lp). Typical value = 0,15.
lpp	1..1	PU	Subtransient reactance (Lpp). Typical value = 0,15.
ra	1..1	PU	Stator resistance (Ra). Typical value = 0.
tpo	1..1	Seconds	Transient rotor time constant (Tpo) (>= 0). Typical value = 1.
tppo	1..1	Seconds	Subtransient rotor time constant (Tppo) (>= 0). Typical value = 0,02.
h	1..1	Seconds	Inertia constant (H) (>= 0). Typical value = 0,4.
d	1..1	Float	Damping factor (D). Unit = delta P/delta speed. Typical value = 2.
vt	1..1	PU	Voltage threshold for tripping (Vt). Typical value = 0,7.
tv	1..1	Seconds	Voltage trip pickup time (Tv) (>= 0). Typical value = 0,1.
tbkr	1..1	Seconds	Circuit breaker operating time (Tbkr) (>= 0). Typical value = 0,08.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1019 shows all association ends of LoadMotor with other classes.

Table 1019 – Association ends of LoadDynamics::LoadMotor with other classes

mult from	name	mult to	type	description
0..1	LoadAggregate	1..1	LoadAggregate	Aggregate load to which this aggregate motor (dynamic) load belongs.

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14.3.3.20 Package HVDCDynamics

14.3.3.20.1 General

High voltage direct current (HVDC) models.

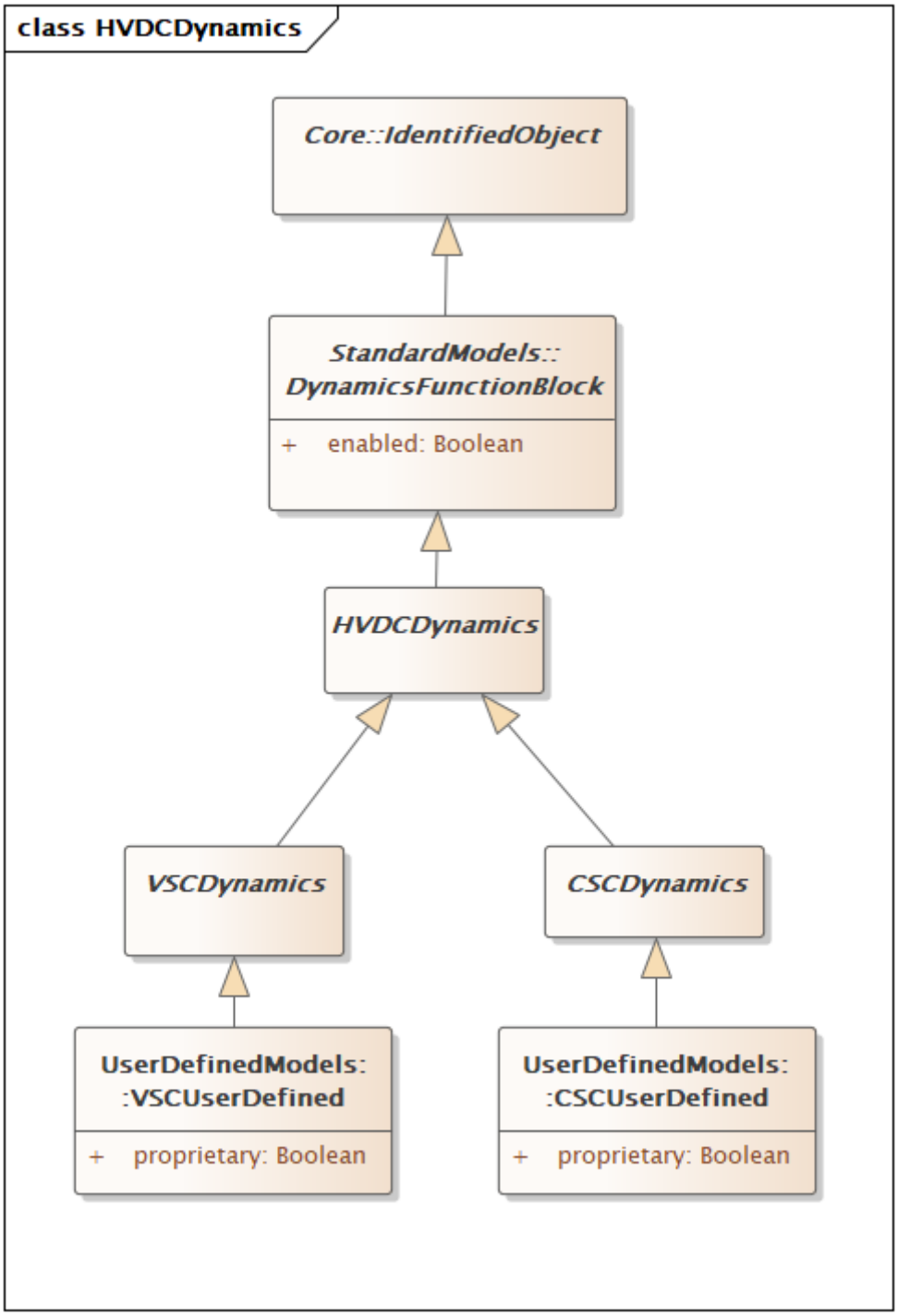

Figure 222 – Class diagram HVDCDynamics::HVDCDynamics

Figure 222: This diagram shows the dynamics model classes related to modelling of the HVDC.

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14.3.3.20.2 (abstract) CSCDynamics

Inheritance path = HVDCDynamics: DynamicsFunctionBlock: IdentifiedObject

CSC function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 1020 shows all attributes of CSCDynamics.

Table 1020 – Attributes of HVDCDynamics::CSCDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1021 shows all association ends of CSCDynamics with other classes.

Table 1021 – Association ends of HVDCDynamics::CSCDynamics with other classes

mult from	name	mult to	type	description
0..1	CSConverter	1..1	CsConverter	Current source converter to which current source converter dynamics model applies.

14.3.3.20.3 (abstract) HVDCDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

HVDC whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 1022 shows all attributes of HVDCDynamics.

Table 1022 – Attributes of HVDCDynamics::HVDCDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.3.20.4 (abstract) VSCDynamics

Inheritance path = HVDCDynamics: DynamicsFunctionBlock: IdentifiedObject

VSC function block whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 1023 shows all attributes of VSCDynamics.

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Table 1023 – Attributes of HVDCDynamics::VSCDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1024 shows all association ends of VSCDynamics with other classes.

Table 1024 – Association ends of HVDCDynamics::VSCDynamics with other classes

mult from	name	mult to	type	description
0..1	VsConverter	1..1	VsConverter	Voltage source converter to which voltage source converter dynamics model applies.

14.3.3.21 Package StaticVarCompensatorDynamics

14.3.3.21.1 General

Static var compensator (SVC) models.

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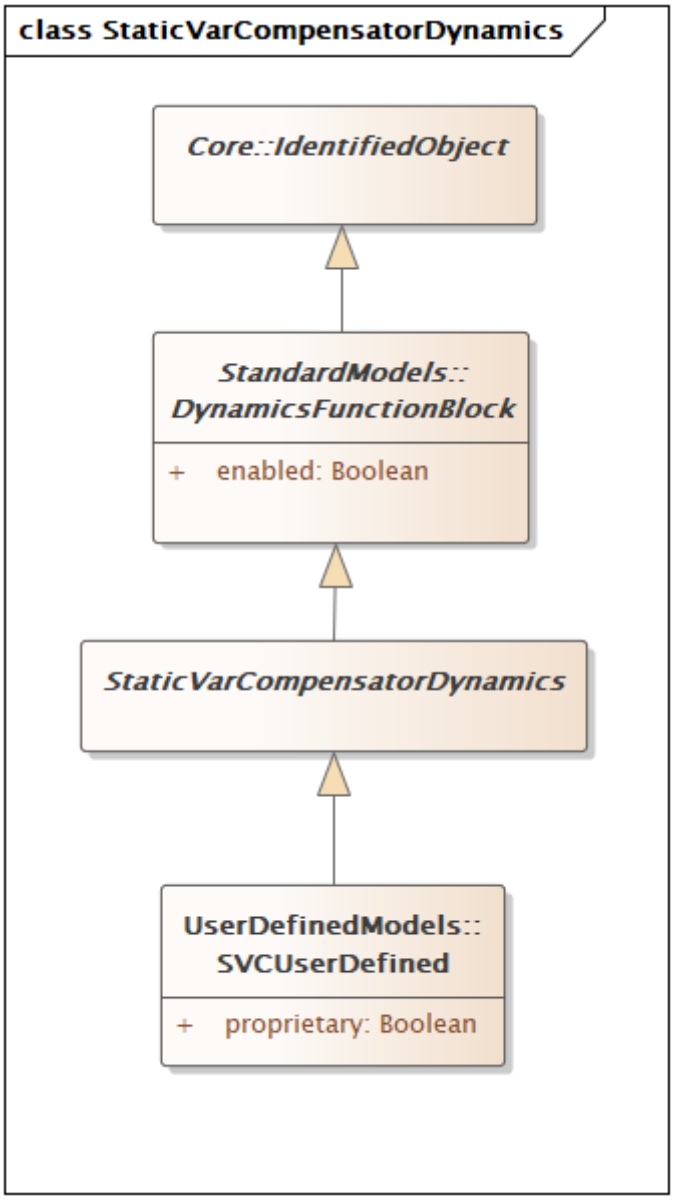

Figure 223 – Class diagram StaticVarCompensatorDynamics:: StaticVarCompensatorDynamics

Figure 223: This diagram shows the dynamics model classes related to modelling of the static var compensator (SVC) models.

14.3.3.21.2 (abstract) StaticVarCompensatorDynamics

Inheritance path = DynamicsFunctionBlock: IdentifiedObject

Static var compensator whose behaviour is described by reference to a standard model or by definition of a user-defined model.

Table 1025 shows all attributes of StaticVarCompensatorDynamics.

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Table 1025 – Attributes of StaticVarCompensatorDynamics:: StaticVarCompensatorDynamics

name	mult	type	description
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1026 shows all association ends of StaticVarCompensatorDynamics with other classes.

Table 1026 – Association ends of StaticVarCompensatorDynamics:: StaticVarCompensatorDynamics with other classes

mult from	name	mult to	type	description
0..1	StaticVarCompensator	1..1	StaticVarCompensator	Static Var Compensator to which Static Var Compensator dynamics model applies.

14.3.4 Package UserDefinedModels

14.3.4.1 General

This subclause contains user-defined dynamic model classes to support the exchange of both proprietary and explicitly defined user-defined models.

Proprietary models represent behaviour which, while not defined by a standard model class, is mutually understood by the sending and receiving applications based on the name passed in the .name attribute of the appropriate xxxUserDefined class. Proprietary model parameters are passed as general attributes using as many instances of the ProprietaryParameterDynamics class as there are parameters.

Explicitly defined models describe dynamic behaviour in detail in terms of control blocks and their input and output signals. Note that the classes to support explicitly defined modelling are not currently defined – it is future work intended to also be supported by the family of xxxUserDefined classes.

Both types of user-defined models use the family of xxxUserDefined classes, which allow a user-defined model to be used:

– as the model for an individual standard function block (such as a turbine-governor or power system stabilizer) in a standard interconnection model whose other function blocks could be either standard or user-defined. For an illustration of this form of usage for a proprietary model, see the ExampleFunctionBlockProprietaryModel diagram in 5.5.
– as the complete representation of a dynamic behaviour model (for an entire synchronous machine, for example) where standard function blocks and standard interconnections are not used at all. For an illustration of this form of usage for a proprietary model, see the ExampleCompleteProprietaryModel diagram in 5.5.

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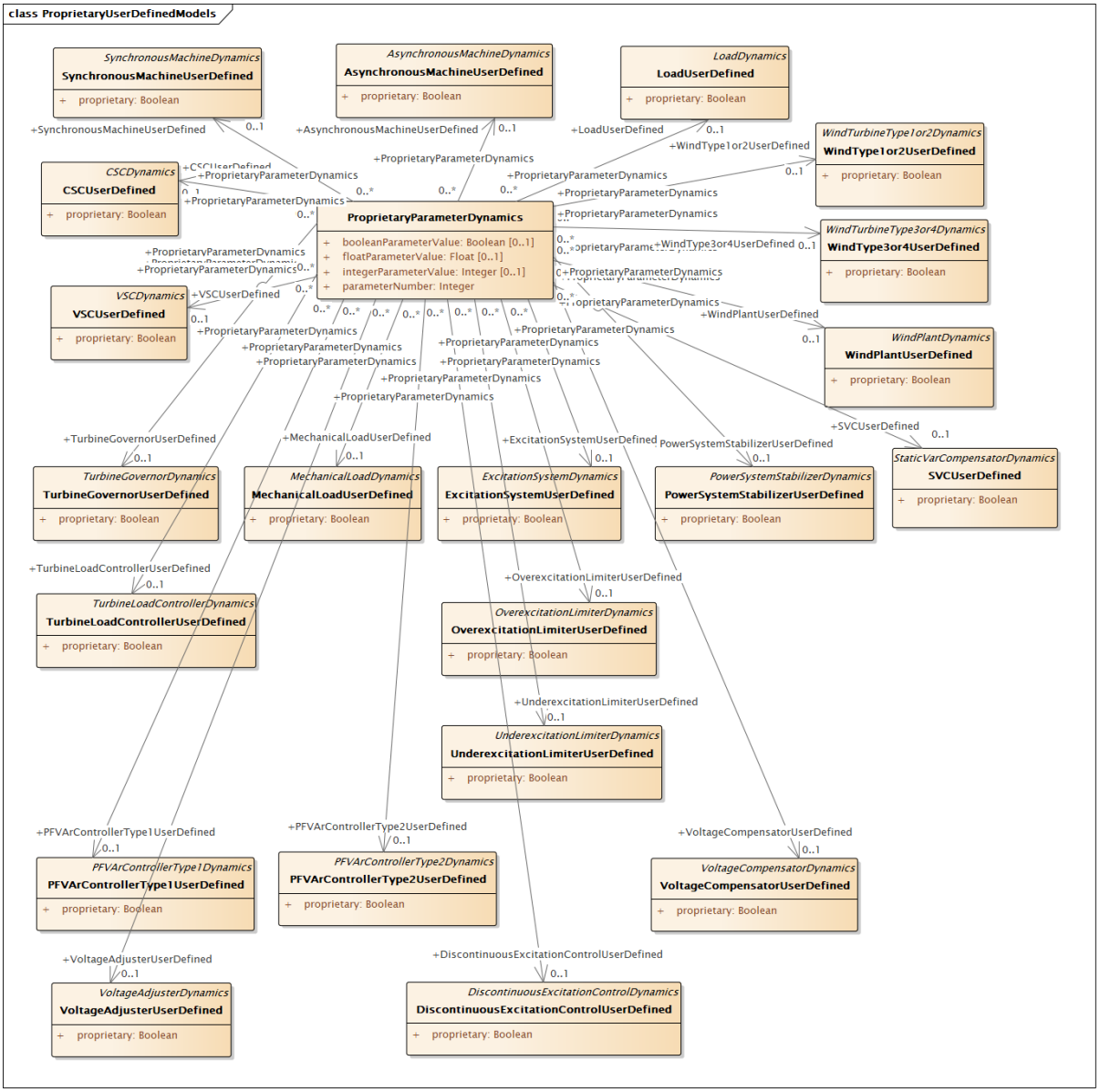

Figure 224 – Class diagram UserDefinedModels::ProprietaryUserDefinedModels

Figure 224: This diagram shows the user-defined model classes and their relationship to the ProprietaryParameterDynamics class that supports the definition of multiple, appropriately data-typed attributes for each user-defined model.

14.3.4.2 CSCUserDefined

Inheritance path = CSCDynamics: HVDCDynamics: DynamicsFunctionBlock: IdentifiedObject

Current source converter (CSC) function block whose dynamic behaviour is described by a user-defined model.

Table 1027 shows all attributes of CSCUserDefined.

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Table 1027 – Attributes of UserDefinedModels::CSCUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1028 shows all association ends of CSCUserDefined with other classes.

Table 1028 – Association ends of UserDefinedModels:: CSCUserDefined with other classes

mult from	name	mult to	type	description
0..1	CSConverter	1..1	CsConverter	inherited from: CSCDynamics

14.3.4.3 SVCUserDefined

Inheritance path = StaticVarCompensatorDynamics: DynamicsFunctionBlock: IdentifiedObject

Static var compensator (SVC) function block whose dynamic behaviour is described by a user-defined model.

Table 1029 shows all attributes of SVCUserDefined.

Table 1029 – Attributes of UserDefinedModels::SVCUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1030 shows all association ends of SVCUserDefined with other classes.

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Table 1030 – Association ends of UserDefinedModels:: SVCUserDefined with other classes

mult from	name	mult to	type	description
0..1	StaticVarCompensator	1..1	StaticVarCompensator	inherited from: StaticVarCompensatorDynamics

14.3.4.4 VSCUserDefined

Inheritance path = VSCDynamics: HVDCDynamics: DynamicsFunctionBlock: IdentifiedObject

Voltage source converter (VSC) function block whose dynamic behaviour is described by a user-defined model.

Table 1031 shows all attributes of VSCUserDefined.

Table 1031 – Attributes of UserDefinedModels::VSCUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1032 shows all association ends of VSCUserDefined with other classes.

Table 1032 – Association ends of UserDefinedModels:: VSCUserDefined with other classes

mult from	name	mult to	type	description
0..1	VsConverter	1..1	VsConverter	inherited from: VSCDynamics

14.3.4.5 WindPlantUserDefined

Inheritance path = WindPlantDynamics: DynamicsFunctionBlock: IdentifiedObject

Wind plant function block whose dynamic behaviour is described by a user-defined model.

Table 1033 shows all attributes of WindPlantUserDefined.

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Table 1033 – Attributes of UserDefinedModels::WindPlantUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1034 shows all association ends of WindPlantUserDefined with other classes.

Table 1034 – Association ends of UserDefinedModels:: WindPlantUserDefined with other classes

mult from	name	mult to	type	description
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindPlantDynamics

14.3.4.6 WindType1or2UserDefined

Inheritance path = WindTurbineType1or2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind type 1 or type 2 function block whose dynamic behaviour is described by a user-defined model.

Table 1035 shows all attributes of WindType1or2UserDefined.

Table 1035 – Attributes of UserDefinedModels::WindType1or2UserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1036 shows all association ends of WindType1or2UserDefined with other classes.

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Table 1036 – Association ends of UserDefinedModels:: WindType1or2UserDefined with other classes

mult from	name	mult to	type	description
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType1or2Dynamics
0..1	AsynchronousMachineDynamics	1..1	AsynchronousMachineDynamics	inherited from: WindTurbineType1or2Dynamics

14.3.4.7 WindType3or4UserDefined

Inheritance path = WindTurbineType3or4Dynamics: DynamicsFunctionBlock: IdentifiedObject

Wind type 3 or type 4 function block whose dynamic behaviour is described by a user-defined model.

Table 1037 shows all attributes of WindType3or4UserDefined.

Table 1037 – Attributes of UserDefinedModels::WindType3or4UserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1038 shows all association ends of WindType3or4UserDefined with other classes.

Table 1038 – Association ends of UserDefinedModels:: WindType3or4UserDefined with other classes

mult from	name	mult to	type	description
0..1	PowerElectronicsConnection	1..1	PowerElectronicsConnection	inherited from: WindTurbineType3or4Dynamics
0..1	RemoteInputSignal	0..1	RemoteInputSignal	inherited from: WindTurbineType3or4Dynamics
1..*	WindPlantDynamics	0..1	WindPlantDynamics	inherited from: WindTurbineType3or4Dynamics

14.3.4.8 SynchronousMachineUserDefined

Inheritance path = SynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

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Synchronous machine whose dynamic behaviour is described by a user-defined model.

Table 1039 shows all attributes of SynchronousMachineUserDefined.

Table 1039 – Attributes of UserDefinedModels::SynchronousMachineUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1040 shows all association ends of SynchronousMachineUserDefined with other classes.

Table 1040 – Association ends of UserDefinedModels:: SynchronousMachineUserDefined with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachine	1..1	SynchronousMachine	inherited from: SynchronousMachineDynamics

14.3.4.9 AsynchronousMachineUserDefined

Inheritance path = AsynchronousMachineDynamics: RotatingMachineDynamics: DynamicsFunctionBlock: IdentifiedObject

Asynchronous machine whose dynamic behaviour is described by a user-defined model.

Table 1041 shows all attributes of AsynchronousMachineUserDefined.

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Table 1041 – Attributes of UserDefinedModels::AsynchronousMachineUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
damping	1..1	Float	inherited from: RotatingMachineDynamics
inertia	1..1	Seconds	inherited from: RotatingMachineDynamics
saturationFactor	0..1	Float	inherited from: RotatingMachineDynamics
saturationFactor120	0..1	Float	inherited from: RotatingMachineDynamics
statorLeakageReactance	1..1	PU	inherited from: RotatingMachineDynamics
statorResistance	1..1	PU	inherited from: RotatingMachineDynamics
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1042 shows all association ends of AsynchronousMachineUserDefined with other classes.

Table 1042 – Association ends of UserDefinedModels:: AsynchronousMachineUserDefined with other classes

mult from	name	mult to	type	description
0..1	AsynchronousMachine	1..1	AsynchronousMachine	inherited from: AsynchronousMachineDynamics

14.3.4.10 TurbineGovernorUserDefined

Inheritance path = TurbineGovernorDynamics: DynamicsFunctionBlock: IdentifiedObject

Turbine-governor function block whose dynamic behaviour is described by a user-defined model.

Table 1043 shows all attributes of TurbineGovernorUserDefined.

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Table 1043 – Attributes of UserDefinedModels::TurbineGovernorUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1044 shows all association ends of TurbineGovernorUserDefined with other classes.

Table 1044 – Association ends of UserDefinedModels:: TurbineGovernorUserDefined with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: TurbineGovernorDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: TurbineGovernorDynamics

14.3.4.11 TurbineLoadControllerUserDefined

Inheritance path = TurbineLoadControllerDynamics: DynamicsFunctionBlock: IdentifiedObject

Turbine load controller function block whose dynamic behaviour is described by a user-defined model.

Table 1045 shows all attributes of TurbineLoadControllerUserDefined.

Table 1045 – Attributes of UserDefinedModels::TurbineLoadControllerUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 1046 shows all association ends of TurbineLoadControllerUserDefined with other classes.

Table 1046 – Association ends of UserDefinedModels:: TurbineLoadControllerUserDefined with other classes

mult from	name	mult to	type	description
0..1	TurbineGovernorDynamics	1..1	TurbineGovernorDynamics	inherited from: TurbineLoadControllerDynamics

14.3.4.12 MechanicalLoadUserDefined

Inheritance path = MechanicalLoadDynamics: DynamicsFunctionBlock: IdentifiedObject

Mechanical load function block whose dynamic behaviour is described by a user-defined model.

Table 1047 shows all attributes of MechanicalLoadUserDefined.

Table 1047 – Attributes of UserDefinedModels::MechanicalLoadUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1048 shows all association ends of MechanicalLoadUserDefined with other classes.

Table 1048 – Association ends of UserDefinedModels:: MechanicalLoadUserDefined with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	0..1	SynchronousMachineDynamics	inherited from: MechanicalLoadDynamics
0..1	AsynchronousMachineDynamics	0..1	AsynchronousMachineDynamics	inherited from: MechanicalLoadDynamics

14.3.4.13 ExcitationSystemUserDefined

Inheritance path = ExcitationSystemDynamics: DynamicsFunctionBlock: IdentifiedObject

Excitation system function block whose dynamic behaviour is described by a user-defined model.

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Table 1049 shows all attributes of ExcitationSystemUserDefined.

Table 1049 – Attributes of UserDefinedModels::ExcitationSystemUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1050 shows all association ends of ExcitationSystemUserDefined with other classes.

Table 1050 – Association ends of UserDefinedModels:: ExcitationSystemUserDefined with other classes

mult from	name	mult to	type	description
0..1	SynchronousMachineDynamics	1..1	SynchronousMachineDynamics	inherited from: ExcitationSystemDynamics

14.3.4.14 OverexcitationLimiterUserDefined

Inheritance path = OverexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Overexcitation limiter system function block whose dynamic behaviour is described by a user-defined model.

Table 1051 shows all attributes of OverexcitationLimiterUserDefined.

Table 1051 – Attributes of UserDefinedModels::OverexcitationLimiterUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 1052 shows all association ends of OverexcitationLimiterUserDefined with other classes.

Table 1052 – Association ends of UserDefinedModels:: OverexcitationLimiterUserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: OverexcitationLimiterDynamics

14.3.4.15 UnderexcitationLimiterUserDefined

Inheritance path = UnderexcitationLimiterDynamics: DynamicsFunctionBlock: IdentifiedObject

Underexcitation limiter function block whose dynamic behaviour is described by a user-defined model.

Table 1053 shows all attributes of UnderexcitationLimiterUserDefined.

Table 1053 – Attributes of UserDefinedModels::UnderexcitationLimiterUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1054 shows all association ends of UnderexcitationLimiterUserDefined with other classes.

Table 1054 – Association ends of UserDefinedModels:: UnderexcitationLimiterUserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: UnderexcitationLimiterDynamics

14.3.4.16 PowerSystemStabilizerUserDefined

Inheritance path = PowerSystemStabilizerDynamics: DynamicsFunctionBlock: IdentifiedObject

Power system stabilizer function block whose dynamic behaviour is described by a user-defined model.

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Table 1055 shows all attributes of PowerSystemStabilizerUserDefined.

Table 1055 – Attributes of UserDefinedModels::PowerSystemStabilizerUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1056 shows all association ends of PowerSystemStabilizerUserDefined with other classes.

Table 1056 – Association ends of UserDefinedModels:: PowerSystemStabilizerUserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PowerSystemStabilizerDynamics

14.3.4.17 DiscontinuousExcitationControlUserDefined

Inheritance path = DiscontinuousExcitationControlDynamics: DynamicsFunctionBlock: IdentifiedObject

Discontinuous excitation control function block whose dynamic behaviour is described by a user-defined model.

Table 1057 shows all attributes of DiscontinuousExcitationControlUserDefined.

Table 1057 – Attributes of UserDefinedModels:: DiscontinuousExcitationControlUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject

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name	mult	type	description
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1058 shows all association ends of DiscontinuousExcitationControlUserDefined with other classes.

Table 1058 – Association ends of UserDefinedModels:: DiscontinuousExcitationControlUserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: DiscontinuousExcitationControlDynamics

14.3.4.18 PFVArControllerType1UserDefined

Inheritance path = PFVArControllerType1Dynamics: DynamicsFunctionBlock: IdentifiedObject

Power factor or VAr controller type 1 function block whose dynamic behaviour is described by a user-defined model.

Table 1059 shows all attributes of PFVArControllerType1UserDefined.

Table 1059 – Attributes of UserDefinedModels::PFVArControllerType1UserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1060 shows all association ends of PFVArControllerType1UserDefined with other classes.

Table 1060 – Association ends of UserDefinedModels:: PFVArControllerType1UserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType1Dynamics

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14.3.4.19 VoltageAdjusterUserDefined

Inheritance path = VoltageAdjusterDynamics: DynamicsFunctionBlock: IdentifiedObject

Voltage adjuster function block whose dynamic behaviour is described by a user-defined model.

Table 1061 shows all attributes of VoltageAdjusterUserDefined.

Table 1061 – Attributes of UserDefinedModels::VoltageAdjusterUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1062 shows all association ends of VoltageAdjusterUserDefined with other classes.

Table 1062 – Association ends of UserDefinedModels:: VoltageAdjusterUserDefined with other classes

mult from	name	mult to	type	description
0..1	PFVArControllerType1Dynamics	1..1	PFVArControllerType1Dynamics	inherited from: VoltageAdjusterDynamics

14.3.4.20 PFVArControllerType2UserDefined

Inheritance path = PFVArControllerType2Dynamics: DynamicsFunctionBlock: IdentifiedObject

Power factor or VAr controller type 2 function block whose dynamic behaviour is described by a user-defined model.

Table 1063 shows all attributes of PFVArControllerType2UserDefined.

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Table 1063 – Attributes of UserDefinedModels::PFVArControllerType2UserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

Table 1064 shows all association ends of PFVArControllerType2UserDefined with other classes.

Table 1064 – Association ends of UserDefinedModels:: PFVArControllerType2UserDefined with other classes

mult from	name	mult to	type	description
0..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: PFVArControllerType2Dynamics

14.3.4.21 VoltageCompensatorUserDefined

Inheritance path = VoltageCompensatorDynamics: DynamicsFunctionBlock: IdentifiedObject

Voltage compensator function block whose dynamic behaviour is described by a user-defined model.

Table 1065 shows all attributes of VoltageCompensatorUserDefined.

Table 1065 – Attributes of UserDefinedModels::VoltageCompensatorUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
enabled	1..1	Boolean	inherited from: DynamicsFunctionBlock
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

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Table 1066 shows all association ends of VoltageCompensatorUserDefined with other classes.

Table 1066 – Association ends of UserDefinedModels:: VoltageCompensatorUserDefined with other classes

mult from	name	mult to	type	description
1..1	ExcitationSystemDynamics	1..1	ExcitationSystemDynamics	inherited from: VoltageCompensatorDynamics

14.3.4.22 LoadUserDefined

Inheritance path = LoadDynamics: IdentifiedObject

Load whose dynamic behaviour is described by a user-defined model.

Table 1067 shows all attributes of LoadUserDefined.

Table 1067 – Attributes of UserDefinedModels::LoadUserDefined

name	mult	type	description
proprietary	1..1	Boolean	Behaviour is based on a proprietary model as opposed to a detailed model. true = user-defined model is proprietary with behaviour mutually understood by sending and receiving applications and parameters passed as general attributes false = user-defined model is explicitly defined in terms of control blocks and their input and output signals.
description	0..1	String	inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	0..1	String	inherited from: IdentifiedObject

14.3.4.23 ProprietaryParameterDynamics root class

Supports definition of one or more parameters of several different datatypes for use by proprietary user-defined models.

This class does not inherit from IdentifiedObject since it is not intended that a single instance of it be referenced by more than one proprietary user-defined model instance.

Table 1068 shows all attributes of ProprietaryParameterDynamics.

Table 1068 – Attributes of UserDefinedModels::ProprietaryParameterDynamics

name	mult	type	description
parameterNumber	1..1	Integer	Sequence number of the parameter among the set of parameters associated with the related proprietary user-defined model.
booleanParameterValue	0..1	Boolean	Boolean parameter value. If this attribute is populated, integerParameterValue and floatParameterValue will not be.
integerParameterValue	0..1	Integer	Integer parameter value. If this attribute is populated, booleanParameterValue and floatParameterValue will not be.

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name	mult	type	description
floatParameterValue	0..1	Float	Floating point parameter value. If this attribute is populated, booleanParameterValue and integerParameterValue will not be.

Table 1069 shows all association ends of ProprietaryParameterDynamics with other classes.

Table 1069 – Association ends of UserDefinedModels:: ProprietaryParameterDynamics with other classes

mult from	name	mult to	type	description
0..*	SynchronousMachineUserDefined	0..1	SynchronousMachineUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	AsynchronousMachineUserDefined	0..1	AsynchronousMachineUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	TurbineGovernorUserDefined	0..1	TurbineGovernorUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	TurbineLoadControllerUserDefined	0..1	TurbineLoadControllerUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	MechanicalLoadUserDefined	0..1	MechanicalLoadUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	ExcitationSystemUserDefined	0..1	ExcitationSystemUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	OverexcitationLimiterUserDefined	0..1	OverexcitationLimiterUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	UnderexcitationLimiterUserDefined	0..1	UnderexcitationLimiterUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	PowerSystemStabilizerUserDefined	0..1	PowerSystemStabilizerUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	DiscontinuousExcitationControlUserDefined	0..1	DiscontinuousExcitationControlUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	PFVArControllerType1UserDefined	0..1	PFVArControllerType1UserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	VoltageAdjusterUserDefined	0..1	VoltageAdjusterUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	PFVArControllerType2UserDefined	0..1	PFVArControllerType2UserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	VoltageCompensatorUserDefined	0..1	VoltageCompensatorUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	LoadUserDefined	0..1	LoadUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	CSCUserDefined	0..1	CSCUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	SVCUserDefined	0..1	SVCUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	VSCUserDefined	0..1	VSCUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	WindPlantUserDefined	0..1	WindPlantUserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	WindType1or2UserDefined	0..1	WindType1or2UserDefined	Proprietary user-defined model with which this parameter is associated.
0..*	WindType3or4UserDefined	0..1	WindType3or4UserDefined	Proprietary user-defined model with which this parameter is associated.

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15 (deprecated) Package EquipmentBoundaryProfile

15.1 General

This profile is deprecated. Use core equipment profile part of the IEC 61970-452 document.

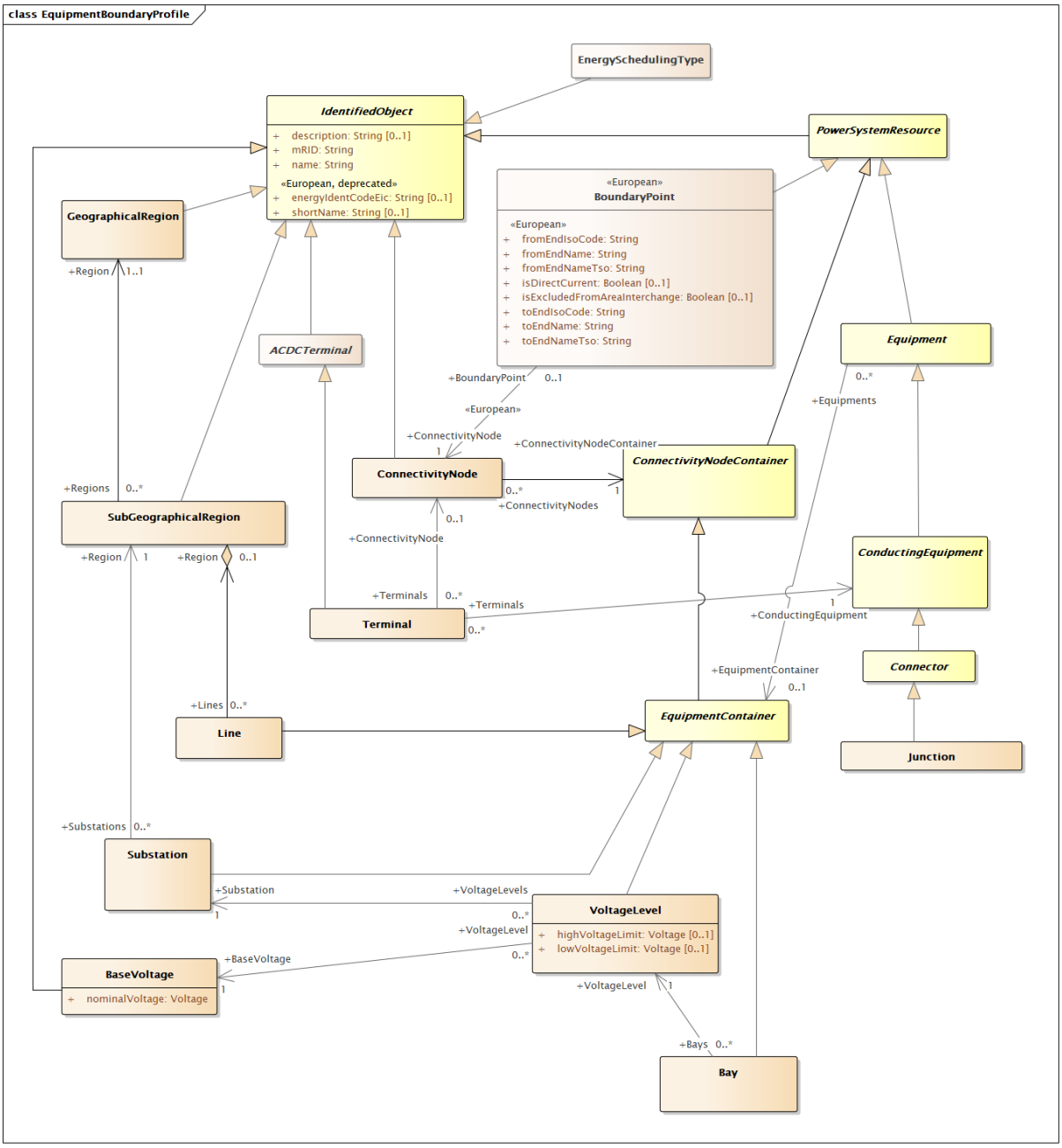

Figure 225 – Class diagram EquipmentBoundaryProfile::EquipmentBoundaryProfile

Figure 225: The diagram shows equipment boundary profile.

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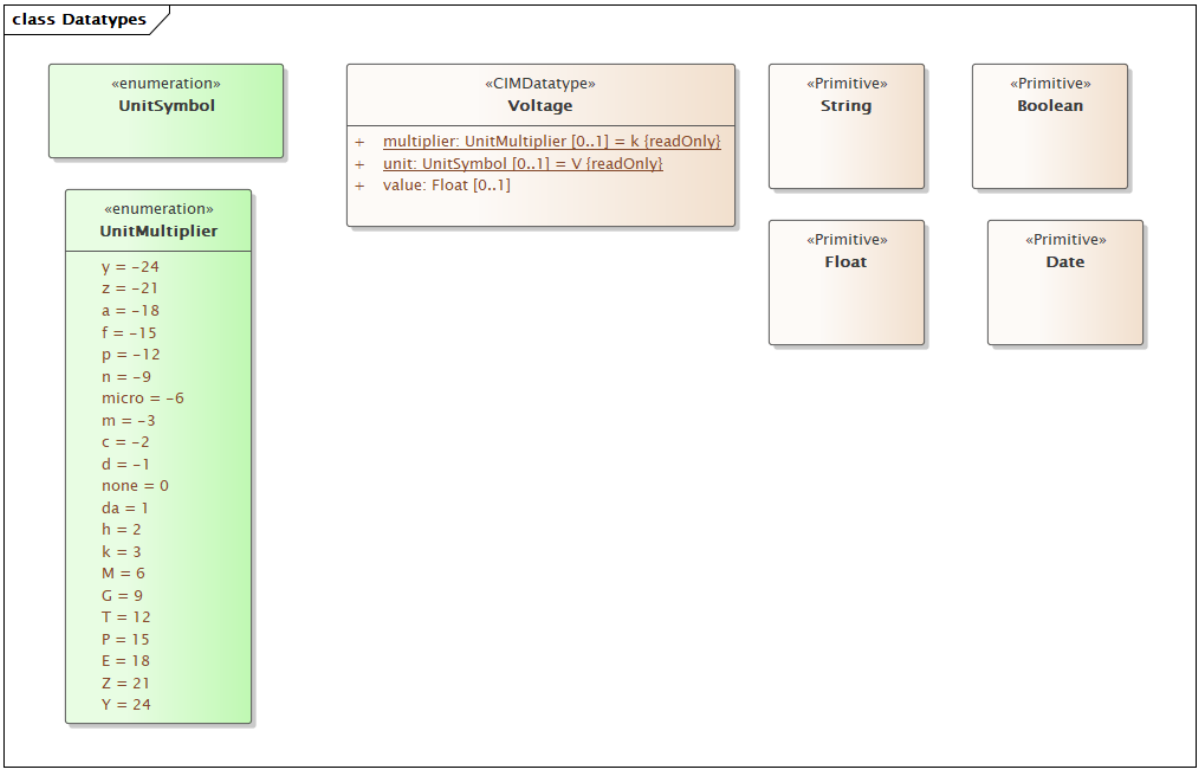

Figure 226 – Class diagram EquipmentBoundaryProfile::Datatypes

Figure 226: The diagram shows datatypes that are used by classes in the profile. Stereotypes are used to describe the datatypes. The following stereotypes are defined:

- <<enumeration>> A list of permissible constant values.
- <<Primitive>> The most basic data types used to compose all other data types.
- <<CIMDatatype>> A datatype that contains a value attribute, an optional unit of measure and a unit multiplier. The unit and multiplier may be specified as a static variable initialized to the allowed value.
- <<Compound>> A composite of Primitive, enumeration, CIMDatatype or other Compound classes, as long as the Compound classes do not recurse.

For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.

15.2 (abstract) ACDCTerminal

Inheritance path = IdentifiedObject

An electrical connection point (AC or DC) to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 1070 shows all attributes of ACDCTerminal.

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Table 1070 – Attributes of EquipmentBoundaryProfile::ACDCTerminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.3 BaseVoltage

Inheritance path = IdentifiedObject

Defines a system base voltage which is referenced.

Table 1071 shows all attributes of BaseVoltage.

Table 1071 – Attributes of EquipmentBoundaryProfile::BaseVoltage

name	mult	type	description
nominalVoltage	1..1	Voltage	The power system resource's base voltage. Shall be a positive value and not zero.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.4 Bay

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A collection of power system resources (within a given substation) including conducting equipment, protection relays, measurements, and telemetry. A bay typically represents a physical grouping related to modularization of equipment.

Table 1072 shows all attributes of Bay.

Table 1072 – Attributes of EquipmentBoundaryProfile::Bay

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject

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name	mult	type	description
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1073 shows all association ends of Bay with other classes.

Table 1073 – Association ends of EquipmentBoundaryProfile::Bay with other classes

mult from	name	mult to	type	description
0..*	VoltageLevel	1..1	VoltageLevel	The voltage level containing this bay.

15.5 (European) BoundaryPoint

Inheritance path = PowerSystemResource: IdentifiedObject

Designates a connection point at which one or more model authority sets shall connect to. The location of the connection point as well as other properties are agreed between organisations responsible for the interconnection, hence all attributes of the class represent this agreement. It is primarily used in a boundary model authority set which can contain one or many BoundaryPoint-s among other Equipment-s and their connections.

Table 1074 shows all attributes of BoundaryPoint.

Table 1074 – Attributes of EquipmentBoundaryProfile::BoundaryPoint

name	mult	type	description
fromEndIsoCode	1..1	String	(European) The ISO code of the region which the "From" side of the Boundary point belongs to or it is connected to. The ISO code is a two-character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.
fromEndName	1..1	String	(European) A human readable name with length of the string 64 characters maximum. It covers the following two cases: – if the Boundary point is placed on a tie-line, it is the name (IdentifiedObject.name) of the substation at which the "From" side of the tie-line is connected to. – if the Boundary point is placed in a substation, it is the name (IdentifiedObject.name) of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc.) at which the "From" side of the Boundary point is connected to.
fromEndNameTso	1..1	String	(European) Identifies the name of the transmission system operator, distribution system operator or other entity at which the "From" side of the interconnection is connected to. The length of the string is 64 characters maximum.

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name	mult	type	description
toEndIsoCode	1..1	String	(European) The ISO code of the region which the "To" side of the Boundary point belongs to or is connected to. The ISO code is a two-character country code as defined by ISO 3166 (http://www.iso.org/iso/country_codes). The length of the string is 2 characters maximum.
toEndName	1..1	String	(European) A human readable name with length of the string 64 characters maximum. It covers the following two cases: – if the Boundary point is placed on a tie-line, it is the name (IdentifiedObject.name) of the substation at which the "To" side of the tie-line is connected to. – if the Boundary point is placed in a substation, it is the name (IdentifiedObject.name) of the element (e.g. PowerTransformer, ACLineSegment, Switch, etc.) at which the "To" side of the Boundary point is connected to.
toEndNameTso	1..1	String	(European) Identifies the name of the transmission system operator, distribution system operator or other entity at which the "To" side of the interconnection is connected to. The length of the string is 64 characters maximum.
isDirectCurrent	0..1	Boolean	(European) If true, this boundary point is a point of common coupling (PCC) of a direct current (DC) interconnection, otherwise the interconnection is AC (default).
isExcludedFromAreaInterchange	0..1	Boolean	(European) If true, this boundary point is on the interconnection that is excluded from control area interchange calculation and consequently has no related tie flows. Otherwise, the interconnection is included in control area interchange and a TieFlow is required at all sides of the boundary point (default).
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1075 shows all association ends of BoundaryPoint with other classes.

Table 1075 – Association ends of EquipmentBoundaryProfile:: BoundaryPoint with other classes

mult from	name	mult to	type	description
0..1	ConnectivityNode	1..1	ConnectivityNode	(European) The connectivity node that is designated as a boundary point.

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15.6 (abstract) ConductingEquipment

Inheritance path = Equipment: PowerSystemResource: IdentifiedObject

The parts of the AC power system that are designed to carry current or that are conductively connected through terminals.

Table 1076 shows all attributes of ConductingEquipment.

Table 1076 – Attributes of EquipmentBoundaryProfile::ConductingEquipment

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1077 shows all association ends of ConductingEquipment with other classes.

Table 1077 – Association ends of EquipmentBoundaryProfile:: ConductingEquipment with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

15.7 ConnectivityNode

Inheritance path = IdentifiedObject

Connectivity nodes are points where terminals of AC conducting equipment are connected together with zero impedance.

Table 1078 shows all attributes of ConnectivityNode.

Table 1078 – Attributes of EquipmentBoundaryProfile::ConnectivityNode

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1079 shows all association ends of ConnectivityNode with other classes.

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Table 1079 – Association ends of EquipmentBoundaryProfile:: ConnectivityNode with other classes

mult from	name	mult to	type	description
0..*	ConnectivityNodeContainer	1..1	ConnectivityNodeContainer	Container of this connectivity node.

15.8 (abstract) ConnectivityNodeContainer

Inheritance path = PowerSystemResource: IdentifiedObject

A base class for all objects that may contain connectivity nodes or topological nodes.

Table 1080 shows all attributes of ConnectivityNodeContainer.

Table 1080 – Attributes of EquipmentBoundaryProfile::ConnectivityNodeContainer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.9 (abstract) Connector

Inheritance path = ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A conductor, or group of conductors, with negligible impedance, that serve to connect other conducting equipment within a single substation and are modelled with a single logical terminal.

Table 1081 shows all attributes of Connector.

Table 1081 – Attributes of EquipmentBoundaryProfile::Connector

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1082 shows all association ends of Connector with other classes.

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Table 1082 – Association ends of EquipmentBoundaryProfile:: Connector with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

15.10 EnergySchedulingType

Inheritance path = IdentifiedObject

Used to define the type of generation for scheduling purposes.

Table 1083 shows all attributes of EnergySchedulingType.

Table 1083 – Attributes of EquipmentBoundaryProfile::EnergySchedulingType

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.11 (abstract) Equipment

Inheritance path = PowerSystemResource: IdentifiedObject

The parts of a power system that are physical devices, electronic or mechanical.

Table 1084 shows all attributes of Equipment.

Table 1084 – Attributes of EquipmentBoundaryProfile::Equipment

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1085 shows all association ends of Equipment with other classes.

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Table 1085 – Association ends of EquipmentBoundaryProfile:: Equipment with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	Container of this equipment.

15.12 (abstract) EquipmentContainer

Inheritance path = ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A modelling construct to provide a root class for containing equipment.

Table 1086 shows all attributes of EquipmentContainer.

Table 1086 – Attributes of EquipmentBoundaryProfile::EquipmentContainer

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.13 GeographicalRegion

Inheritance path = IdentifiedObject

A geographical region of a power system network model.

Table 1087 shows all attributes of GeographicalRegion.

Table 1087 – Attributes of EquipmentBoundaryProfile::GeographicalRegion

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.14 (abstract) IdentifiedObject root class

This is a root class to provide common identification for all classes needing identification and naming attributes.

Table 1088 shows all attributes of IdentifiedObject.

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Table 1088 – Attributes of EquipmentBoundaryProfile::IdentifiedObject

name	mult	type	description
description	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
energyIdentCodeEic	0..1	String	(deprecated,European) The attribute is used for an exchange of the EIC code (Energy identification Code). The length of the string is 16 characters as defined by the EIC code. For details on EIC scheme please refer to ENTSO-E web site.
mRID	1..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name	1..1	String	The name is any free human readable and possibly non unique text naming the object.
shortName	0..1	String	(deprecated,European) The attribute is used for an exchange of a human readable short name with length of the string 12 characters maximum.

15.15 Junction

Inheritance path = Connector: ConductingEquipment: Equipment: PowerSystemResource: IdentifiedObject

A point where one or more conducting equipments are connected with zero resistance.

Table 1089 shows all attributes of Junction.

Table 1089 – Attributes of EquipmentBoundaryProfile::Junction

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1090 shows all association ends of Junction with other classes.

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Table 1090 – Association ends of EquipmentBoundaryProfile:: Junction with other classes

mult from	name	mult to	type	description
0..*	EquipmentContainer	0..1	EquipmentContainer	inherited from: Equipment

15.16 Line

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

Contains equipment beyond a substation belonging to a power transmission line.

Table 1091 shows all attributes of Line.

Table 1091 – Attributes of EquipmentBoundaryProfile::Line

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1092 shows all association ends of Line with other classes.

Table 1092 – Association ends of EquipmentBoundaryProfile::Line with other classes

mult from	name	mult to	type	description
0..*	Region	0..1	SubGeographicalRegion	The sub-geographical region of the line.

15.17 (abstract) PowerSystemResource

Inheritance path = IdentifiedObject

A power system resource (PSR) can be an item of equipment such as a switch, an equipment container containing many individual items of equipment such as a substation, or an organisational entity such as sub-control area. Power system resources can have measurements associated.

Table 1093 shows all attributes of PowerSystemResource.

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Table 1093 – Attributes of EquipmentBoundaryProfile::PowerSystemResource

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

15.18 SubGeographicalRegion

Inheritance path = IdentifiedObject

A subset of a geographical region of a power system network model.

Table 1094 shows all attributes of SubGeographicalRegion.

Table 1094 – Attributes of EquipmentBoundaryProfile::SubGeographicalRegion

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1095 shows all association ends of SubGeographicalRegion with other classes.

Table 1095 – Association ends of EquipmentBoundaryProfile:: SubGeographicalRegion with other classes

mult from	name	mult to	type	description
0..*	Region	1..1	GeographicalRegion	The geographical region which this sub-geographical region is within.

15.19 Substation

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A collection of equipment for purposes other than generation or utilization, through which electric energy in bulk is passed for the purposes of switching or modifying its characteristics.

Table 1096 shows all attributes of Substation.

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Table 1096 – Attributes of EquipmentBoundaryProfile::Substation

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1097 shows all association ends of Substation with other classes.

Table 1097 – Association ends of EquipmentBoundaryProfile:: Substation with other classes

mult from	name	mult to	type	description
0..*	Region	1..1	SubGeographicalRegion	The SubGeographicalRegion containing the substation.

15.20 Terminal

Inheritance path = ACDCTerminal: IdentifiedObject

An AC electrical connection point to a piece of conducting equipment. Terminals are connected at physical connection points called connectivity nodes.

Table 1098 shows all attributes of Terminal.

Table 1098 – Attributes of EquipmentBoundaryProfile::Terminal

name	mult	type	description
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1099 shows all association ends of Terminal with other classes.

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Table 1099 – Association ends of EquipmentBoundaryProfile:: Terminal with other classes

mult from	name	mult to	type	description
0..*	ConductingEquipment	1..1	ConductingEquipment	The conducting equipment of the terminal. Conducting equipment have terminals that may be connected to other conducting equipment terminals via connectivity nodes or topological nodes.
0..*	ConnectivityNode	0..1	ConnectivityNode	The connectivity node to which this terminal connects with zero impedance.

15.21 VoltageLevel

Inheritance path = EquipmentContainer: ConnectivityNodeContainer: PowerSystemResource: IdentifiedObject

A collection of equipment at one common system voltage forming a switchgear. The equipment typically consists of breakers, busbars, instrumentation, control, regulation and protection devices as well as assemblies of all these.

Table 1100 shows all attributes of VoltageLevel.

Table 1100 – Attributes of EquipmentBoundaryProfile::VoltageLevel

name	mult	type	description
highVoltageLimit	0..1	Voltage	The bus bar's high voltage limit. The limit applies to all equipment and nodes contained in a given VoltageLevel. It is not required that it is exchanged in pair with lowVoltageLimit. It is preferable to use operational VoltageLimit, which prevails, if present.
lowVoltageLimit	0..1	Voltage	The bus bar's low voltage limit. The limit applies to all equipment and nodes contained in a given VoltageLevel. It is not required that it is exchanged in pair with highVoltageLimit. It is preferable to use operational VoltageLimit, which prevails, if present.
description	0..1	String	inherited from: IdentifiedObject
energyIdentCodeEic	0..1	String	(deprecated,European) inherited from: IdentifiedObject
mRID	1..1	String	inherited from: IdentifiedObject
name	1..1	String	inherited from: IdentifiedObject
shortName	0..1	String	(deprecated,European) inherited from: IdentifiedObject

Table 1101 shows all association ends of VoltageLevel with other classes.

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Table 1101 – Association ends of EquipmentBoundaryProfile:: VoltageLevel with other classes

mult from	name	mult to	type	description
0..*	BaseVoltage	1..1	BaseVoltage	The base voltage used for all equipment within the voltage level.
0..*	Substation	1..1	Substation	The substation of the voltage level.

15.22 UnitMultiplier enumeration

The unit multipliers defined for the CIM. When applied to unit symbols, the unit symbol is treated as a derived unit. Regardless of the contents of the unit symbol text, the unit symbol shall be treated as if it were a single-character unit symbol. Unit symbols should not contain multipliers, and it should be left to the multiplier to define the multiple for an entire data type.

For example, if a unit symbol is "m2Pers" and the multiplier is "k", then the value is k(m**2/s), and the multiplier applies to the entire final value, not to any individual part of the value. This can be conceptualized by substituting a derived unit symbol for the unit type. If one imagines that the symbol "Þ" represents the derived unit "m2Pers", then applying the multiplier "k" can be conceptualized simply as "kÞ".

For example, the SI unit for mass is "kg" and not "g". If the unit symbol is defined as "kg", then the multiplier is applied to "kg" as a whole and does not replace the "k" in front of the "g". In this case, the multiplier of "m" would be used with the unit symbol of "kg" to represent one gram. As a text string, this violates the instructions in IEC 80000-1. However, because the unit symbol in CIM is treated as a derived unit instead of as an SI unit, it makes more sense to conceptualize the "kg" as if it were replaced by one of the proposed replacements for the SI mass symbol. If one imagines that the "kg" were replaced by a symbol "Þ", then it is easier to conceptualize the multiplier "m" as creating the proper unit "mÞ", and not the forbidden unit "mkg".

Table 1102 shows all literals of UnitMultiplier.

Table 1102 – Literals of EquipmentBoundaryProfile::UnitMultiplier

literal	value	description
y	-24	Yocto 10**-24.
z	-21	Zepto 10**-21.
a	-18	Atto 10**-18.
f	-15	Femto 10**-15.
p	-12	Pico 10**-12.
n	-9	Nano 10**-9.
micro	-6	Micro 10**-6.
m	-3	Milli 10**-3.
c	-2	Centi 10**-2.
d	-1	Deci 10**-1.
none	0	No multiplier or equivalently multiply by 1.
da	1	Deca 10**1.
h	2	Hecto 10**2.
k	3	Kilo 10**3.
M	6	Mega 10**6.
G	9	Giga 10**9.

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literal	value	description
T	12	Tera 10**12.
P	15	Peta 10**15.
E	18	Exa 10**18.
Z	21	Zetta 10**21.
Y	24	Yotta 10**24.

15.23 UnitSymbol enumeration

The derived units defined for usage in the CIM. In some cases, the derived unit is equal to an SI unit. Whenever possible, the standard derived symbol is used instead of the formula for the derived unit. For example, the unit symbol Farad is defined as "F" instead of "CPerV". In cases where a standard symbol does not exist for a derived unit, the formula for the unit is used as the unit symbol. For example, density does not have a standard symbol and so it is represented as "kgPerm3". With the exception of the "kg", which is an SI unit, the unit symbols do not contain multipliers and therefore represent the base derived unit to which a multiplier can be applied as a whole.

Every unit symbol is treated as an unparseable text as if it were a single-letter symbol. The meaning of each unit symbol is defined by the accompanying descriptive text and not by the text contents of the unit symbol.

To allow the widest possible range of serializations without requiring special character handling, several substitutions are made which deviate from the format described in IEC 80000-1. The division symbol "/" is replaced by the letters "Per". Exponents are written in plain text after the unit as "m3" instead of being formatted as "m" with a superscript of 3 or introducing a symbol as in "m^3". The degree symbol "°" is replaced with the letters "deg". Any clarification of the meaning for a substitution is included in the description for the unit symbol.

Non-SI units are included in list of unit symbols to allow sources of data to be correctly labelled with their non-SI units (for example, a GPS sensor that is reporting numbers that represent feet instead of meters). This allows software to use the unit symbol information correctly convert and scale the raw data of those sources into SI-based units.

The integer values are used for harmonization with IEC 61850.

Table 1103 shows all literals of UnitSymbol.

Table 1103 – Literals of EquipmentBoundaryProfile::UnitSymbol

literal	value	description
none	0	Dimension less quantity, e.g. count, per unit, etc.
m	2	Length in metres.
kg	3	Mass in kilograms. Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
s	4	Time in seconds.
A	5	Current in amperes.
K	6	Temperature in kelvins.
mol	7	Amount of substance in moles.
cd	8	Luminous intensity in candelas.
deg	9	Plane angle in degrees.
rad	10	Plane angle in radians (m/m).

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literal	value	description
sr	11	Solid angle in steradians (m2/m2).
Gy	21	Absorbed dose in grays (J/kg).
Bq	22	Radioactivity in becquerels (1/s).
degC	23	Relative temperature in degrees Celsius. In the SI unit system the symbol is °C. Electric charge is measured in coulomb that has the unit symbol C. To distinguish degree Celsius from coulomb the symbol used in the UML is degC. The reason for not using °C is that the special character ° is difficult to manage in software.
Sv	24	Dose equivalent in sieverts (J/kg).
F	25	Electric capacitance in farads (C/V).
C	26	Electric charge in coulombs (A·s).
S	27	Conductance in siemens.
H	28	Electric inductance in henrys (Wb/A).
V	29	Electric potential in volts (W/A).
ohm	30	Electric resistance in ohms (V/A).
J	31	Energy in joules (N·m = C·V = W·s).
N	32	Force in newtons ($kg \cdot m/s^2$).
Hz	33	Frequency in hertz (1/s).
lx	34	Illuminance in lux (lm/m^2).
lm	35	Luminous flux in lumens (cd·sr).
Wb	36	Magnetic flux in webers (V·s).
T	37	Magnetic flux density in teslas (Wb/m2).
W	38	Real power in watts (J/s). Electrical power may have real and reactive components. The real portion of electrical power (I^2R or VIcos(phi)), is expressed in Watts. See also apparent power and reactive power.
Pa	39	Pressure in pascals (N/m^2). Note: the absolute or relative measurement of pressure is implied with this entry. See below for more explicit forms.
m2	41	Area in square metres (m^2).
m3	42	Volume in cubic metres (m^3).
mPers	43	Velocity in metres per second (m/s).
mPers2	44	Acceleration in metres per second squared (m/s^2).
m3Pers	45	Volumetric flow rate in cubic metres per second (m^3/s).
mPerm3	46	Fuel efficiency in metres per cubic metres (m/m^3).
kgm	47	Moment of mass in kilogram metres (kg·m) (first moment of mass). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
kgPerm3	48	Density in kilogram/cubic metres (kg/m^3). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
m2Pers	49	Viscosity in square metres / second (m^2/s).
WPermK	50	Thermal conductivity in watt/metres kelvin.
JPerK	51	Heat capacity in joules/kelvin.

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literal	value	description
ppm	52	Concentration in parts per million.
rotPers	53	Rotations per second (1/s). See also Hz (1/s).
radPers	54	Angular velocity in radians per second (rad/s).
WPerm2	55	Heat flux density, irradiance, watts per square metre.
JPerm2	56	Insulation energy density, joules per square metre or watt second per square metre.
SPerm	57	Conductance per length (F/m).
KPers	58	Temperature change rate in kelvins per second.
PaPers	59	Pressure change rate in pascals per second.
JPerkgK	60	Specific heat capacity, specific entropy, joules per kilogram Kelvin.
VA	61	Apparent power in volt amperes. See also real power and reactive power.
VAr	63	Reactive power in volt amperes reactive. The “reactive” or “imaginary” component of electrical power (VIsin(phi)). (See also real power and apparent power). NOTE Different meter designs use different methods to arrive at their results. Some meters may compute reactive power as an arithmetic value, while others compute the value vectorially. The data consumer should determine the method in use and the suitability of the measurement for the intended purpose.
cosPhi	65	Power factor, dimensionless. NOTE 1 This definition of power factor only holds for balanced systems. See the alternative definition under code 153. NOTE 2 Beware of differing sign conventions in use between the IEC and EEI. It is assumed that the data consumer understands the type of meter in use and the sign convention in use by the utility.
Vs	66	Volt seconds (Ws/A).
V2	67	Volt squared (W^2/A^2).
As	68	Ampere seconds (A·s).
A2	69	Amperes squared (A^2).
A2s	70	Ampere squared time in square amperes (A^2s).
VAh	71	Apparent energy in volt ampere hours.
Wh	72	Real energy in watt hours.
VArh	73	Reactive energy in volt ampere reactive hours.
VPerHz	74	Magnetic flux in volt per hertz.
HzPers	75	Rate of change of frequency in hertz per second.
character	76	Number of characters.
charPers	77	Data rate (baud) in characters per second.
kgm2	78	Moment of mass in kilogram square metres (kg·m²) (Second moment of mass, commonly called the moment of inertia). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
dB	79	Sound pressure level in decibels. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.

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literal	value	description
WPers	81	Ramp rate in watts per second.
lPers	82	Volumetric flow rate in litres per second.
dBm	83	Power level (logarithmic ratio of signal strength, Bel-mW), normalized to 1mW. Note: multiplier “d” is included in this unit symbol for compatibility with IEC 61850-7-3.
h	84	Time in hours, hour = 60 min = 3600 s.
min	85	Time in minutes, minute = 60 s.
Q	100	Quantity power, Q.
Qh	101	Quantity energy, Qh.
ohmm	102	Resistivity, ohm metres, (rho).
APerm	103	A/m, magnetic field strength, amperes per metre.
V2h	104	Volt-squared hour, volt-squared-hours.
A2h	105	Ampere-squared hour, ampere-squared hour.
Ah	106	Ampere-hours, ampere-hours.
count	111	Amount of substance, Counter value.
ft3	119	Volume, cubic feet.
m3Perh	125	Volumetric flow rate, cubic metres per hour.
gal	128	Volume in gallons, US gallon (1 gal = 231 in3 = 128 fl ounce).
Btu	132	Energy, British Thermal Units.
l	134	Volume in litres, litre = dm3 = m3/1000.
lPerh	137	Volumetric flow rate, litres per hour.
lPerl	143	Concentration, The ratio of the volume of a solute divided by the volume of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µL/L’.
gPerg	144	Concentration, The ratio of the mass of a solute divided by the mass of the solution. Note: Users may need use a prefix such a ‘µ’ to express a quantity such as ‘µg/g’.
molPerm3	145	Concentration, The amount of substance concentration, (c), the amount of solvent in moles divided by the volume of solution in m³.
molPermol	146	Concentration, Molar fraction, the ratio of the molar amount of a solute divided by the molar amount of the solution.
molPerkg	147	Concentration, Molality, the amount of solute in moles and the amount of solvent in kilograms.
sPers	149	Time, Ratio of time. Note: Users may need to supply a prefix such as ‘µ’ to show rates such as ‘µs/s’.
HzPerHz	150	Frequency, rate of frequency change. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mHz/Hz’.
VPerV	151	Voltage, ratio of voltages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mV/V’.
APerA	152	Current, ratio of amperages. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mA/A’.

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literal	value	description
VPerVA	153	Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
rev	154	Amount of rotation, revolutions.
kat	158	Catalytic activity, katal = mol / s.
JPerkg	165	Specific energy, Joules / kg.
m3Uncompensated	166	Volume, cubic metres, with the value uncompensated for weather effects.
m3Compensated	167	Volume, cubic metres, with the value compensated for weather effects.
WPerW	168	Signal Strength, ratio of power. Note: Users may need to supply a prefix such as ‘m’ to show rates such as ‘mW/W’.
therm	169	Energy, therms.
onePerm	173	Wavenumber, reciprocal metres, (1/m).
m3Perkg	174	Specific volume, cubic metres per kilogram, v.
Pas	175	Dynamic viscosity, pascal seconds.
Nm	176	Moment of force, newton metres.
NPerm	177	Surface tension, newton per metre.
radPers2	178	Angular acceleration, radians per second squared.
JPerm3	181	Energy density, joules per cubic metre.
VPerm	182	Electric field strength, volts per metre.
CPerm3	183	Electric charge density, coulombs per cubic metre.
CPerm2	184	Surface charge density, coulombs per square metre.
FPerm	185	Permittivity, farads per metre.
HPerm	186	Permeability, henrys per metre.
JPermol	187	Molar energy, joules per mole.
JPermolK	188	Molar entropy, molar heat capacity, joules per mole kelvin.
CPerkg	189	Exposure (x rays), coulombs per kilogram.
GyPers	190	Absorbed dose rate, grays per second.
WPersr	191	Radiant intensity, watts per steradian.
WPerm2sr	192	Radiance, watts per square metre steradian.
katPerm3	193	Catalytic activity concentration, katals per cubic metre.
d	195	Time in days, day = 24 h = 86400 s.
anglemin	196	Plane angle, minutes.
anglesec	197	Plane angle, seconds.
ha	198	Area, hectares.
tonne	199	Mass in tons, “tonne” or “metric ton” (1000 kg = 1 Mg).
bar	214	Pressure in bars, (1 bar = 100 kPa).
mmHg	215	Pressure, millimetres of mercury (1 mmHg is approximately 133.3 Pa).

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literal	value	description
M	217	Length, nautical miles (1 M = 1852 m).
kn	219	Speed, knots (1 kn = 1852/3600) m/s.
Mx	276	Magnetic flux, maxwells (1 Mx = 10-8 Wb).
G	277	Magnetic flux density, gausses (1 G = 10-4 T).
Oe	278	Magnetic field in oersteds, (1 Oe = (103/4p) A/m).
Vh	280	Volt-hour, Volt hours.
WPerA		Active power per current flow, watts per Ampere.
onePerHz		Reciprocal of frequency (1/Hz).
VPerVAr		Power factor, PF, the ratio of the active power to the apparent power. Note: The sign convention used for power factor will differ between IEC meters and EEI (ANSI) meters. It is assumed that the data consumers understand the type of meter being used and agree on the sign convention in use at any given utility.
ohmPerm	86	Electric resistance per length in ohms per metre ((V/A)/m).
kgPerJ		Weight per energy in kilograms per joule (kg/J). Note: multiplier “k” is included in this unit symbol for compatibility with IEC 61850-7-3.
JPers		Energy rate in joules per second (J/s).

15.24 Voltage datatype

Electrical voltage, can be both AC and DC.

Table 1104 shows all attributes of Voltage.

Table 1104 – Attributes of EquipmentBoundaryProfile::Voltage

name	mult	type	description
value	0..1	Float	
multiplier	0..1	UnitMultiplier	(const=k)
unit	0..1	UnitSymbol	(const=V)

15.25 Boolean primitive

A type with the value space "true" and "false".

15.26 Date primitive

Date as "yyyy-mm-dd", which conforms with ISO 8601. UTC time zone is specified as "yyyy-mm-ddZ". A local timezone relative UTC is specified as "yyyy-mm-dd(+/-)hh:mm".

15.27 Float primitive

A floating point number. The range is unspecified and not limited.

15.28 String primitive

A string consisting of a sequence of characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

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16 Profile constraints

This clause defines requirements and constraints that shall be fulfilled by applications that conform to this document. The naming of the rules shall not be used for machine processing. The rule names are just a string. The naming convention of the constraints is as follows.

“{rule.Type}:{rule.Standard}:{rule.Profile}:{rule.Property}:{rule.Name}”

where

rule.Type: C – for constraint; R – for requirement

rule.Standard: the number of the standard e.g. 301 for 61970-301, 456 for 61970-456, 13 for 61968-13. 61970-600 specific constraints refer to 600 although they are related to one or combination of the 61970-450 series profiles.

rule.Profile: the abbreviation of the profile, e.g. TP for Topology profile. If set to “ALL” the constraint is applicable to all IEC 61970-600 profiles.

rule.Property: for UML classes, the name of the class, for attributes and associations, the name of the class and attribute or association end, e.g. EnergyConsumer, IdentifiedObject.name, etc. If set to “NA” the property is not applicable to a specific UML element.

rule.Name: the name of the rule. It is unique for the same property.

Example: C:600:ALL:IdentifiedObject.name:stringLength

- R:452:ALL:NA:exchange

  Optional and required attributes and associations must be imported and exported if they are in the model file prior to import.

- R:452:ALL:NA:exchange1

  If an optional attribute does not exist in the imported file, it does not have to be exported in case exactly the same data set is exported, i.e. the tool is not obliged to automatically provide this attribute. If the export is resulting from an action by the user performed after the import, e.g. data processing or model update the export can contain optional attributes.

- R:452:ALL:NA:exchange2

  In most of the profiles the selection of optional and required attributes is made so as to ensure a minimum set of required attributes without which the exchange does not fulfil its basic purpose. Business processes governing different exchanges can require mandatory exchange of certain optional attributes or associations. Optional and required attributes and associations shall therefore be supported by applications which claim conformance with certain functionalities of the IEC 61970-452. This provides flexibility for the business processes to adapt to different business requirements and base the exchanges on IEC 61970-452 compliant applications.

- R:452:ALL:NA:exchange3

  An exporter may, at his or her discretion, produce a serialization containing additional class data described by the CIM Schema but not required by this document provided these data adhere to the conventions established in Clause 5.

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- R:452:ALL:NA:exchange4

  From the standpoint of the model import used by a data recipient, the document describes a subset of the CIM that importing software shall be able to interpret in order to import exported models. Data providers are free to exceed the minimum requirements described herein as long as their resulting data files are compliant with the CIM Schema and the conventions established in Clause 5. The document, therefore, describes additional classes and class data that, although not required, exporters will, in all likelihood, choose to include in their data files. The additional classes and data are labelled as required (cardinality 1..1) or as optional (cardinality 0..1) to distinguish them from their required counterparts. Please note, however, that data importers could potentially receive data containing instances of any and all classes described by the CIM Schema.

- R:452:ALL:NA:cardinality

  The cardinality defined in the CIM model shall be followed, unless a more restrictive cardinality is explicitly defined in this document. For instance, the cardinality on the association between VoltageLevel and BaseVoltage indicates that a VoltageLevel shall be associated with one and only one BaseVoltage, but a BaseVoltage can be associated with zero to many VoltageLevels.

- R:452:ALL:NA:associations

  Associations between classes referenced in this document and classes not referenced here are not required regardless of cardinality.

- R:452:ALL:IdentifiedObject.name:rule

  The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. It must be a human readable identifier without additional embedded information that would need to be parsed. The attribute is used for purposes such as User Interface and data exchange debugging. The MRID defined in the data exchange format is the only unique and persistent identifier used for this data exchange. The attribute IdentifiedObject.name is, however, always required for CoreEquipment profile and Short Circuit profile.

- R:452:ALL:IdentifiedObject.description:rule

  The attribute “description” inherited by many classes from the abstract class IdentifiedObject must contain human readable text without additional embedded information that would need to be parsed.

- R:452:ALL:NA:uniqueIdentifier

  All IdentifiedObject-s shall have a persistent and globally unique identifier (Master Resource Identifier – mRID).

- R:452:ALL:ConductingEquipment:connectivity

  All subtypes of ConductingEquipment are required to have associations to Terminals. The number of associated Terminals is specified in IEC 61970-301 in section 4.8.2 “Number of terminals for ConductingEquipment objects.” The associated Terminal(s) are not require to have associations to ConnectivityNodes, for instance a ShuntCompensator whose Terminal is not associated to a ConnectivityNode.

- R:452:ALL:NA:classes

  Instance data to be exchanged must make use of the most detailed class possible within a profile, i.e. using sub-typed classes rather than general classes e.g. NuclearGeneratingUnit instead of GeneratingUnit.

- R:452:ALL:NA:ground

  There shall not be any ground voltage level. The ConnectivityNode connected to the Ground instance belongs to the same containment instance as the grounding device. This means that the Ground instances are contained in the same voltage level where other switching devices are contained with a non-zero BaseVoltage.

- R:452:ALL:NA:unitMultiplier

  For exchange of attributes defined using CIM Data Types (ActivePower, Susceptance, etc.) a unit multiplier of 1 is used if the UnitMultiplier specified in this document is “none”.

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- R:600:ALL:NA:float

  An attribute that is defined as float (e.g. has a type Float or a type which is a Datatype with .value attribute of type Float) shall support ISO/IEC 60559:2020 for floating-point arithmetic using single precision floating point. A single precision float supports 7 significant digits where the significant digits are described as an integer, or a decimal number with 6 decimal digits. Two float values are equal when the significant with 7 digits are identical, e.g. 1234567 is equal 1.234567E6 and so are 1.2345678 and 1.234567E0.

- R:600:EQ:ReactiveCapabilityCurve:curve

  The two different styles of ReactiveCapabilityCurves can be define for a SynchronousMachine. It is defined by Curve.curveStyle attribute as shown in Annex B.

- R:456:SV:SvPowerFlow:instance

  As a minimum there shall be an instance of SvPowerFlow associated with the Terminal for all the following classes and their specialisation (subclass) that is part of an energized TopologicalIsland: RotatingMachine, EnergyConsumer, EquivalentInjection, ShuntCompensator, StaticVarCompensator, ExternalNetworkInjection, PowerElectronicsConnection and EnergySource. Additional instances of SvPowerFlow may optionally be available for other in service (SvStatus.inService=true) and energised equipment (equipment connected to a TopologicalNode part of the TopologicalIsland).

- R:600:NA:constraintFloat

  An attribute defined as xsd:float can be constrained to integer in given cases. This means that the value can include leading zero, but no decimal value or can fully conform to xsd:integer. Example of valid values are: "1", "001", "1.0", "1.000000" (as positive or negative value). Example of invalid value are: "1.", "1.1", "1.000000001".

- C:13:GL:CoordinateSystem.crsUrn:epsg

  CoordinateSystem.crsUrn: If not specified elsewhere, the CoordinateSystem.crsUrn uses WGS84 (latitude, longitude), i.e. urn:ogc:def:crs:EPSG::4326.

- C:13:GL:PositionPoint:coordinates

  In order to support various coordinate systems, position attributes are defined as strings. It is up to the implementation to ensure to correctly parse coordinate positions as defined by the coordinate system type.

- C:452:ALL:IdentifiedObject.description:stringLength

  The string IdentifiedObject.description is maximum 256 characters.

- C:452:ALL:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name has a maximum of 128 characters.

- C:452:EQ:ACDCConverter:containment

  For ACDCConverter (CsConverter, VsConverter) the association Equipment.EquipmentContainer is required and shall point to DCEquipmentContainer of type DCConverterUnit. In this case the association DCConverterUnit.Substation is required.

- C:452:EQ:ACLineSegment.BaseVoltage:calculations

  All implementations shall use association to a BaseVoltage for the purpose of any per unit calculations and shall not rely on the voltages (neither nominal nor actual values obtained by previous or current solution) at the nodes, which the ACLineSegment connects to.

- C:452:EQ:ACLineSegment.BaseVoltage:calculations2

  In case there are interconnected ACLineSegments with different BaseVoltage for different parts of the networks (when assembling different model authority sets) the application needs to handle this to ensure accurate physical units.

- C:452:EQ:ACLineSegment.r:valueRange

  ACLineSegment.r shall be a positive value or zero.

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- C:452:EQ:ACLineSegment.x:valueRange

  The attribute shall be a positive value. As negative reactance values are not allowed for ACLineSegment-s it is recommended to model series compensators explicitly.

- C:452:EQ:AsynchronousMachine.ratedS:dynamicSimulation

  RotatingMachine.ratedS is required in case of data exchange to support dynamic simulations.

- C:452:EQ:AsynchronousMachine:aggregate

  If one AsynchronousMachine is associated with one GeneratingUnit the flag Equipment.aggregate shall be consistent in case it is provided at both AsynchronousMachine and GeneratingUnit.

- C:452:EQ:AuxilaryEquipment:containment

  For AuxilaryEquipment (CurrentTransformer, PotentialTransformer, PostLineSensor, SurgeArrester, WaveTrap, FaultIndicator) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay or Line.

- C:452:EQ:BusbarSection:containment

  For BusbarSection the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel or Bay (when a disconnector is splitting a busbar section in two).

- C:452:EQ:Clamp:containment

  For Clamp the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay or Line when outside substation.

- C:452:EQ:ConductingEquipment.BaseVoltage:whereRequired

  The ConductingEquipment.BaseVoltage association is required for the following ConductingEquipment: ACLineSegment, EquivalentBranch and SeriesCompensator.

  For all other Equipment-s, not contained in a VoltageLevel, the association ConductingEquipment.BaseVoltage can be provided (as it is optional), however the association to BaseVoltage coming from the container or transformer ends takes precedence.

- C:452:EQ:Conductor:containment

  For Conductor (ACLineSegment) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Line.

- C:452:EQ:ConnectivityNode:containment

  The association ConnectivityNode.ConnectivityNodeContainer is required however the type of EquipmentContainer the association shall point to is not specified. Therefore the association ConnectivityNode.ConnectivityNodeContainer shall point to any type of EquipmentContainer given by the connecting equipment (that is linked to the associated Terminal). Machine based validation is not performed. It is required that import and export shall not make any changes to this association.

- C:452:EQ:CurveData.Curve:equationY1

  If CurveData.Curve is a ReactiveCapabilityCurve, each CurveData shall satisfy the following relation:

  – CurveData.xvalue^2+CurveData.y1value^2 shall be less than or equal to RotatingMachine.ratedS^2.

- C:452:EQ:CurveData.Curve:equationY2

  If CurveData.Curve is a ReactiveCapabilityCurve, each CurveData shall satisfy the following relation:

  – CurveData.xvalue^2+CurveData.y2value^2 shall be less than or equal to RotatingMachine.ratedS^2.

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- C:452:EQ:CurveData.Curve:reactive

  If CurveData.Curve is a ReactiveCapabilityCurve, the CurveData.y2value shall be greater than or equal to CurveData.y1value.It is not allowed that all CurveData.y2value values are equal to CurveData.y1value values.

- C:452:EQ:CurveData.Curve:reactiveCountP

  For a SynchronousMachine with a ReactiveCapabilityCurve the number of CurveData instances depends on the attribute SynchronousMachine.type as follows:

  – condenser, ReactiveCapabilityCurve is given by SynchronousMachine.maxQ and SynchronousMachine.minQ. There shall not be any association to ReactiveCapabilityCurve.

  – generator or generatorOrCondenser, at least two CurveData instances with CurveData.xvalue greater than or equal to 0.

  – motor or motorOrCondenser, at least two CurveData instances with CurveData.xvalue less than or equal to 0.

  – generatorOrMotor or generatorOrCondenserOrMotor, at least three CurveData instances with at least one having CurveData.xvalue greater than or equal to 0 and one having CurveData.xvalue less than or equal to 0.

  In general, it is required that the points defined by the ReactiveCapabilityCurve form an area of reactive capability different than 0.

- C:452:EQ:CurveData.Curve:VsCapabilityCurve

  If CurveData.Curve is a VsCapabilityCurve, the CurveData.y2value shall be greater than CurveData.y1value.

- C:452:EQ:CurveData.Curve:VsCapabilityCurveCount

  If CurveData.Curve is a VsCapabilityCurve at least two CurveData shall be associated.

- C:452:EQ:CurveData.xvalue:value

  The GeneratingUnit.minOperatingP shall be equal of the minimum of the CurveData.xvalue among all points defined for a ReactveCapabilityCurve. The GeneratingUnit.maxOperatingP shall be equal of the maximum of the CurveData.xvalue among all points defined for a ReactveCapabilityCurve.

- C:452:EQ:Cut:containment

  For Cut the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel or DCConverterUnit or Line when outside substation.

- C:452:EQ:DayType.name:validValues

  For DayType the name attribute indicates the days of the week that a given DayType represents. The name attribute is restricted to the following names: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday, Weekday, Weekend, All. If the name attribute is All, it represents all seven days of the week. If the name attribute is Weekday, it represents Monday through Friday. If the name attribute is Weekend, it represents Saturday and Sunday.

- C:452:EQ:DCBusbar:containment

  For DCBusbar the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

- C:452:EQ:DCChopper:containment

  For DCChopper the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

- C:452:EQ:DCGround:containment

  For DCGround the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

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- C:452:EQ:DCGround:instance

  At least one DCGround shall be defined for each dc circuit.

- C:452:EQ:DCLineSegment.resistance:valueRange

  The attribute DCLineSegment.resistance shall be a positive value.

- C:452:EQ:DCLineSegment:containment

  For DCLineSegment the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCLine. In the case of modelling back to back configuration the association shall point to EquipmentContainer of type Substation.

- C:452:EQ:DCSeriesDevice:containment

  For DCSeriesDevice the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

- C:452:EQ:DCShunt:containment

  For DCShunt the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

- C:452:EQ:DCSwitch:containment

  For DCSwitch (DCDisconnector, DCBreaker) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type DCConverterUnit.

- C:452:EQ:Disconnector:containment

  For Disconnector the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation.

- C:452:EQ:EarthFaultCompensator:containment

  For EarthFaultCompensator (GroundingImpedance, PetersenCoil) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel.

- C:452:EQ:EnergyConnection:containment

  For EnergyConnection (EnergySource, EnergyConsumer, NonConformLoad, ConformLoad, LinearShuntCompensator, NonlinearShuntCompensator, ExternalNetworkInjection, StaticVarCompensator, SynchronousMachine, AsynchronousMachine) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel.

- C:452:EQ:EnergyConsumer:constantP

  If the association EnergyConsumer.LoadResponse to LoadResponseCharacteristic is missing, this load is assumed to be constant power.

- C:452:EQ:Equipment:details

  Instance data to be exchanged shall make use of the most detailed class possible based on the level of detailed information currently available about the equipment at the time of the exchange. The classes GeneratingUnit, Switch, and EnergyConsumer should only be used if the information to determine the more detailed class (ThermalGeneratingUnit, HydroGeneratingUnit, Breaker, Disconnector, etc.) is not available.

  If in the future the more detailed information becomes available, the instance data shall be exchanged with the more specific specialisation class. The more specific specialization class in the newer serialization shall be exchanged with the same MRID as the more generic class exchanged previously. (Note: Rules such as this pertaining to serialization shall be moved to a document pertaining to all IEC 61970 profiles when that document becomes available.)

- C:452:EQ:EquivalenInjection:instance

  Using EquivalentInjection to model a distribution network equivalent is recommended practice instead of using ExternalNetworkInjection(s).

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- C:452:EQ:EquivalentBranch:containment

  For EquivalentBranch the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel, Line or Substation.

- C:452:EQ:EquivalentInjection:containment

  For EquivalentInjection the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel or Line.

- C:452:EQ:EquivalentShunt:containment

  For EquivalentShunt the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel.

- C:452:EQ:Fuse:containment

  For Fuse the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation.

- C:452:EQ:GeneratingUnit.minOperatingP:valueRangePair

  GeneratingUnit.maxOperatingP shall be greater than or equal to GeneratingUnit.minOperatingP.

- C:452:EQ:GeneratingUnit:containment

  For GeneratingUnit (SolarGeneratingInit, NuclearGeneratingUnit, ThermalGeneratingUnit, HydroGeneratingUnit, WindGeneratingUnit) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Substation.

- C:452:EQ:GeneratingUnit:maxOperatingP:ratedS

  GeneratingUnit.maxOperatingP shall be less than or equal to the sum of RotatingMachine.ratedS for RotatingMachine-s associated to the GeneratingUnit.

- C:452:EQ:GeneratingUnit:realPowerLimits

  The net real power limits can be defined in three ways; 1) with the attributes “maxOperatingP” and “minOperatingP”, or 2) with the attribute “ratedNetMaxP” or 3) with the attributes “ratedGrossMinP” and “ratedGrossMaxP” used in conjunction with an associated GrossToNetActivePowerCurve.

- C:452:EQ:GeneratingUnit:typeDependency

  GeneratingUnit.maxOperatingP and GeneratingUnit.minOperatingP shall depend on the SynchronousMachine.type of the associated SynchronousMachine as follows:

  – condenser: GeneratingUnit.minOperatingP and GeneratingUnit.maxOperatingP shall equal zero.

  – generator or generatorOrCondenser: GeneratingUnit.minOperatingP shall be greater than or equal to zero and GeneratingUnit.maxOperatingP shall be greater than zero.

  – motor or motorOrCondenser: GeneratingUnit.minOperatingP shall be a negative value and GeneratingUnit.maxOperatingP shall be a negative value or zero.

  – generatorOrMotor or generatorOrCondenserOrMotor: GeneratingUnit.minOperatingP shall be a negative value and GeneratingUnit.maxOperatingP shall be greater than zero.

- C:452:EQ:GrossToNetActivePowerCurve:values

  For GrossToNetActivePowerCurve used in transmission system data exchanges where the x and y values will always be specified in MW, the xMultiplier and y1Multiplier attributes do not need to be supplied thus not included in the profile.

- C:452:EQ:Ground:containment

  For Ground the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay or VoltageLevel.

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- C:452:EQ:GroundDisconnector:containment

  For GroundDisconnector the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation.

- C:452:EQ:HydroGeneratingUnit.energyConversionCapability:typeConsistency

  If HydroGeneratingUnit.energyConversionCapability is "generator" the associated SynchronousMachine shall have SynchronousMachine.type set to "generator" or "generatorOrCondenser". If HydroGeneratingUnit.energyConversionCapability is "pumpAndGenerator" the associated SynchronousMachine shall have SynchronousMachine.type set to "motor", "generatorOrMotor" or "generatorOrCondenserOrMotor".

- C:452:EQ:HydroPump:containment

  For HydroPump the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Substation.

- C:452:EQ:Jumper:containment

  For Jumper the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel, DCConverterUnit or Line when outside substation.

- C:452:EQ:Junction:containment

  For Junction the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Line (in case they model T-junction of a Line), or VoltageLevel or Bay (in case they model named join locations within a Substation).

- C:452:EQ:LinearShuntCompensator.gPerSection:valueRange

  LinearShuntCompensator.gPerSection shall be a positive value or zero.

- C:452:EQ:NA:negativeImpedance

  Except for series capacitors, negative branch reactance shall not represent real devices. Negative impedance is allowed for EquivalentBranch. Negative value limits the selection of load flow solution techniques and shall be avoided, if possible.

- C:452:EQ:NA:zeroImpedance

  Zero impedance lines are permitted if they represent real elements in the model.

- C:452:EQ:NonlinearShuntCompensatorPoint.g:valueRange

  NonlinearShuntCompensatorPoint.g shall be a positive value or zero.

- C:452:EQ:OperationalLimitSet:limits

  In case the OperationalLimitSet is the operational limit of the AuxiliaryEquipment, then the association end OperationalLimitSet.Equipment is also required. In the case where OperationalLimitSet.Equipment is associated with an instance of ConductingEquipment, OperationalLimitSet.Terminal shall be one of the ConductingEquipment's Terminal-s.

- C:452:EQ:PhaseTapChanger:controlModeP

  The association TapChanger.TapChangerControl for PhaseTapChanger-s shall only point to a TapChangerControl that has the following control modes for RegulatingControl.mode: activePower or voltage.

- C:452:EQ:PowerTransformer:containment

  For PowerTransformer the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Substation or DCConverterUnit. For the case of a transformer that connects two substations, the terminal of one of the PowerTransformerEnd-s can be connected to a ConnectivityNode defined in another substation.

- C:452:EQ:PowerTransformerEnd.b:valueRange

  PowerTransformerEnd.b shall be negative value or zero. Negative magnetising branch susceptance (PowerTransformerEnd.b) means inductive reactive power losses in no load.

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- C:452:EQ:PowerTransformerEnd.g:valueRange

  PowerTransformerEnd.g shall be positive value or zero. Positive magnetising branch conductance (PowerTransformerEnd.g) means positive active power losses in no load.

- C:452:EQ:PowerTransformerEnd.x:value

  Transformers with zero series reactance do not exist. PowerTransformerEnd.x of high voltage end in case of a two winding transformer shall be a positive value. In case of a three winding transformer the PowerTransformerEnd.x shall not be zero.

- C:452:EQ:PowerTransformerEnd:instance

  Negative reactance values are valid for transformers.

- C:452:EQ:PowerTransformerEnd:pu

  The parameters r, x, g and b are specified for each end and are not related to the overall base voltage. These values are specified in engineering units. Any PU calculations are internal to particular tools and are not part of the data exchange.

- C:452:EQ:ProtectedSwitch:containment

  For ProtectedSwitch (Breaker, DisconnectingCircuitBreaker, LoadBreakSwitch) the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type Bay, VoltageLevel or DCConverterUnit.

- C:452:EQ:RatioTapChangerl:controlMode

  The association TapChanger.TapChangerControl for RatioTapChanger-s shall only point to a TapChangerControl the has the following control modes for RegulatingControl.mode: voltage, reactivePower and powerFactor.

- C:452:EQ:ReactiveCapabiltyCurve.CurveData:xvalue

  All CurveData.xvalue for a given ReactiveCapabilityCurve shall be unique, e.g. it is not allowed for two or more .xvalue to have the same float value for a given ReactiveCapabilityCurve.

- C:452:EQ:RegularTimePoint:values

  RegularTimePoint is associated with types of schedules (ConformLoadSchedule, NonConformLoadSchedule) where value1 will always be specified in MW and value2 will always be specified in MVAr. In these cases, the value1Multiplier and value2Multiplier attributes of BasicIntervalSchedule do not need to be specified thus not included in the profile.

- C:452:EQ:RegulatingControl.targetValue:tapChanger

  In cases where RequlatingControl.mode is "voltage" and both TapChanger.controlEnabled and RequlatingControl.enabled are "true",

  – The RegulatingControl.targetValue in per unit value is calculated by RegulatingControl.targetValue/BaseVoltage.nominalVoltage.

  shall be within the regulating capability limits:

  – The tap changer upper capability limit in per unit value is calculated by 1+RatioTapChanger.stepVoltageIncrement/100*(TapChanger.highStep-TapChanger.neutralStep).

  – The tap changer lower capability limit in per unit value is calculated by 1-RatioTapChanger.stepVoltageIncrement/100*(TapChanger.neutralStep-TapChanger.lowStep).

- C:452:EQ:RegulatingControl:point

  In cases where the controlling point is a ConnectivityNode multiple instances of RegulatingControl maybe be needed for proper modelling and connectivity of the grid.

- C:452:EQ:RegulatingControl:RegulatingEquipment

  A RegulatingControl that is not a TapChangerControl must have at least one regulating equipment associated through the RegulatingCondEq.RegulatingControl. That is, a RegulatingControl cannot exist without some equipment using it for regulating.

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- C:452:EQ:RegulatingControl:samePoint

  A RegulatingControl will have associations to one or more instances of RegulatingCondEq and an association to a Terminal. The ConnectivityNode associated with the Terminal is the regulated point. It is common to have cases where multiple pieces of equipment regulate ConnectivityNodes that under normal network topology are associated with the same TopologicalNode. In this case, the same instance of RegulatingControl should be used by all of those regulating equipment if possible. If it is not possible, such as the case where a SynchronousMachine and a RatioTapChanger are regulating the same point using associations to instances of RegulatingControl and TapChangerControl, the number of instances of RegulatingControl and TapChangerControl should be minimized. Additionally, the target and deadband values for the same regulated point should not be contradictory.

  Profile restriction: If multiple instances of RegulatingControl control the same regulation point, the targetValues must not be contradictory.

- C:452:EQ:RegulationSchedule:values

  By convention, for RegulationSchedule the attribute ”value1” represents the target voltage or real power. “value2” is the deviation. For instance, a value1 of 100 and value2 of 1 means regulating to 100 kV plus or minus 1 kV, if the units are set to kV. In this case the range would be from 99 kV to 101 kV.

- C:452:EQ:SeriesCompensator:containment

  For SeriesCompensator the association Equipment.EquipmentContainer is required and shall point to EquipmentContainer of type VoltageLevel when in substation, DCConverterUnit or Line when outside substation.

- C:452:EQ:ShuntCompensator.voltageSensitivity:valueRange

  The ShuntCompensator.voltageSensitivity attribute shall be greater than zero.

- C:452:EQ:ShuntCompensator:controlMode

  For ShuntCompensator, the association RegulatingCondEq.RequlatingControl shall only point to a RequlatingControl that has the following control modes for RegulatingControl.mode: voltage, reactivePower and powerFactor.

- C:452:EQ:StaticVarCompensator:controlMode

  For StaticVarCompensator, the association RegulatingCondEq.RequlatingControl is required and shall only point to a RequlatingControl that has the following control modes for RegulatingControl.mode: voltage and reactivePower.

- C:452:EQ:Switch:connection

  Switch and its subclasses shall only connect to ConnectivityNode-s that are contained in either the same VoltageLevel or in different VoltageLevel-s which have the same BaseVoltage.

- C:452:EQ:Switch:usage

  Busbar couplers shall be modelled explicitly if they are to be retained. The substitution of busbar couplers with zero impedance branches shall not be permitted for exchanges.

- C:452:EQ:SynchronousMachine.maxQ:valueRangePair

  SynchronousMachine.maxQ shall be greater than or equal to SynchronousMachine.minQ.

- C:452:EQ:SynchronousMachine.qPercent:usage

  The attribute SynchronousMachine.qPercent in the EQ profile specifies a default value. The participation of the devices regulating particular node can be set in other profiles such as SSH. Therefore, the implementations shall consider this in the load flow calculation and normalize the participation based on what devices are participating in the regulation of a particular node.

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- C:452:EQ:SynchronousMachine.type:condenser

  The SynchronousMachine is not required to be associated with a GeneratingUnit via the association RotatingMachine.GeneratingUnit in cases where a synchronous condenser is being modelled as there is no capability for real power output. In this case, the SynchronousMachine.type shall be set to “condenser”. The following SynchronousMachine.type-s: "motorOrCondenser", "generatorOrCondenser", and "generatorOrCondenserOrMotor" are not allowed to use.

- C:452:EQ:SynchronousMachine:aggregate

  If only one SynchronousMachine is associated with the GeneratingUnit then the Equipment.aggregate flag shall be consistent between the SynchronousMachine and GeneratingUnit if it exists in both.

- C:452:EQ:SynchronousMachine:controlMode

  For SynchronousMachine, the association RegulatingCondEq.RequlatingControl shall only point to a RequlatingControl that has the following control modes for RegulatingControl.mode: voltage, reactivePower and powerFactor.

- C:452:EQ:SynchronousMachine:ratedS

  For SynchronousMachine, the attribute RotatingMachine.ratedS is required in case of data exchange to support dynamic simulations.

- C:452:EQ:SynchronousMachine:reactiveLimits

  ReactiveCapabilityCurve-s are not required if the reactive power limits of the SynchronousMachine do not vary with real power output. SynchronousMachine.minQ and SynchronousMachine.maxQ are required if SynchronousMachine.InitialReactiveCapabilityCurve is missing. If the association SynchronousMachine.InitialReactiveCapabilityCurve is provided it takes precedence to the information provided by the attributes SynchronousMachine.minQ and SynchronousMachine.maxQ. However, if both SynchronousMachine.minQ, SynchronousMachine.maxQ and ReactiveCapabilityCurve are present, the SynchronousMachine.minQ shall be equal to min of CurveData.y1value-s and SynchronousMachine.maxQ shall be equal to max CurveData.y2value-s.

- C:452:EQ:TapChangerControl:remoteQcontrol

  A power transformer cannot efficiently control reactive power flow on remote terminals. Therefore, a TapChangerControl controlling reactive power flow shall only control the flow at one of the Terminal-s associated with PowerTransformerEnd-s of the PowerTransformer where the TapChanger is located. Control of a remote Terminal not associated with the PowerTransformer that has the TapChanger is not allowed. This constraint defines that multiple TapChanger-s cannot be controlled by the same TapChangerControl.

- C:452:EQ:Terminal:connection

  Terminal-s of the two sides of a two-terminal ConductingEquipment (or any of its subclasses) shall not be connected to the same ConnectivityNode.

- C:452:OP:Measurement.measurementType:accumulatorValues

  For Accumulator, Measurement.measurementType is restricted to the following valid values: ApparentEnergy, ReactiveEnergy, ActiveEnergy.

- C:452:OP:Measurement.measurementType:analogValues

  For Analog, Measurement.measurementType is restricted to the following valid values: ThreePhasePower, ThreePhaseActivePower, ThreePhaseReactivePower, LineCurrent, PhaseVoltage, Voltage, Angle, Frequency and TapPosition.

- C:452:OP:Measurement.measurementType:discreteValues

  For Discrete, Measurement.measurementType is restricted to the following valid values: SwitchPosition.

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- C:452:OP:Measurement.Terminal:requiredCases

  The association Measurement.Terminal shall reference a Terminal of the Equipment referenced by Measurement.PowerSystemResource except in cases where Measurement.measurementType is either TapPosition or SwitchPosition in which the association is not exchanged.

- C:452:OP:Measurement.unitSymbol:accumulatorValues

  For Accumulator, Measurement.unitSymbol is restricted to the following valid values: VAh, VArh, Wh.

- C:452:OP:Measurement.unitSymbol:analogValues

  For Analog, Measurement.unitSymbol is restricted to the following valid values: W, deg, VA, A, VAr, V, Hz.

- C:452:OP:Measurement.unitSymbol:discreteValues

  For Discrete, Measurement.unitSymbol is restricted to the following valid values: none.

- C:452:OP:MeasurementValueSource:name

  For MeasurementValueSource, attribute IdentifiedObject.name is restricted to the following strings for MeasurementValueSource: ICCP, SCADA.

- C:452:OP:NA:measurement

  In the context of this profile the following classes: AccumulatorValue, AnalogValue, DiscreteValue, StringMeasurementValue are only used to define measurements that are available via ICCP and to define the location of SCADA measurements for use by state estimator. It is not used to supply values for those measurements. Consequently, the value attribute is not included in this profile.

- C:452:SC:BusbarSection.ipMax:optional

  The attribute ipMax is an optional attribute even if short circuit data is exchanged, as not always entered by the user (e.g. the IEC 60909-4 example test model described in Chapter 6.2 of the standard does not include these values).

- C:452:SC:PowerTransformerEnd.grounded:grounding

  If TransformerEnd.grounded is true, then TransformerEnd.rground and TransformerEnd.xground are required.

- C:452:SC:SynchronousMachine.earthing:attributes

  If SynchronousMachine.earthing is true, then SynchronousMachine.earthingStarPointR and SynchronousMachine.earthingStarPointX are required.

- C:453:DL:Diagram:units

  Coordinate units used in the exchange are mm (millimetres).

- C:453:DL:DiagramObject:SynchronousMachine

  The DiagramObject should link to SynchronousMachine and not GeneratingUnit.

- C:453:DL:DiagramStyle:name

  The inherited IdentifiedObject.name shall have one of the following names: "node-breaker", "bus-branch", "hybrid (node-breaker and bus-branch)" or "geoschematic".

- C:453:DL:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name has a maximum of 128 characters.

- C:456:SSH:ACDCConverter:targets

  The target values and related attributes for ACDCConverter and its subclasses are optional in SSH. However, depending on the control mode of the converter some of the attributes shall be considered as required. The description of the control modes in the enumerations CsPpccControlKind, VsPpccControlKind and VsQpccControlKind provide information on necessary attributes which are then considered required attributes for each control mode in SSH.

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- C:456:SSH:EnergyConsumer.p:ValueRange

  Negative active power loads shall not be exchanged. In cases where this is needed EquivalentInjection is used instead.

- C:456:SSH:EnergyConsumer.q:ValueRange

  Negative reactive power loads shall not be exchanged. In cases where this is needed EquivalentInjection is used instead.

- C:456:SSH:EnergySource:EnergySourcePQ

  The attributes voltageAngle and voltageMagnitude shall not be used when the EnergySource is representing a constant active and reactive power injection (PQ injection), i.e. they shall only be used when the EnergySource is modelling a voltage source.

- C:456:SSH:EquivalentInjection.p:limits

  The negated value (necessary due to sign convention) of EquivalentInjection.p shall be less than or equal to EquivalentInjection.maxP and shall be greater than or equal to EquivalentInjection.minP.

- C:456:SSH:EquivalentInjection.q:limits

  The negated value (necessary due to sign convention) of EquivalentInjection.q shall be less than or equal to EquivalentInjection.maxQ and shall be greater than or equal to EquivalentInjection.minQ.

- C:456:SSH:EquivalentInjection:regulation

  If EquivalentInjection.regulationCapability in EQ is true, then EquivalentInjection.regulationStatus and EquivalentInjection.regulationTarget are required in SSH. If EquivalentInjection.regulationCapability in EQ is false, then EquivalentInjection.regulationStatus and EquivalentInjection.regulationTarget are not exchanged in SSH.

- C:456:SSH:ExternalNetworkInjection.p:limits

  The negated value (necessary due to sign convention) of ExternalNetworkInjection.p shall be less than or equal to ExternalNetworkInjection.maxP and shall be greater than or equal to ExternalNetworkInjection.minP.

- C:456:SSH:ExternalNetworkInjection.q:limits

  The negated value (necessary due to sign convention) of ExternalNetworkInjection.q shall be less than or equal to ExternalNetworkInjection.maxQ and shall be greater than or equal to ExternalNetworkInjection.minQ.

- C:456:SSH:GeneratingUnit.normalPF:values

  GeneratingUnit.normalPF is normally zero for generators which are non-energized (do not participate in the load flow). It can have a number in case of activating the GeneratingUnit is a remedial action. As .normalPF values are not normalized across different MAS, the application that imports a model shall have a power flow calculation logic setup to be able to normalize normalPF values for the purpose of the distributed slack and preserve the original values of GeneratingUnit.normalPF in case of export.

- C:456:SSH:NA:angleReference

  Angle reference: In cases where it is required to exchange a solved model and compare power flow results (assuming identical/comparable calculation methods and initial conditions) it is required that the angle reference slack is placed within the exported model. The slack generator is the SynchronousMachine connected to the TopologicalNode referenced by TopologicalIsland.AngleRefTopologicalNode and which has the highest SynchronousMachine.referencePriority in the exported MAS’s SSH.

- C:456:SSH:NA:distributedActivePowerSlack

  Active power slack by multiple generators per ControlArea: all GeneratingUnit(s) have non-zero normalPF, but there must be one GeneratingUnit per ControlArea that have maximum (the highest) participation factor (GeneratingUnit.normalPF) so that this GeneratingUnit is used as a single active power slack in case the tooling does not support active power slack by multiple generators per ControlArea.

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- C:456:SSH:NA:notSolvedModel

  In case of exchange of non-solved model (i.e. SV profile is not exchanged) the tools shall assign the angle reference node using the information from SSH (the highest SynchronousMachine.referencePriority or GeneratingUnit.normalPF in the exported MAS's SSH).

- C:456:SSH:NA:singleActivePowerSlack

  Active power slack by a single generator per ControlArea: one generator has GeneratingUnit.normalPF set to a highest value (non-zero) and all other generating units have a zero GeneratingUnit.normalPF.

- C:456:SSH:RegulatingControl.targetValue:value

  RegulatingControl.targetValue shall be positive value in cases where the RegulatingControl.mode is set to voltage in EQ profile.

- C:456:SSH:RotatingMachine.p:limits

  The negated value (necessary due to sign convention) of RotatingMachine.p shall be less than or equal to GeneratingUnit.maxOperatingP and shall be greater than or equal to GeneratingUnit.minOperatingP of the associated GeneratingUnit.

- C:456:SSH:RotatingMachine.q:limits

  In case there is no ReactiveCapabilityCurve associated with the SynchronousMachine, the negated value (necessary due to sign convention) of RotatingMachine.q shall be less than or equal to SynchronousMachine.maxQ and shall be greater than or equal to SynchronousMachine.minQ.

- C:456:SSH:RotatingMachine:pAndQcapabilityCurve

  In cases where a ReactiveCapabilityCurve is associated, the RotatingMachine.p shall be less than or equal to the maximum active power value defined by the curve and it shall be greater than or equal to the minimum active power value defined by the curve. The RotatingMachine.q shall be less than or equal to the maximum reactive power value defined by the curve and it shall be greater than or equal to the minimum reactive power value defined by the curve.

- C:456:SSH:ShuntCompensator.sections:value

  In cases where RegulatingControl.discrete is true and RegulatingControl.enabled is true, ShuntCompensator.sections shall be integer.

- C:456:SSH:SynchronousMachine.operatingMode:matchType

  The SynchronousMachine.operatingMode shall be consistent with the SynchronousMachine.type.

  – SynchronousMachine.operatingMode = “motor” shall be provided for SynchronousMachine.type in ["motor", "generatorOrMotor", "motorOrCondenser", "generatorOrCondenserOrMotor"],

  – SynchronousMachine.operatingMode = “condenser" shall be provided for SynchronousMachine.type in ["condenser", "generatorOrCondenser", "motorOrCondenser", "generatorOrCondenserOrMotor"], and

  – SynchronousMachine.operatingMode = “generator” shall be provided for SynchronousMachine.type in ["generator", "generatorOrMotor", "generatorOrCondenser", "generatorOrCondenserOrMotor"].

- C:456:SSH:TapChanger.step:value

  In cases where RegulatingControl.discrete is true and RegulatingControl.enabled is true, TapChanger.step shall be integer.

- C:456:SV:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name has a maximum of 128 characters.

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- C:456:SV:SvPowerFlow.p:synchronousMachine

  The SvPowerFlow.p that is given by the steady state power flow solution for a SynchronousMachine shall normally be within the capability of the machine defined in the ReactiveCapabilityCurve or GeneratingUnit.maxOperatingP and GeneratingUnit.minOperatingP when ReactiveCapabilityCurve is not present. Active power can be outside the capability as part of start-up or shutdown of the generator. CIM cannot represent different operation mode so this constraint will only give a warning. Note that different data exchange processes can assign more restrictive severity depending on the business needs and power flow algorithm applied by the business process.

- C:456:SV:SvPowerFlow.q:synchronousMachine

  The SvPowerFlow.q that is given by the steady state power flow solution for a SynchronousMachine shall normally be within the capability of the machine defined in the ReactiveCapabilityCurve or SynchronousMachine.maxQ, SynchronousMachine.minQ when ReactiveCapabilityCurve is not present. Reactive power can be outside the capability if powerflow excludes reactive restriction. CIM cannot represent this so this contains will also give a warning. Note that different data exchange processes can assign more restrictive severity depending on the business needs and power flow algorithm applied by the business process.

- C:456:SV:SvShuntCompensatorSections.sections:value

  In cases where RegulatingControl.discrete is true and RegulatingControl.enabled is true, SvShuntCompensatorSections.sections shall be integer.

- C:456:SV:SvSwitch:instance

  SvSwitch shall be exchanged for all switching devices. It is expected that in most cases the SvSwitch.open in SV instance data will be identical with Switch.open in SSH instance data. However, in cases where a regulating control is modifying the connection state of the controlled device a difference between SvSwitch.open and Switch.open can occur.

- C:456:SV:SvTapStep.position:value

  In cases where RegulatingControl.discrete is true and RegulatingControl.enabled is true, SvTapStep.position shall be integer.

- C:456:SV:SvVoltage.v:absoluteLimit

  SvVoltage.v shall be greater than 0.4 pu in cases where associated voltage limits are not defining different limit range.

- C:456:SV:SvVoltage.v:limits

  The SvVoltage.v shall be less than or equal to VoltageLimit.value associated with OperationalLimitType.limitType=highVoltage and greater than or equal to VoltageLimit.value associated with OperationalLimitType.limitType=lowVoltage.

- C:456:SV:TopologicalIsland:instance

  At least one TopologicalIsland instance shall be present per SV instance.

  The TopologicalIsland-s for a merged model which are defined in the state variables instance file for the merged model are created with the solving on the power flow of the merged model, i.e. there are no TopologicalIsland-s defined per MAS in a merged model. In case a solved merged model is exchanged for a single MAS the state variables instance file shall include at least one instance of TopologicalIsland.

- C:456:TP:IdentifiedObject.description:stringLength

  The string IdentifiedObject.description is maximum 256 characters.

- C:456:TP:IdentifiedObject.name:instance

  Name is required for all classes in the profile except ACDCTerminal.

- C:456:TP:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name has a maximum of 128 characters.

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- C:456:TP:Terminal:switch

  The Terminal(s) of the two sides of a retained Switch (Switch.retained=true) or any of its subclasses shall not be connected to the same TopologicalNode.

- C:456:TP:TopologicalNode:containment

  The association TopologicalNode.ConnectivityNodeContainer is required however CGMES does not specify to which type of EquipmentContainer the association shall point to. Therefore the association TopologicalNode.ConnectivityNodeContainer shall point to any type of EquipmentContainer given by the connecting equipment (that is linked to the associated Terminal). Machine based validation is not performed. It is required that import and export shall not make any changes to this association.

- C:457:DY:ExcitationSystemDynamics.SynchronousMachineDynamics:reference

  ExcitationSystemDynamics.SynchronousMachineDynamics cannot point to an object of type SynchronousMachineSimplified.

- C:457:DY:GovSteamSGO.pmin:datatype

  There is a mistake in the IEC 61970-302. The datatype of the attribute GovSteamSGO.pmin is Seconds. The value exchanged with this attribute shall be PU as it is for GovSteamSGO.pmax.

- C:457:DY:IdentifiedObject.description:stringLength

  The string IdentifiedObject.description has a maximum of 256 characters.

- C:457:DY:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name has a maximum of 128 characters.

- C:457:DY:mwbase:equation

  The vast majority of turbine-governor models have mwbase as a parameter. The parameter is specified explicitly in the turbine-governor models to support use of a different value by transient analysis tools, if needed. If this parameter is present it shall correspond to RotatingMachine.ratedPowerFactor * RotatingMachine.ratedS.

- C:457:DY:TurbineGovernorDynamics:deltaomega

  Some turbine governors diagrams have deltaomega as an input, others just omega. In cases where deltaomega is an input the following relation with the speed is defined: deltaomega = (speed – 1) in pu.

- C:600:All:NA:datatypes

  According to 61970-501, datatypes are not exchanged in the instance data. The UnitMultiplier is 1 in cases none value is specified in the profile.

- C:600:ALL:NA:minMaxValues

  Maximum values shall always be greater than or equal to minimum values if not specified otherwise explicitly.

- C:600:EQ:ACLineSegment:BaseVoltageDiff

  CGMES exchanges allow 10 % difference of the BaseVoltage.nominalVoltage at the two ends of an ACLineSegment representing a complete tie-line or connecting to a boundary node.

- C:600:EQ:EnergyConsumer:junction

  Line containment is not allowed in cases where instances of an EnergyConsumer or any of its subclasses (NonConformLoad or ConformLoad) is connected to an ACLineSegment.

  In order to cover such cases a T-Junction shall be modelled, and Substation classes instantiated. For instance, the number of Substation-s is two plus the number of T-Junctions that need to be modelled.

- C:600:EQ:EquivalentInjection.regulationCapability:notHvdc

  If EquivalentInjection connects to a BoundaryPoint with flag isDirectCurrent=false (meaning this is not HVDC), the EquivalentInjection.regulationCapability in EQ shall be set to false and there shall not be a ReactiveCapabilityCurve associated.

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- C:600:EQ:Line:ACLineSegment

  As in the containment rules it is defined that ACLineSegment is contained in Line (also Line can have multiple ACLineSegment-s), the ACLineSegment-s that are parts of a tie-line and exchanged with a MAS of an individual grid model are contained in the Line that is in the Boundary MAS. The Line in Boundary MAS contains at least 2 ACLineSegment-s and a ConnectivityNode.

- C:600:EQ:ReactiveCapabilityCurve:units

  For a ReactiveCapabilityCurve associated with SynchronousMachine, the Curve.xUnit shall be set to UnitSymbol.W and both Curve.y1Unit and Curve.y2Unit shall be set to UnitSymbol.VAr. As the multiplier is not included in the profile it is defined the same as the multiplier used for datatype ActivePower and ReactivePower, i.e. UnitMultiplier.M.

- C:600:EQ:RegulatingControl:point

  The controlled point and the controlling equipment shall be located in the same TopologicalIsland.

  In cases where the controlling point is a TopologicalNode only one RegulatingControl shall be instantiated following the terms of constraint C:452:EQ:RegulatingControl:samePoint.

- C:600:EQ:RotatingMachine.ratedS:required

  RotatingMachine.ratedS is required.

- C:600:EQ:Substation:count

  The number of Substation-s shall reflect the design of the power system. Cases of a single Substation in a power system model or having a Substation per VoltageLevel are reported as warnings.

- C:600:EQ:TapChanger.neutralU:ValueRangePair

  The TapChanger.neutralU shall be the same as PowerTransformerEnd.ratedU.

- C:600:EQBD:IdentifiedObject.description:stringLength

  The string is maximum 256 characters.

- C:600:EQBD:IdentifiedObject.name:stringLength

  The string IdentifiedObject.name is maximum 128 characters.

- C:600:EQBD:NA:constraints

  All constraints defined for IEC 61970-600-2 EQ profile are applied to EQBD profile as well.

- C:600:GL:NA:position

  IEC 61968-13 allows both PositionPoint and mainAddress to be exchanged. CGMES requires that PositionPoint is exchanged and mainAddress is not exchanged.

- C:600:SC:SeriesCompensator.varistorRatedCurrent:required

  The SeriesCompensator.varistorRatedCurrent is required if SeriesCompensator.varistorPresent is true.

- C:600:SC:SeriesCompensator.varistorVoltageThreshold:required

  The SeriesCompensator.varistorVoltageThreshold is required if SeriesCompensator.varistorPresent is true.

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Annex A
(informative)

Summary of changes applied to this edition

A.1 General

This annex summarizes the changes applied to this edition compared to IEC TS 61970-600-2:2017. The changes are summarized in the following categories:

- Category 1 (High). Structural change of data model and/or relations.

  Changes in this category require substantial adaptation due to a structural change of the data model.

- Category 2 (Moderate): Changes that have moderate implementation impact

  Changes in this category require some adaptation of the tools in order to enable parsing of the data or data conversion to internal data models of the tools.

- Category 3 (Low): Changes of clarification character with some implementation impact

  Changes in this category require consideration of the implementation as in some cases a change in a description of a profile specific note could have an impact on the interpretation.

- Category 4 (No): Changes that do not have implementation impact

  Changes on this category are mostly profiling bug fixes and alignment with the structure of CIM17, hence should not, normally, impact implementation. This category also includes changes due to integration of defined rules and requirements in various documents (implementation guides, quality rules documents, etc.) related to the CGMES. It is assumed that these changes are already implemented in order to comply with exchanges in business processes.

Note that in cases where the document refers to CIM17 it should be understood as UML of canonical CIM model in which this CGMES release is based on and it is documented in IEC 61970-301:2020.

A.2 Overview of changes

This specification aims at consolidating information from various documents used when implementing business processes in Europe. Compared to IEC TS 61970-600-2:2017 this specification includes the following main improvements:

- This version of the CGMES implements necessary modifications requirements of major processes in Europe, namely CGM (Common Grid Model) and TYNDP (Ten Year Network Development Plan) which are defined and referenced in multiple EU regulations or guidelines, e.g. Regulation (EU) No 347/2013, Regulation (EU) 2015/1222, Regulation (EU) 2017/1485. The focus of this version was on CGM where agreements from various implementation guides, or supportive documents were integrated in main specification of CGMES. Issues were recorded during the implementation of the profiles from 2014 and CGMES Conformity Assessment process were resolved and implemented in the new version.

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- The following number of changes were introduced in CGMES profiles:
  - There are no changes categorized in category “High” impact as in general CIM17 and CGMES remain backwards compatible on functional level[8].
  - There are 7 changes in category “Moderate” impact.
  - There are 148 changes in category “Low” impact.
  - There are 82 changes in category “No” impact.
- This document uses CIM17 (IEC 61970-301:2020) as a basis for the profiles. CIM17 is in principle backwards compatible with CIM16. There might be some limited amount of breaking changes due to the fact that the version of the profiles from 2014 was not based on the last version of CIM16. All important changes are recorded in the following clauses.
- As the profiling tools and methods improved in the last years it was possible to apply an extensive set of validations in order to ensure consistency among the profiles. This comes together with necessary update of descriptions, profile constraints and cardinality of attributes and associations with the ultimate goal to enable seamless data exchange which satisfies the needs to the utilities and EU legislation. Changes were also applied having in mind the exchanges between transmission and distribution system operators.
- Dynamics profile is added to the scope of the specification as the IEC 61970-457 is not published yet. This profile is practically the same as IEC 61970-457 which is based on IEC 61970-302:2018. In order to have dynamics profile fully in line with CIM17 the class EnergyConnection was added as an abstract class and some descriptions were updated. This does not change instance data.

A.3 General changes

The following summarises the general changes.

Category 1: Moderate impact

- The namespace is changed from “http://iec.ch/TC57/2013/CIM-schema-cim16#” to http://iec.ch/TC57/CIM100# due to the decision to use CIM17 as a base model.
- The stereotype of all extended classes and attributes is changed from “Entsoe” to “European”. In RDFS the prefix used for the extension namespace is “eu” instead of “entsoe”. The namespace of the extensions is changed from “http://entsoe.eu/CIM/SchemaExtension/3/1” to http://iec.ch/TC57/CIM100-European#
- Profile constraints are defined in Clause 16 “Profile constraints” and aligned with IEC 61970-400 series and IEC 61970-301 standards for CIM17. Therefore, all profile notes defined in UML of IEC TS 61970-600-2:2017 were deleted. The change is marked as moderate only because the rules needs more detail review when implementing. However, many of these rules are the same as previous editions. Many of the former profile notes/constraints were modified to implement CGM requirements and issues resolutions.

Category 2: Low impact

- A change of descriptions in UML or a profile constraint which was added to specify sign convention or value constraint. This kind of changes are related to the integration of Clause E7 of IEC TS 61970-600-1:2017 into this document and IEC 61970-301 or IEC 61970-400 series of standards (which is considered to have no impact on implementations) as well as constraints proposed by the implementation of EU legislation related to the creation and exchange of common grid models.

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8 A version of the CGMES is considered as functional backwards compatible with its predecessor if the data exchanged can be transformed to fully satisfy the functional requirements for both new and previous version. For example a change of a name of class or attribute, or deletion and substitution of elements.

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- Explicit equipment boundary profile definition (EQBD) has been deprecated in this edition in favour of using its full profile counterpart (EQ) for exchange of boundary datasets. Topology boundary profile (TPBD) is no longer part of the CGMES as boundary TP is not exchanges due to the arrangements made to ensure that TP is an output.
- The descriptions of many attributes in different profiles were amended to include a constraint related to the allowed values. These changes were introduced based on CGM requirements and interoperability testing efforts. The following list contains may not be complete: DiagramObjectPoint.sequenceNumber, ActivePowerLimit.value, ActivePowerLimit.normalValue, ApparentPowerLimit.normalValue, ApparentPowerLimit.value, CurrentLimit.normalValue, CurrentLimit.value, VoltageLimit.normalValue, VoltageLimit.value, ControlArea.pTolerance, EnergySource.voltageAngle, EnergySource.voltageMagnitude, EquivalentInjection.regulationTarget, GeneratingUnit.normalPF, VsConverter.droop, VsConverter.droopCompensation, VsConverter.qShare, VsConverter.targetPWMfactor, VsConverter.targetPowerFactorPcc, VsConverter.targetPhasePcc, VsConverter.delta, VsConverter.uv, VsConverter.targetUpcc, CsConverter.targetAlpha, CsConverter.targetGamma, CsConverter.minAlpha, CsConverter.maxAlpha, CsConverter.alpha, CsConverter.gamma, CsConverter.maxGamma, CsConverter.minGamma, CsConverter.targetIdc, ACDCConverter.poleLossP, ACDCConverter.idleLoss, ACDCConverter.resistiveLooss, ACDCConverter.switchingLoss, ACDCConverter.targetUdc, RegulatingControl.targetDeadband, BatteryUnit.storedE, ShuntCompensator.sections, SvShuntCompensatorSections.sections, SvVoltage.v, ACDCConverter.maxUdc, ACDCConverter.minUdc, ACDCConverter.uc, ACDCConverter.udc, CsConverter.maxIdc, CsConverter.minIdc, VsConverter.targetUpcc, RotatingMachine.ratedPowerFactor, ShuntCompensator.normalSections, PowerTransformerEnd.ratedU, StaticVarCompensator.slope, PowerElectronicsConnection.ratedU, RotatingMachine.ratedU, SeriesCompensator.varistorRatedCurrent, Switch.ratedCurrent, HydroTurbine.speedRating, BatteryUnit.ratedE, GeneratingUnit.ratedGrossmaxP, GeneratingUnit.ratedGrossminP, GeneratingUnit.ratedNetMaxP, Length.
- The descriptions of many classes or attributes in different profiles were modified to include clarifications or constraints. These changes were introduced based on CGM requirements and interoperability testing efforts. The following list contains may not be complete and some listed items may duplicate the detailed list of changes provided in A.4: PhaseTapChangerAsymmetrical, PhaseTapChangerAsymmetrical.windingConnectionAngle, Equipment.inService, ShuntCompensator.aVRDelay, ControlArea, TieFlow.positiveFlowIn, TieFlow, OperationalLimit, PhaseCode, UnitSymbol, Switch, SeriesCompensator.varistorPresent, RegularTimePoint.sequenceNumber, ACDCConverter.PccTerminal, DCPolarityKind.positive, DCPolarityKind.middle, DCPolarityKind.negative, ACDCConverterDCTerminal.polarity, ACDCConverter.ratedUdc, ACDCConverter.baseS, CsConverter.ratedIdc, DCConductingEquipment.ratedUdc, ACDCConverter.minUdc, EnergyConsumer, EnergyConsumer.pfixed, EnergyConsumer.qfixed, LoadResponseCharacteristic, RegulatingControl.targetdeadband, NonlinearShuntCompensator, EquivalentNetwork, NonlinearShuntCompenstorPoint, ShuntCompensator.sections, CsConverter, CsConverter.alpha, CsConverter.gamma, CsOperatingModeKind.inverter, CsOperatingModeKind.rectifier, EquivalentBranch, TopologicalIsland, DCTopologicalIsland, Equipment.aggregate, ExternalNetworkInjection.minQ, ExternalNetworkInjection.maxQ, VsPpccControlKind.pPcc, VsPpccControlKind.udc, VsPpccControlKind.pPccAndUdcDroop, VsPpccControlKind.pPccAndUdcDroopWithCompensation, VsPpccControlKind.pPccAndUdcDroopPilot, VsQpccControlKind.reactivePcc, VsQpccControlKind.voltagePcc, VsQpccControlKind.powerFactorPcc, CsPpccControlKind.activePower, CsPpccControlKind.dcVoltage, CsPpccControlKind.dcCurrent, ACDCConverter.maxP, ACDCConverter.minP, VsConverter.maxValveCurrent, OperationalLimitType.acceptableDuration, Terminal.phases, VoltageLevel.highVoltageLimit, VoltageLevel.lowVoltageLimit, VoltageLimit, EquivalentInjection.regulationStatus, EquivalentInjection.regulationCapability, PhaseTapChangerLinear, PhaseTapChangerSymmetrical, PhaseTapChangerLinear.xMax, PhaseTapChangerNonLinear.xMax, PhaseTapChangerTablePoint.angle,

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PhaseTapChangerLinear.stepPhaseShiftIncrement, PhaseTapChangerNonLinear.voltageStepIncrement, RegulatingControl.

Category 3: No impact

- Editorial changes in various notes of classes, attributes, associations, UML packages, UML diagrams and profile specific notes. Such changes are due to proofreading of the documentation.
- Changes of the descriptions of the associations. The reason for such changes is that in previous releases the role notes of associations were inversed (due to profiling issue) compared to the canonical CIM.
- Tagged value IsRdfAbout=True added to classes that are stereotyped “Description”. it is used in the following profiles: SV, SSH, TP and Boundary TP.
- The following clarification statement is added to Domain related diagrams in UML: “For all datatypes both positive and negative values are allowed unless stated otherwise for a particular datatype.”. Former profile notes related to this statement are deleted.
- Annex D.9 of IEC TS 61970-600-1:2017 is included as an annex of IEC 61970-301:2020.
- IEC TS 61970-600-1:2017, Clause E.18 defines rules regarding the exchange of UnitMultiplier. This was integrated in IEC 61970-400 series standards by defining that “For exchange of attributes defined using CIM Data Types (ActivePower, Susceptance, etc.) the UnitMultiplier is 1 if the UnitMultiplier specified in this document is none.” IEC 61970-600-2 also has a constraint (C:600:All:::datatypes) on this.
- All profiles now use Float instead of Simple_Float. The profile rule R:600:ALL:NA:float was added in Clause 16.
- In the UML equipment profile is split in three profiles in an explicit way, i.e. not relying of the stereotypes Operation and ShortCircuit. Stereotypes Operation and ShortCircuit are deleted and profile packages renamed to CoreEquipmentProfile, OperationProfile and ShortCircuitProfile to match IEC 61970-452.
- A flat structure (no subpackages) is applied in each profile package in UML except Dynamics profile. Profile packages are renamed
- Changes in Dynamics profile to align with CIM17
  - EnergyConnection was added as an abstract class and the inheritance path of EnergyConsumer and RegulatingCondEy was modified.
  - Editorial changes in the following descriptions were introduced:
    i) for classes: PowerSystemResource, AsynchronousMachine, ExcRQB
    ii) for association role notes: AsynchronousMachine -> AsynchronousMachineDynamics, EnergyConsumer -> LoadDynamics, StaticVarCompensator -> StaticVarCompensatorDynamics, SynchronousMachine -> SynchronousMachineDynamics, CSConverter -> CSCDynamics, VsConverter -> VSCDynamics
  - The following attributes were changed to optional: SynchronousMachineTimeConstantReactance.xQuadTrans, .tpqo, and SynchronousMachineDetailed.saturationFactor120QAxis, .saturationFactorQAxis and SynchronousMachineTimeConstantReactance.rotorType.
  - Diagram profile nodes were deleted, and profile constraints were added.

A.4 Specific changes

A.4.1 Equipment Boundary profile

Table A.1 summarises the changes on Equipment Boundary profile.

Table A.1 – Changes on Equipment Boundary profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/EquipmentBoundary/3/1	http://iec.ch/TC57/ns/CIM/EquipmentBoundary-EU/3.0	As it is new version.	Low
2	Profile URI	http://entsoe.eu/CIM/EquipmentBoundaryOperation/3/1	-	Not needed anymore as there are no operation specific classes in this profile.	Low
3	Namespace prefix of EnergySchedulingType	entsoe	cim	Issue resolution alignment with IEC 61970-452 and CIM17	Low
4	Namespace prefix of Association EnergySource-EnergySchedulingType	entsoe	cim	Issue resolution alignment with IEC 61970-452 and CIM17	Low
5	Association Terminal – ConnectivityNode		Added with the same cardinality and direction as in the EQ profile.	Issue resolution. Bug fix. Note that tools complaint with regular EQ profile of IEC TS 61970-600:2017 have this implemented for the EQ profile.	Low
6	Substation		Added to the profile including the associations with VoltageLevel and SubGeographicalRegion.	Issue resolution. Bug fix. IEC TS 61970-600:2017 was meant to support boundary point placed in a substation. Note that tools complaint with regular EQ profile of IEC TS 61970-600:2017 have this implemented for the EQ profile.	Low
7	VoltageLevel		Added to the profile including the association with BaseVoltage	Issue resolution. Bug fix. IEC TS 61970-600:2017 was meant to support boundary point placed in a substation. Note that tools complaint with regular EQ profile of IEC TS 61970-600:2017 have this implemented for the EQ profile.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
8	Bay		Added to the profile including the association with VoltageLevel	Issue resolution. Bug fix. IEC TS 61970-600:2017 was meant to support boundary point placed in a substation. Note that tools complaint with regular EQ profile of IEC TS 61970-600:2017 have this implemented for the EQ profile.	Low
9	ConnectivityNode	Profile note: “The following is defined: – If the model is a TSO EQ, the ConnectivityNodes should be grouped under VoltageLevel; – If the model is a Boundary EQ, the ConnectivityNodes should be grouped under Line; – If the model is an assembled one, the ConnectivityNodes can be grouped under either VoltageLevel or Line. With this approach the Line is also in the Boundary set. Instances of ACLineSegment can be in the Boundary set instance of Line or in another instance of Line. Consequently there can be instances of Line that contain only ConnectivityNodes, but no ACLineSegments.”	Profile note deleted.	Issue resolution. Bug fix. IEC TS 61970-600:2017 was meant to support boundary point placed in a substation. Note that tools complaint with regular EQ profile of IEC TS 61970-600:2017 have this implemented for the EQ profile. Boundary profile applies the same constraints as main equipment and topology profiles.	Low
10	Line	Profile note: “Lines in the boundary are only allowed to have ConnectivityNodes and TopologicalNodes”	Profile note deleted.	Issue resolution. Bug fix. NOTE Boundary profile applies the same constraints as main equipment and topology profiles.	Low
11	EnergySource		Inheritance path changed due to insertion of the EnergyConnection as an abstract class. The class does not have any attributes.	Alignment with CIM17.	No
12	Bay	Stereotype: Operation	Remove the stereotype which places this class in the EQ Core profile.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
13	ConnectivityNode	Stereotype: Operation	Remove the stereotype which places this class in the EQ Core profile.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low
14	Association Terminal ConnectivityNode	Cardinality 1..1	0..1.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low
15	BoundaryPoint		New class added to the profile. Attributes that were extending ConnectivityNode were moved to BoundaryPoint. Descriptions of the attributes were updated, and character limitations changed. Two additional attributes were added .isDirectCurrent and .isExcludedFromAreaInterchange.	Issue resolution. CGM requirement.	Moderate
Count:					**1 Moderate, 13 Low, 1 No**

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A.4.2 Equipment profiles: Core Equipment, Operation and ShortCircuit

Table A.2 summarises the changes on Core Equipment, Operation and ShortCircuit profiles.

Table A.2 – Changes on Equipment profile and sub-profiles

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/EquipmentCore/3/1	http://iec.ch/TC57/ns/CIM/CoreEquipment-EU/3.0	As it is new version.	Low
2	Profile URI	http://entsoe.eu/CIM/EquipmentOperation/3/1	http://iec.ch/TC57/ns/CIM/Operation-EU/3.0	As it is new version.	Low
3	Profile URI	http://entsoe.eu/CIM/EquipmentShortCircuit/3/1	http://iec.ch/TC57/ns/CIM/ShortCircuit-EU/3.0	As it is new version.	Low
4	Namespace prefix of EnergySchedulingType	entsoe	cim	Issue resolution alignment with IEC 61970-452 and CIM17.	Moderate
5	Namespace prefix of Association EnergySource-EnergySchedulingType	entsoe	cim	Issue resolution alignment with IEC 61970-452 and CIM17.	Moderate
6	ActivePowerLimit.value	.value	.normalValue Also ActivePowerLimit is in Core Equipment profile instead of Operation	Issue resolution CGM requirement and IEC 61970-452 alignment. Note: ActivePowerLimit.value attribute has been added to SSH profile. With this change exchange of diff EQ for limits is not required anymore.	Low
7	VoltageLimit.value	.value	.normalValue	Issue resolution CGM requirement and IEC 61970-452 alignment. Note: VoltageLimit.value attribute has been added to SSH profile. With this change exchange of diff EQ for limits is not required anymore.	Low
8	CurrentLimit.value	.value	.normalValue	Issue resolution CGM requirement and IEC 61970-452 alignment. Note: CurrentLimit.value attribute has been added to SSH profile. With this change exchange of diff EQ for limits is not required anymore.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
9	ApparentPowerLimit.value	.value	.normalValue Also ApparentPowerLimit is in Core Equipment profile instead of Operation	Issue resolution CGM requirement and IEC 61970-452 alignment. Note: ApparentPowerLimit.value attribute has been added to SSH profile. With this change exchange of diff EQ for limits is not required anymore.	Low
10	EnergySource.pMax	-	Added as an optional attribute	Issue resolution CGM requirement and IEC 61970-452 alignment.	Low
11	EnergySource.pMin	-	Added as an optional attribute	Issue resolution CGM and TYNDP requirement.	Low
12	BatteryUnit	-	Added	Issue resolution TYNDP requirement.	Low
13	PhotoVoltaicUnit	-	Added	Issue resolution TYNDP requirement.	Low
14	PowerElectronicsConnection	-	Added. Optional attributes in Core: .maxQ, .minQ, .ratedS, .ratedU and association with PowerElectronicsUnit	Issue resolution ensure linkage with DY profile and compatibility with IEC 61400-27-1.	Low
15	PowerElectronicsWindUnit	-	Added	Issue resolution ensure linkage with DY profile and compatibility with IEC 61400-27-1.	Low
16	PowerElectronicsUnit	-	Added Optional attributes in Core: .maxP, .minP,	Issue resolution ensure linkage with DY profile and compatibility with IEC 61400-27-1.	Low
17	FuelType	-	Added: brownCoalLignite, coalDerivedGas, peat, other	Issue resolution CGM requirement.	Low
18	HydroGeneratingUnit.dropHeight	-	Added as an optional attribute	Issue resolution CGM and TYNDP requirement.	Low
19	HydroGeneratingUnit.turbineType	-	Added as an optional attribute. Enumeration HydroTurbineKind with attributes francis, pelton, Kaplan is added.	Issue resolution CGM and TYNDP requirement.	Low
20	WindPowerPlant	-	Added as an extension. Association with WindGeneratingUnit added as an extension.	Issue resolution CGM and TYNDP requirement.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
21	SolarPowerPlant	-	Added as an extension. Association with SolarGeneratingUnit added as an extension.	Issue resolution CGM and TYNDP requirement.	Low
22	CogenerationPlant	-	Added including the association to ThermalGeneratingUnit.	Issue resolution CGM and TYNDP requirement.	Low
23	CombinedCyclePlant	-	Added including the association to ThermalGeneratingUnit.	Issue resolution CGM and TYNDP requirement.	Low
24	CAESPlant	-	Added including the association to ThermalGeneratingUnit.	Issue resolution CGM and TYNDP requirement.	Low
25	DCConductingEquipment.ratedUdc	-	Added as a required attribute. .ratedUdc deleted from DCShunt and DCSeriesDevice	Issue resolution. Alignment with CIM17.	Moderate
26	TapChanger		Added in IEC 61970-301 Nominal quantities are not related to the equipment but to the system nominal voltage in the grid. Rated quantities such as ratedU are related to the nameplate data.	Integration of E2.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
27	TapChanger.neutralStep		Description modified with: "…It is the step position where the voltage is neutralU when the other terminals of the transformer are at the ratedU. If there are other tap changers on the transformer those taps are kept constant at their neutralStep…."	Integration of Annex E2.2 from IEC TS 61970-600-1:2017 into IEC 61970-301.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
28	TapChanger.neutralU		Description modified with: "...This attribute is not relevant for PhaseTapChangerAsymmetrical, PhaseTapChangerSymmetrical and PhaseTapChangerLinear...." "...It is the voltage at the terminal of the PowerTransformerEnd associated with the tap changer when all tap changers on the transformer are at their neutralStep position. Normally neutralU of the tap changer is the same as ratedU of the PowerTransformerEnd, but it can differ in special cases such as when the tapping mechanism is separate from the winding more common on lower voltage transformers...."	Integration of E2.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
29	PhaseTapChangerNonLinear.voltageStepIncrement		Description modified with: "....specified in percent of rated voltage...."	Integration of E2.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
30	RatioTapChanger		Description modified with: "...Angle sign convention (general): Positive value indicates a positive phase shift from the winding where the tap is located to the other winding (for a two-winding transformer)...."	Integration of D.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
31	RegulatingControl		Description modified with: "...The specified terminal shall be associated with the connectivity node of the controlled point...."	Integration of E.12 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
32	TapChanger		Profile constraint added: C:600:EQ:TapChanger.neutralU:ValueRangePair: The TapChanger.neutralU shall be the same as PowerTransformerEnd.ratedU.	Integration of E2.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
33	ACLineSegment		IEC TS 61970-600-1:2017 Section E10.2 imposes rules related to ACLineSegment. The following profile constraint (C:452:EQ:ACLineSegment:BaseVoltage) was added: “All implementations shall use association to a BaseVoltage for the purpose of any per unit calculations and shall not rely on the voltages (neither nominal nor actual values obtained by previous or current solution) at the nodes, which the ACLineSegment connects to. In case there are interconnected ACLineSegments with different BaseVoltage for different parts of the networks (when assembling different model authority sets) the application needs to handle this to ensure accurate physical units.”	Integration of E10.2 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
34	PowerTransformerEnd		Description modified with: “…Parameters are always provided, even if the PowerTransformerEnds have the same rated voltage. In this case, the parameters are provided at the PowerTransformerEnd which has TransformerEnd.endNumber equal to 1.….”	Integration of E15 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
35	EnergySource.voltageAngle		Removed from the profile. Note that it is added to SSH profile.	Integration of E19 of IEC TS 61970-600-1:2017 into IEC 61970-456.	No
36	EnergySource.voltageMagnitude		Removed from the profile. Note that it is added to SSH profile.	Integration of E19 of IEC TS 61970-600-1:2017 into IEC 61970-456.	No
37	GeneratingUnit		Deleted the following profile note ”GeneratingUnit.normalPF is used for representing distributed slack participation factor.” as 1st it does not belong to EQ and 2nd it is added to .normalPF in CIM17	Alignment with CIM17. Integration of Subclause 6.6.5 of IEC TS 61970-600-1:2017 into IEC 61970-301 (description of GeneratingUnit.normalPF which is in SSH).	No

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No	Item	Changed From	Changed To	Reason	Impact classification
38	StaticVarCompensator		Deleted the following profile note: “-The value of the inductiveRating attribute must always be negative. The description of the inductiveRating should be understood as – Inductive reactance at maximum inductive reactive power. -The value of the capactiveRating attribute must always be positive. The description of the capacitiveRating should be understood as – Capacitive reactance at maximum capacitive reactive power.”	Issue resolution. Alignment with CIM17. Integration of Subclause 6.10.44 of IEC TS 61970-600-1:2017 into IEC 61970-301. Integrated in description of StaticVarCompensator.inductiveRating and .capactiveRating.	No
39	PowerTransformer		Deleted profile note “PowerTransformer.isPartOfGenerating Unit is related to the IEC 60909. It has an impact on how the correction factors are calculated for transformers, since the transformer is not necessarily part of a generating unit, while a generator typically is. It is not always possible to derive this information from the model. This is why the attribute is necessary.”	Issue resolution. Alignment with CIM17. Integrated in description of PowerTransformer.isPartOfGeneratingUnit.	No
40	CsConverter		Delete profile note “For CSC HVDC the transformer must be modelled explicitly. The way to validate this is not resolved.”	Issue resolution. Integrated in IEC 61970-301.	No
41	PowerTransformerEnd		Delete profile note “PowerTransformerEnd.x shall be consistent with PhaseTapChangerLinear.xMin and PhaseTapChangerNonLinear.xMin. In case of inconsistency, PowerTransformerEnd.x shall be used. PhaseTapChangerLinear.xMin and PhaseTapChangerNonLinear.xMin will be removed in future versions of CIM.”	Issue resolution. Alignment with CIM17. Integrated in description of PhaseTapChangerLinear.xMin and PhaseTapChangerNonLinear.xMin.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
42	VoltageLevel		Delete profile note “DCConductingEquipment and DCConverterUnit is not allowed in VoltageLevels that is intended for AC equipment only. DCConductingEquipment are allowed in DCEquipmentContainers only. The ACDCConverter is ConductingEquipment. Other AC equipment in a converter unit are the power transformer and converter reactor. All three shall be located in a DCConverterUnit and not a VoltageLevel.”	Integration of Subclause 6.7.22 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
43	GeneratingUnit		Delete profile note “The control status of the unit is defined with the attribute “genControlSource”, but it is not required. The participation factor attributes “longPF”, “normalPF”, and “shortPF” are not required in 61970-452. For ENTSO-E "normalPF" is used as defined in the separate Note to the GeneratingUnit.”	Integration of Subclause 6.6.5 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
44	Measurement		Delete profile note “Measurement.phases: if the attribute is not specified the Measurement is understood as three phase (ABC).”	Integration of Subclause 6.5.18 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
45	BaseVoltage		Delete profile note “nominalVoltage must be a positive value and not zero.”	Integration of Subclause 6.7.3 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
46	EnergyConsumer		Delete profile note “Attributes (pfixed, qfixed, pfixedPct and qfixedPct) are used for load allocation. Attributes (pfixed and qfixed) represent base load, while attributes (pfixedPct and qfixedPct) represent the time-varying components”	Integration of Subclause 6.10.10 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No

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No	Item	Changed From	Changed To	Reason	Impact classification
47	PowerTransformerEnd		Delete profile note "Each PowerTransformerEnd must be contained by a PowerTransformer. Because a PowerTransformerEnd (or any other object) cannot be contained by more than one parent, a PowerTransformerEnd cannot have an association to an EquipmentContainer (Substation, VoltageLevel, etc)."	Integration of Subclause 6.10.32 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
48	ExternalNetworkInjection		Delete profile note "The attributes ikSecond and voltageFactor are optional attributes even if short circuit data is exchanged. These attributes are used only if short circuit calculations are done according to superposition method."	Integration of Subclause 6.10.11 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No

No	Item	Changed From	Changed To	Reason	Impact classification
49	EarthFaultCompensator		Delete profile note “The GroundingImpedance and PetersenCoil connect the same way as other devices using Terminal. The phases attribute on Terminal indicates the connection is at the neutral phase. If a PowerTransformer has a Y connection on one terminal (secondary) and in case you wish to connect a PetersenCoil. The PowerTransformer's secondary Terminal where the coil is connected should be specified with phases attribute equal to ABCN and the PetersenCoil's Terminal should be specified with phase N. This allows the ACLineSegment to connect with phases ABC (the default) and the PetersenCoil to connect to the same TransformerTerminal but only on the N phase. It is expected all EarthFaultCompensators (GroundingImpedance and PetersenCoil) to have only one Terminal and be solidly connected to ground on the other unmodelled terminal. It is expected all PetersenCoil terminals to have only phase N. Basically a normal modelling of non-neutral network is done then add to the Petersen coil model which includes: 1) Adding the phasing of ABCN to the transformer secondary terminal so you have a place to connect neutral 2) Adding the PetersenCoil object with its associated Terminal, which is phase N, since only one Terminal is modelled the PetersenCoil is assumed solidly grounded 3) Associate the PetersenCoil's single Terminal to the ConnectivityNode at transformer secondary.”	Integration of Subclause 6.10.9 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
50	Season		Delete profile note "To specify a relative date as the startDate or endDate for a Season, the year component of the ISO 8601 date format (YYYY-MM-DD) can be omitted. The resulting format would be MM-DD."	Integration of Subclause 6.11.13 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
51	ControlArea		Delete profile note "In case the ContolArea is used for exchanging interchange values for a model the ControlArea is defined on the basis of the terminals on the border of the control area. Therefore, all generation and load within the area defined by the terminals on the border are considered in the area interchange control."	Integration of Subclause 6.13.2 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
52	SynchronousMachine.r0, .r2, .x0, .x2	Datatypes: PU	Datatypes: .r0 and .r2 changed to Resistance; .x0 and .x2 changed to Reactance	Alignment with CIM17. Bug fix.	Low

No	Item	Changed From	Changed To	Reason	Impact classification
53	DCConductingEqiopment, ACDCConverter, DCLine, DCConverterUnit		Delete profile note “The ACDCConverter is ConductingEquipment and hence may be located in any of the AC EquipmentContainers (Substation, VoltageLevel…). AC equipment in a converter unit are power transformer, converter reactor (a SeriesCompensator) and Switches. According to the UML DCNodes may only be contained by a DCEquipmentContainer. Hence it is not possible to describe DC connectivity outside a DCEquipmentContainer. The containment rules for DC related equipment is as follows: 1) DCConductingEquipment are allowed in DCEquipmentContainers only. 2) A DCConverterUnit shall be contained by a Substation. 3) A DCLine shall have no superior container. 4) All AC equipment related to a converter shall be located in a DCConverterUnit, not a VoltageLevel. 5) ACDCConverters can only be located in DCConverterUnit.”	Integration of 6.3.2, 6.3.9, 6.3.10, 6.3.14 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
54	DCConductingEquipment	Concrete class	Abstract class	Bug fix. Alignment with IEC 61970-452.	No
55	EnergySource, EnergyConsumer, RegulatingCondEq		Inheritance path changed due to insertion of the EnergyConnection as an abstract class. The class does not have any attributes.	Alignment with CIM17.	No
56	Control, MeasurementValue		Inheritance path changed due to insertion of the IOPoint as an abstract class. The class does not have any attributes.	Alignment with CIM17.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
57	PhaseCode		The following additional attributes: none, X, XY,XN,XYN are added.	Alignment with CIM17.	Low
58	ActivePowerPerCurrentFlow		Deleted denominatorUnit and denominatorMultiplier. The resulting datatype is MWPerA.	Alignment with CIM17.	No
59	PerLengthDCLineParameter		Deleted the class and related datatypes CapacitancePerLength, InductancePerLength, ResistancePerLength	Alignment with CIM17 and IEC 61970-452.	No
60	RotationSpeed	1/s	Deleted denominatorUnit and denominatorMultiplier. The resulting datatype is Hz.	Alignment with CIM17.	No
61	DCLineSegment.capacitance	Cardinality: 0..1	Cardinality: 1..1	Issue resolution. Minimising inconsistency.	Low
62	DCLineSegment.inductance	Cardinality: 0..1	Cardinality: 1..1	Issue resolution. Minimising inconsistency.	Low
63	DCLineSegment.resistance	Cardinality: 0..1	Cardinality: 1..1	Issue resolution. Minimising inconsistency.	Low
64	Ground	Stereotypes: ShortCircuit and Operation	Remove the stereotypes which places this class in the EQ Core profile.	Issue resolution.	Low
65	GeneratingUnit		GeneratingUnit.startupTime added as an optional attribute.	Issue resolution to address CGM requirement: "TSOs shall provide the redispatch potential in both directions for generators that can be started within two hours for the D-1 scenarios".	Low
66	GrossToNetActivePowerCurve	Stereotype: Operation	Stereotype removed as the class should be in Core profile of Equipment profile	Issue resolution. Alignment between profiles.	Low
67	CurveData	Cardinality of association with Curve: 0..*	Changed to 1..*	Issue resolution.	Low
68	SubLoadArea	Stereotype: Operation	Stereotype removed as the class should be in Core profile of Equipment profile	Issue resolution. Alignment between profiles	Low
78	LoadArea	Stereotype: Operation	Stereotype removed as the class should be in Core profile of Equipment profile	Issue resolution. Alignment between profiles	Low
79	Terminal	Association with RegulatingControl, cardinality: 0..1	The cardinality changed to 0..* at the side of RegulatingControl	Issue resolution alignment with IEC 61970-452.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
80	LimitTypeKind		The enumeration LimitTypeKind is extended with operationalVoltageLimit, alarmVoltage, warningVoltage, stability. The class is renamed to LimitKind. The attribute OperationalLimitType.limitType is renamed to kind.	Issue resolution. CGM requirement. Naming convention in CIM modelling guidelines.	Low
81	DisconnectingCircuitBreaker		Added to the Core profile.	Issue resolution. CGM requirement.	Low
82	Fuse		Added to the profile	Issue resolution.	Low
83	Jumper		Added to the profile	Issue resolution.	Low
84	EnergySource		Attributes: r, r0, rn, x, xo, xn stereotyped with ShortCircuit	Issue resolution. Alignment with IEC 61970-452.	Low
85	PowerTransformerEnd		The following profile constraint (C:452:EQ:PowerTransformerEnd:pu) is added: The parameters r, x, g and b are specified for each end and are not related to the overall base voltage. These values are specified in engineering units. Any PU calculations are internal to particular tools and are not part of the data exchange.	Issue resolution.	Low
86	PowerTransformerEnd		The following profile constraint (C:452:EQ:PowerTransformerEnd:instance) is added: Negative reactance values are valid for transformers.	Issue resolution.	Low
87	ACLineSegment		The following profile constraint is integrated in C:452:EQ:ACLineSegment.x:value: As negative reactance values are not allowed for ACLineSegment-s it is recommended to model series compensators explicitly..	Issue resolution.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
88	RegulatingControl	Profile note: RegulatingControl targetrange and targetvalue are required if a RegulationSchedule is not provided	The note is deleted.	Issue resolution.	Low
89	ACDCTerminal.sequenceNumber	Cardinality: 0..1	Cardinality: 1..1 The following related profile note is deleted. “ACLineSegment in case of MutualCoupling and for EquivalentBranch. The attribute sequenceNumber is optional in the profile for all other classes, but it is recommended to exchange the sequenceNumber.”	Alignment with IEC 61970-452.	Low
90	PhaseTapChangerLinear.xMin, PhaseTapChangerNonLinear.xMin		These attributes are stereotyped “deprecated” as they are marked as deprecated in CIM17. They will be deleted in next CGMES release.	Alignment with IEC 61970-301.	No
91	RatioTapChanger.tculControlMode		The deprecated attribute is deleted including the enumeration TransformerControlMode.	Alignment with IEC 61970-452.	Low
92	PowerTransformer		The following profile note is deleted “Association between PowerTransformer and BaseVoltage is not exchanged.”	Integration of 6.10.31 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17)	No
93	Datatype ActivePowerPerFrequency	MW/Hz	Wpers (Watt per second)	Alignment with IEC 61970-301.	Low
94	Datatype VoltagePerReactivePower	kV/Mvar	VperVar (Volt per Var)	Alignment with IEC 61970-301.	Low
95	Datatype Money	EUR	Not fixed to EUR	In order to allow multiple currencies to be used.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
96	IdentifiedObject	Profile note	The following profile note is deleted: “The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.”	This is already specified in IEC TS 61970-600-1 as a general rule.	No
97	Clamp and Cut		The classes Clamp and Cur including the associations to ACLineSegment are added to the EQ Core profile.	Alignment with IEC 61970-452.	Low
98	Association ControlArea.EnergyArea		The following profile note is deleted:” ControlArea.EnergyArea is required for Operation, not for Planning (bus-branch model)”. The association is set as required in EQ Core profile.	The note is implemented in UML by changing the cardinality of the association.	No
99	CurveData		The following profile note is deleted: “The CurveData class is used to represent points for various curves that derive from the Curve class. The curves defined in this profile are: • GrossToNetActivePowerCurve • ReactiveCapabilityCurve.”	The note is considered not relevant anymore.	No
100	Terminal		The following profile note is deleted: “Each ConductingEquipment could have Terminals. All ConductingEquipment is associated to two Terminals, 1634 except PowerTransformer that could have 2 ou 3 Terminals, and BusBarSection that has only one Terminal.”	The note is considered not relevant anymore.	No
101	Association PhaseTapChangerTabular-PhaseTapChangerTable	Cardinality 0..1	1 (required)	Alignment with IEC 61970-452. It does not make sense to have tabular exchange without a table.	No
102	Association AccumulatorLimitSet-Accumulator	Cardinality 0..* at Measurements side	1..*	Alignment with IEC 61970-452. It does not make sense to have AccumulatorLimitSet without Accumulator.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
103	Association AnalogLimitSet-Analog	Cardinality 0..* at Measurements side	1..*	Alignment with IEC 61970-452. It does not make sense to have AnalogLimitSet without Analog.	No
104	GroundDisconnector	Stereotypes: ShortCircuit and Operation	Remove the stereotypes which places this class in the EQ Core profile.	Issue resolution.	Low
105	Bay	Stereotype: Operation	Remove the stereotype which places this class in the EQ Core profile.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low
106	ConnectivityNode	Stereotype: Operation	Remove the stereotype which places this class in the EQ Core profile.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low
107	Association Terminal ConnectivityNode	Cardinality 1..1	0..1.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation. Also related to integration of E17 of IEC TS 61970-600-1:2017 into this document.	Low
108	PetersenCoil, GroundingImpedance and EarthFaultCompensator	Stereotype. ShortCircuit	Stereotype of the class deleted but all attributes stereotyped with ShortCurcuit and placed in EQ ShortCircuit profile.	Necessary to manage the split of the equipment profile. All equipment shall be in EQ core.	No
109	StationSupply	Stereotype. ShortCircuit	Stereotype removed, which places the class in EQ core profile.	Necessary to manage the split of the equipment profile. All equipment shall be in EQ core.	Low
110	NonConformLoadSchedule		Class placed in Operation profile.	Necessary to manage the split of the equipment profile. All schedule related classes shall be in EQ operation profile.	Low
111	ConformLoadSchedule		Class placed in Operation profile.	Necessary to manage the split of the equipment profile. All schedule related classes shall be in EQ operation profile.	Low
112	AuxiliaryEquipment		AuxiliaryEquipment and Sensor added in EQ as abstract. CurrentTransformer, PotentialTransformer, WaveTrap, PostLineSensor, SurgeArrester, FaultIndicator in added in EQ as concrete. The association AuxiliaryEquipment to Terminal added in EQ core profile.	Issue resolution.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
113	GeneratingUnit.initialP		This required attribute is deleted as there is a general agreement that it is not expected to be able to run power flow on the EQ alone. SSH is required to be able to execute power flow. Therefore the following note of GeneratingUnit in EQ is deleted: To define a GeneratingUnit requires defining the initial real power injection, net real power limits, and the status of the unit. The initial injection is defined using the attribute “initialP”.	Issue resolution. The role of the attribute is fulfilled by and attribute in SSH profile.	Low
114	OperationalLimitType	Profile note: “Since the terminal assignment is always explicitly exchanged as OperationalLimitSet.Terminal and given that the exchange of a Terminal.sequenceNumber is not necessary.”	Profile note deleted.	Fix of inconsistency as Terminal.sequenceNumber is required attribute.	No
115	ControlArea	Profile note: “In case multiple generators all these GeneratingUnit(s) have non-zero normalPF, but there must be one GeneratingUnit per control area that have maximum participation factor (GeneratingUnit.normalPF)”	Profile note deleted. Profile constraint (C:456:SSH:GeneratingUnit::SlackRules) added.	Clarify that the maximum is the highest among other participation factors.	No
116	ShuntCompensator		The attributes ShuntCompensator.switchOnCount and ShuntCompensator.switchOnDate are deleted.	Alignment with IEC 61970-452. The attributes were not used and were not consistent with similar attributes on Switch, which are neither in IEC 61970-452 nor in CGMES.	Low
117	BaterryUnit.ratedE		Added to EQ core as a required attribute. RealEnergy datatype was added to the profile with multiplier M.	Alignment with IEC 61970-452. Support of battery unit enhanced.	Low
118	EnergyArea		The class is placed in the Core Equipment profile	Alignment with IEC 61970-452.	Low
119	ACDCConverter		maxP and minP attributes added.	Issue resolution. Enable explicit modelling of limits for HVDC converters. Alignment with CIM17.	Low
120	VsConverter		Attribute maxValveCurrent deleted.	Issue resolution. Alignment with CIM17.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
121	OperationalLimitType		Attribute isInfiniteDuration is added as required attribute.	Issue resolution. CGM requirement. Alignment with CIM17.	Low
122	LimitKind.patl		The description of the attribute is modified to include constraint on the usage and consider the new attribute OperationalLimitType.isInfiniteDuration	Issue resolution. CGM requirement.	Low
123	ACDCConverterDCTerminal		Set the attribute .polarity to required.	Issue resolution. Improvements related to HVDC modelling.	Low
124	RegulatingControl, TapChangerControl		Change the cardinalities of RegulatingControl[0..1]-[0..*] RegulatingCondEq and TapChangerControl[0..1]-[0..*] TapChanger from 0..* to 1..*.	Issue resolution. CGM requirement. Alignment with CIM17.	Low
125	StaticVarCompensator		The attributes StaticVarCompensator.sVCControlMode , StaticVarCompensator.voltageSetPoint and the enumeration SVCControlMode are deprecated in this edition as they duplicate RegulatingControl hence shall not be used.	Issue resolution. CGM requirement.	Low
126	Equipment		The attribute .normallyInService is added to the EQ Core profile as optional attribute.	Alignment with IEC 61970-452.	Low
127	RegulatingControl.RegulatingCondEq	Cardinality 1..*	Cardinality 0..*.	Implementation of country comments related to this document.	Low
128	OperationalLimitType.direction	Cardinality 0..1.	Cardinality 1..1.	Implementation of country comments related to this document.	Low
129	ActivePower and ReactivePower		Datatypes deleted in OP profile	Implementation of country comments related to this document.	Low
130	MonthDay (Primitive)		Moved to core equipment profile	Implementation of country comments related to this document.	Low
131	Seconds (datatype)		Deleted in OP profile	Implementation of country comments related to this document.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
132	LoadGroup, NonConformLoadGroup, ConformLoadGroup, RegulatingControl, Switch, TapChanger, StationSupply, ConformLoad, NonConformLoad, EnergyConnection, ConductingEquipment, Equipment		Deleted in OP profile. These classes are already in the EQ core.	Implementation of country comments related to this document.	Low
133	EnergyConsumer.pfixed, .pfixedPct, .qfixed, .qfixedPct		Added to equipment profile and deleted the class in OP profile.	Implementation of country comments related to this document.	Low
134	SeasonDayTypeSchedule, Season, DayType, TapSchedule, RegulationSchedule, SwitchSchedule, ConformLoadSchedule, NonConformLoadSchedule, BasicIntervalSchedule, RegularIntervalSchedule, RegularTimePoint		These classes are moved to EQ. Associations as well.	Implementation of country comments related to this document.	Low
135	Attributes shortName and energyIdentCodeEic		Deleted in OP profile	Implementation of country comments related to this document.	Low
136	version class		Created the configuration with ontology (version) classes and populate the initial values in each profile package.	Implementation of country comments related to this document.	Low
137	AccumulatorValue, AnalogValue, DiscreteValue, and StringMeasurementValue		Removed the value attribute.	Implementation of country comments related to this document.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
138	Series compensator		3 attributes: (SeriesCompensator.varistorPresent, SeriesCompensator. varistorVoltageThreshold, SeriesCompensator.varistorRatedCurrent) were moved to SC.	Implementation of country comments related to this document.	Low
139	StaticVarCompensator. sVCControlMode and .voltageSetPoint		Made StaticVarCompensator.sVCControlMode and .voltageSetPoint optional	Implementation of country comments related to this document.	Low
140	OperationalLimitSet.Terminal	Cardinality 0..1.	Cardinality 1..1.	Implementation of country comments related to this document.	Low
Count:					**3 Moderate, 83 Low, 45 No**

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A.4.3 Topology profile

Table A.3 summarises the changes on Topology profile.

Table A.3 – Changes on Topology profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/Topology/4/1	http://iec.ch/TC57/ns/CIM/Topology-EU/3.0	It is a new version.	Low
2	IdentifiedObject	Profile note	The following profile note is deleted: “The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.”	This is already specified in IEC TS 61970-600-1 as a general rule.	No
3	BaseVoltage	Profile note	The following profile note is deleted: “BaseVoltage is an abstract class. It is used here to generate the TopologicalNode.BaseVoltage association that is a pointer to BaseVoltage concrete class in the Equipment Profile”	This note is not necessary as the class is marked as abstract.	No
4	ConnectivityNodeContainer	Profile note	The following profile note is deleted: “ConnectivityNodeContainer is an abstract class. It is used here to generate the TopologicalNode.ConnectivityNodeContainer association that is a pointer to ConnectivityNodeContainer concrete subclasses in the Equipment Profile”	This note is not necessary as the class is marked as abstract.	No
5	Association ConnectivityNode TopologicalNode	Cardinality: 0..1	1 (required)	Issue resolution. Alignment with IEC 61970-456. Enables better handling of hybrid bus branch and node breaker modelling representation. Also related to Integration of E17 of IEC TS 61970-600-1:2017 into this document.	Low
6	ConnectivityNode	Stereotype: Operation	Remove the stereotype which places this class in the EQ Core profile.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation.	Low

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No	Item	Changed From	Changed To	Reason	Impact classification
7	Association DCNode DCTopologicalNode	Cardinality: 0..1	1 (required)	Issue resolution. Alignment with IEC 61970-456.	Low
8	Association DCTopologicalNode DCEquipmentContainer	Cardinality: 0..1	1 (required)	Issue resolution. Alignment with IEC 61970-456.	Low
9	Association Terminal.TopologicalNode	Cardinality: 1	0..1 (optional) at the side of TopologicalNode and from 0..* to 1..* at the side of Terminal.	Issue resolution. Enables better handling of hybrid bus branch and node breaker modelling representation. Also relates to Integration of E17 of IEC TS 61970-600-1:2017 into this document. See also rule EXCH8 in IEC 61970-600-1.	Low
10	Asociation BoundaryPoint-TopologicalNode		Deleted BoundaryPoint and the association to TopologicalNode in TP profile.	Alignment with IEC 61970-301:2020. Implementation of country comments related to this document.	Low
11	BoundaryPoint.toEndName	Profile note	Fixed the description of BoundaryPoint.toEndName to refer to 64 characters instead of 32	Alignment with IEC 61970-301:2020. Implementation of country comments related to this document.	Low
12	BoundaryPoint		Made BoundaryPoint inherit from PowerSystemResource	Alignment with IEC 61970-301:2020. Implementation of country comments related to this document.	Low
Count:					**9 Low, 3 No**

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A.4.4 Steady State Hypothesis profile

Table A.4 summarises the changes Steady State Hypothesis profile.

Table A.4 – Changes on Steady State Hypothesis profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/SteadyStateHypothesis/1/1	http://iec.ch/TC57/ns/CIM/SteadyStateHypothesis-EU/3.0	It is a new version.	Low
2	Equipment.inService	-	Added as a required attribute	Issue resolution CGM requirement.	Low
3	ActivePowerLimit.value	-	Added as a required attribute. Stereotype “Description” added.	Issue resolution CGM requirement and IEC 61970-456 alignment.	Low
4	VoltageLimit.value	-	Added as a required attribute. Stereotype “Description” added.	Issue resolution CGM requirement and IEC 61970-456 alignment.	Low
5	CurrentLimit.value	-	Added as a required attribute. Stereotype “Description” added.	Issue resolution CGM requirement and IEC 61970-456 alignment.	Low
6	ApparentPowerLimit.value	-	Added as a required attribute. Stereotype “Description” added.	Issue resolution CGM requirement and IEC 61970-456 alignment.	Low
7	PowerElectronicsConnection.p and .q	-	Added as required attributes	Issue resolution ensure linkage with DY profile and compatibility with IEC 61400-27-1.	Low
8	GeneratingUnit.normalPF		Description modified with: The sum of the participation factors across generating units does not have to sum to one. It is used for representing distributed slack participation factor.	Integration of E14 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
9	GeneratingUnit.normalPF		Profile constraint related to normalPF and slack (C:456:SSH:GeneratingUnit::SlackRules) defined	Integration of E14 of IEC TS 61970-600-1:2017 into this document.	No
10	EnergySource.voltageAngle		Added to the profile. Note that it is removed from EQ profile.	Integration of E19 of IEC TS 61970-600-1:2017 into this document.	No
11	EnergySource.voltageMagnitude		Added to the profile. Note that it is removed from EQ profile.	Integration of E19 of IEC TS 61970-600-1:2017 into this document.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
12	EnergySource.voltageAngle	Description: Phase angle of a-phase open circuit.	Description: Phase angle of a-phase open circuit used when voltage characteristics need to be imposed at the node associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network.	Integration of E19 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
13	EnergySource.voltageMagnitude	Description: Phase-to-phase open circuit voltage magnitude.	Description: Phase-to-phase open circuit voltage magnitude used when voltage characteristics need to be imposed at the node associated with the terminal of the energy source, such as when voltages and angles from the transmission level are used as input to the distribution network.	Integration of E19 of IEC TS 61970-600-1:2017 into IEC 61970-301.	No
14	ACDCConverter		Deleted profile note “Note that for ACDCConverter.targetPpcc load sign convention is used, i.e. positive sign means flow out from a node.”	Issue resolution. Integration of Subclause 7.5.2 of IEC TS 61970-600-1:2017 into IEC 61970-301 (CIM17 description of ADDCConverter.targetPpcc).	No
15	VsConverter		Deleted profile note “Note that for ACDCConverter.targetPpcc load sign convention is used, i.e. positive sign means flow out from a node. Note that for VsConverter.targetQpcc load sign convention is used, i.e. positive sign means flow out from a node.”	Issue resolution. Integration of Subclause 7.5.7 of IEC TS 61970-600-1:2017 into IEC 61970-301 (CIM17 description of VsConverter.targetQpcc).	No

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No	Item	Changed From	Changed To	Reason	Impact classification
16	RegulatingControl		Deleted profile note “Note that RegulatingControl.targetDeadband is primarily used if the RegulatingControl.discrete is set to "true". Tools should handle cases in which RegulatingControl.targetDeadband has a value if RegulatingControl.discrete is set to "false" or cases in which RegulatingControl.targetDeadband equals zero. Note that for instance, if the targetValue is 100 kV and the targetDeadband is 2 kV the range is from 99 to 101 kV.”	Integration of Subclause 7.8.26 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
17	EnergySource, EnergyConsumer, RegulatingCondEq		Inheritance path changed due to insertion of the EnergyConnection as an abstract class. The class does not have any attributes.	Alignment with CIM17.	No
18	DisconnectingCircuitBreaker, Switch		DisconnectingCircuitBreaker added. Attribute Switch.locked added to the profile as required attribute.	Issue resolution. CGM requirement.	Low
19	Fuse		Added to the profile	Issue resolution.	Low
20	Jumper		Added to the profile	Issue resolution.	Low
21	RegulatingControl		Attributes maxAllowedTargetValue and minAllowedTargetValue added as optional attributes. The description of RegulatingControl is updated.	Issue resolution. CGM requirement.	Low
22	Equipment.inService		The description of the attribute is updated..	Issue resolution.	Low
23	GroundDisconnector		Remove stereotype Operation and ShortCircuit	Profile fix. Only Description stereotype is used in SSH profile.	No
24	StationSupply		Remove stereotype Operation	Profile fix. Only Description stereotype is used in SSH profile.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
25	IdentifiedObject		Deleted identifiedObject.name attribute Deleted the following note: The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model. Deleted the following note: IdentifiedObject.name in the SSH profile is not used for classes marked with the stereotype "Description".	Issue resolution. Alignment with IEC 61970-456. Name is exchanged in EQ core and does not need to be exchanged in SSH profile.	Low
26	BaterryUnit		PowerElectronicsUnit added as abstract class. BaterryUnit added as a concrete class with the following required attributes: batteryState and .storedE. RealEnergy datatype and enumeration BatteryStateKind also added.	Alignment with IEC 61970-456. Support of battery unit enhanced.	Low
27	VsPpccControlKind		Attribute phasePcc added.	Issue resolution. Enable better modelling of HVDC control modes. Alignment with CIM17.	Low
28	VsQpccControlKind		Attribute pulseWidthModulation added.	Issue resolution. Enable better modelling of HVDC control modes. Alignment with CIM17.	Low
29	VsConverter		The following attributes were added: . targetPowerFactorPcc, .targetPhasePcc , .targetPWMfactor.	Issue resolution. Enable better modelling of HVDC control modes. Alignment with CIM17.	Low
30	ACDCConverter, VsConverter, CsConverter		The following attributes were set to optional: ACDCConverter.targetPpcc, and .targetUdc VsConverter. targetQpcc, targetUpcc, droop, droopCompensation, .qShare CsConverter.targetIdc, .targetAlpha, .targetGamma	Issue resolution. Enable better modelling of HVDC control modes. Alignment with CIM17.	Low
Count:					**18 Low, 12 No**

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A.4.5 State Variables profile

Table A.5 summarises the changes on State Variables profile.

Table A.5 – Changes on State Variables profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/StateVariables/4/1	http://iec.ch/TC57/ns/CIM/StateVariable-EU/3.0	It is a new version.	Low
2	TopologicalIsland		Profile constraint (C:456:SV:TopologicalIsland::instance) added.	Integration of E.6 of IEC TS 61970-600-1:2017 into this document.	No
3	SvStatus		Deleted the note “The description of SvStatus.inService should be understood as "It indicated if the equipment is energized by the power flow. It reflects if the equipment is connected within a solvable island".”	Issue resolution. Alignment with CIM17. Integrated in description of SvStatus.inService.	No
4	VsConverter.uf		The attribute is renamed to .uv.	Alignment with CIM17.The change occurred in CIM16 however it was agreed to keep old name for IEC TS 61970-600-2:2017. The change occurred due to alignment of CIM with IEC HVDC related standards.	Low
5	IdentifiedObject	Profile note	The following profile note is deleted: “The attribute “name” inherited by many classes from the abstract class IdentifiedObject is not required to be unique. Software developers should not count on this to link the power system model.”	This is already specified in IEC TS 61970-600-1 as a general rule.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
6	SvPowerFlow	Profile note: “SvPowerFlow class is required to be instantiated for the following classes: subclasses of the RotatingMachine, subclasses of the EnergyConsumer, EquivalentInjection, ShuntCompensator, StaticVarCompensator and EnergySource.”	Profile constraint (C:456:SV:SvPowerFlow::instance) added: “SvPowerFlow class is required to be instantiated for the following classes in case the equipment is energized: subclasses of the RotatingMachine, EnergyConsumer and its subclasses, EquivalentInjection, ShuntCompensator, StaticVarCompensator and EnergySource. Applications shall support export of SvPowerFlow for all Terminals (except for BusbarSection) for all other elements of the model in order to be able to exchange a complete solved model.”	Include EnergyConsumer as it is allowed to instantiate EnergyConsumer.	No
7	SvStatus		The stereotype Operation removed.	The SV profile does not distinguish operation profile.	No
8	SvSwitch		Added to the profile with required attribute .open.	Issue resolution. CGM requirement related to the usage of RequlatingControl.	Low
Count:					**3 Low, 5 No**

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A.4.6 Diagram Layout profile

Table A.6 summarises the changes on Diagram Layout profile.

Table A.6 – Changes on Diagram Layout profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/DiagramLayout/3/1	http://iec.ch/TC57/ns/CIM/DiagramLayout-EU/3.0	It is a new version.	Low
2	DiagramObject		Deleted profile note "DiagramObject.rotation=0 has the following meaning: The connection point of an element which has one terminal is pointing to the top side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the top side of the diagram. DiagramObject.rotation=90 has the following meaning: The connection point of an element which has one terminal is pointing to the right hand side of the diagram. The connection point "From side" of an element which has more than one terminal is pointing to the right hand side of the diagram."	Integration of Subclause10.3.4 of IEC TS 61970-600-1:2017 into IEC 61970-301 (added to CIM17).	No
3	OrientationKind		Added .positive	Alignment with IEC 61970-453.	Low
4	VisibilityLayer.VisibleObjects	0..*	1..*	Alignment with IEC 61970-453.	Low
5	IdentifiedObject		The attribute description is added.	Alignment with IEC 61970-453.	Low
6	Diagram		The descriptions of Diagram and Diagram.orientation is modified.	Alignment with IEC 61970-453.	No
7	VisibilityLayer		The description is modified.	Alignment with IEC 61970-453.	No
Count:					**4 Low, 3 No**

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A.4.7 Geographical Location profile

Table A.7 summarises the changes on Geographical Location profile.

Table A.7 – Changes on Geographical Location profile

No	Item	Changed From	Changed To	Reason	Impact classification
1	Profile URI	http://entsoe.eu/CIM/GeographicalLocation/2/1	http://iec.ch/TC57/ns/CIM/GeographicalLocation-EU/3.0	It is a new version.	Low
2	Association CoordinateSystem – Location	Source role name: Location	Source role name: Locations	Alignment with CIM17.	Low
3	PositionPoint		The following profile constraint (C:13:GL:PositionPoint::coordinates) is added to explain position attributes: “In order to support various coordinate systems, position attributes are defined as strings. It is up to the implementation to ensure to correctly parse coordinate positions as defined by the coordinate system type.”	Clarification constraint needed due to an issue request to change the datatypes of the position related attributes.	No
4	WorkLocation		Class added as an abstract class.	Alignment with IEC 61968-13.	Low
5	ServiceLocation		Class added as a concrete class with no attributes.	Alignment with IEC 61968-13.	Low
6	PositionPoint		The following profile constraint (C:600:GL:::position) is added: IEC 61968-13 allows both PositionPoint and mainAddress to be exchanged. CGMES requires that PositionPoint is exchanged and mainAddress is not exchanged.	Alignment with IEC 61968-13.	Low
7	Association PositionPoint-Location	Cardinality at the side of PositionPoint: 1..*.	0..*	Alignment with IEC 61968-13.	Low
8	Location.mainAddress		Added to the profile as optional attribute. The compounds StreetAddress, Status, StreetDetail and TownDetail were also added.	Alignment with IEC 61968-13.	Low
9	Profile note		The following profile note is deleted: Please pay attention: the multiplicity of this association is different with the distribution profile. The change was made to fit transmission needs.	Alignment with IEC 61968-13.	No

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No	Item	Changed From	Changed To	Reason	Impact classification
10	BoundaryPoint		Association with Location was added.	Issue resolution. Enable exchanging of GIS for BoundaryPoint. Artificial usage of Junction is not applied anymore.	Low
11	Asociation BoundaryPoint-Location		Deleted BoundaryPoint and the association to Location	Implementation of country comments related to this document.	Low
Count:					**9 Low, 2 No**

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Annex B
(normative)

Clarification on ReactiveCapabilityCurve styles

B.1 General

The annex summarizes the use of ReactiveCapabilityCurve styles.

B.2 Reactive capability curve styles

All available styles of ReactiveCapabilityCurve are defined by the attribute Curve.curveStyle. The type of the attribute is an enumeration CurveStyle which has two values: constantYValue and straightLineYValues as shown in Figure B.1.

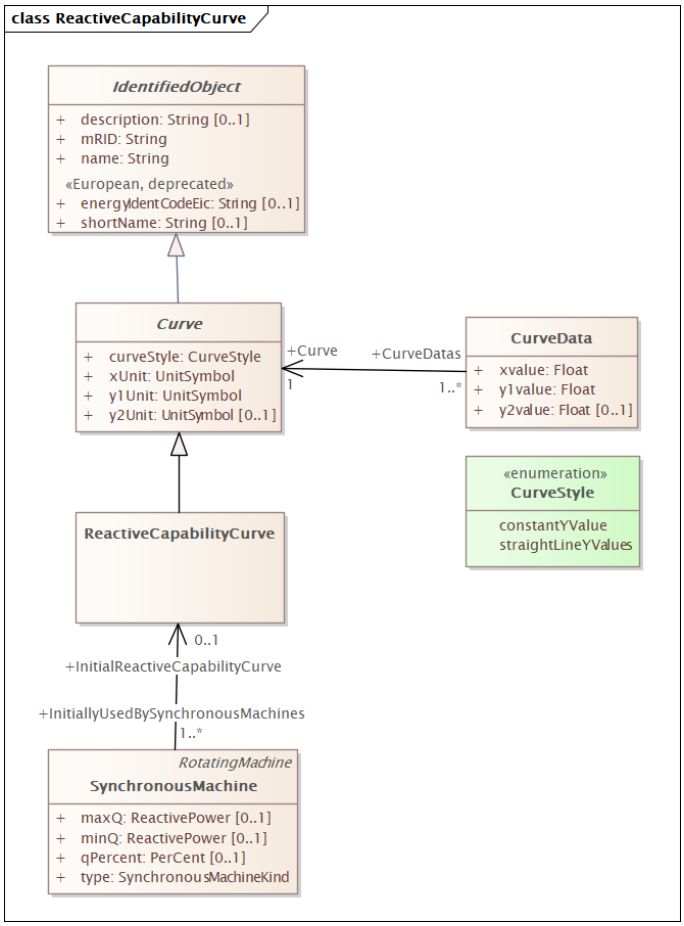

Figure B.1 – ReactiveCapabilityCurve and associated classes

The two styles of ReactiveCapabilityCurve modelling available are presented in Figure B.2 and Figure B.3 as:

1) CurveStyle.straightLineYValues style of ReactiveCapabilityCurve,
2) CurveStyle.constantYValue style of ReactiveCapabilityCurve

In case of CurveStyle.straightLineYValues the Y-axis values (q_1,… q_n) are assumed to be a straight line between them. It is also known as linear interpolation.

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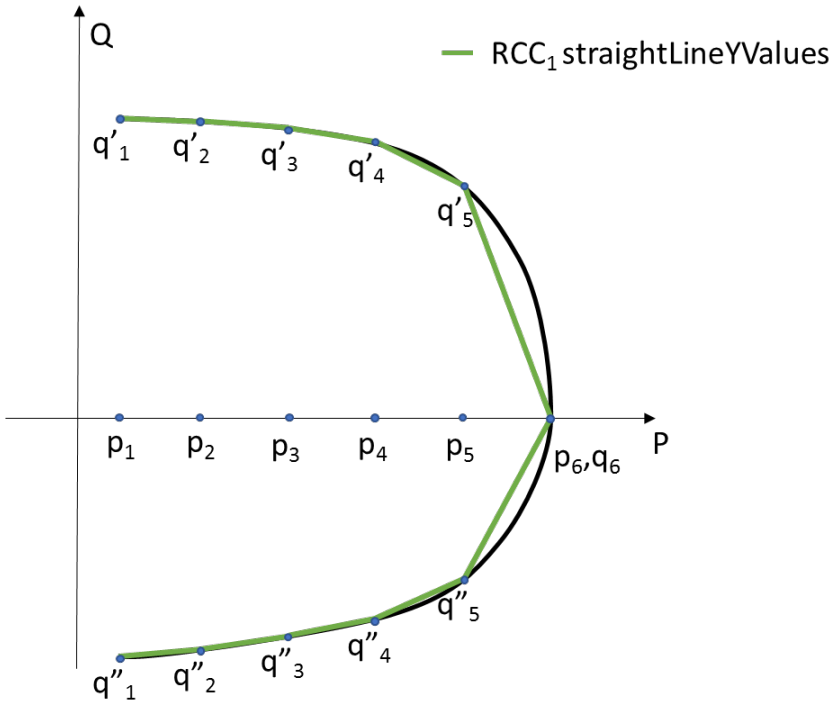

Figure B.2 – straightLineYValues style of of ReactiveCapabilityCurve (RCC)

NOTE $S2 \geq P2 + Q2$, where S is RotatingMachine.ratedS while P and Q are RotatingMachine.p and RotatingMachine.q respectively.

In case of CurveStyle.constantYValues use the Y-axis values (q_1,... q_n) are assumed constant until the next curve point and prior to the first curve point.

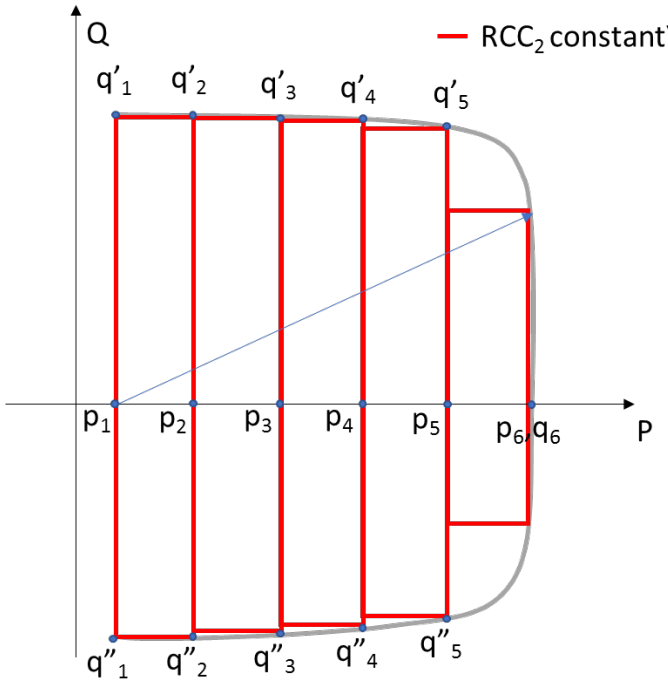

Figure B.3 – constantYValues style of of ReactiveCapabilityCurve

NOTE $S2 \geq P2 + Q2$, where S is RotatingMachine.ratedS while P and Q are RotatingMachine.p and RotatingMachine.q respectively.

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General note: While CurveStyle.straightLineYValues shall be used for RCC showing circular shapes or significant local changes in reactive power values, the CurveStyle.constantYValue allows relatively precise description of RCC following rectangular like shapes.

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Bibliography

[1] IEC 61970-501:2006, *Energy management system application program interface (EMS-API) – Part 501: Common Information Model Resource Description Framework (CIM RDF) schema*

[2] IEC 61970-552:2016, *Energy management system application program interface (EMS-API) – Part 552: CIMXML Model exchange format*

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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
info@iec.ch
www.iec.ch